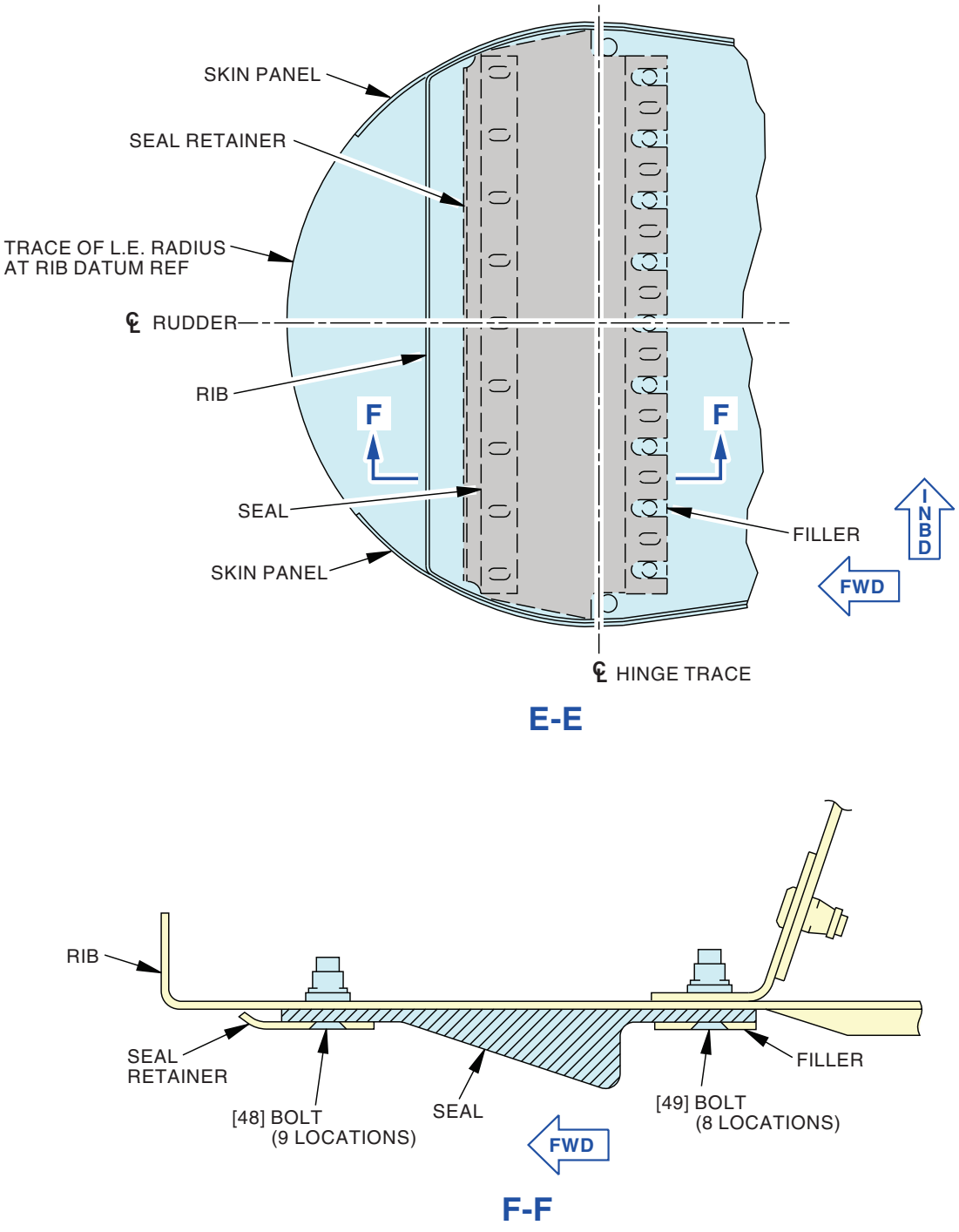


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Rudder Installation
Figure 402/27-21-11-990-802 (Sheet 9 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-11-400-801

3. Rudder Installation

(Figure 401, Figure 402)

A. References

Reference	Title
27-21-00-700-802	Rudder Hinge Bearing and PCU Bearing Freeplay (P/B 501)
27-21-00-700-803	Rudder Centering Test (P/B 501)
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-700-809	Rudder Hinge Friction Test (P/B 501)
27-21-24-400-801	Standby Rudder Power Control Unit Installation (P/B 401)
27-21-91-400-801	Main Rudder Power Control Unit Installation (P/B 401)
31-31-41-700-801	Rudder Position Transmitter Installation Test (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
SPL-2033	Sling Equipment - Rudder Removal/Installation Part #: C27035-33 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D00633	Grease - Aircraft General Purpose	BMS3-33
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Pin	55-40-00-01A-005	PTF ALL
22	Pin	55-40-00-01A-005	PTF ALL
35	Pin	55-40-00-01A-005	PTF ALL
43	Pin	55-40-00-01A-005	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge
325	Vertical Fin - Rudder

F. Access Panels

Number	Name/Location
324AAL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324ABL	Panel Assy - Trailing Edge, Beam Seal, Vertical Fin
324ACL	Rudder Nose Cover/Rudder Hinge No. 9 At Rudder Station 276.24
324ADL	Rudder Nose Cover/Rudder Hinge No. 8 At Rudder Station 257.93
324AEL	Rudder Nose Cover/Rudder Hinge No. 7 At Rudder Station 239.61
324AFL	Rudder Nose Cover/Rudder Hinge No. 6 At Rudder Station 184.67
324AHL	Rudder Nose Fairing/Rudder Hinge No. 5 At Rudder Station 129.74
324AJL	Rudder Nose Fairing/Rudder Hinge No. 4 At Rudder Station 74.80
324ALL	Rudder Nose Cover/Rudder Hinge No. 1 At Rudder Station 6.96
324QL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324SL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324UL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324XL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324ZL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin

G. Rudder Installation

CAUTION

BE CAREFUL WHEN YOU MOVE THE RUDDER. IF THE CONTROL ROD HITS THE VERTICAL STABILIZER, DAMAGE CAN OCCUR.

SUBTASK 27-21-11-820-004

WARNING

MAKE SURE THAT YOU HAVE THE RUDDER BALANCED. IF NOT BALANCED, THE RUDDER CAN CAUSE UNWANTED FLUTTER AND INSTABILITY OF THE AIRPLANE. THIS CAN CAUSE AN IN-FLIGHT EMERGENCY. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Balance and stamp the rudder assembly with the new static balance hinge moment if you install a reworked, repaired, repainted, or new rudder assembly (SRM PROCEDURE 51-60-07-0G-0).

SUBTASK 27-21-11-860-003

(2) Loosen all the seals at the trailing edge of the fin and move them fully forward.

SUBTASK 27-21-11-420-001

(3) Do these steps to put the rudder [1] in its position and connect the hinges:

NOTE: A maximum of two washers are allowed under the bolt head or two washers under the nut. No more than three washers are allowed on any individual bolt unless relief for the bolt is necessary. A maximum of four washers are allowed on an individual bolt if one each is used for fillet distortion compensation, fillet relief, finish protection, and grip adjustment or counterbore substitution.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Apply a thin layer of grease, D00633, to the smooth surfaces of all the hinge bolts (outer diameter), bushings (inner diameter), and bearings before you install them.

(b) Put the sling equipment, SPL-2033, in its position and connect it to the rudder [1] (Figure 401).

1) Make sure that a load cell with a minimum capacity of 500 lb (227 kg) is attached to the master link.

WARNING DO NOT LIFT THE RUDDER WITH A FORCE THAT IS MORE THAN 257-LBS. IF YOU USE A FORCE THAT IS MORE THAN 257-LBS, IT CAN CAUSE DAMAGE TO THE AIRPLANE AND INJURY TO PERSONNEL.

(c) Lift the rudder [1] to its position on the vertical fin.

1) Use the load cell to make sure that the force used to lift the rudder [1] is not more than 257 lb (117 kg).

2) Use the ropes to make sure that the rudder does not hit the airplane.

(d) Install the bolts, washers, spacers, and bushings that follow:

1) At hinges 1, 4, 5, 6, 8, and 9, install the bushings [32], washers [31], and bolts [30] to the hinge fittings.

2) At hinge 2, install the spacer [5] (under the fitting), bushing [6], washer [7], and bolt [8] to the hinge fitting.

3) At hinge 3, install the bushing [18], washer [17], and bolt [16] to the hinge fitting.

4) At hinge 7, install bushing [38], washer [37], and bolt [36] to the hinge fitting.

(e) Release the rudder [1] slowly until the load is on its hinges.

(f) Make sure that the torque required to rotate the bolts is less than 100 in-lb (11.3 N·m).

(g) Install the nuts, washers, and pins in the order that follow:

1) At hinge 2, install the washer [4] (above the fitting) and nut [3].

a) Tighten the nut [3] to the lower end of the torque range 150 in-lb (16.9 N·m) - 200 in-lb (22.6 N·m).

<1> Continue to tighten the nut until the next castellation aligns with the hole in the bolt [8].

b) Install the new pin [2].

2) At hinges 1, 4, 5, 6, 8, and 9, install the washers [33] and nuts [34].

a) Tighten the nuts [34] to the lower end of the torque range 150 in-lb (16.9 N·m) - 200 in-lb (22.6 N·m).

<1> Continue to tighten the nut until the next castellation aligns with the hole in the bolt [30].

b) Install the new pin [35].

3) At hinge 3, do these steps:

a) Install the adjustment bushing [19] at hinge 3.

b) Adjust the adjustment bushing [19] until it is against the fin bearing with finger tight pressure only.

c) Install the washer [20] and nut [21].

d) Tighten the nut [21] to the lower end of the torque range 150 in-lb (16.9 N·m) - 200 in-lb (22.6 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<1> Continue to tighten the nut until the next castellation aligns with the hole in the bolt [16].

e) Install the new pin [22].

f) Install the MS20995C32 lockwire, G01048, from the adjustment bushing [19] to hinge fitting.

g) Fill the holes of the MS20995C32 lockwire, G01048, with Cor-Ban 27L Compound, G50237.

4) At hinge 7, install the washer [39], washer [41], and nut [42].

a) Tighten the nut [42] to the lower end of the torque range 150 in-lb (16.9 N·m) - 200 in-lb (22.6 N·m).

<1> Continue to tighten the nut until the next castellation aligns with the hole in the bolt [36].

b) Install the new pin [43].

SUBTASK 27-21-11-620-001

(4) Apply sealant, A00247, to both ends of the bonding jumper [47] and their fasteners.

SUBTASK 27-21-11-420-002

(5) Connect the control rod for the rudder position transmitter to the vertical fin:

(a) Install the washer [25], washer [24], and bolt [23].

SUBTASK 27-21-11-420-003

(6) Install the bolt [44], washers [45], and nut [46] to connect the bonding jumper [47] to the vertical fin at hinge 9.

(a) Measure the electrical bonding resistance between the vertical fin and rudder [1].

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.01 ohm (10 milliohm) or less.

SUBTASK 27-21-11-080-002

(7) Remove the sling equipment, SPL-2033.

(a) Remove the load cell from the master link.

SUBTASK 27-21-11-220-001

(8) Make sure that there is clearance between all the seals and the forward part of the rudder and the balance weight.

(a) Manually move the rudder approximately seven degrees in each direction to examine the clearance.

SUBTASK 27-21-11-710-001

(9) Move the rudder to the left and right from the neutral position.

(a) Make sure that rudder moves left and right a minimum of 30 degrees from the neutral position.

SUBTASK 27-21-11-420-004

(10) Do this step to connect the rudder standby power control unit (PCU) to the rudder [1]:

(a) Install bolt [26], washer [27], bushing [50], washer [28], and nut [29].

(b) Tighten the nut [29] to 350 in-lb (39.5 N·m) - 400 in-lb (45.2 N·m) .

(c) Install the rudder standby PCU control rod [52] (TASK 27-21-24-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-11-420-005

(11) Do these steps to connect the rudder main PCU to the rudder [1]:

(a) Install the bushing [11], bolt [10], bolt [9], washer [12], and nut [13].

1) Tighten the nut [13] to 900 in-lb (102 N·m) - 1050 in-lb (119 N·m).

CAUTION

DO NOT TIGHTEN THE NUT [13] AFTER YOU TIGHTEN THE NUT[15]. YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(b) Install washer [14] and nut [15].

1) Tighten the nut [15] to 450 in-lb (51 N·m) - 550 in-lb (62 N·m).

(c) Install the control rods [51] that you removed between the torque tube and the main PCU (TASK 27-21-91-400-801).

SUBTASK 27-21-11-220-002

(12) Make sure that the mismatch between the rudder [1] and the vertical fin is correct at each of rib locations (View B-B, Figure 402).

SUBTASK 27-21-11-820-001

(13) Adjust the seals at the trailing edge of the vertical fin and tighten them (Views J and C-C, Figure 402).

SUBTASK 27-21-11-820-002

(14) Adjust the balance weight seals on the vertical fin and tighten the seal retainers (Views I and A-A, Figure 402).

SUBTASK 27-21-11-820-003

(15) Adjust the seal compression for the lower (rudder to body) seal (Views K, D-D, and E-E, Figure 402):

(a) Move the rudder [1] to its neutral position.

(b) Make sure that the seals contact and that the seal compression is 0.00 in. (0.00 mm) - -0.04 in. (-1.02 mm).

(c) If it is necessary, do these steps to adjust the seal:

1) Loosen the bolts [48] and bolts [49] from the seal.

2) Adjust the seal until they both contact and the seal compression is 0.00 in. (0.00 mm) - -0.04 in. (-1.02 mm).

3) Tighten the bolts [48] and bolts [49].

(d) Make sure that the seal does not cause a blockage at the left and right rudder drain holes.

SUBTASK 27-21-11-860-004

(16) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	16	C00788	FLIGHT CONTROL TRIM IND RUD
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

SUBTASK 27-21-11-220-003

(17) Do this task: Rudder Hinge Bearing and PCU Bearing Freeplay, TASK 27-21-00-700-802.

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SUBTASK 27-21-11-220-004

(18) Do this task: Rudder Travel Test, TASK 27-21-00-700-805.

SUBTASK 27-21-11-220-005

(19) Do this task: Rudder Centering Test, TASK 27-21-00-700-803.

SUBTASK 27-21-11-220-006

(20) Do this task: Rudder Hinge Friction Test, TASK 27-21-00-700-809.

SUBTASK 27-21-11-220-007

(21) Do this task: Rudder Position Transmitter Installation Test, TASK 31-31-41-700-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-11-410-001

(1) Install these covers:

Number	Name/Location
324AAL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324ABL	Panel Assy - Trailing Edge, Beam Seal, Vertical Fin
324ACL	Rudder Nose Cover/Rudder Hinge No. 9 At Rudder Station 276.24
324ADL	Rudder Nose Cover/Rudder Hinge No. 8 At Rudder Station 257.93
324AEL	Rudder Nose Cover/Rudder Hinge No. 7 At Rudder Station 239.61
324AFL	Rudder Nose Cover/Rudder Hinge No. 6 At Rudder Station 184.67
324AHL	Rudder Nose Fairing/Rudder Hinge No. 5 At Rudder Station 129.74
324AJL	Rudder Nose Fairing/Rudder Hinge No. 4 At Rudder Station 74.80
324ALL	Rudder Nose Cover/Rudder Hinge No. 1 At Rudder Station 6.96
324QL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324SL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324UL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324XL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324ZL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER INDEX PLATE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the rudder index plate.

(2) An installation of the rudder index plate.

TASK 27-21-17-000-801

2. Rudder Index Plate Removal

(Figure 401)

A. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

B. Rudder Index Plate Removal

SUBTASK 27-21-17-930-001

(1) Put a tape on the airplane adjacent to the rudder index plate [1].

(a) Accurately make a mark of the position of the rudder index plate [1] on the tape.

SUBTASK 27-21-17-020-001

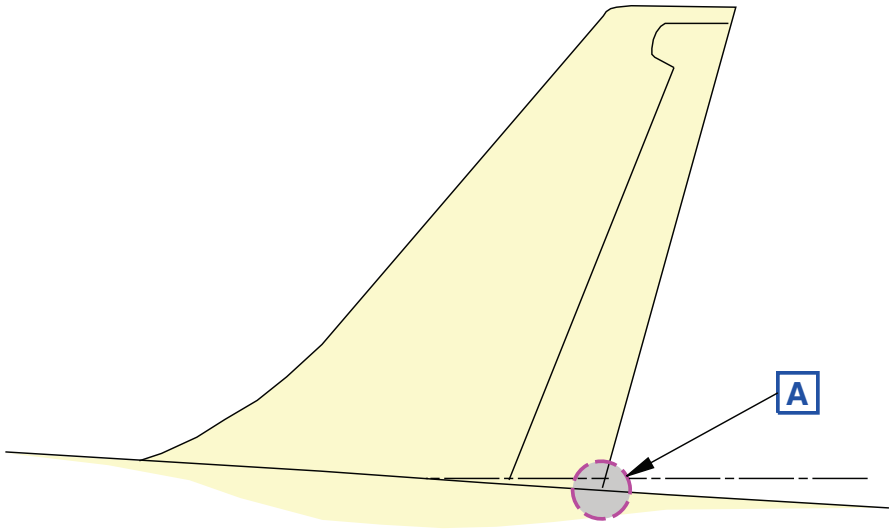
(2) Remove the screws [2] and washers [3] to remove the rudder index plate [1].

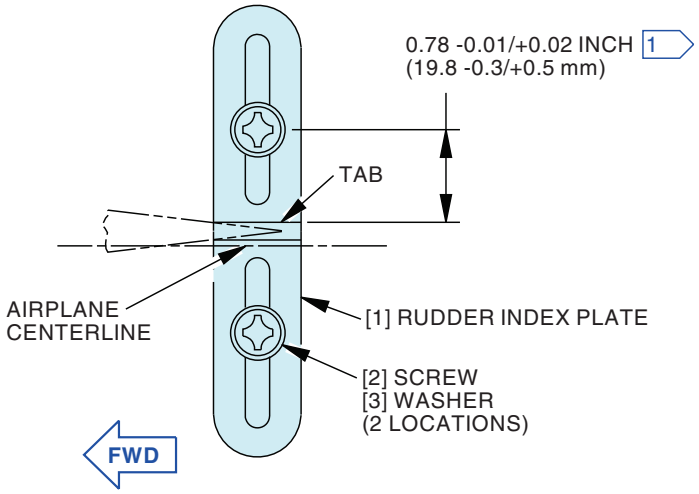
SUBTASK 27-21-17-020-002

(3) Remove the rudder index plate [1] from the airplane.

——— END OF TASK ———

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RUDDER INDEX PLATE

A

NOTE:

1 THIS DIMENSION MAY BE DIFFERENT IF THE INDEX PLATE HAS BEEN ADJUSTED FOR LATERAL TRIM.

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Rudder Index Plate Installation
Figure 401/27-21-17-990-801

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TASK 27-21-17-400-801

3. Rudder Index Plate Installation

(Figure 401)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Rudder index plate	27-21-17-01-025	PTF ALL

B. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

C. Rudder Index Plate Installation

SUBTASK 27-21-17-420-001

(1) If there is tape with a mark that shows the correct position, then do these steps to install the rudder index plate [1]:

(a) Put the rudder index plate [1] in its position and align it with the mark.

(b) Install the screws [2] and washers [3] to install the rudder index plate [1].

(c) Remove the tape (that has the mark on it) from the airplane.

SUBTASK 27-21-17-420-002

(2) If there is not a tape with a mark that shows the correct position, then do these steps to install the rudder index plate [1]:

(a) Put the rudder index plate [1] in its position (Figure 401).

(b) Install the screws [2] and washers [3] to install the rudder index plate [1].

(c) Make sure the rudder index plate [1] is set at 0.78 +0.02 / -0.01 in. (19.8 +0.5 / -0.3 mm) (Figure 401).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STANDBY RUDDER POWER CONTROL UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the standby rudder power control unit (PCU)

(2) An installation of the standby rudder PCU.

TASK 27-21-24-000-801

2. Standby Rudder Power Control Unit Removal

(Figure 401)

A. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-203	Container - Oil Resistant, 1 U.S Gal (3.8 L)

C. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

D. Access Panels

Number	Name/Location
324DL	Trailing Edge Access

E. Prepare for the Removal

SUBTASK 27-21-24-860-001

(1) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-24-860-002

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	7	C00285	YAW DAMPER AC
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

SUBTASK 27-21-24-010-001

(3) Remove this access panel:

Number	Name/Location
324DL	Trailing Edge Access

F. Standby Rudder PCU Removal

SUBTASK 27-21-24-020-001

(1) Disconnect the hydraulic lines [12] from the power control unit [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

MAKE SURE THAT YOU CAREFULLY DRAIN HYDRAULIC FLUID INTO A SUITABLE CONTAINER. CAREFULLY CLEAN, DRY, AND DISPOSE OF ALL LEAKAGE. HYDRAULIC FLUID CAUSES DAMAGE TO RUBBER, COPPER, PLASTICS, PAINTS, AND MANY OTHER MATERIALS.

(a) Put a oil resistant container, STD-203, below the hydraulic connections.

(b) Disconnect the hydraulic lines [12] from the power control unit [4].

(c) Install the plugs and caps to the hydraulic lines [12] and ports.

(d) Install a tag on the hydraulic lines [12] to make sure that you install them correctly.

1) Discard the packing [3] and packing [11].

SUBTASK 27-21-24-020-002

(2) Disconnect the electrical connector [1] from the power control unit [4].

SUBTASK 27-21-24-020-003

CAUTION

MAKE SURE THAT YOU REMOVE THE VERNIER CONTROL ROD FROM THE PCU BEFORE YOU DISCONNECT THE PCU FROM THE RUDDER. IF YOU DO NOT REMOVE THE VERNIER CONTROL ROD FIRST, IT CAN CAUSE DAMAGE TO THE VERNIER CONTROL ROD.

(3) Disconnect the vernier control rod [9] from the power control unit [4].

(a) Remove the nut [8], washer [7], washer [6], and bolt [5] to disconnect the control rod [9] from the power control unit [4].

(b) Remove the nut [28], washer [27], washer [26], and bolt [25] to disconnect the control rod [9] from the power control unit [4].

(c) Remove the control rod [9] from the airplane.

SUBTASK 27-21-24-020-004

CAUTION

MAKE SURE THAT YOU MOVE THE PCU CAREFULLY DURING THE REMOVAL OR INSTALLATION OF THE PCU. THE PCU CAN HIT THE BOLTS. THIS CAN CAUSE DAMAGE TO THE BOLTS.

(4) Remove the power control unit [4].

(a) Remove the nut [24], washers [23], bushing [22], washer [21], and bolt [20] to disconnect the power control unit [4] from the rudder.

CAUTION

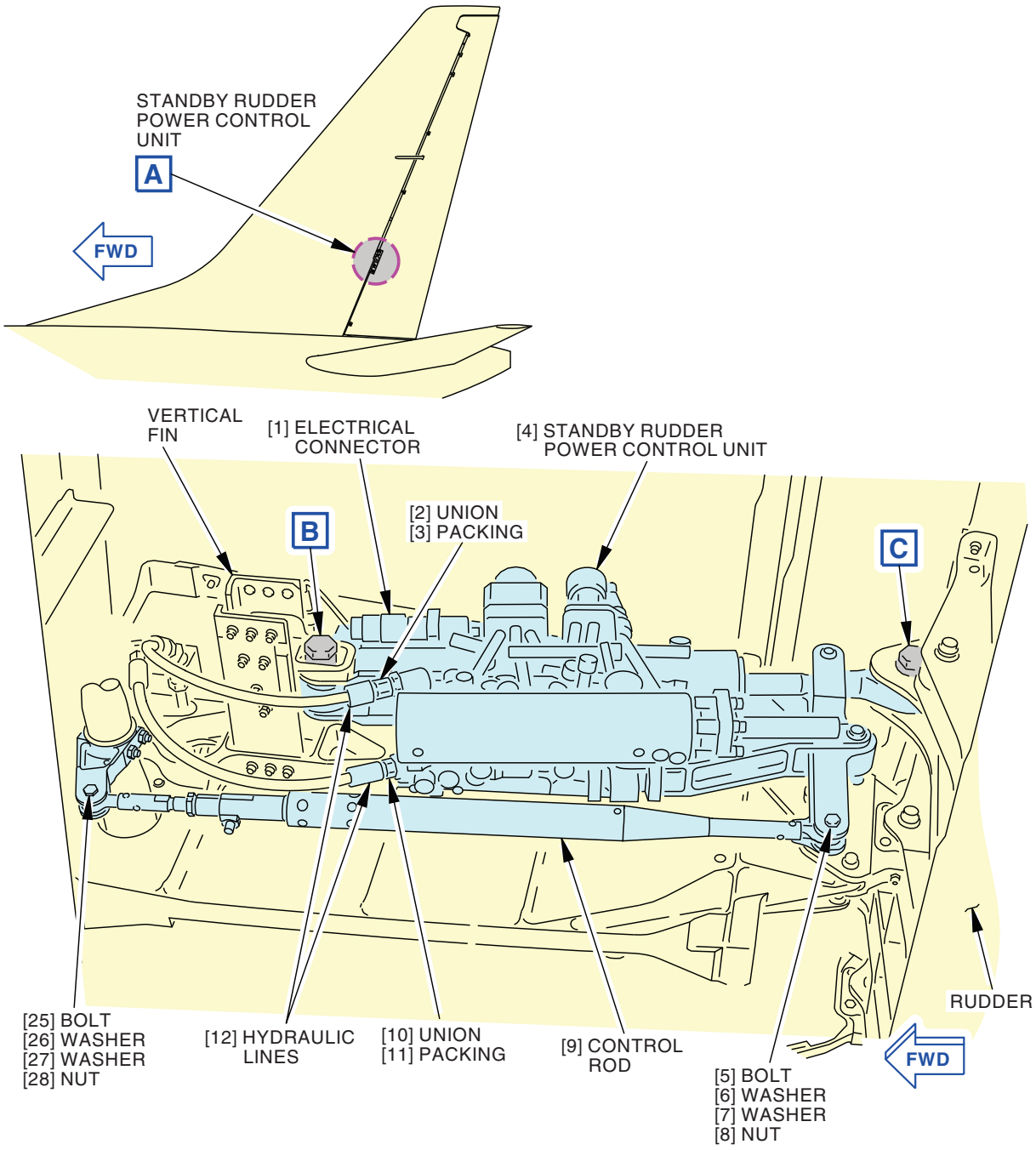
MAKE SURE THAT YOU HOLD THE STANDBY RUDDER PCU DURING THE REMOVAL OF THE STANDBY RUDDER PCU. THE STANDBY RUDDER PCU CAN FALL AND CAUSE DAMAGE TO THE STANDBY RUDDER PCU.

(b) Remove the nut [19], washer [18], washer [17], bushing [16], bushing [15], washer [14], and bolt [13] that attach the power control unit [4] to the airplane structure.

(c) Remove the power control unit [4] from the airplane.

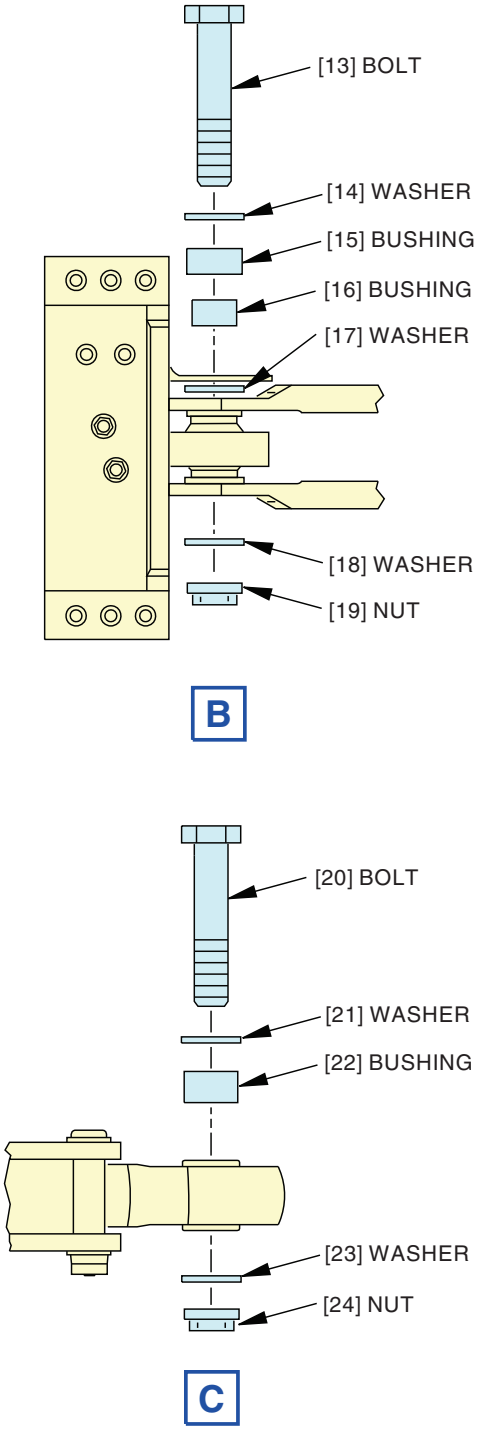
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Standby Rudder Power Control Unit Installation
Figure 401/27-21-24-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Standby Rudder Power Control Unit Installation
Figure 401/27-21-24-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-24-400-801

3. Standby Rudder Power Control Unit Installation

(Figure 401)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-700-806	Standby Rudder Power Control Unit Deadband Test (P/B 501)
27-21-00-820-805	Control Rod for Standby Rudder Power Control Unit Adjustment (P/B 501)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-21-24-210-801	Standby Rudder Power Control Unit (with Hydraulic Power on) General Visual Inspection (P/B 601)

B. Tools/Equipment

Reference	Description
STD-203	Container - Oil Resistant, 1 U.S Gal (3.8 L)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Power control unit	27-21-24-01-125	PTF ALL
9	Control rod	27-21-24-01-066	PTF ALL

E. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

F. Access Panels

Number	Name/Location
324DL	Trailing Edge Access

G. Standby Rudder PCU Installation

SUBTASK 27-21-24-220-001

(1) Examine the standby rudder PCU and all connections to the standby rudder PCU for worn areas, do this task: Standby Rudder Power Control Unit (with Hydraulic Power on) General Visual Inspection, TASK 27-21-24-210-801.

SUBTASK 27-21-24-420-001

(2) If it the unions and packings are not installed in the power control unit [4], do these steps:

(a) Lubricate the unions and packings with hydraulic fluid, D00153.

(b) Install the union [2] and new packing [3] in the power control unit [4].

(c) Install the union [10] and new packing [11] in the power control unit [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-24-420-002

CAUTION: MAKE SURE THAT YOU MOVE THE PCU CAREFULLY DURING THE REMOVAL OR INSTALLATION OF THE PCU. THE PCU CAN HIT THE BOLTS. THIS CAN CAUSE DAMAGE TO THE BOLTS.

(3) Install the power control unit [4].

NOTE: Follow the instructions to install the PCU carefully and do the procedure in the specified sequence. Damage to the control rod [9] may occur if you do not follow the instructions correctly.

(a) Put the power control unit [4] in its position.

(b) Install the bolt [13], washers [14], bushing [15], bushing [16], washer [17], washer [18], and nut [19] to connect the power control unit [4] to the airplane structure.

1) Tighten the nut [19] to 450 ±50 in-lb (51 ±6 N·m) or tighten the bolt [13] to 525 ±50 in-lb (59 ±6 N·m).

(c) Install the bolt [20], washer [21], bushing [22], washer [23], and nut [24] to connect the power control unit [4] to the rudder.

1) Tighten the nut [24] to 350 ±25 in-lb (40 ±3 N·m) or tighten the bolt [20] to 413 ±25 in-lb (47 ±3 N·m).

SUBTASK 27-21-24-420-003

(4) Connect the vernier control rod [9] to the power control unit [4].

(a) Install the bolt [25], washer [26], washer [27], and nut [28] to connect the control rod [9] to the power control unit [4].

(b) Install the bolt [5], washer [6], washer [7], and nut [8] to connect the control rod [9] to the power control unit [4].

SUBTASK 27-21-24-020-005

CAUTION: MAKE SURE THAT YOU CORRECTLY INSTALL THE HYDRAULIC LINES TO THE STANDBY RUDDER PCU. MAKE SURE THAT THE CORRECT UNIONS ARE INSTALLED IN THE HYDRAULIC PORTS. YOU CAN CAUSE DAMAGE TO THE STANDBY RUDDER PCU.

(5) Connect the hydraulic lines [12] to the power control unit [4].

CAUTION: MAKE SURE THAT YOU CAREFULLY DRAIN HYDRAULIC FLUID INTO A SUITABLE CONTAINER. CAREFULLY CLEAN, DRY, AND DISPOSE OF ALL LEAKAGE. HYDRAULIC FLUID CAUSES DAMAGE TO RUBBER, COPPER, PLASTICS, PAINTS, AND MANY OTHER MATERIALS.

(a) Remove the caps and plugs from the hydraulic lines [12] and ports.

(b) Remove the identification tags from the hydraulic lines [12].

(c) Remove the oil resistant container, STD-203, used to catch fluids from the hydraulic lines [12].

(d) Connect the hydraulic lines [12] to the power control unit [4].

(e) Tighten the B-nuts on the hydraulic lines [12] to 140 ±7 in-lb (16 ±1 N·m).

SUBTASK 27-21-24-020-006

(6) Connect the electrical connector [1] to the power control unit [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-24-860-003

(7) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	7	C00285	YAW DAMPER AC
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

SUBTASK 27-21-24-820-001

(8) Do this task: Control Rod for Standby Rudder Power Control Unit Adjustment, TASK 27-21-00-820-805.

SUBTASK 27-21-24-860-004

(9) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

SUBTASK 27-21-24-710-001

(10) Do this task: Standby Rudder Power Control Unit Deadband Test, TASK 27-21-00-700-806.

SUBTASK 27-21-24-710-002

(11) Do this task: Rudder Travel Test, TASK 27-21-00-700-805.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-24-410-001

(1) Install this access panel:

Number	Name/Location
324DL	Trailing Edge Access

SUBTASK 27-21-24-410-002

(2) If it is necessary, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STANDBY RUDDER POWER CONTROL UNIT - INSPECTION/CHECK

1. General

A. This procedure contains these tasks:

(1) Standby Rudder Power Control Unit General Visual Inspection

(2) Standby Rudder Power Control Unit Detail Visual Inspection.

TASK 27-21-24-210-801

2. Standby Rudder Power Control Unit (with Hydraulic Power on) General Visual Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)

B. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

C. Access Panels

Number	Name/Location
324FL	Vertical Fin, Access

D. Standby Rudder Power Control Unit (with Hydraulic Power on) General Visual Inspection

SUBTASK 27-21-24-010-002

(1) Open this access panel:

Number	Name/Location
324FL	Vertical Fin, Access

SUBTASK 27-21-24-860-005

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize the standby hydraulic system, do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

SUBTASK 27-21-24-210-001

(3) Do a general visual inspection of the standby rudder power control unit.

SUBTASK 27-21-24-210-002

(4) Do a general visual inspection to make sure there are no hydraulic leaks from the standby rudder power control unit.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-24-860-006

(5) Remove pressure from the standby hydraulic system, do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-24-410-003

(6) Close this access panel:

Number	Name/Location
324FL	Vertical Fin, Access

END OF TASK

TASK 27-21-24-210-802

3. **Standby Rudder Power Control Unit Detail Visual Inspection**

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

B. Access Panels

Number	Name/Location
324DL	Trailing Edge Access

C. Standby Rudder Power Control Unit Detail Visual Inspection

SUBTASK 27-21-24-010-003

(1) Open this access panel:

Number	Name/Location
324DL	Trailing Edge Access

SUBTASK 27-21-24-210-003

(2) Do a detailed visual inspection of the standby rudder power control unit attachment points.

(a) Examine the attachment points for security and damage.

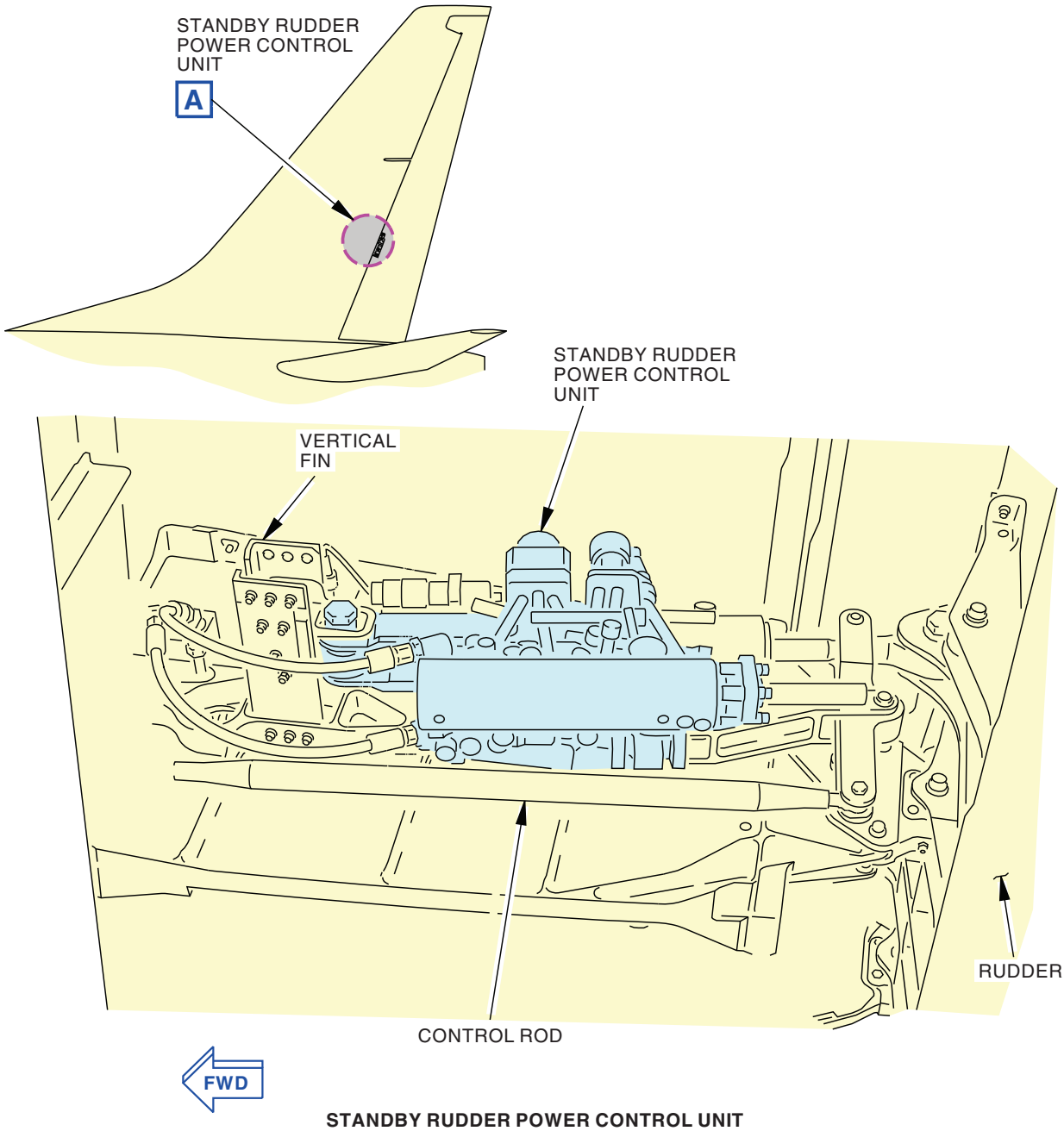
SUBTASK 27-21-24-410-004

(3) Close this access panel:

Number	Name/Location
324DL	Trailing Edge Access

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Standby Rudder Power Control Unit Visual Inspection
Figure 601/27-21-24-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER PEDALS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the rudder pedals

(2) An installation of the rudder pedals.

B. This procedure is applicable for the captain's rudder pedals and first officer's rudder pedals.

TASK 27-21-31-000-801

2. Rudder Pedals Removal

(Figure 401, Figure 402)

A. References

Reference	Title
25-11-01-000-801	Captain and First Officer Seat Removal (P/B 401)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-31-00-000-801	Elevator and Tab Control System - Deactivation (P/B 201)
33-11-00-990-801	Figure: Lighting Modules (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

E. Prepare for the Removal

SUBTASK 27-21-31-040-001

(1) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-31-040-002

(2) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-31-040-003

(3) Do this task: Elevator and Tab Control System - Deactivation, TASK 27-31-00-000-801

SUBTASK 27-21-31-020-011

(4) If necessary, remove the Captain and First Officer's seats. Do this task: Captain and First Officer Seat Removal, TASK 25-11-01-000-801.

SUBTASK 27-21-31-010-002

(5) Open this access door:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-21-31-010-001

(6) Remove these access panels to get access to the captains's and first officer's rudder forward quadrant [31]. (TASK 53-14-01-020-801)

(a) Remove these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

SUBTASK 27-21-31-480-001

(7) Install the rig pin R-3 from the rigging pin kit, SPL-1585 in the forward rig pin hole of the captain's rudder forward quadrant [31] (Figure 401).

SUBTASK 27-21-31-480-002

(8) Install the rig pin R-4 from the rigging pin kit, SPL-1585 in the forward rig pin hole of the first officer's rudder forward quadrant [31] (Figure 401).

F. Rudder Pedals Removal

SUBTASK 27-21-31-020-010

(1) Remove top cover and pedal shrouds. Do the following steps:

(a) Remove the lighting module [33]:

1) Loosen the fasteners [36] on the front of the lighting module [33]. (Figure 33-11-00-990-801)

2) Pull the lighting module [33] aft and clear of the top cover [23].

NOTE: Do not disconnect the electrical connectors from the lighting module [33].

(b) Remove the screws [24], screws [29] and washers [25] that attach the top cover [23] to the lower cover [32].

(c) Remove the screws [27] that attach the outboard pedal shroud [21] and inboard pedal shroud [22] to the stanchion walls.

(d) Remove the screws [28] that attach the angle brackets [30] to the top cover [23].

NOTE: Ensure the angle brackets remain attached to the pedal shrouds.

(e) Remove the outboard pedal shroud [21] and the inboard pedal shroud [22] to gain access to the bolt [34] that attaches the pedal adjustment cable [35] to the top cover [23].

NOTE: Ensure that no ducting is attached to the pedal shrouds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Remove the bolt [34] that attaches the pedal adjustment cable [35] to the cable standoff on the top cover [23].

(f) Remove the top cover [23].

1) If it is necessary, pull the control column aft, move the pedals, and turn the top cover [23] as it clears adjacent structure.

SUBTASK 27-21-31-020-005

(2) Remove the nuts [10], washers [9], bushings [8], and bolts [7] to disconnect the two brake pushrods [11] from the bellcranks [6].

SUBTASK 27-21-31-020-006

(3) Remove the screws [19] and brackets [20] from the rudder pedals [12].

SUBTASK 27-21-31-020-007

(4) Remove the pin [3A], nuts [3], washers [2], and bolts [1] to disconnect the two rudder pedal tension rods [4]from the rudder pedals [12].

SUBTASK 27-21-31-020-009

(5) Remove the rudder pedals [12] from the support structure:

(a) Remove the nut [13] and washer [17].

(b) Hold the rudder pedals [12] and pull the support shaft [16] out.

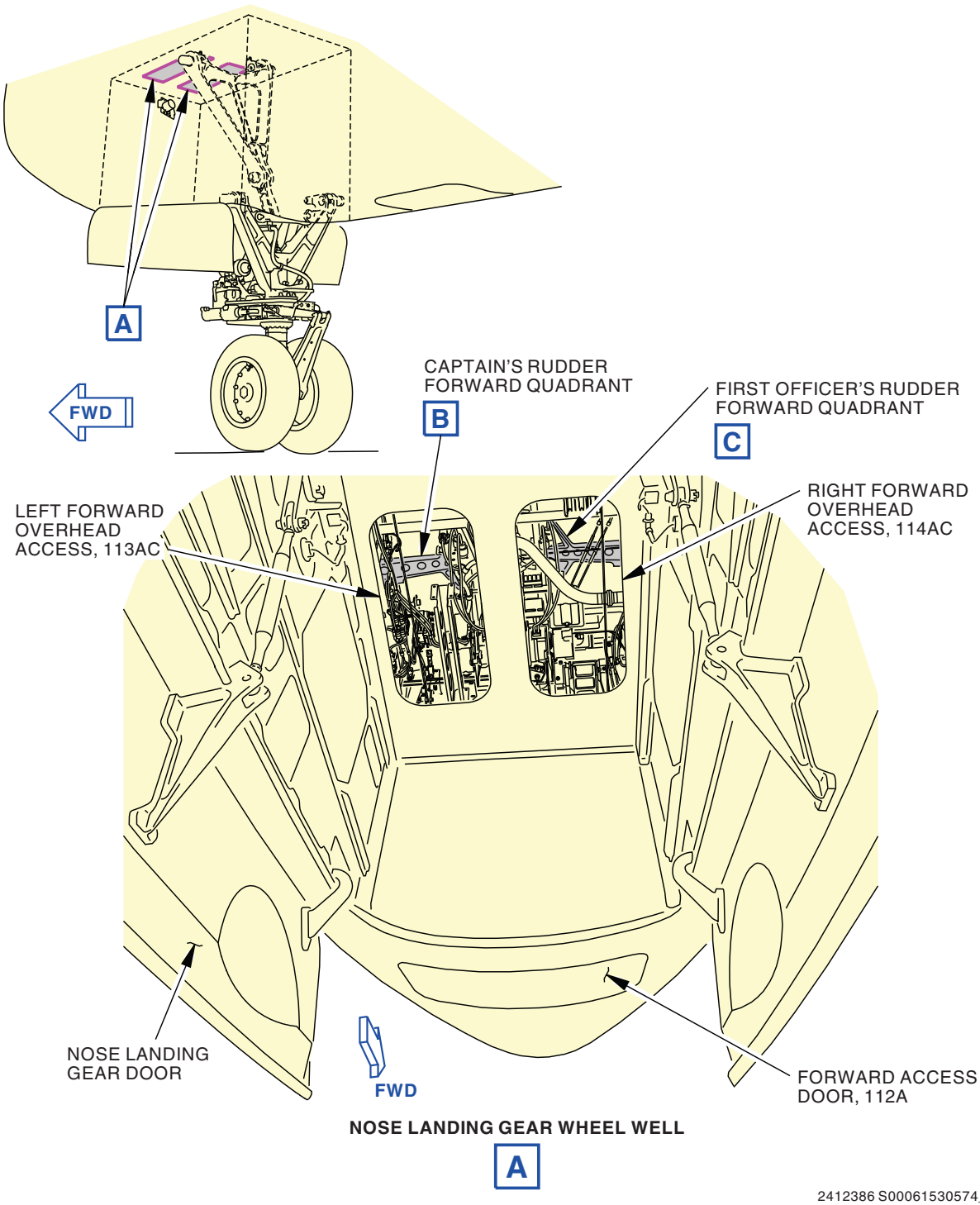
NOTE: The bushings, bearings, and washers will be loosed as the support shaft [16] is removed.

(c) Pull each rudder pedal [12] through the floor beams, into the flight compartment.

(d) Remove the rudder pedals [12] from the airplane.

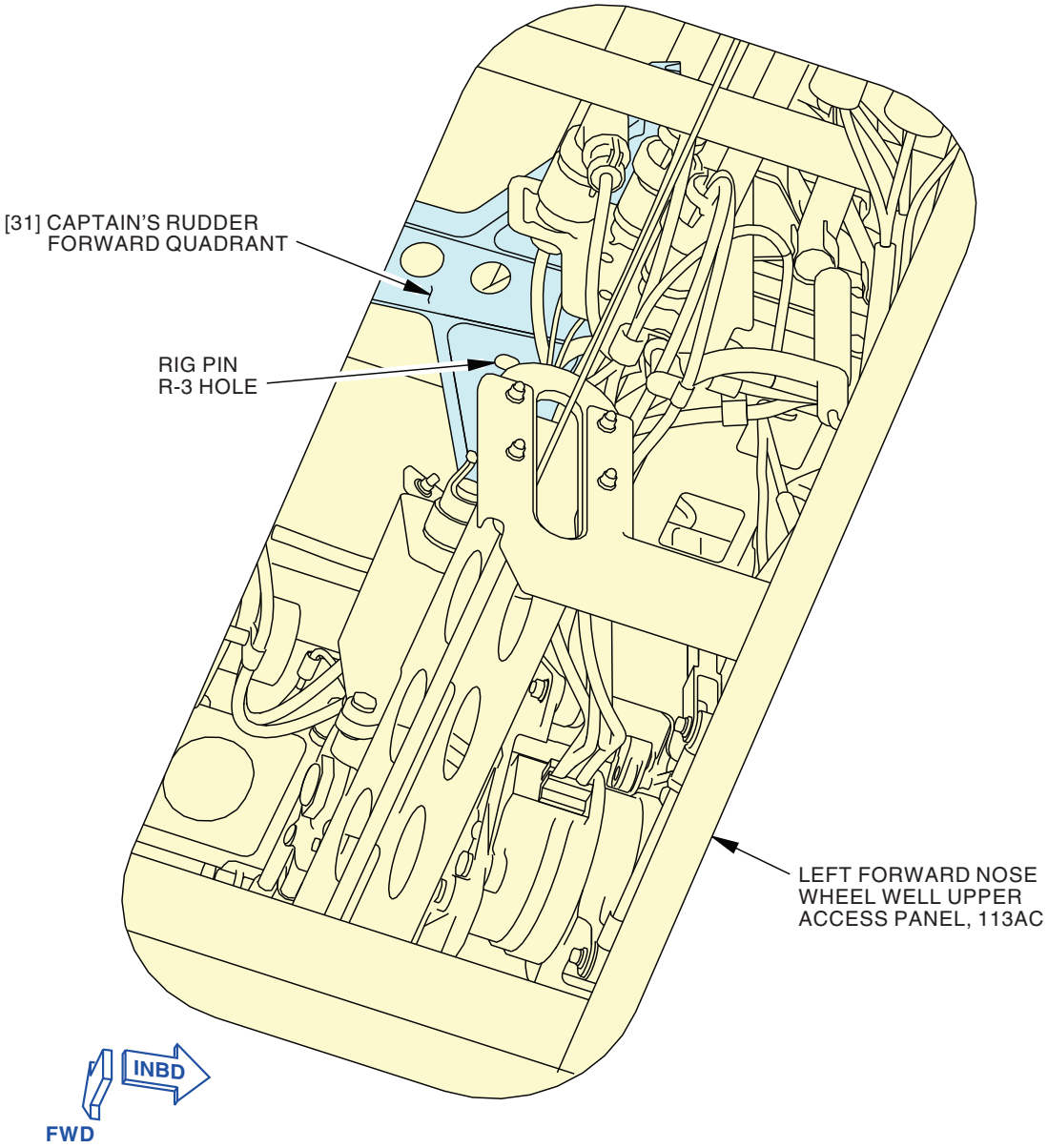
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Forward Quadrant Rig Pin Installation
Figure 401/27-21-31-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



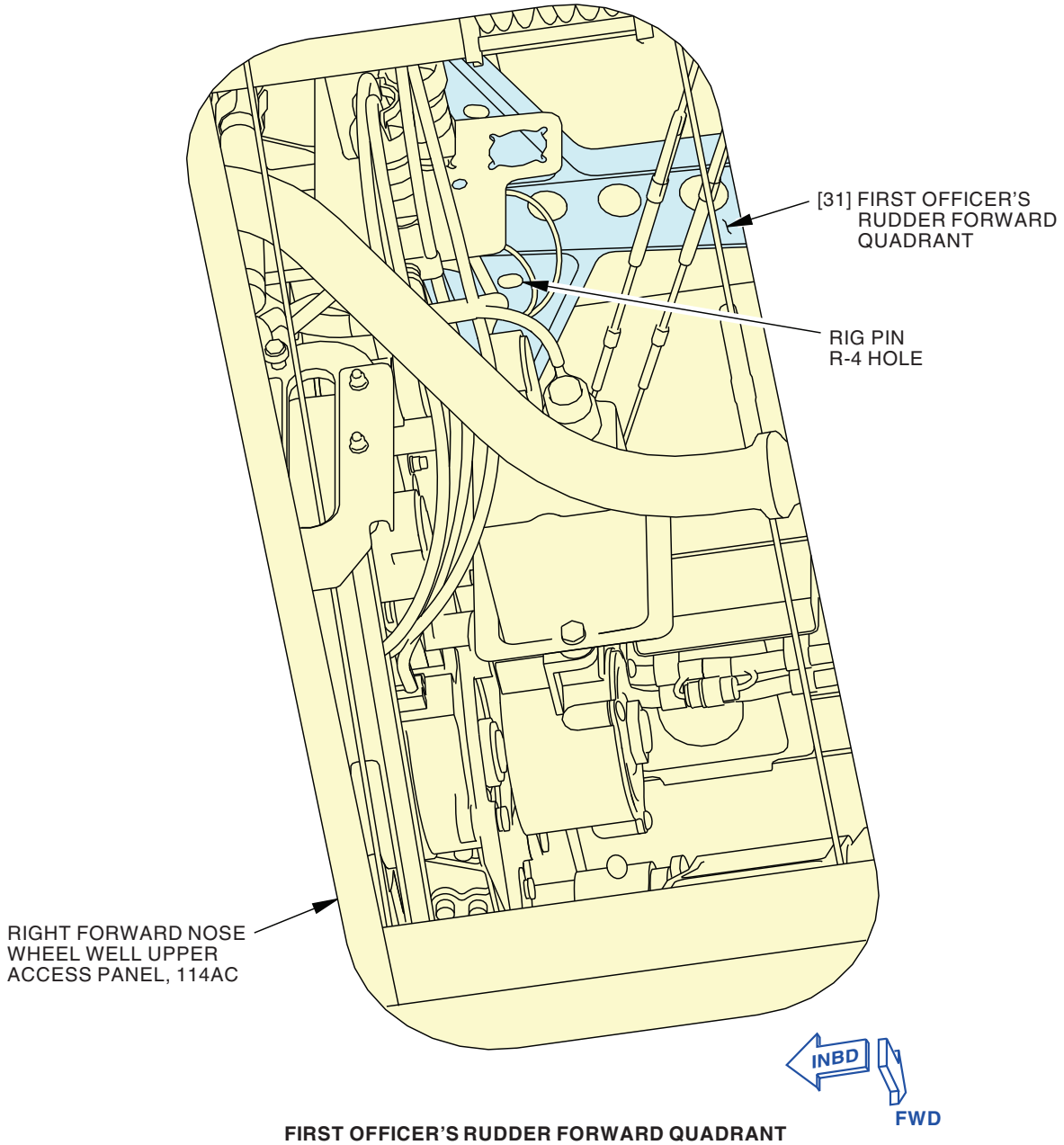
CAPTAIN'S RUDDER FORWARD QUADRANT

B

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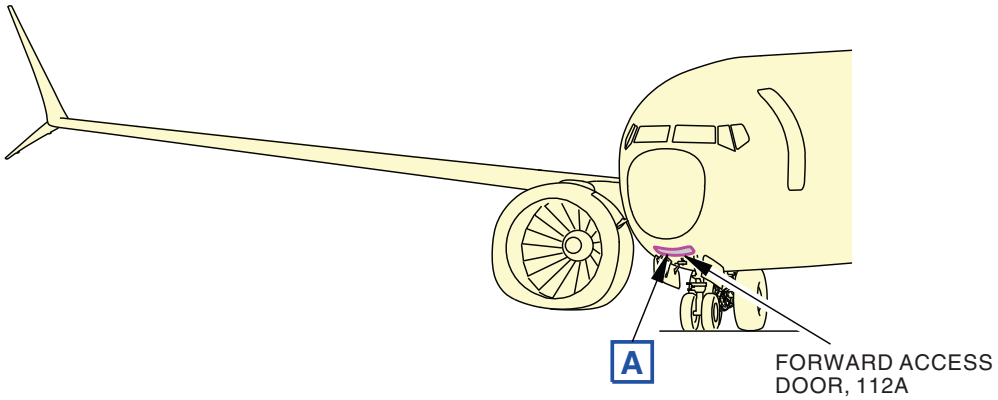
Rudder Forward Quadrant Rig Pin Installation
Figure 401/27-21-31-990-801 (Sheet 2 of 3)

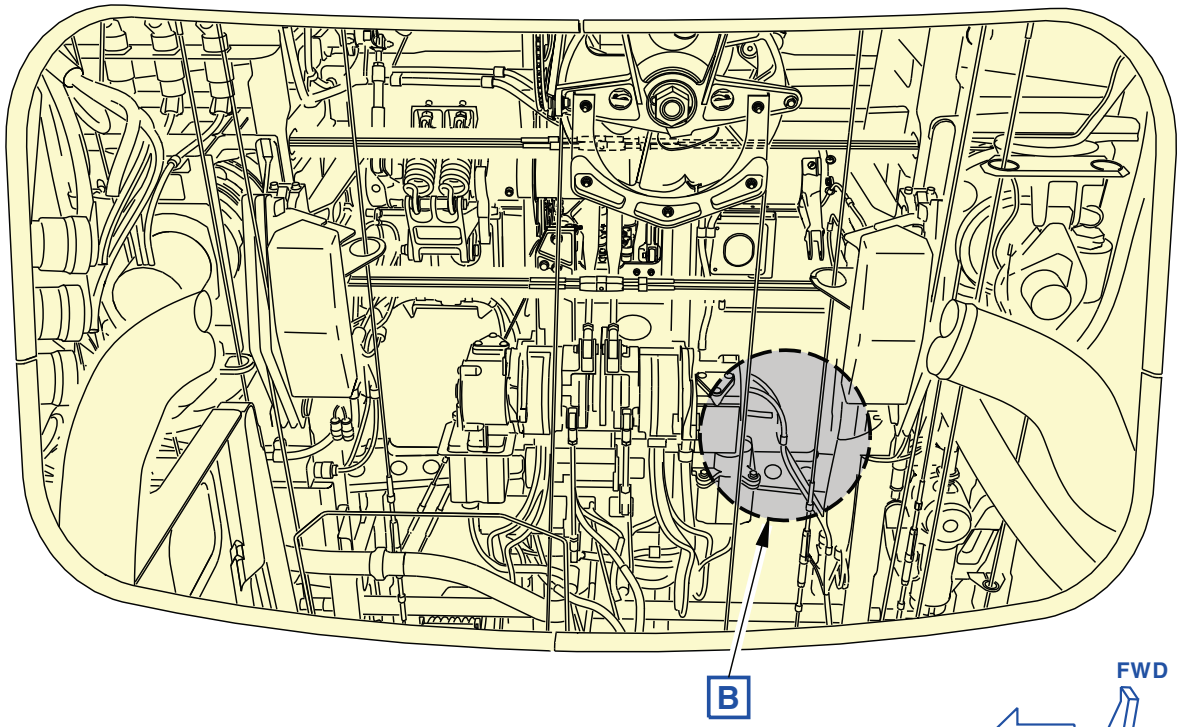
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Forward Quadrant Rig Pin Installation
Figure 401/27-21-31-990-801 (Sheet 3 of 3)

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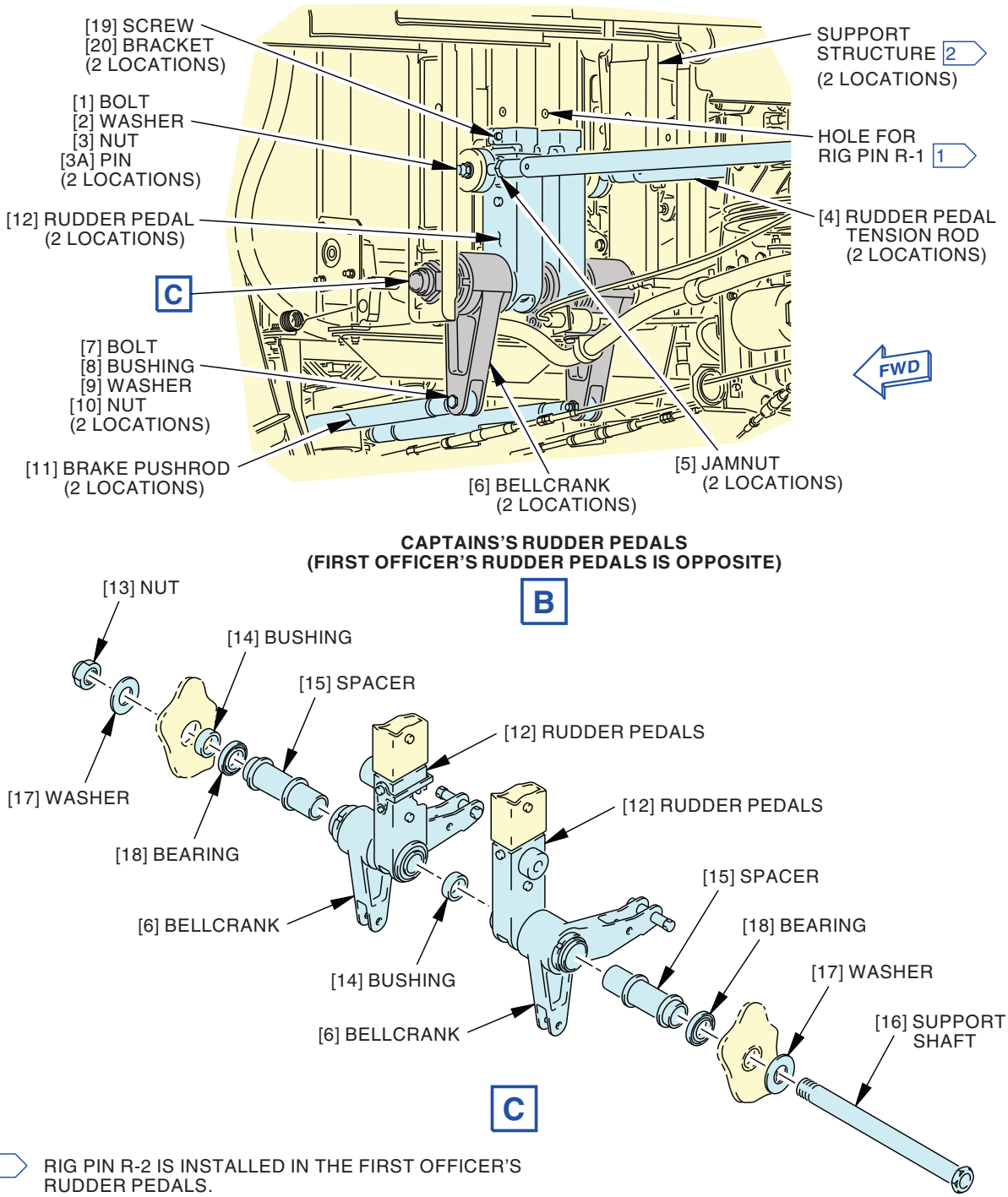




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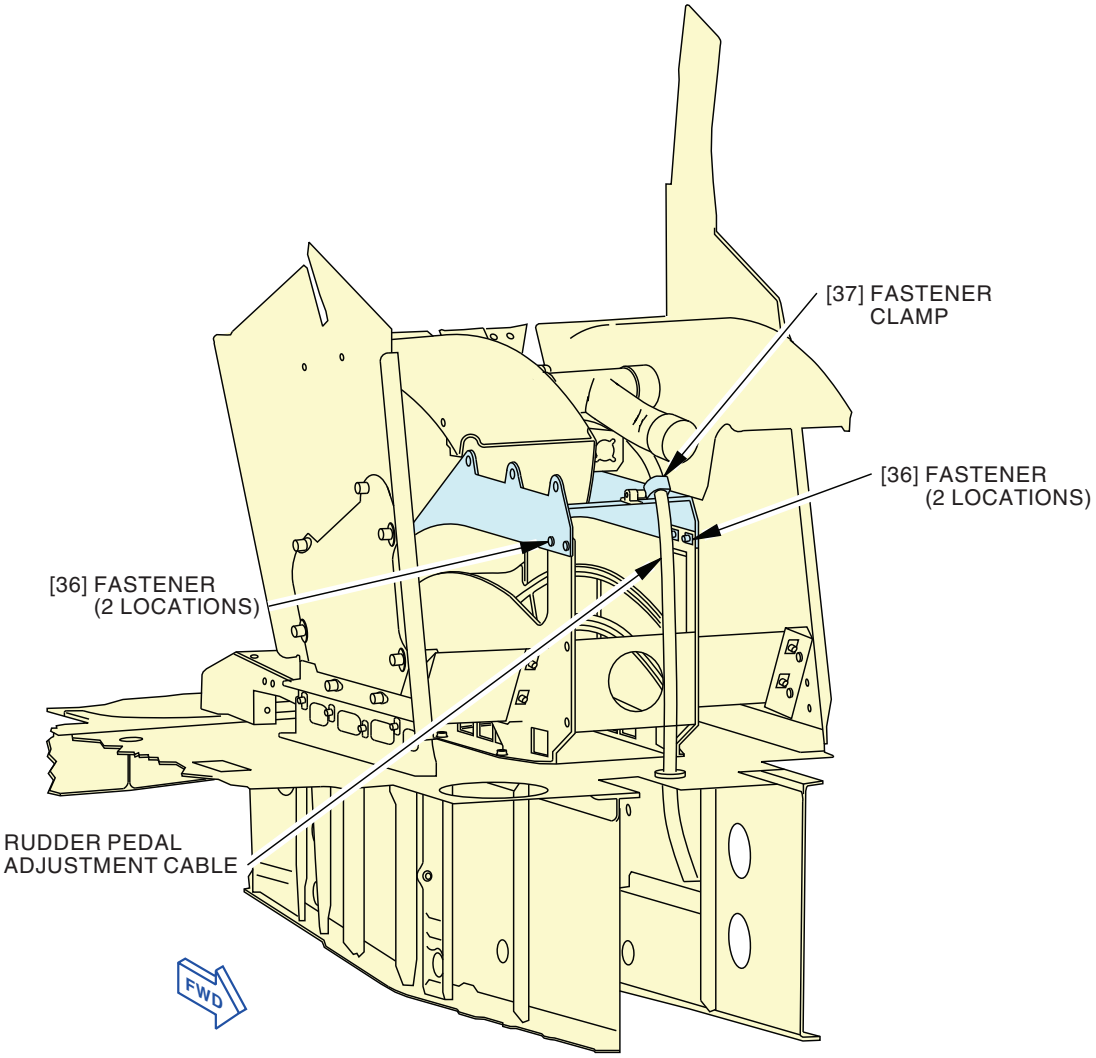
Rudder Pedals Installation
Figure 402/27-21-31-990-802 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Pedals Installation
Figure 402/27-21-31-990-802 (Sheet 2 of 5)

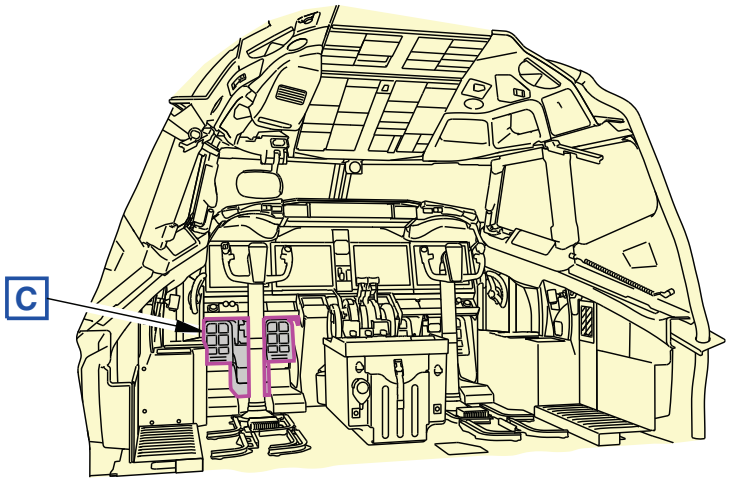
ECCN 9E991 BOEING PROPRIETARY - See title page for details



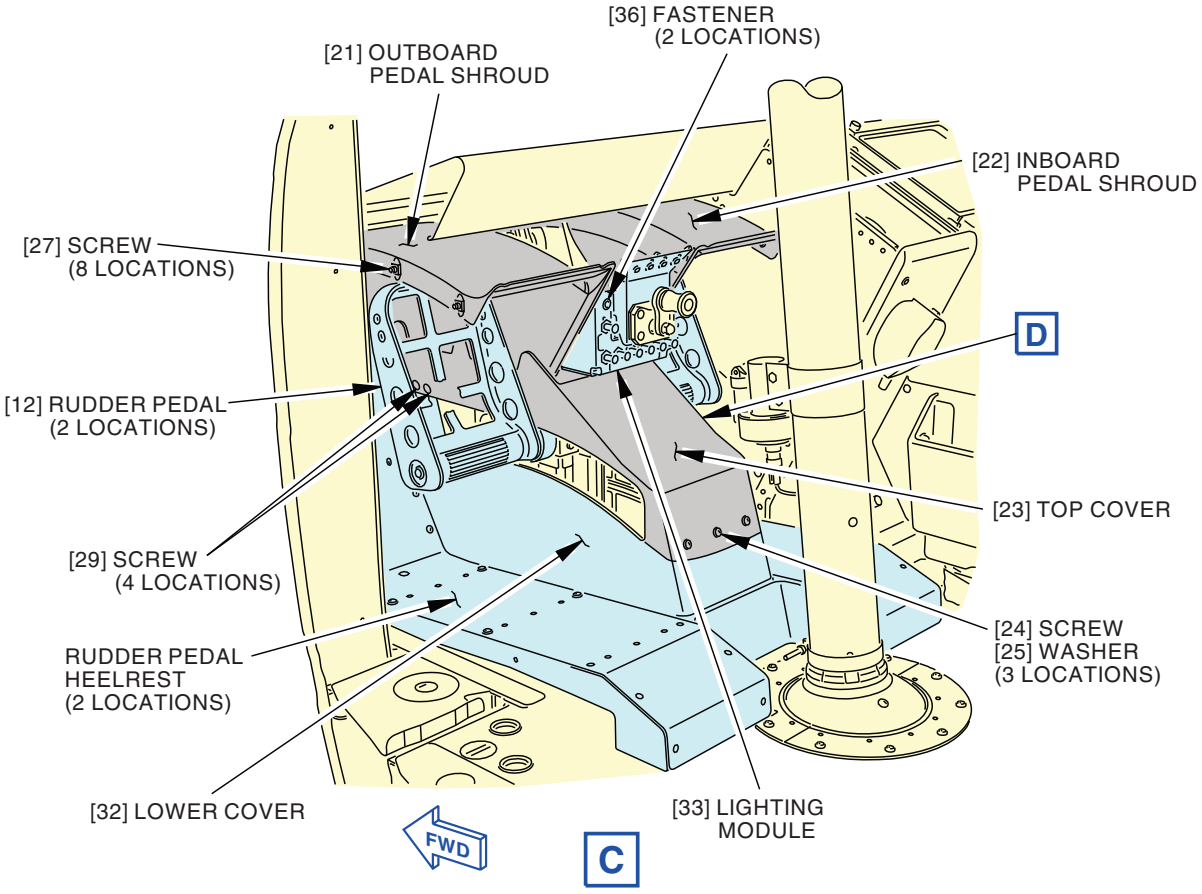
2988226 S0000765089_V1

Rudder Pedals Installation
Figure 402/27-21-31-990-802 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



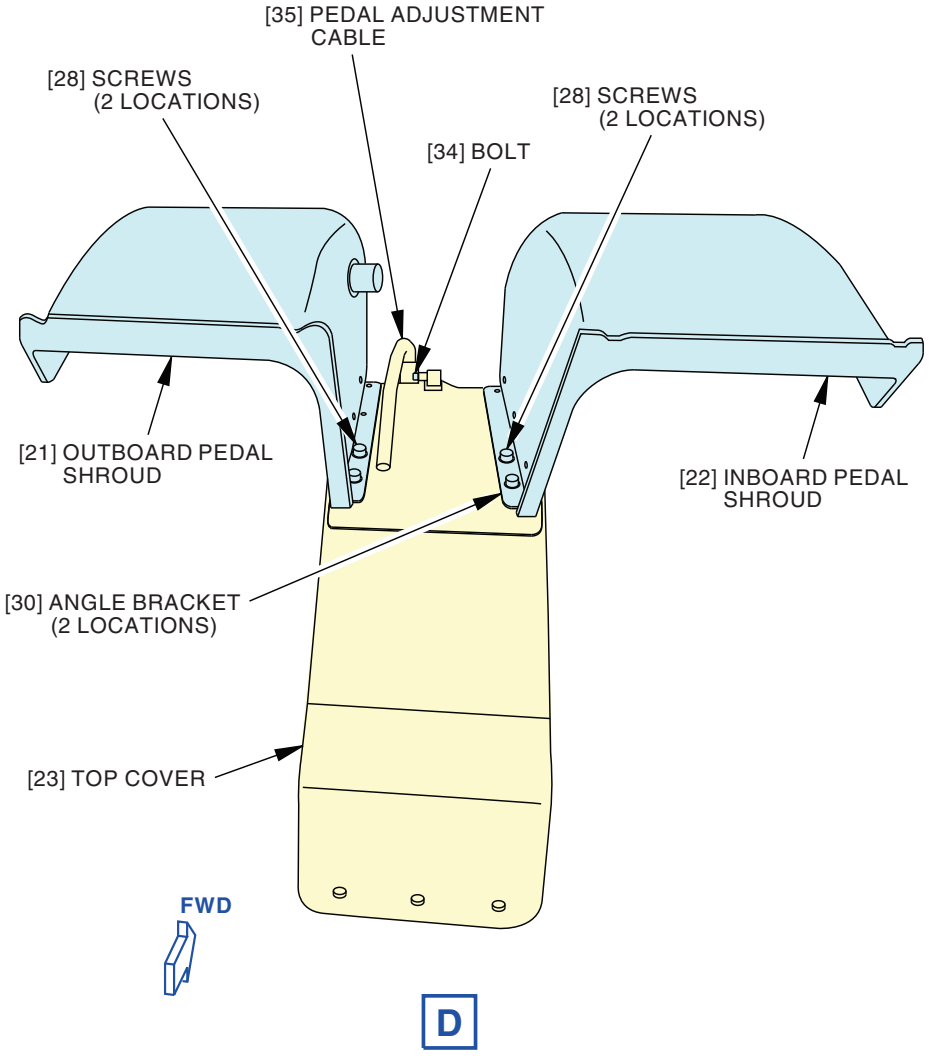
FLIGHT COMPARTMENT



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Rudder Pedals Installation
Figure 402/27-21-31-990-802 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2919763 S0000702950_V2

Rudder Pedals Installation
Figure 402/27-21-31-990-802 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-31-400-801

3. Rudder Pedals Installation

(Figure 401, Figure 402)

A. References

Reference	Title
25-11-01-400-801	Captain and First Officer Seat Installation (P/B 401)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-21-00-700-801	Rudder Pedal Adjustment and Limit Travel Test (P/B 501)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-31-00-400-801	Elevator and Tab Control System - Activation (P/B 201)
33-11-00-990-801	Figure: Lighting Modules (P/B 201)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

E. Rudder Pedals Installation

SUBTASK 27-21-31-480-003

(1) Make sure that the rig pin R-3 from the rigging pin kit, SPL-1585, is installed in the captain's rudder forward quadrant [31].

SUBTASK 27-21-31-480-004

(2) Make sure that the rig pin R-4 from the rigging pin kit, SPL-1585, is installed in the first officer's rudder forward quadrant [31].

SUBTASK 27-21-31-420-001

(3) Install the rudder pedals [12]:

(a) Put the spacers [15] and bearings [18] into the rudder pedals [12].

(b) Put the inboard rudder pedal [12] between the support structure and align the holes.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Put the washer [17] on the support shaft [16].

(d) Install the support shaft [16] from the inboard side through the hole in the support structures and through the bushing [14] and rudder pedal [12].

(e) Install the bushing [14] on the support shaft [16].

(f) Install the washer [17] and nut [13] on the support shaft [16].

1) Tighten the nut [13] to 200 in-lb (22.6 N·m) - 500 in-lb (56.5 N·m).

(g) Make sure the support shaft [16] is not free to turn in the support structures.

SUBTASK 27-21-31-820-001

(4) Do these steps to adjust the rudder pedal position:

(a) Turn the rudder pedal adjustment crank in the full clockwise direction for the installed rudder pedals [12] until the jackshaft is in its full forward position.

(b) Turn the rudder pedal adjustment crank 4 turns in the counterclockwise direction.

(c) If you install the captain's rudder pedals [12], then install the rig pin R-1 from the test set rigging pin kit, SPL-1585, in the captain's rudder pedals [12].

(d) If you install the first officer's rudder pedals [12], then install the rig pin R-2 from the test set rigging pin kit, SPL-1585, in the first officer's rudder pedals [12].

SUBTASK 27-21-31-820-002

(5) Do these steps to adjust the two rudder pedal tension rods [4]:

(a) Loosen the jamnuts [5] on the two rudder pedal tension rods [4].

(b) Turn the rudder pedal tension rods [4] until you can attach them to the rudder pedals [12].

(c) Make sure you can see the rod end threads through the inspection holes at two ends of the rudder pedal tension rods [4] after the adjustment.

(d) Make sure that the threads show in at least one-half of the inspection hole.

(e) Tighten the jamnuts [5] on the rudder pedal tension rods [4].

SUBTASK 27-21-31-420-002

(6) Install the bolts [1], washers [2], nuts [3] and pin [3A] to connect the two rudder pedal tension rods [4]to the rudder pedals [12].

SUBTASK 27-21-31-420-003

(7) Install the brackets [20] and the screws [19] on the rudder pedals [12].

SUBTASK 27-21-31-420-004

(8) Install the bolts [7], bushings [8], washers [9], and nuts [10] to connect the two brake pushrods [11] to the bellcranks [6].

(a) Adjust the length of the brake pushrods [11], if it is necessary.

SUBTASK 27-21-31-420-010

(9) Install top cover and shrouds. Do the following steps:

(a) Put the top cover [23] on the lower cover [32].

(b) Attach the pedal adjustment cable [35] to the top cover [23].

1) Install bolt [34] through clamp on pedal adjustment cable [35] and into cable standoff on top cover [23].

(c) Install the screws [24], screws [29] and washers [25] to attach the top cover [23] to the lower cover [32].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Put the outboard pedal shroud [21] in its position.

NOTE: If necessary ensure that all ducting is properly attached and positioned.

1) Install screws [28] into the angle bracket [30].
2) Tighten the screws [28] to a torque of 20 in-lb (2 N·m).

(e) Put the inboard pedal shroud [22] in its position.

NOTE: If necessary ensure that all ducting is properly attached and positioned.

1) Install screws [28] into the angle bracket [30].
2) Tighten the screws [28] to a torque of 20 in-lb (2 N·m).

(f) Install the screws [27] to attach the outboard pedal shroud [21] and inboard pedal shroud [22] to the stanchion walls.

(g) Install the lighting module [33].

1) Put the lighting module [33] into its position. (Figure 33-11-00-990-801)
2) Align the pins on the back of the lighting module [33] with the holes in the bracket below the rudder pedal adjustment crank.
3) Tighten the fasteners [36].

SUBTASK 27-21-31-080-001

(10) Remove rig pins R-3 and R-4 (Figure 401).

SUBTASK 27-21-31-080-002

(11) Remove rig pin R-1 or R-2 (Figure 402).

SUBTASK 27-21-31-730-001

(12) Do this task: Rudder Pedal Adjustment and Limit Travel Test, TASK 27-21-00-700-801.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-31-410-001

(1) Install the access panels for the captain's and first officer's rudder forward quadrant [31]. (TASK 53-14-01-420-801)

(a) Install these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

SUBTASK 27-21-31-410-002

(2) Close this access door:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-21-31-420-011

(3) If removed, install the Captain and First Officer's seats. Do this task: Captain and First Officer Seat Installation, TASK 25-11-01-400-801.

SUBTASK 27-21-31-440-001

(4) Do this task: Elevator and Tab Control System - Activation, TASK 27-31-00-400-801.

SUBTASK 27-21-31-440-002

(5) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-31-440-003

(6) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER PEDAL ADJUSTMENT CRANK - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of a rudder pedal adjustment crank

(2) An installation of a rudder pedal adjustment crank.

B. The rudder pedal adjustment crank is referred to as the "crank assembly" in this procedure.

C. The crank assembly includes the crank lever [14] and cable [1].

D. The procedure is applicable for the captain's and first officer's crank assembly.

E. It is not necessary to remove the cable to remove the crank lever.

TASK 27-21-41-000-801

2. Rudder Pedal Adjustment Crank Removal

(Figure 401)

A. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
222	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Right

C. Access Panels

Number	Name/Location
112A	Forward Access Door

D. Prepare for the Removal

SUBTASK 27-21-41-010-001

(1) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-21-41-010-002

(2) Open the main instrument panel shroud to get access to the rudder pedal adjustment cable [1].

SUBTASK 27-21-41-010-003

(3) Open the main instrument panel module assembly [14].

E. Crank Assembly Removal

SUBTASK 27-21-41-020-001

(1) Do these steps to remove the crank lever [10]:

(a) Remove the nut [8] and washer [9] from the crank lever [10].

(b) Remove the crank lever [10] from the bearing retainer [12].

(c) Remove the spacer [11].

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SUBTASK 27-21-41-020-002

(2) Do these steps to remove the rudder pedal adjustment cable [1] from the crank lever [10]:

(a) Remove the MS20995C32 lockwire, G01048, from the cable retaining nut [13].

(b) Loosen the cable retaining nut [13].

(c) Pull the rudder pedal adjustment cable [1] forward until it disconnects from the crank support bracket [7].

SUBTASK 27-21-41-020-003

(3) Do these steps to disconnect the rudder pedal adjustment cable [1] from the rudder jackshaft [4]:

(a) Loosen the retaining nut [3] at the rudder jackshaft [4].

(b) Hold the retaining nut [3] and pull the rudder pedal adjustment cable [1] rudder jackshaft [4].

(c) Remove the retaining nut [3] and washer [2].

SUBTASK 27-21-41-020-004

(4) Do these steps to remove the rudder pedal adjustment cable [1]:

(a) Remove the bolts [5] from the cable clamps [6].

(b) Remove the cable clamp [6] and keep a record of their locations.

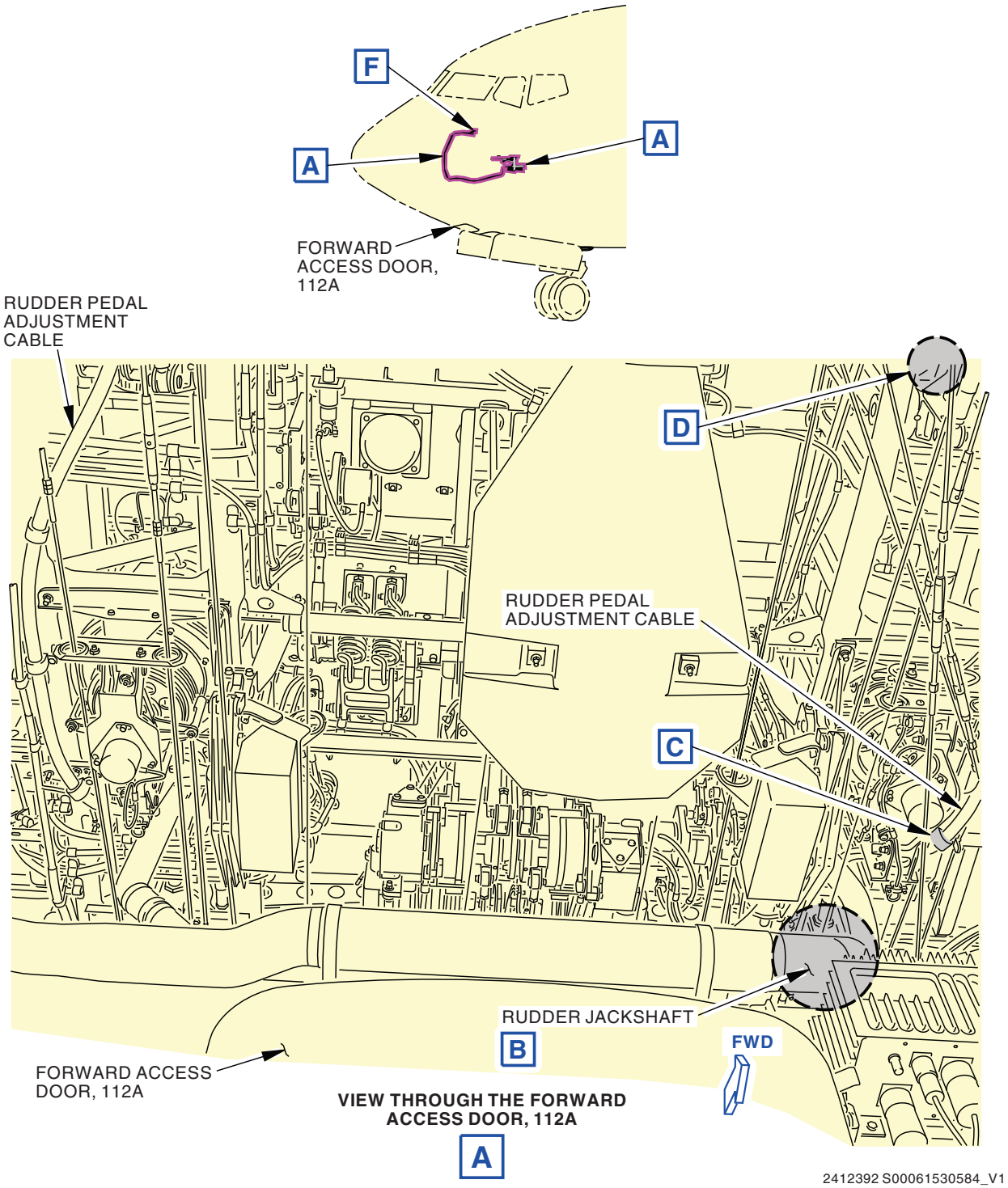
(c) Remove the elastic grommet.

(d) Pull the rudder pedal adjustment cable [1] through the grommet hole.

(e) Remove the rudder pedal adjustment cable [1].

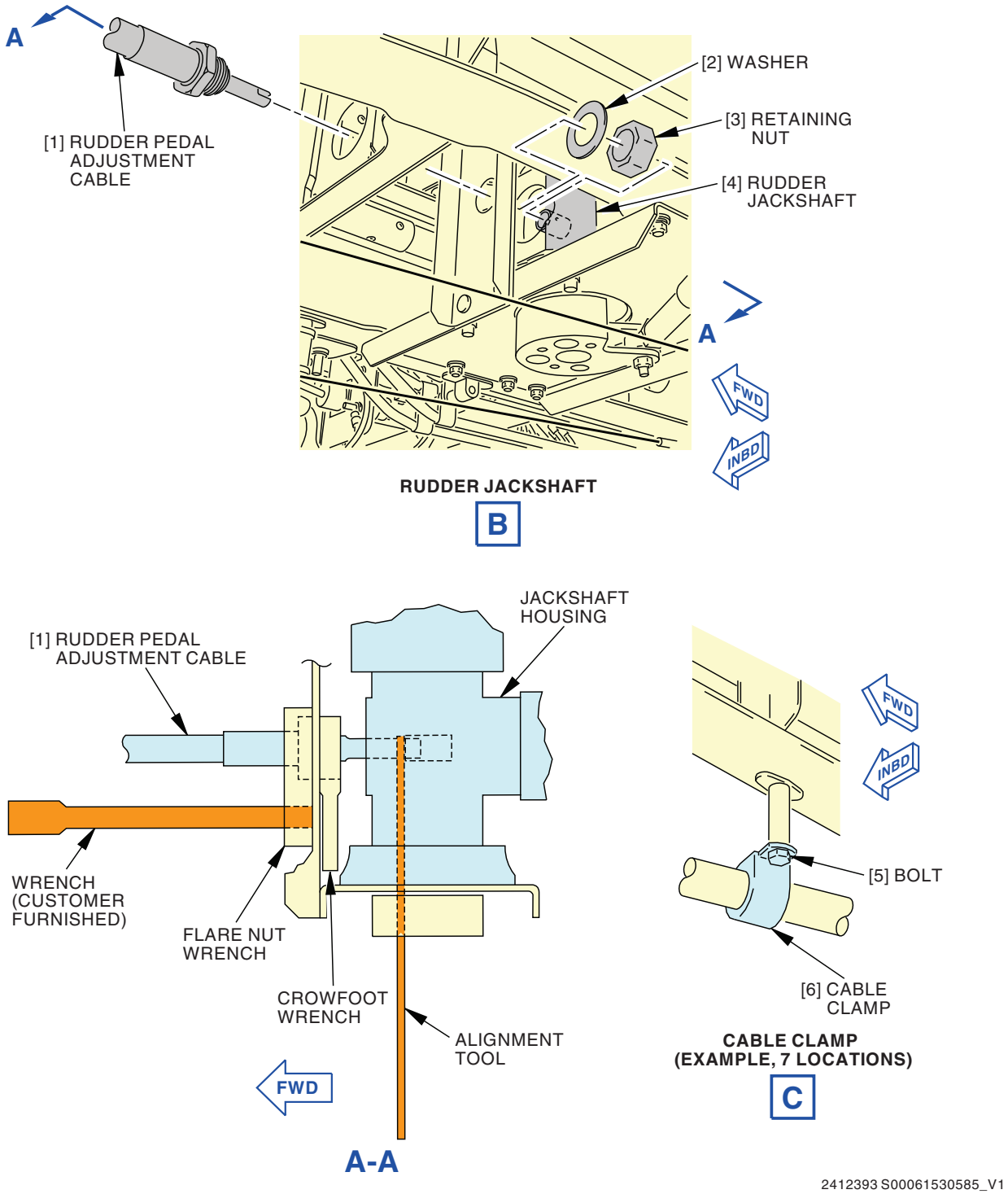
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ECCN 9E991 BOEING PROPRIETARY - See title page for details



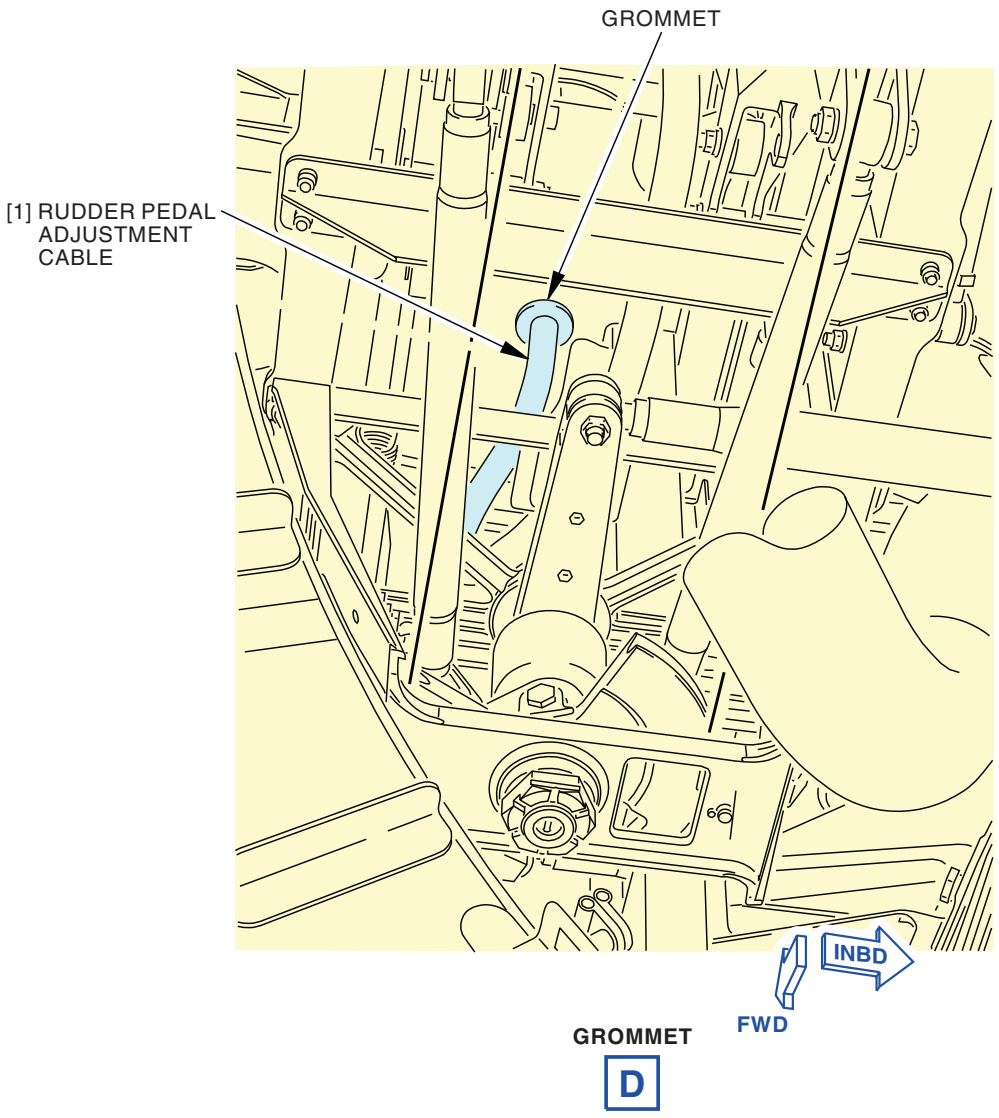
Rudder Pedal Adjustment Crank Installation
Figure 401/27-21-41-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Pedal Adjustment Crank Installation
Figure 401/27-21-41-990-801 (Sheet 2 of 4)

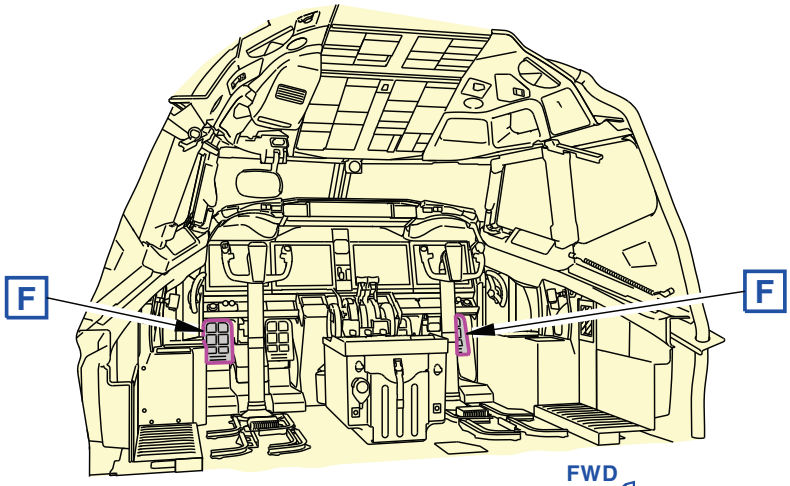
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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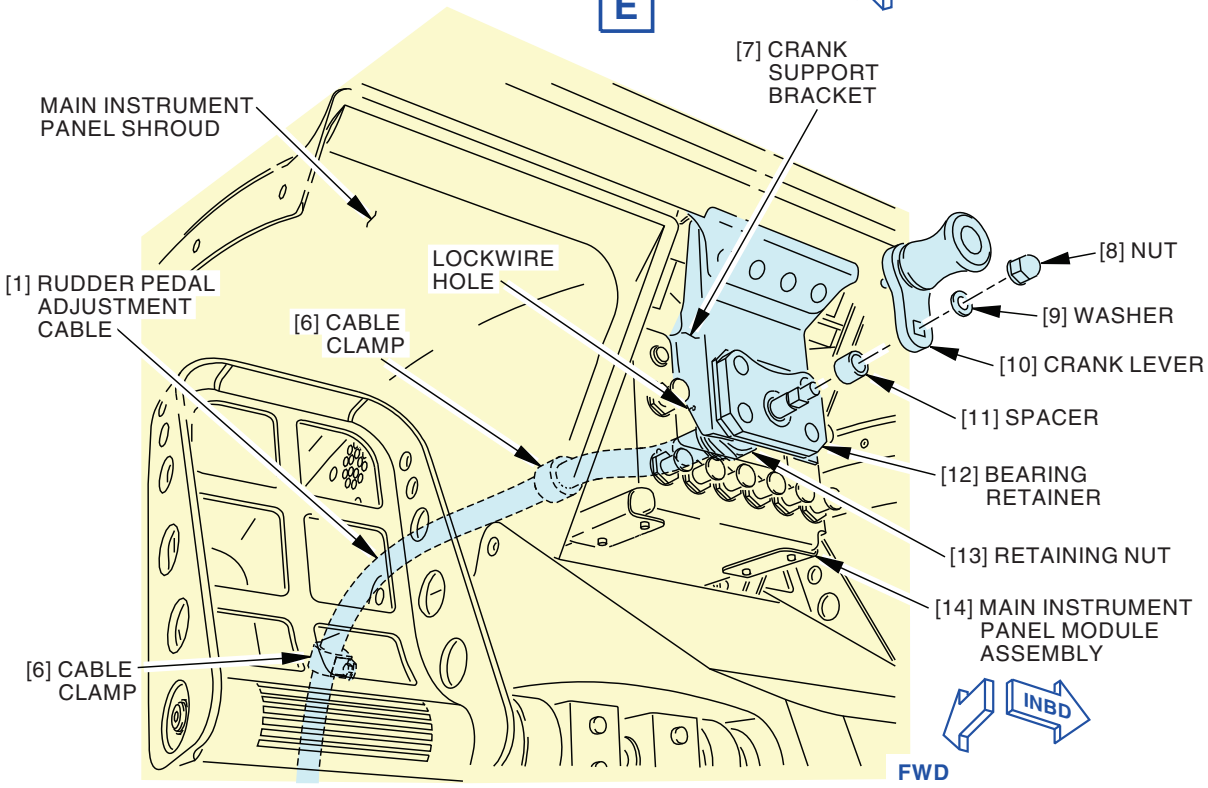
Rudder Pedal Adjustment Crank Installation
Figure 401/27-21-41-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT

E



LEFT MAIN INSTRUMENT PANEL MODULE ASSEMBLY
(RIGHT MAIN INSTRUMENT PANEL MODULE ASSEMBLY IS EQUIVALENT)

F

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Rudder Pedal Adjustment Crank Installation
Figure 401/27-21-41-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-41-400-801

3. Rudder Pedal Adjustment Crank Installation

(Figure 401)

A. References

Reference	Title
27-21-00-700-801	Rudder Pedal Adjustment and Limit Travel Test (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1689	Alignment Tool, B27064-5 (included in B27064 Torque Equipment) Part #: B27064-1 Supplier: 81205 Part #: B27064-2 Supplier: 81205
SPL-11751	1-1/16 Crowfoot Wrench, B27064-13 (included in B27064-1 Torque Equipment) Part #: B27064-1 Supplier: 81205
STD-24936	Crowfoot Wrench - 1-1/16 inch

C. Consumable Materials

Reference	Description	Specification
D00013	Grease - Aircraft And Instrument Grease	MIL-PRF-23827 (NATO G-354) (Supersedes MIL-G-23827)
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Rudder pedal adjustment cable	27-21-41-37-025	PTF ALL
10	Crank lever	27-21-41-37-025	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left
222	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Right

F. Access Panels

Number	Name/Location
112A	Forward Access Door

G. Rudder Pedal Adjustment Crank Installation

SUBTASK 27-21-41-420-001

(1) Do these steps to install the rudder pedal adjustment cable [1]:

(a) Install the grommet on the rudder pedal adjustment cable [1].

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(b) Push the rudder pedal adjustment cable [1] through the grommet hole.

NOTE: One end of the cable should reach the crank lever [10] and the other end should reach the rudder jackshaft [4].

(c) Install the cable clamps [6] and bolts [5] on the rudder pedal adjustment cable [1].

1) Make sure that the rudder pedal adjustment cable [1] radius is more than 5.0 inches (127 millimeters) for all the curves in the rudder pedal adjustment cable [1].

NOTE: If the rudder pedal adjustment cable [1] radius is less than 5.0 inches (127 millimeters), the rudder pedal adjustment cable [1] will not operate correctly.

SUBTASK 27-21-41-420-002

(2) Connect the rudder pedal adjustment cable [1] to the rudder jackshaft [4]:

(a) Put the washer [2] and retaining nut [3] on the rudder pedal adjustment cable [1].

(b) Put the rudder pedal adjustment cable [1] through the rudder jackshaft [4].

(c) Apply a thin layer of grease, D00013 to the outside diameter of the rudder pedal adjustment cable [1] before it is installed in the rudder jackshaft [4].

(d) Do these steps:

1) Install the alignment tool alignment tool, B27064-5, SPL-1689 through the hole in the bottom of the rudder jackshaft [4] to hold the jackshaft universal joint in its position.

2) Turn the rudder pedal adjustment cable [1] end until it engages the jackshaft pin.

3) Use the crowfoot wrench, B27064-13, SPL-11751, or crowfoot wrench, STD-24936, crowfoot wrench to hold the retaining nut [3].

4) Use the crowfoot wrench, B27064-13, SPL-11751, or crowfoot wrench, STD-24936, to hold the sleeve of the rudder pedal adjustment cable [1].

(e) Tighten the retaining nut [3].

1) Tighten the retaining nut [3] to 90 +310 / -0 in-lb (10.2 +35.1 / -0.0 N·m) before it touches the jackshaft support.

2) Turn the retaining nut [3] 1/4 to 1/2 of a turn more when it touches the jackshaft support.

SUBTASK 27-21-41-420-003

(3) Do these steps to connect the rudder pedal adjustment cable [1] to the crank lever [10]:

(a) Put the cable retaining nut [13] on one end of the rudder pedal adjustment cable [1].

(b) Push the rudder pedal adjustment cable [1] through the crank support bracket [7] and bearing retainer [12].

(c) Tighten the cable retaining nut [13].

1) Tighten the cable retaining nut [13] finger tight plus one quarter turn.

(d) Install the MS20995C32 lockwire, G01048 on the retaining nut [13].

NOTE: To get the best MS20995C32 lockwire, G01048, angle, make sure that the aft cable assembly can turn easily and smoothly.

SUBTASK 27-21-41-420-004

(4) Do these steps to install the crank lever [10]:

(a) Install the spacer [11] on the rudder pedal adjustment cable [1].

(b) Install the crank lever [10], washer [9], and nut [8].

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SUBTASK 27-21-41-410-001

(5) Close the main instrument panel module assembly [14].

SUBTASK 27-21-41-410-002

(6) Close the main instrument panel module shroud.

SUBTASK 27-21-41-710-001

(7) Do this task to test the rudder pedal adjustment cranks: Rudder Pedal Adjustment and Limit Travel Test, TASK 27-21-00-700-801.

SUBTASK 27-21-41-410-003

(8) Close this access panel:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

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RUDDER FORWARD QUADRANT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the rudder forward quadrant

(2) An installation of the rudder forward quadrant.

B. The rudder forward quadrant includes the quadrant and jackshaft.

C. This procedure is applicable for the captain's and first officer's rudder forward quadrants.

TASK 27-21-51-000-801

2. Rudder Forward Quadrant Removal

(Figure 401, Figure 402, Figure 403)

A. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
31-31-42-000-801	Rudder Pedal Position Sensor Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
311	Area Aft of Pressure Bulkhead - Left
324	Vertical Fin - Rear Spar To Trailing Edge

D. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
114AC	Fwd Nose Wheel Well Upper Access Panel
114AW	Forward Nose Wheel Well Panel
311BL	Stabilizer Trim Access Door
324CL	Vertical Fin, Access

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E. Prepare for the Removal

SUBTASK 27-21-51-480-006

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-21-51-860-001

(2) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-51-010-001

(3) To get access to the rudder feel and centering unit, open this access panel:

Number	Name/Location
324CL	Vertical Fin, Access

SUBTASK 27-21-51-080-001

(4) Install rig pin R-5 from the rigging pin kit, SPL-1585, in the rudder feel and centering unit (Figure 401).

SUBTASK 27-21-51-010-002

(5) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-21-51-010-003

(6) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-21-51-010-004

(7) Remove these access panels:

(TASK 53-14-01-020-801)

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
114AC	Fwd Nose Wheel Well Upper Access Panel
114AW	Forward Nose Wheel Well Panel

SUBTASK 27-21-51-020-011

(8) If it is necessary, remove the duct [107] as follows (Figure 402, View L):

(a) Make sure that the L and R PACK switches, on the P5-10 air conditioning panel, are set to OFF and install the DO NOT OPERATE tag, STD-858.

(b) Make sure that the L and R RECIRC FAN switches, on the P5-10 air conditioning panel, are set to OFF and install the DO NOT OPERATE tag, STD-858.

(c) Remove the plastic tie straps [110] from the flexible sleeve hoses [112].

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(d) Remove the nut [115], washer [114], screw [113],clamp [111], and wiring installation clamp [117].

NOTE: The clamp shares a support with wiring installation.

(e) Disconnect duct [107] from duct [108], duct [109], and duct [116].

(f) Remove the flexible sleeve hoses [112] from duct [107].

(g) Remove the duct [107].

(h) Put protective covers on the open ends of duct [108], duct [109], and duct [116].

SUBTASK 27-21-51-020-001

(9) For the first officer's rudder forward quadrant, do this task:Rudder Pedal Position Sensor Removal, TASK 31-31-42-000-801.

F. Rudder Forward Quadrant Removal

SUBTASK 27-21-51-010-005

WARNING

YOU MUST LOOSEN AND DISCONNECT THE RUDDER CABLES FROM THE RUDDER FORWARD QUADRANTS BEFORE YOU DISCONNECT THE BUS ROD. RELIEVE TENSION UNIFORMLY IN THE TWO RUDDER CABLES. IF YOU DO NOT LOOSEN THE RUDDER CABLES EQUALLY, IT CAN CAUSE INJURY TO PERSONNEL OR DAMAGE TO EQUIPMENT.

(1) Loosen the turnbuckles [91] for the rudder cable RB [21] and rudder cable RA [22] (Figure 403).

SUBTASK 27-21-51-020-002

(2) Disconnect rudder cable RB [21] or rudder cable RA [22] from the applicable rudder forward quadrant [39] (Figure 402).

(a) Remove the bolt [30], washer [19], washer [31] (if applicable), and nut [20] from the cable retainer [32].

SUBTASK 27-21-51-020-012

(3) Disconnect the bus rod [50] from the rudder forward quadrant [39].

(a) Remove the pin [49], nut [48], washer [47], spacers [46], and bolt [45].

(b) Discard the pin [49].

SUBTASK 27-21-51-020-013

(4) Disconnect the rudder pedal tension rods [40] from the rudder forward quadrant [39].

(a) Remove the pins [44], nuts [43], washers [42], and bolts [41].

(b) Discard the pins [44].

SUBTASK 27-21-51-020-005

(5) Remove the nuts [90], washers [89], bolts [88], washers [92], nuts [93], bolts [94], and cover [87] from the floor beam structure.

SUBTASK 27-21-51-020-006

(6) Remove the nuts [73], washers [72], and bolts [71] from the adjustment shaft support bracket [74].

SUBTASK 27-21-51-020-008

(7) Remove the nuts [54], washers [53], screws [52], and cable guard [51] from the jackshaft support channel [70].

SUBTASK 27-21-51-020-009

(8) For the captain's rudder forward quadrant [39], do these steps:

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(a) Disconnect the rudder pedal steering arm [24] from the rudder forward quadrant [39].

1) Remove the lockwire from the rudder pedal steering arm [24].

2) Remove the screws [68] and washers [69].

3) Move the rudder pedal steering arm [24] out of the way.

SUBTASK 27-21-51-020-010

(9) For the first officer's rudder forward quadrant [39], do these steps:

(a) Remove the rudder pedal position sensor quadrant [23].

1) Remove the lockwire from the rudder pedal position sensor quadrant [23].

2) Remove the screws [77] and washers [78].

(b) Remove the nuts [85], washers [84], bolts [83], and support [86] from the jackshaft support channel [70].

SUBTASK 27-21-51-020-016

(10) Remove nut [118], washer [119], washer [120] and the adjustment shaft [75] from the adjustment shaft support bracket [74].

NOTE: The adjustment shaft [75] will disengage from the rudder forward quadrant [39].

SUBTASK 27-21-51-020-014

(11) Remove the jackshaft support channel [70].

(a) Hold the rudder forward quadrant [39].

NOTE: The rudder forward quadrant [39] is free to fall from the upper bearing housing after you remove the screws.

(b) Remove the nuts [60], washers [59], and screws [58].

(c) Remove the nut [63], washer [62], and screw [61].

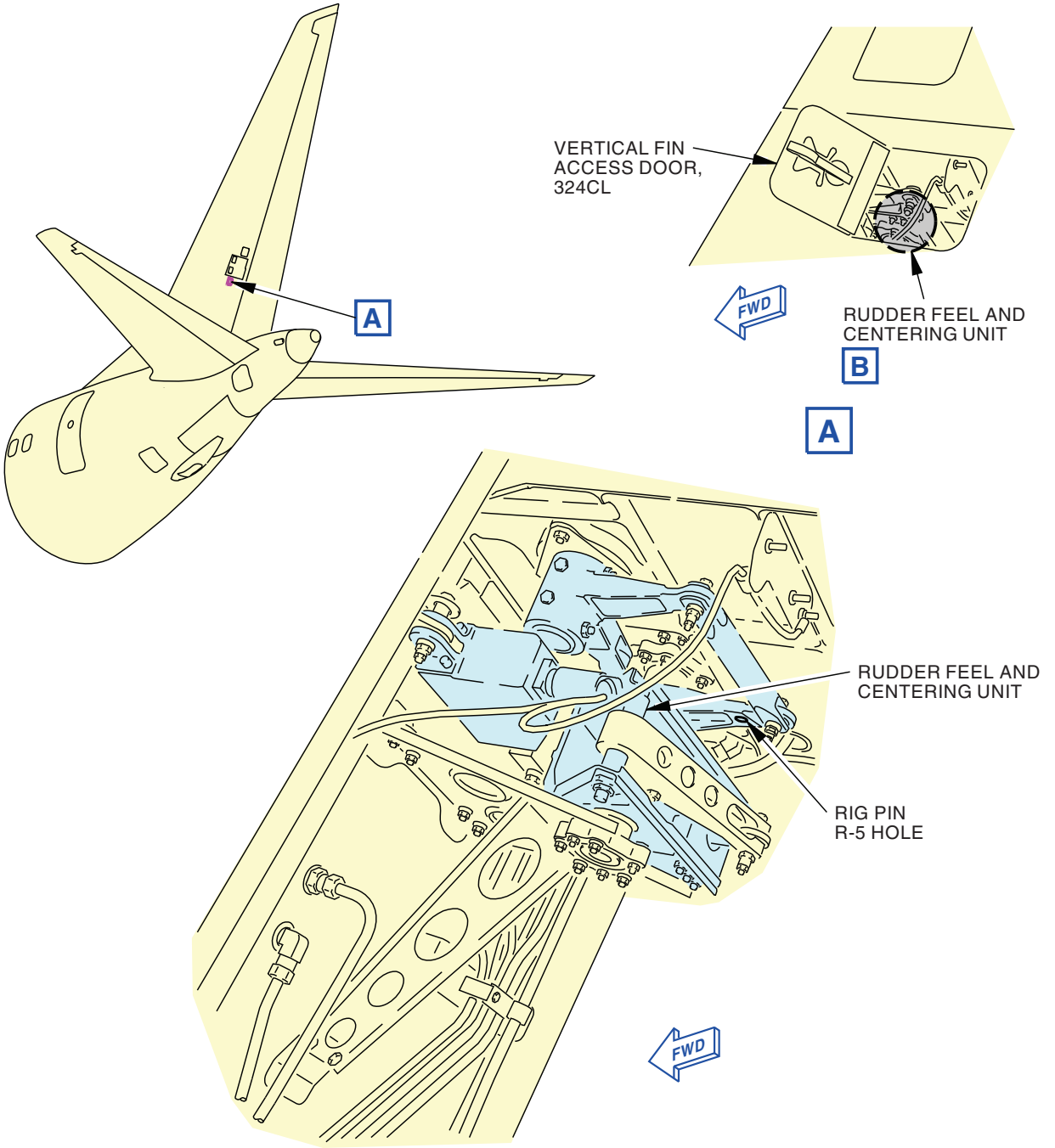
SUBTASK 27-21-51-020-015

(12) Remove the rudder forward quadrant [39].

(a) If necessary, move the rudder forward quadrant [39] in an aft and outboard direction to clear the floor beam structure and control cables.

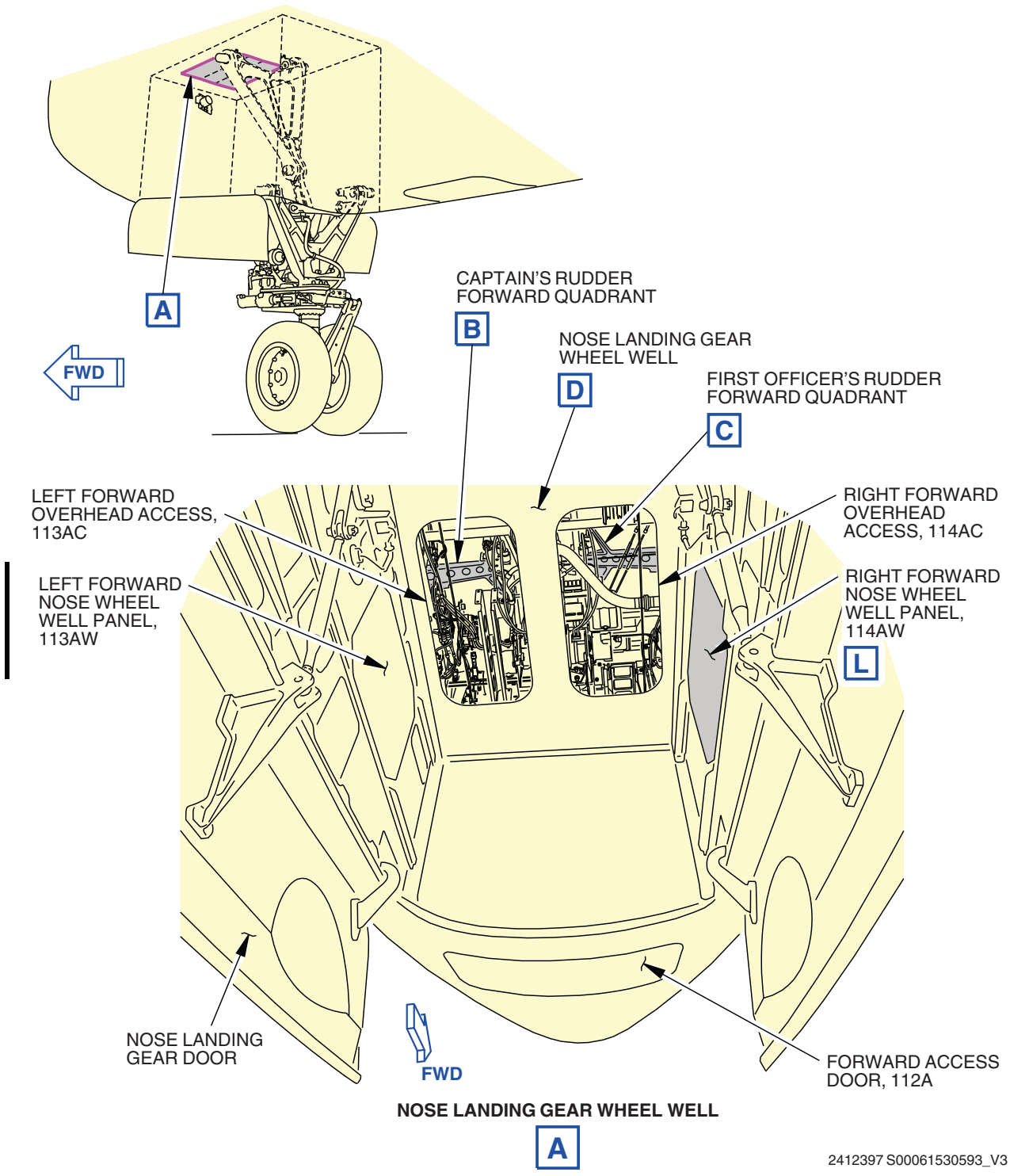
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



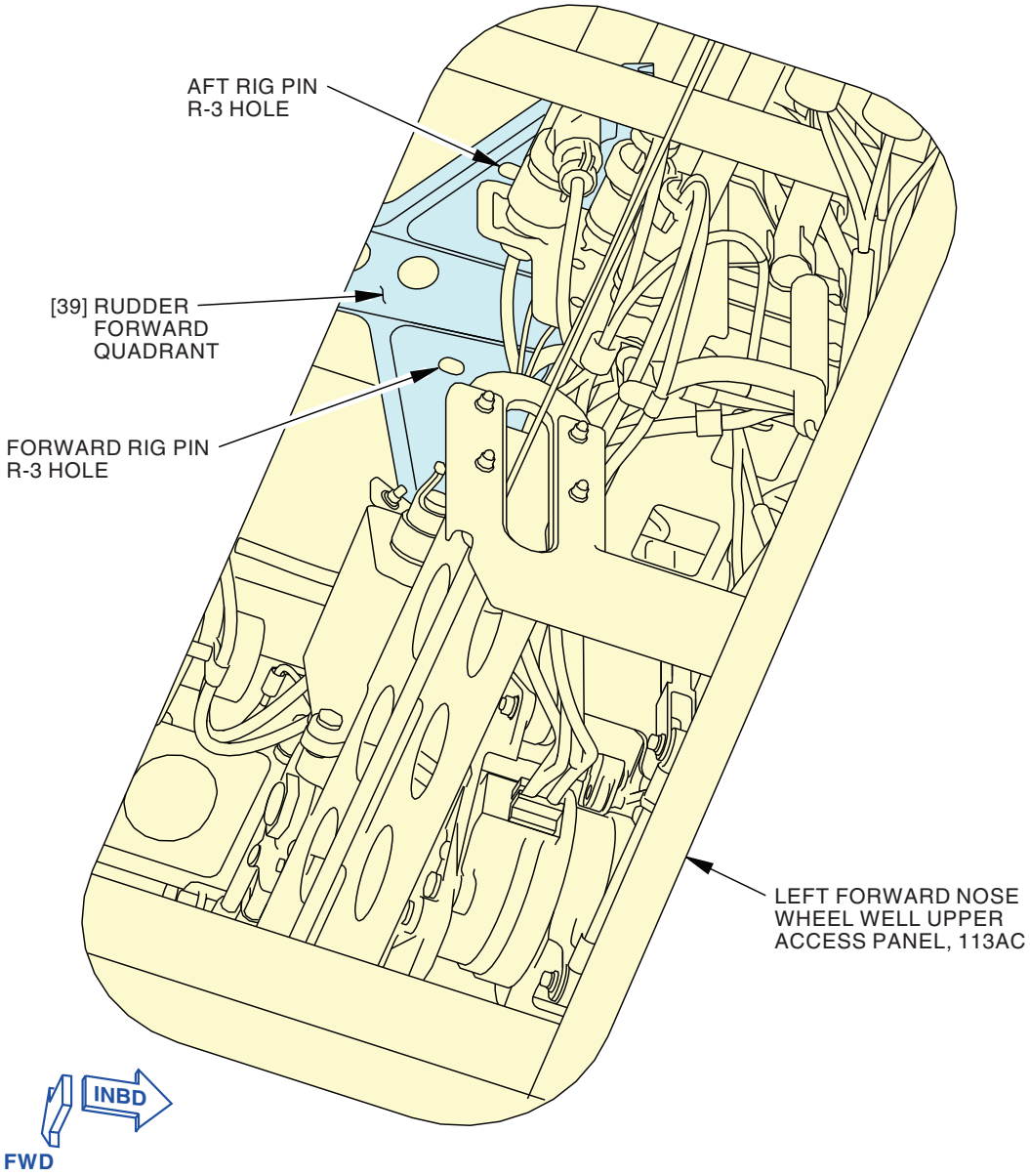
Rudder Rig Pin Installation
Figure 401/27-21-51-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Forward Quadrant Installation
Figure 402/27-21-51-990-802 (Sheet 1 of 11)

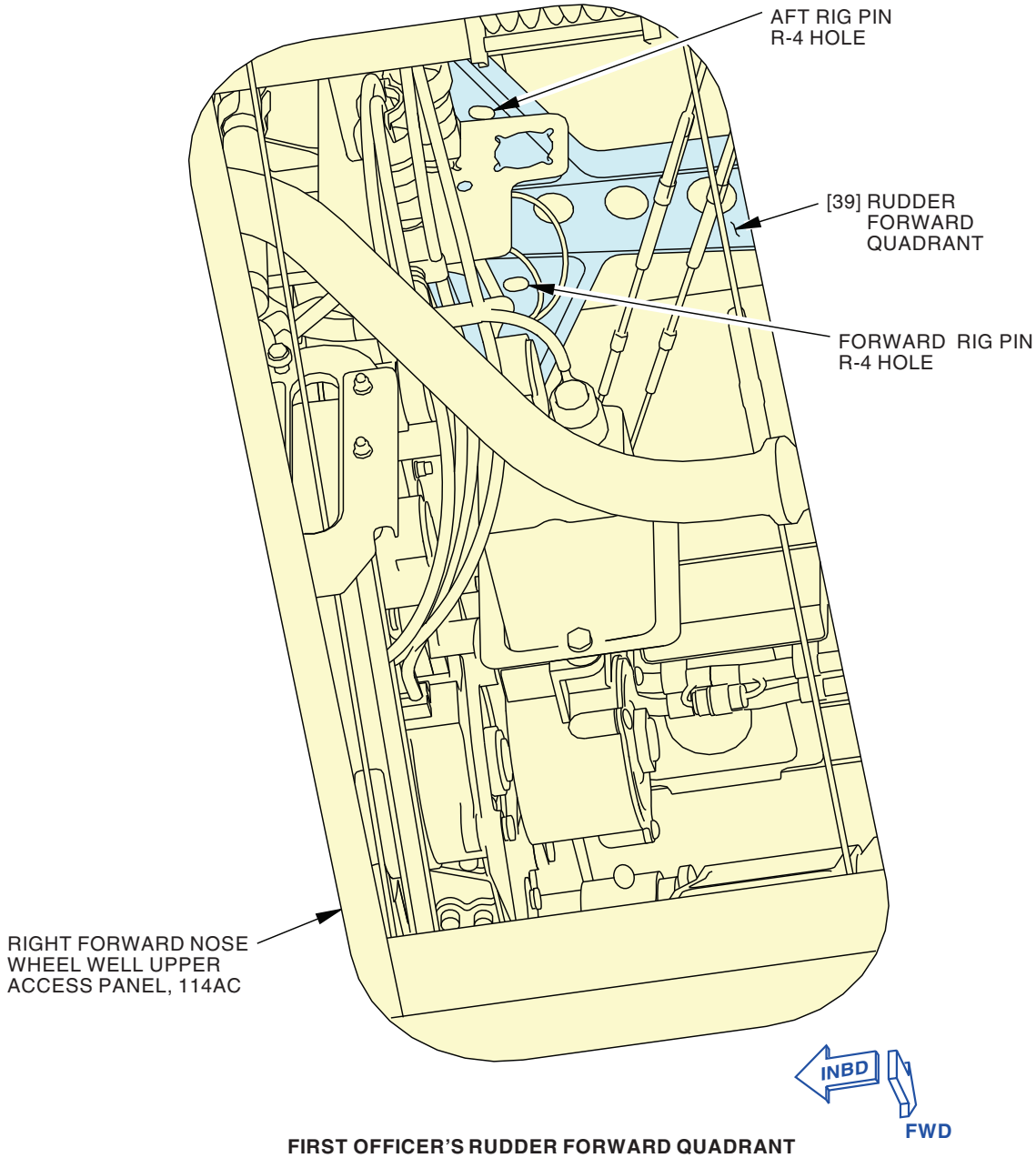
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Rudder Forward Quadrant Installation
Figure 402/27-21-51-990-802 (Sheet 2 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



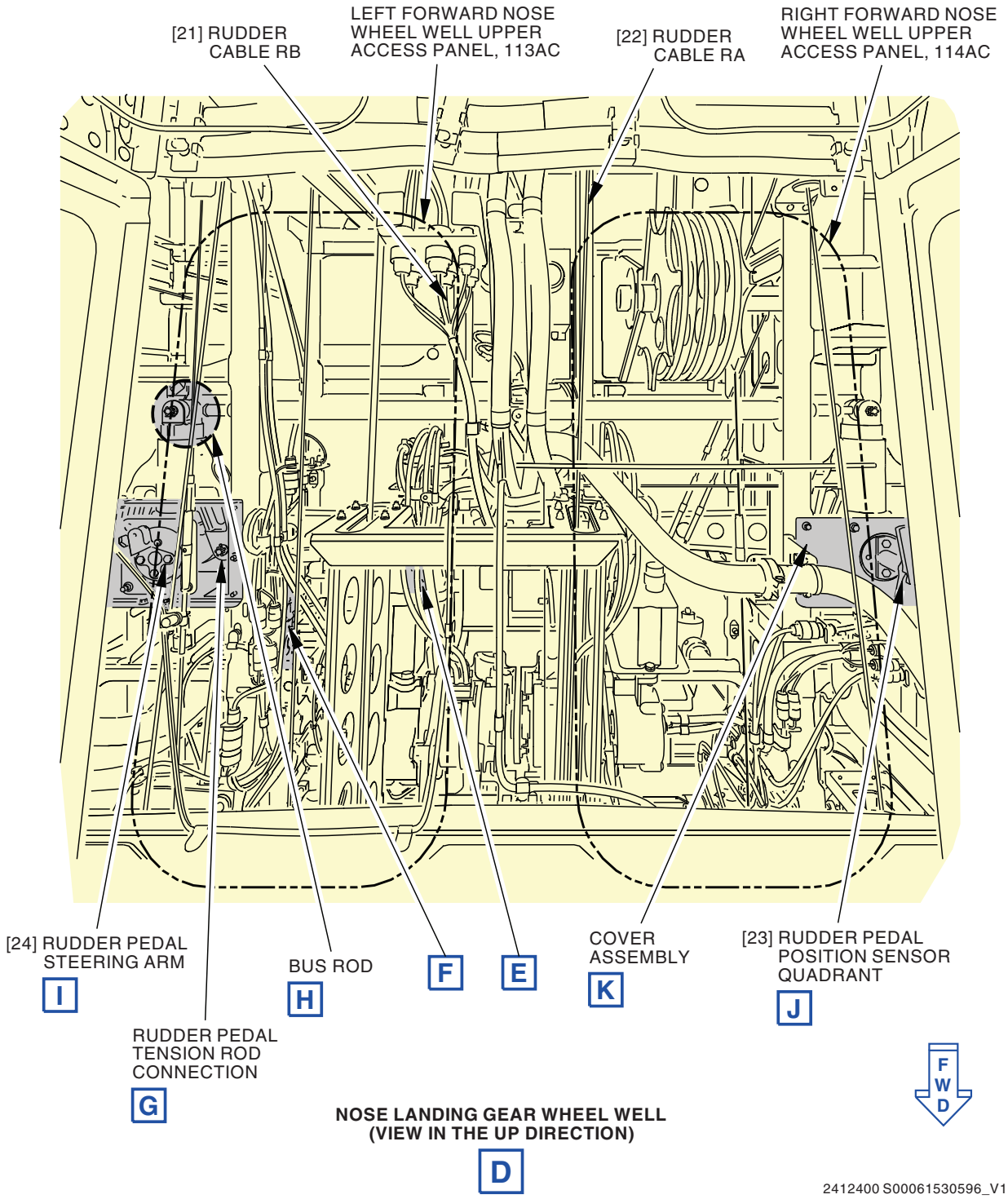
FIRST OFFICER'S RUDDER FORWARD QUADRANT

C

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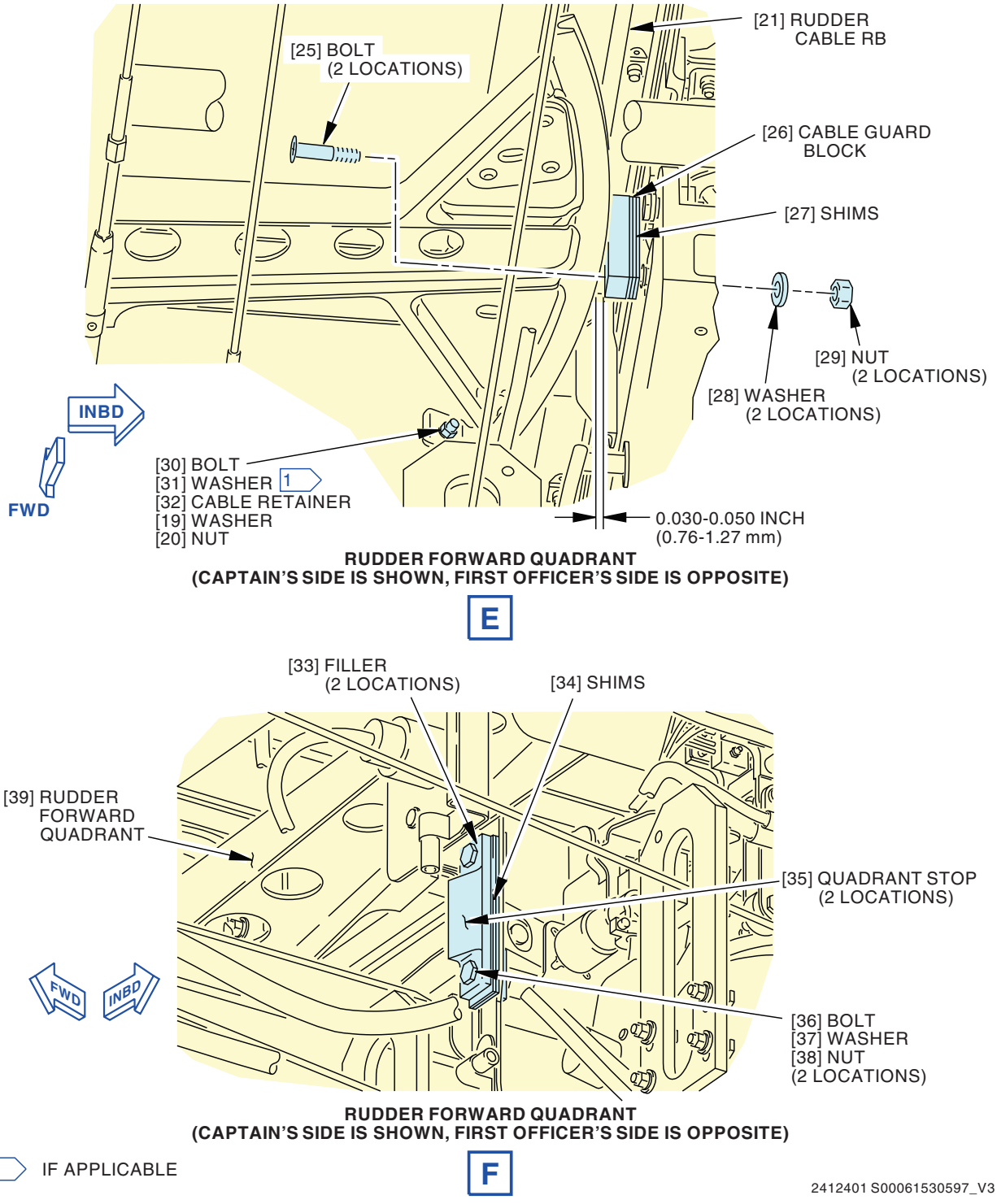
Rudder Forward Quadrant Installation
Figure 402/27-21-51-990-802 (Sheet 3 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



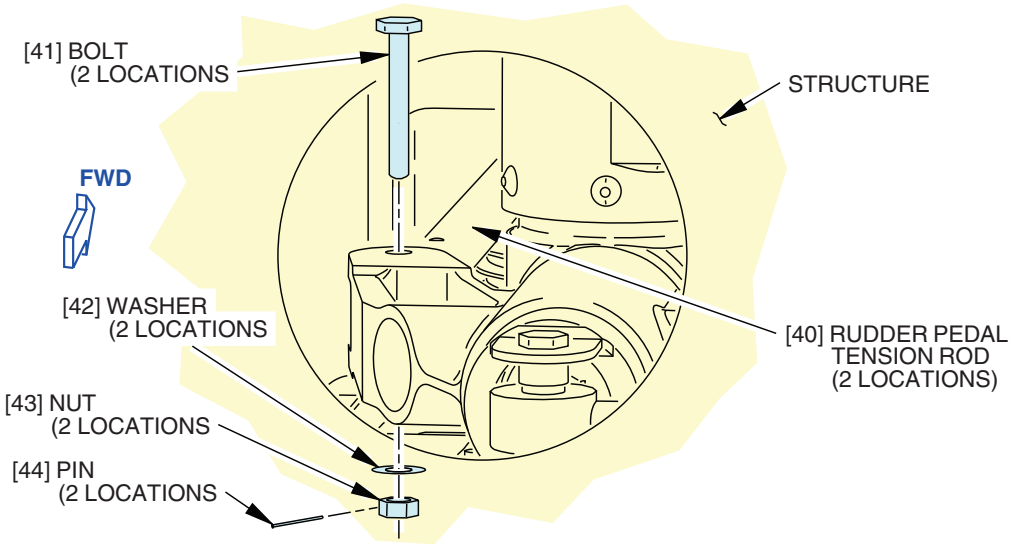
Rudder Forward Quadrant Installation
Figure 402/27-21-51-990-802 (Sheet 4 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



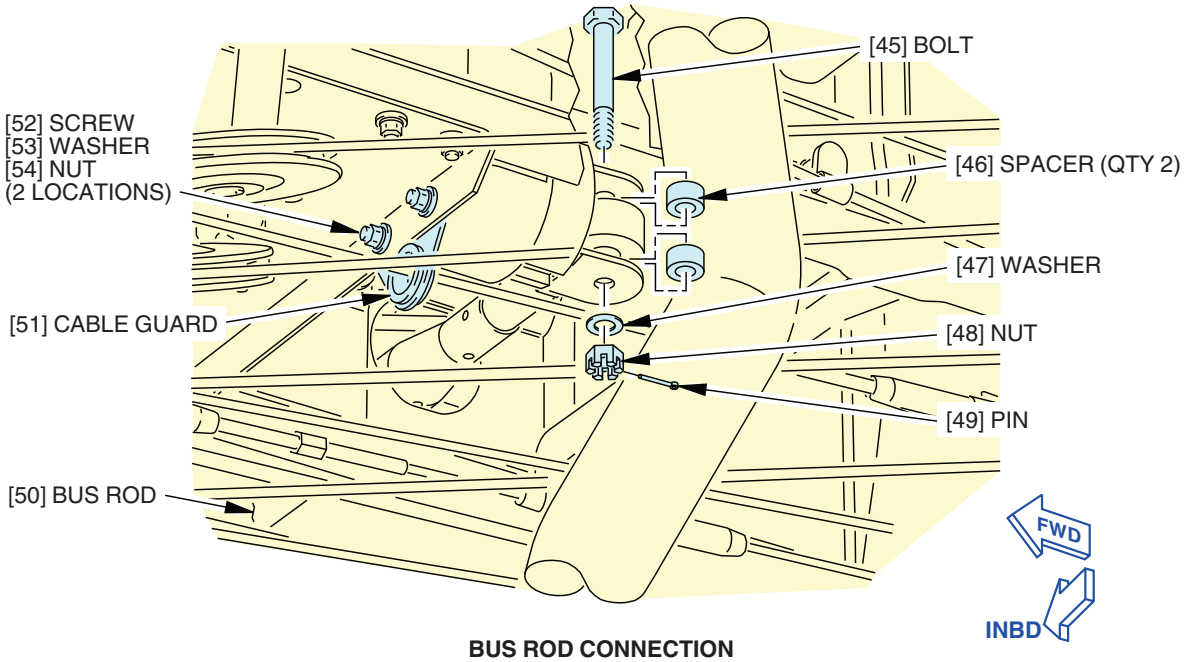
Rudder Forward Quadrant Installation
Figure 402/27-21-51-990-802 (Sheet 5 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RUDDER PEDAL TENSION ROD CONNECTION
(RUDDER FORWARD QUADRANT)
(CAPTAIN'S SIDE IS SHOWN, FIRST OFFICER'S SIDE IS OPPOSITE)

G



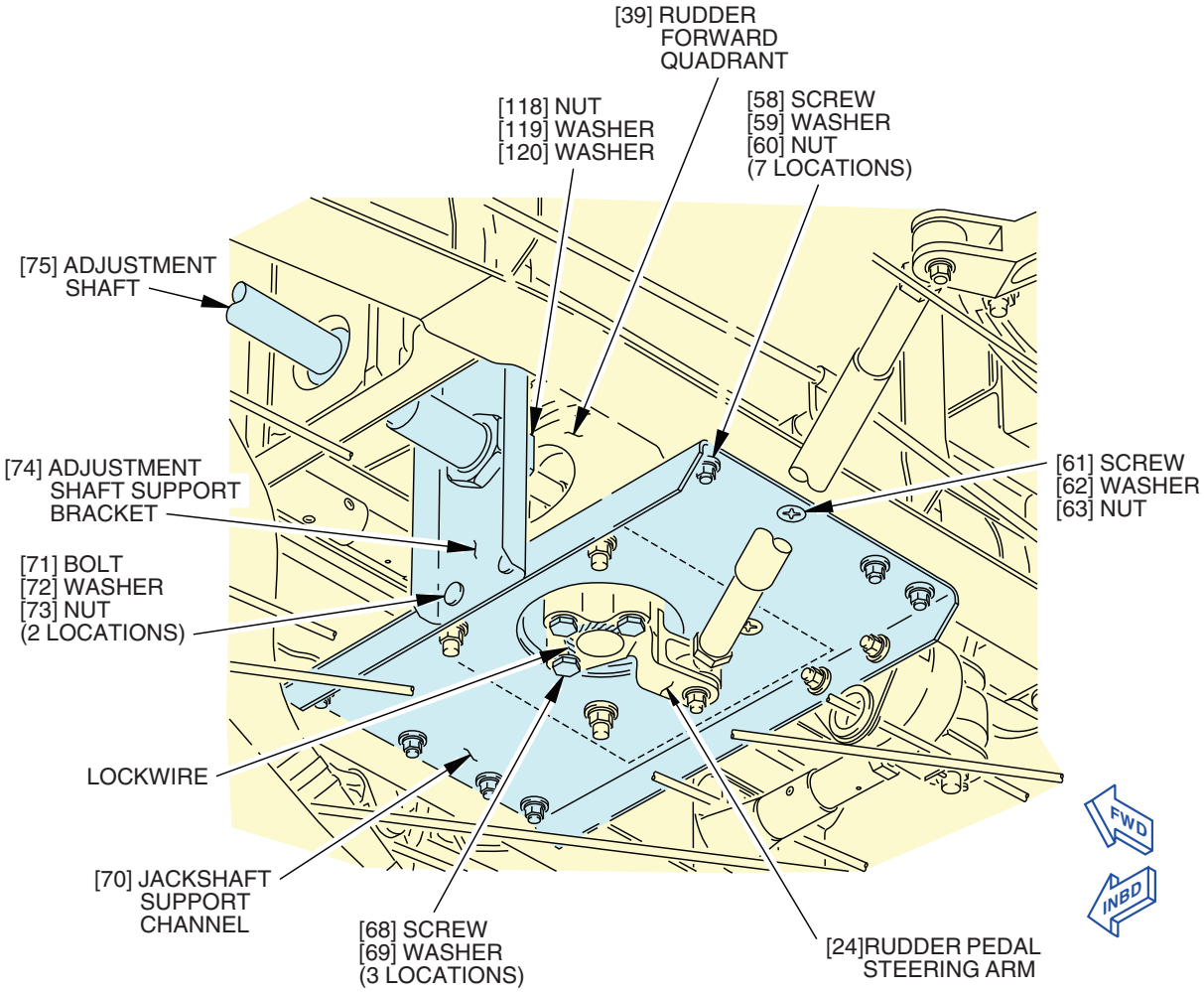
BUS ROD CONNECTION
(RUDDER FORWARD QUADRANT)
(CAPTAIN'S SIDE IS SHOWN, FIRST OFFICER'S SIDE IS OPPOSITE)

H

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Rudder Forward Quadrant Installation
Figure 402/27-21-51-990-802 (Sheet 6 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



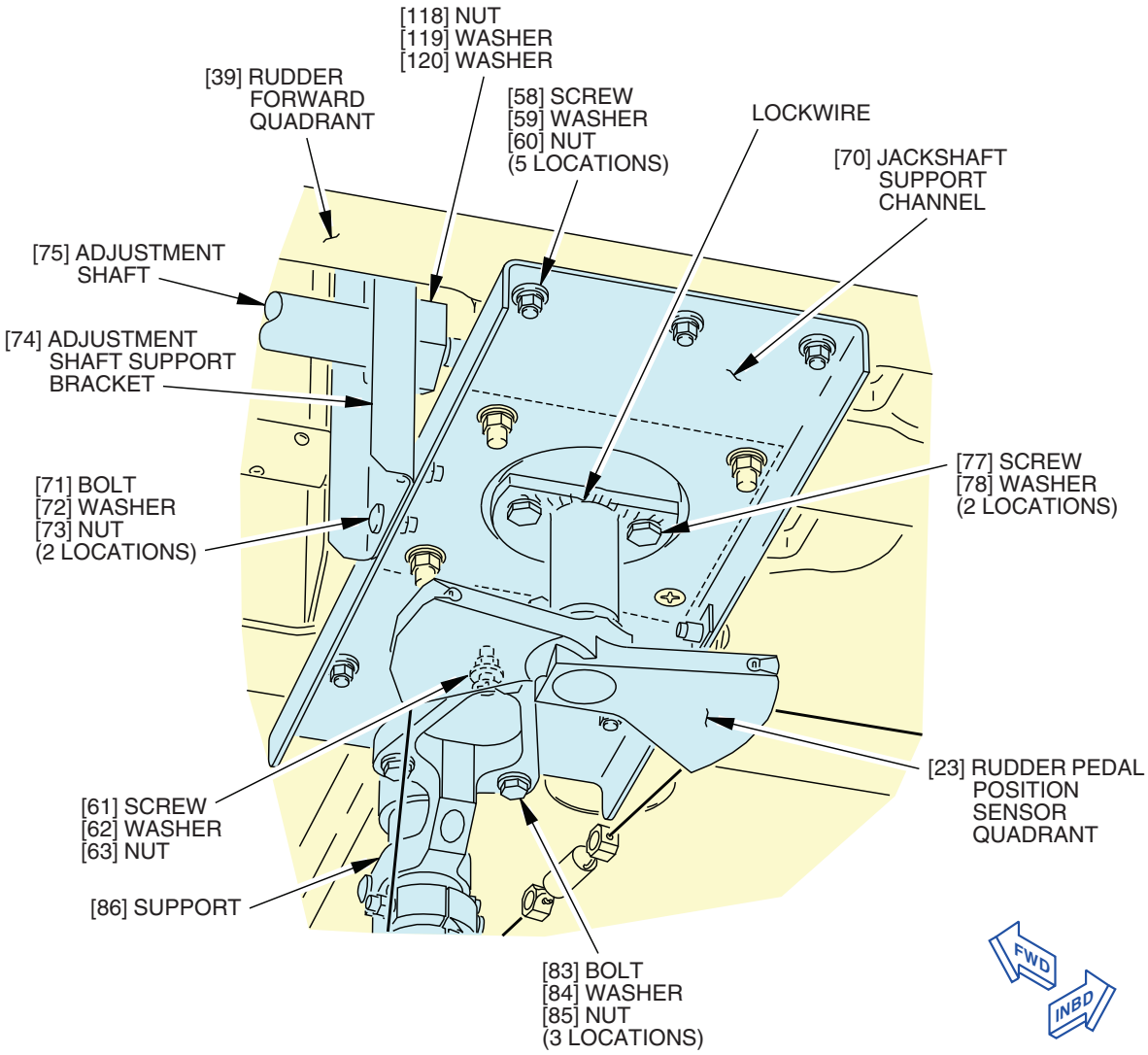
CAPTAIN'S RUDDER FORWARD QUADRANT, RUDDER PEDAL STEERING ARM AND ADJUSTMENT

I

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Rudder Forward Quadrant Installation
Figure 402/27-21-51-990-802 (Sheet 7 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



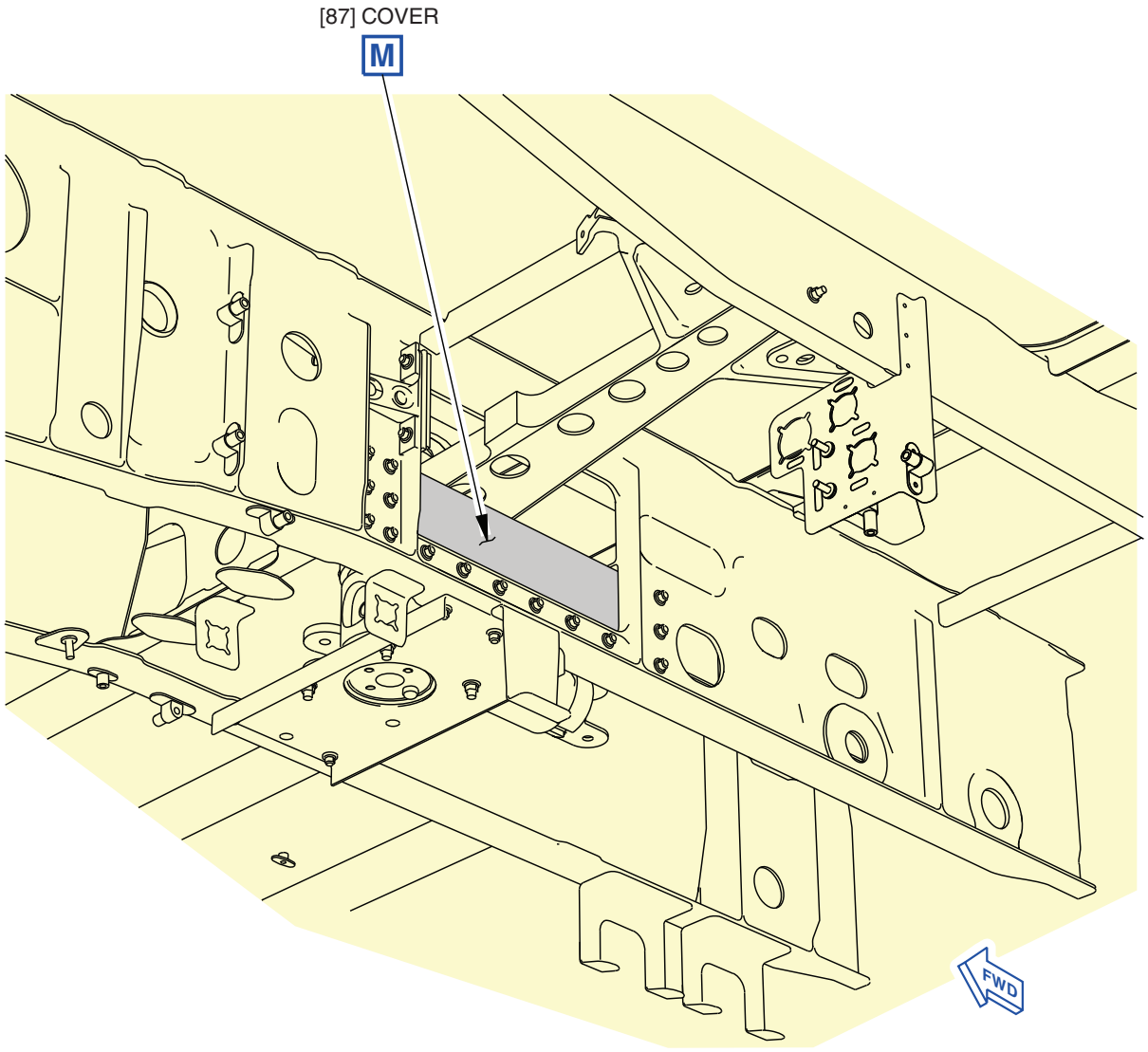
FIRST OFFICER'S RUDDER FORWARD QUADRANT, RUDDER PEDAL POSITION SENSOR QUADRANT

J

2412405 S00061530601_V2

Rudder Forward Quadrant Installation
Figure 402/27-21-51-990-802 (Sheet 8 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



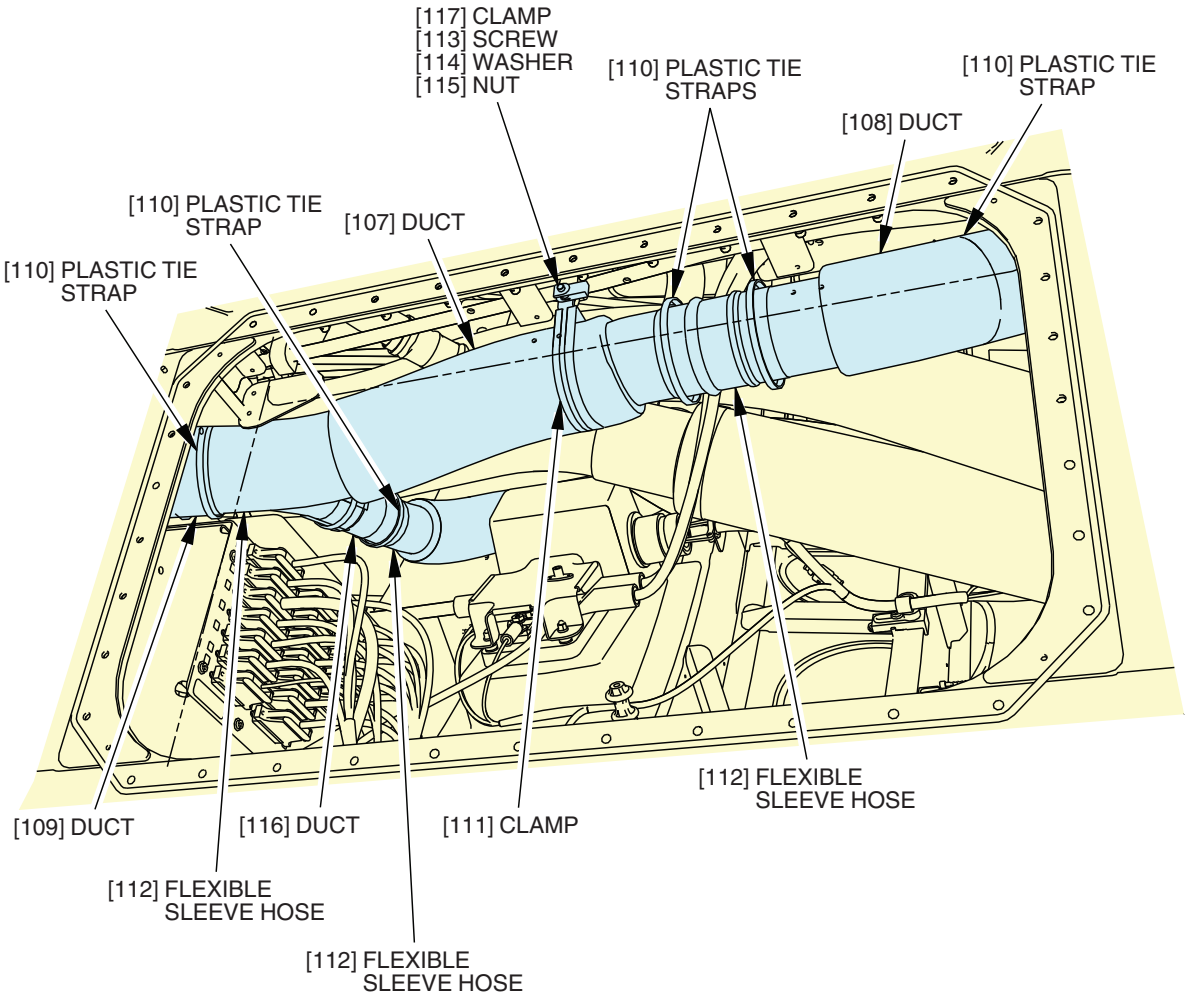
COVER ASSEMBLY
(FIRST OFFICER'S SIDE IS SHOWN, CAPTAIN'S SIDE IS OPPOSITE)

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2412406 S00061530602_V2

Rudder Forward Quadrant Installation
Figure 402/27-21-51-990-802 (Sheet 9 of 11)

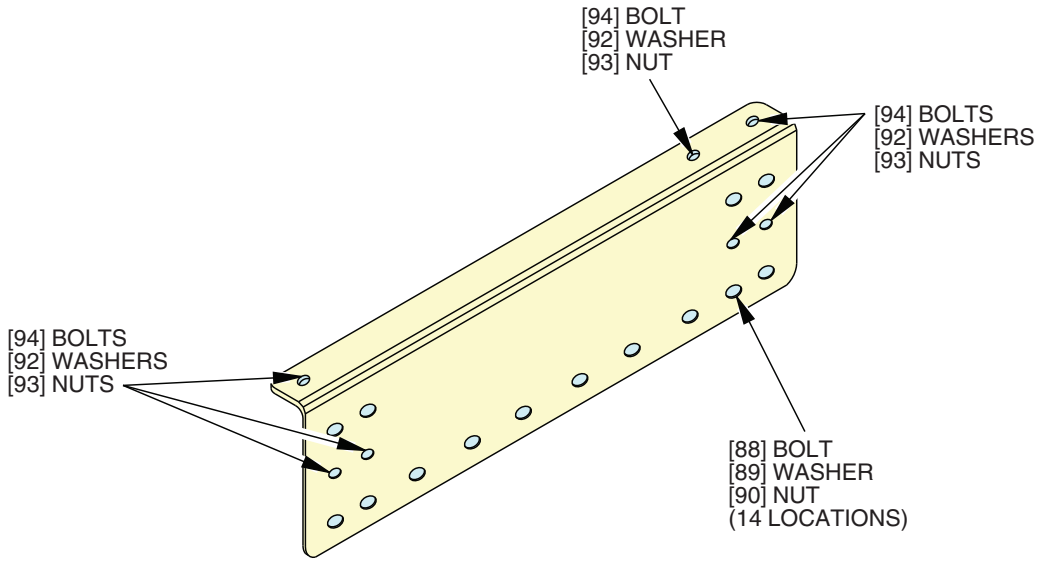
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Rudder Forward Quadrant Installation
Figure 402/27-21-51-990-802 (Sheet 10 of 11)

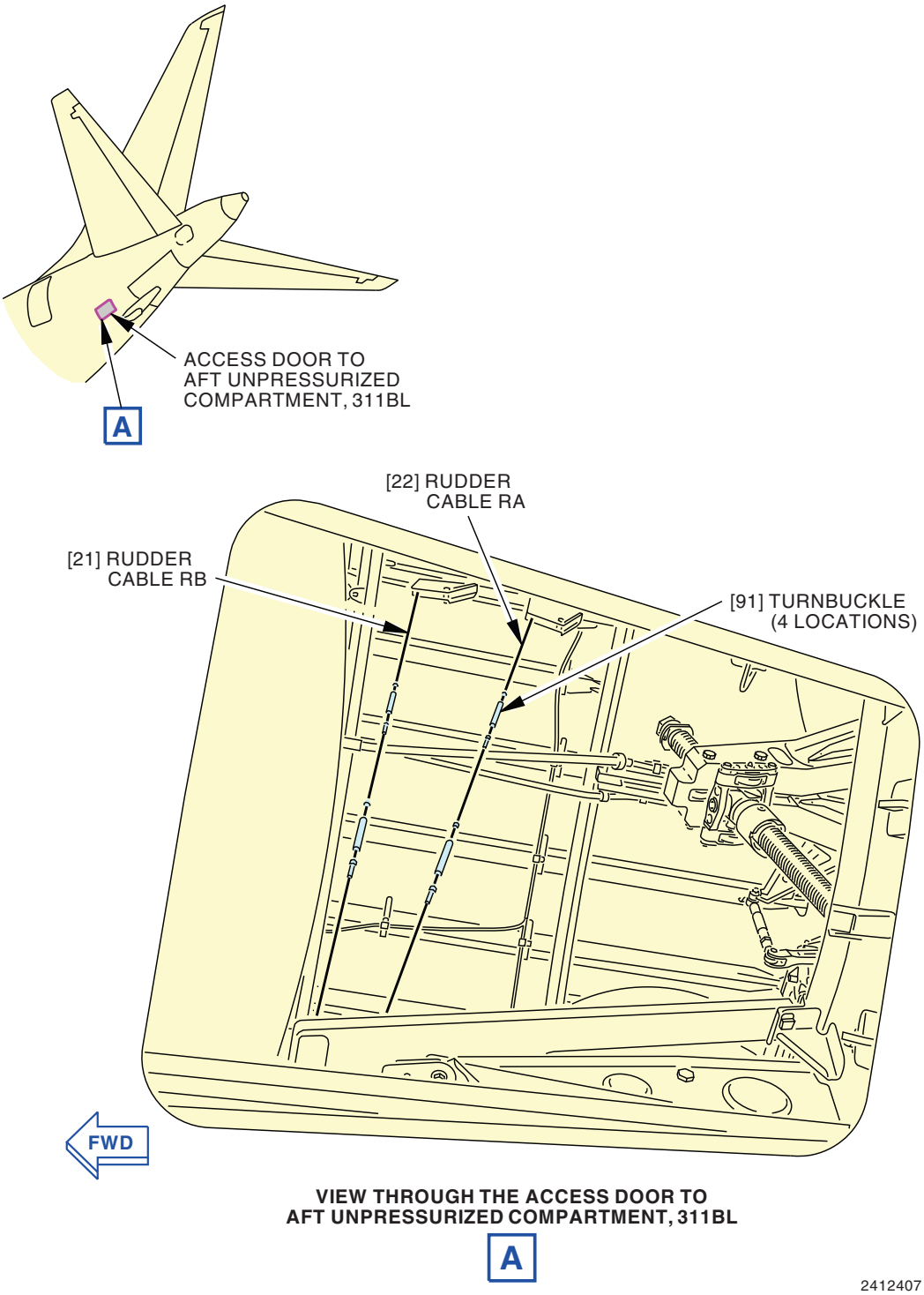
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Rudder Forward Quadrant Installation
Figure 402/27-21-51-990-802 (Sheet 11 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cables and Turnbuckles Location
Figure 403/27-21-51-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-51-400-801

3. Rudder Forward Quadrant Installation

(Figure 401, Figure 402, Figure 403)

A. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
20-10-93-400-801	Control Cable End Retaining Plate - Installation (P/B 401)
21-00-00-800-801	Supply Conditioned Air to the Airplane (P/B 201)
27-21-00-700-807	Rudder Trim System Test (P/B 501)
27-21-00-820-802	Rudder Pedals Adjustment (P/B 501)
27-21-00-820-803	Rudder Control Cables RA and RB Adjustment (P/B 501)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
31-31-42-400-801	Rudder Pedal Position Sensor Installation (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-3923	Tool - Installation, Tie Strap Part #: GS2B-E Supplier: 06383 Part #: GS4H-E Supplier: 06383 Part #: GTS-E Supplier: 06383 Part #: PPTS Supplier: 06383 Opt Part #: GS2B Supplier: 06383 Opt Part #: GS2BL Supplier: 06383 Opt Part #: GS4H Supplier: 06383 Opt Part #: GTS Supplier: 06383
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

C. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
23	Rudder pedal position sensor quadrant	31-31-42-02-065	PTF ALL
24	Rudder pedal steering arm	32-51-21-01-225	PTF ALL
27	Shim	27-21-51-02-610	PTF ALL
		27-21-51-03-380	PTF ALL
39	Rudder forward quadrant	27-21-51-02-450	PTF ALL
		27-21-51-03-140	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
49	Pin	27-21-51-02-010	PTF ALL
		27-21-51-03-005	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
311	Area Aft of Pressure Bulkhead - Left
324	Vertical Fin - Rear Spar To Trailing Edge

F. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
114AC	Fwd Nose Wheel Well Upper Access Panel
114AW	Forward Nose Wheel Well Panel
311BL	Stabilizer Trim Access Door
324CL	Vertical Fin, Access

G. Rudder Forward Quadrant Installation

SUBTASK 27-21-51-420-001

(1) Put the rudder forward quadrant [39] in its position.

(a) Put the rudder forward quadrant [39] in the upper bearing housing and the quadrant arm through the floor beam structure.

1) If necessary, move the rudder forward quadrant [39] in a forward and inboard direction to clear the floor beam structure and control cables.

SUBTASK 27-21-51-420-019

(2) Install the jackshaft support channel [70].

(a) Install the screws [58], washers [59], and nut [60].

(b) Install the screw [61], washer [62], and nut [63].

SUBTASK 27-21-51-420-008

(3) Install the bolts [71], washers [72], nuts [73], to the adjustment shaft support bracket [74].

SUBTASK 27-21-51-420-007

(4) Connect the adjustment shaft [75] to the rudder forward quadrant [39].

(a) Turn the adjustment crank for the rudder pedals clockwise to the full forward position.

(b) Install the nut [118] on adjustment shaft [75].

1) Make sure that the nut [118] run-on torque is 90 in-lb (10 N·m) - 400 in-lb (45 N·m).

(c) Remove the nut [118] from the adjustment shaft [75].

(d) Install nut [118], washer [119], washer [120], and adjustment shaft [75] to the adjustment shaft support bracket [74], and the universal joint of the rudder forward quadrant [39].

1) Make sure that the slot in the adjustment shaft [75] engages the pin on the universal joint.

2) Tighten nut [118] until it stops, plus an additional 1/4 - 1/2 turn.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Turn the adjustment crank counterclockwise to the full aft position, then clockwise to the full forward position.

1) Make sure movement of the pedals and forward quadrant assembly is smooth.

SUBTASK 27-21-51-420-020

(5) For the captain's rudder forward quadrant [39], do these steps (Figure 402):

(a) Connect the rudder pedal steering arm [24] to the rudder forward quadrant [39].

1) Install the screws [68], andwashers [69].

2) Install MS20995NC32 lockwire, G01912.

SUBTASK 27-21-51-420-002

(6) For the first officer's rudder forward quadrant [39], do these steps:

(a) Install the bolts [83], washers [84], nuts [85], and support [86] to the jackshaft support channel [70].

(b) Install the rudder pedal position sensor quadrant [23].

1) Install the screws [77] and washers [78].

2) Install MS20995NC32 lockwire, G01912.

SUBTASK 27-21-51-420-016

(7) Install the screws [52], washers [53], nuts [54], and cable guard [51] to the jackshaft support channel [70].

SUBTASK 27-21-51-480-001

(8) Install the rig pin R-3 from the rigging pin kit, SPL-1585, in the forward rig pin hole of the captain's rudder forward quadrant [39].

SUBTASK 27-21-51-480-002

(9) Install the rig pin R-4 from the rigging pin kit, SPL-1585, in the forward rig pin hole of the first officer's rudder forward quadrant [39].

SUBTASK 27-21-51-820-001

(10) Make sure that the clearance between the cable guard block [26] and rudder forward quadrant [39] is 0.030 in. (0.762 mm) - 0.050 in. (1.270 mm).

(a) If it is necessary, add or remove the shims [27].

1) Remove the nuts [29], washers [28], and bolts [25] from the cable guard block [26].

2) Add or remove the shims [27] laminations.

3) Install the bolts [25], washers [28], and nuts [29] to the cable guard block [26].

SUBTASK 27-21-51-420-018

(11) Connect the rudder pedal tension rods [40] to the rudder forward quadrant [39].

(a) If it is necessary, adjust the length of the rudder pedal tension rods [40].

(b) Install the bolts [41], washers [42], nuts [43], and new pins [44].

SUBTASK 27-21-51-420-003

(12) Temporarily install the bolt [45] to connect the bus rod [50] and rudder forward quadrant [39].

(a) If it is necessary, adjust the bus rod [50] until the rig pins R-3 and R-4 move freely.

SUBTASK 27-21-51-080-002

(13) Remove the rig pins R-3 and R-4.

SUBTASK 27-21-51-420-005

(14) For the captain's rudder forward quadrant, do these steps:

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(a) Move the captain's rudder forward quadrant arm until it touches the quadrant stop [35].

(b) Make sure that the rig pin R-3 installs easily in the aft rig pin hole.

1) If it is necessary, add or remove the shims [34].

a) Remove the nuts [38], washers [37], bolts [36], and filler [33] from the quadrant stop [35].

b) Add or remove the shims [34] laminations.

c) Install the filler [33], bolts [36], washers [37], and nuts [38] to the quadrant stop [35].

(c) Remove the rig pin R-3.

SUBTASK 27-21-51-420-006

(15) For the first officer's rudder forward quadrant, do these steps:

(a) Move the first officer's rudder forward quadrant arm until it touches the quadrant stop [35].

(b) Make sure that the rig pin R-4 installs easily in the aft rig pin hole.

1) If it is necessary, add or remove the shims [34].

a) Remove the nuts [38], washers [37], bolts [36], and filler [33] from the quadrant stop [35].

b) Add or remove the shims [34] laminations.

c) Install the filler [33], bolts [36], washers [37], and nuts [38] to the quadrant stop [35].

(c) Remove the rig pin R-4.

SUBTASK 27-21-51-480-003

(16) Install the rig pin R-3 in the forward rig pin hole of the captain's rudder forward quadrant [39].

SUBTASK 27-21-51-480-004

(17) Install the rig pin R-4 in the forward rig pin hole of the first officer's rudder forward quadrant [39].

SUBTASK 27-21-51-420-009

(18) Install the bolts [88], washers [89], nuts [90], bolts [94], washers [92], nuts [93], and cover [87] to the floor beam structure.

SUBTASK 27-21-51-860-002

(19) Put the rudder pedals to the neutral position.

(a) Turn the adjustment crank clockwise for the rudder pedals to the full forward position.

(b) Turn the adjustment crank back four turns from the full forward position, to get to the neutral position.

SUBTASK 27-21-51-820-002

(20) Do this task: Rudder Pedals Adjustment, TASK 27-21-00-820-802.

SUBTASK 27-21-51-420-010

(21) Connect the bus rod [50] to the rudder forward quadrant [39].

(a) Remove the bolt [45] from the bus rod [50].

(b) Turn the bus rod two full turns to make it shorter.

(c) Make sure that the threads of the two bus rod ends show through the holes in the bus rod.

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(d) Install the bolt [45], spacers [46], washer [47], nut [48], and new pin [49].

SUBTASK 27-21-51-420-017

(22) Connect the rudder cable RB [21] or rudder cable RA [22] to the applicable rudder forward quadrant [39].

(a) Install the bolt [30], washer [19], washer [31] (if applicable), nut [20], and cable retainer [32](TASK 20-10-93-400-801).

SUBTASK 27-21-51-420-012

(23) Do this task: Rudder Control Cables RA and RB Adjustment, TASK 27-21-00-820-803.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-51-420-013

(1) For the first officer's rudder forward quadrant, do this task: Rudder Pedal Position Sensor Installation, TASK 31-31-42-400-801.

SUBTASK 27-21-51-420-015

(2) If removed, install the duct [107] as follows (Figure 402, View L):

(a) Remove the protective covers on the open ends of duct [108], duct [109], and duct [116].

(b) Install the duct [107] to duct [108], duct [109] and duct [116].

(c) Install the flexible sleeve hoses [112] with a minimum of 0.25 in. (0.64 cm) gap between beaded ducts.

(d) Install the plastic tie straps [110] to the flexible sleeve hoses [112] using tie strap installation tool, COM-3923.

1) Strap should be 0.20 in. (0.51 cm)- 0.40 in. (1.02 cm) measured from the edge of flexible sleeve hose [112] to the adjacent edge of plastic tie strap [110].

(e) Install the clamp [111], wiring installation clamp [117], screw [113], washer [114], and nut [115].

(f) Supply conditioned air to the flight compartment (TASK 21-00-00-800-801).

(g) Make sure that there are no leaks at the joints of duct [107], duct [108], duct [109] and duct [116].

1) If there are any leaks, repair duct [107], duct [108], duct [109] or duct [116].

(h) Remove the DO NOT OPERATE tag, STD-858 and make sure that the L and R PACK switches, on the P5-10 air conditioning panel, are set to OFF.

(i) Remove the DO NOT OPERATE tag, STD-858 and make sure that the L and R RECIRC FAN switches, on the P5-10 air conditioning panel, are set to AUTO.

SUBTASK 27-21-51-080-003

(3) Remove the rig pin R-5 from the rudder feel and centering unit (Figure 401).

SUBTASK 27-21-51-080-004

(4) Make sure that the rig pins R-3 and R-4 are removed.

SUBTASK 27-21-51-860-003

(5) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

SUBTASK 27-21-51-730-001

(6) Do this task: Rudder Trim System Test, TASK 27-21-00-700-807.

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SUBTASK 27-21-51-010-006

(7) Close this access panel:

Number	Name/Location
324CL	Vertical Fin, Access

SUBTASK 27-21-51-410-001

(8) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-21-51-410-002

(9) Close this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-21-51-010-007

(10) Install these access panels:

(TASK 53-14-01-420-801)

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
114AC	Fwd Nose Wheel Well Upper Access Panel
114AW	Forward Nose Wheel Well Panel

SUBTASK 27-21-51-790-001

(11) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER FORWARD QUADRANT - INSPECTION/CHECK

1. General

A. This procedure has this inspection task:

(1) Rudder Mechanical Control Path

TASK 27-21-51-210-801

2. Rudder Mechanical Control Path Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

B. Access Panels

Number	Name/Location
112A	Forward Access Door

C. Rudder Mechanical Control Path Inspection

SUBTASK 27-21-51-010-008

(1) To get access to the rudder mechanical control path, do this task:

Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-21-51-210-001

(2) Do a detailed inspection of the rudder mechanical control path.

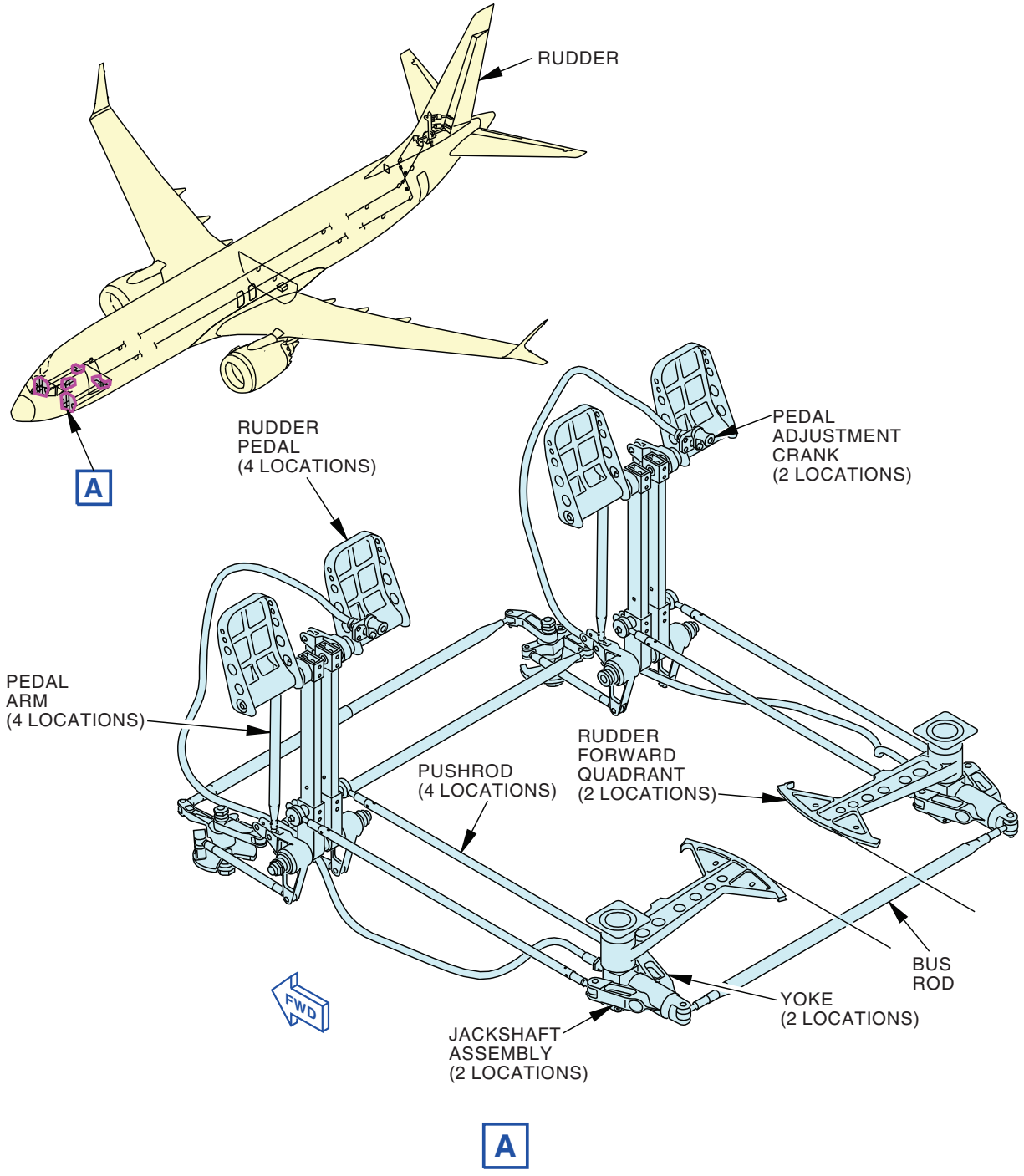
SUBTASK 27-21-51-010-009

(3) Close this access panel:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

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Rudder and Rudder Trim Control System - Rudder Pedals and Forward Quadrants
Figure 601/27-21-51-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER AFT QUADRANT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Rudder Aft Quadrant Removal

(2) Rudder Aft Quadrant

B. The rudder aft quadrant [20] includes the quadrant, the quadrant shaft [17], and the two bearing housings.

TASK 27-21-61-000-801

2. Rudder Aft Quadrant Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
311	Area Aft of Pressure Bulkhead - Left
324	Vertical Fin - Rear Spar To Trailing Edge

D. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel
311BL	Stabilizer Trim Access Door
323FL	Vertical Fin, Access
323GL	Vertical Fin, Access
324CL	Vertical Fin, Access

E. Prepare for the Removal

SUBTASK 27-21-61-860-001

(1) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

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SUBTASK 27-21-61-010-001

(2) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-21-61-010-002

(3) Remove these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

(a) Do this task: Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801.

SUBTASK 27-21-61-480-001

(4) Install the rig pin R-3 or R-4 from the rigging pin kit, SPL-1585, in the forward rig pin hole of the captain's or first officer's rudder forward quadrant (Figure 401).

SUBTASK 27-21-61-010-003

(5) Do these steps to get access to the rudder aft quadrant [20]:

Remove these access panels:

Number	Name/Location
323FL	Vertical Fin, Access
323GL	Vertical Fin, Access

SUBTASK 27-21-61-010-004

(6) Do these steps to get access to the cable turnbuckles [3]:

Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-21-61-010-005

(7) Open this access panel:

Number	Name/Location
324CL	Vertical Fin, Access

F. Rudder Aft Quadrant Removal

(Figure 402)

SUBTASK 27-21-61-020-001

(1) Loosen the cable RA [2] and cable RB [1] at the turnbuckles [3].

SUBTASK 27-21-61-020-002

(2) Remove the bolts [9], washer [28] (if applicable), washers [11], and nuts [12] to remove the cable retainer [10] from the rudder aft quadrant [20].

SUBTASK 27-21-61-020-003

(3) Disconnect the cable RA [2] and cable RB [1] from the rudder aft quadrant [20].

SUBTASK 27-21-61-020-004

(4) Remove the nut [15], washer [14], and bolt [13] which connect the control rod [16] to the rudder aft quadrant [20].

SUBTASK 27-21-61-020-005

(5) Do these steps to remove the rudder aft quadrant [20]:

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(a) Remove the bolts [25] and washers [26] which attach the left bearing housing [19] to the airplane structure.

(b) Slide the quadrant shaft [17] towards the left until the end of the quadrant shaft [17] is clear of the hole in the right side mounting surface.

(c) Remove the left bearing housing [19] and plate [18] from the quadrant shaft [17].

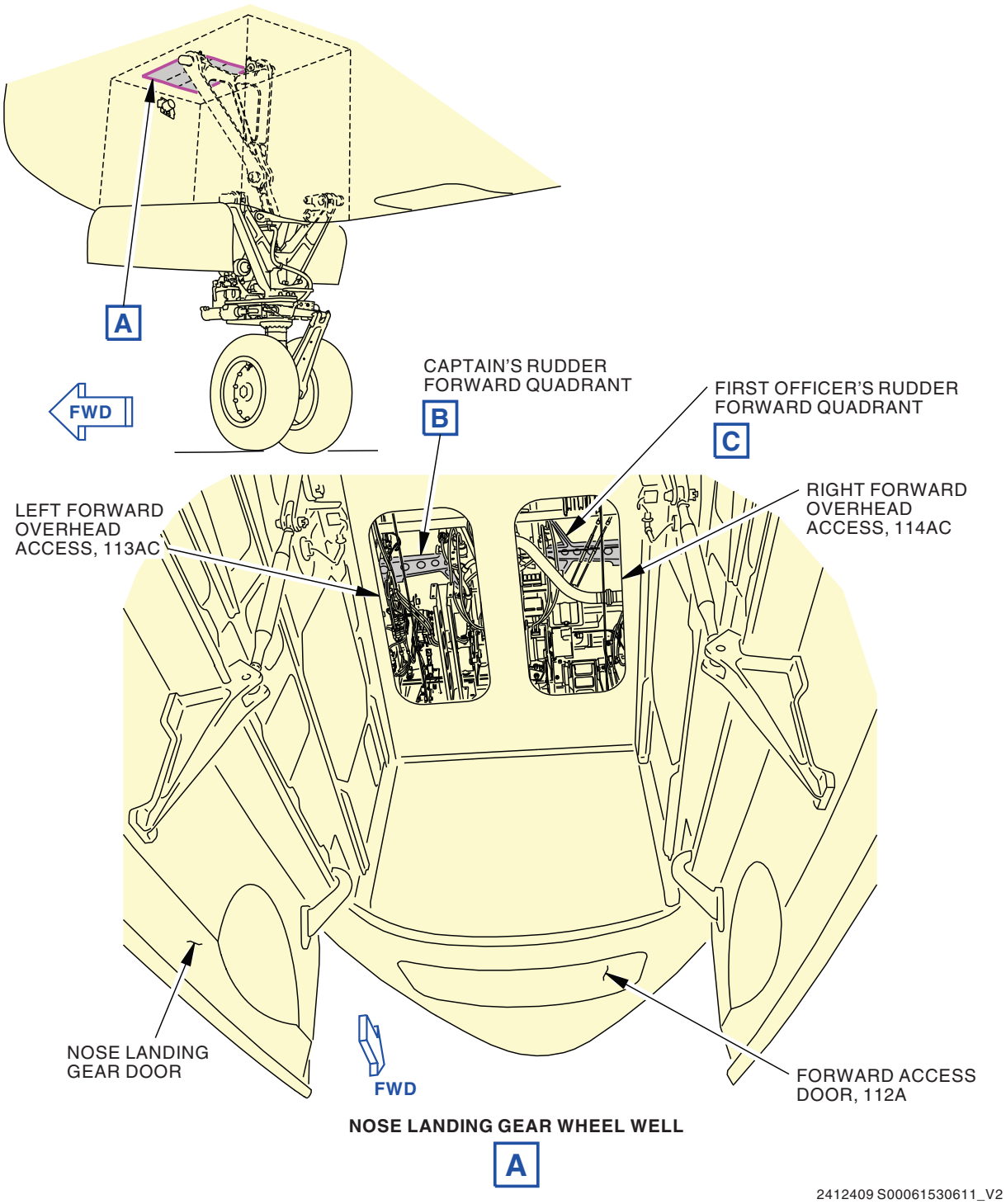
(d) Remove the nuts [7], washers [6], washers [5], bolts [4], and spacers [8] which attach the rudder aft quadrant [20] to the quadrant shaft [17].

(e) Remove the rudder aft quadrant [20] from the quadrant shaft [17].

(f) Remove the rudder aft quadrant [20], quadrant shaft [17], right bearing housing [27], and plate [18] from the airplane.

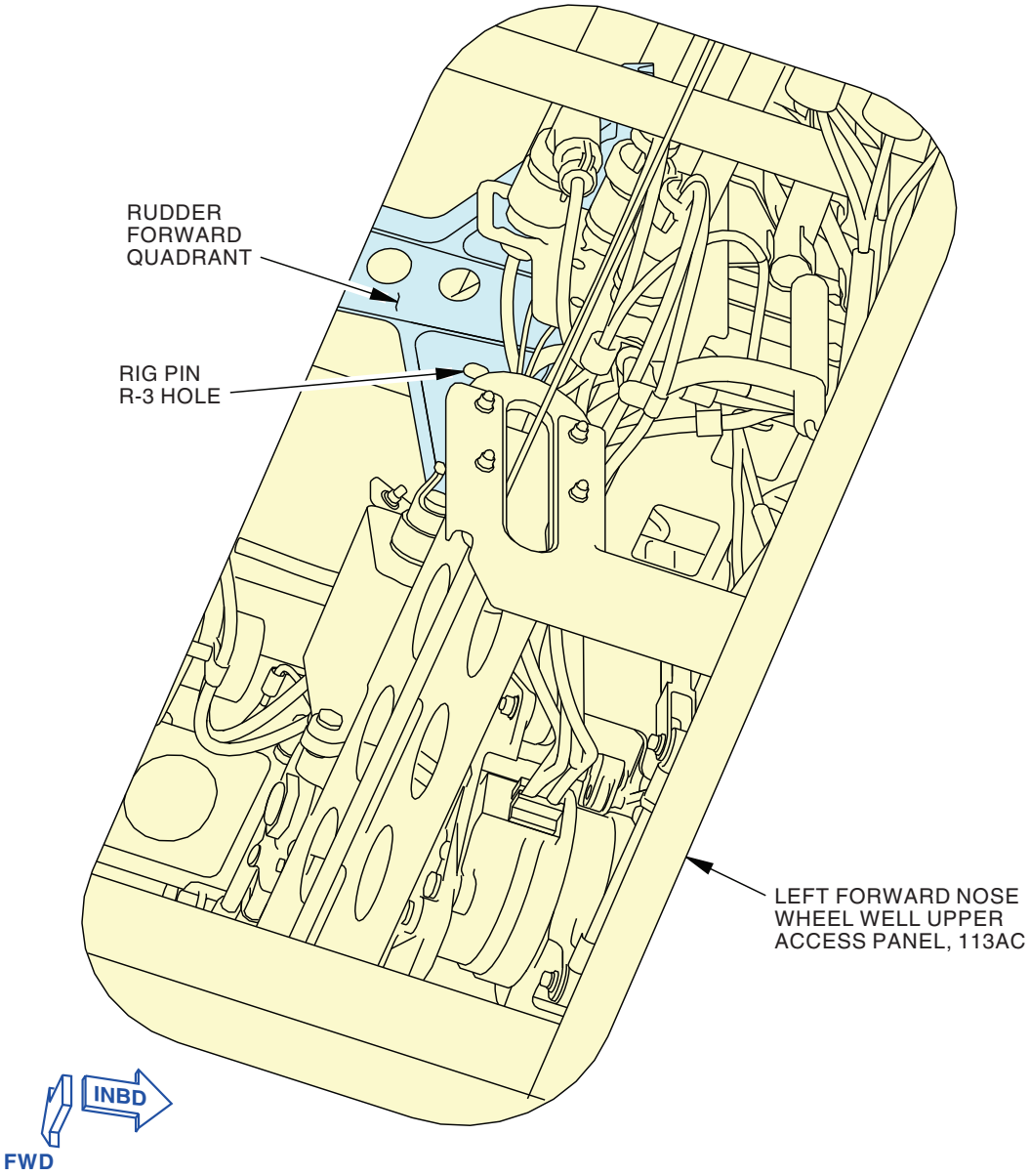
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Forward Quadrant Rig Pin Installation
Figure 401/27-21-61-990-801 (Sheet 1 of 3)

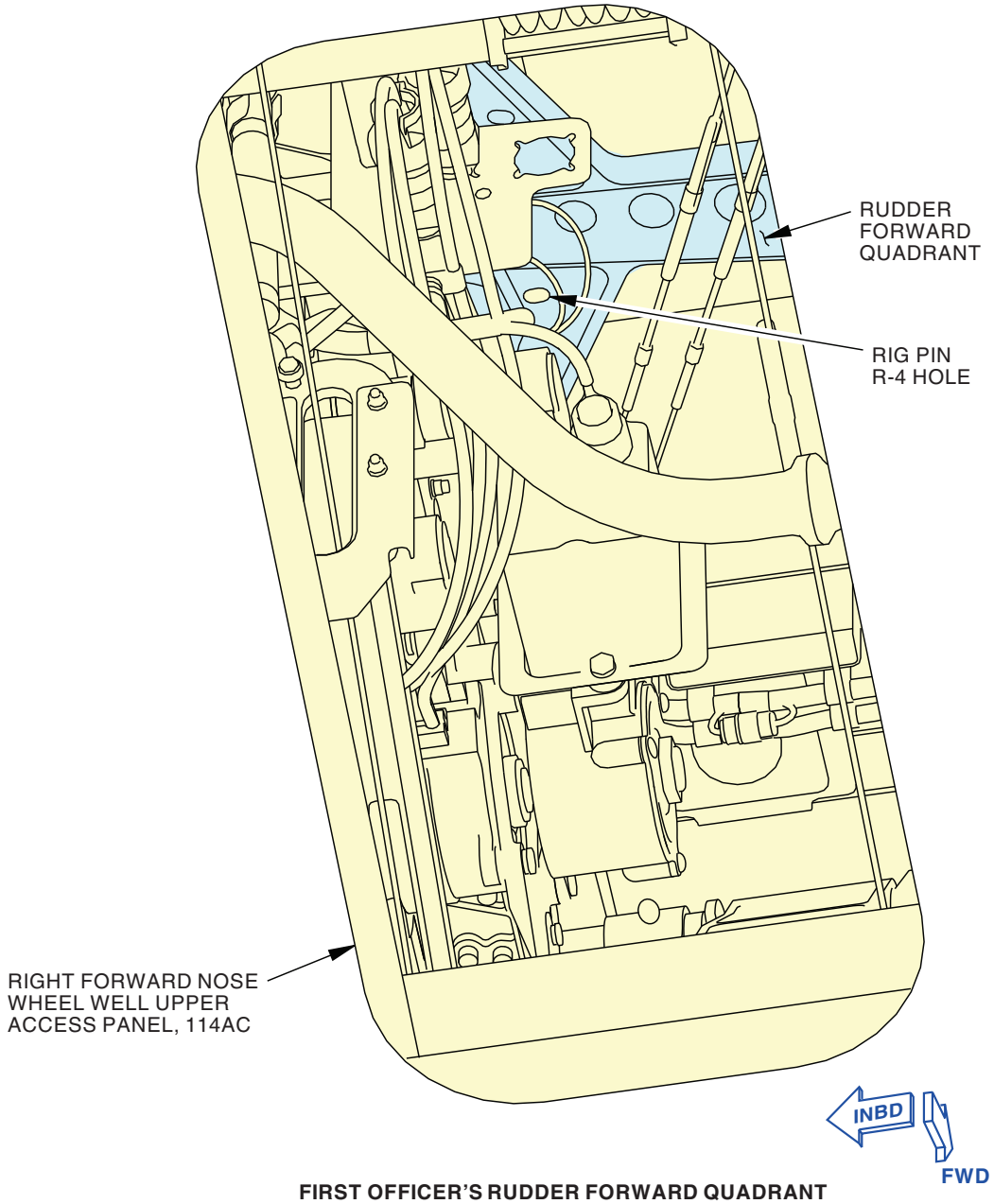
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Rudder Forward Quadrant Rig Pin Installation
Figure 401/27-21-61-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



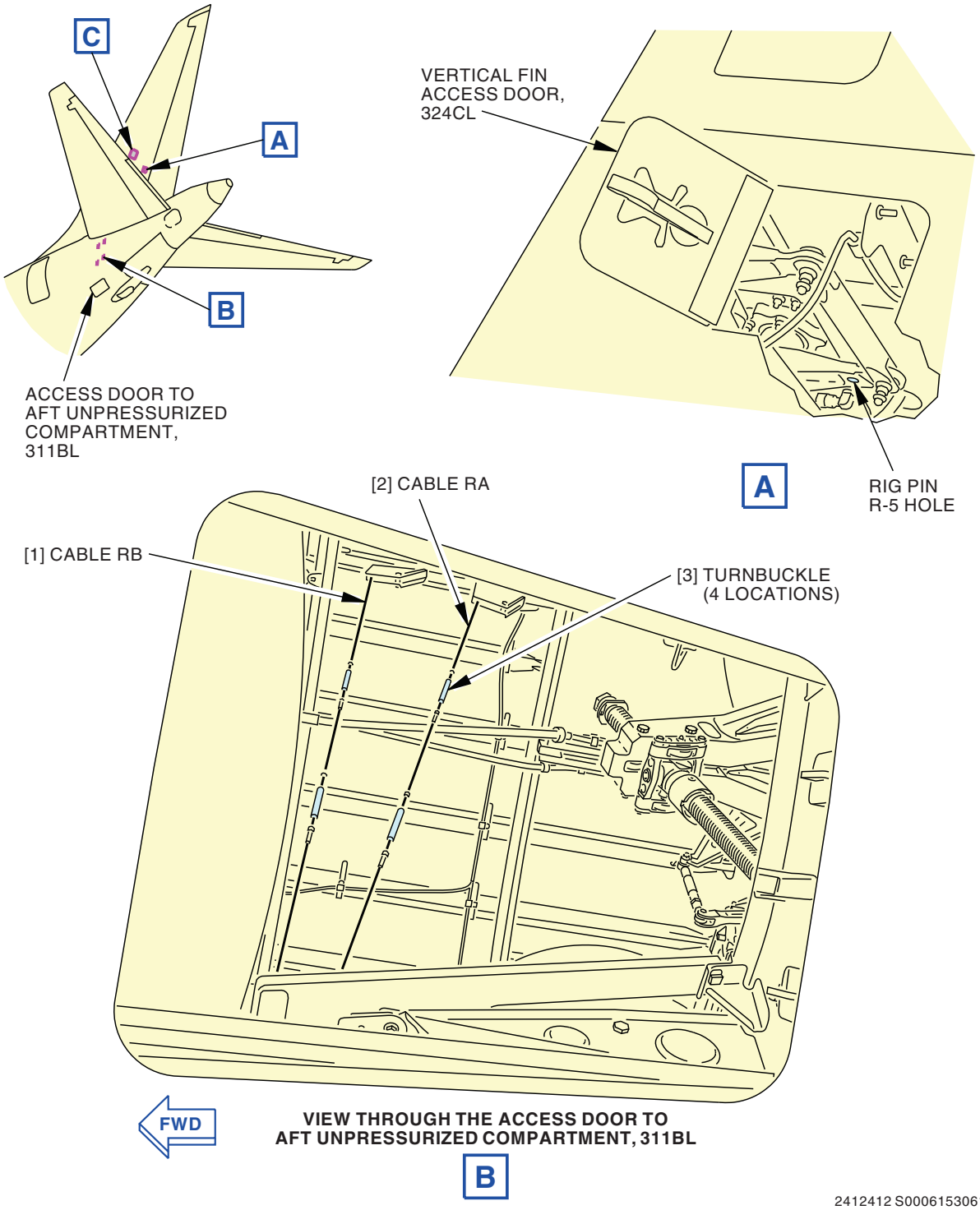
FIRST OFFICER'S RUDDER FORWARD QUADRANT

C

2412411 S00061530613_V2

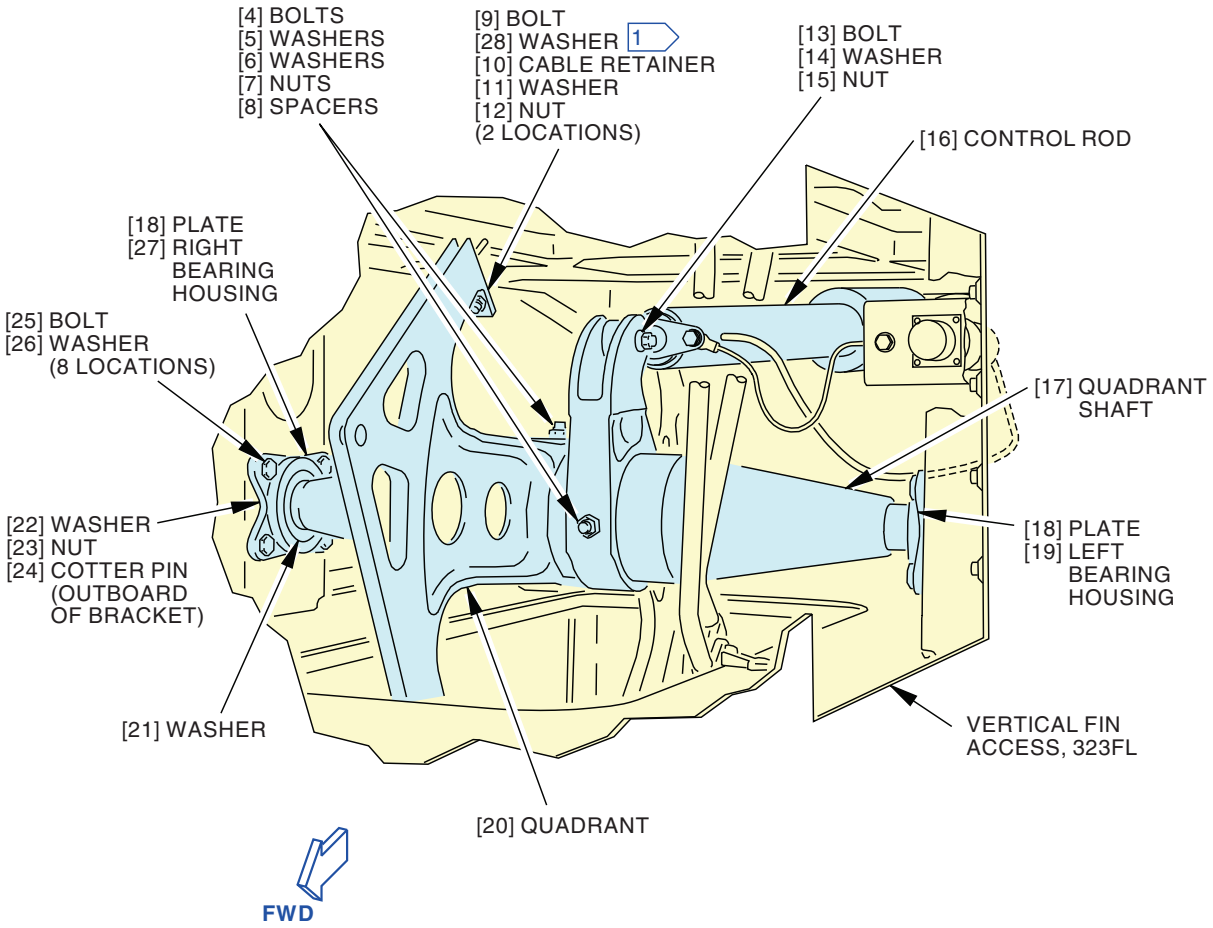
Rudder Forward Quadrant Rig Pin Installation
Figure 401/27-21-61-990-801 (Sheet 3 of 3)

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Rudder Aft Quadrant Installation
Figure 402/27-21-61-990-802 (Sheet 1 of 2)

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RUDDER AFT QUADRANT

C

1 IF APPLICABLE

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Rudder Aft Quadrant Installation
Figure 402/27-21-61-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-61-400-801

3. Rudder Aft Quadrant Installation

(Figure 401, Figure 402)

A. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
20-10-93-400-801	Control Cable End Retaining Plate - Installation (P/B 401)
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-820-803	Rudder Control Cables RA and RB Adjustment (P/B 501)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
311	Area Aft of Pressure Bulkhead - Left
324	Vertical Fin - Rear Spar To Trailing Edge

E. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel
311BL	Stabilizer Trim Access Door
323FL	Vertical Fin, Access
323GL	Vertical Fin, Access
324CL	Vertical Fin, Access

F. Rudder Aft Quadrant Installation

(Figure 402)

SUBTASK 27-21-61-860-002

(1) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-61-420-001

(2) Do these steps to install the rudder aft quadrant [20]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put the washer [21], plate [18], and the right bearing housing [27] on the right side of the quadrant shaft [17].

1) Make sure the countersunk side of the washer [21] is toward the center of the quadrant shaft [17].

(b) Install the washer [22] and nut [23] on the right side of the quadrant shaft [17].

1) Tighten the nut [23] to 190 in-lb (21 N·m) - 280 in-lb (32 N·m).

(c) Install a new cotter pin [24].

(d) Put all the parts through the access hole.

(e) Slide the rudder aft quadrant [20] onto the quadrant shaft [17].

(f) Install the bolts [4], spacers [8], washers [5], washers [6], and nuts [7] to attach the rudder aft quadrant [20] to the quadrant shaft [17].

(g) Put the left bearing housing [19] on the left side of the quadrant shaft [17].

(h) Put the quadrant shaft [17] and plate [18] in its position.

(i) Install the bolts [25] and washers [26] to attach the left bearing housing [19] and right bearing housing [27] to the airplane structure.

SUBTASK 27-21-61-420-002

(3) Turn the quadrant shaft [17].

(a) Make sure that the quadrant shaft [17] moves freely in the left bearing housing [19] and right bearing housing [27].

SUBTASK 27-21-61-420-003

(4) Connect the cable RA [2] and cable RB [1] to the rudder aft quadrant [20].

(a) Install the cable retainer [10], bolts [9], washer [28] (if applicable), washers [11], and nuts [12] (TASK 20-10-93-400-801).

SUBTASK 27-21-61-420-004

(5) Tighten the turnbuckles [3] until the cable RA [2] and cable RB [1] are not loose.

SUBTASK 27-21-61-420-005

(6) Install the bolt [13], washer [14], and nut [15] to connect the control rod [16] to the rudder aft quadrant [20].

SUBTASK 27-21-61-480-002

(7) Install the rig pin R-5 from the rigging pin kit, SPL-1585, to the rudder feel and centering unit (Figure 402).

SUBTASK 27-21-61-820-001

(8) Adjust the cable RA [2] and cable RB [1] (TASK 27-21-00-820-803).

SUBTASK 27-21-61-080-001

(9) Remove the rig pins R-3 and R-4 from the rudder forward quadrants (Figure 401).

SUBTASK 27-21-61-080-002

(10) Remove the rig pins R-5 from the rudder feel and centering unit.

SUBTASK 27-21-61-710-001

(11) Do this task: Rudder Travel Test, TASK 27-21-00-700-805.

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G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-61-410-001

(1) Close this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-21-61-940-001

(2) Install these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

(a) Do this task: Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801.

(b) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

SUBTASK 27-21-61-410-002

(3) Close this access panel:

Number	Name/Location
324CL	Vertical Fin, Access

SUBTASK 27-21-61-410-003

(4) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-21-61-410-004

(5) Install these access panels:

Number	Name/Location
323FL	Vertical Fin, Access
323GL	Vertical Fin, Access

SUBTASK 27-21-61-390-001

(6) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER AFT QUADRANT - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Detail Visual Inspection of the Rudder Aft Quadrant components

(2) General Visual Inspection of the Rudder Aft Quadrant components.

TASK 27-21-61-210-801

2. Rudder Aft Quadrant Detail Visual Inspection

(Figure 601, Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II

B. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

C. Access Panels

Number	Name/Location
323FL	Vertical Fin, Access
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access

D. Rudder Aft Quadrant Detail Visual Inspection

SUBTASK 27-21-61-010-006

(1) Remove these access panels:

Number	Name/Location
323FL	Vertical Fin, Access
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access

SUBTASK 27-21-61-210-001

(2) Do a detail visual inspection of the Rudder Aft Quadrant, Torque Tube Assembly, Feel and Centering Unit, and associated Input/Output rods.

(a) Aft quadrant

(b) Aft quadrant output rod

(c) Aft quadrant bearings

(d) Aft quadrant shaft

(e) Rudder torque tube

(f) Rudder torque tube bearings

(g) Rudder torque tube cranks

(h) Rudder torque tube input rods

(i) Feel and centering unit

(j) Feel and centering unit output rod

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(k) Trim Actuator attachment points.

SUBTASK 27-21-61-010-007

(3) Install these access panels:

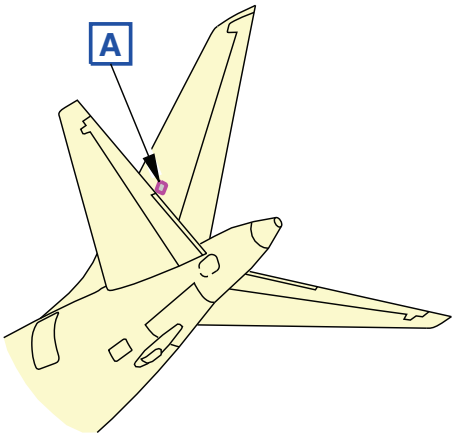
Number	Name/Location
323FL	Vertical Fin, Access
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access

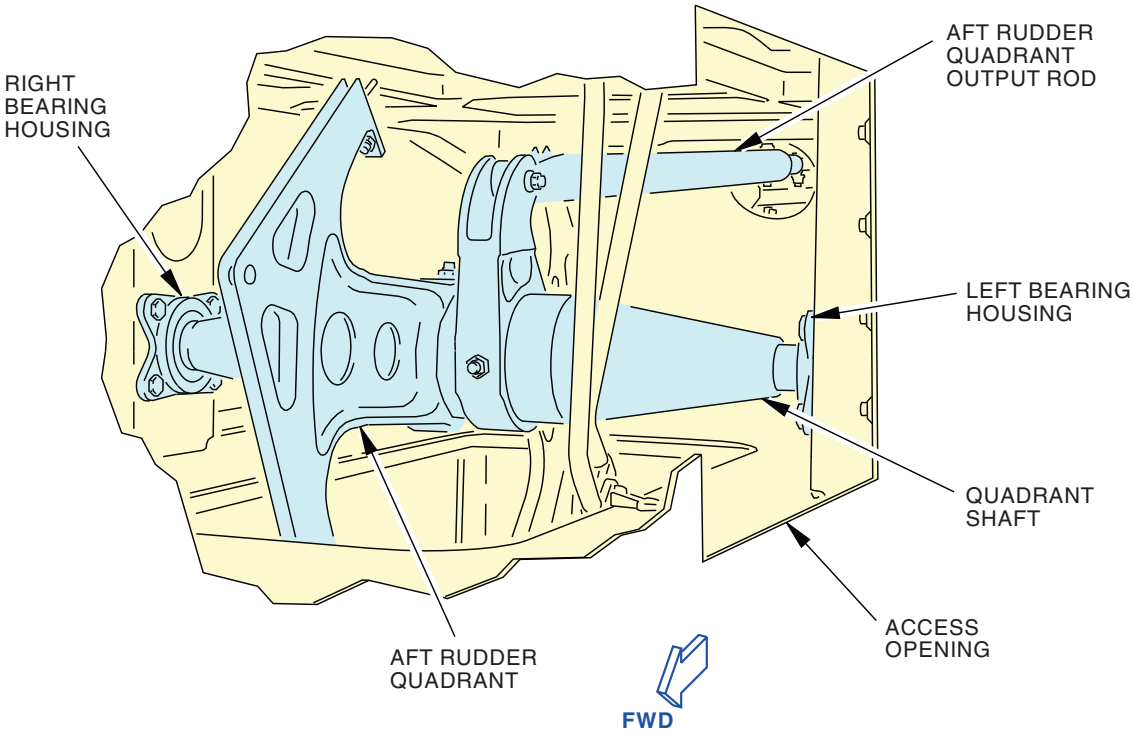
SUBTASK 27-21-61-390-002

(4) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

——— END OF TASK ———

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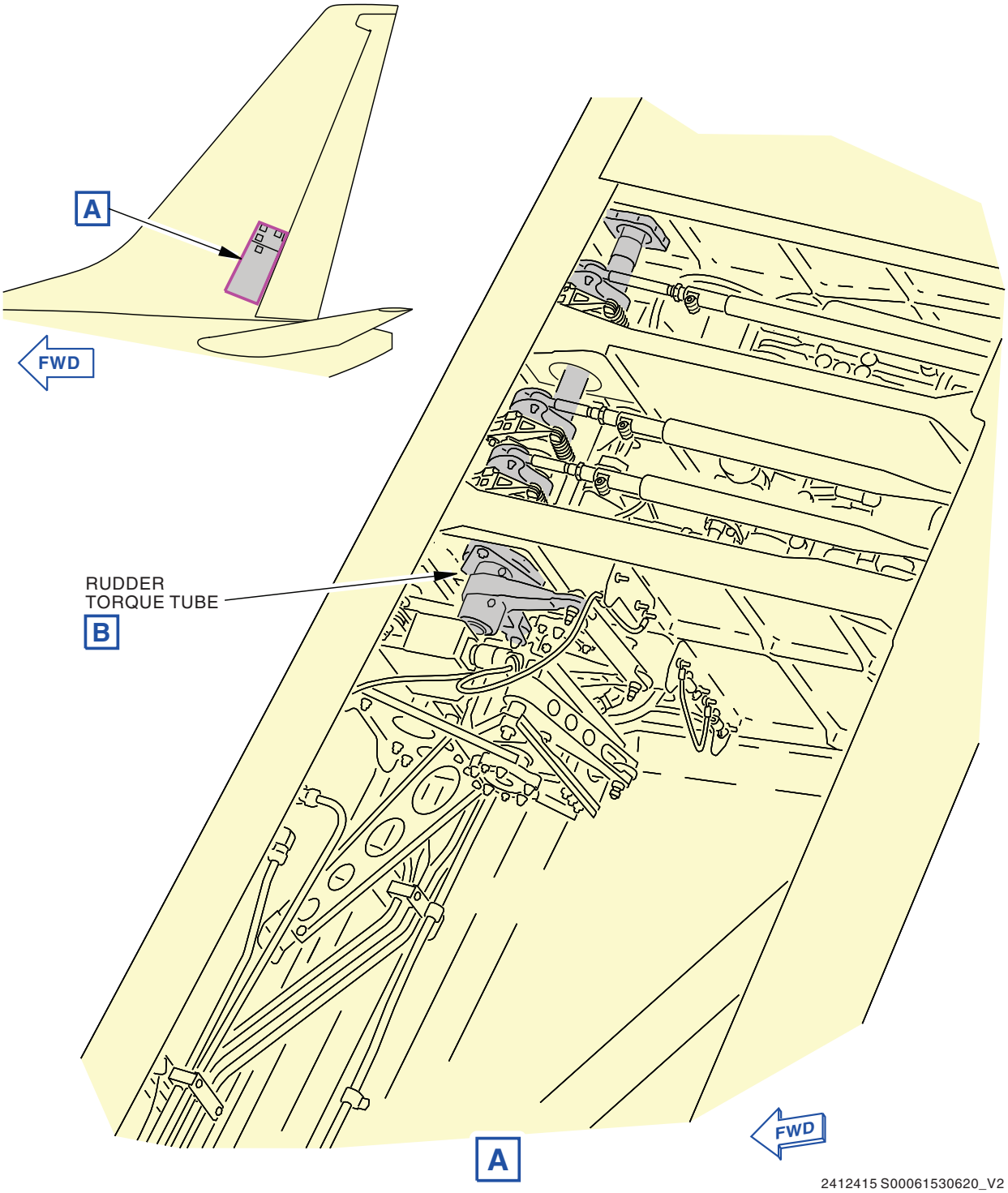




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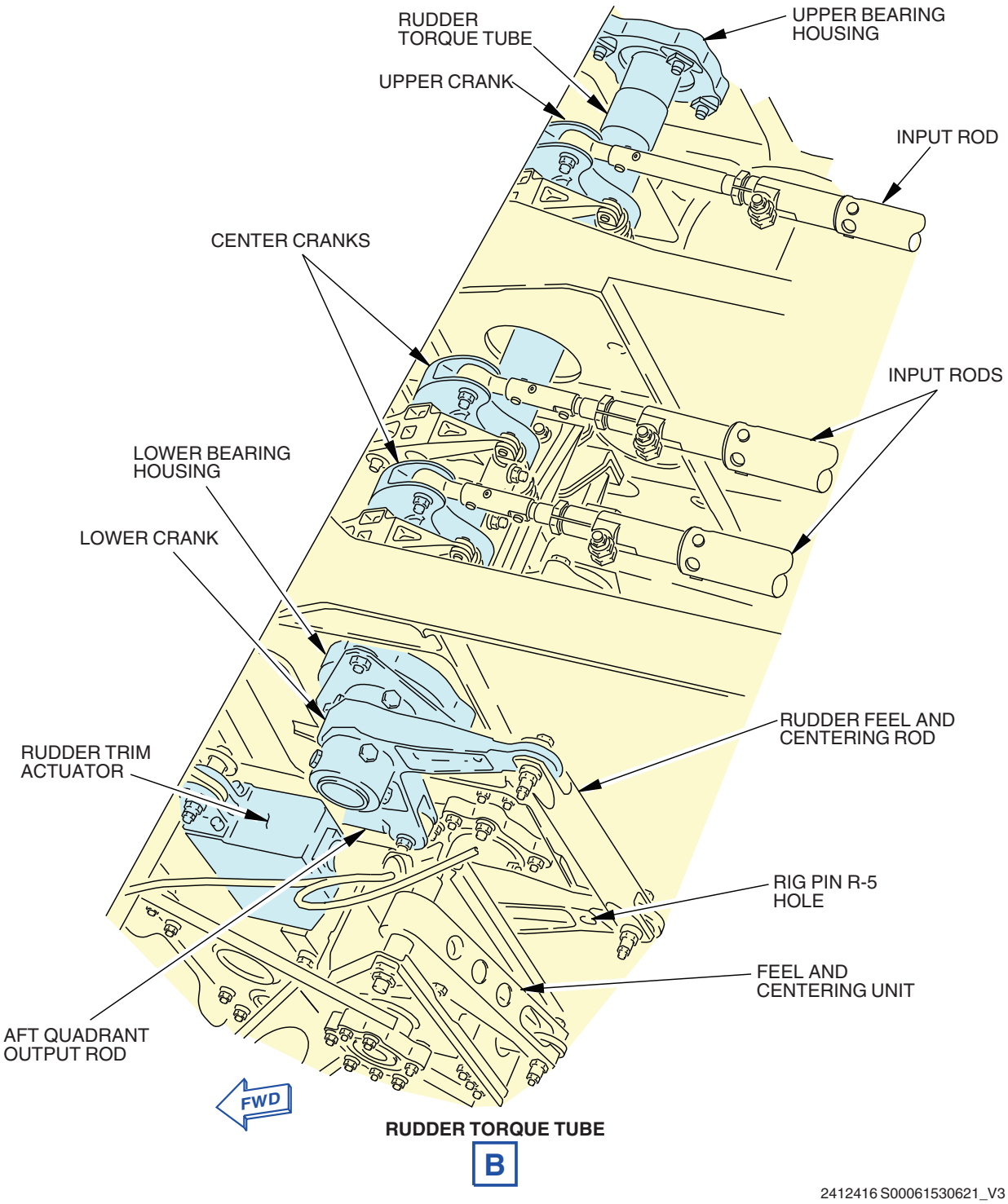
Aft Rudder Quadrant Visual Inspection
Figure 601/27-21-61-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Torque Tube Visual Inspection
Figure 602/27-21-61-990-804 (Sheet 1 of 2)

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Rudder Torque Tube Visual Inspection
Figure 602/27-21-61-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-61-210-802

3. Rudder Aft Quadrant General Visual Inspection

(Figure 601, Figure 602)

A. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II

B. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

C. Access Panels

Number	Name/Location
323CL	Vertical Fin, Rear Spar Access Door
323FL	Vertical Fin, Access
324DL	Trailing Edge Access

D. Rudder Aft Quadrant General Visual Inspection

SUBTASK 27-21-61-010-008

(1) Open these access panels:

Number	Name/Location
323CL	Vertical Fin, Rear Spar Access Door
323FL	Vertical Fin, Access
324DL	Trailing Edge Access

SUBTASK 27-21-61-210-002

(2) Do a general visual inspection of the Rudder Aft Quadrant, Torque Tube Assembly, Feel and Centering Unit, and associated Input/Output rods.:

(a) Aft quadrant

(b) Aft quadrant output rod

(c) Aft quadrant bearings

(d) Aft quadrant shaft

(e) Rudder torque tube

(f) Rudder torque tube bearings

(g) Rudder torque tube cranks

(h) Rudder torque tube input rods

(i) Feel and centering unit.

(j) Feel and centering unit output rod.

SUBTASK 27-21-61-010-009

(3) Close these access panels:

Number	Name/Location
323CL	Vertical Fin, Rear Spar Access Door
323FL	Vertical Fin, Access
324DL	Trailing Edge Access

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-61-390-003

(4) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER CONTROL TORQUE TUBE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the rudder control torque tube

(2) An installation of the rudder control torque tube.

TASK 27-21-71-000-801

2. Rudder Control Torque Tube Removal

(Figure 401)

A. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-87-000-801	Rudder Trim Actuator Removal (P/B 401)

B. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

C. Access Panels

Number	Name/Location
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access

D. Prepare for the Removal

SUBTASK 27-21-71-860-001

(1) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-71-010-001

(2) Remove these access panels:

Number	Name/Location
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access

E. Rudder Control Torque Tube Removal

SUBTASK 27-21-71-020-001

(1) Remove the nut [33], washer [32], washer [31], and bolt [30] to disconnect the standby rudder PCU control rod [2] from the upper crank [34].

SUBTASK 27-21-71-020-002

(2) Remove the nuts [33], washers [32], washers [31], and bolts [30] to disconnect the main rudder PCU control rod [3] from the center cranks [29].

SUBTASK 27-21-71-020-003

(3) Remove the nut [11], washer [10], and bolt [9] to disconnect the aft quadrant control rod [12] from the lower crank [20].

SUBTASK 27-21-71-020-004

(4) Remove the nut [7], nut [6], washer [5], and bolt [4] to disconnect the control rod [8] from the lower crank [20].

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SUBTASK 27-21-71-020-005

(5) Remove the rudder trim actuator [18] (TASK 27-21-87-000-801).

SUBTASK 27-21-71-020-006

(6) Do these steps to remove the rudder control torque tube [1]:

(a) Use a marker or pen to make marks to identify the installed positions of the center cranks [29] and upper crank [34].

(b) Remove the nuts [27], washers [26], washers [25], bolt [24], and bolt [28] that attach the upper crank [34] to the rudder control torque tube [1].

(c) Remove the nuts [27], washers [26], washers [25], bolts [24], and bolts [28] that attach the center cranks [29] to the rudder control torque tube [1].

(d) Remove the nuts [39], washers [38], washers [37], and bolts [36] that attach the lower bearing housing [21] to the airplane structure.

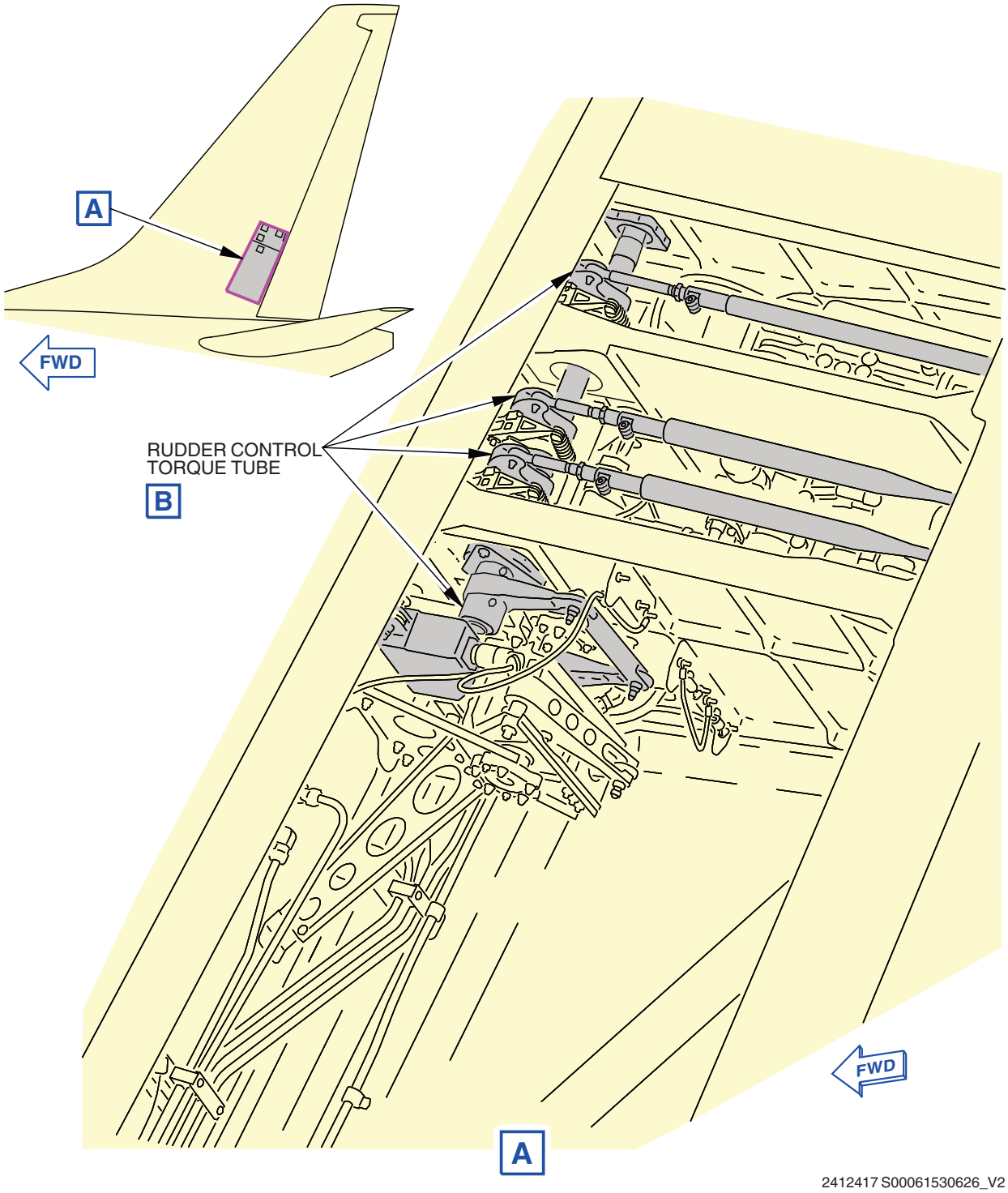
NOTE: Do not remove the lower crank from the rudder control torque tube.

(e) Pull the rudder control torque tube [1] down through the upper bearing housing [35], upper crank [34], and center crank [29].

(f) Remove the rudder control torque tube [1], upper crank [34], and center cranks [29] from the airplane.

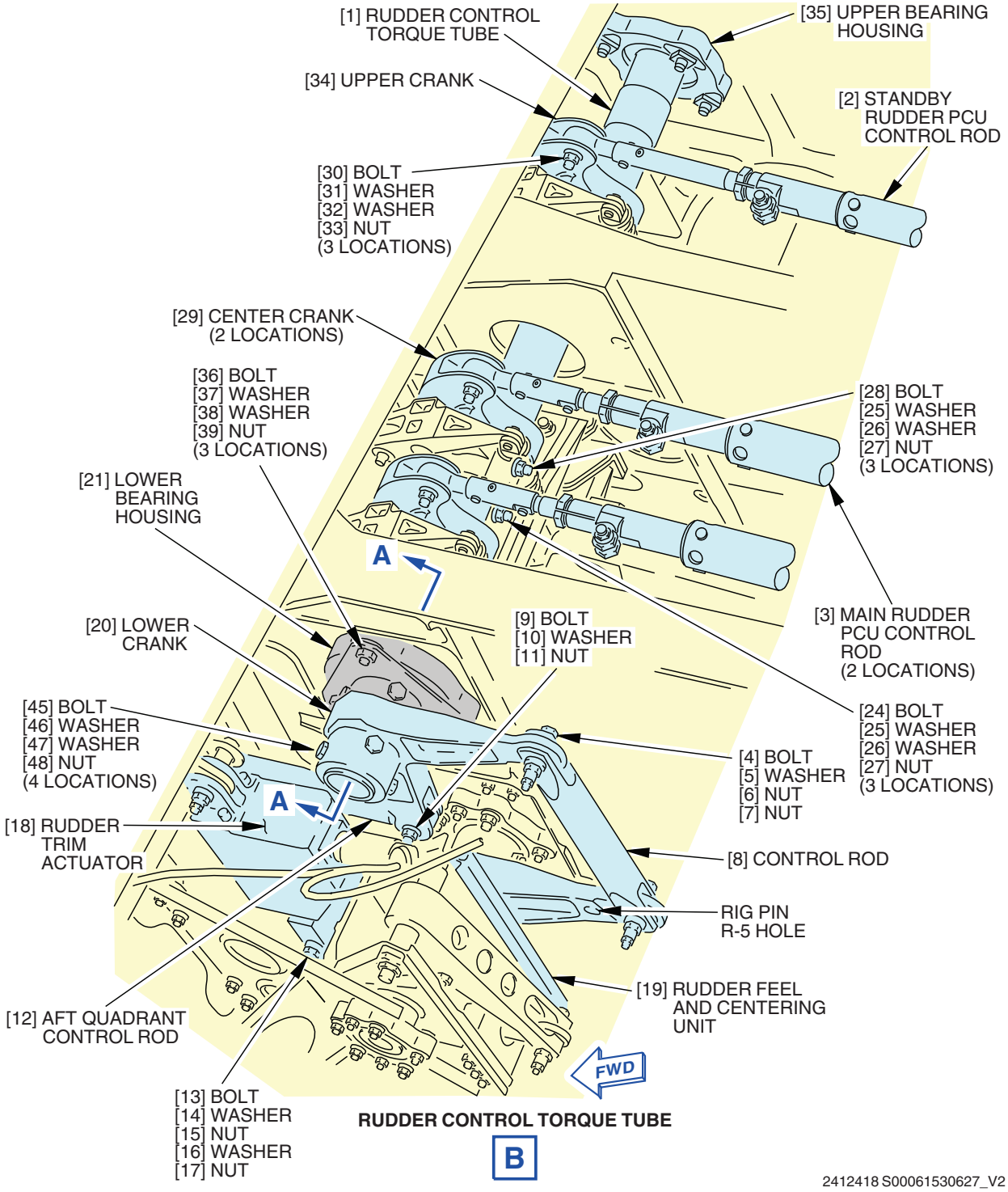
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



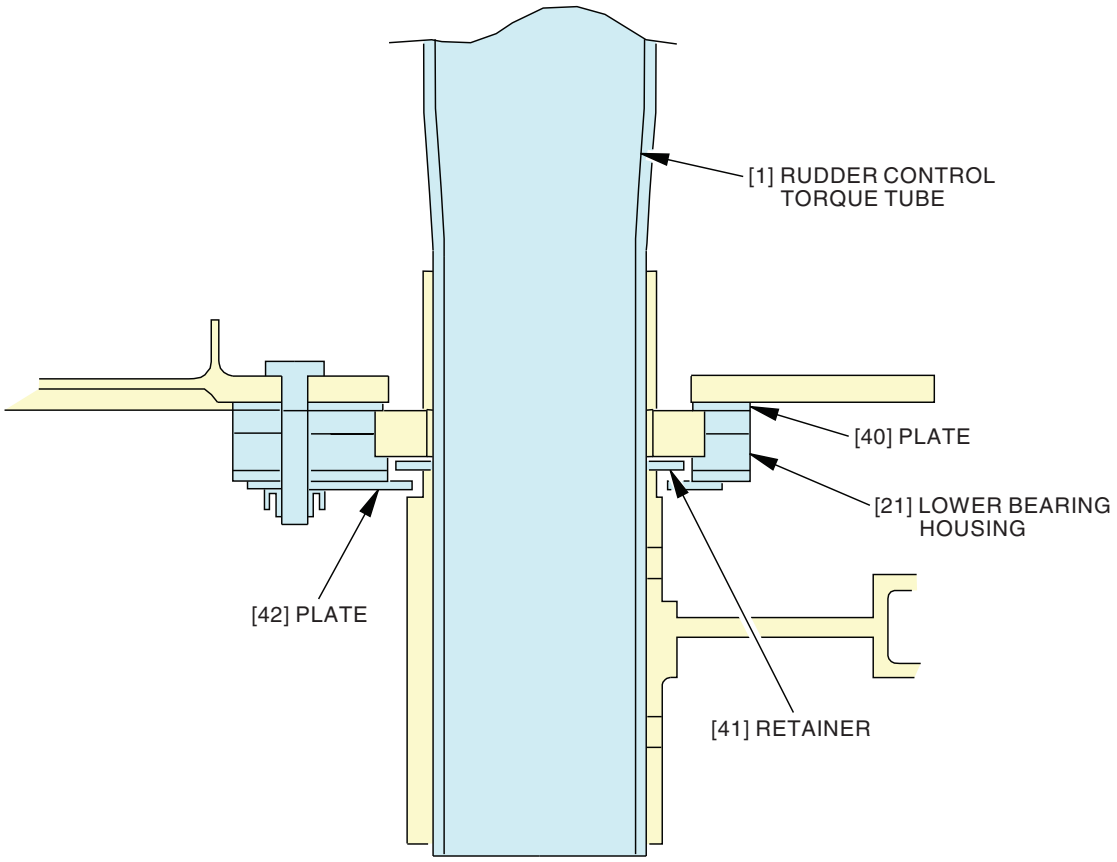
Rudder Control Torque Tube Installation
Figure 401/27-21-71-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Control Torque Tube Installation
Figure 401/27-21-71-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A

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Rudder Control Torque Tube Installation
Figure 401/27-21-71-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-71-400-801

3. Rudder Control Torque Tube Installation

(Figure 401)

A. References

Reference	Title
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-820-804	Control Rod for Main Rudder Power Control Unit Adjustment (P/B 501)
27-21-00-820-805	Control Rod for Standby Rudder Power Control Unit Adjustment (P/B 501)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-21-87-400-801	Rudder Trim Actuator Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Rudder control torque tube	27-21-71-01-055	PTF ALL

D. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

E. Access Panels

Number	Name/Location
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access

F. Rudder Control Torque Tube Installation

SUBTASK 27-21-71-420-001

(1) Do these steps to install the rudder control torque tube [1]:

(a) Make sure that the plate [40], lower bearing housing [21], retainer [41], and plate [42] are in their positions on the rudder control torque tube [1].

(b) Put the rudder control torque tube [1] through the structural cutouts, center cranks [29], upper crank [34], and upper bearing housing [35].

(c) Install the bolts [36], washers [37], washers [38], and nuts [39] to attach the lower bearing housing [21] to the airplane structure.

(d) Do these steps to install the center cranks [29] on the rudder control torque tube [1]:

1) Place the center cranks [29] in the original position on the rudder control torque tube [1].

2) Install the bolts [24], bolts [28], washers [25], washers [26], and nuts [27] to attach the center cranks [29] to the rudder control torque tube [1].

3) Tighten the nuts [27] to 15 in-lb (1.7 N·m) - 25 in-lb (2.8 N·m) above the run-on torque.

(e) Do these steps to install the upper crank [34] on the rudder control torque tube [1]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Place the upper crank [34] in the original position on the rudder control torque tube [1].
2) Install the bolt [24], bolt [28], washers [25], washers [26], and nuts [27] to attach the upper cranks [34] to the rudder control torque tube [1].
3) Tighten the nuts [27] to 15 in-lb (1.7 N·m) - 25 in-lb (2.8 N·m) above the run-on torque.

SUBTASK 27-21-71-710-001

(2) Make sure that the rudder control torque tube [1] turns freely and that there is no play in the horizontal and vertical directions.

(a) If the rudder control torque tube [1] does not turn freely, do a check of the lower bearing housing [21] and upper bearing housing [35] installation.

SUBTASK 27-21-71-420-002

(3) Install the bolt [30], washer [31] (under the bolt), washer [32] (under the nut), and nut [33] to connect the standby rudder PCU control rod [2] to the upper crank [34].

SUBTASK 27-21-71-420-003

(4) Install the bolts [30], washers [31] (under the bolt), washers [32] (under the nut), and nuts [33] to connect the main rudder PCU control rods [3] to the center crank [29].

SUBTASK 27-21-71-420-004

(5) Install the rudder trim actuator [18] (TASK 27-21-87-400-801).

SUBTASK 27-21-71-420-005

(6) Install the bolt [9], washer [10], and nut [11] to connect the aft quadrant control rod [12] to the lower crank [20].

SUBTASK 27-21-71-420-006

(7) Do these steps to connect the control rod [8] to the lower crank [20]:

(a) Install the bolt [4], washer [5], and nut [6].

1) Tighten the nut [6] to 45 in-lb (5.1 N·m) - 60 in-lb (6.8 N·m).

CAUTION

DO NOT TRY TO TIGHTEN THE NUT [6] AGAIN AFTER YOU TIGHTEN THE NUT [7]. YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(b) Install the nut [7].

1) Tighten the nut [7] to 12.0 in-lb (1.4 N·m) - 15.0 in-lb (1.7 N·m).

SUBTASK 27-21-71-640-001

(8) Apply grease, D00633, to the upper bearing housing [35] and lower bearing housing [21].

(a) Fill the upper bearing housing [35] and lower bearing housing [21] with grease, D00633, until the grease is level with the surrounding structure.

SUBTASK 27-21-71-860-002

(9) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

G. Rudder Control Torque Tube Installation Test

SUBTASK 27-21-71-820-001

(1) Do this task: Control Rod for Main Rudder Power Control Unit Adjustment, TASK 27-21-00-820-804.

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SUBTASK 27-21-71-820-002

(2) Do this task: Control Rod for Standby Rudder Power Control Unit Adjustment, TASK 27-21-00-820-805.

SUBTASK 27-21-71-710-002

(3) Do this task: Rudder Travel Test, TASK 27-21-00-700-805.

SUBTASK 27-21-71-410-001

(4) Install these access panels:

Number	Name/Location
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER FEEL AND CENTERING UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the rudder feel and centering unit.

(2) An installation of the rudder feel and centering unit.

(3) A removal of the rudder feel and centering unit spring assembly.

(4) An installation of the rudder feel and centering unit spring assembly.

TASK 27-21-82-000-801

2. Rudder Feel and Centering Unit Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
324	Vertical Fin - Rear Spar To Trailing Edge

D. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access
324DR	Vertical Fin, Trailing Edge Access

E. Prepare for the Removal

SUBTASK 27-21-82-860-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-82-010-001

(2) Open these access panels:

Number	Name/Location
112A	Forward Access Door
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access
324DR	Vertical Fin, Trailing Edge Access

SUBTASK 27-21-82-010-002

(3) Remove these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

(a) Do this task: Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801.

SUBTASK 27-21-82-480-001

(4) Install the rig pin R-3 or R-4 from the rigging pin kit, SPL-1585 in the forward rig pin hole of the captain's or first officer's rudder forward quadrants (Figure 401).

F. Rudder Feel and Centering Unit Removal

(Figure 402)

SUBTASK 27-21-82-020-001

(1) Remove the rudder feel and centering unit [7]:

(a) Remove the nut [27], washer [26], nut [25], washer [24], and bolt assembly [23] to disconnect the feel and rudder feel and centering unit [7] from the rudder trim actuator [1].

(b) Remove the nut [33], nut [32], washer [31], washer [30], bushing [29], and bolt assembly [28] to disconnect the rudder feel and centering unit crank [8] from the rudder feel output rod [9].

(c) Remove the nuts [6], washers [5], and bolts [4] which attach the upper bearing housing [2] and the plate [3] to the airplane structure.

(d) Hold the rudder feel and rudder feel and centering unit [7], and remove these parts from the support bracket [18]:

1) Remove the nuts [22], washers [21], washers [20], and bolts [19].

2) Remove the nuts [17], washers [16], washers [15], and bolts [14].

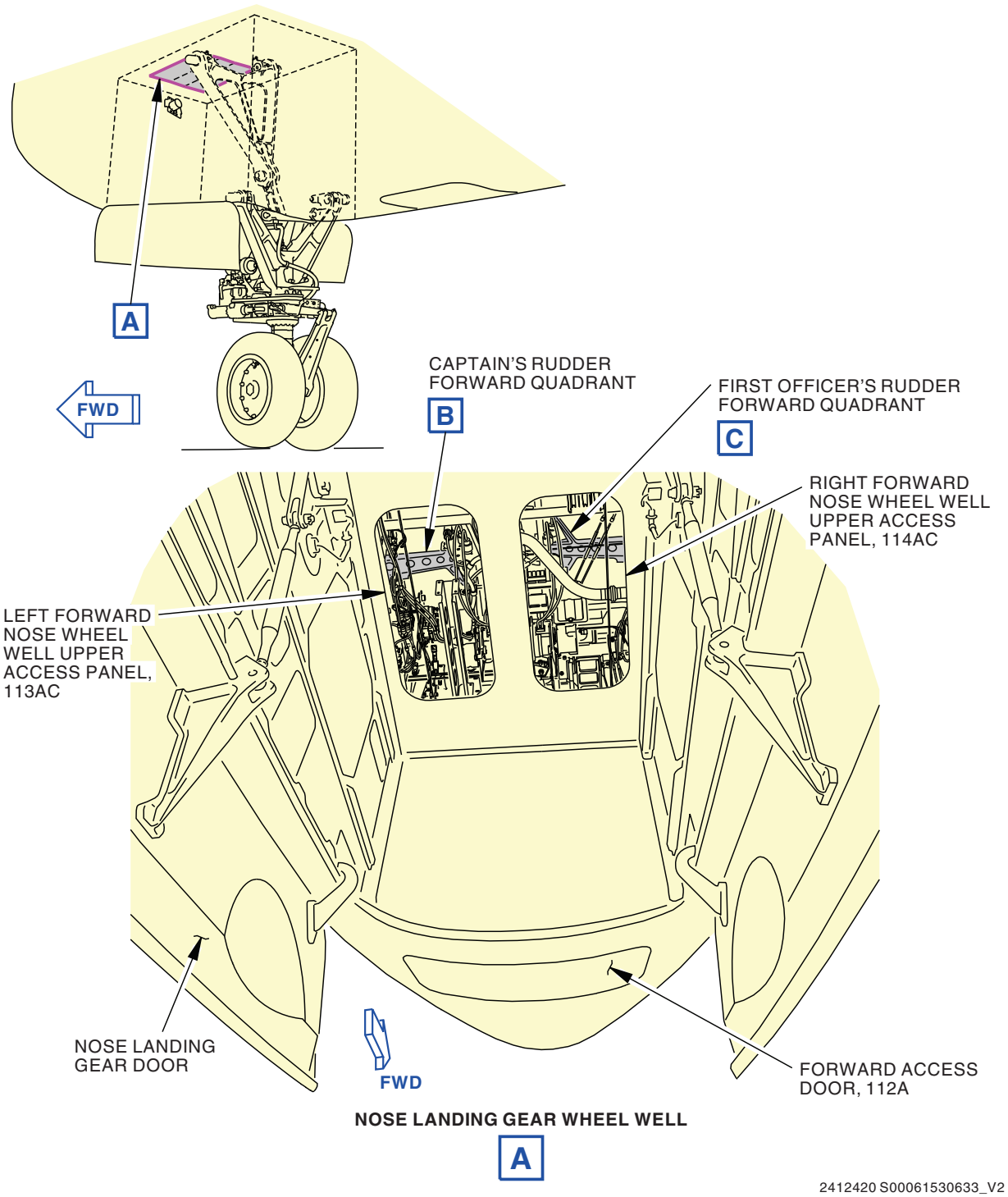
(e) Remove the nuts [13], washers [12], and bolts [11] from the lower bearing housing [10].

NOTE: If it is necessary, turn the rudder feel and centering unit [7] to get access to the bolts.

(f) Remove the rudder feel and centering unit [7] and support bracket [18].

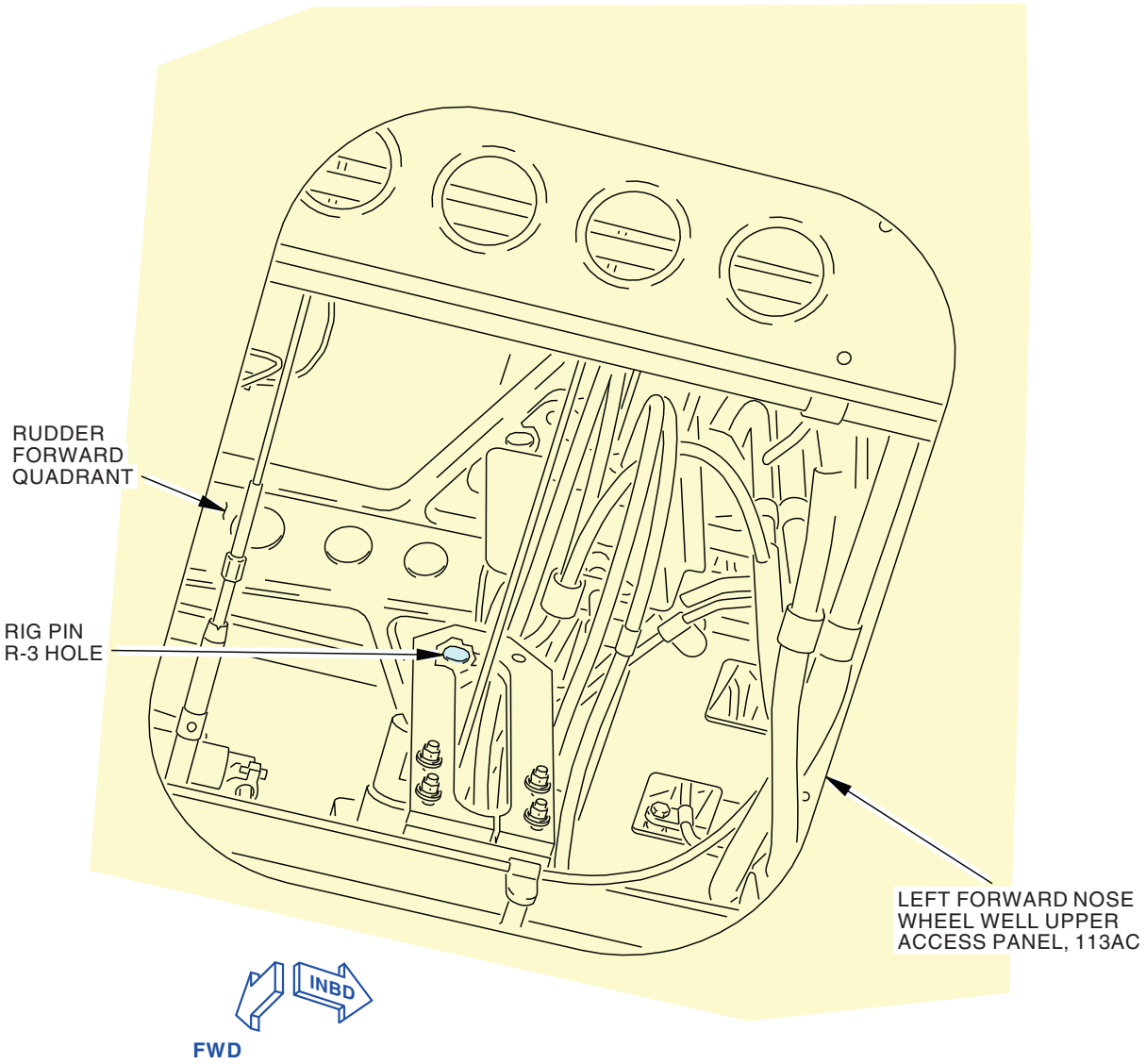
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Forward Quadrant Rig Pin Installation
Figure 401/27-21-82-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



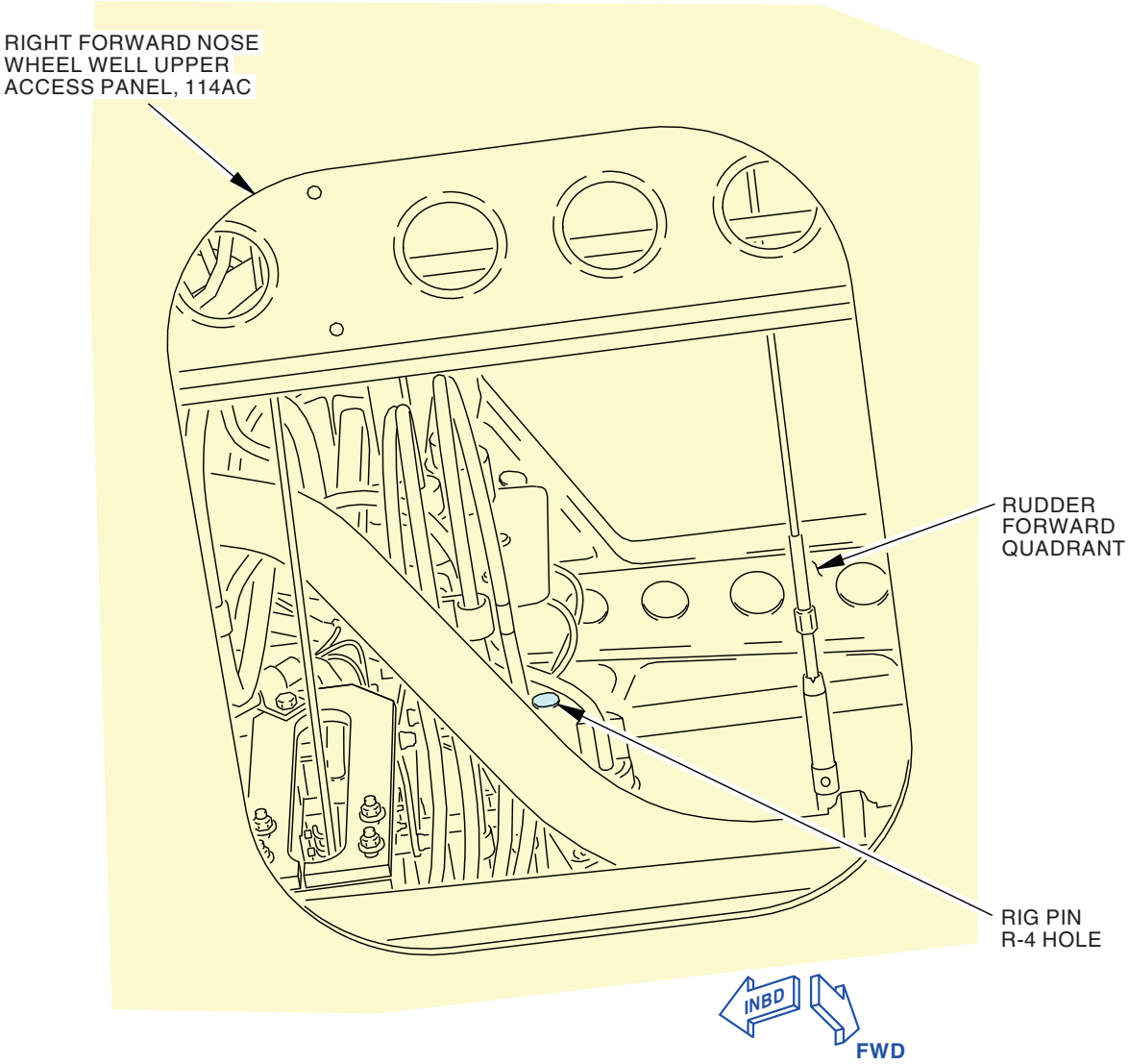
CAPTAIN'S RUDDER FORWARD QUADRANT

B

2412421 S00061530634_V1

Rudder Forward Quadrant Rig Pin Installation
Figure 401/27-21-82-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



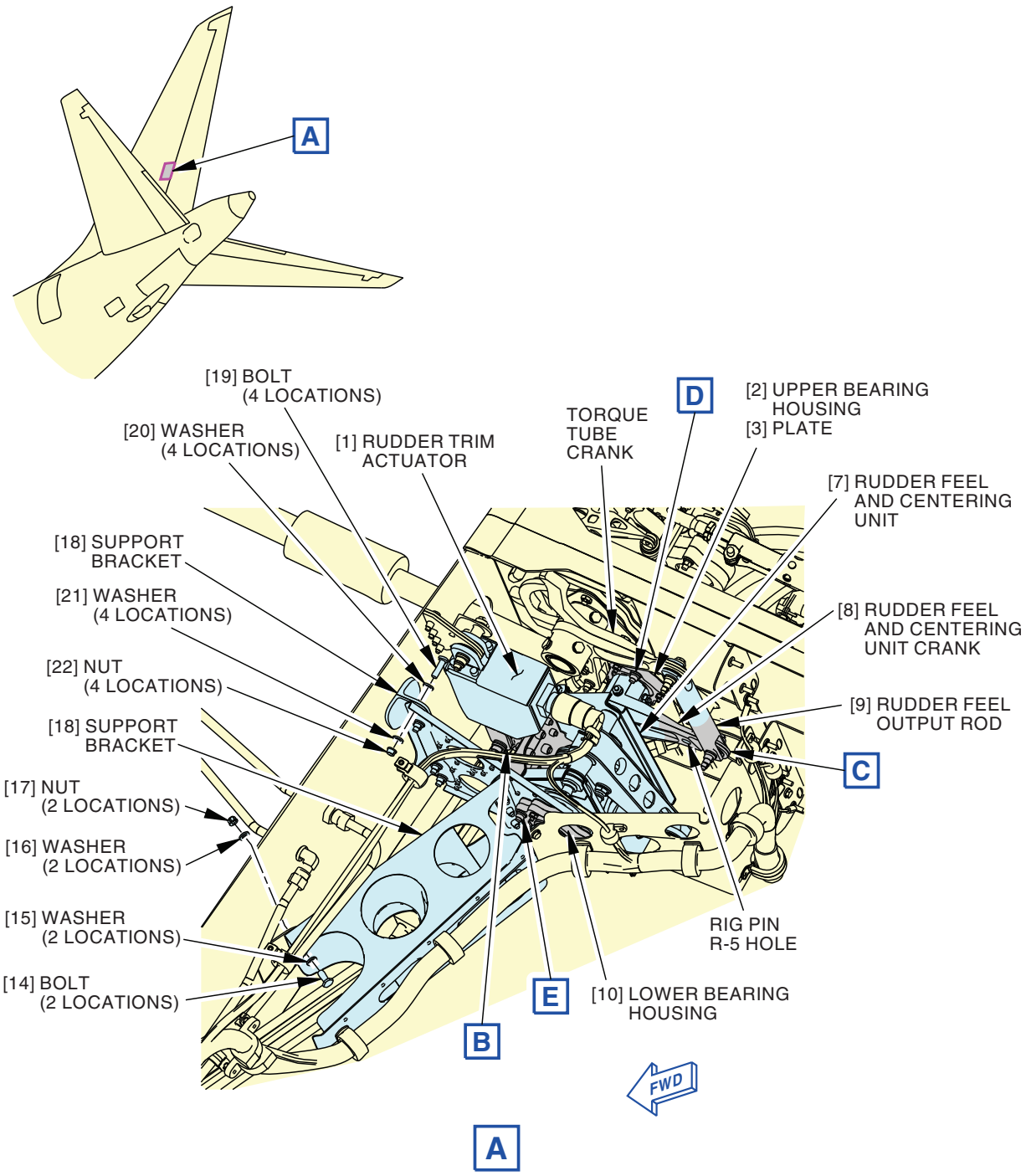
FIRST OFFICER'S RUDDER FORWARD QUADRANT

C

2412422 S00061530635_V1

Rudder Forward Quadrant Rig Pin Installation
Figure 401/27-21-82-990-801 (Sheet 3 of 3)

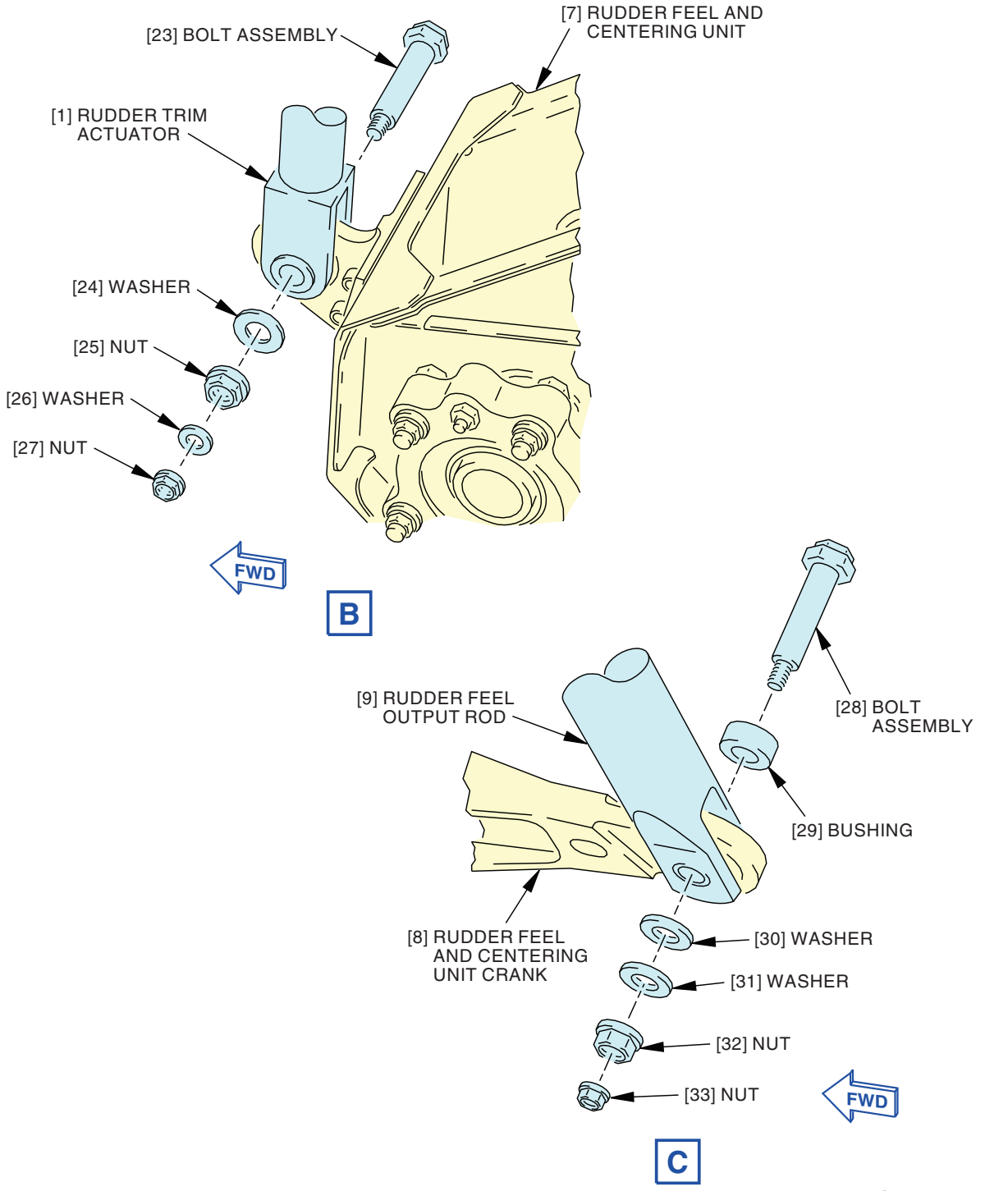
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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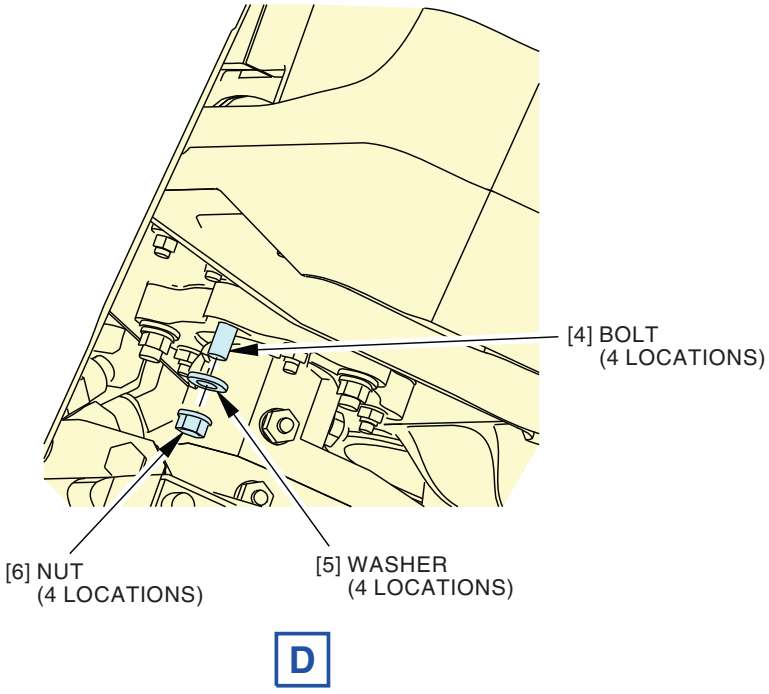
Rudder Feel and Centering Unit Installation
Figure 402/27-21-82-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Feel and Centering Unit Installation
Figure 402/27-21-82-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Feel and Centering Unit Installation
Figure 402/27-21-82-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-82-400-801

3. Rudder Feel and Centering Unit Installation

(Figure 401, Figure 402)

A. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
12-22-21-600-802	Spring Slider Shaft Lubrication (P/B 301)
27-21-00-700-804	Rudder Pedal Forces Test (P/B 501)
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
32-51-00 P/B 501	NOSE WHEEL STEERING SYSTEM - ADJUSTMENT/TEST
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Rudder feel and centering unit	27-21-82-01-155	PTF ALL

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
324	Vertical Fin - Rear Spar To Trailing Edge

E. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access
324DR	Vertical Fin, Trailing Edge Access

F. Rudder Feel and Centering Unit Installation

SUBTASK 27-21-82-640-001

(1) Lubricate the spring slider shaft (Spring Slider Shaft Lubrication, TASK 12-22-21-600-802).

SUBTASK 27-21-82-420-001

(2) Install the rudder feel and centering unit [7]:

(a) Install the bolts [11], washers [12], and nuts [13] to attach the lower bearing housing [10] to the support bracket [18].

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(b) Put the rudder feel and centering unit [7] and the support bracket [18] in their positions.

(c) Install the support bracket [18]:

1) Install the bolts [19], washers [20] (under the bolt), washers [21] (under the nut), and nuts [22].

2) Install the bolts [14], washers [15] (under the bolt), washers [16] (under the nut), and nuts [17].

(d) Install the bolts [4], washers [5], and nuts [6] to attach the upper bearing housing [2] and the plate [3] to the airplane structure.

SUBTASK 27-21-82-210-001

(3) Examine the rudder feel output rod [9] for damage.

NOTE: The feel output rod has two load paths.

SUBTASK 27-21-82-420-002

(4) Connect the rudder feel and centering unit crank [8] to the rudder feel output rod [9]:

(a) Install the bolt assembly [28], bushing [29], washer [30], washer [31], and nut [32].

1) Tighten the nut [32] 45 in-lb (5.1 N·m) to 60 in-lb (6.8 N·m).

CAUTION

DO NOT TRY TO TIGHTEN THE NUT [32] AGAIN AFTER YOU TIGHTEN THE NUT [33]. IF YOU DO, YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(b) Install the nut [33].

1) Tighten the nut [33] 12 in-lb (1.4 N·m) to 15 in-lb (1.7 N·m).

SUBTASK 27-21-82-480-002

(5) Install the rig pin R-5 of the rigging pin kit, SPL-1585 from the rudder feel and centering unit [7].

SUBTASK 27-21-82-420-003

(6) Connect the rudder trim actuator [1] to the rudder feel and centering unit [7]:

(a) Install the bolt assembly [23], washer [24], and nut [25].

1) Tighten the nut [25] 85 in-lb (9.6 N·m) to 100 in-lb (11.3 N·m).

CAUTION

DO NOT TRY TO TIGHTEN THE NUT [25] AGAIN AFTER YOU TIGHTEN THE NUT [27]. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(b) Install the washer [26] and nut [27].

(c) Tighten the nut [27] 30 in-lb (3.4 N·m) to 35 in-lb (4.0 N·m).

SUBTASK 27-21-82-080-001

(7) Remove the rig pins R-3 or R-4 of the rigging pin kit, SPL-1585 from the rudder forward quadrants.

SUBTASK 27-21-82-080-002

(8) Remove the rig pin R-5 of the rigging pin kit, SPL-1585 from the rudder feel and centering unit [7].

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G. Rudder Feel and Centering Unit Test

SUBTASK 27-21-82-020-002

(1) Disconnect the nose gear steering from the rudder system by rotating the towing handle, located on the left-hand side of the nose gear, to the TOW Position (NOSE WHEEL STEERING SYSTEM - ADJUSTMENT/TEST, PAGEBLOCK 32-51-00/501).

NOTE: It is not necessary to disconnect the steering if the airplane is on jacks and the nose landing gear is fully extended.

SUBTASK 27-21-82-860-002

(2) Do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

SUBTASK 27-21-82-700-001

(3) Test the Feel and Centering System:

(a) Operate the rudder pedals through their full travel range three times.

(b) Make sure that the rudder trim indicator shows zero units of trim.

(c) Push the captain's left rudder pedal forward approximately 0.5 in. (13 mm) and release it.

1) Make sure that each side of the rudder trailing edge aligns with the tab on the rudder index plate.

(d) Push the captain's right rudder pedal forward approximately 0.5 in. (13 mm) and release it.

1) Make sure that each side of the rudder trailing edge aligns with the tab on the rudder index plate.

SUBTASK 27-21-82-730-001

(4) Do this task: Rudder Pedal Forces Test, TASK 27-21-00-700-804.

SUBTASK 27-21-82-730-002

(5) Do this task: Rudder Travel Test, TASK 27-21-00-700-805.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-82-860-003

(1) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

SUBTASK 27-21-82-410-001

(2) Close these access panels:

Number	Name/Location
112A	Forward Access Door
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access
324DR	Vertical Fin, Trailing Edge Access

SUBTASK 27-21-82-410-002

(3) Install these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

(a) Do this task: Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801.

(b) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-82-420-004

(4) Connect steering to the rudder system by rotating the towing handle located on the left-hand side of the nose gear to the NORMAL position, if necessary.

——— END OF TASK ———

TASK 27-21-82-000-802

4. Rudder Feel and Centering Unit Spring Assembly Removal

(Figure 403)

A. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
324	Vertical Fin - Rear Spar To Trailing Edge

C. Access Panels

Number	Name/Location
324BR	Vertical Fin, Trailing Edge Access

D. Prepare for the Removal

SUBTASK 27-21-82-860-004

(1) Open this access panel:

Number	Name/Location
324BR	Vertical Fin, Trailing Edge Access

SUBTASK 27-21-82-860-005

(2) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-82-860-006

(3) Move the rudder trim knob 5 to 7 degrees NOSE LEFT, and wait until the rudder trim indicator stops.

E. Spring Assembly Removal

SUBTASK 27-21-82-860-007

(1) Do the steps to decrease the preload on the spring assembly [50]:

(a) Make sure that the roller that is in the arm assembly of the rudder feel and centering unit [7] is centered within the detent of the cam.

NOTE: The cam is attached to the primary shaft of the rudder feel and centering unit [7].

(b) If necessary, move the rudder in the LEFT direction to center the roller in the cam.

SUBTASK 27-21-82-020-003

(2) Remove the spring assembly [50] as follows:

(a) Remove the nut [57], washer [56], nut [55], and washer [54].

1) Keep the washer [54] and the washer [56] for installation if they are not damaged.

2) Discard the nut [55] and the nut [57].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

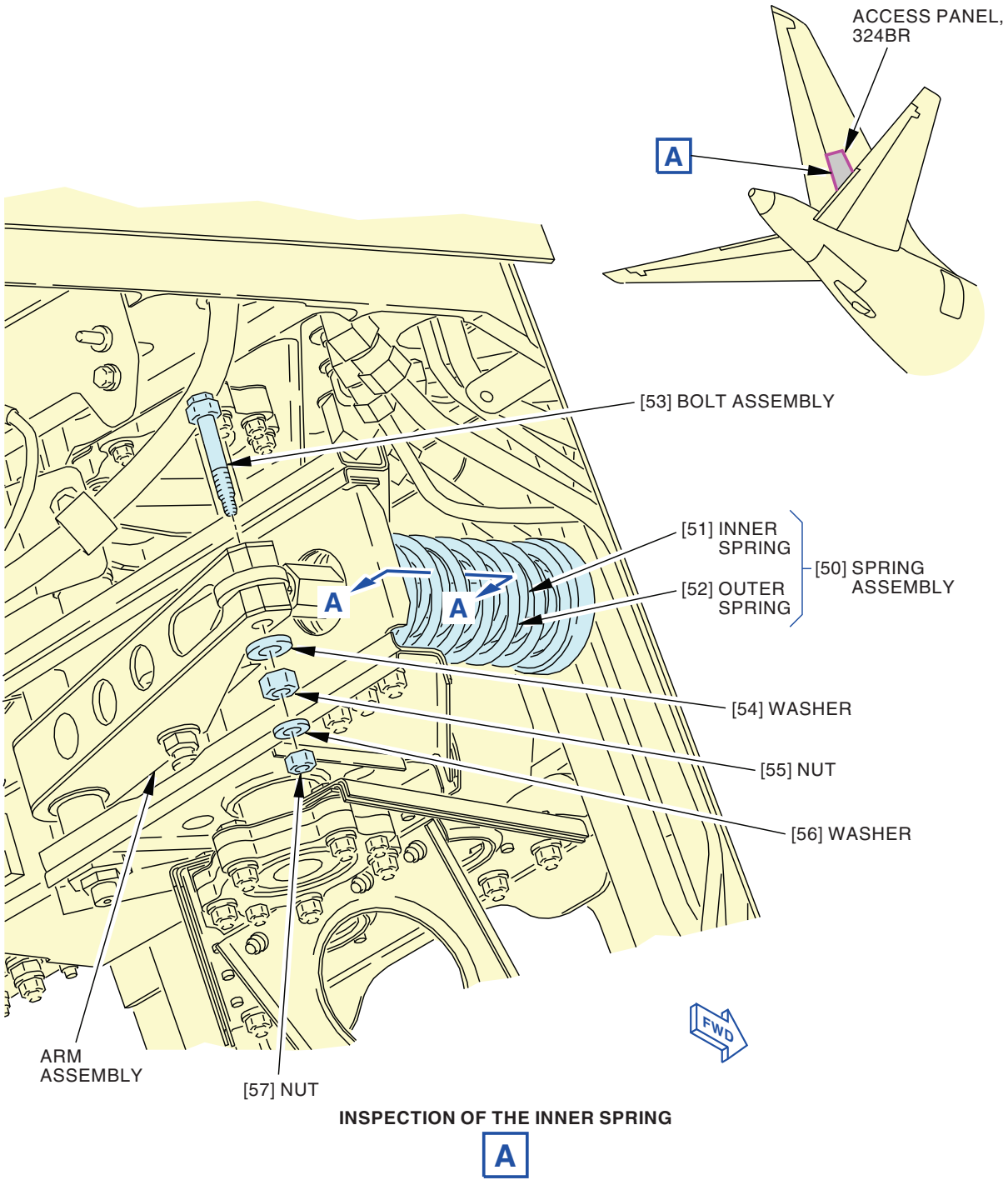
(b) Remove the bolt [53].

1) If necessary to remove the load on the bolt [53], use hand pressure to compress the spring assembly [50].

(c) Remove the spring assembly [50].

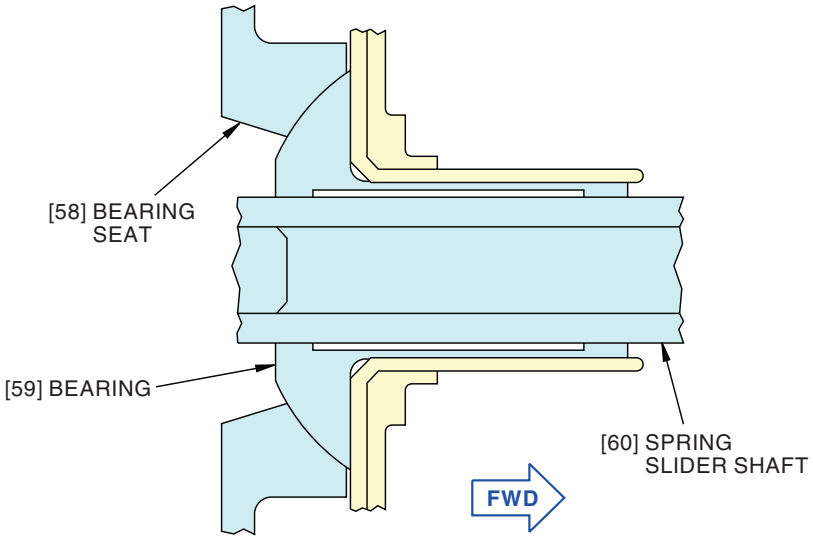
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Feel and Centering Unit Spring Assembly Installation
Figure 403/27-21-82-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A

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Rudder Feel and Centering Unit Spring Assembly Installation
Figure 403/27-21-82-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-82-400-802

5. Rudder Feel and Centering Unit Spring Assembly Installation

(Figure 403)

A. References

Reference	Title
12-22-21-600-802	Spring Slider Shaft Lubrication (P/B 301)
27-21-00-700-804	Rudder Pedal Forces Test (P/B 501)
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
C00032	Coating - Protective Enamel, General Use	BMS10-60 Type I
D00633	Grease - Aircraft General Purpose	BMS3-33

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
324	Vertical Fin - Rear Spar To Trailing Edge

D. Access Panels

Number	Name/Location
324BR	Vertical Fin, Trailing Edge Access

E. Spring Assembly Installation

SUBTASK 27-21-82-420-005

(1) Do these steps to install the spring assembly [50]:

(a) Lubricate the bearing [59] with grease, D00633.

1) Make sure to apply grease, D00633 on the bearing [59] surface that touches the bearing seat [58].

(b) Lubricate the spring slider shaft [60]. To lubricate it, do this task: Spring Slider Shaft Lubrication, TASK 12-22-21-600-802.

(c) Insert the spring assembly [50] through the bearing seat [58].

(d) Install the bolt [53], washer [54], and nut [55].

1) Tighten the nut [55] to 55 ±5 in-lb (6 ±1 N·m).

(e) Install the washer [56] and nut [57].

CAUTION DO NOT TRY TO TIGHTEN THE NUT [55] AGAIN AFTER YOU TIGHTEN THE NUT [57], YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

1) Tighten the nut [57] to 13.5 ±1.5 in-lb (1.5 ±0.2 N·m).

(f) On the rudder feel and centering unit, use a black marker or rubber stamp to mark the R after the serial number.

NOTE: The suffix R indicates the springs have been changed out.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Apply protective coating, C00032 or equivalent that is resistant to hydraulic fluid.

(g) Move the rudder trim knob until the rudder trim indicator returns to 0.00 +/-0.25 units of trim.

F. Spring Assembly Post-Installation Test

SUBTASK 27-21-82-730-003

(1) Test the Feel and Centering System:

(a) Operate the rudder pedals through their full travel range three times.

(b) Make sure the rudder trim indicator shows zero units of trim.

(c) Push the captain's left rudder pedal forward approximately 0.5 inch and release it.

1) Make sure each side of the rudder trailing edge aligns with the tab on the rudder index plate.

(d) Push the captain's right rudder pedal forward approximately 0.5 inch and release it.

1) Make sure that each side of the rudder trailing edge aligns with the tab on the rudder index plate.

SUBTASK 27-21-82-420-006

(2) Connect the steering for the nose landing gear to the rudder system if it disconnected.

SUBTASK 27-21-82-730-004

(3) Do this task: Rudder Pedal Forces Test, TASK 27-21-00-700-804.

SUBTASK 27-21-82-730-005

(4) Do this task: Rudder Travel Test, TASK 27-21-00-700-805.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-82-860-008

(1) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

SUBTASK 27-21-82-860-009

(2) Close this access panel:

Number	Name/Location
324BR	Vertical Fin, Trailing Edge Access

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER FEEL AND CENTERING UNIT SPRING ASSEMBLY - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) Inspection of the rudder feel and centering unit inner spring.

TASK 27-21-82-211-801

2. Rudder Feel and Centering Unit Inner Spring - Inspection

Figure 27-21-82-990-803

A. General

(1) This task does a detailed visual inspection of the inner spring in the rudder feel and centering unit spring assembly.

B. References

Reference	Title
27-21-82 P/B 401	RUDDER FEEL AND CENTERING UNIT - REMOVAL/INSTALLATION
27-21-82-990-803	Figure: Rudder Feel and Centering Unit Spring Assembly Installation (P/B 401)

C. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

D. Access Panels

Number	Name/Location
324BR	Vertical Fin, Trailing Edge Access

E. Prepare for the Inspection

SUBTASK 27-21-82-860-010

(1) Open this access panel:

Number	Name/Location
324BR	Vertical Fin, Trailing Edge Access

F. Rudder Feel and Centering Unit Inner Spring Inspection

SUBTASK 27-21-82-211-001

(1) Do the inspection of the inner spring as follows:

(a) Do a close visual inspection of the inner spring.

(b) Use a soft tool to push on the side of the inner spring, make sure that it is not loose or broken.

(c) If the inner spring is loose or broken, replace the spring assembly. (RUDDER FEEL AND CENTERING UNIT - REMOVAL/INSTALLATION, PAGEBLOCK 27-21-82/401)

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-82-860-011

(1) Close this access panel:

Number	Name/Location
324BR	Vertical Fin, Trailing Edge Access

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER TRIM ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Rudder Trim Actuator Removal

(2) Rudder Trim Actuator Installation

TASK 27-21-87-000-801

2. Rudder Trim Actuator Removal

(Figure 401)

A. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
324	Vertical Fin - Rear Spar To Trailing Edge

C. Access Panels

Number	Name/Location
324BL	Vertical Fin, Trailing Edge Access

D. Prepare for the Removal

SUBTASK 27-21-87-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	16	C00788	FLIGHT CONTROL TRIM IND RUD
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

SUBTASK 27-21-87-860-002

(2) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-87-010-001

(3) Open this access panel:

Number	Name/Location
324BL	Vertical Fin, Trailing Edge Access

E. Rudder Trim Actuator Removal

SUBTASK 27-21-87-020-001

(1) Use these steps to remove the rudder trim actuator [1]:

(a) Disconnect the electrical connector [9] from the rudder trim actuator [1].

(b) Remove the nut [4], washer [5], self locking nut [6], washer [7], and bolt assembly [3] to disconnect the rudder feel and centering unit [2] from the rudder trim actuator [1].

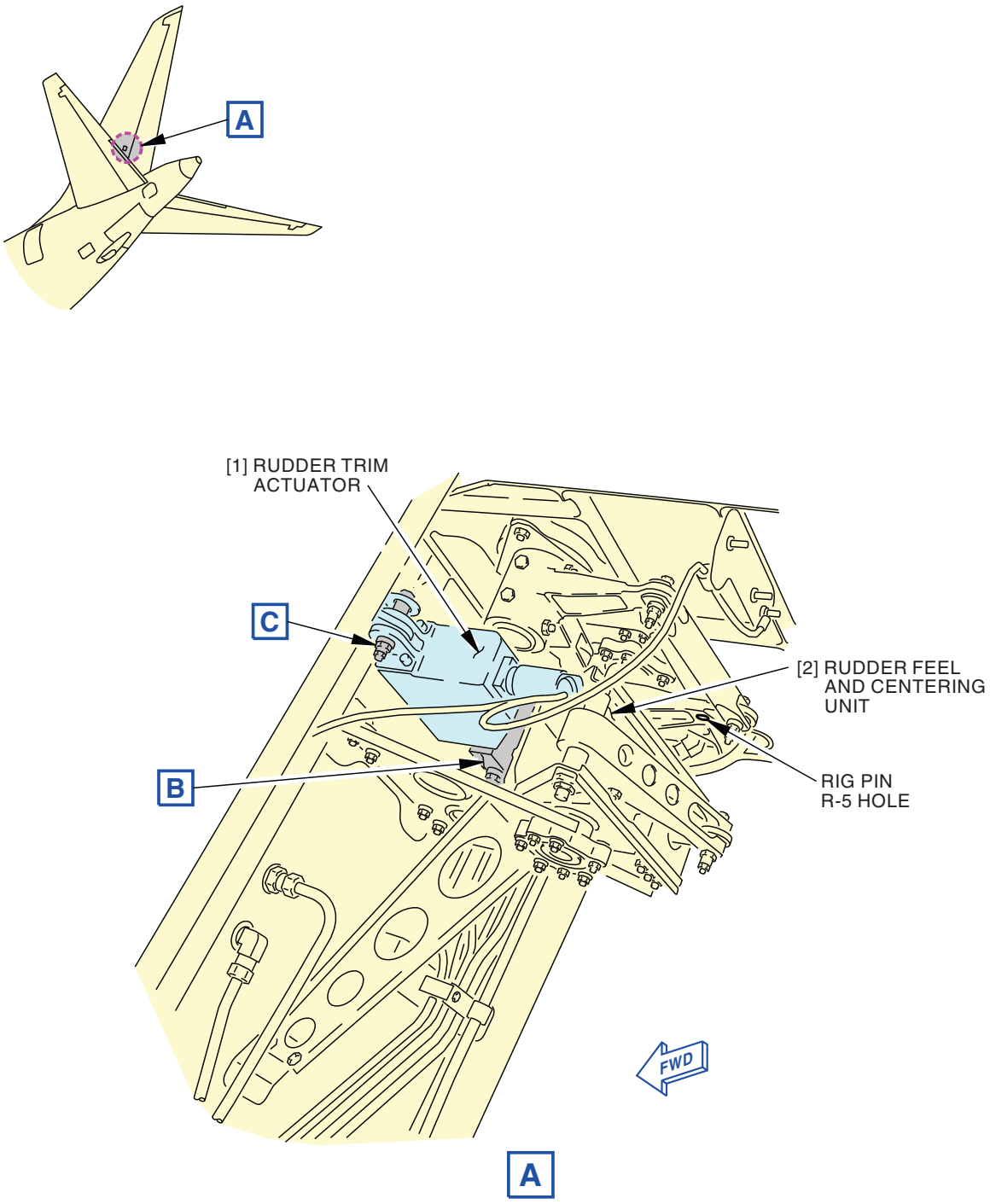
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Remove the nut [10], washer [11], self locking nut [12], washer [13], bushing [14], and bolt [8] that attach the rudder trim actuator [1] to the airplane structure.

(d) Remove the rudder trim actuator [1] from the airplane.

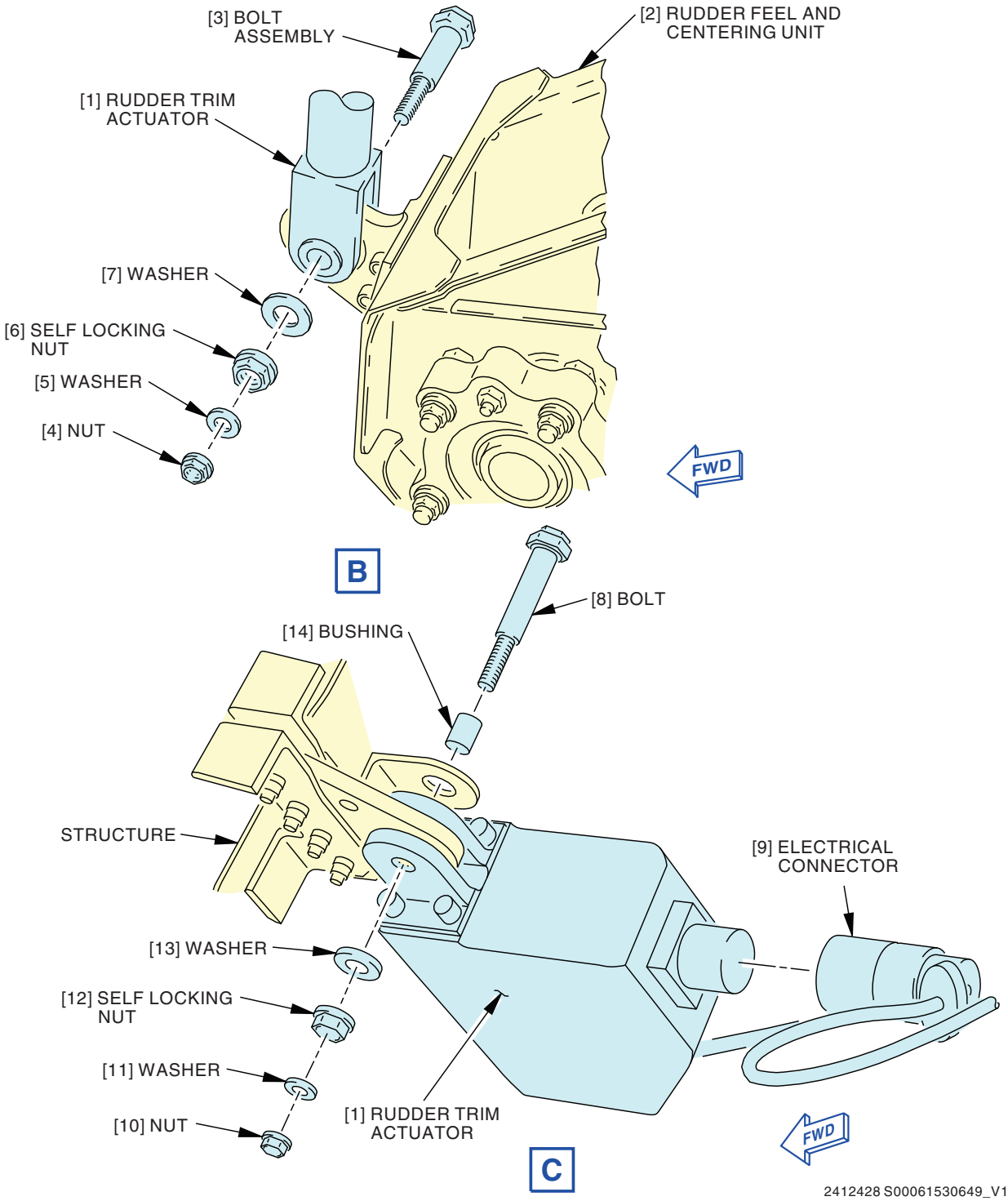
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Trim Actuator Installation
Figure 401/27-21-87-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Trim Actuator Installation
Figure 401/27-21-87-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-87-400-801

3. Rudder Trim Actuator Installation

(Figure 401)

A. References

Reference	Title
27-11-94-990-801	Figure: Aileron/Rudder Trim Control Module (P/B 401)
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-700-807	Rudder Trim System Test (P/B 501)
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
324	Vertical Fin - Rear Spar To Trailing Edge

D. Access Panels

Number	Name/Location
324BL	Vertical Fin, Trailing Edge Access

E. Rudder Trim Actuator Installation

(Figure 401)

SUBTASK 27-21-87-860-003

(1) Make sure that these circuit breakers are open:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	16	C00788	FLIGHT CONTROL TRIM IND RUD
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

SUBTASK 27-21-87-860-004

(2) Make sure that you remove pressure from the rudder hydraulic systems A and B (TASK 27-21-00-800-802).

SUBTASK 27-21-87-420-001

(3) Use these steps to install the rudder trim actuator [1]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install the bolt [8], bushing [14], washer [13], and self locking nut [12] to connect the rudder trim actuator [1] to the airplane structure.

1) Tighten the self locking nut [12] to 85 in-lb (10 N·m)-100 in-lb (11 N·m).

CAUTION DO NOT TRY TO TIGHTEN THE SELF LOCKING NUT AFTER YOU TIGHTEN THE OUTER NUT. YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(b) Install the washer [11] and nut [10].

1) Tighten the nut [10] to 30 in-lb (3 N·m)-35 in-lb (4 N·m).

(c) Install the bolt assembly [3], washer [7], and self locking nut [6] to connect the rudder trim actuator [1] to the rudder feel and centering unit [2].

1) Tighten the self locking nut [6] to 85 in-lb (10 N·m)-100 in-lb (11 N·m).

CAUTION DO NOT TRY TO TIGHTEN THE SELF LOCKING NUT AFTER YOU TIGHTEN THE OUTER NUT. YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(d) Install the washer [7] and nut [4].

1) Tighten the nut [4] to 30 in-lb (3 N·m)-35 in-lb (4 N·m).

(e) Connect the electrical connector [9] to the rudder trim actuator [1].

SUBTASK 27-21-87-860-005

(4) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

SUBTASK 27-21-87-860-006

(5) Do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

SUBTASK 27-21-87-860-007

(6) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	16	C00788	FLIGHT CONTROL TRIM IND RUD
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

SUBTASK 27-21-87-820-001

(7) Operate the rudder trim knob until you can install the rig pin R-5 from the rigging pin kit, SPL-1585 to the rudder feel and centering unit [2].

SUBTASK 27-21-87-820-002

(8) If the rudder trim indicator reading is not 0.0 +/- 0.1 unit, turn the adjustment screw on the bottom of the rudder trim indicator until the pointer reads 0.0 +/- 0.1 unit (Figure 27-11-94-990-801).

SUBTASK 27-21-87-480-001

(9) Remove the rig pin R-5 from the rudder feel and centering unit [2].

SUBTASK 27-21-87-730-001

(10) Do this task: Rudder Travel Test, TASK 27-21-00-700-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-87-730-002

(11) Do this task: Rudder Trim System Test, TASK 27-21-00-700-807.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-87-860-008

(1) If it is necessary, do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-87-410-001

(2) Install this access panel:

Number	Name/Location
324BL	Vertical Fin, Trailing Edge Access

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN RUDDER POWER CONTROL UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the main rudder Power Control Unit (PCU)

(2) An installation of the main rudder PCU

(3) A removal of the main rudder PCU load limiter solenoid valve

(4) An installation of the main rudder PCU load limiter solenoid valve.

TASK 27-21-91-000-801

2. Main Rudder Power Control Unit Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1563	Hoist - Fishpole, 500 lb Safe Working Limit (SWL) Part #: 10/4180 Supplier: K1425 Part #: IA5101-1 Supplier: 053H3 Part #: IA5101-501 Supplier: 053H3 Part #: IA5119-1 Supplier: 053H3 Part #: IA5119-501 Supplier: 053H3 Part #: PF51-003-1 Supplier: 1YRX6 Part #: PF51-009-1 Supplier: 1YRX6 Opt Part #: 10/3641 Supplier: K1425 Opt Part #: 10/3641C1 Supplier: K1425 Opt Part #: MINILIFT Supplier: K1425
SPL-1727	Lock - Ground, Rudder, PCU Removed Part #: C27057-1 Supplier: 81205
SPL-1732	Equipment - Removal/Installation, Rudder PCU Part #: C27049-36 Supplier: 81205 Opt Part #: C27049-1 Supplier: 81205
STD-203	Container - Oil Resistant, 1 U.S Gal (3.8 L)
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

D. Access Panels

Number	Name/Location
192DR	ECS High Pressure Access Door
324DL	Trailing Edge Access

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
324FR	Panel Assy - Trailing Edge, Beam Seal, Vert Fin

E. Prepare for the Removal

SUBTASK 27-21-91-210-001

CAUTION

CARE MUST BE TAKEN TO PROPERLY SUPPORT THE PCU DURING REMOVAL, STORAGE, AND INSTALLATION TO PREVENT CONTACT WITH THE SUMMING LEVER BOLTS, WHICH COULD BEND OR BREAK THE PROTRUDING END OF THE INNER BOLT. HANDLING DAMAGE TO THE THREADED END OF THE SUMMING LEVER INNER BOLT WILL LIKELY CAUSE DAMAGE TO THE THREADS OF THE SUMMING LEVER OUTER BOLT.

(1) Do the visual inspection of the external summing levers [3] inner bolt (View C-C, Figure 401).

NOTE: You can see the bolt through the space between the rudder fairings and the vertical fin blade seal from the right side when the rudder turned to the left with hydraulic power OFF.

(a) Make sure that the bolt have the nut and collar.

(b) If the bolt has damage, then replace the bolt.

SUBTASK 27-21-91-860-001

(2) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-91-860-002

(3) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	7	C00285	YAW DAMPER AC
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

SUBTASK 27-21-91-860-003

(4) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	15	C01628	FORCE FIGHT MONITOR

SUBTASK 27-21-91-010-001

(5) To get access to the main rudder Power Control Unit (PCU), do this step:

(a) Remove these access panels:

Number	Name/Location
192DR	ECS High Pressure Access Door
324DL	Trailing Edge Access
324FR	Panel Assy - Trailing Edge, Beam Seal, Vert Fin

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Main Rudder PCU Removal

SUBTASK 27-21-91-020-001

> **CAUTION** REMOVE THE CONTROL ROD FROM THE PCU BEFORE YOU DISCONNECT THE PCU FROM THE RUDDER. IF YOU DO NOT REMOVE THE CONTROL ROD FIRST, DAMAGE TO THE CONTROL ROD CAN OCCUR DURING THE CHECK.

(1) Remove the control rods [8]:

(a) Remove the nuts [7], washers [6], washers [5], and bolts [4] to disconnect the control rods [8] from the external summing levers [3].

(b) Remove the nuts [44], washers [43], washers [42], and bolts [41] to disconnect the control rods [8] from the input cranks [10].

(c) Remove the control rods [8] from the airplane.

SUBTASK 27-21-91-480-001

(2) Install the equipment, SPL-1732 (Figure 402):

(a) Install the bracket assembly [56] with screws [51] and washers [52] to the vertical fin.

1) Tighten the screws [51] to 18 in-lb (2.0 N·m) - 25 in-lb (2.8 N·m).

(b) Install the beam assembly [55] with screws [53] and washers [54] to the main rudder power control unit [1].

1) Tighten the screws [53] to 45 in-lb (5.1 N·m) - 60 in-lb (6.8 N·m)..

(c) Install the arm assembly [57] to the bracket assembly [56].

SUBTASK 27-21-91-020-002

(3) Remove the nut [32], washer [31], nut [30], washer [29], and bolt assembly [28] to disconnect the rod assembly [2] from the rudder.

SUBTASK 27-21-91-020-003

> **CAUTION** MAKE SURE THAT YOU HOLD THE MAIN RUDDER PCU DURING THE REMOVAL OF THE FORWARD MANIFOLD. IF THE MAIN RUDDER PCU FALLS, IT CAN CAUSE DAMAGE.

(4) Remove the nut [27], washer [26], nut [25], washer [24], washer [33], bushing [23], and bolt assembly [22] to disconnect the forward manifold [11] from the vertical fin structure.

SUBTASK 27-21-91-020-004

(5) Disconnect the electrical connectors [9] from the main rudder power control unit [1].

SUBTASK 27-21-91-020-005

(6) Disconnect the hydraulic lines from the main rudder power control unit [1].

> **CAUTION** MAKE SURE THAT YOU CAREFULLY DRAIN HYDRAULIC FLUID INTO A SUITABLE CONTAINER. CAREFULLY CLEAN, DRY, AND DISPOSE OF ALL LEAKAGE. HYDRAULIC FLUID CAUSES DAMAGE TO RUBBER, COPPER, PLASTICS, PAINTS, AND MANY OTHER MATERIALS.

(a) Put a oil resistant container, STD-203, below the hydraulic connections.

(b) Disconnect the hydraulic system A return line [12] from the main rudder power control unit [1].

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(c) Disconnect the hydraulic system A pressure line [21] from the main rudder power control unit [1].

(d) Disconnect the hydraulic system B return line [14] from the main rudder power control unit [1].

(e) Disconnect the hydraulic system B pressure line [13] from the main rudder power control unit [1].

(f) Install the caps/plugs, STD-14686, to the hydraulic lines and ports.

(g) Install a tag on the hydraulic lines to make sure you install them correctly.

SUBTASK 27-21-91-020-006

CAUTION

MAKE SURE THAT YOU MOVE THE PCU CAREFULLY DURING THE REMOVAL OR INSTALLATION OF THE PCU. THE PCU CAN HIT THE BOLTS. THIS CAN CAUSE DAMAGE TO THE BOLTS.

CAUTION

THE PCU SHOULD NEVER BE SET DIRECTLY ON A HARD SURFACE WITHOUT SUPPORTS OR OTHER METHODS TO PREVENT CONTACT WITH THE SUMMING LEVER BOLTS. IF NOT, IT CAN CAUSE DAMAGE.

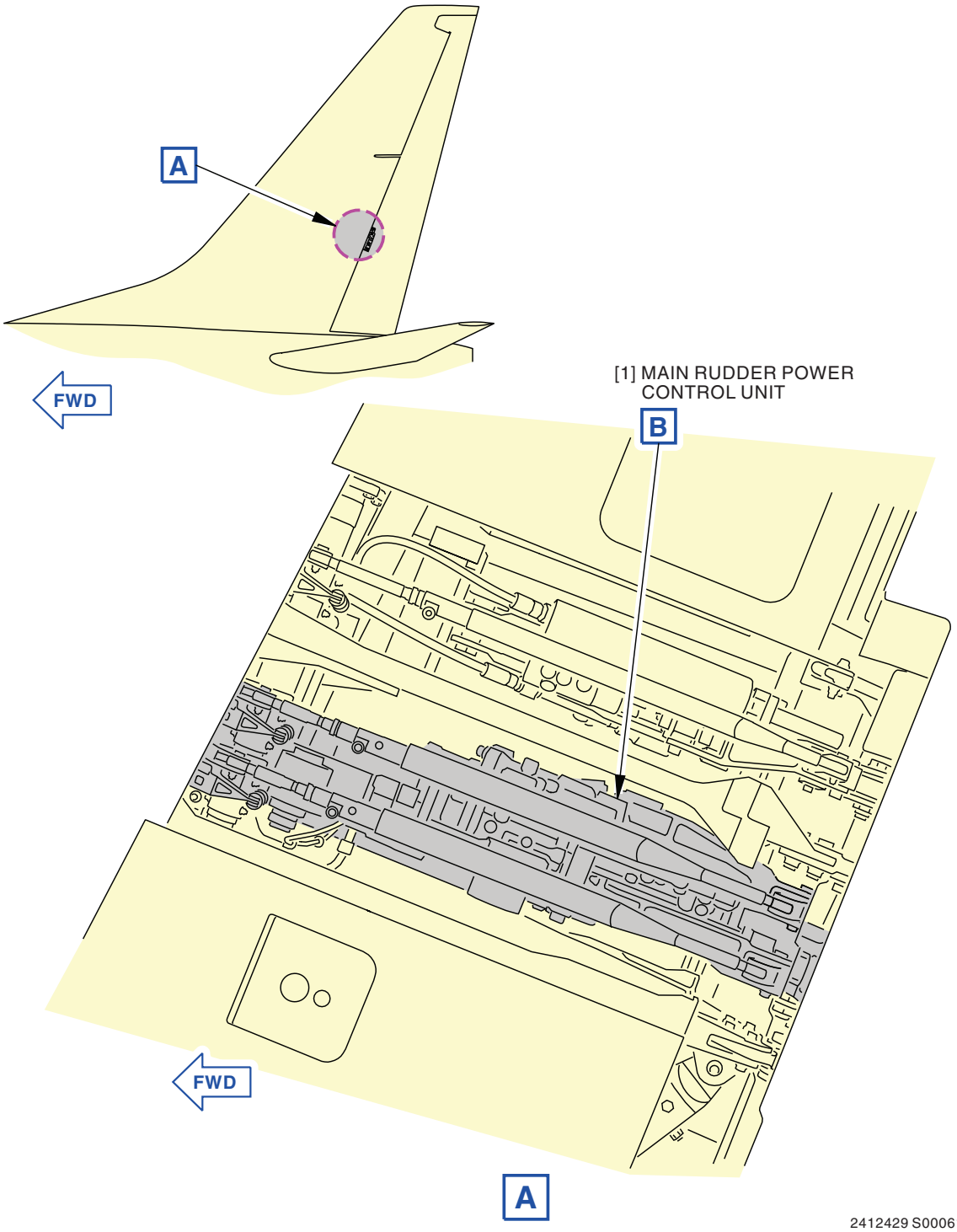
(7) Use a fishpole hoist, COM-1563, to remove the main rudder power control unit [1] from the airplane.

SUBTASK 27-21-91-480-002

(8) Install the lock, SPL-1727, to hold the rudder in its position.

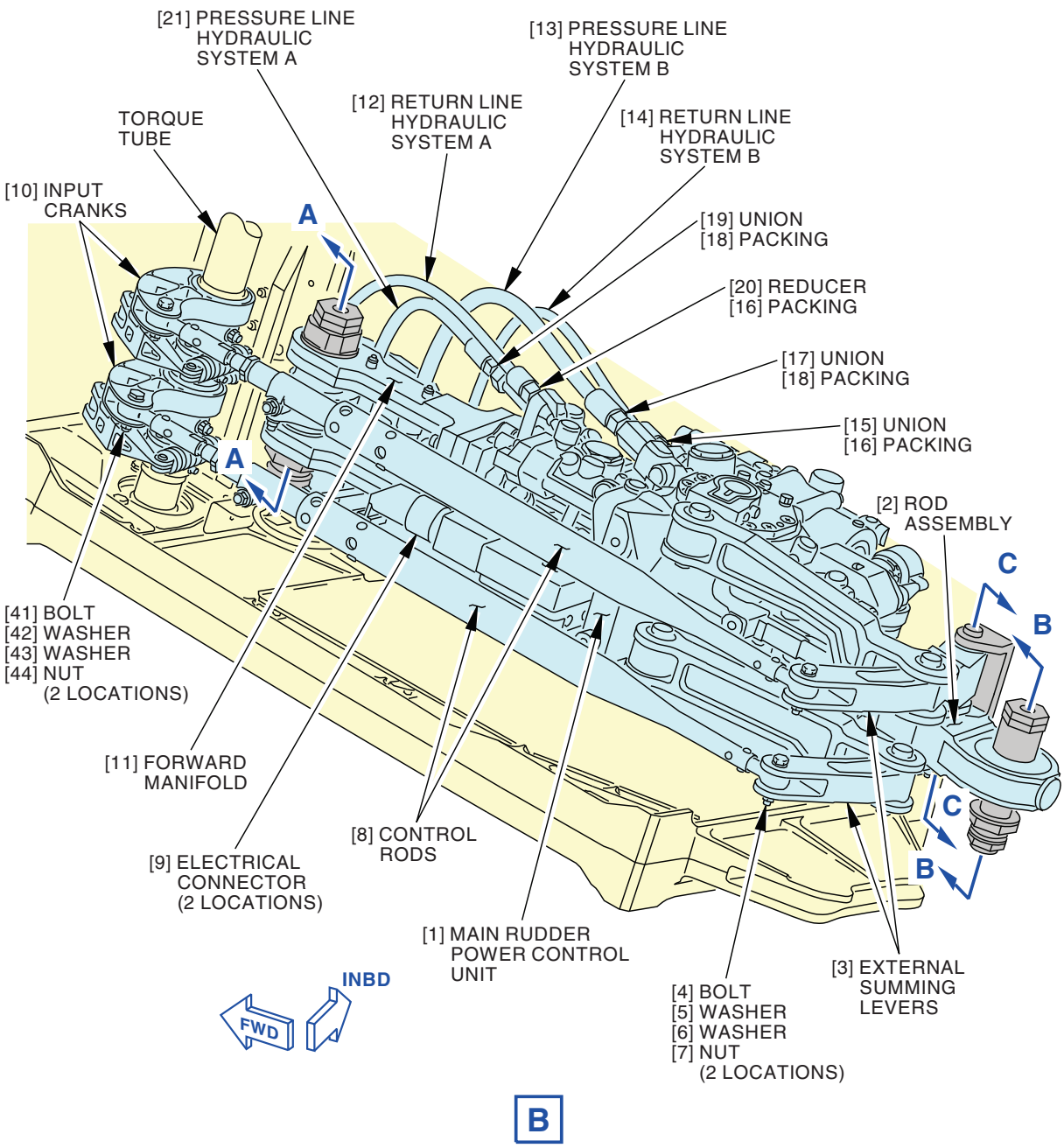
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Rudder Power Control Unit Installation
Figure 401/27-21-91-990-801 (Sheet 1 of 4)

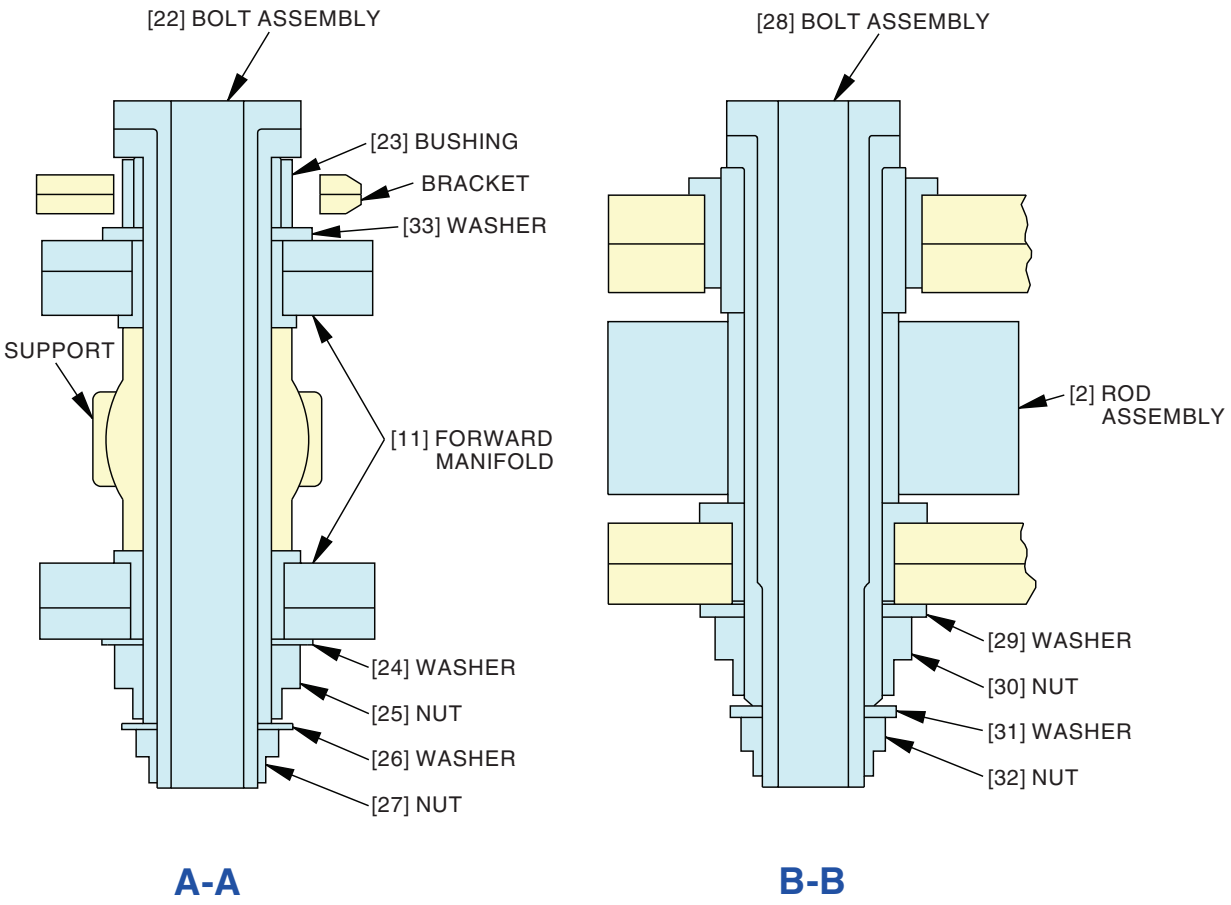
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412430 S00061530655_V2

Main Rudder Power Control Unit Installation
Figure 401/27-21-91-990-801 (Sheet 2 of 4)

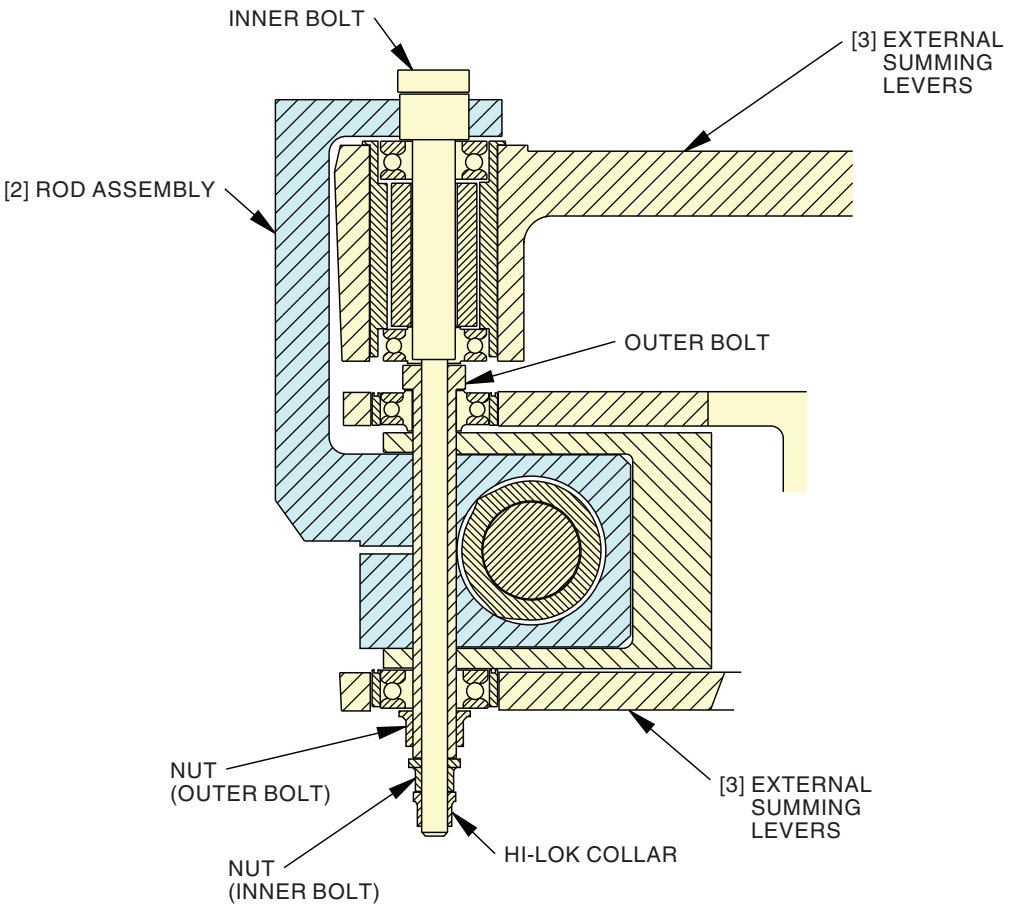
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412431 S00061530656_V1

Main Rudder Power Control Unit Installation
Figure 401/27-21-91-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

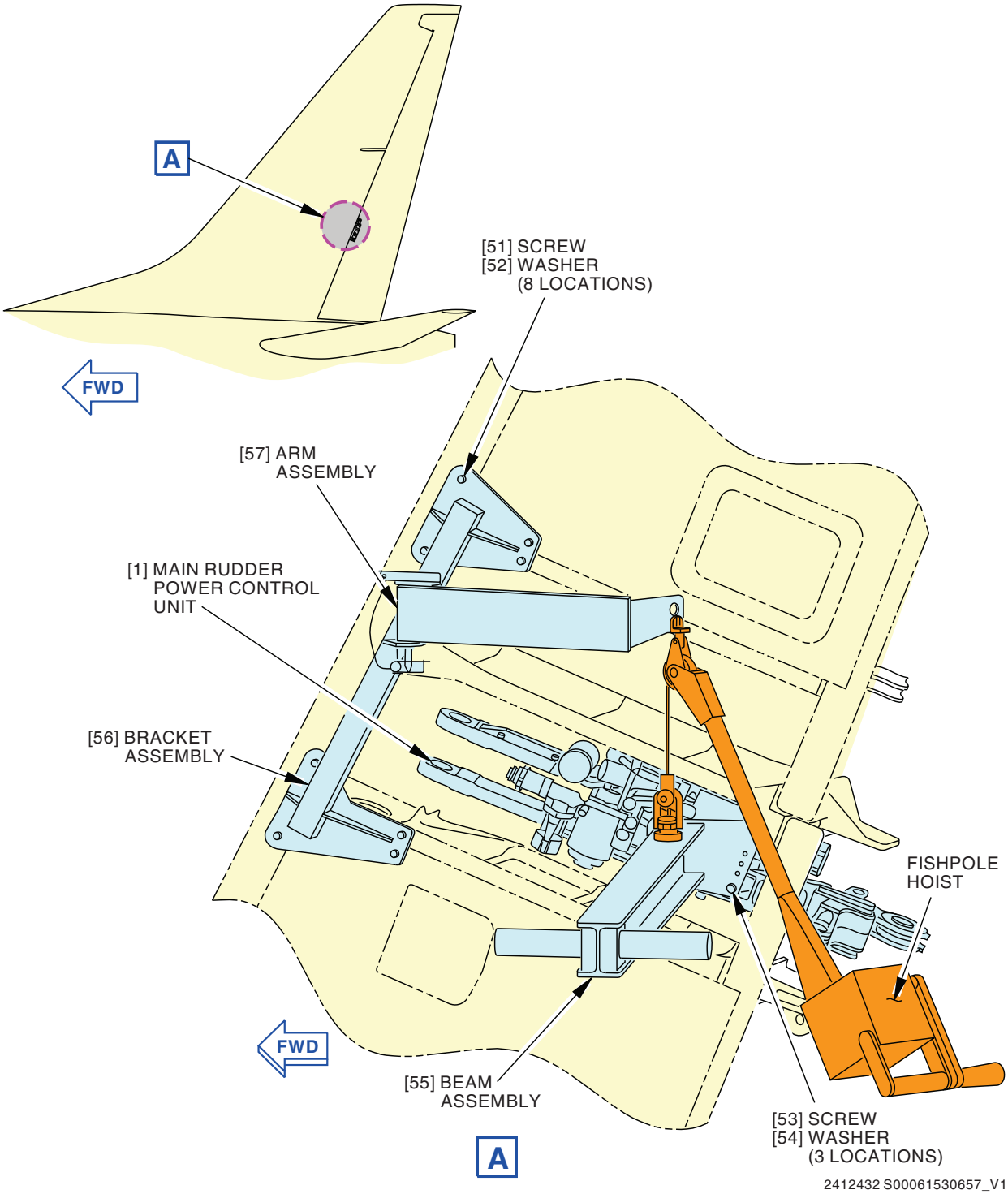


C-C

2787622 S0000632908_V1

Main Rudder Power Control Unit Installation
Figure 401/27-21-91-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Rudder PCU Tool Installation
Figure 402/27-21-91-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-91-400-801

3. Main Rudder Power Control Unit Installation

(Figure 401, Figure 402)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
20-10-20-800-801	Cap and Plug Hydraulic Fittings (P/B 201)
27-21-00-700-802	Rudder Hinge Bearing and PCU Bearing Freeplay (P/B 501)
27-21-00-700-803	Rudder Centering Test (P/B 501)
27-21-00-700-804	Rudder Pedal Forces Test (P/B 501)
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-700-809	Rudder Hinge Friction Test (P/B 501)
27-21-00-700-810	Main Rudder PCU Internal Hydraulic Leakage Test (P/B 501)
27-21-00-700-812	Rudder Main PCU Force Fight Monitor Test (P/B 501)
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-820-804	Control Rod for Main Rudder Power Control Unit Adjustment (P/B 501)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1563	Hoist - Fishpole, 500 lb Safe Working Limit (SWL) Part #: 10/4180 Supplier: K1425 Part #: IA5101-1 Supplier: 053H3 Part #: IA5101-501 Supplier: 053H3 Part #: IA5119-1 Supplier: 053H3 Part #: IA5119-501 Supplier: 053H3 Part #: PF51-003-1 Supplier: 1YRX6 Part #: PF51-009-1 Supplier: 1YRX6 Opt Part #: 10/3641 Supplier: K1425 Opt Part #: 10/3641C1 Supplier: K1425 Opt Part #: MINILIFT Supplier: K1425
SPL-1727	Lock - Ground, Rudder, PCU Removed Part #: C27057-1 Supplier: 81205
SPL-1731	Torque Wrench Eqpt - Rudder PCU Attachment Part #: C27042-1 Supplier: 81205
SPL-1732	Equipment - Removal/Installation, Rudder PCU Part #: C27049-36 Supplier: 81205 Opt Part #: C27049-1 Supplier: 81205
STD-203	Container - Oil Resistant, 1 U.S Gal (3.8 L)
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Main rudder power control unit	27-21-91-02-190	PTF ALL
8	Control rod	27-21-91-02-045	PTF ALL
16	Packing	27-21-91-02-200	PTF ALL
18	Packing	27-21-91-02-200	PTF ALL

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
324	Vertical Fin - Rear Spar To Trailing Edge

F. Access Panels

Number	Name/Location
324DL	Trailing Edge Access
324DR	Vertical Fin, Trailing Edge Access
324FR	Panel Assy - Trailing Edge, Beam Seal, Vert Fin

G. Main Rudder PCU Installation

SUBTASK 27-21-91-080-001

(1) Remove the lock, SPL-1727.

SUBTASK 27-21-91-480-003

(2) Install the equipment, SPL-1732, to the main rudder power control unit [1] (Figure 402).

SUBTASK 27-21-91-420-001

CAUTION

MAKE SURE THAT YOU FOLLOW THE INSTRUCTIONS TO INSTALL THE MAIN RUDDER CAREFULLY AND DO THE PROCEDURE IN THE SPECIFIED SEQUENCE. DAMAGE TO THE VERNIER CONTROL ROD [9] MAY OCCUR IF YOU DO NOT FOLLOW THE INSTRUCTIONS CORRECTLY.

CAUTION

MAKE SURE THAT YOU MOVE THE PCU CAREFULLY DURING THE REMOVAL OR INSTALLATION OF THE PCU. THE PCU CAN HIT THE BOLTS. THIS CAN CAUSE DAMAGE TO THE BOLTS.

CAUTION

MAKE SURE THE SUMMING LEVER BOLTS WERE NOT DAMAGED OR BENT DURING REMOVAL, STORAGE OR INSTALLATION OF THE PCU.

(3) Install the main rudder power control unit [1] (Figure 401):

(a) Use a fishpole hoist, COM-1563, to put the main rudder power control unit [1] in its position with the forward manifold [11] in the forward direction.

(b) Connect the forward manifold [11] to the vertical fin structure:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Install the washer [33], bushing [23], bolt assembly [22], washer [24], and nut [25].
2) Use the torque wrench equipment, SPL-1731, to tighten the nut [25] to 1000 in-lb (113.0 N·m) - 1200 in-lb (135.6 N·m).
3) Install the washer [26] and nut [27].

CAUTION

DO NOT TIGHTEN THE NUT [25] AFTER YOU TIGHTEN THE NUT [27]. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

4) Tighten the nut [27] to 650 in-lb (73.4 N·m) - 800 in-lb (90.4 N·m)..

(c) Connect the rod assembly [2] to the rudder:

1) Install the bolt assembly [28], washer [29], and nut [30].
2) Tighten the nut [30] to 900 in-lb (101.7 N·m) - 1050 in-lb (118.6 N·m).
3) Install the washer [31] and nut [32].

CAUTION

DO NOT TIGHTEN THE NUT [30] AFTER YOU TIGHTEN THE NUT [32]. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

4) Tighten the nut [32] 450 in-lb (50.8 N·m) - 550 in-lb (62.1 N·m).

SUBTASK 27-21-91-080-002

(4) Remove the equipment, SPL-1732, from the main rudder power control unit [1]:

(a) Remove the screws [53] and washers [54] from the beam assembly [55].
(b) Remove the screws [51] and washers [52] from the bracket assembly [56].
(c) Remove the bracket assembly [56] and beam assembly [55].

SUBTASK 27-21-91-020-007

CAUTION

MAKE SURE THAT YOU CORRECTLY INSTALL THE HYDRAULIC LINES TO THE MAIN RUDDER PCU. MAKE SURE THAT YOU INSTALL THE CORRECT UNIONS IN THE HYDRAULIC PORTS. YOU CAN CAUSE DAMAGE TO THE MAIN RUDDER PCU.

(5) Connect the hydraulic lines to the main rudder power control unit [1]:

CAUTION

MAKE SURE THAT YOU CAREFULLY DRAIN HYDRAULIC FLUID INTO A SUITABLE CONTAINER. CAREFULLY CLEAN, DRY, AND DISPOSE OF ALL LEAKAGE. HYDRAULIC FLUID CAUSES DAMAGE TO RUBBER, COPPER, PLASTICS, PAINTS, AND MANY OTHER MATERIALS.

(a) Remove the caps/plugs, STD-14686, from the hydraulic lines and ports (TASK 20-10-20-800-801).
(b) If the main rudder power control unit [1] does not have the unions and packings installed, do these steps:

1) Apply hydraulic fluid, D00153, to the unions, reducer and packings.
2) Install the union [15] and packing [16].
3) Install the union [17] and packing [18].
4) Install the union [19] and packing [18].

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5) Install the reducer [20] and packing [16].

(c) Connect the hydraulic system A return line [12] to the main rudder power control unit [1].

1) Tighten the B-nut on hydraulic system A return line [12] to 110 ±6 in-lb (12.4 ±0.7 N·m).

(d) Connect the hydraulic system A pressure line [21] to the main rudder power control unit [1].

1) Tighten the B-nut on hydraulic system A pressure line [21] to 110 ±6 in-lb (12.4 ±0.7 N·m).

(e) Connect the hydraulic system B return line [14] to the main rudder power control unit [1].

1) Tighten the B-nut on hydraulic system B return line [14] to 110 ±6 in-lb (12.4 ±0.7 N·m).

(f) Connect the hydraulic system B pressure line [13] to the main rudder power control unit [1].

1) Tighten the B-nut on hydraulic system B pressure line [13] to 170 ±9 in-lb (19.2 ±1.0 N·m).

(g) Remove the oil resistant container, STD-203, put below the hydraulic connections.

SUBTASK 27-21-91-420-002

(6) Install the control rods [8]:

(a) Put the control rods [8] in its position with the adjustment end in the forward direction.

(b) Connect the control rods [8] to the external summing lever [3]:

1) Install the bolts [4], washers [5], washers [6], and nuts [7].

(c) Connect the control rods [8] to the input cranks [10]:

1) Install the bolts [41], washers [42], washers [43], and nuts [44].

SUBTASK 27-21-91-020-008

(7) Connect the electrical connectors [9] to the main rudder power control unit [1].

SUBTASK 27-21-91-860-004

(8) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	7	C00285	YAW DAMPER AC
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

SUBTASK 27-21-91-860-005

(9) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	15	C01628	FORCE FIGHT MONITOR

SUBTASK 27-21-91-860-006

(10) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-91-860-007

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(11) Do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

SUBTASK 27-21-91-870-001

(12) Move the rudder pedals to bleed the main rudder Power Control Unit (PCU) until the movement of the rudder is smooth.

SUBTASK 27-21-91-790-001

(13) Examine the hydraulic connections for leakage.

SUBTASK 27-21-91-860-008

(14) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-91-820-001

(15) Do this task: Control Rod for Main Rudder Power Control Unit Adjustment, TASK 27-21-00-820-804.

SUBTASK 27-21-91-820-002

(16) Do this task: Rudder Main PCU Force Fight Monitor Test, TASK 27-21-00-700-812.

SUBTASK 27-21-91-820-003

(17) Do this task: Main Rudder PCU Internal Hydraulic Leakage Test, TASK 27-21-00-700-810.

SUBTASK 27-21-91-730-001

(18) Do this task: Rudder Pedal Forces Test, TASK 27-21-00-700-804.

SUBTASK 27-21-91-730-002

(19) Do this task: Rudder Hinge Friction Test, TASK 27-21-00-700-809.

SUBTASK 27-21-91-730-003

(20) Do this task: Rudder Hinge Bearing and PCU Bearing Freeplay, TASK 27-21-00-700-802.

SUBTASK 27-21-91-730-004

(21) Do this task: Rudder Centering Test, TASK 27-21-00-700-803.

SUBTASK 27-21-91-730-005

(22) Do this task: Rudder Travel Test, TASK 27-21-00-700-805.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-91-410-001

(1) Install these access panels:

Number	Name/Location
324DL	Trailing Edge Access
324DR	Vertical Fin, Trailing Edge Access
324FR	Panel Assy - Trailing Edge, Beam Seal, Vert Fin

SUBTASK 27-21-91-410-002

(2) If it is necessary, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-91-000-802

4. Main Rudder PCU Load Limiter Solenoid Valve Removal

(Figure 403)

A. General

(1) There is two load limiter solenoid valves, one yaw damper solenoid valve, and one transfer valve installed on the main rudder Power Control Unit (PCU). For the transfer valve and yaw damper solenoid, refer to PAGEBLOCK 22-23-21/401.

(2) Each of the two main rudder PCU load limiter solenoid valves is referred to as the "solenoid valve [71]" in this procedure. This procedure is applicable to both solenoid valves.

B. References

Reference	Title
22-23-21 P/B 401	MAIN RUDDER POWER CONTROL UNIT (PCU) SOLENOID AND ELECTROHYDRAULIC SERVO VALVES - REMOVAL/INSTALLATION
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
324	Vertical Fin - Rear Spar To Trailing Edge

D. Access Panels

Number	Name/Location
324DR	Vertical Fin, Trailing Edge Access

E. Prepare for the Removal

SUBTASK 27-21-91-860-009

(1) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-91-860-010

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	7	C00285	YAW DAMPER AC
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

SUBTASK 27-21-91-010-002

(3) To get access to the main rudder power control unit [1], do this step:

Remove this access panel:

Number	Name/Location
324DR	Vertical Fin, Trailing Edge Access

F. Solenoid Valve Removal

SUBTASK 27-21-91-020-009

(1) Do these steps to remove the solenoid valve [71]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the bolts [72] that attach the solenoid valve [71] to the main rudder power control unit [1].

(b) Remove the seal plate [73].

(c) Remove the O-ring [74].

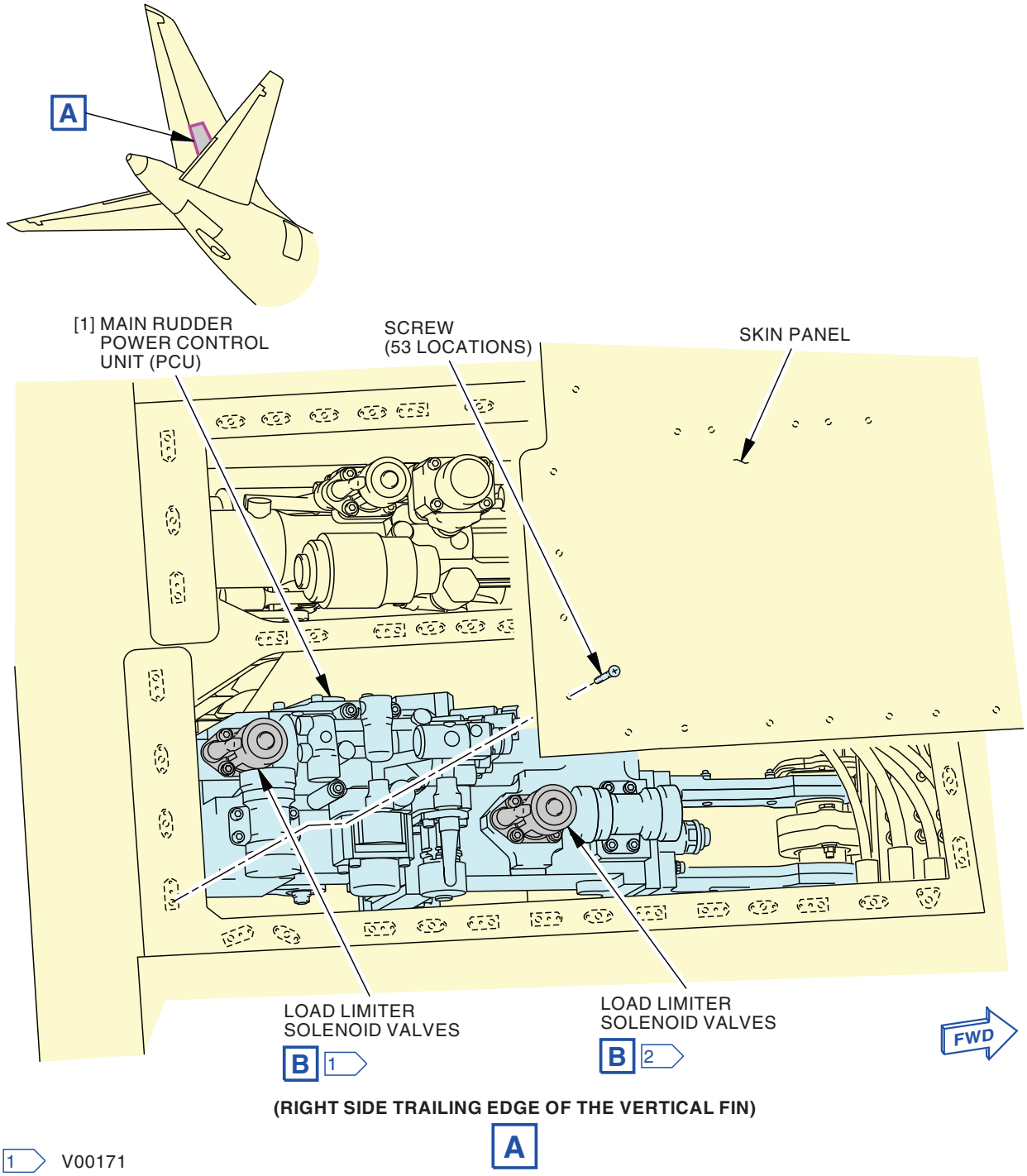
1) Discard the O-ring [74].

(d) Remove the solenoid valve [71].

(e) Put plugs on the main rudder power control unit [1] ports.

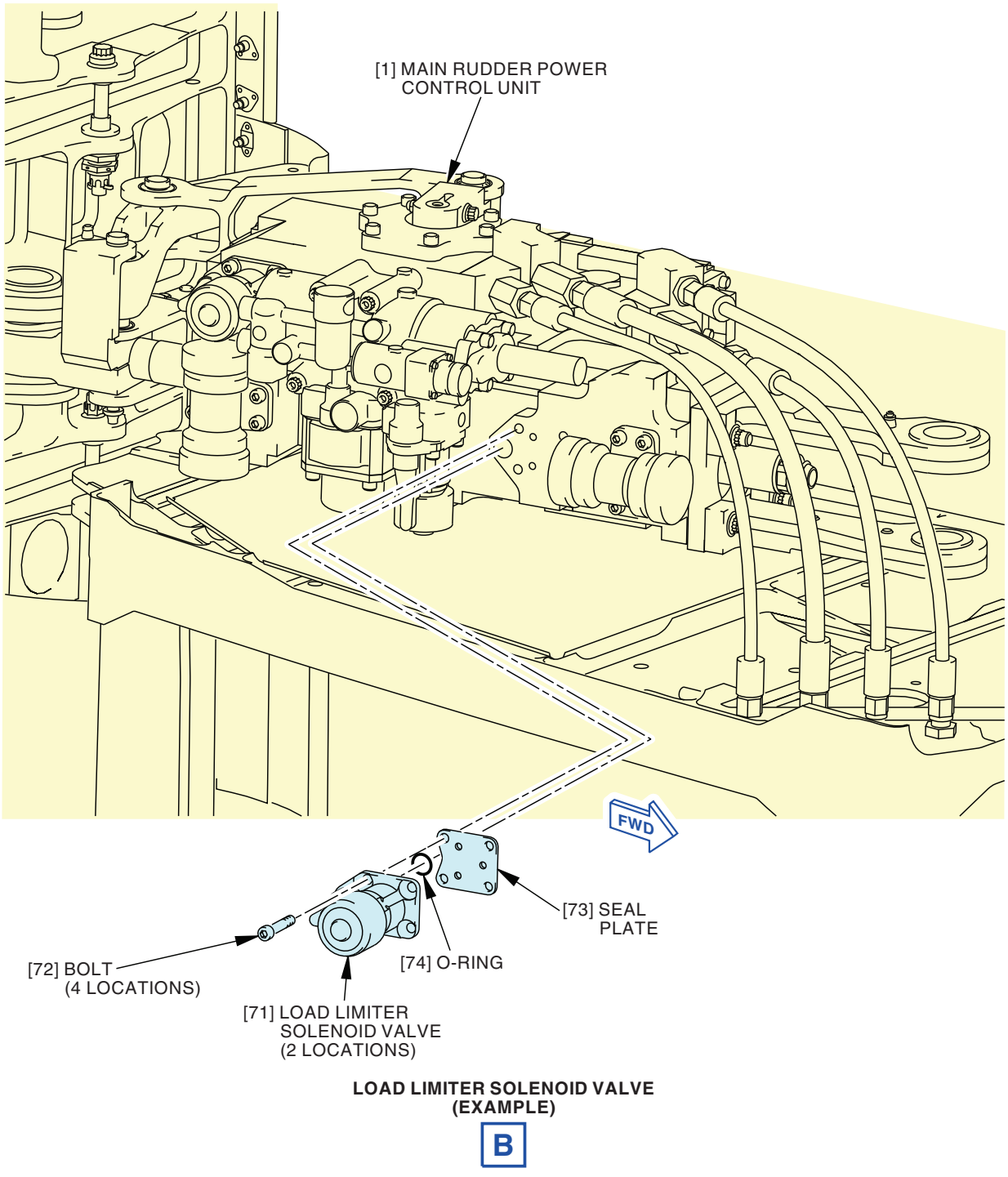
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Rudder PCU Load Limiter Solenoid Valve Installation
Figure 403/27-21-91-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Rudder PCU Load Limiter Solenoid Valve Installation
Figure 403/27-21-91-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-91-400-802

5. Main Rudder PCU Load Limiter Solenoid Valve Installation

(Figure 403)

A. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
71	Solenoid valve	27-21-91-02-232	PTF ALL
		27-21-91-02-332	PTF ALL
74	O-ring	27-21-91-02-328	PTF ALL

D. Location Zones

Zone	Area
211	Flight Compartment - Left
324	Vertical Fin - Rear Spar To Trailing Edge

E. Access Panels

Number	Name/Location
324DR	Vertical Fin, Trailing Edge Access

F. Prepare for the Installation

SUBTASK 27-21-91-860-011

(1) Make sure that you remove pressure from the Rudder Hydraulic Systems A and B.

(a) To remove it, do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

G. Solenoid Valve Installation

SUBTASK 27-21-91-020-010

(1) Do these steps to install the solenoid valve [71]:

(a) Remove the plugs on the main rudder power control unit [1] ports.

(b) Clean the seal plate [73].

(c) Lubricate the new O-ring [74] with hydraulic fluid, D00153.

(d) Install the new O-ring [74].

(e) Install the seal plate [73].

(f) Install the solenoid valve [71].

(g) Install the bolts [72] that attach the solenoid valve [71] to the seal plate [73].

(h) Tighten the bolts [72] to 50 in-lb (5.6 N·m) - 70 in-lb (7.9 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-91-860-012

(2) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

SUBTASK 27-21-91-860-013

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	7	C00285	YAW DAMPER AC
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

SUBTASK 27-21-91-410-003

(4) Install this access panel:

Number	Name/Location
324DR	Vertical Fin, Trailing Edge Access

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STANDBY RUDDER POWER CONTROL UNIT SHUTOFF VALVE - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal of the rudder valve motor from the standby rudder power control unit (PCU) shutoff valve

(2) An installation of the rudder valve motor to the standby rudder PCU shutoff valve

(3) A removal of the standby rudder power control unit (PCU) shutoff valve

(4) An installation of the standby rudder PCU shutoff valve

(5) An installation test of the standby rudder PCU shutoff valve.

B. The standby rudder PCU shutoff valve is referred to as the "shutoff valve" in this procedure.

C. You do not have to remove the shutoff valve from the modular package assembly to remove the rudder valve motor.

TASK 27-21-94-000-801

2. Rudder Valve Motor Removal

(Figure 201)

A. General

(1) This task includes the steps to remove the rudder valve motor.

B. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

D. Prepare for the Removal

SUBTASK 27-21-94-860-001

(1) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-94-860-002

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

E. Rudder Valve Motor Removal

SUBTASK 27-21-94-020-001

(1) Disconnect the electrical connector [5] from the rudder valve motor [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) After you disconnect the electrical connector(s), examine the connector(s) for corrosion.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(b) If you find corrosion, correct the problem (SWPM 20-60-03).

SUBTASK 27-21-94-020-002

(2) Move the manual override lever [2] on the rudder valve motor [4] to position 2.

SUBTASK 27-21-94-020-003

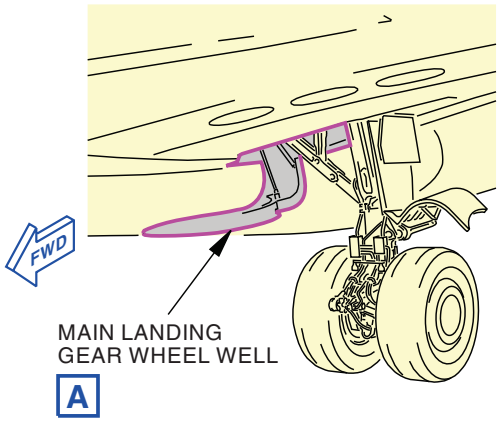
(3) Remove the lockwire and screws [6] from the rudder valve motor [4].

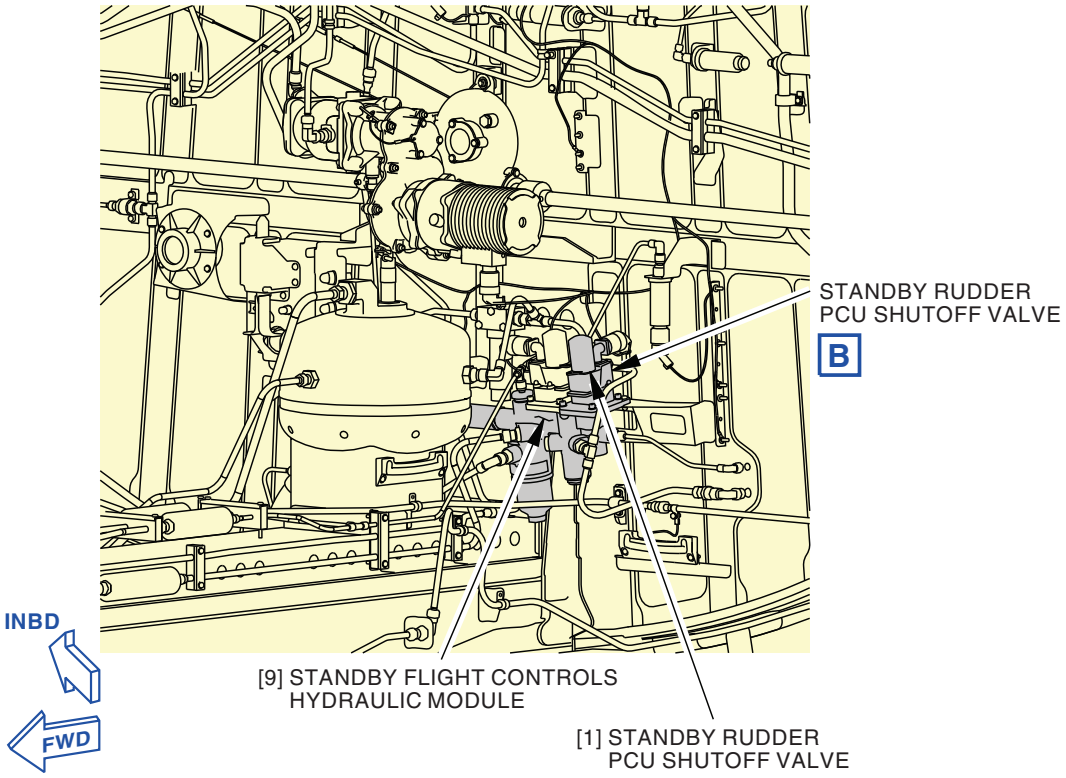
SUBTASK 27-21-94-020-004

(4) Remove the rudder valve motor [4] from the standby rudder PCU shutoff valve [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





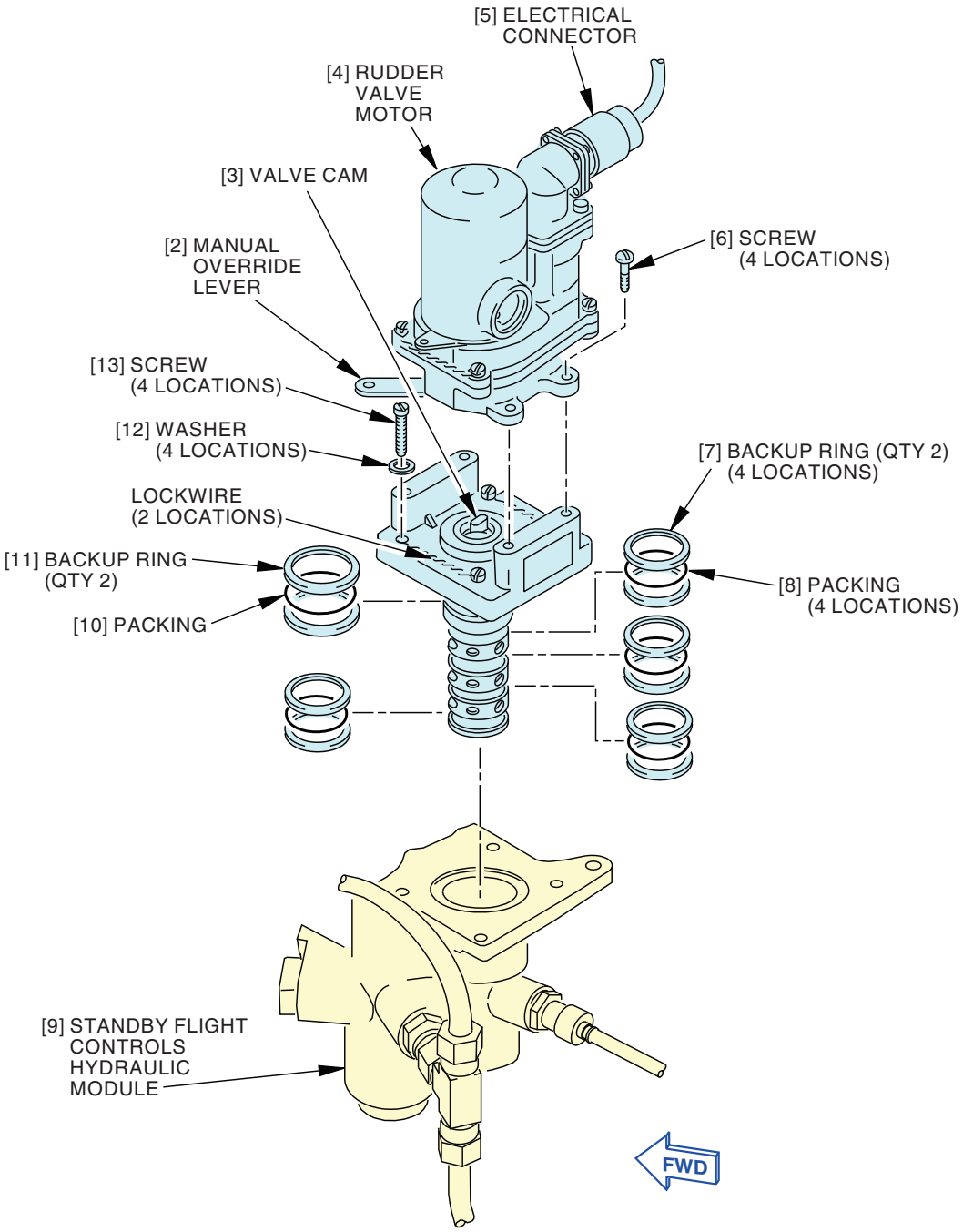
**MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE)**

A

2412435 S00061530666_V2

**Standby Rudder PCU Shutoff Valve Installation
Figure 201/27-21-94-990-801 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Standby Rudder PCU Shutoff Valve Installation
Figure 201/27-21-94-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-94-400-801

3. Rudder Valve Motor Installation

(Figure 201)

A. General

(1) This task includes the steps to install the rudder valve motor.

B. References

Reference	Title
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Consumable Materials

Reference	Description	Specification
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	
G50225	Lockwire - MS20995C20, Corrosion Resistant Steel - 0.020 Inch (0.508 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Rudder valve motor	27-21-94-02-035	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

F. Rudder Valve Motor Installation

SUBTASK 27-21-94-020-005

(1) Make sure that the manual override lever [2] on the rudder valve motor [4] is in position 2.

SUBTASK 27-21-94-420-001

(2) Put the rudder valve motor [4] on the standby rudder PCU shutoff valve [1].

(a) Make sure that the rudder valve motor [4] is engaged with the valve cam [3].

SUBTASK 27-21-94-420-002

(3) Install the screws [6] that attach the rudder valve motor [4] to the standby rudder PCU shutoff valve [1].

(a) Tighten the screws [6] to 17 in-lb (1.92 N·m) - 20 in-lb (2.26 N·m).

(b) Install MS20995C20 lockwire, G50225, on the screws [6].

SUBTASK 27-21-94-210-001

(4) If you did not examine the electrical connector(s) at the removal step, do these steps:

NOTE: You can inspect the electrical connector(s) after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(a) Before you connect the electrical connector(s), examine the connector(s) for corrosion.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES AT AIRPORTS WHERE DEICING FLUID THAT CONTAIN POTASSIUM FORMATE IS USED. ALSO DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTOR IS PART OF A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(b) If you find corrosion, correct the problem (SWPM 20-60-03).

SUBTASK 27-21-94-800-001

(5) Apply D5026NS or ZC-026 compound, G50171, to the electrical connector [5] (SWPM 20-60-03).

SUBTASK 27-21-94-420-003

(6) Connect the electrical connector [5] to the rudder valve motor [4].

SUBTASK 27-21-94-860-003

(7) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

SUBTASK 27-21-94-860-004

(8) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-21-94-730-001

(9) Do this task: Standby Rudder PCU Shutoff Valve Installation Test, TASK 27-21-94-710-801.

——— END OF TASK ———

TASK 27-21-94-000-802

4. Standby Rudder PCU Shutoff Valve Removal

(Figure 201)

A. General

(1) This task provides instructions to remove the shutoff valve (with the rudder valve motor attached) from the standby flight controls hydraulic module.

B. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

D. Prepare for the Removal

SUBTASK 27-21-94-860-005

(1) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-94-860-006

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

E. Shutoff Valve Removal

SUBTASK 27-21-94-020-006

(1) Disconnect the electrical connector [5] from the rudder valve motor [4].

(a) After you disconnect the electrical connector(s), examine the connector(s) for corrosion.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(b) If there was corrosion, refer to (SWPM 20-60-03) to correct the problem.

SUBTASK 27-21-94-020-007

(2) Remove the lockwires from the screws [13].

SUBTASK 27-21-94-020-008

(3) Remove the screws [13] and washers [12] which attach the standby rudder PCU shutoff valve [1] to the standby flight controls hydraulic module [9].

SUBTASK 27-21-94-020-009

CAUTION

DO NOT LET HYDRAULIC FLUID GET ON EQUIPMENT. IF HYDRAULIC FLUID GETS ON EQUIPMENT, IMMEDIATELY CLEAN IT FROM THE EQUIPMENT. HYDRAULIC FLUID CAN CAUSE DAMAGE TO EQUIPMENT.

(4) Put a container below the standby rudder PCU shutoff valve [1] to catch hydraulic fluid.

SUBTASK 27-21-94-020-010

(5) Carefully turn and pull directly up on the standby rudder PCU shutoff valve [1] to remove it from the standby flight controls hydraulic module [9].

SUBTASK 27-21-94-020-011

(6) Put a cover on the standby flight controls hydraulic module [9] to keep dirt out.

END OF TASK

TASK 27-21-94-400-802

5. Standby Rudder PCU Shutoff Valve Installation

(Figure 201)

A. General

(1) This task provides instructions to install the shutoff valve (with the rudder valve motor attached) to the standby flight controls hydraulic module.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Packing	27-21-94-02-025	PTF ALL
10	Packing	27-21-94-02-015	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

F. Standby Rudder PCU Shutoff Valve Installation

SUBTASK 27-21-94-020-012

(1) Remove the cover from the standby flight controls hydraulic module [9].

SUBTASK 27-21-94-420-004

(2) If the shutoff valve does not have the packings [8], packing [10], backup rings [7], and backup rings [11] installed, do these steps:

(a) Lubricate the packings [8], packing [10], backup rings [7], and backup rings [11] with hydraulic fluid, D00153.

(b) Install the packings [8], packing [10], backup rings [7], and backup rings [11] to the standby rudder PCU shutoff valve [1].

SUBTASK 27-21-94-420-005

(3) Carefully install the standby rudder PCU shutoff valve [1] in the standby flight controls hydraulic module [9].

SUBTASK 27-21-94-420-006

(4) Install the washers [12] and screws [13] which attach the standby rudder PCU shutoff valve [1] to the standby flight controls hydraulic module [9].

(a) Tighten the screws [13] to 78 in-lb (8.8 N·m) – 82 in-lb (9.3 N·m).

SUBTASK 27-21-94-420-007

(5) Install the MS20995C32 lockwire, G01048, on the screws [13].

SUBTASK 27-21-94-210-002

(6) If you did not examine the electrical connector(s) at the removal step, do these steps:

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(a) Before you connect the electrical connector(s), examine the connector(s) for corrosion.

(b) If there was corrosion, refer to (SWPM 20-60-03) to correct the problem.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-94-800-002

(7) Apply the D5026NS or ZC-026 compound, G50171, to the electrical connector [5] (SWPM 20-60-03).

SUBTASK 27-21-94-420-008

(8) Connect the electrical connector [5] to the rudder valve motor [4].

SUBTASK 27-21-94-860-007

(9) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

SUBTASK 27-21-94-860-008

(10) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-21-94-710-001

(11) Do this task: Standby Rudder PCU Shutoff Valve Installation Test, TASK 27-21-94-710-801.

SUBTASK 27-21-94-860-009

(12) If it is necessary, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

——— END OF TASK ———

TASK 27-21-94-710-801

6. Standby Rudder PCU Shutoff Valve Installation Test

(Figure 201, Figure 202)

A. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Standby Rudder PCU Shutoff Valve Installation Test

SUBTASK 27-21-94-860-011

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

SUBTASK 27-21-94-860-012

(2) Make sure that the FLIGHT CONTROL A and B switches are in the ON position.

NOTE: The FLIGHT CONTROL A and B switches are on the upper left corner of the pilot's overhead panel.

SUBTASK 27-21-94-860-013

(3) Make sure that the manual override lever on the rudder valve motor is in position 1.

SUBTASK 27-21-94-860-014

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Put the FLIGHT CONTROL A or B switch to the STDBY RUD position.

SUBTASK 27-21-94-860-015

(5) Make sure that the manual override lever on the rudder valve motor is in position 2.

SUBTASK 27-21-94-860-016

(6) Do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

NOTE: Nose landing gear will move when the rudder pedals are moved. To prevent NLG movement, install a lockout pin in the nose wheel steering mechanism.

SUBTASK 27-21-94-710-002

(7) Push the rudder pedals until they touches the forward quadrant stops.

(a) Have a second maintenance personnel visually make sure that the rudder moves correctly.

SUBTASK 27-21-94-790-001

(8) Make sure that there is no hydraulic fluid leakage from the shutoff valve.

SUBTASK 27-21-94-860-017

(9) Put the FLIGHT CONTROL A or B switch to the ON position.

SUBTASK 27-21-94-860-018

(10) Make sure that the manual override lever on the rudder valve motor is in position 1.

SUBTASK 27-21-94-860-019

(11) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

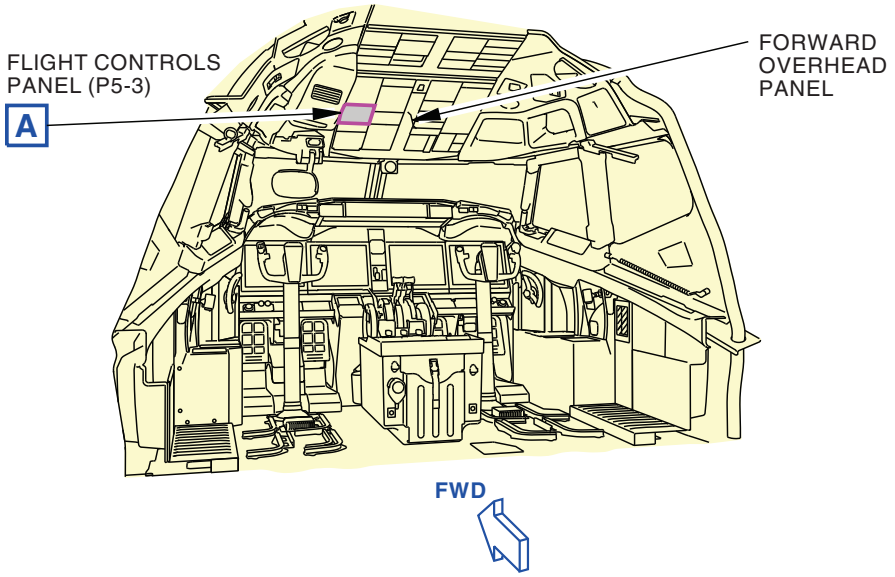
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-94-860-020

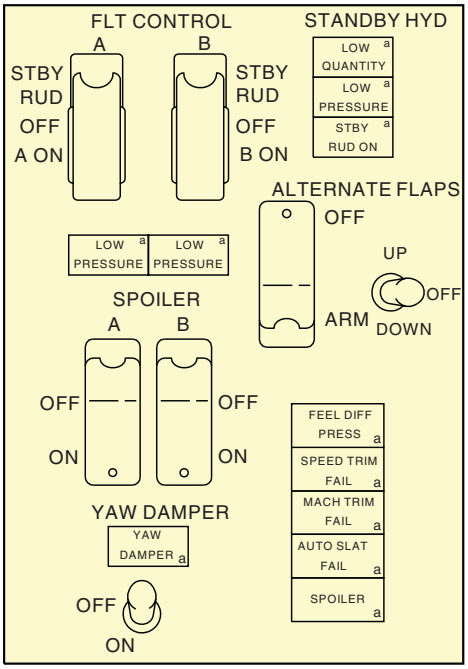
(12) Do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



FLIGHT CONTROLS PANEL (P5-3)

A

2412437 S00061530668_V2

Flight Control Switches
Figure 202/27-21-94-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER HYDRAULIC FUSE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the rudder hydraulic fuse

(2) An installation of the rudder hydraulic fuse.

TASK 27-21-95-000-801

2. Rudder Hydraulic Fuse Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-41-00-800-801	Horizontal Stabilizer Trim Control System - Deactivation (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1672	Lock Assembly - Stabilizer Trim Part #: F71336-501 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Prepare for the Removal

SUBTASK 27-21-95-040-001

(1) Do this task: Horizontal Stabilizer Trim Control System - Deactivation, TASK 27-41-00-800-801.

(a) If the lock assembly, SPL-1672, is not available, install the DO NOT OPERATE tag, STD-858, on the stabilizer trim wheel.

SUBTASK 27-21-95-860-001

(2) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-95-860-002

(3) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-95-010-001

(4) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Rudder Hydraulic Fuse Removal

SUBTASK 27-21-95-020-003

(1) Remove the bolt [1], washer [2], and nut [3].

SUBTASK 27-21-95-020-002

(2) Remove the bolt [8], washer [9], nut [10], and clamp [4].

(a) Leave the clamp [11] around the hydraulic line [12].

SUBTASK 27-21-95-020-001

(3) Do these steps to remove the rudder hydraulic fuse [7]:

(a) Disconnect the hydraulic line [5] and hydraulic line [6] from the rudder hydraulic fuse [7].

(b) Remove the rudder hydraulic fuse [7].

G. Rudder Hydraulic Fuse Test

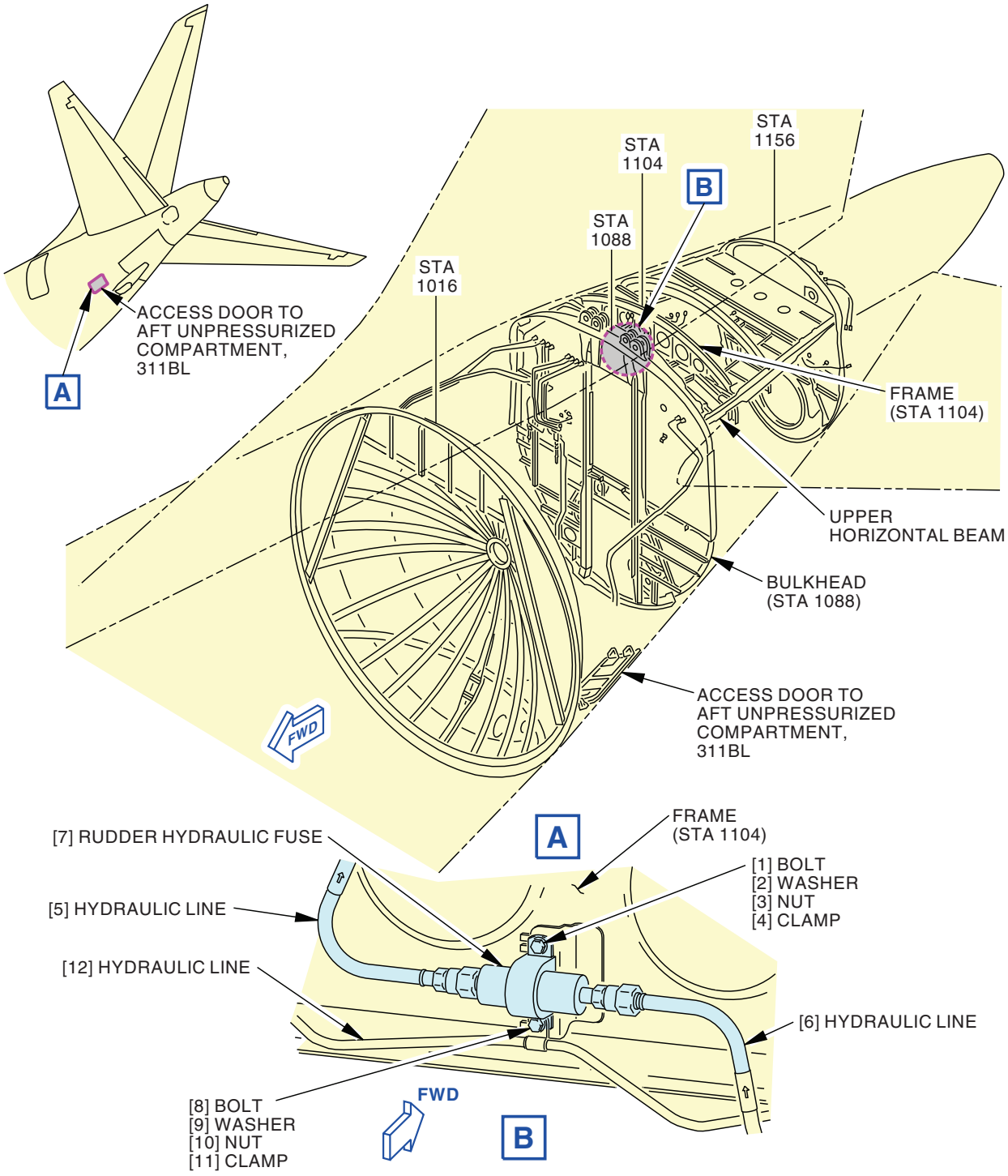
SUBTASK 27-21-95-860-008

(1) Do the functional test of the rudder hydraulic fuse with the suppliers recommended component maintenance test instructions and test equipment.

NOTE: The suppliers instructions give a proof pressure test, fusing and reset test, pressure drop test, closing time test, internal leakage test. This is an off-airplane bench test.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Hydraulic Fuse Installation
Figure 401/27-21-95-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-95-400-801

3. Rudder Hydraulic Fuse Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
27-41-00-800-802	Horizontal Stabilizer Trim Control System - Activation (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Rudder hydraulic fuse	27-21-95-01-040	PTF ALL

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Rudder Hydraulic Fuse Installation

SUBTASK 27-21-95-420-001

(1) Do these steps to install the rudder hydraulic fuse [7]:

(a) Put the rudder hydraulic fuse [7] in its position.

NOTE: The flow direction arrow on the rudder hydraulic fuse must agree with the arrow on the airplane.

(b) Connect the hydraulic line [5] and hydraulic line [6] to the rudder hydraulic fuse [7].

(c) Tighten the B-nuts on hydraulic line [5] and hydraulic line [6] to 270 ±14 in-lb (30.5 ±1.6 N·m).

SUBTASK 27-21-95-420-002

(2) Install the clamp [4] with the bolt [1], washer [2], and nut [3].

SUBTASK 27-21-95-420-003

(3) Install the clamp [11] with the bolt [8], washer [9], and nut [10].

SUBTASK 27-21-95-860-005

(4) Do these steps to do a test of the rudder hydraulic fuse [7]:

(a) Activate the rudder hydraulic system B (TASK 27-21-00-840-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Pressurize the rudder hydraulic system B (TASK 27-21-00-800-801).

(c) Move the rudder pedals through their full travel three times.

(d) Make sure that the rudder moves correctly.

SUBTASK 27-21-95-210-001

(5) Visually examine the rudder hydraulic fuse [7] for hydraulic leaks.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-95-440-001

(1) Do this task: Horizontal Stabilizer Trim Control System - Activation, TASK 27-41-00-800-802.

(a) If installed, remove the DO NOT OPERATE tag from the stabilizer trim wheel.

SUBTASK 27-21-95-860-006

(2) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-95-410-001

(3) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR AND TAB CONTROL SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Elevator Hydraulic System A and B - Pressurization

(2) Put the Elevator Hydraulic Systems A and B Back to the Condition before the Pressurization

(3) Remove Pressure from the Elevator Hydraulic Systems A and B

(4) Put the Elevator Hydraulic Systems A and B Back to the Condition before the Pressure Removal

(5) Null Procedure - Mach Trim Actuator

(6) Elevator Control Pushrod - Visual Inspection

(7) Elevator Feel and Centering Unit - Detailed Inspection

(8) Elevator Feel and Centering Unit Output Rod - Detailed Visual Inspection

(9) Elevator Tab and Tab Mechanism - Detailed Visual Inspection

(10) Elevator and Tab Control System - Deactivation

(11) Elevator and Tab Control System - Activation.

B. Hydraulic systems A, B must have pressure removed before maintenance is done on the elevator control system. This is to prevent injury to persons because of accidental operation of the control system while you do maintenance. You must be careful to put maintenance stands and equipment away from control surface travel.

C. Electric pump 1 pressurizes the hydraulic system B and electric pump 2 pressurizes the hydraulic system A. Electric pump 1 heat exchanger is installed in the fuel tank 2. Electric pump 2 heat exchanger is installed in the fuel tank 1.

D. This procedure operates the electric hydraulic system A or B pumps to pressurize the hydraulic systems A and B. Pressurization of the hydraulic systems with the engine-driven hydraulic pumps is not included in this procedure.

TASK 27-31-00-800-801

2. Elevator Hydraulic System A and B - Pressurization

(Figure 201)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Pressurization of the Elevator Hydraulic System A and B

SUBTASK 27-31-00-860-002

CAUTION

DO NOT OPERATE SYSTEM A OR B HYDRAULIC PUMPS FOR MORE THAN TWO MINUTES WITHOUT FUEL IN THE FUEL TANKS. THE NO. 1 (FOR SYSTEM A) AND NO. 2 (FOR SYSTEM B) FUEL TANKS MUST HAVE A MINIMUM OF 250 GALLONS (1675 POUNDS/760 KILOGRAMS) OF FUEL IN THEM. IF THERE IS NOT SUFFICIENT FUEL IN THE FUEL TANKS, THE HYDRAULIC FLUID WILL BECOME TOO HOT.

(1) Make sure that the No. 1 and No. 2 fuel tanks each have 250 gallons (1675 pounds/760 kilograms) of fuel in them.

SUBTASK 27-31-00-210-001

(2) Make sure that the hydraulic reservoirs are full.

(a) To fill the hydraulic reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-31-00-860-003

(3) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-31-00-860-004

(4) Set the parking brake.

SUBTASK 27-31-00-480-001

WARNING

MAKE SURE THAT THE GROUND LOCKS ARE INSTALLED IN ALL THE LANDING GEARS BEFORE YOU PRESSURIZE THE HYDRAULIC SYSTEM. WITHOUT THE GROUND LOCKS, THE LANDING GEARS CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-31-00-860-005

WARNING

KEEP PERSONNEL AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Pressurize the elevator hydraulic system A:

(a) Set the ELEC 2, HYD PUMPS A switch on the forward overhead panel to the ON position.

(b) To pressurize the hydraulic system A, set the FLT CONTROL A switch on the forward overhead panel to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-780-001

WARNING

KEEP PERSONNEL AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

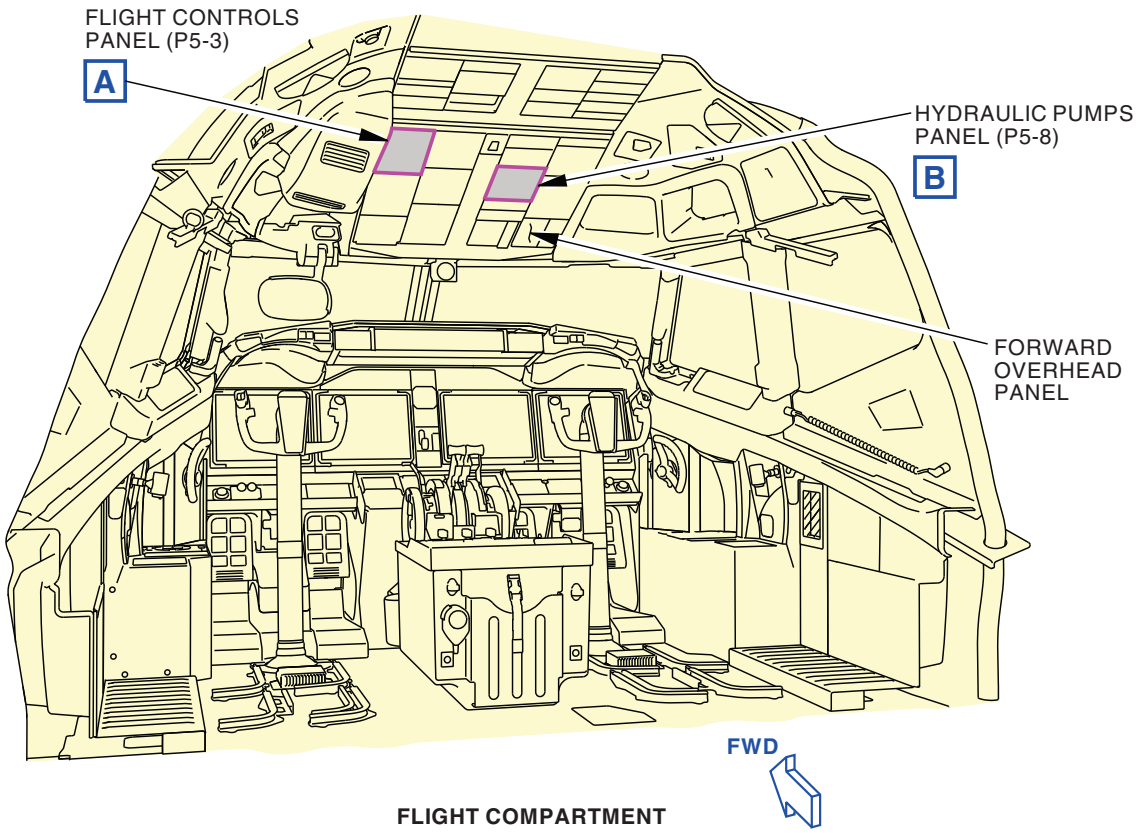
(7) Pressurize the elevator hydraulic system B:

(a) Set the ELEC 1, HYD PUMPS B switch on the forward overhead panel to the ON position.

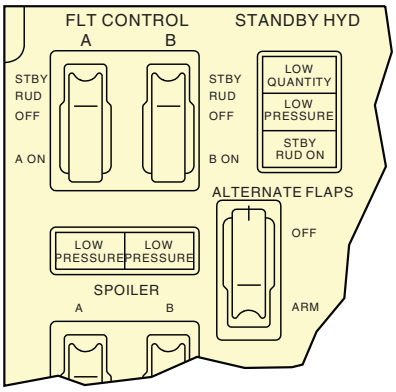
(b) To pressurize the hydraulic system B, set the FLT CONTROL B switch on the forward overhead panel to the ON position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

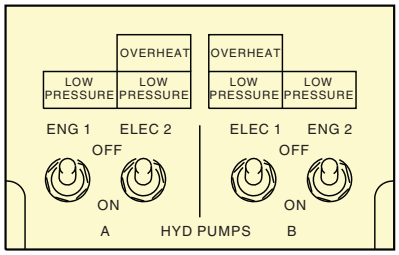


FLIGHT COMPARTMENT



FLIGHT CONTROLS PANEL (P5-3)

A



HYDRAULIC PUMPS PANEL (P5-8)

B

2412439 S00061530682_V2

Elevator Power Control Switches
Figure 201/27-31-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-840-801

3. Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization

(Figure 201)

A. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

B. Putting the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization

SUBTASK 27-31-00-860-006

(1) Set the ELEC 2, HYD PUMPS A and the ELEC 1, HYD PUMPS B switches on the forward overhead panel to the OFF position.

SUBTASK 27-31-00-860-007

(2) Set the FLT CONTROL A and B switches on the forward overhead panel to the OFF position.

——— END OF TASK ———

TASK 27-31-00-800-802

4. Pressure from the Elevator Hydraulic Systems A and B Removal

(Figure 201, Figure 202)

A. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Pressure from the Elevator Hydraulic Systems A and B Removal

SUBTASK 27-31-00-860-008

(1) Set the FLT CONTROL A and B switches on the forward overhead panel to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-860-009

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-31-00-860-010

(3) Set the ELEC 2, HYD PUMPS A and the ELEC 1, HYD PUMPS B switches on the forward overhead panel to the OFF position.

SUBTASK 27-31-00-870-001

(4) Operate the rudder pedals to decrease the remaining hydraulic system pressure.

SUBTASK 27-31-00-860-011

(5) If it is necessary to open a hydraulic line, you must release pressure from the reservoir for the applicable hydraulic system (TASK 29-09-00-860-802).

SUBTASK 27-31-00-480-002

(6) Install the flight controls hydraulic module locks (Figure 202):

(a) Disconnect the electrical connectors from the flight controls shutoff valve at the flight controls hydraulic modules A and B.

(b) Examine the connectors for corrosion.

1) If corrosion is found, refer to SWPM 20-60-03 to correct the problem.

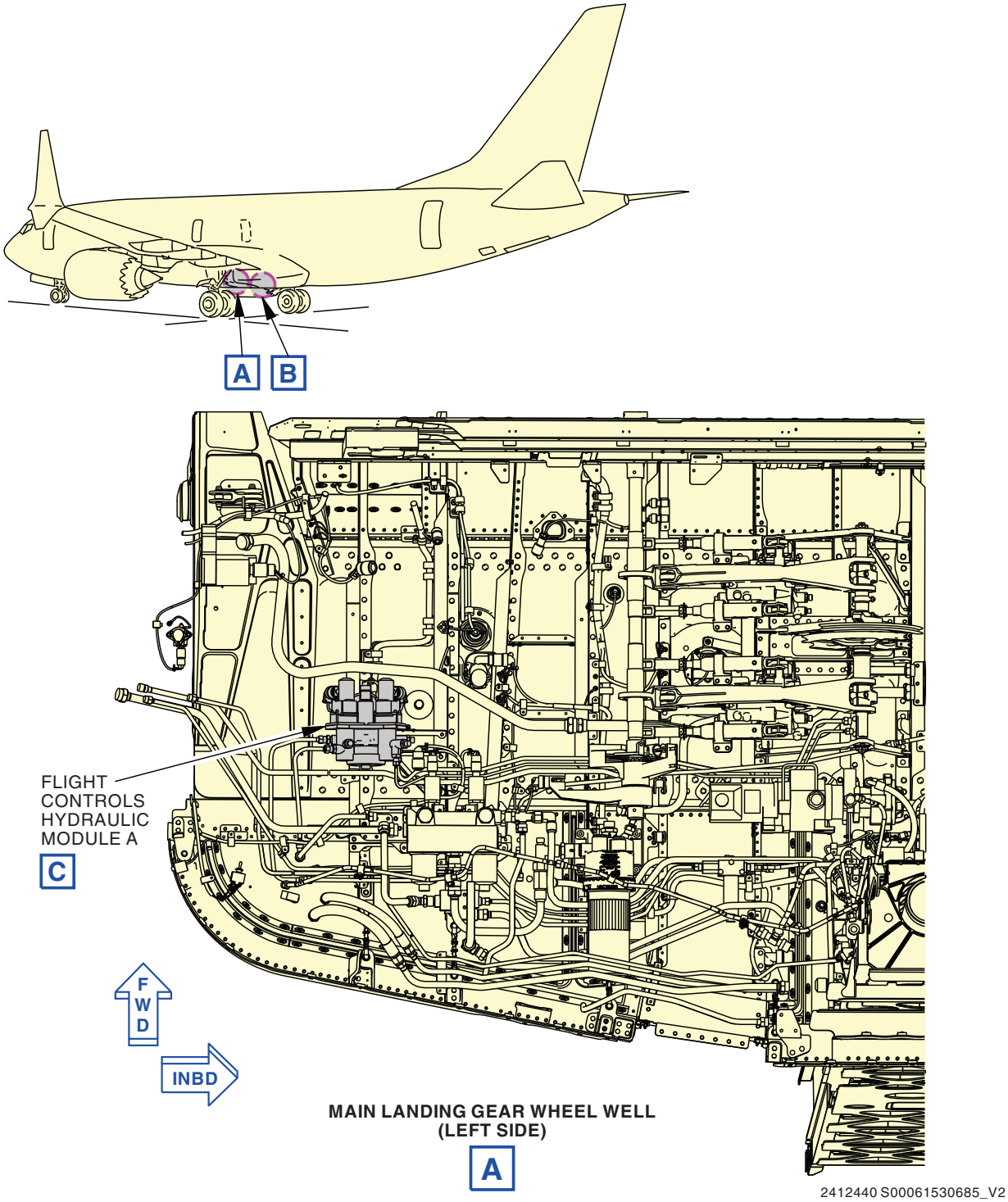
(c) Put the manual override lever at the two flight controls shutoff valves in the position 2 (OFF position).

(d) Install the lock, SPL-1675, on the two flight controls shutoff valves.

1) Install the detent pins.

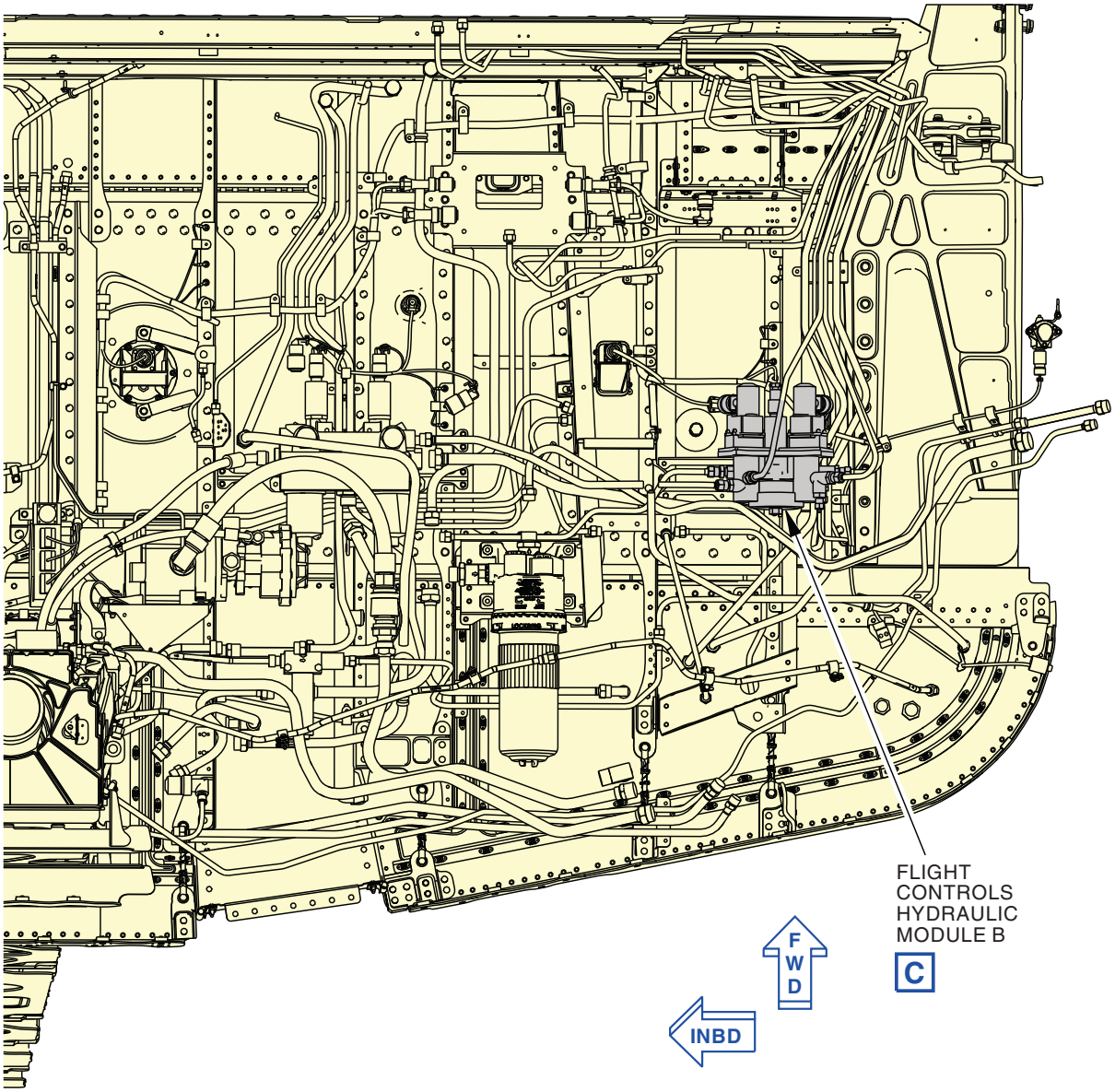
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



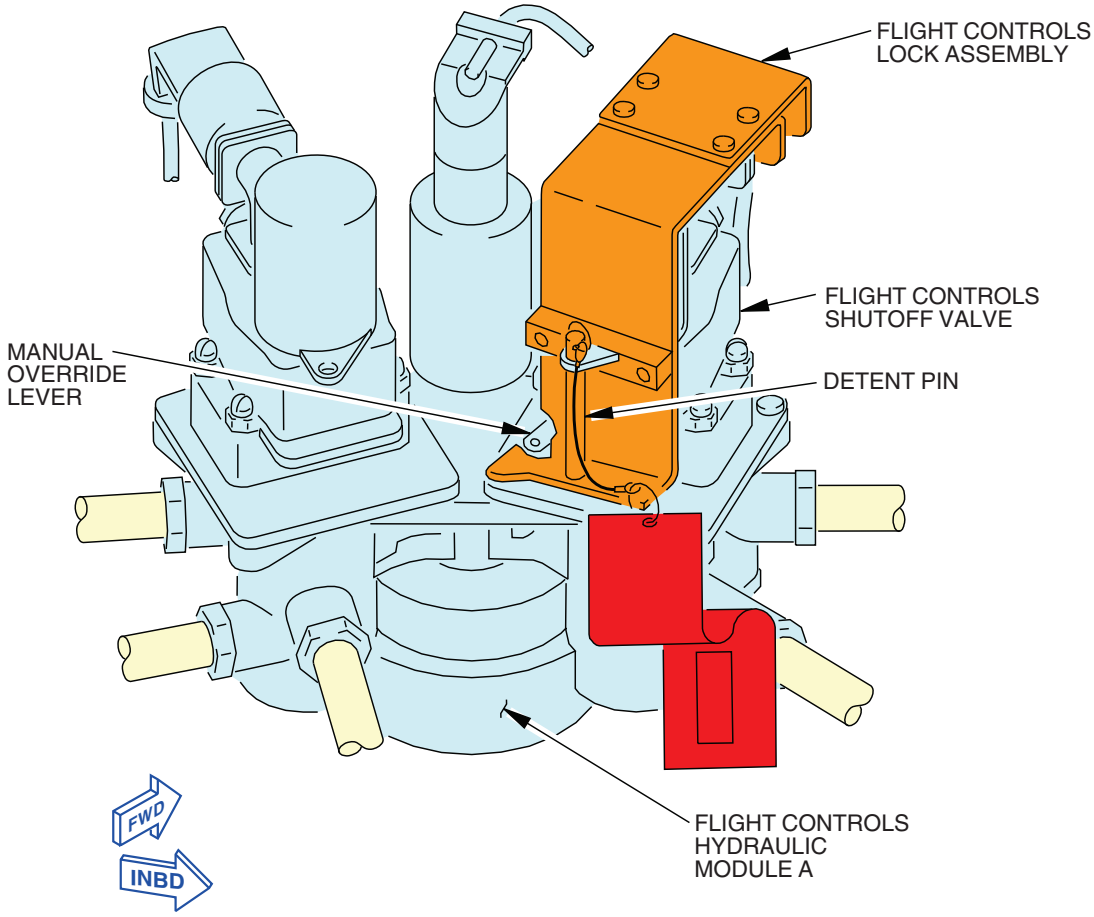
Flight Controls Hydraulic Module Lock Installation
Figure 202/27-31-00-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Controls Hydraulic Module Lock Installation
Figure 202/27-31-00-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT CONTROLS HYDRAULIC MODULE A
(MODULE B IS EQUIVALENT)

C

2412442 S00061530687_V4

Flight Controls Hydraulic Module Lock Installation
Figure 202/27-31-00-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-840-802

5. Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal

(Figure 201, Figure 202)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-09-31-710-801	Flight Controls Shutoff Valve Test (P/B 501)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

E. Put the Elevator Systems A and B Back to the Condition Before the Pressure Removal

SUBTASK 27-31-00-080-002

(1) Remove the flight controls hydraulic module locks from the flight controls shutoff valves.

(a) Remove the detent pins.

(b) Remove the lock, SPL-1675, from the two flight controls shutoff valves.

(c) Put the manual override lever at the two flight controls shutoff valves in the position 1 (ON position).

(d) Examine the connectors for corrosion.

1) If there is corrosion, refer to SWPM 20-60-03 to correct the problem.

(e) Apply the D5026NS or ZC-026 compound, G50171, to the connectors (SWPM 20-60-03).

(f) Connect the electrical connectors to the flight controls shutoff valve at the flight controls hydraulic modules A and B.

SUBTASK 27-31-00-860-012

(2) Set the FLT CONTROL A and B switches on the forward overhead panel to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-860-013

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-31-00-700-006

(4) Do this task: Flight Controls Shutoff Valve Test, TASK 27-09-31-710-801.

SUBTASK 27-31-00-860-014

(5) If it is necessary, fill the hydraulic reservoirs (TASK 12-12-00-610-801).

——— END OF TASK ———

TASK 27-31-00-700-801

6. Null Procedure - Mach Trim Actuator

A. References

Reference	Title
22-11-00-820-802	Elevator Rigging Test (P/B 501)
22-11-00-993-825	Table: (P/B 501)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Prepare for the Test

SUBTASK 27-31-00-860-016

(1) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand, to the CUTOUT position.

SUBTASK 27-31-00-860-017

(2) Make sure that the VHF NAV and IRS switches, on the P5 forward overhead panel, are in the NORMAL positions.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-860-018

(3) Set the left and right IRS select switches, on the P5 aft overhead panel, to the ALIGN or NAV position.

SUBTASK 27-31-00-860-019

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Supply hydraulic power to hydraulic systems A and B (TASK 29-11-00-860-801).

SUBTASK 27-31-00-010-001

(5) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-980-001

(6) Do these steps to set the B dimension:

NOTE: The B dimension is measured between the center of the upper and lower Gimbal pins on the stabilizer trim jackscrew. To do this, one person must be in the tail of the airplane with a headset and the other person in the flight compartment.

(a) Turn the stab trim wheel handle on the control stand to set the B dimension.

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(b) Use the trammel trammel bar, SPL-1677, to measure the B dimension.

(c) Make sure that the B dimension is 39.17 ±0.01 in. (994.92 ±0.26 mm).

F. Null Procedure for Mach Trim Actuator

SUBTASK 27-31-00-740-001

(1) Push the INIT REF key on the Control Display Unit (CDU) keyboard.

NOTE: It is necessary to follow all the on-screen instructions for this test.

(a) Push the Line Select Key (LSK) that is adjacent to each selection:

1) INDEX
2) MAINT
3) DFCS

NOTE: Whenever the DFCS LSK is selected the DFCS BITE test will run.

4) EXTENDED MAINTENANCE
5) RIGGING

SUBTASK 27-31-00-740-002

(2) Push the LSK adjacent to <ELEVATOR on the CDU.

SUBTASK 27-31-00-740-003

(3) Push the LSK adjacent to <ELEV RIG on the CDU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-740-012

(4) Push CONTINUE.

SUBTASK 27-31-00-740-004

(5) Follow the instructions on the CDU and push the LSK adjacent to <CONTINUE until Test 51.02 MT ACT POS menu shows on the CDU.

NOTE: The DFCS BITE Rigging Test will move the mach trim actuator to the fully extended position.

SUBTASK 27-31-00-740-005

(6) Use Test 51.02 and 51.03 to make sure that the MT ACT POS for FCC channel A is within limits (TASK 22-11-00-820-802).

NOTE: You can find the permitted value limits in Table 22-11-00-993-825.

NOTE: For Tests 51.02 and 51.03, there is no value for the B side; N/A.

SUBTASK 27-31-00-740-006

(7) Continue with test 51.02 until test 51.03 is displayed.

(a) Make sure that the value limits are within an acceptable range (TASK 22-11-00-820-802).

NOTE: You can find the permitted value limits in Table 22-11-00-993-825.

(b) Look for “A” adjacent to the applicable LSK and make sure that the Z value shown is between the upper and lower limit.

NOTE: The null procedure for the mach trim actuator is complete.

SUBTASK 27-31-00-860-020

(8) Select PREV MENU (LSK 6R) on the CDU, and do the steps that follow:

(a) Select EXIT (LSK 6L) on the RIGGING INDEX screen.

(b) Complete the instructions on the END OF BITE RIGGING screen.

(c) Select CONTINUE (LSK 6L).

(d) Select EXIT (LSK 6L) on the TEST COMPLETE screen.

(e) Complete the instructions on the END OF DFCS screen.

(f) Select EXIT (LSK 6L).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-410-033

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-840-004

(2) Remove hydraulic power from hydraulic systems A and B (TASK 29-11-00-860-805).

SUBTASK 27-31-00-860-165

(3) Set the STAB TRIM PRI switch, S272, on the control stand, to the NORMAL position.

SUBTASK 27-31-00-860-166

(4) Set the left and right IRS select switches, on the P5 aft overhead panel, to the OFF position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-200-801

7. Elevator Control Pushrod - Detailed Visual Inspection

(Figure 203)

NOTE: This procedure is a scheduled maintenance task.

A. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II

B. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

C. Access Panels

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09

D. Elevator Control Pushrod Detailed Visual Inspection

SUBTASK 27-31-00-010-002

(1) Open these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09

SUBTASK 27-31-00-860-021

(2) If hydraulic systems A and B have pressure, do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-200-001

(3) Do a detailed visual inspection of the elevator control pushrods (on both elevators) (Figure 203).

(a) Make sure that all hardware is installed and secure.

(b) Check for unusual wear and loose parts at the two connection points of the elevator control pushrod.

NOTE: Try to wiggle or move connection bolts with hand pressure only.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-420-001

(4) Tighten and properly secure any loose items.

SUBTASK 27-31-00-960-001

(5) If replacement is necessary, replace any worn or loose parts with new hardware.

SUBTASK 27-31-00-410-001

(6) Close these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09

SUBTASK 27-31-00-390-002

(7) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

SUBTASK 27-31-00-741-001

(8) If hydraulic pressure in systems A and B is necessary, do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

——— END OF TASK ———

TASK 27-31-00-200-802

8. Elevator Feel and Centering Unit Detailed Inspection

(Figure 203)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

B. Access Panels

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

C. Detailed Inspection of the Elevator Feel and Centering Unit

SUBTASK 27-31-00-780-002

(1) If hydraulic systems A and B have pressure, remove pressure from the elevator hydraulic systems A and B (TASK 27-31-00-800-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-010-003

(2) Open these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-00-200-002

(3) Do a detailed inspection of the Elevator Feel and Centering Unit.

(a) Check for any hydraulic leaks at the actuator seals and hydraulic line connections.

(b) Check for unusual wear or loose parts at the Feel and Centering Unit connection points.

1) Make sure that all of the hardware is installed and secure.

a) Tighten and properly secure any loose items.

2) If replacement is necessary, replace any loose or worn hardware with new hardware.

SUBTASK 27-31-00-410-002

(4) Close these access panels:

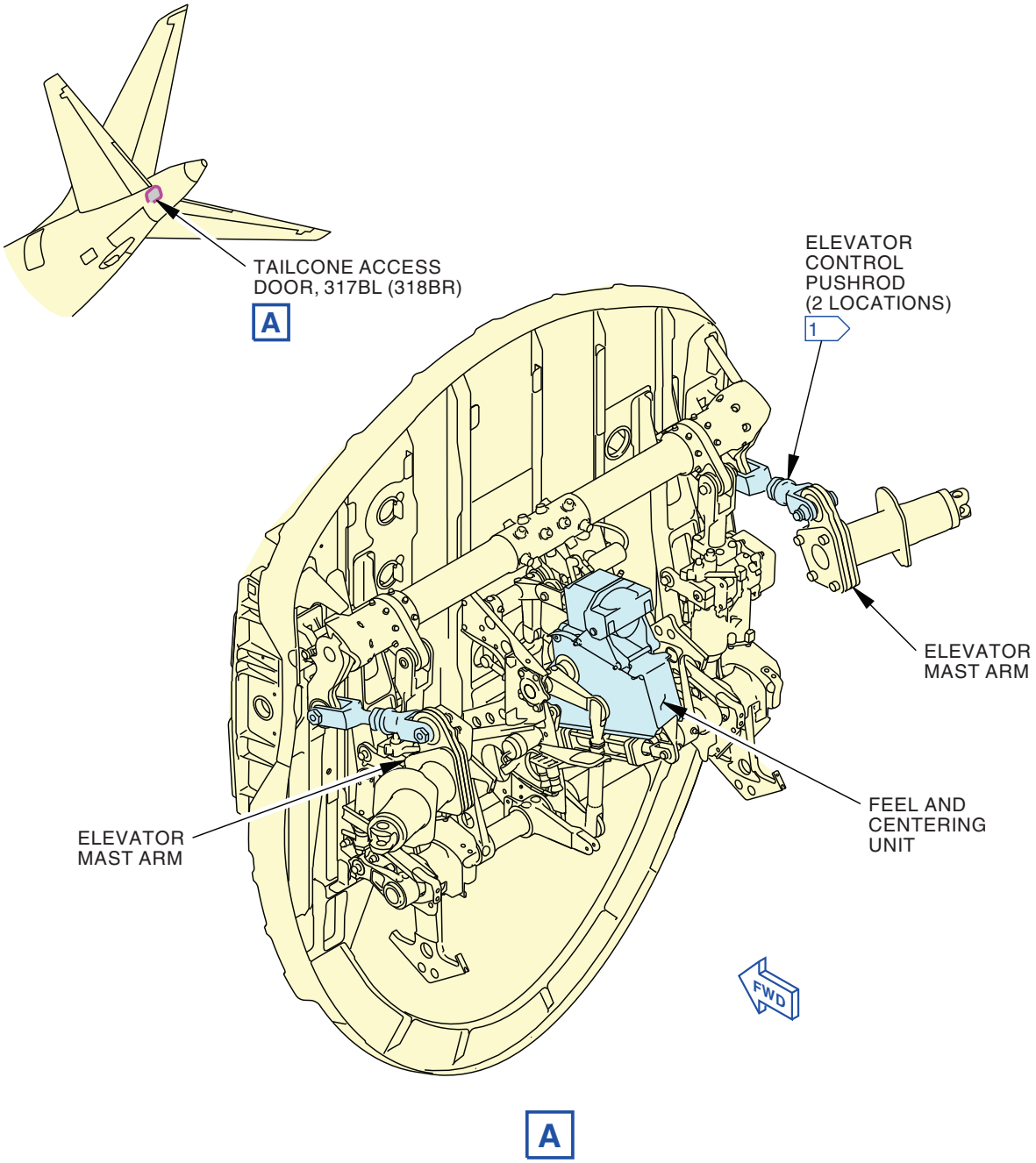
Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-00-840-001

(5) If you need hydraulic pressure in systems A and B, do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 ACCESS TO THE FORWARD ELEVATOR CONTROL PUSHROD CONNECTION POINT IS THROUGH THE TAILCONE ACCESS DOOR 317BL AND 318BR. ACCESS TO THE ELEVATOR CONTROL PUSHROD AFT CONNECTION POINT IS AT THE TOP OF THE ELEVATOR AT THE ELEVATOR MAST ARM.

2412443 S00061530692_V2

Elevator Feel and Centering Unit/Elevator Control Pushrod
Figure 203/27-31-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-211-801

9. Elevator Feel and Centering Unit Output Rod - Detailed Visual Inspection

Figure 204

A. General

(1) This procedure provides instructions for the detailed visual inspection of the elevator feel and centering unit output rod.

B. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

C. Access Panels

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

D. Detailed Visual Inspection of the Elevator Feel and Centering Unit Output Rod

SUBTASK 27-31-00-780-003

(1) If hydraulic systems A and B have pressure, remove pressure from the elevator hydraulic systems A and B (TASK 27-31-00-800-802).

SUBTASK 27-31-00-010-004

(2) Open these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-00-211-001

(3) Do a detailed visual inspection of the Elevator Feel and Centering Unit Output Rod:

(a) Check for unusual wear or loose parts at the Output Rod connection points.

1) Make sure that all of the hardware is installed and secure.

a) Tighten and properly secure any loose items.

2) If replacement is necessary, replace any loose or worn hardware with new hardware.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-410-003

(1) Close these access panels:

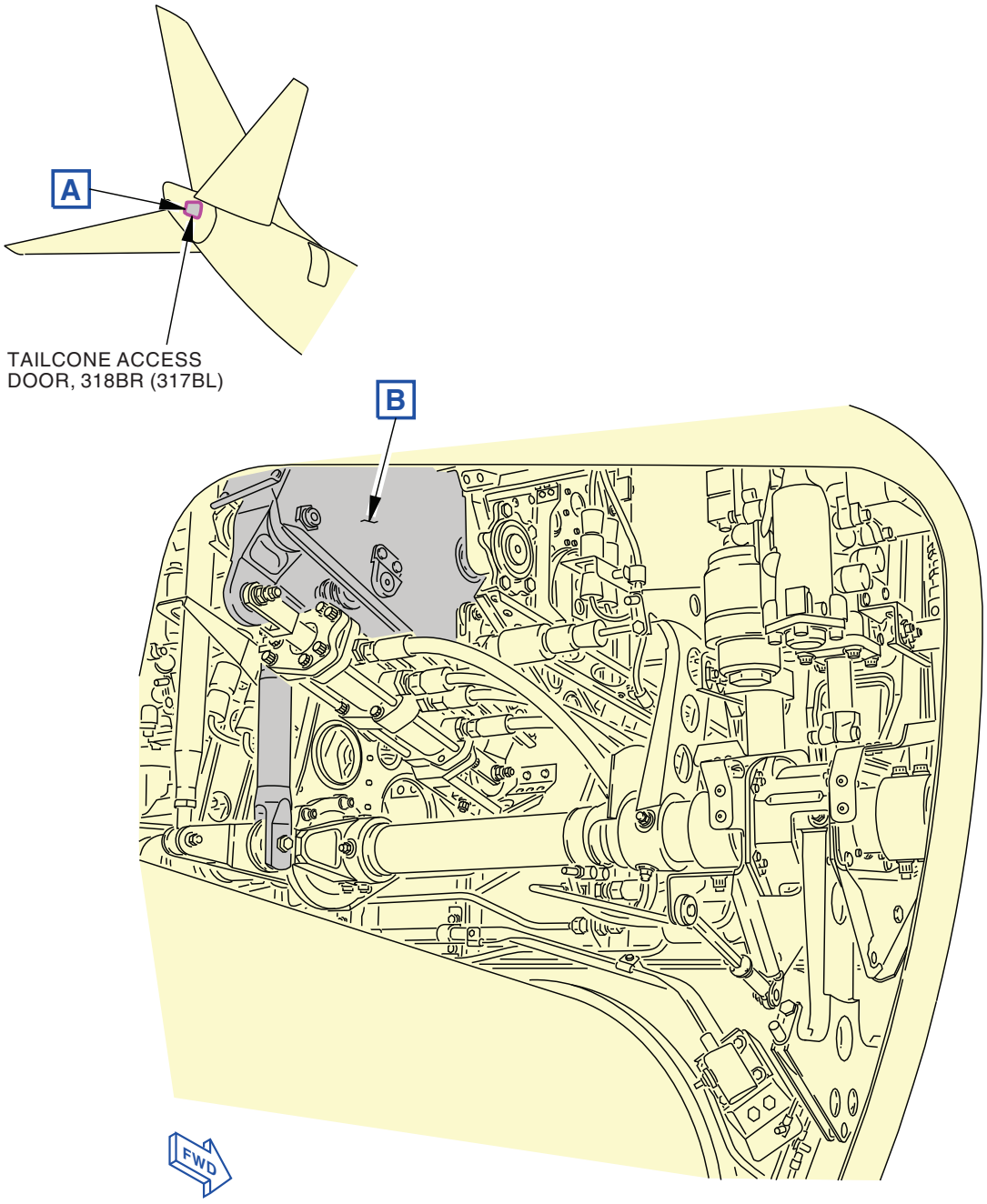
Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-00-840-002

(2) If necessary, apply hydraulic pressure in systems A and B (TASK 27-31-00-840-802).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



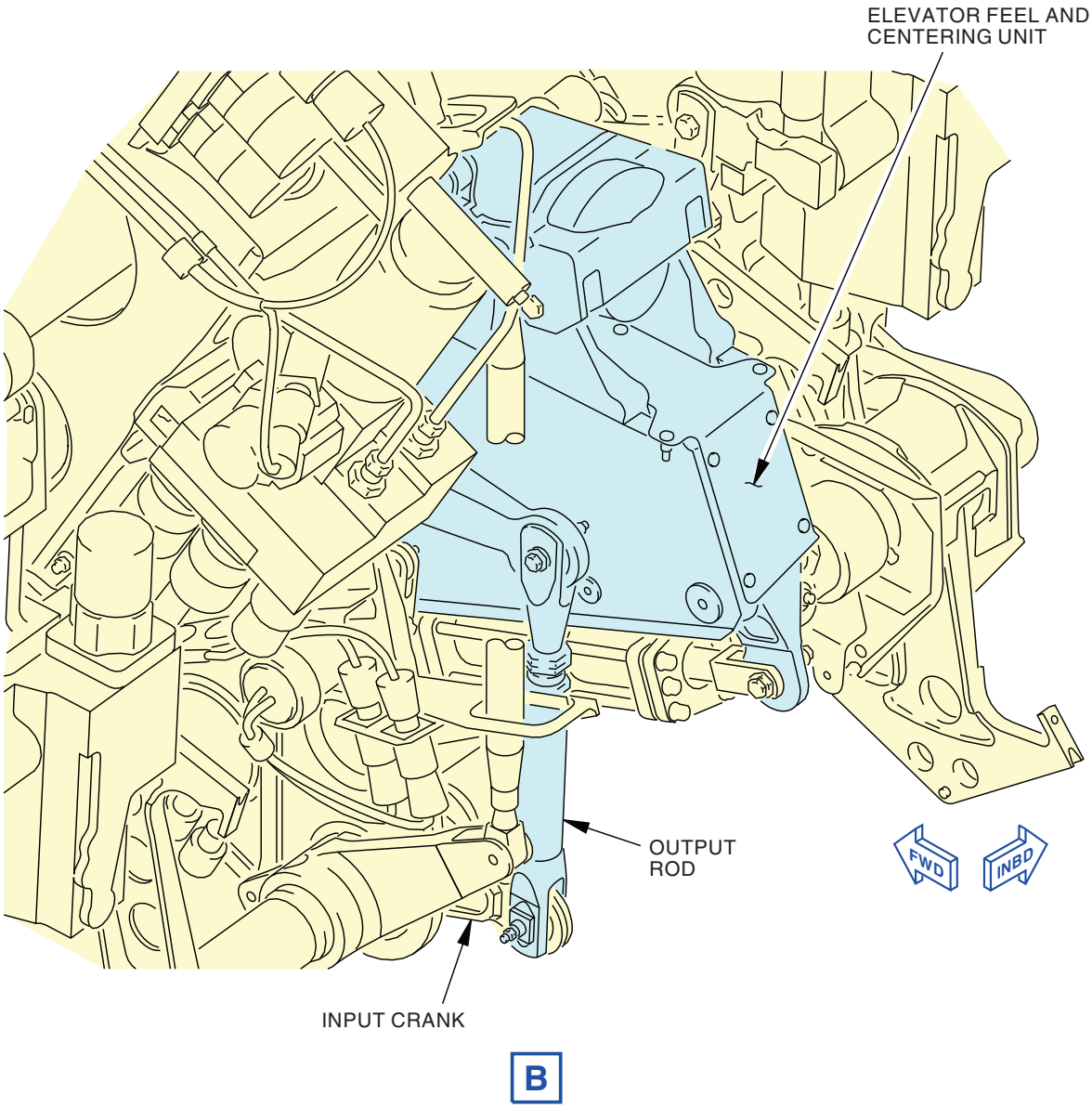
VIEW THROUGH THE TAILCONE ACCESS
DOOR, 318BR

A

2412444 S00061530694_V2

Elevator Feel and Centering Unit Output Rod - Detailed Visual Inspection
Figure 204/27-31-00-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412445 S00061530695_V1

Elevator Feel and Centering Unit Output Rod - Detailed Visual Inspection
Figure 204/27-31-00-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

27-CMR-01

TASK 27-31-00-220-801

10. Elevator Tab and Tab Mechanism - Detailed Visual Inspection

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure is a Certification Maintenance Requirement (CMR).

B. References

Reference	Title
27-31-31-400-801	Elevator Tab Installation (P/B 401)
27-31-34-400-801	Elevator Tab Control Mechanism - Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II

D. Location Zones

Zone	Area
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
334	Left Horizontal Stabilizer - Elevator
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

F. Detailed Visual Inspection of the Elevator Tab and Tab Mechanism

SUBTASK 27-31-00-860-022

(1) Remove pressure from the elevator hydraulic systems A and B (TASK 27-31-00-800-802).

SUBTASK 27-31-00-010-005

(2) For the left elevator tab and tab mechanism, open these access panels:

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
334PT	Horizontal Stabilizer, Tab Control Rod Fairing

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-010-040

(3) For the right elevator tab and tab mechanism, open these access panels:

Number	Name/Location
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-00-200-003

(4) Do a detailed visual inspection of the elevator tab hinges (Figure 205).

(a) Make sure that all hardware is installed and secure (TASK 27-31-31-400-801).

(b) Check for unusual wear and loose parts.

NOTE: Try to wiggle or move hinge bolts with hand pressure only.

(c) Inspect the lug assembly bearings to ensure the bearings are lubricated and in good condition.

SUBTASK 27-31-00-420-002

(5) Tighten and properly secure any loose items.

SUBTASK 27-31-00-960-002

(6) Replace, as necessary, with any new hardware.

SUBTASK 27-31-00-200-004

(7) Do a detailed inspection for security and condition of the elevator tab control mechanism and elevator tab pushrod attachment (Figure 206).

(a) Make sure that all hardware is installed and secure (TASK 27-31-34-400-801).

(b) Check for unusual wear or loose parts by lightly wiggling the pushrods.

SUBTASK 27-31-00-200-005

(8) Do a detailed visual inspection for security and condition of the elevator tab pushrod and elevator tab mast fitting to pushrod attachment (Figure 207).

(a) Turn the tab assembly trailing edge down so that the installation of the tab pushrods to the tab mast fittings are visible (TASK 27-31-34-400-801).

NOTE: This can also be done in the balanced mode with the trailing edge down.

SUBTASK 27-31-00-420-003

(9) Tighten and properly secure any loose items.

SUBTASK 27-31-00-960-003

(10) Replace, as necessary, with any new hardware.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-410-004

(1) For the left elevator tab and tab mechanism, close these access panels:

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
334PT	Horizontal Stabilizer, Tab Control Rod Fairing

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-410-034

(2) For the right elevator tab and tab mechanism, close these access panels:

Number	Name/Location
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-00-390-004

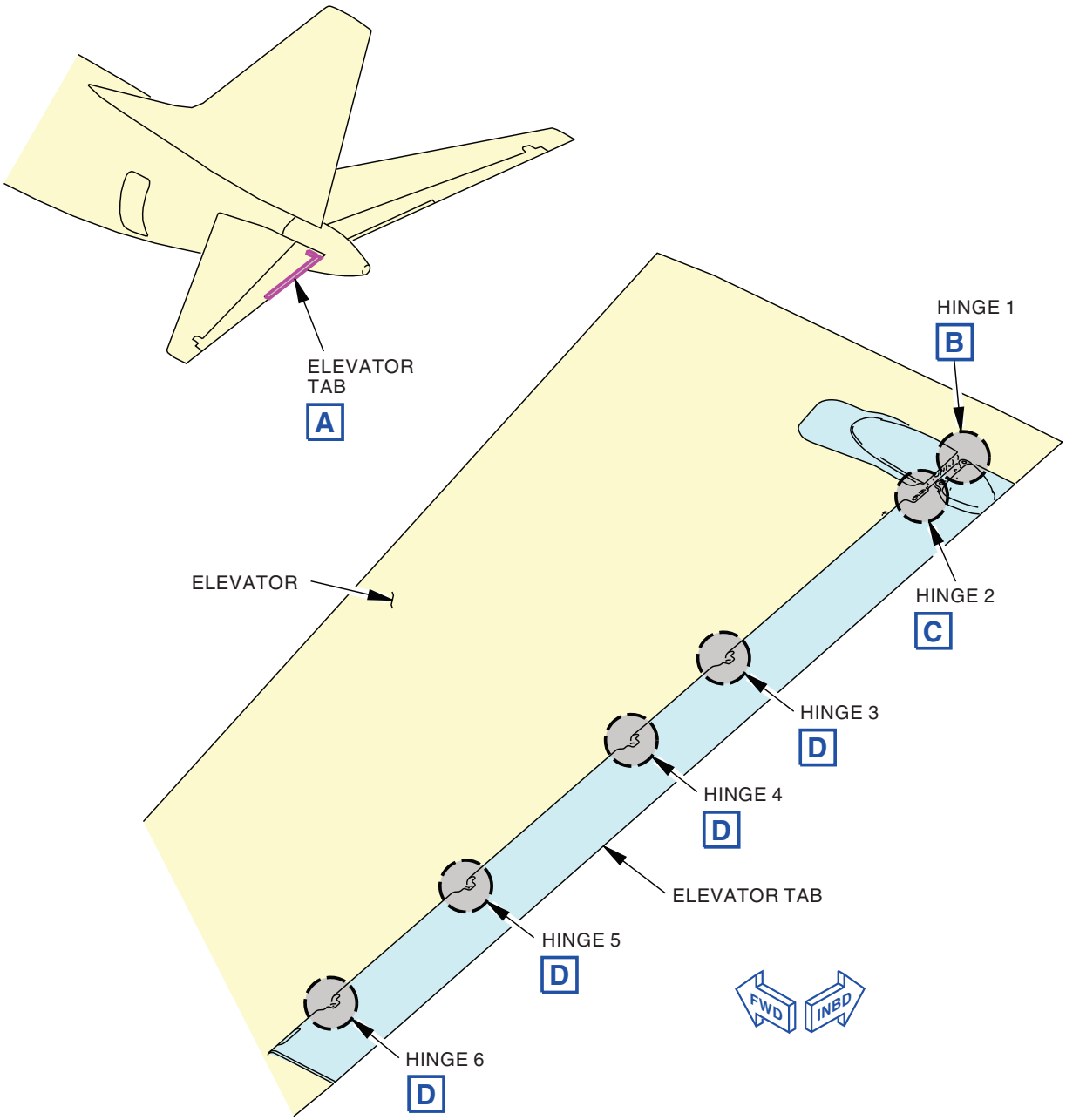
(3) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

SUBTASK 27-31-00-860-023

(4) Put the Elevator Hydraulic Systems A and B back to the condition before the pressure removal (TASK 27-31-00-840-802).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



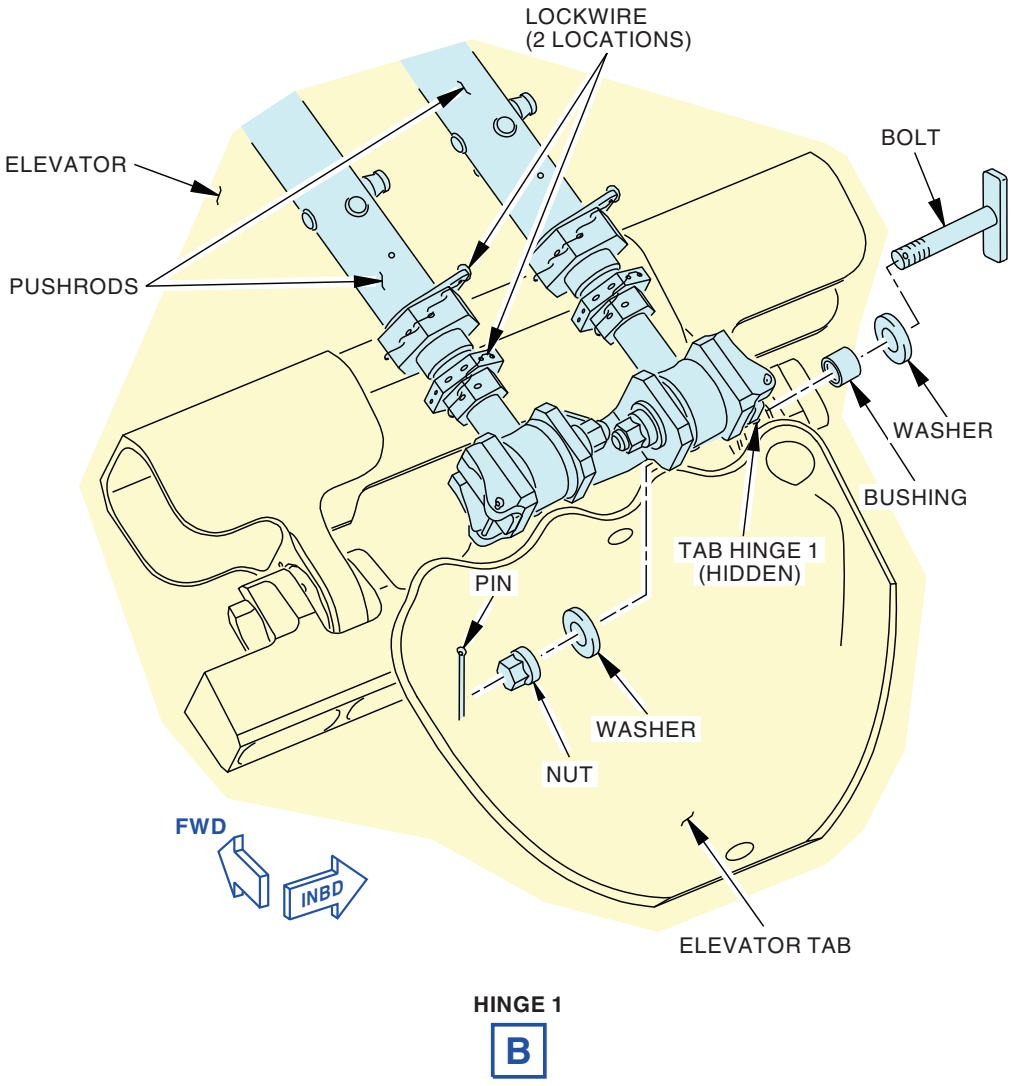
LEFT ELEVATOR TAB
(RIGHT ELEVATOR TAB IS OPPOSITE)

A

2412446 S00061530697_V1

Elevator Tab Hinges - Inspection
Figure 205/27-31-00-990-805 (Sheet 1 of 3)

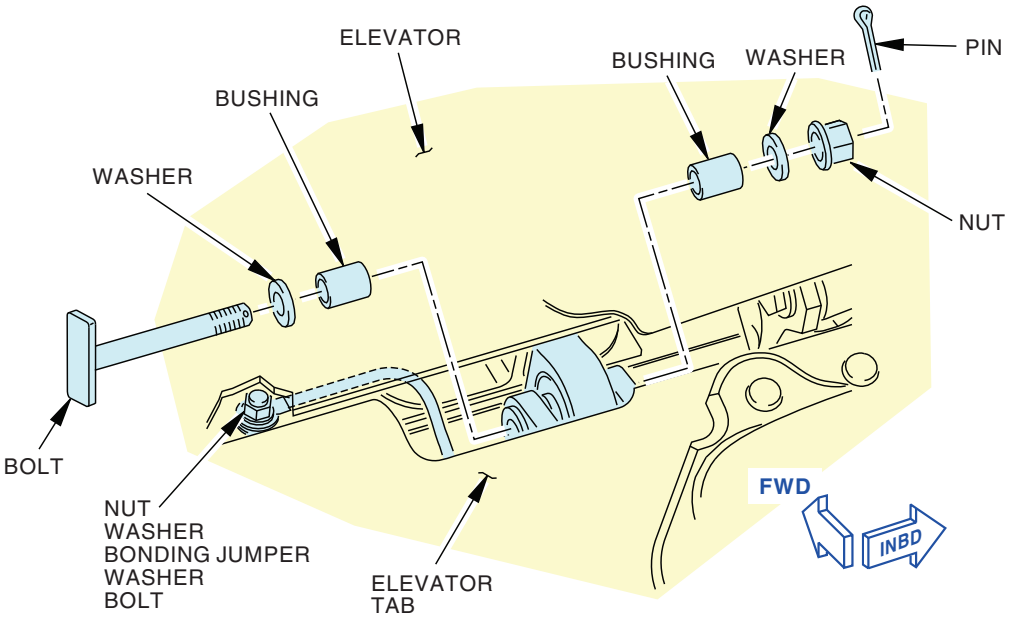
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412447 S00061530698_V2

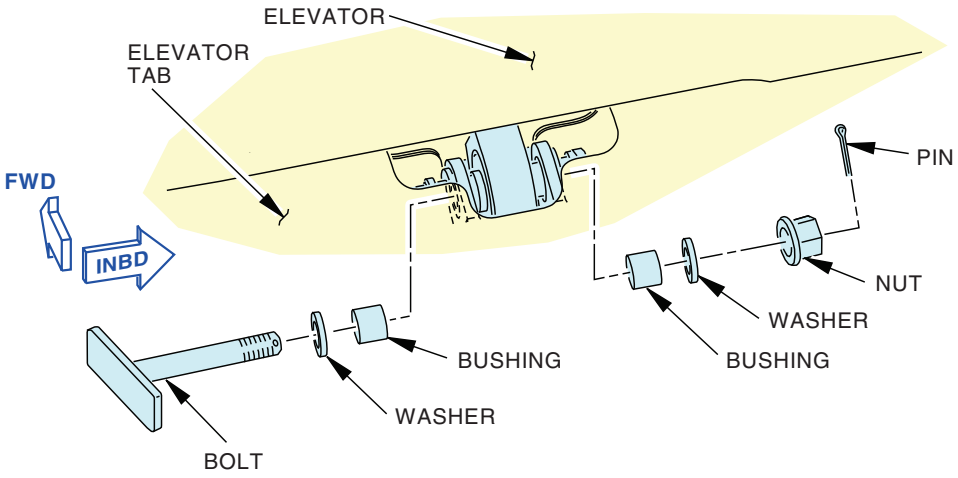
Elevator Tab Hinges - Inspection
Figure 205/27-31-00-990-805 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HINGE 2

C



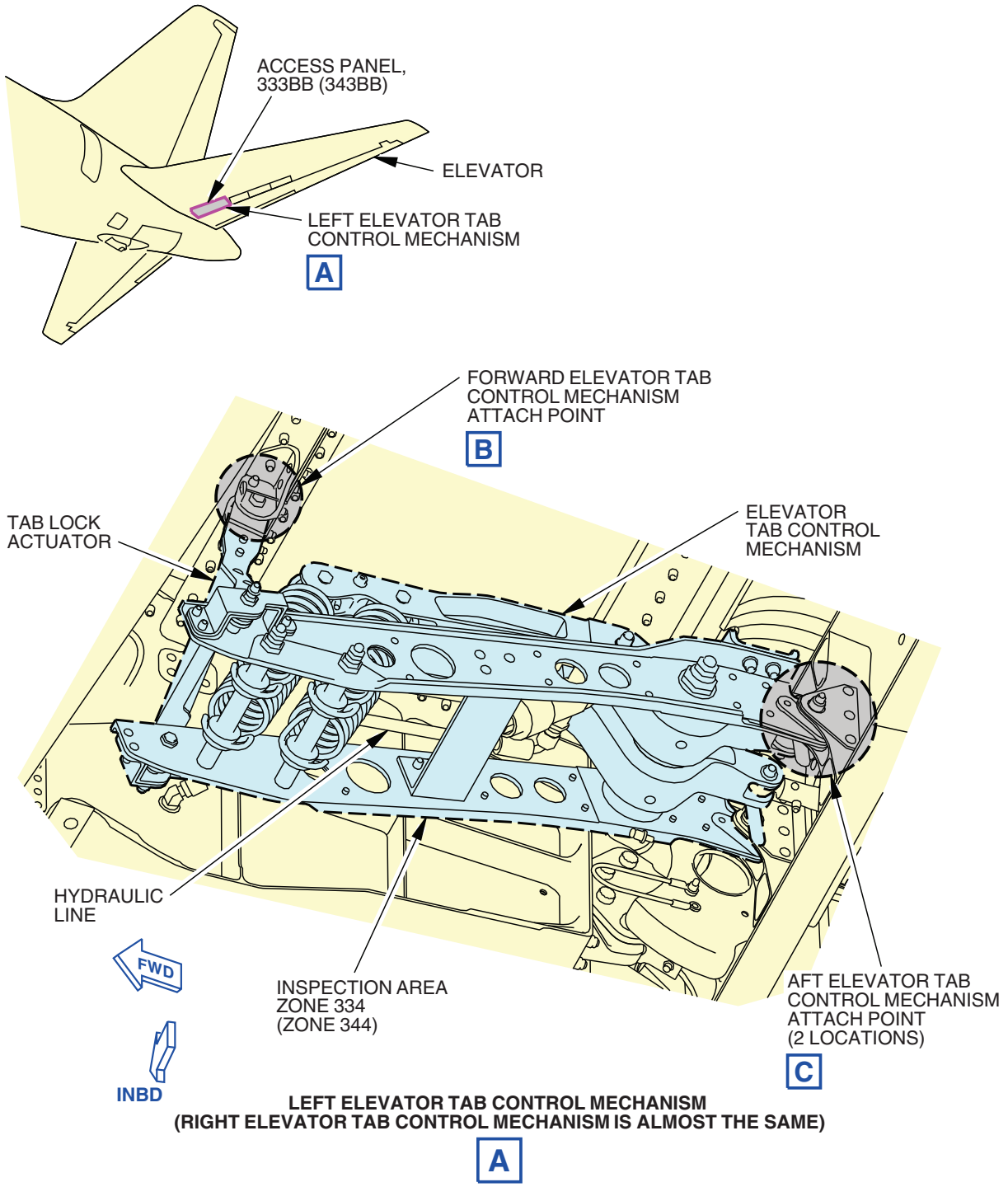
HINGE 3
(HINGES 4, 5, AND 6 ARE EQUIVALENT)

D

2412448 S00061530699_V1

Elevator Tab Hinges - Inspection
Figure 205/27-31-00-990-805 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



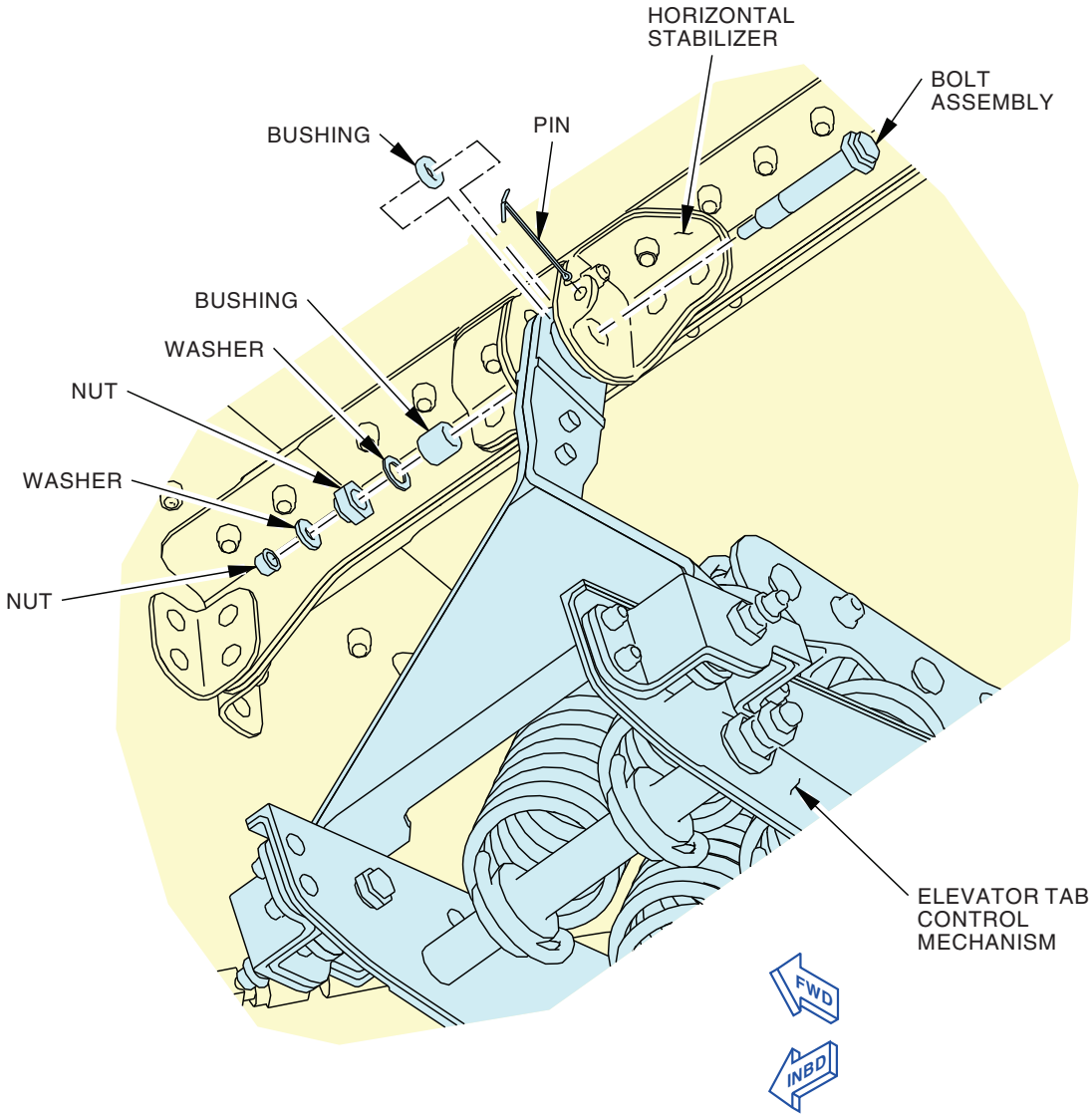
LEFT ELEVATOR TAB CONTROL MECHANISM
(RIGHT ELEVATOR TAB CONTROL MECHANISM IS ALMOST THE SAME)

A

2412449 S00061530700_V3

Elevator Tab Control Mechanism and Elevator Tab Pushrod Attachment - Inspection
Figure 206/27-31-00-990-806 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



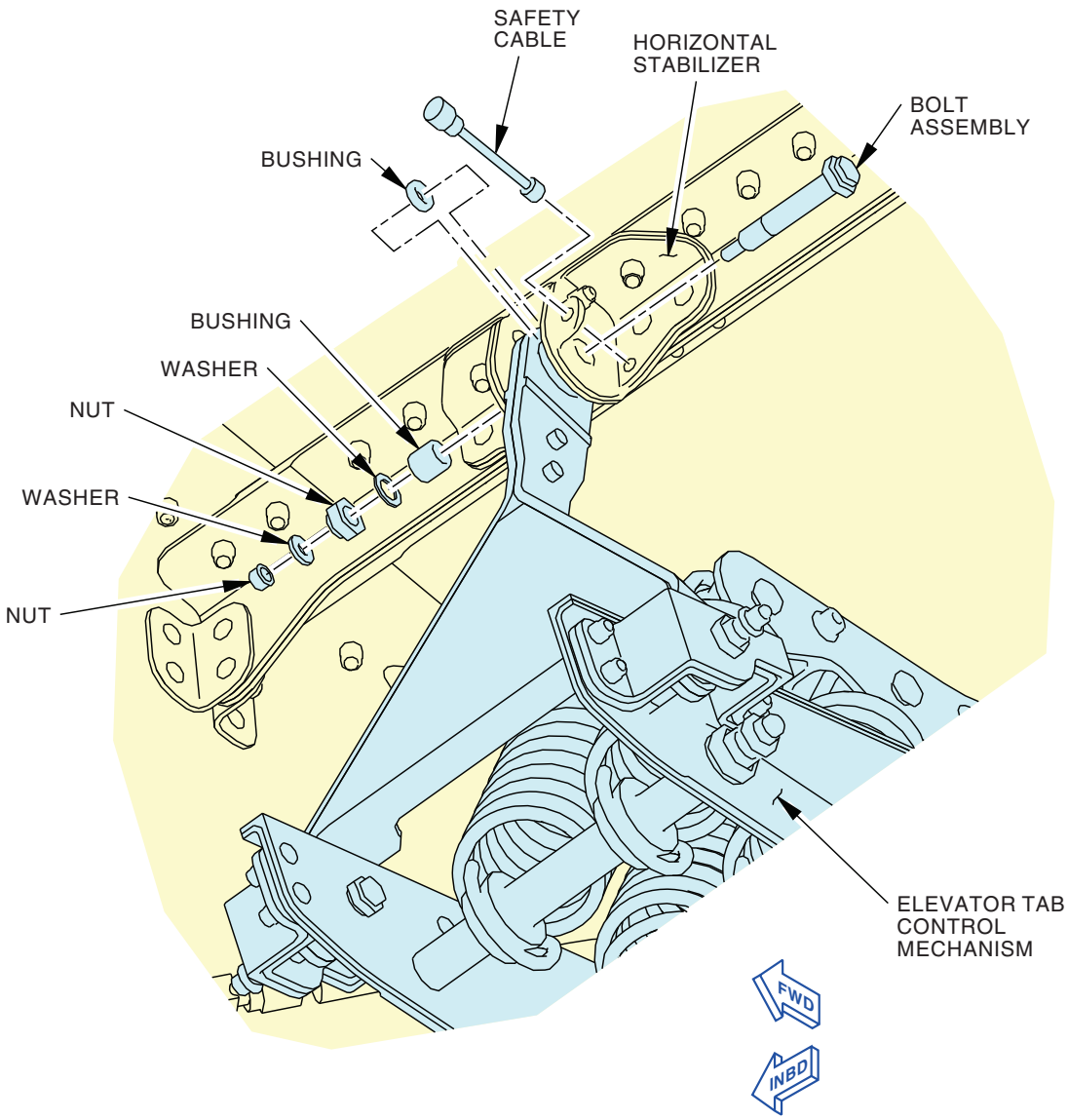
FORWARD ELEVATOR TAB CONTROL MECHANISM ATTACH POINT

B

2412450 S00061530701_V3

Elevator Tab Control Mechanism and Elevator Tab Pushrod Attachment - Inspection
Figure 206/27-31-00-990-806 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



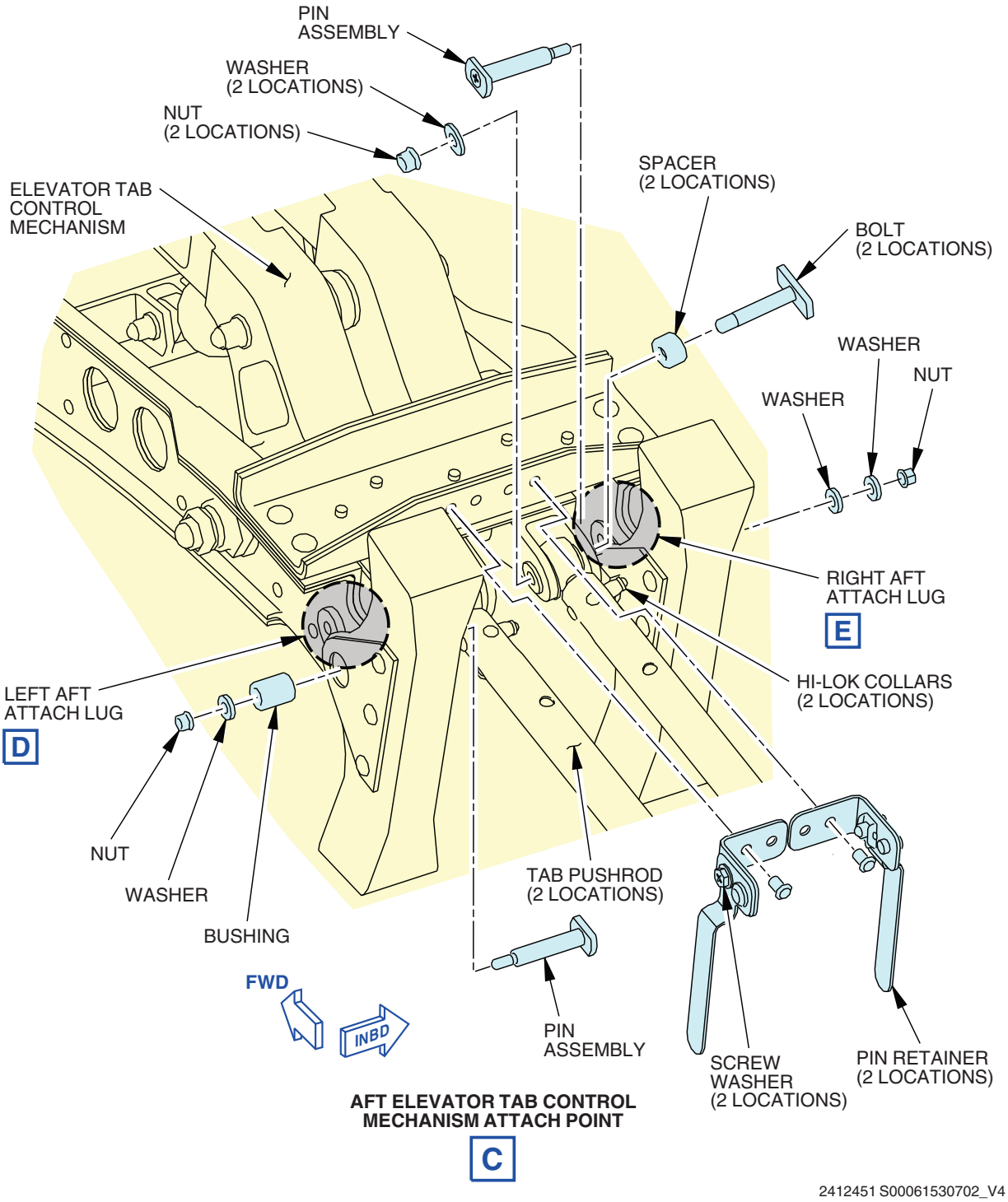
FORWARD ELEVATOR TAB CONTROL MECHANISM ATTACH POINT

B

2978808 S0000757155_V1

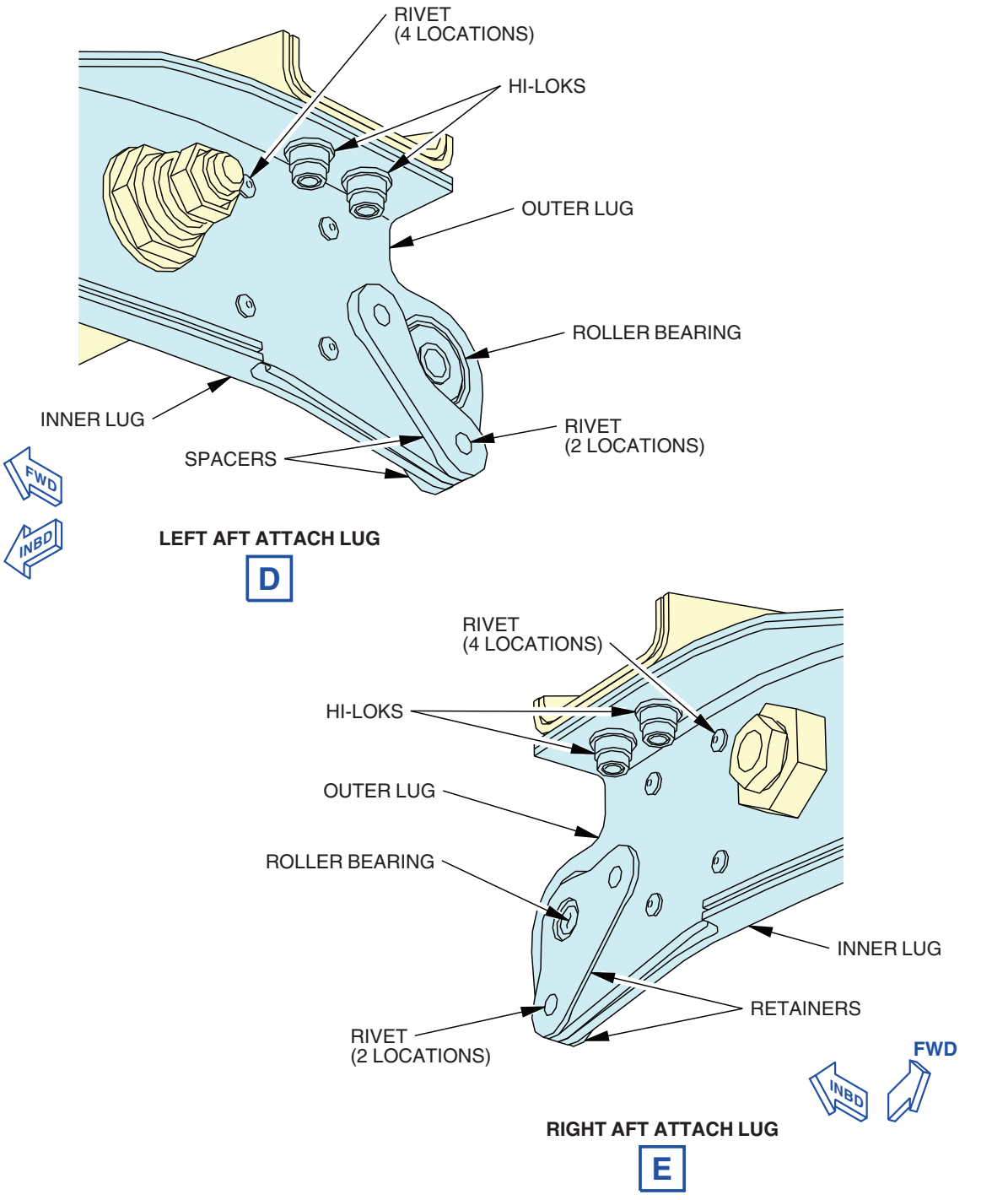
Elevator Tab Control Mechanism and Elevator Tab Pushrod Attachment - Inspection
Figure 206/27-31-00-990-806 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



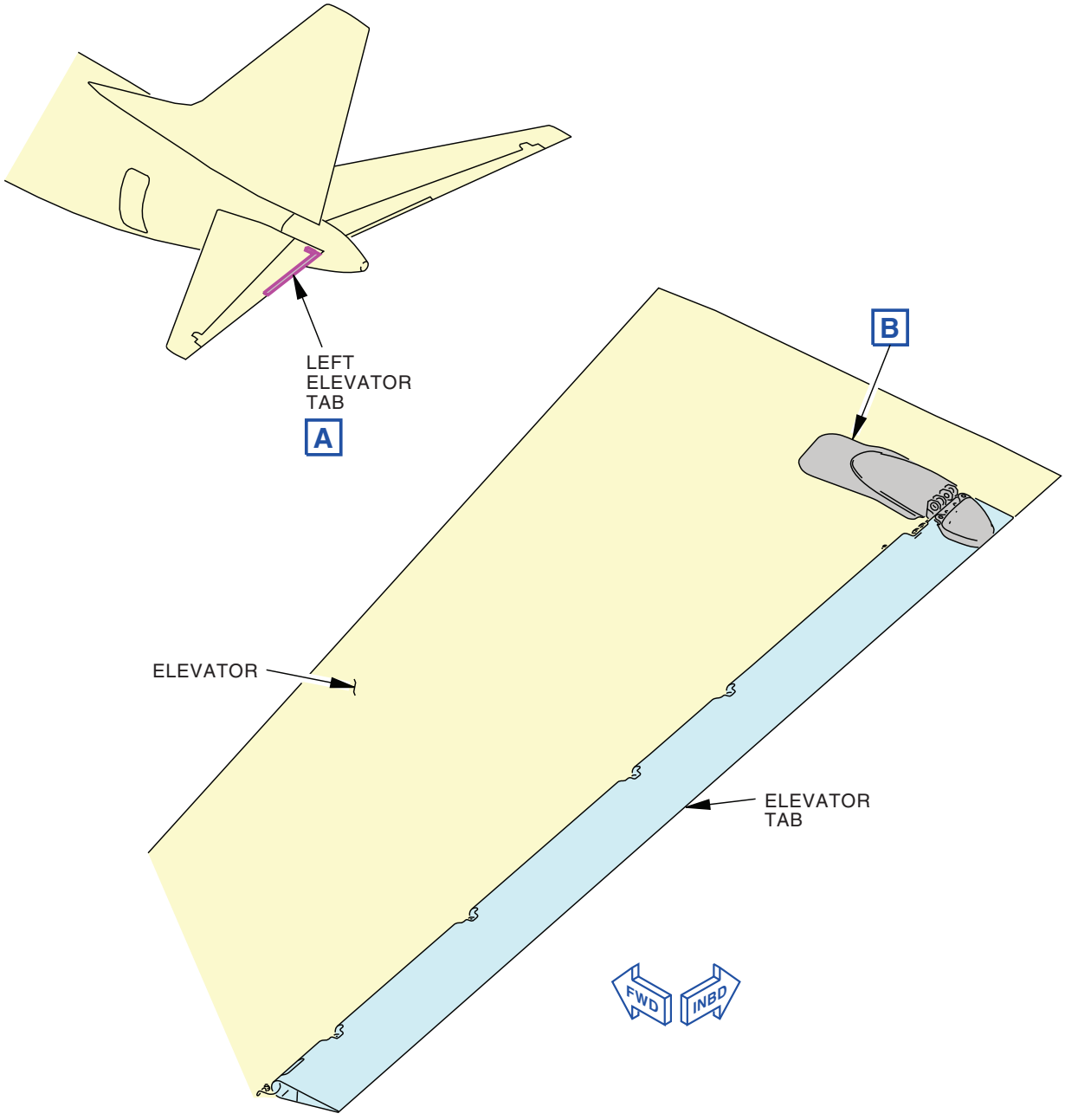
Elevator Tab Control Mechanism and Elevator Tab Pushrod Attachment - Inspection
Figure 206/27-31-00-990-806 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Tab Control Mechanism and Elevator Tab Pushrod Attachment - Inspection
Figure 206/27-31-00-990-806 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



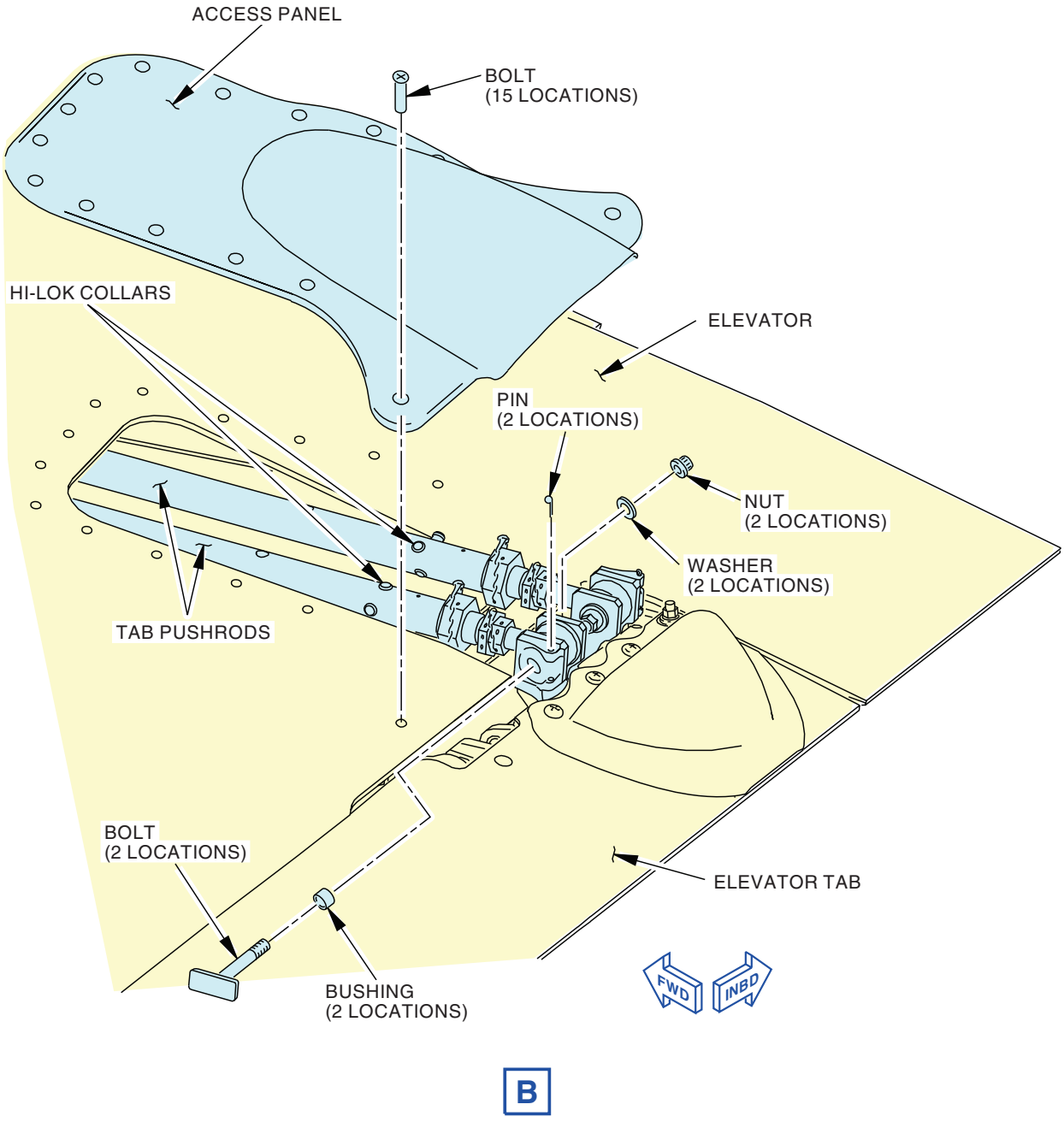
LEFT ELEVATOR TAB
(RIGHT ELEVATOR TAB IS OPPOSITE)

A

2412453 S00061530704_V1

Elevator Tab Pushrod and Elevator Tab Mast Fitting to Pushrod Attachment - Inspection
Figure 207/27-31-00-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Tab Pushrod and Elevator Tab Mast Fitting to Pushrod Attachment - Inspection
Figure 207/27-31-00-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-000-801

11. Elevator and Tab Control System - Deactivation

(Figure 208)

A. General

(1) Use this task to deactivate the elevator and tab control system.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Deactivation of the Elevator and Tab Control System

SUBTASK 27-31-00-860-025

(1) Set the FLT CONTROL A and B switches on the forward overhead panel to the OFF position.

SUBTASK 27-31-00-860-026

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-31-00-860-027

(3) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	4	C00456	AFCS SYS A MACH TRIM AC

SUBTASK 27-31-00-040-001

(4) Install the flight controls lock assembly:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Disconnect the electrical connector from the flight controls shutoff valve at the flight controls hydraulic modules A and B.

NOTE: For hydraulic system A, the flight control shut off valve is mounted on the inboard side of the hydraulic module A in the left main wheel well. For hydraulic system B, the flight control shutoff valve is mounted on the outboard side of the hydraulic module B in the right main wheel well.

(b) Put the manual override lever at the two flight controls shutoff valves in the position 2 (OFF position).

(c) Install the lock, SPL-1675, on the two flight control shut off valves.

1) Install the detent pins.

SUBTASK 27-31-00-780-004

(5) Move the control column forward and aft to decrease the remaining hydraulic system pressure until the display unit shows less than 60 psi (414 kPa) for System Hydraulics A and B.

NOTE: You can view the hydraulic pressure, for the applicable system, on the display unit for the engine display.

SUBTASK 27-31-00-210-002

WARNING

MAKE SURE THAT THERE IS NO PRESSURE IN THE HYDRAULIC SYSTEM. PRESSURE IN THE HYDRAULIC SYSTEM CAN MOVE THE CONTROL SURFACES, CABLES, LANDING GEARS, AND LANDING GEAR DOORS QUICKLY. THEIR MOVEMENT CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(6) Do the elevator and tab control system tryout.

NOTE: This tryout is to make sure that the elevator and tab control system is in a zero energy state.

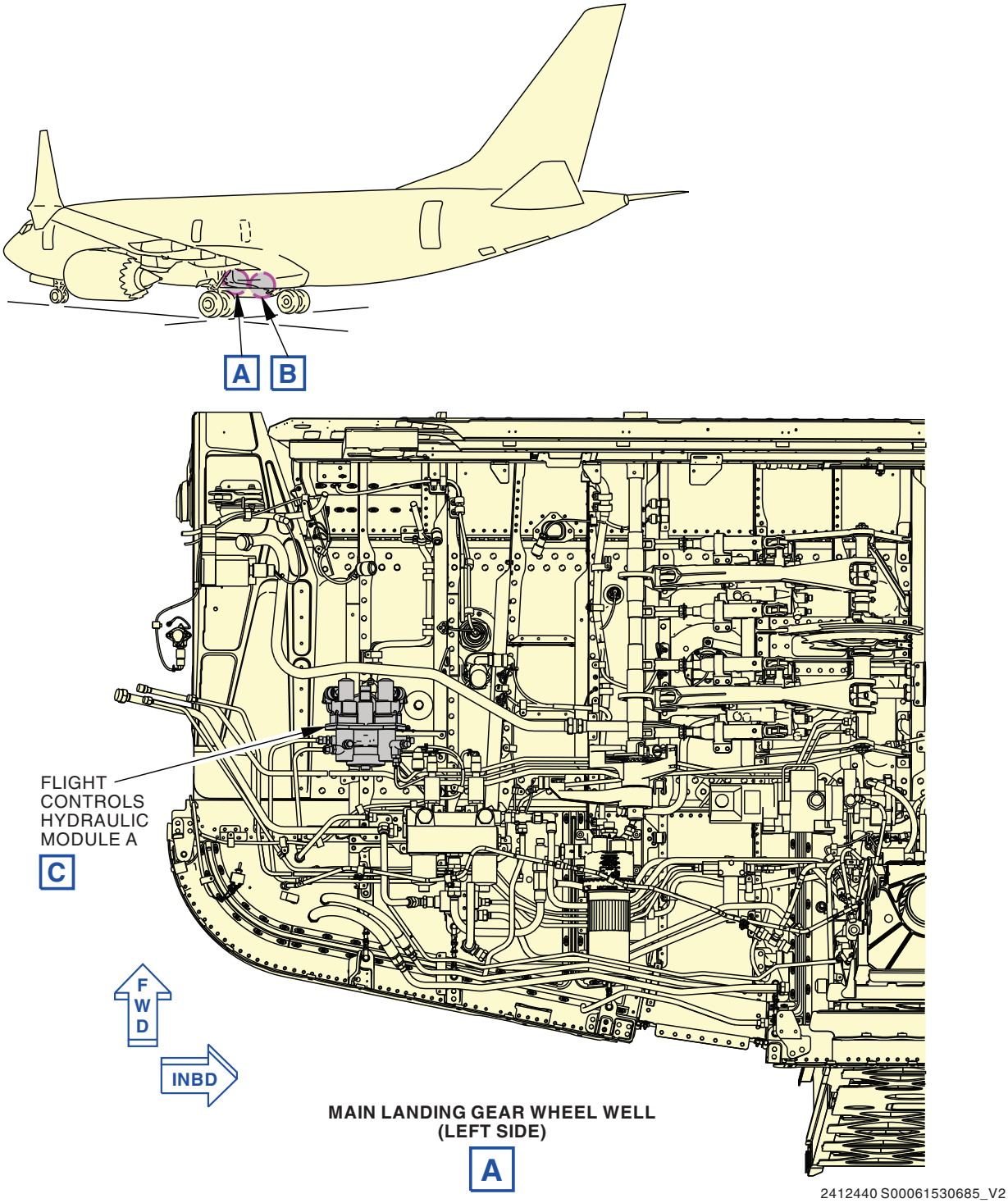
(a) Move the captain's control column forward and aft.

1) Make sure that the column forces are much higher than usual and that a small dead band condition exist when the direction is changed.

(b) Install a DO NOT OPERATE tag, STD-858, on the captain's and first officer's control wheel.

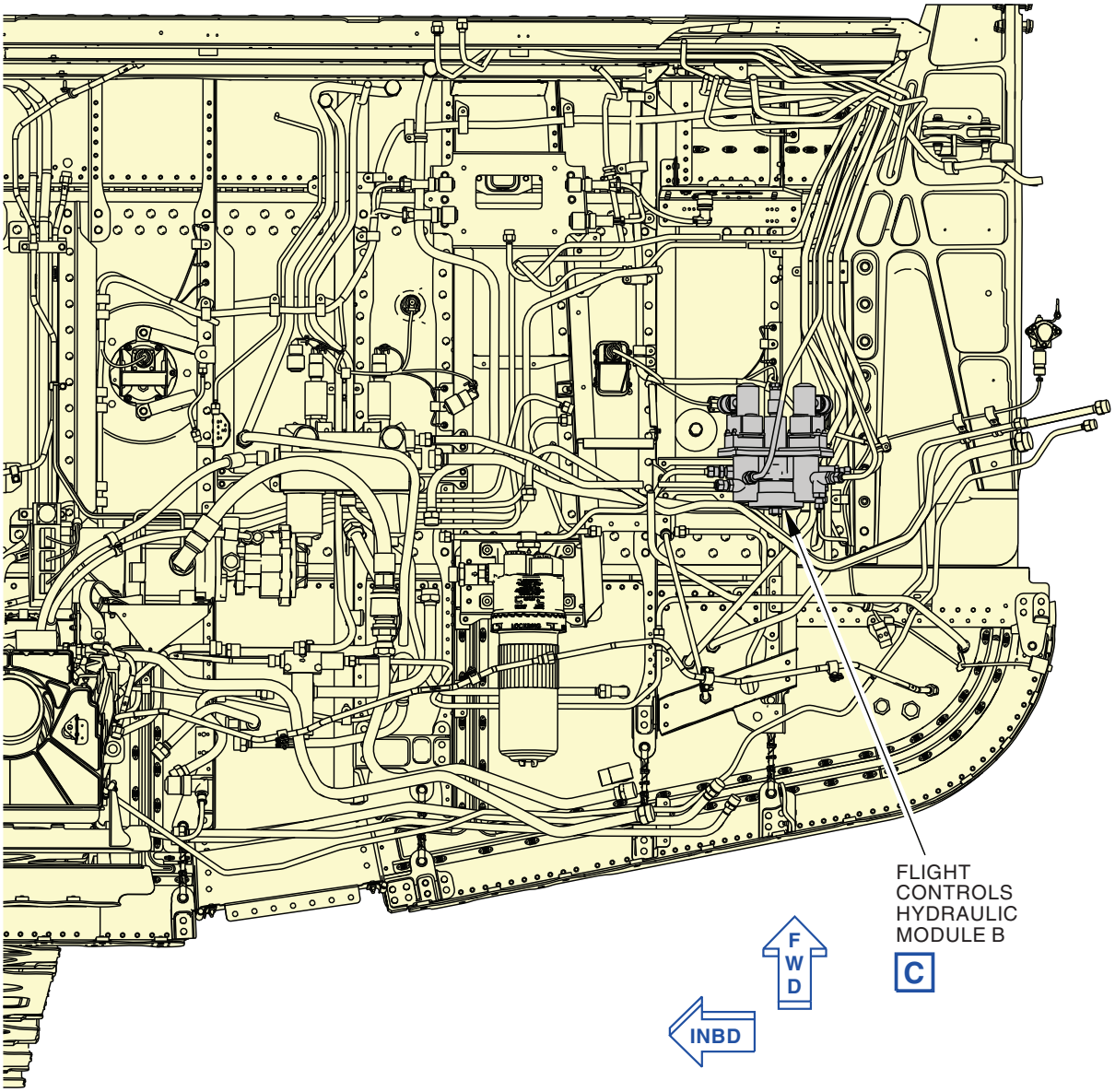
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



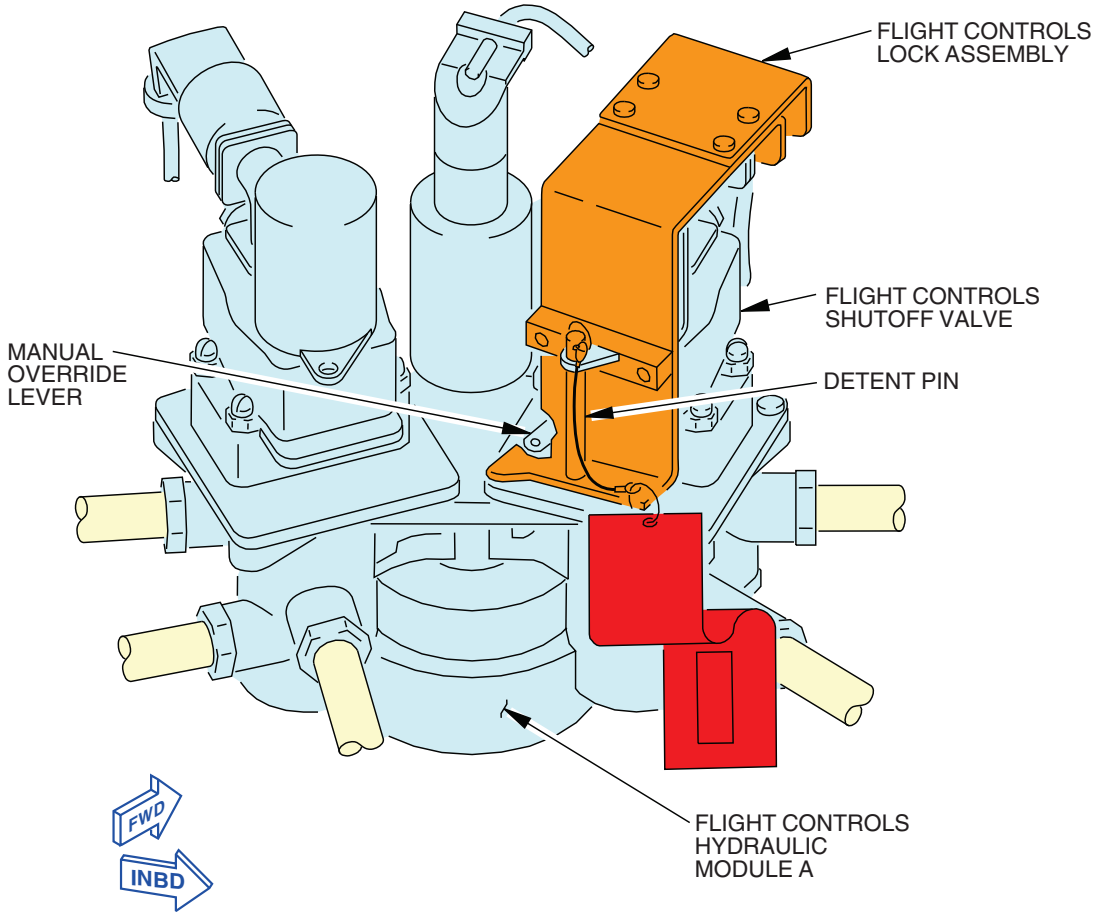
Flight Controls Hydraulic Module Lock Installation
Figure 208/27-31-00-990-808 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Controls Hydraulic Module Lock Installation
Figure 208/27-31-00-990-808 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT CONTROLS HYDRAULIC MODULE A
(MODULE B IS EQUIVALENT)

C

2412442 S00061530687_V4

Flight Controls Hydraulic Module Lock Installation
Figure 208/27-31-00-990-808 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-400-801

12. Elevator and Tab Control System - Activation

(Figure 208)

A. General

(1) Use this task to activate the elevator and tab control system.

B. References

Reference	Title
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Activation of the Elevator and Tab Control System

SUBTASK 27-31-00-860-028

(1) Remove the flight controls lock assembly from the flight controls shutoff valves:

(a) Remove the detent pins.

(b) Remove the lock, SPL-1675, from the two flight controls shutoff valves.

(c) Put the manual override lever at the two flight controls shutoff valves in the position 1 (ON position).

(d) Examine the electrical connectors for corrosion.

1) If there is corrosion, refer to SWPM 20-60-03 to correct the problem.

(e) Apply D5026NS or ZC-026 compound, G50171 to the connectors (SWPM 20-60-03).

(f) Connect the electrical connector to the flight controls shutoff valve at the flight controls hydraulic modules A and B.

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SUBTASK 27-31-00-440-001

(2) Set the FLT CONTROL A and B switches on the P5 forward overhead panel to the ON position.

SUBTASK 27-31-00-410-005

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-31-00-410-006

(4) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	4	C00456	AFCS SYS A MACH TRIM AC

SUBTASK 27-31-00-080-003

(5) Remove the DO NOT OPERATE tag, STD-858, on the captain's and first officer's control wheel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR AND ELEVATOR TRIM CONTROL SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Operational Test - Elevator and Elevator Trim Control System - Use this task to do a quick check of the elevator control system.1

(2) Operational Test - Elevator Tab Control System - Use this task to do a quick check of the elevator tab control system.

(3) Operational Test - Elevator Control Column Override - Use this task to do a quick check of the control column override assembly.

(4) Functional Test - Elevator Control Column Override - Use this task to do a functional check of the control column override assembly.

(5) Elevator Control Cables EA and EB and Pitch Force Transducers Adjustment - Use this task to adjust the elevator control cable and pitch force transducers.

(6) Pitch Force Transducer Operational Test - Use this task to do an operational test of the pitch control wheel steering force transducers.

(7) Elevator Control Pushrods Adjustment - Use this task to adjust the control pushrods which connect the elevator to the output torque tube.

(8) Input Rods of the Elevator Power Control Unit - Use this task to adjust the two input rods (primary and secondary) of the right and left elevator power control units (PCUs).

(9) Input Rods of the Elevator Neutral Shift Mechanism and Output Rod of the Feel and Centering Unit Adjustment - Use this task to adjust the input rods which connect the neutral shift mechanism to the stabilizer. It also includes steps to adjust the output rod which connect the feel and centering unit to the input torque tube.

(10) Elevator Tab Pushrods Adjustment - Use this task to adjust the elevator tab pushrods.

(11) Input Rod of the Elevator Feel Computer Adjustment - Use this test to adjust the input rod (stabilizer pushrod) of the elevator feel computer.

(12) Elevator Surface Friction and Balance Test - Use this task to make sure the elevators are balanced.

(13) Elevator Balance Panels Test - Use this test to make sure the elevator balance panels are installed properly.

(14) Elevator Limit Travel Test - Use this test to make sure the elevator operates correctly.

(15) Elevator Output Torque Tube Linkage Test - Use this task to do a check of the elevator output torque tube bus linkage.

(16) Elevator Tab Linkage Test - Use this task to do a check of the elevator tab control system.

(17) Manual Mode Test - Use this task to make sure the control columns and control wheels operate freely.

(18) Control Column Travel and Centering Test - Use this test to make sure the control columns can move through their full travel. This test also includes the centering test of the control columns.

(19) Control Column Power Force Test - Use this test to make sure the control columns move with a correct force.

(20) Neutral Shift Test - Use this test to make sure the neutral shift mechanism moves the elevator correctly when the stabilizer moves. When the stabilizer moves, it provides input to the elevator PCUs.

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(21) Feel Differential Pressure Light Test - Use this test to make sure the FEEL DIFF PRESS light and the trailing edge flaps switch, S1051 operate correctly.

(22) Feel Differential Pressure Light Test - Use this test to make sure the FEEL DIFF PRESS light operates correctly.

(23) Pitot Probe Leak Test - Use this test to make sure there is no leakage at the pitot probes.

(24) Feel Force System Test - Use this test to make sure the elevator feel computer operates correctly. The elevator feel computer changes the control columns forces as the airspeed changes.

(25) Sensor Null Test - Use this test to make sure the elevator is in the correct rig position with DFCS Computer BITE.

(26) Elevator PCU Input Rod (Pogo) Functional Test - Use this task to do a check of force required to compress and extend each input rod (pogo) of the elevator power control unit (PCU).

(27) Elevator Power Off Flight Test

(28) Elevator Tab Pushrod Adjustment after the Elevator Power Off Flight Test

(29) Elevator Jam Landing Assist Test

TASK 27-31-00-710-801

2. Elevator and Elevator Trim Control System - Operational Test

A. General

(1) Use this task to do a quick check of the elevator control system.

B. References

Reference	Title
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Elevator and Elevator Trim Control System Operational Test

SUBTASK 27-31-00-860-029

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-31-00-710-001

(2) Operate the control column with the systems A and B hydraulic power:

(a) Set the FLT CONTROL A and B switches, on the forward overhead panel, P5, to the ON position.

(b) Move the control column full aft.

1) Make sure that the elevators move full up and the system operates smoothly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Move the control column full forward.

1) Make sure that the elevators move full down and the system operates smoothly.

SUBTASK 27-31-00-710-002

(3) Operate the control column with the system B hydraulic power:

(a) Set the FLT CONTROL A switch, on the forward overhead panel, P5, to the OFF position.

(b) Move the control column full aft.

1) Make sure that the elevators move full up and the system operates smoothly.

(c) Move the control column full forward.

1) Make sure that the elevators move full down and the system operates smoothly.

SUBTASK 27-31-00-710-003

(4) Operate the control column with the system A hydraulic power:

(a) Set the FLT CONTROL A switch, on the forward overhead panel, P5, to the ON position.

(b) Set the FLT CONTROL B switch, on the forward overhead panel, P5, to the OFF position.

(c) Move the control column full aft.

1) Make sure that the elevators move full up and the system operates smoothly.

(d) Move the control column full forward.

1) Make sure that the elevators move full down and the system operates smoothly.

SUBTASK 27-31-00-860-030

(5) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

——— END OF TASK ———

TASK 27-31-00-700-802

3. Elevator Tab Control System - Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this test to do a quick check of the elevator tab control system.

B. References

Reference	Title
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Test

SUBTASK 27-31-00-860-031

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize elevator hydraulic systems A and B (TASK 27-31-00-800-801).

SUBTASK 27-31-00-860-032

(2) Make sure that the flaps are in the full up position.

SUBTASK 27-31-00-860-033

(3) Make sure that the control column is in the neutral position.

SUBTASK 27-31-00-760-001

(4) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	2	C01045	AFCS SYS A FCC DC
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC
E	6	C01205	SMYD-1 SNSR EXC AC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	4	C01207	SMYD-2 SNSR EXC AC
B	5	C01206	SMYD-2 CMPTR DC
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	3	C01046	AFCS SYS B FCC DC

E. Operational Test - Elevator Tab Control System

SUBTASK 27-31-00-730-001

(1) Do the operational test of the elevator tab control system:

(a) Move and hold the control column full aft.

1) Make sure that the elevator tabs move down.

(b) Move and hold the control column full forward.

1) Make sure that the elevator tabs move up.

(c) Release the control column.

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(d) Cycle the control column and return it to neutral.

(e) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	8	C01464	FLIGHT CONTROL ELEV TAB VLV RIGHT

(f) Set the flaps to the 5 degree position.

(g) Move and hold the control column full forward.

1) Make sure that the left elevator tab is below the elevator and the right elevator tab is above the elevator.

(h) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	8	C01464	FLIGHT CONTROL ELEV TAB VLV RIGHT

1) Make sure that there is an 8-12 second time delay before the right elevator tab begins to move.

(i) Cycle the control column full forward and full aft.

1) Make sure that both elevator tabs move in the same direction as the elevators.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-034

(1) Set the flaps to the full up position.

SUBTASK 27-31-00-860-035

(2) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

——— END OF TASK ———

TASK 27-31-00-710-802

4. Elevator Control Column Override - Operational Test

A. General

(1) Use this task to do a quick check of the elevator control column override assembly.

B. References

Reference	Title
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Elevator Control Column Override - Operational Test

SUBTASK 27-31-00-860-036

> **WARNING** MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-31-00-710-004

(2) Do an operational test of the elevator control column override:

(a) Override Test of the First Officer's Control Column:

1) Push the captain's and first officer's control columns forward.
2) Hold the captain's control column forward (against the forward stop).
3) Pull the first officer's control column aft.
4) Make sure that you can pull the first officer's control column aft.
5) Release the captain's and first officer's control columns go to the neutral position.

(b) Override Test of the Captain's Control Column:

1) Push the captain's and first officer's control columns forward.
2) Hold the first officer's control column forward (against the forward stop).
3) Pull the captain's control column aft.
4) Make sure that you can pull the captain's control column aft.
5) Release the captain's and first officer's control columns go to the neutral position.

SUBTASK 27-31-00-860-037

(3) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

——— END OF TASK ———

TASK 27-31-00-720-801

5. Elevator Control Column Override - Functional Test

(Figure 501, Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this task to do a functional check of the elevator control column override.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

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C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1674	Assembly - Adapter, Control Wheel, Torque and Force Test Part #: C27060-22 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205 Opt Part #: F72867-1 Supplier: 81205
STD-755	Scale - Spring, Push/Pull Type, 0 - 150lbs (0 - 68kg) Capacity

D. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

E. Elevator Control Column Override Functional Test

SUBTASK 27-31-00-860-038

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-31-00-710-005

(2) Do the functional test of the elevator control column override:

(a) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

1) Put a 6 in. (152 mm) strip of adhesive tape vertically on the right elevator index plate (Figure 502).

2) Make a mark on the tape where the upper surface of the right elevator trailing edge would extend to the tape.

3) Attach the spring scale, STD-755, and adapter, SPL-1674, to the captain's control column.

NOTE: The force gage is a push pull scale for measuring control column loads.

4) Block the first officer's control column in the neutral position.

NOTE: One mechanic can hold the first officer's control column steady in the neutral position.

5) Slowly pull the captain's control column aft and hold at a minimum right elevator position of 1.95 ±0.05 in. (49.5 ±1.3 mm) (Figure 502).

6) Make sure that the captain's control column force is 40 lbf (177.9 N) - 100 lbf (444.8 N).

7) Release the captain's control column slowly to center the system.

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(b) Push the captain's control column slowly forward and hold at a minimum right elevator position of 1.95 ±0.05 in. (49.5 ±1.3 mm) (Figure 502).

(c) Make sure that the captain's control column force is 32 lbf (142.3 N) - 102 lbf (453.7 N).

(d) Remove the spring scale, STD-755, and adapter, SPL-1674, from the captain's control column.

(e) Remove the strip of adhesive tape from the right elevator index plate.

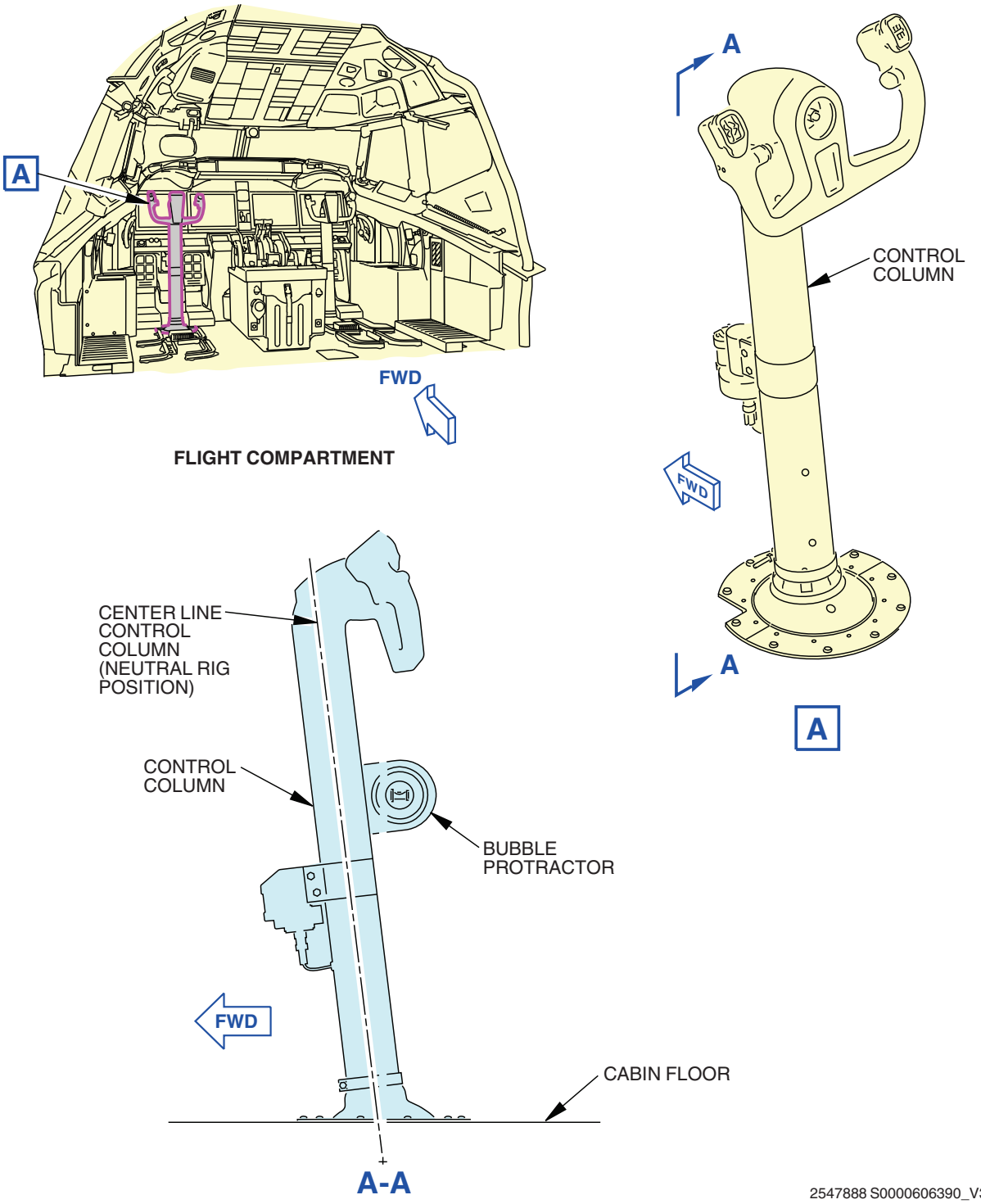
F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-039

(1) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

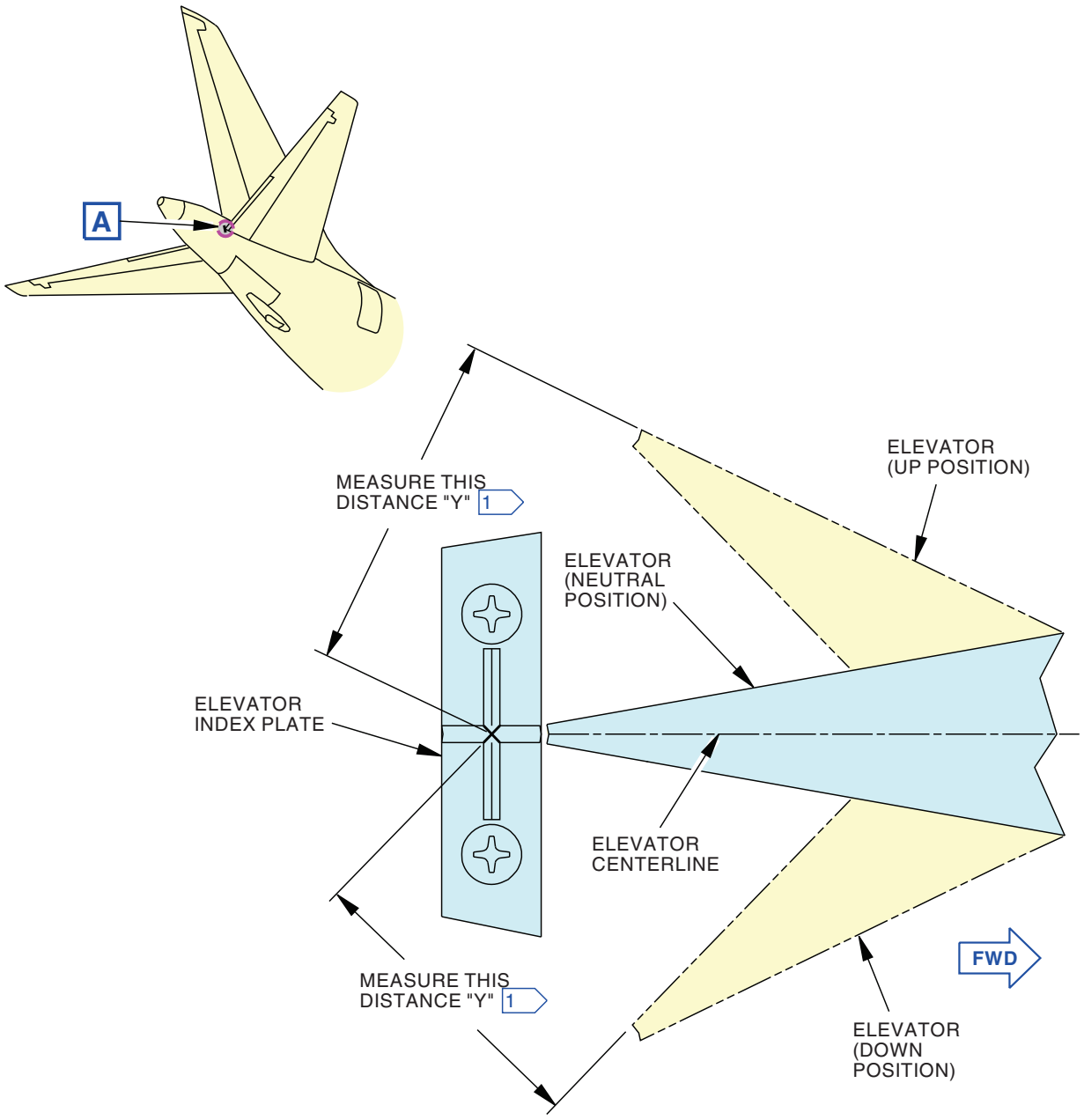
——— END OF TASK ———

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Elevator Control Column Override - Functional Test
Figure 501/27-31-00-990-828

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 THE DISTANCE "Y" IS MEASURED FROM THE MIDDLE OF THE ELEVATOR INDEX PLATE TO THE UPPER SURFACE OF THE INBOARD EDGE OF THE ELEVATOR.

MEASUREMENT LOCATION (RIGHT SIDE)

A

2547889 S0000606391_V1

Elevator Full Travel Measurements
Figure 502/27-31-00-990-830

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-820-801

6. Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment

(Figure 503, Figure 504, Figure 505)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this task to adjust the elevator control cables and pitch force transducers.

(2) Before you do cable adjustment, permit a minimum of one hour at a constant ambient temperature of ± 5 degrees F (± 2.8 degrees C) for the airframe temperature to become stable.

(a) It is recommended to do the cable adjustments in a controlled environment such as a hangar.

NOTE: Direct sunlight on the airframe will cause the cable readings to be inaccurate.

(b) All temperature readings will be taken as ambient outside but adjacent to the applicable area of the airframe.

(c) The cable tension values will not be correct when there are temperature differences along the cable run.

(d) It is recommended that airplanes supplied with the cables that contain higher zinc coating (Grade D or Grade K) be re-tensioned 3 to 4 months into service. This will bring the tension of the control cables within the desired range. After the cables are re-tensioned, the high zinc cables (Grade D or Grade K) will continue to relax at a rate that is equivalent to the Grade B cables. To identify the grade of the cable, refer to the table below:

Table 501/27-31-00-993-810 Control Cable Grades and Diameters

CONTROL CABLE DIAMETER (TERMINAL SIZE REF)	CONSTRUCTION FIRST DASH NUMBER *[1]									
	GRADE A					GRADE B		GRADE D		GRADE K
	ZINC			TIN OVER ZINC		ZINC	TIN OVER ZINC	ZINC	TIN OVER ZINC	ZINC
	7 X 7	7 X 7 (RT)	7 X 19	7 X 7	7 X 7 (RT)	7 X 19	7 X 19	7 X 19	7 X 19	7 X 19
.094 (3/32)	3B	3G*[2]	3A	3D	3E *[2]	- — —	- — —	- — —	- — —	- — —
.125 (1/8)	- — —	- — —	4A	- — —	- — —	4F	4C	4J*[3]	4H*[3]	4K*[3]
.188 (3/16)	- — —	- — —	- — —	- — —	- — —	- — —	- — —	6J*[4]	- — —	- — —

*[1] First dash number in the part number of the cable. Example: BACC2C-3D-01085-CE is Grade A, Tin Over Zinc, 7 x 7.

*[2] Code 3E and 3G are specified only where there is a specific requirement for REDUCED TOLERANCE (RT) percent elongation wire rope, 7 X 7 RT, per BMS7-265.

*[3] Only Grade D and Grade K .125 (1/8) cables (4J, 4H and 4K) are high zinc cables.

*[4] It is not necessary to adjust the Grade D 0.188 (3/16) cables (6J) 3 to 4 months after installation.

(e) It is recommended that cable assemblies on each side of a cable loop be replaced at the same time using the same cable grade material in order to minimize asymmetric relaxation.

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B. References

Reference	Title
22-11-92-820-801	Pitch Force Transducer Adjustment (DFCS BITE Test) (P/B 501)
27-09-14-860-802	Cable Control - Tension Measurement (P/B 201)
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1553	Tensiometer - Cable, Low Tension (200 lbs and below) Part #: 102-03120 Supplier: 21844 Part #: ACM-200 Supplier: 13331 Part #: ACX-250-1 Supplier: 13331 Part #: T60-1001-C8-1A Supplier: 0N8U4
COM-1554	Tensiometer - Cable, High Tension (200 lbs and above) Part #: 102-03108 Supplier: 21844 Part #: 102-03120 Supplier: 21844 Part #: ACM-300 Supplier: 13331 Part #: ACM-600 Supplier: 13331 Part #: ACX-500-1 Supplier: 13331 Part #: T5-8008-106-00 Supplier: 0N8U4 Part #: T60-1001-C9-1A Supplier: 0N8U4
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Adjustment

SUBTASK 27-31-00-860-041

(1) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-860-040

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-010-006

(3) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-31-00-010-007

(4) Open these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-00-860-042

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(5) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677 and not the stabilizer indicator in the flight deck.

SUBTASK 27-31-00-480-003

(6) Use a low tension cable tensiometer, COM-1553, or high tension cable tensiometer, COM-1554, to measure the cable tension (TASK 27-09-14-860-802).

NOTE: If the tension measurement is in the limit and new control cables are not installed, the following adjustment procedure is not necessary.

G. Elevator Control Cables EA and EB and Pitch Force Transducers Adjustment

SUBTASK 27-31-00-020-004

(1) Loosen the bolts [4] near the side of the adjustable end of the two pitch force transducers (Figure 503).

SUBTASK 27-31-00-020-001

(2) Remove the nuts [3], washers [2], and bolts [1] from the adjustable end of the two pitch force transducers (Figure 503).

NOTE: The bolts will be installed when you complete the adjustment of the pitch force transducers.

SUBTASK 27-31-00-710-006

(3) Adjust the elevator control cables EA and EB and the pitch force transducers:

(a) Install the rig pin E-5 from the rigging pin kit, SPL-1585, in the elevator aft quadrant (Figure 505).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that you can move the rig pin freely.
2) If it is necessary, move the elevator by hand until you can install the rig pin E-5.

(b) Install the rig pin E-1 from the rigging pin kit, SPL-1585, in the elevator left forward quadrant (Figure 503).
1) Make sure that you can move the rig pin E-1 freely.

(c) Remove the nuts [6], washers [5], and bolts [4] from the two elevator forward quadrants.
NOTE: The bolts will be installed when you complete the adjustment of the pitch force transducers.

(d) Install the rig pins E-2 and E-3 from the rigging pin kit, SPL-1585, through the installation holes of the bolts [4] (Figure 503).
1) Make sure that you can move the rig pins E-2 and E-3 freely.

(e) Use a low tension cable tensiometer, COM-1553, or high tension cable tensiometer, COM-1554, to measure the cable tension (TASK 27-09-14-860-802).

(f) If you install new elevator control cables EA and EB, then do these steps to stretch the cables:
1) Adjust the turnbuckles on the EAL, EAR, EBL, and EBR cables to 300 ±20 lbf (1334 ±89 N) or 300 ±20 lb (136 ±9 kg).
2) Adjust the seals in the pressure bulkhead, if it is necessary, to keep the cables straight.
3) Remove the rig pin E-1 from the elevator left forward quadrant.
4) Remove the rig pin E-5 from the elevator aft quadrant.
5) Move the control column 25 times through full travel.
6) Install the rig pin E-5 to the elevator aft quadrant.
a) Make sure that you can move the rig pin E-5 freely.
b) If it is necessary, move the elevator by hand until the rig pin holes align.
7) Install the rig pin E-1 to the elevator left forward quadrant.
a) Make sure that you can move the rig pin E-1 freely.
8) Make sure that the rig pins E-2 and E-3 move freely through the installation holes of the bolts [4].

(g) Adjust the control cables EA and EB to their correct cable rigging tension:
1) Adjust the turnbuckles to get the cable tension (TASK 27-09-14-860-802):
NOTE: Adjust the tensions close to the maximum values in the table. The cables will stretch over a period of time.

(h) Adjust the pitch force transducers (TASK 22-11-92-820-801).

(i) Make sure that the tension of control cables EA and EB are at the correct tension (TASK 27-09-14-860-802).

(j) Make sure that you can move the rig pins E-1, E-2, E-3, and E-5 freely.

(k) Remove the rig pins E-2 and E-3.

(l) Install the bolts [1], washers [2], and nuts [3].

(m) Install the bolts [4], washers [5], and nuts [6].
1) Torque the nuts [6] to 200 ±40 in-lb (23 ±5 N·m).

(n) Install the locking clips on the turnbuckles.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(o) Remove the rig pins E-1 and E-5.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-940-001

(1) Close these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-00-940-002

(2) Close this access panel:

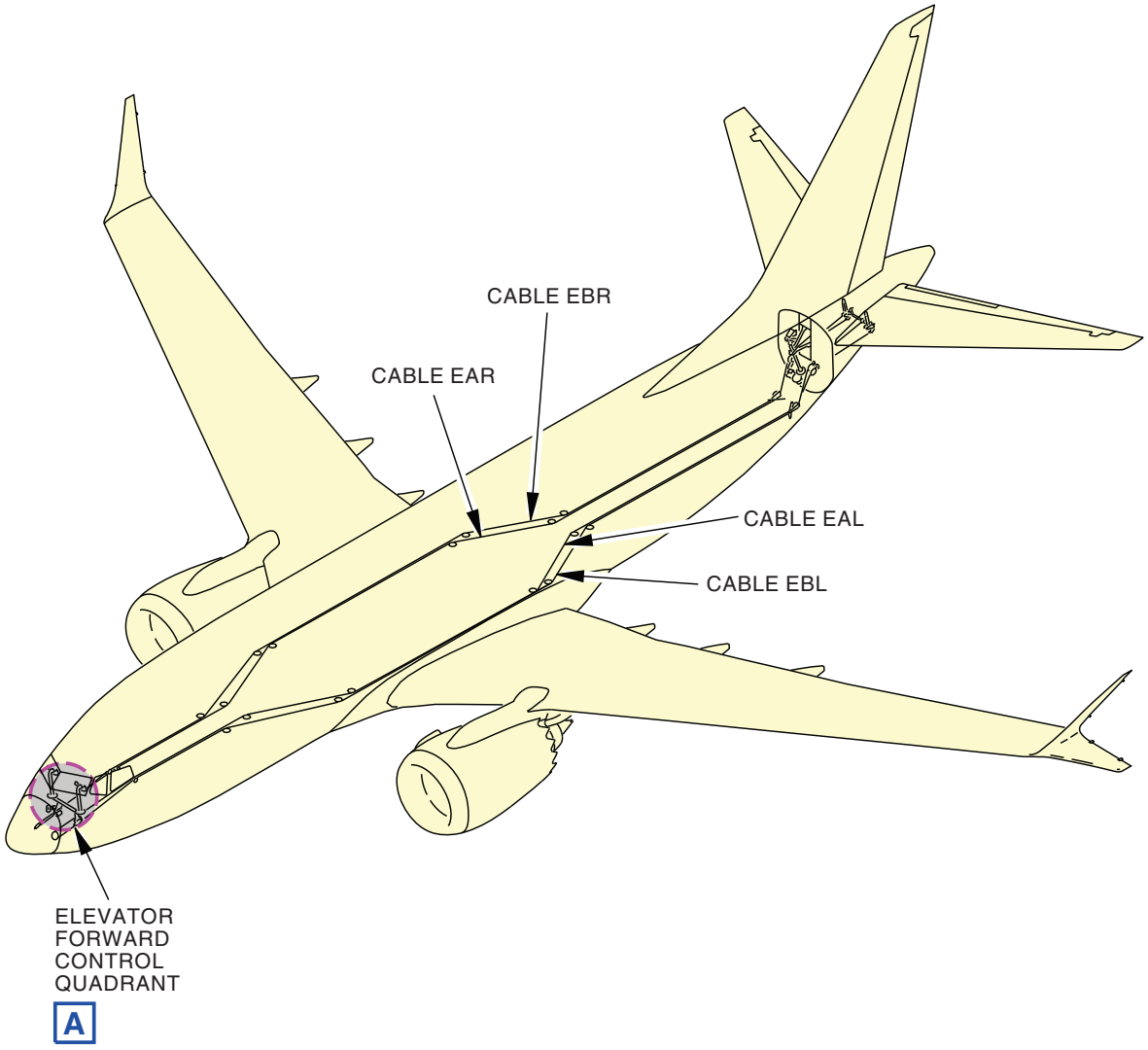
Number	Name/Location
112A	Forward Access Door

SUBTASK 27-31-00-860-043

(3) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

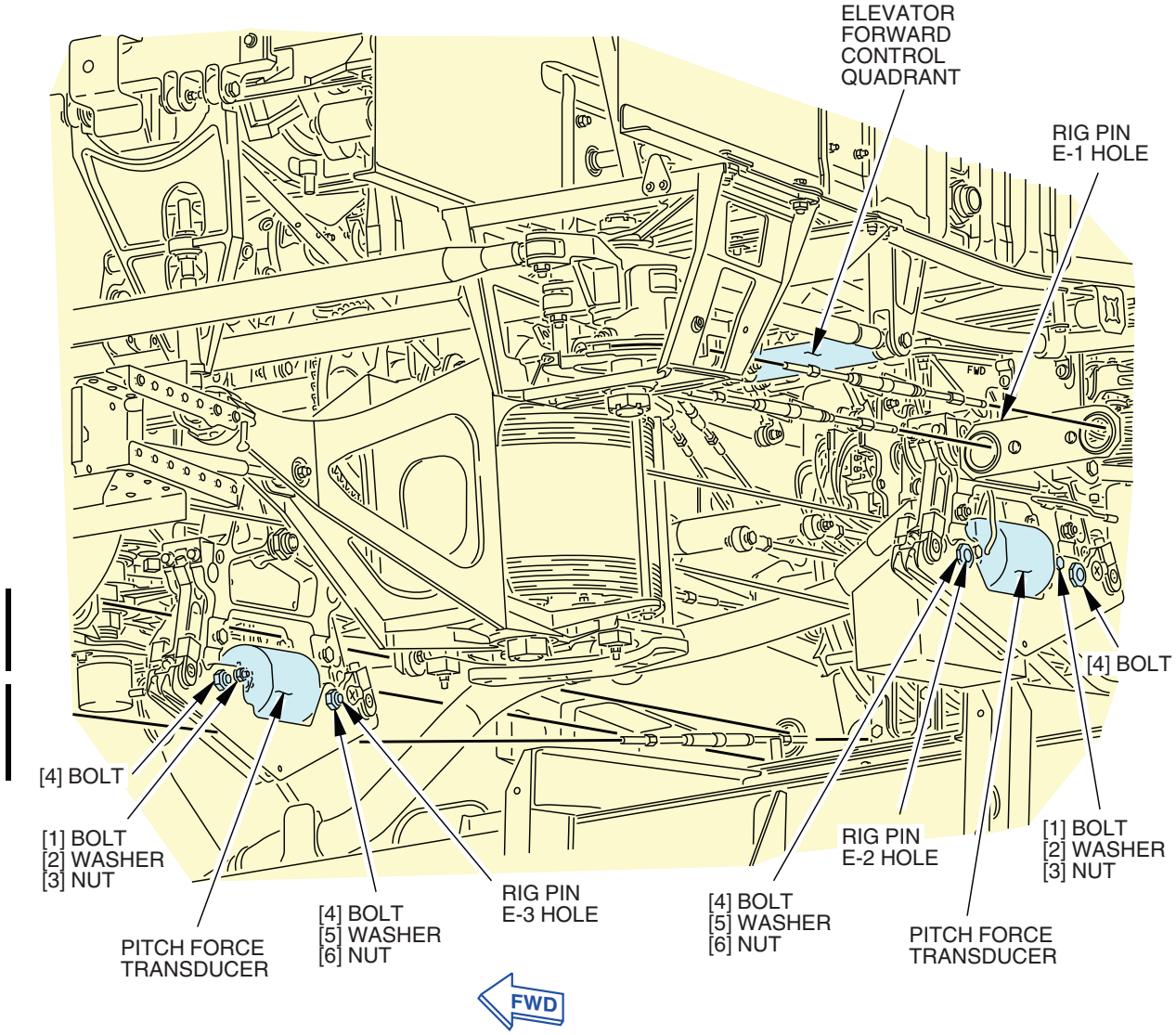


**ELEVATOR AND TAB CONTROL CABLES,
CONTROL COLUMN AND FORWARD QUADRANT**

2412455 S00061530714_V2

**Elevator Control Cables Adjustment
Figure 503/27-31-00-990-809 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



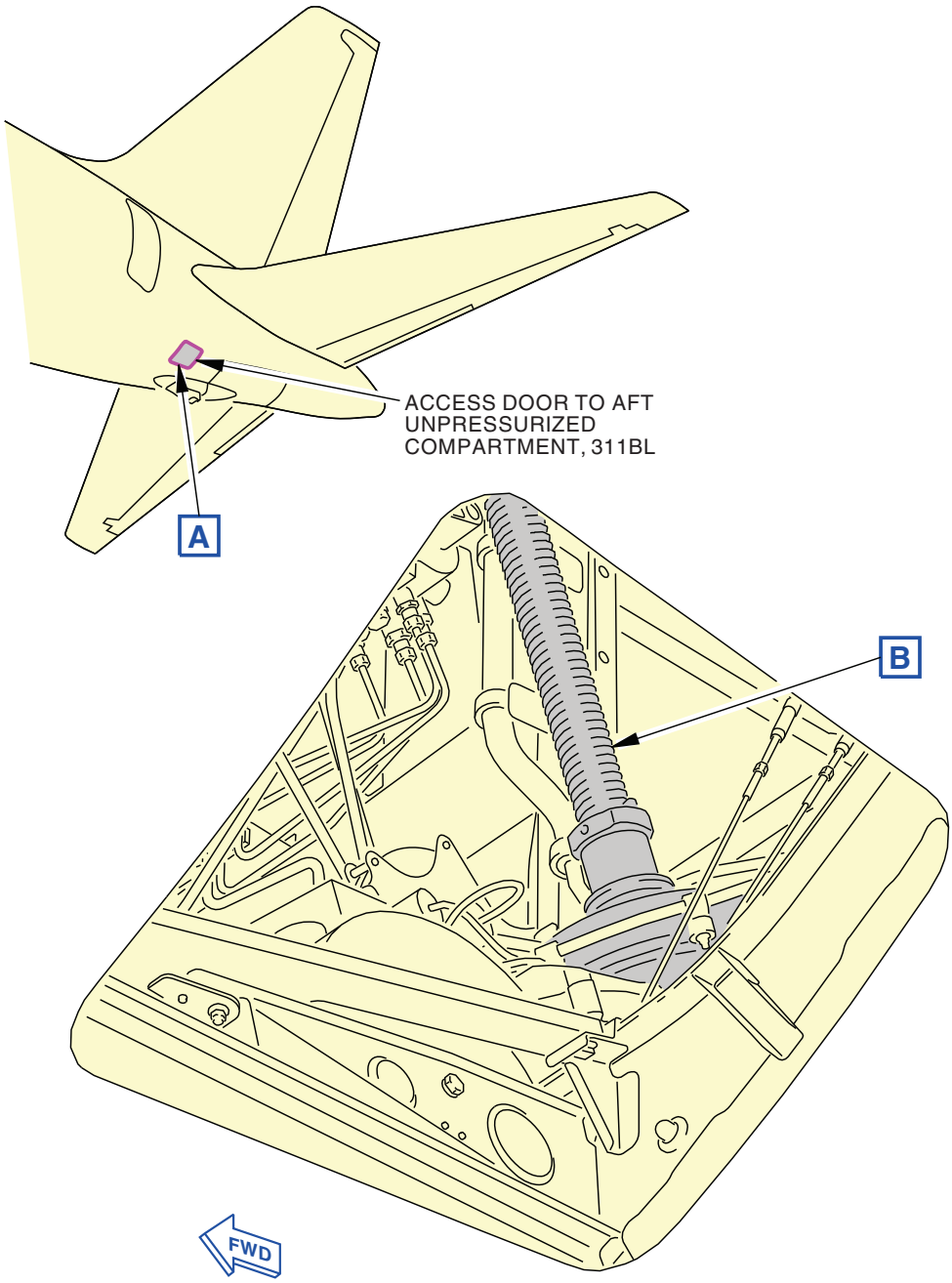
ELEVATOR FORWARD CONTROL QUADRANT ASSEMBLY

A

2412456 S00061530715_V2

Elevator Control Cables Adjustment
Figure 503/27-31-00-990-809 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



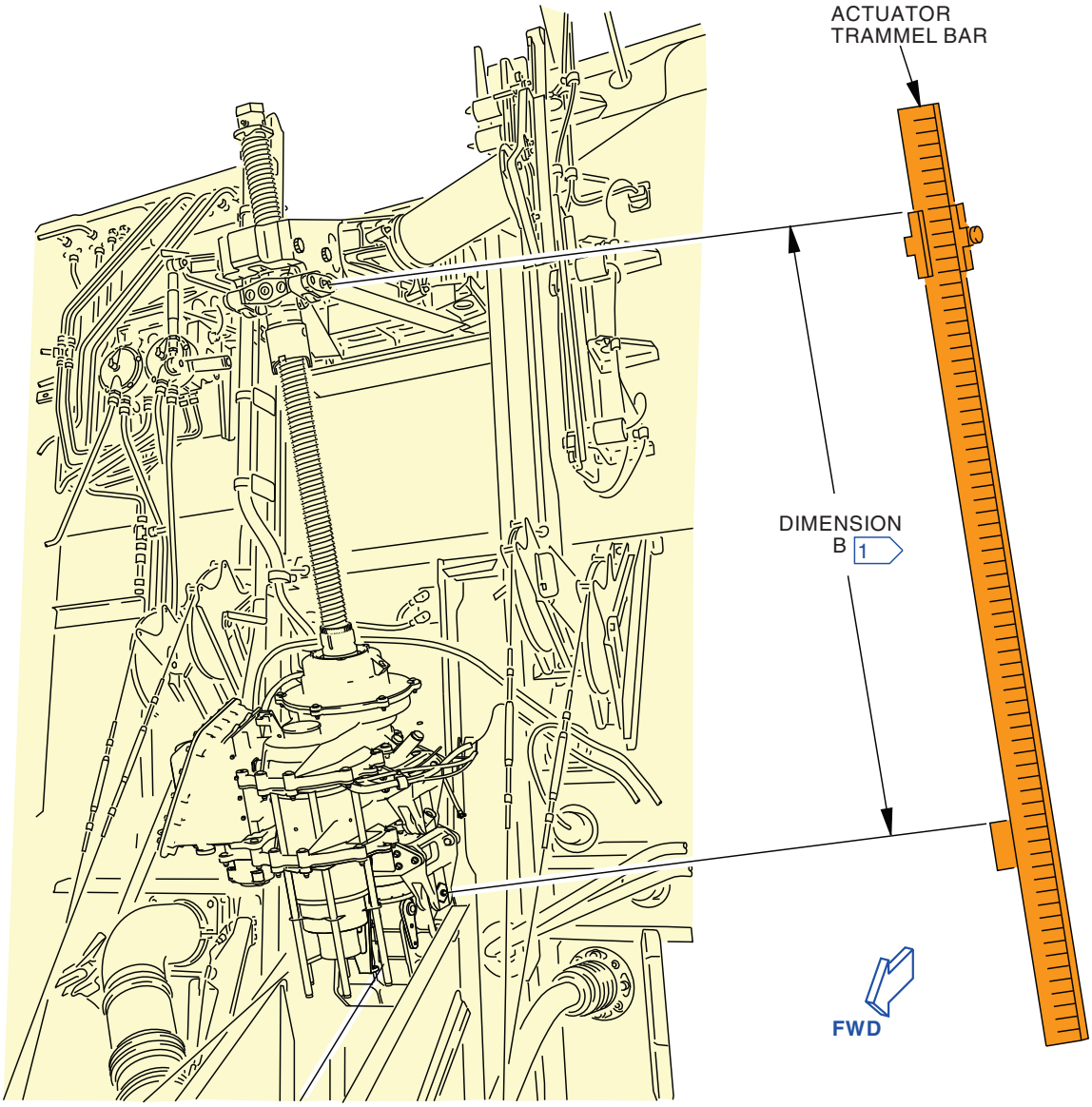
**VIEW THROUGH THE ACCESS DOOR TO
AFT UNPRESSURIZED COMPARTMENT, 311BL**

A

2412457 S00061530716_V1

**Stabilizer Trim Jackscrew Setting
Figure 504/27-31-00-990-810 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACTUATOR

B

NOTE:

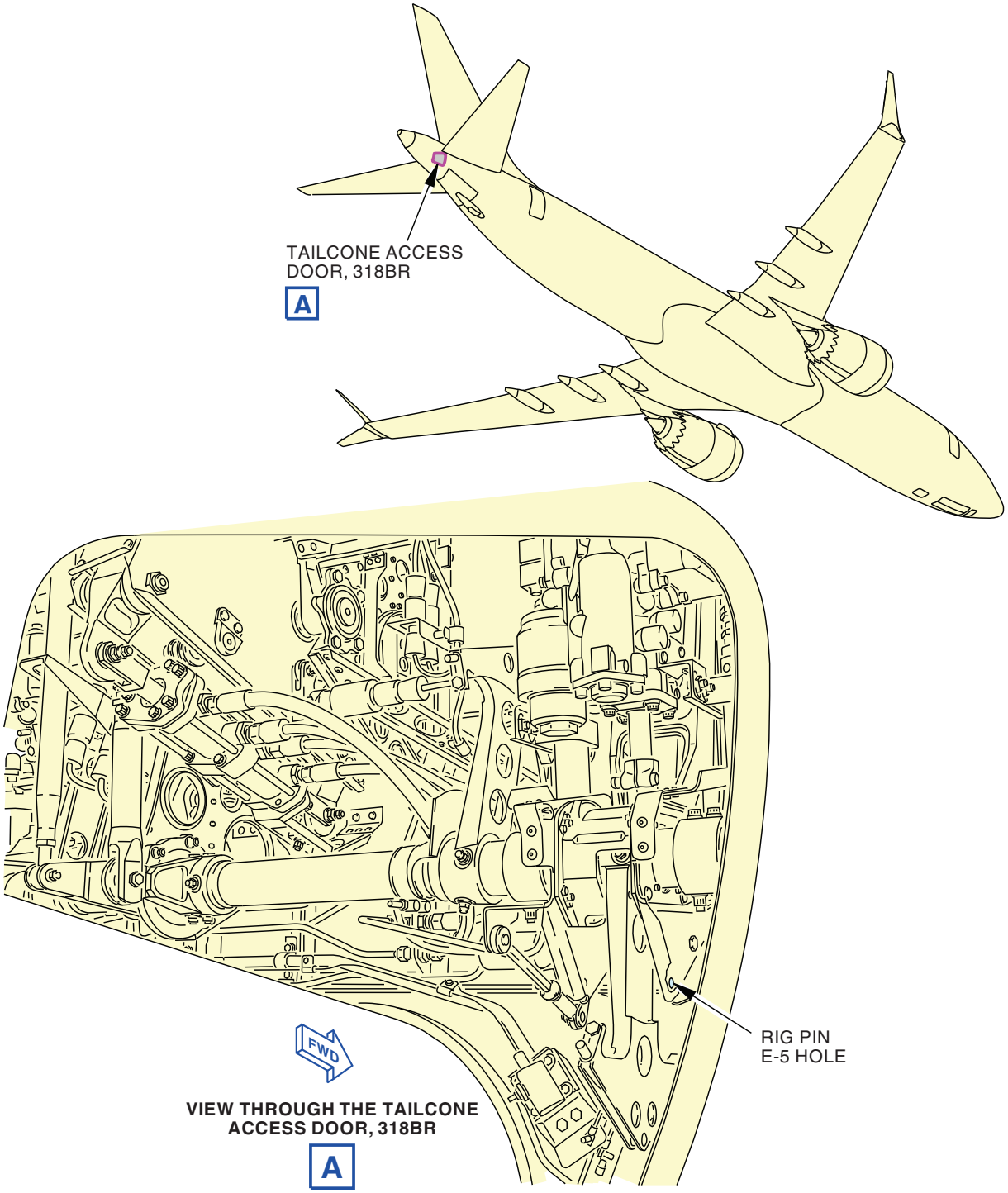
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

2412458 S00061530717_V2

Stabilizer Trim Jackscrew Setting
Figure 504/27-31-00-990-810 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Aft Control Quadrant Rig Pin Location
Figure 505/27-31-00-990-811

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-710-805

7. Pitch Force Transducer Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This test makes sure that the pitch force transducers operate correctly.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
31-65-00-970-801	System Maintenance Data Pages Function (P/B 201)
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare For the Operational Test

SUBTASK 27-31-00-860-142

> **WARNING** KEEP PERSONS AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES. THE AILERONS, ELEVATORS, RUDDER, FLAPS, SLATS, SPOILERS, STABILIZER, AND NOSE GEAR CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task to supply hydraulic power to Hydraulic Systems A and B: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

F. Pitch Force Transducer Test - DFCS BITE

SUBTASK 27-31-00-980-004

(1) Do these steps to put the control wheel to the center:

NOTE: The power control units must be serviceable for the correct adjustment of the cables in relation to the roll force transducer.

(a) Turn the control wheel in the clockwise direction 30 degrees.

(b) Turn the control wheel in the counterclockwise direction 30 degrees.

(c) Slowly turn the control wheel to the center.

SUBTASK 27-31-00-860-139

(2) Attach the DO NOT OPERATE tags, STD-858, to the control columns.

SUBTASK 27-31-00-860-140

(3) Set the aileron trim to 0.0 unit.

SUBTASK 27-31-00-860-141

(4) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand to the CUTOUT position.

SUBTASK 27-31-00-860-143

(5) Make sure that the VHF NAV and IRS switches, on the forward overhead panel, P5, are in the NORMAL positions.

SUBTASK 27-31-00-860-144

(6) Set the left and right IRS select switches, on the aft overhead panel, P5, to the ALIGN or NAV position.

SUBTASK 27-31-00-740-010

(7) Make these selections to get access to the PITCH CWS test:

(a) Push the INIT REF key on the CDU keyboard.

(b) <INDEX

(c) MAINT>

(d) <DFCS

(e) <EXTENDED MAINTENANCE

(f) <RIGGING

(g) <CWS

(h) Do the preparation steps as shown on the CDU display.

(i) <PITCH CWS

(j) <CONTINUE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Pitch Force Transducer - Manual Test

SUBTASK 27-31-00-480-014

(1) Connect the force gauge, COM-1557, to the center hub of the captains control column.

SUBTASK 27-31-00-730-015

(2) Do the steps shown on the CDU to complete the test:

(a) Make sure that the values for the A side and B side are between the limits that show on the CDU display.

SUBTASK 27-31-00-080-011

(3) Remove the force gauge, COM-1557, from the control wheels.

H. Pitch Force Transducer Interface Test - DFCS BITE

SUBTASK 27-31-00-740-011

(1) Make these selections on the CDU to do a transducer interface test:

(a) Push the INIT REF key on the CDU keyboard.

(b) <INDEX

(c) MAINT>

(d) <DFCS

(e) <LRU REPLACEMENT TESTS

(f) <CHANNEL A AND B

(g) Push the NEXT PAGE key on the CDU keyboard.

(h) <PITCH SENS

(i) Do the preparation steps as shown on the CDU.

(j) <CONTINUE

(k) Make sure that the test is completed with no failure.

(l) Push the INIT REF key on the CDU keyboard to get out of the BITE.

I. Existing Faults Check

SUBTASK 27-31-00-480-015

(1) Do these steps if you use the universal maintenance device, SPL-11321 (ML), to get access to the Existing Faults function:

(a) Connect the maintenance laptop (TASK 46-13-00-480-801).

(b) Use the universal maintenance device, SPL-11321, to get access to the Onboard Network System (ONS) maintenance browser.

SUBTASK 27-31-00-860-145

(2) Or, do these steps if you use the MAX Display System (MDS) Multi Function Display (MFD) to get access to the Existing Faults function:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If the MAINT or ONBD MAINT button does not show on the MFD, push the SYS key on the Multi-Function Panel (P9–82)

NOTE: To navigate/tab through the selectable elements on the Multi-Function Display (MFD), turn the Rotary Selector knob on the P9-82 panel or the MAINT SELECTOR knob on the P61-4 panel in the clockwise or counterclockwise direction until the desired element is highlighted.

NOTE: To make a selection on the MFD, push the center selection button SEL of the Rotary Selector knob on the P9-82 panel or the MAINT SELECTOR knob on the P61-4 panel.

(b) When the MAINT button shows, select it.

(c) Select the ONBD MAINT button from the Maintenance Menu bar.

SUBTASK 27-31-00-700-005

(3) From the ONS maintenance browser main menu, go to the 27 Flight Controls display in the EXTENDED MAINT>EXISTING FAULTS (TASK 31-65-00-970-801).

(a) Make sure that these Flight Control Computer (FCC) or Spoiler Control Electronics (SCE) maintenance messages do not show:

1) 27-02320
2) 27-02330
3) 27-02060
4) 27-02070

J. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-710-014

(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-31-00-860-155

(2) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand to the NORMAL position.

——— END OF TASK ———

TASK 27-31-00-820-802

8. Elevator Control Pushrods - Adjustment

(Figure 504, Figure 506, Figure 520)

A. General

(1) Use this test to adjust the elevator control pushrods that connect the output torque tube to the elevators.

(2) The elevator power-off flight test is required after adjustment of an elevator control pushrod.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

E. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

G. Prepare for the Adjustment

SUBTASK 27-31-00-860-045

(1) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-860-044

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-010-009

(3) Open these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-860-046

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

H. Elevator Control Pushrods Adjustment

SUBTASK 27-31-00-860-047

(1) Push down on the trailing edge of the right elevator [30], to bottom out the elevator PCU [31] in the extended position.

SUBTASK 27-31-00-710-007

(2) Make sure that the right elevator [30] is in the full down position with the inboard end of the elevator trailing edge 10.43 +0.09 / -0.32 in. (264.92 +2.29 / -8.13 mm) below the index mark (Figure 520).

(a) Remove the nut [27], nut [26], washer [25], bushing [24], and bolt [23] to disconnect the right elevator control pushrod [28] from the elevator buss crank [22].

(b) Loosen the checknut [29] on the right elevator control pushrod [28].

(c) Adjust the elevator control pushrod [28] until the elevator [30] is in the full down position with the inboard end of the elevator trailing edge at 10.43 +0.09 / -0.32 in. (264.92 +2.29 / -8.13 mm) below the index mark (Figure 520).

NOTE: One half turn of the elevator control pushrod is equal to 0.20 in. (5.1 mm) at the elevator trailing edge.

(d) Lubricate the bolt [23] and bushing [24] with grease, D00633.

1) Wipe off the excess of grease, D00633.

(e) Install the bolt [23], bushing [24], washer [25], and nut [26] (Figure 506).

1) Tighten the nut [26] by hand.

SUBTASK 27-31-00-980-002

(3) Push down on the trailing edge of the left elevator [30], to bottom out of the elevator PCU [31] in the extended position.

SUBTASK 27-31-00-820-001

(4) Make sure that the left elevator [30] is in the full down position with the inboard end of the elevator trailing edge 10.43 +0.09 / -0.32 in. (264.92 +2.29 / -8.13 mm) from the elevator neutral index mark (Figure 520).

(a) Remove the nut [27], nut [26], washer [25], bushing [24], and bolt [23] to disconnect the left elevator control pushrod [28] from the elevator buss crank [22].

(b) Loosen the checknut [29] on the left elevator control pushrod [28].

(c) Adjust the elevator control pushrod [28] until the elevator [30] is in the full down position with the inboard end of the elevator trailing edge at 10.43 +0.09 / -0.32 in. (264.92 +2.29 / -8.13 mm) from the elevator neutral index mark (Figure 520).

NOTE: One half turn of the elevator control pushrod is equal to 0.20 in. (5.1 mm) at the elevator trailing edge.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Lubricate the bolt [23] and bushing [24] with grease, D00633.

1) Wipe off the excess of grease, D00633.

(e) Install the bolt [23], bushing [24], washer [25], and nut [26] (Figure 506).

1) Tighten the nut [26] by hand.

SUBTASK 27-31-00-970-001

(5) Make sure that the final dimension in the full down position of the left elevator [30] is 0.00 ±0.12 in. (0.0 ±3.0 mm) of the final dimension in the full down position of the right elevator [30].

(a) If it is necessary, re-adjust the right elevator [30] to 10.43 +0.09 / -0.32 in. (264.92 +2.29 / -8.13 mm) from the index plate (Figure 520).

SUBTASK 27-31-00-420-004

(6) Tighten the checknuts [29] on the left and right elevator control pushrods [28] to 750 in-lb (85 N·m) - 900 in-lb (102 N·m).

(a) Install MS20995NC32 lockwire, G01912, on the checknuts [29].

SUBTASK 27-31-00-420-005

(7) Tighten the nuts [26] to 600 in-lb (67.8 N·m) - 700 in-lb (79.1 N·m).

SUBTASK 27-31-00-420-006

I

CAUTION

DO NOT TRY TO TIGHTEN THE NUT [26] AGAIN AFTER YOU TIGHTEN THE NUT [27]. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(8) Install the nuts [27].

(a) Tighten the nuts [27] to 170 in-lb (19.2 N·m) - 240 in-lb (27.1 N·m).

NOTE: Bolts protrusion through the nuts can be 0.01 in. (0.25 mm) minimum.

SUBTASK 27-31-00-860-048

(9) Release the elevator PCUs [31] from their extended position.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-940-003

(1) Close these access panels:

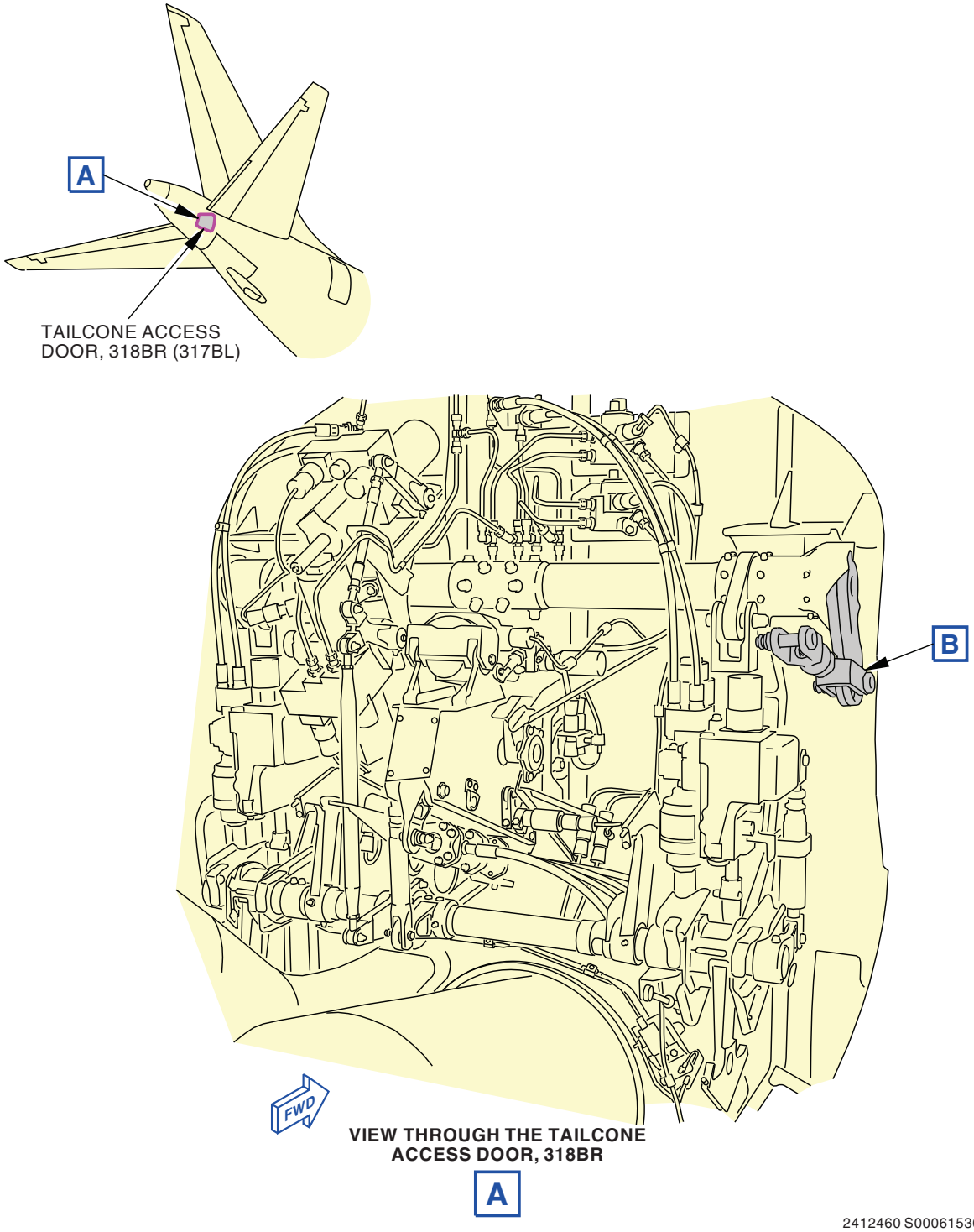
Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-00-750-001

(2) Do this task: Elevator Power-Off Flight Test, TASK 27-31-00-710-803.

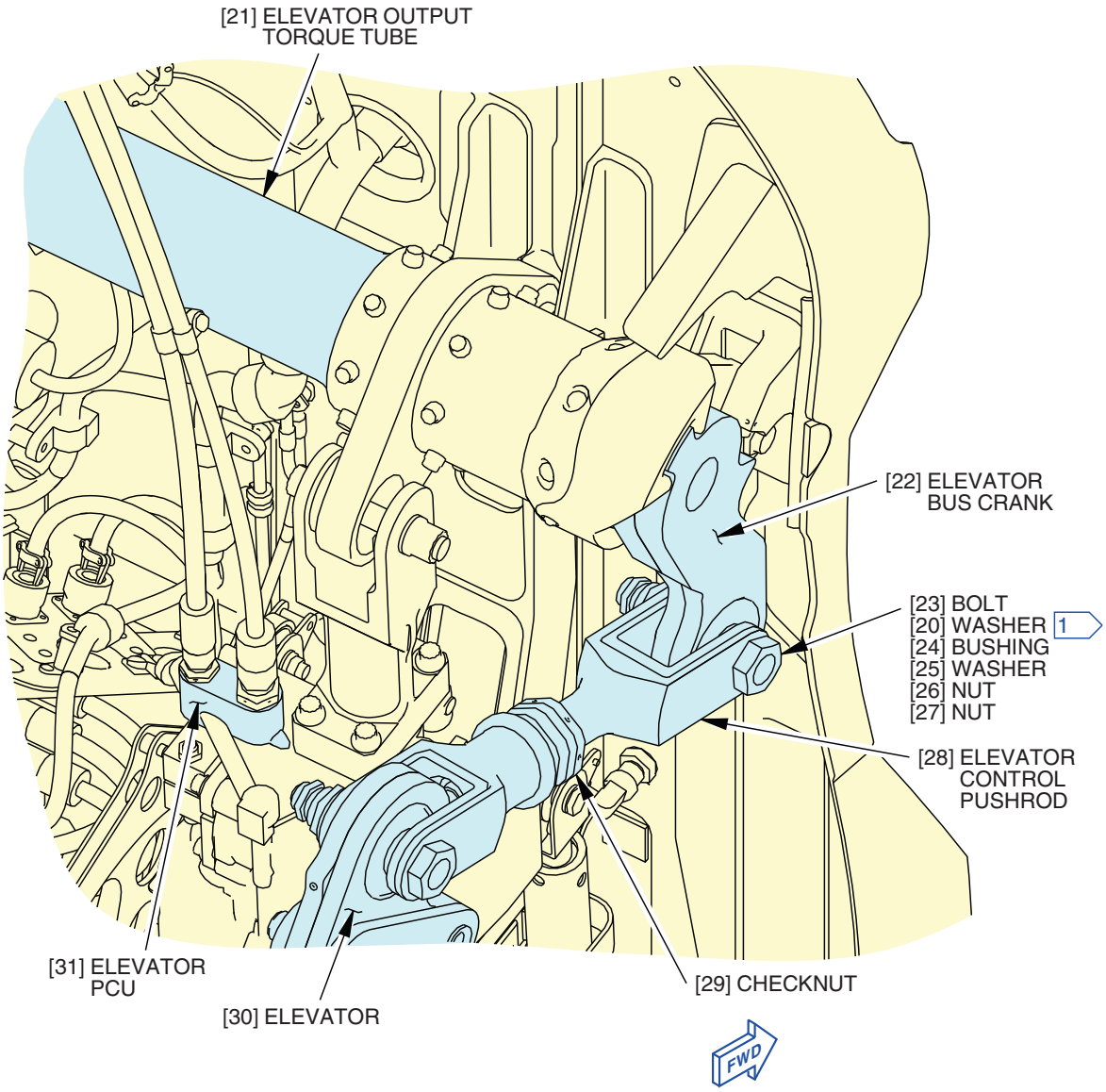
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Pushrods Adjustment
Figure 506/27-31-00-990-812 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)
(EXAMPLE)

B

1 IF INSTALLED.

2412461 S00061530721_V4

Elevator Pushrods Adjustment
Figure 506/27-31-00-990-812 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-820-803

9. Input Rods of the Elevator Power Control Unit - Adjustment

(Figure 504, Figure 505, Figure 507, Figure 508)

A. General

(1) Use this test to adjust the two input rods (primary and secondary) of the right and left elevator power control units (PCUs).

NOTE: For this procedure, the primary input rod is on the left side of the PCUs and the secondary input rod is on the right side of the PCUs when looking forward. This applies to both left and right elevator PCUs.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

D. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

F. Prepare for the Adjustment

SUBTASK 27-31-00-860-050

(1) Make sure that the adjustment of elevator control pushrods is satisfactory (TASK 27-31-00-820-802).

SUBTASK 27-31-00-860-051

(2) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

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SUBTASK 27-31-00-010-011

(3) Open these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-00-860-052

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

SUBTASK 27-31-00-860-053

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Pressurize the elevator hydraulic systems A and B, do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-31-00-480-004

(6) Install the rig pin E-5 from the rigging pin kit, SPL-1585, to the elevator aft quadrant (Figure 508).

(a) Make sure that the rig pin E-5 fits freely in the rig pin hole.

G. Input Rods for the Elevator PCU Adjustment

SUBTASK 27-31-00-820-002

(1) Adjust the input rods for the elevator PCU (Figure 508):

NOTE: These steps adjust both the right and left elevator PCU input rods.

(a) If adjusting the right elevator PCU, do the following:

1) Set the FLT CONTROL A switch to the OFF position and FLT CONTROL B switch to the ON position.

(b) If adjusting the left elevator PCU, do the following:

1) Set the FLT CONTROL B switch to the OFF position and FLT CONTROL A switch to the ON position.

(c) Measure the distance from the upper trailing edge surface of the right elevator to the neutral index mark of the index plate (Figure 507).

(d) If the distance is greater than 0.06 in. (1.5 mm), do the following adjustment:

1) Remove the nut [55], washer [54], washer [53], and bolt [52] to disconnect the secondary input rod [58] from the upper attach point (Figure 508).

2) Loosen the checknut [57] on the primary input rod [59].

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3) Rotate the primary input rod [59] tube until the distance from the trailing edge of the elevator to the index plate is no more than 0.06 in. (1.5 mm) (Figure 507).

4) Tighten the checknut [57] to 195 in-lb (22 N·m) - 235 in-lb (27 N·m) (Figure 508).

   a) Check the elevator trailing edge to index plate dimension after tightening the checknut [57].

      NOTE: The dimension can change after the checknut has been tightened.

5) Loosen the checknut [57] on the secondary input rod [58].

6) Rotate the secondary input rod [58] tube until the upper attach bolt [52] can be freely installed through the clevis and the PCU crank arm.

7) Install the bolt [52], washer [53], washer [54], and nut [55].

8) Tighten the nut [55].

9) Tighten the checknut [57] to 195 in-lb (22 N·m) - 235 in-lb (27 N·m).

   a) After tightening the checknut [57], check the elevator trailing edge to index plate dimension with FLT CONTROL A and FLT CONTROL B switches in the ON position.

      NOTE: The dimension can change after the checknut has been tightened.

10) Install lockwire on the checknut [57] of the input rods.

    NOTE: Install lockwire from the checknut to the input rod tube, and then to the small hole in the upper clevis.

11) If the right elevator PCU input rods have been adjusted, set the FLT CONTROL B switch to the OFF position.

12) If the left elevator PCU input rods have been adjusted, set the FLT CONTROL A switch to the OFF position.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-080-004

(1) Remove the rig pin E-5 from the elevator aft quadrant.

SUBTASK 27-31-00-940-004

(2) Close these access panels:

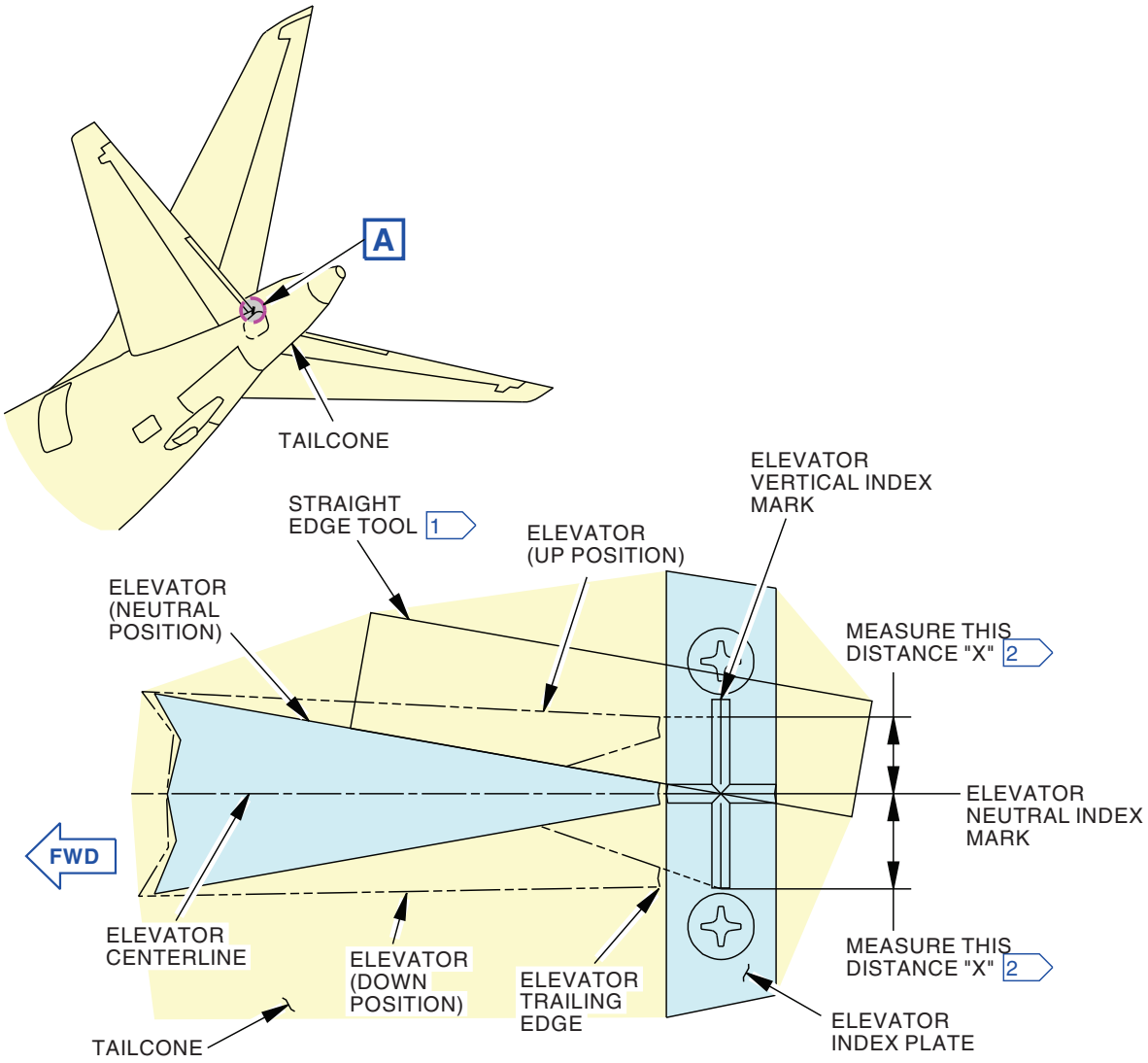
Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-00-860-054

(3) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

——— END OF TASK ———

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ELEVATOR TAB PUSHROD ADJUSTMENT MEASUREMENTS (LEFT SIDE)

A

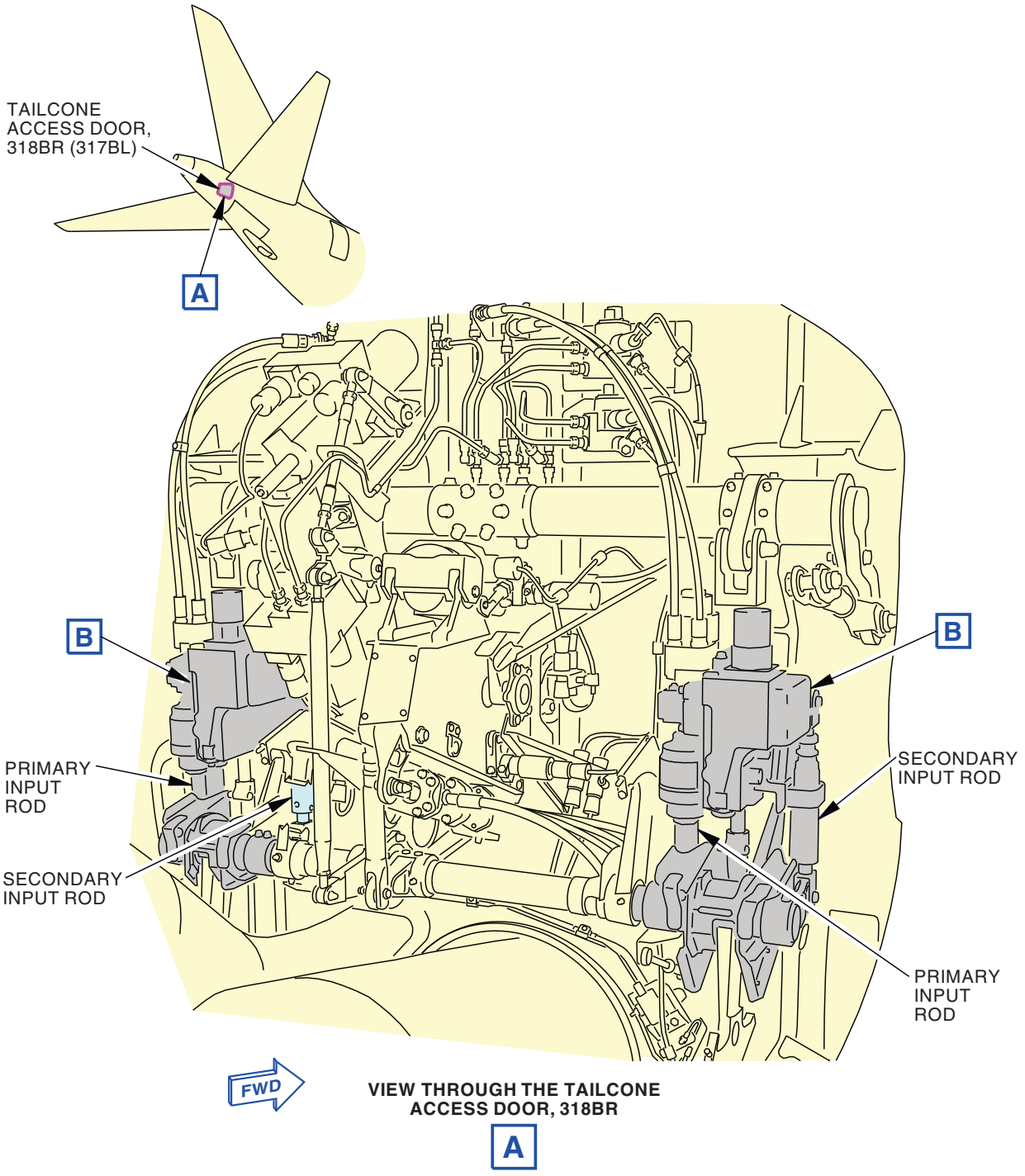
1 ELEVATOR IS AT NEUTRAL WHEN THE BOTTOM EDGE OF A STRAIGHT EDGE TOOL PLACED ON THE UPPER ELEVATOR SURFACE PASSES THROUGH THE MIDDLE OF THE ELEVATOR INDEX PLATE, AS SHOWN.

2 THE DISTANCE "X" IS MEASURED USING A STRAIGHT EDGE TOOL ON THE UPPER SURFACE OF THE ELEVATOR TO INTERSECT WITH THE ELEVATOR VERTICAL INDEX MARK AT THE ELEVATOR NEUTRAL INDEX MARK.

2412462 S00061530722_V1

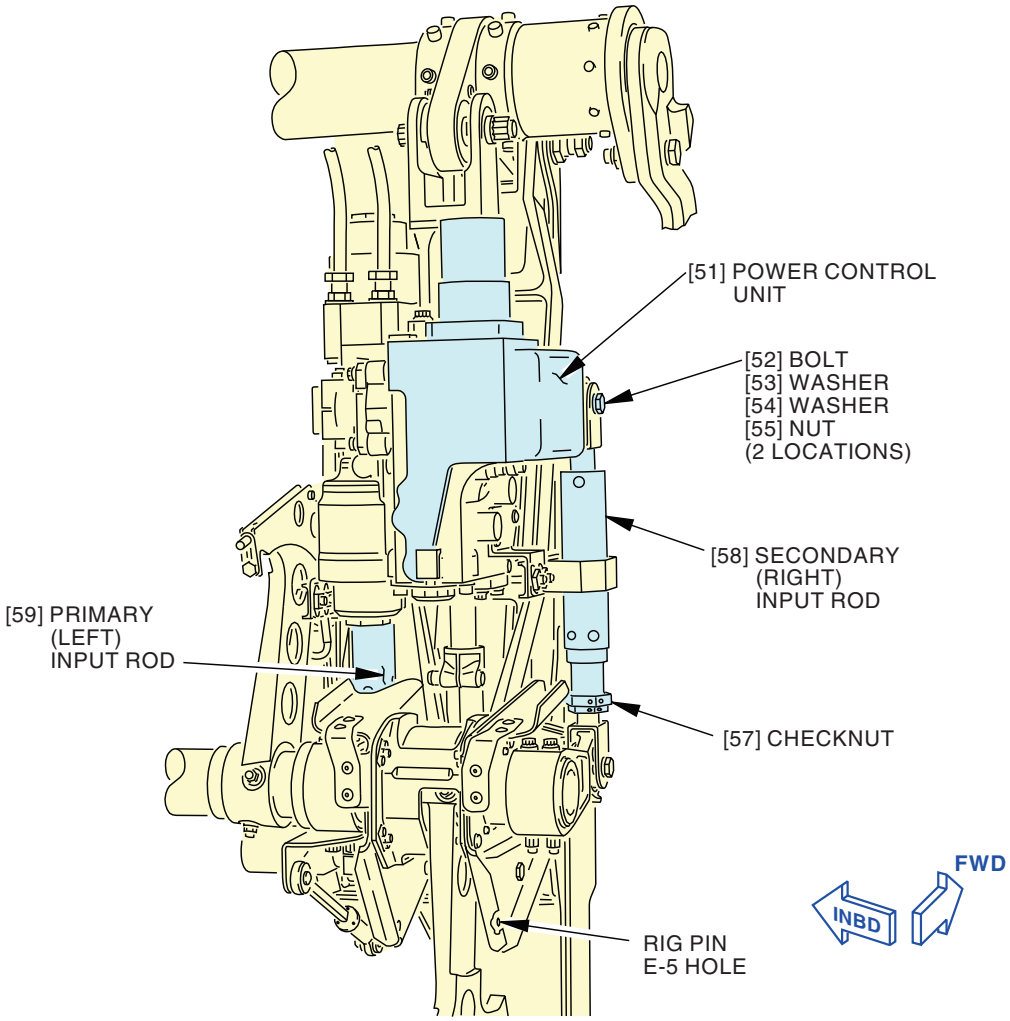
Elevator Tab Adjustment
Figure 507/27-31-00-990-829

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator PCU Input Rods Adjustment
Figure 508/27-31-00-990-814 (Sheet 1 of 2)

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RIGHT POWER CONTROL UNIT
(LEFT POWER CONTROL UNIT IS EQUIVALENT)

B

2412465 S00061530726_V1

Elevator PCU Input Rods Adjustment
Figure 508/27-31-00-990-814 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-820-804

10. Input Rods of the Elevator Neutral Shift Mechanism and Output Rod of the Feel and Centering Unit - Adjustment

(Figure 504, Figure 509)

A. General

(1) Use this test to adjust the input rods which connect the elevator neutral shift mechanism to the stabilizer. This test also includes steps to adjust the output rod which connects the elevator feel and centering unit to the input torque tube.

B. References

Reference	Title
22-11-14-820-801	Elevator Neutral Shift Sensor Adjustment and Test (P/B 501)
22-11-31-820-801	Stabilizer Position Sensor Adjustment and Test (P/B 501)
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

E. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Adjustment

SUBTASK 27-31-00-860-056

(1) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-860-055

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-010-013

(3) Open these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-00-860-057

CAUTION MAKE SURE TO PREVENT DAMAGE TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CHAF OR CAN CAUSE DAMAGE TO THE WIRES OR BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)), (View B, Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

SUBTASK 27-31-00-480-005

(5) Install the rig pin E-5 from the rigging pin kit, SPL-1585, to the elevator aft quadrant (Figure 509).

H. Input Rods of the Elevator Neutral Shift Mechanism and Output Rod of the Elevator Feel and Centering Unit Adjustment

SUBTASK 27-31-00-220-001

(1) Measure the distance "Z" between the centerline of the left neutral shift crank bolt [82] and the aft side of the vertical bulkhead (View D, Figure 509).

(a) Make sure that the distance is 1.057 ±0.030 in. (26.85 ±0.76 mm).

(b) If the distance is not 1.057 ±0.030 in. (26.85 ±0.76 mm), adjust the neutral shift input rods [84] of the elevator elevator neutral shift mechanism bellcrank [87]:

1) Remove the nut [79], washer [78], washer [77], and bolt [76] to disconnect the output rod of the elevator feel and centering unit [71] from the input crank [75].

2) Loosen the checknuts on the neutral shift input rods [84].

3) Remove the nuts [85], washers [86], washers [83], washers [81], and bolts [82] to disconnect the neutral shift input rods [84] from the elevator neutral shift mechanism bellcrank [87].

4) Adjust the neutral shift input rods [84] until you get a "Z" dimension of 1.057 ±0.030 in. (26.85 ±0.76 mm) at the left neutral shift input rod [84].

5) Install the bolts [82], washers [81], washers [83], washers [86], and nuts [85] to connect the stabilizer position sensor input rods [80] and neutral shift input rods [84] to the elevator neutral shift mechanism bellcrank [87].

a) Install the MS20995NC32 lockwire, G01912.

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SUBTASK 27-31-00-820-004

(2) Adjust the output rod of the elevator feel and centering unit [71] (View B, Figure 509):

(a) If it is not disconnected, remove the nut [79], washer [78], washer [77], and bolt [76] to disconnect the output rod [74] of the elevator feel and centering unit [71] from the input crank [75].

(b) Measure the distance between the bolt hole in the output rod [74] and the bolt hole in the input crank [75]:

(c) If the distance is greater than 0.06 in. (1.52 mm), then do these steps:

1) Turn the clevis [131] in the sleeve [73] half turn and/or sleeve [73] in the output rod [74] end half turn.

NOTE: Use a combination of clevis/sleeve rotations in order to keep the number of threads exposed approximately equal in both parts.

2) Make sure that the distance of the two bolt holes is less than 0.06 in. (1.52 mm).

3) Keep the number of threads that you can see in the two parts approximately the same.

(d) If the distance is less than 0.06 in. (1.52 mm), then do these steps:

1) Turn the sleeve [73] until you can install the bolt [76] easily.

a) Turn the output rod [74] end and sleeve [73] in the same direction.

2) Keep the number of threads that you can see in the two parts are approximately same.

CAUTION

DO NOT TRY TO TIGHTEN THE BOLT AGAIN AFTER YOU TIGHTEN THE NUT. YOU CAN CAUSE DAMAGE TO EQUIPMENT.

3) Install the bolt [76], washer [77], washer [78], and nut [79] to connect the output rod [74] to the input crank [75].

a) Tighten the bolt [76] to 75 in-lb (8.5 N·m) - 85 in-lb (9.6 N·m).

b) Tighten the nut [79] to 25 in-lb (2.8 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 27-31-00-820-005

(3) If it is necessary, do this task: Elevator Neutral Shift Sensor Adjustment and Test, TASK 22-11-14-820-801.

SUBTASK 27-31-00-820-006

(4) If it is necessary, do this task: Stabilizer Position Sensor Adjustment and Test, TASK 22-11-31-820-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-080-005

(1) Remove the rig pin E-5 from the elevator aft quadrant.

SUBTASK 27-31-00-940-005

(2) Close these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

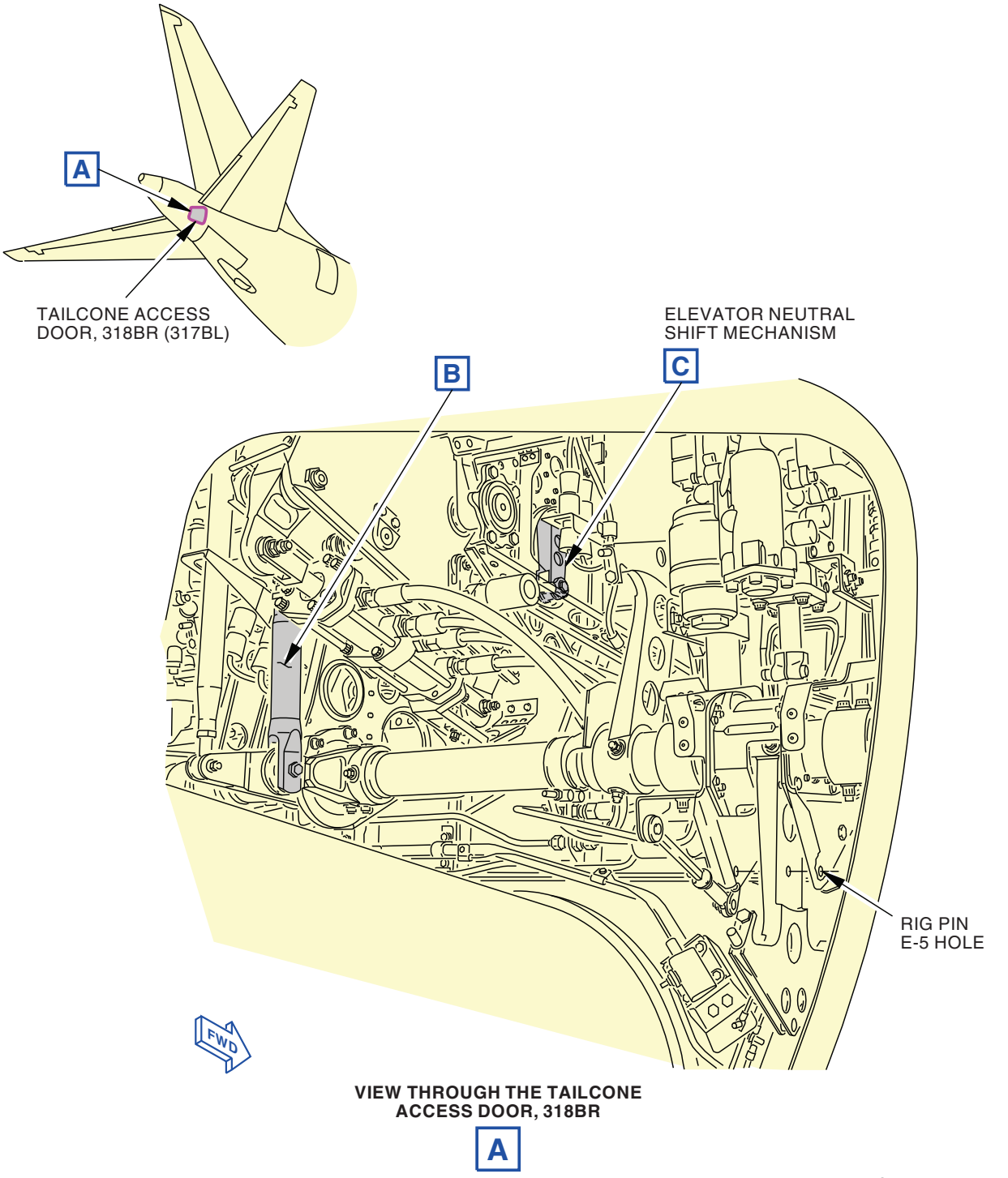
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-860-058

(3) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

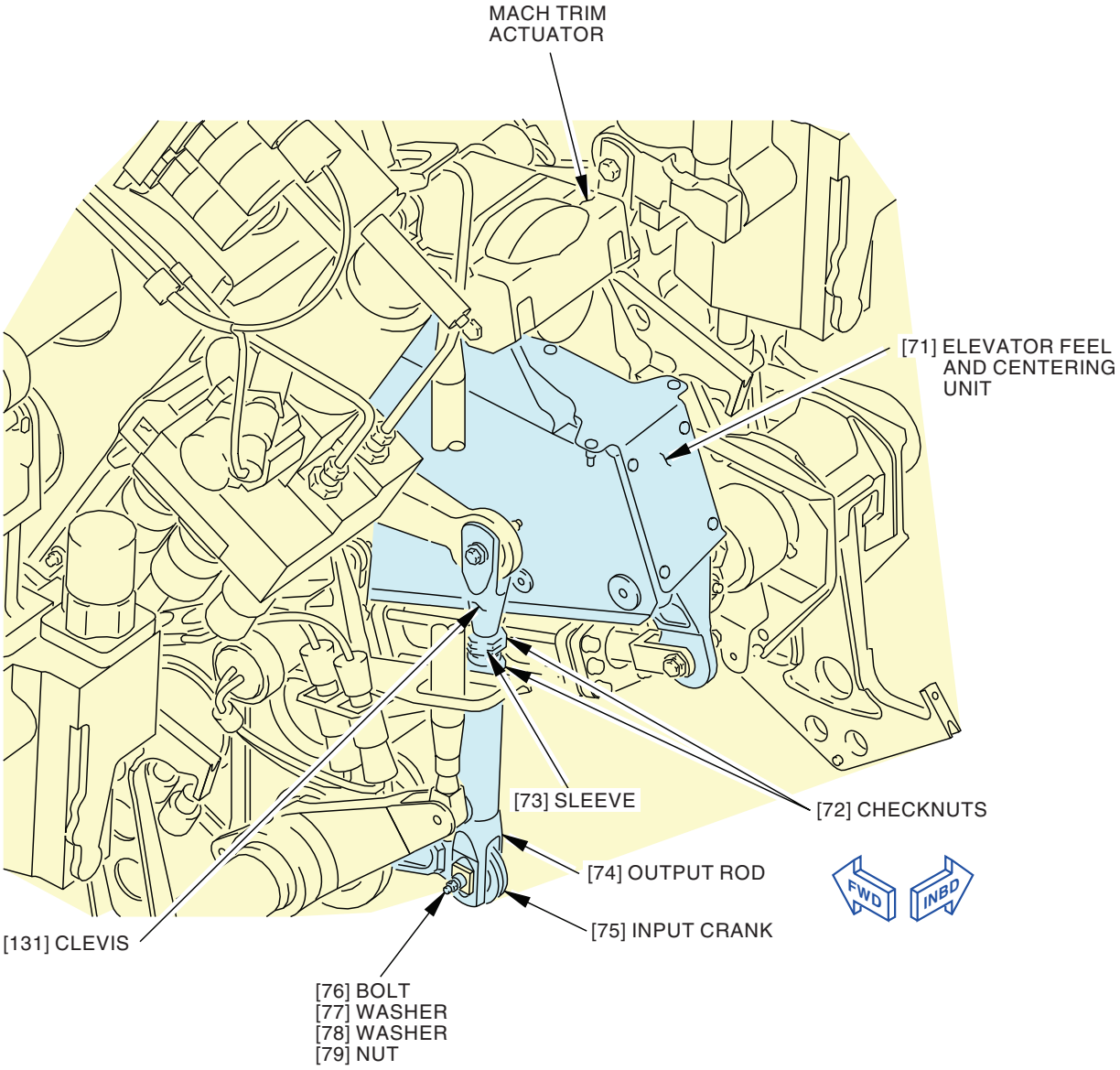
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



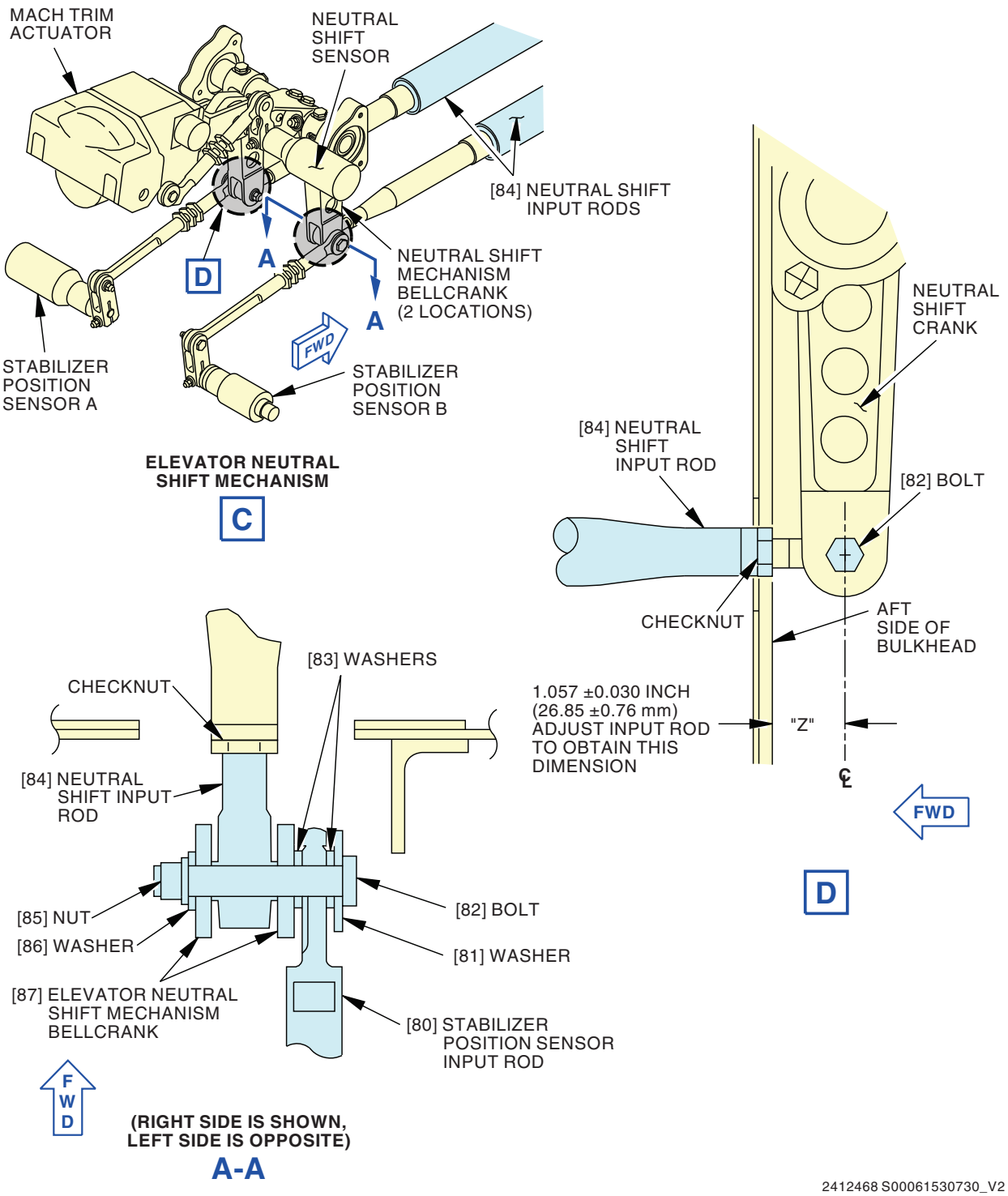
Elevator Neutral Shift Mechanism Input Rods and Elevator Feel and Centering Unit Output Rod Adjustment
Figure 509/27-31-00-990-815 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Neutral Shift Mechanism Input Rods and Elevator Feel and Centering Unit Output Rod Adjustment
Figure 509/27-31-00-990-815 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Neutral Shift Mechanism Input Rods and Elevator Feel and Centering Unit Output Rod Adjustment
Figure 509/27-31-00-990-815 (Sheet 3 of 3)

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TASK 27-31-00-820-805

11. Elevator Tab Pushrods - Adjustment

(Figure 504, Figure 507, Figure 510, Figure 511)

NOTE: A flight test is required if either or both of the elevator tab pushrods are adjusted (TASK 27-31-00-710-803).

A. General

(1) Use this test to adjust the elevator tab pushrods. This task is applicable for the left and right elevator tab pushrods.

NOTE: Adjustment of the elevator tab pushrods to a dimension recorded subsequent to the last successful flight test will reduce the probability that a second flight test will be required.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
33-44-04-040-801	Tailcone Anti-Collision Light Deactivation (P/B 201)
33-44-04-440-801	Tailcone Anti-Collision Light Activation (P/B 201)
49-11-00-040-801	APU System - Deactivation (P/B 201)
49-11-00-440-801	APU System - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-10345	Rigging Bar - Elevator Tab Part #: C27083-18 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
D50004	Compound - Antiseize	BMS3-28

E. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

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F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

G. Prepare for the Adjustment

SUBTASK 27-31-00-860-147

(1) Do this task: APU System - Deactivation, TASK 49-11-00-040-801.

SUBTASK 27-31-00-860-148

(2) Do this task: Tailcone Anti-Collision Light Deactivation, TASK 33-44-04-040-801.

SUBTASK 27-31-00-860-060

(3) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-860-059

(4) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-010-015

(5) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-010-016

(6) For the left elevator tab pushrods, remove this access panel:

Number	Name/Location
334PT	Horizontal Stabilizer, Tab Control Rod Fairing

SUBTASK 27-31-00-010-017

(7) For the right elevator tab pushrods, remove this access panel:

Number	Name/Location
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-00-860-061

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(8) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

H. Elevator Tab Pushrods Adjustment

SUBTASK 27-31-00-820-007

(1) Do the adjustment of the pushrods on one elevator at a time, starting with the right elevator.

SUBTASK 27-31-00-020-002

(2) Loosen the checknuts [103] and checknuts [105] from the tab pushrods [110].

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SUBTASK 27-31-00-860-062

(3) Push the elevator trailing edge full up by hand.

SUBTASK 27-31-00-860-063

(4) When you move the elevator trailing edge down, align the center of the elevator trailing edge with the index mark in ±0.04 in. (1.0 mm), or less (Figure 507).

(a) Keep the elevators at this position when you measure and adjust the tab pushrods.

(b) If the elevator position is changed, you must put the elevator trailing edge at the full up position and do this step again before you continue.

SUBTASK 27-31-00-480-006

(5) Install the rigging bar, SPL-10345 (Figure 511).

(a) Install the rigging bar, SPL-10345, below the elevator and elevator tab at the third elevator tab hinge (from inboard).

1) Make sure that the stand offs for the rigging bar, SPL-10345, rest on the skin fastener heads to prevent skin variation.

(b) Make sure that the cross bar on the rigging bar, SPL-10345, touches the lower surface of the elevator trailing edge.

(c) Push the rigging bar, SPL-10345, FWD until elevator stop indexes on the AFT edge of elevator near elevator rear spar.

(d) Slide step gage FWD under elevator tab to obtain reading.

SUBTASK 27-31-00-820-008

(6) Adjust the tab pushrods [110] until the gap measured from the elevator tab faired position to the elevator tab rigged up position with the rigging bar, SPL-10345, (dimension X) is the same dimension that you measured and recorded prior to the last successful flight test and before any pushrod adjustment.

NOTE: Adjustment of the elevator tab pushrods to a dimension recorded subsequent to the last successful flight test will reduce the probability that a second flight test will be required.

NOTE: You possibly need to do a coarse adjustment of the elevator tab pushrods if the gap between the elevator tab trailing edge and fixture is too much.

You must always do a fine adjustment of the elevator tab pushrods if dimension X is not in the tolerance.

NOTE: When adjusting the elevator tab pushrods, the number of threads showing on the elevator tab pushrod ends should be approximately equal to the number of threads showing on the bushing [104].

(a) After tab pushrod adjustment, make sure that the distance between the left elevator tab and rigging bar, SPL-10345, is within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm) (Figure 511).

(b) After tab pushrod adjustment, make sure that the distance between the right elevator tab and rigging bar, SPL-10345, is within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm) (Figure 511).

(c) If the distance between the left elevator tab and rigging bar, SPL-10345, is not within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm), then adjust to 0.36 ±0.01 in. (9.14 ±0.26 mm) (Figure 511).

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(d) If the distance between the right elevator tab and rigging bar, SPL-10345, is not within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm), then adjust to 0.36 ±0.01 in. (9.14 ±0.26 mm) (Figure 511).

SUBTASK 27-31-00-820-009

(7) If the prior tab location was not measured or recorded, then adjust to 0.36 ±0.01 in. (9.14 ±0.26 mm) (Figure 511).

(a) After adjustment, make sure that the distance between the left elevator tab and rigging bar, SPL-10345, is within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm) (Figure 511).

(b) After adjustment, make sure that the distance between the right elevator tab and rigging bar, SPL-10345, is within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm) (Figure 511).

SUBTASK 27-31-00-820-010

(8) If it is necessary, do this coarse adjustment of the elevator tab pushrods by turning the tab rod ends (Figure 510):

(a) Remove the pins [111], nuts [109], washers [108], bolts [106], and bushings [107] from the tab pushrods [110].

(b) Turn the tab pushrod ends until the distance between the elevator tab trailing edge and rigging bar, SPL-10345, (dimension X) is the same dimension that you measured and recorded subsequent to the last successful flight test and before any pushrod adjustment.

1) If the prior tab location was not measured or recorded, then adjust to 0.36 ±0.01 in. (9.14 ±0.26 mm).

NOTE: One half turn of the tab pushrod end is equal to 0.10 in. (2.54 mm) at the elevator tab trailing edge.

(c) Apply compound, D50004, on the threads of the nuts [109].

(d) Install the bushings [107], bolts [106], washers [108], and nuts [109] to connect the tab pushrods [110] to the elevator tab [102].

NOTE: The inboard bolt is installed with the head inboard and outboard bolt is installed with the head outboard

1) If you loosened or removed the nuts [109], they need to be replaced and not re-installed.

a) Using new nuts with compound, D50004, will ensure that they have the required run-on torque of 3.5 in-lb (0.4 N·m).

SUBTASK 27-31-00-820-011

(9) Do this fine adjustment of the elevator tab pushrods (Figure 510):

(a) Turn the bushing [104] to adjust the length of the tab pushrods [110].

NOTE: One half turn of the bushing [104] is equal to 0.01 in. (0.3 mm) at the elevator tab trailing edge.

(b) Loosen the nuts [109] that attach the tab pushrods [110] to the elevator tabs [102].

(c) Remove the pins [111] from in front of the head of the bolts [106].

(d) Move the rectangle bolt heads out of the anti-rotation bolt bosses and make sure that you can easily turn the bolts [106] by hand.

1) If it is necessary, adjust the bushing [104] on the tab pushrods [110].

NOTE: One half turn of the bushing/rod end assembly is equal to 0.11 in. (2.79 mm).

(e) Apply compound, D50004, on the threads of the nuts [109].

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CAUTION

TIGHTEN THE NUTS [109] CORRECTLY. IF YOU TIGHTEN THEM INCORRECTLY THEN THE PUSHRODS CAN BECOME DISCONNECTED. THIS CAN CAUSE DAMAGE TO THE ELEVATOR, AND STABILIZER.

(f) Tighten the nuts [109] to 70 in-lb (8 N·m) - 80 in-lb (9 N·m) and install pins [111].

1) If you loosened or removed the old nuts [109], they need to be replaced with compound, D50004, and not re-installed.

a) Using new nuts [109] with compound, D50004, will ensure that they have the required run-on torque of 3.5 in-lb (0.4 N·m).

(g) Tighten the checknuts [103] and checknuts [105] on the tab pushrods [110] to 150 in-lb (17 N·m) - 200 in-lb (23 N·m).

(h) Install lockwire on the checknut [103] and between the bushing [104] and checknut [105].

SUBTASK 27-31-00-220-002

(10) Make sure that the gap measured from the elevator tab faired position to the elevator tab rigged up position with the rigging bar, SPL-10345, (dimension X) is the same dimension that you measured and recorded subsequent to the last successful flight test and before any pushrod adjustment.

(a) If the prior tab location was not measured or recorded, then adjust to 0.36 ±0.01 in. (9.14 ±0.26 mm).

SUBTASK 27-31-00-220-003

(11) Do a check of the thread engagement of the tab pushrod [110] ends and bushings [104]:

(a) Make sure that the end of the bushing [104] covers the inspection hole on each tab pushrod [110].

(b) Make sure that the inspection hole of each tab pushrod end bearing is not shown.

(c) If the thread engagement is not correct, you must do the coarse adjustment again.

SUBTASK 27-31-00-200-006

(12) Verify that the forward and aft tab rod end bearings are aligned in the correct alignment (Figure 510):

(a) Rotate each tab rod counterclockwise (when facing forward) until contact is made between the tab rod end and surrounding structure.

1) This initial contact point is the start of the measurement of the rod end bearing alignment.

(b) Rotate each tab rod clockwise (when facing forward) until contact is made between the tab rod end and surrounding structure.

1) Measure the distance travelled between the contact points as the tab rod was rotated clockwise from contact point to contact point.

(c) There should be a minimum of 0.040 in. (1.016 mm) travel from contact point to contact point.

1) If there is less than 0.040 in. (1.016 mm) of travel from contact point to contact point, adjust the tab rod end until a minimum travel of 0.040 in. (1.016 mm) is achieved.

a) If it was necessary to adjust the tab rods, make sure that you tighten the checknuts [103] and checknuts [105] to the proper torque and do a check of the rod end rotation from contact point to contact point.

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SUBTASK 27-31-00-420-007

(13) Install lockwire on the checknuts [103] and checknuts [105] (Figure 510).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-149

(1) Do this task: APU System - Activation, TASK 49-11-00-440-801.

SUBTASK 27-31-00-860-150

(2) Do this task: Tailcone Anti-Collision Light Activation, TASK 33-44-04-440-801.

SUBTASK 27-31-00-410-011

NOTE: Depending on which systems have been disturbed, you possibly not have to do a power-off flight test. To see if you need to do a power-off flight test, check the requirements in the ‘‘General’’ section at the beginning of the power-off flight test task (TASK 27-31-00-710-803).

(3) Install these access panels:

Number	Name/Location
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-00-390-003

(4) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

SUBTASK 27-31-00-940-006

(5) Close this access panel:

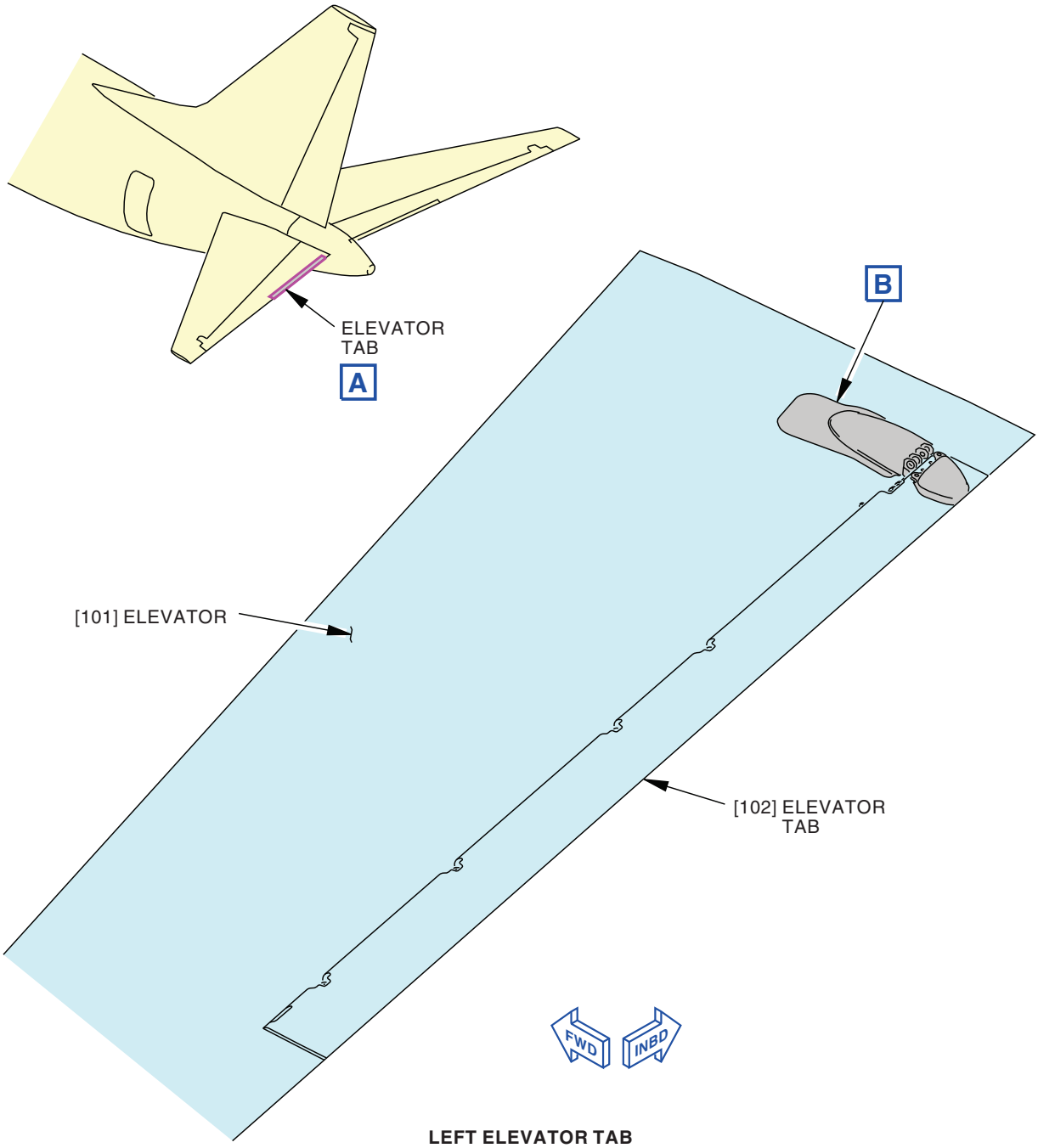
Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-860-064

(6) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

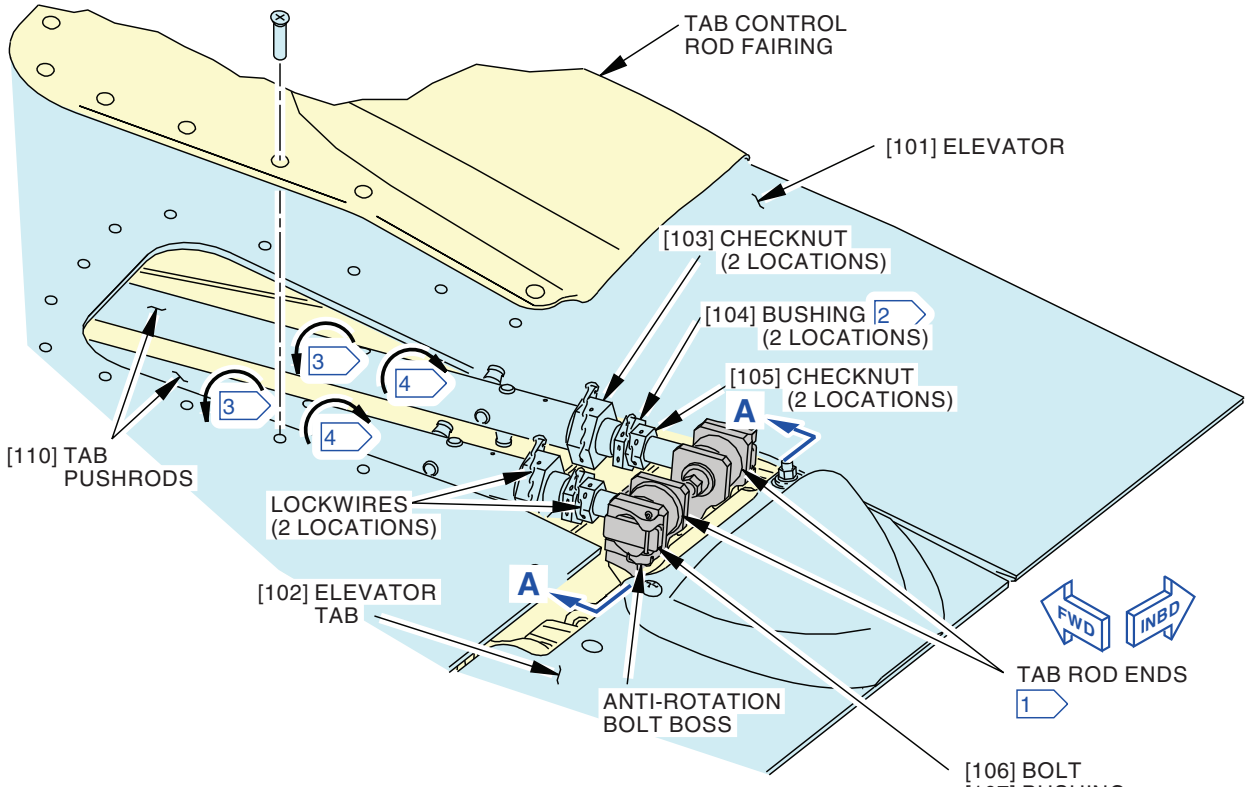
——— END OF TASK ———

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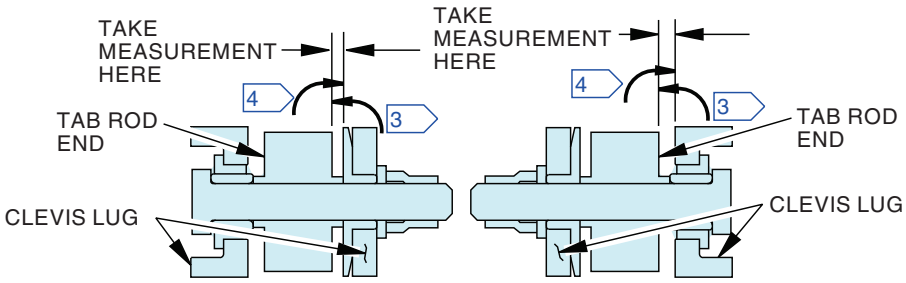
Elevator Tab Pushrods Adjustment
Figure 510/27-31-00-990-816 (Sheet 1 of 2)

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LEFT ELEVATOR TAB PUSHRODS
(RIGHT ELEVATOR TAB PUSHRODS ARE OPPOSITE)

B



A-A

1 TURN THE ELEVATOR TAB ROD ENDS TO MAKE COARSE ADJUSTMENTS OF THE ELEVATOR TAB TRAILING EDGE.

2 TURN THE BUSHING [104] (VERNIER COLLAR) TO MAKE FINE ADJUSTMENTS OF THE ELEVATOR TAB TRAILING EDGE.

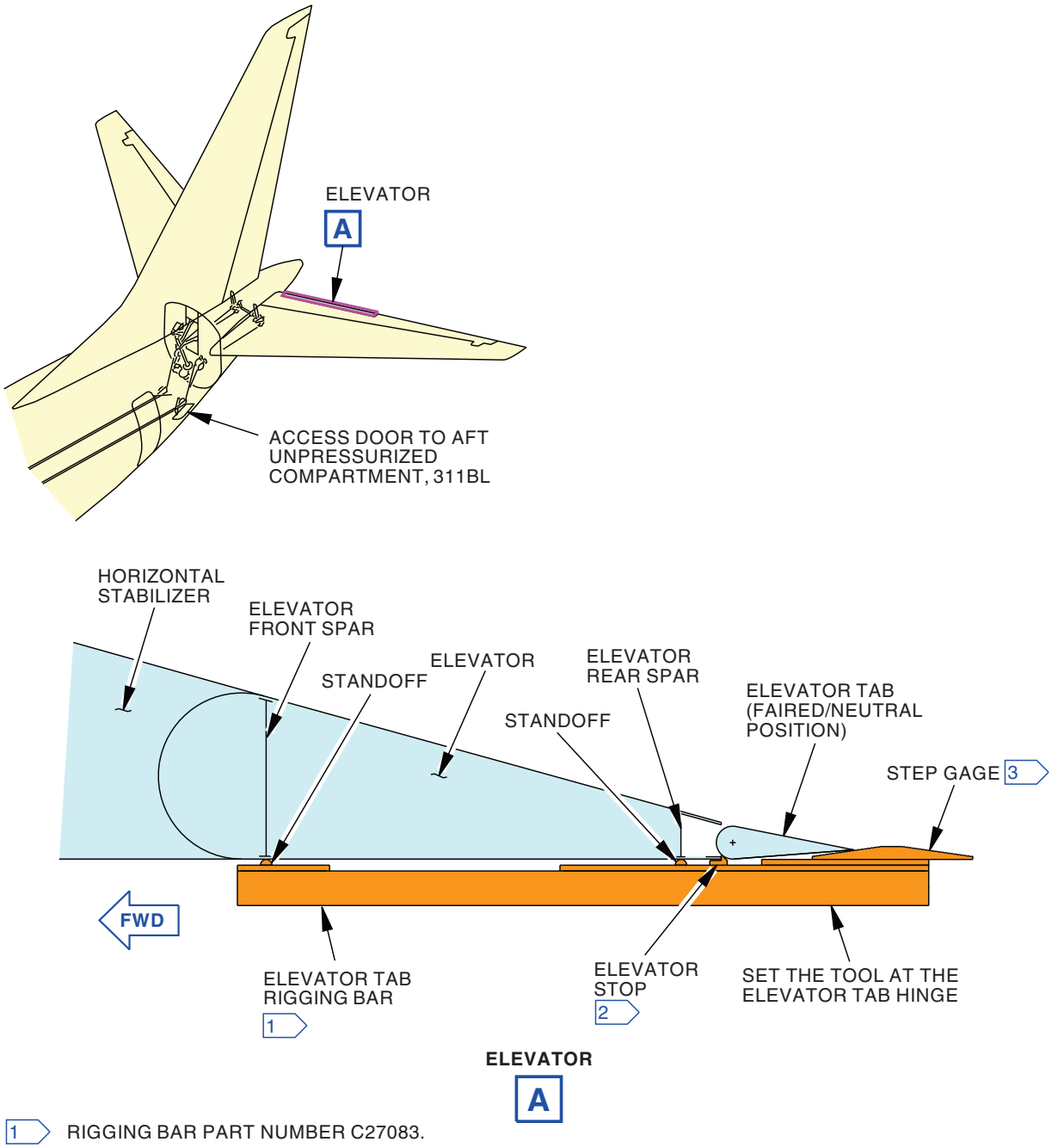
3 ROTATE THE TAB PUSHRODS COUNTERCLOCKWISE UNTIL CONTACT IS MADE. START THE MEASUREMENT.

4 ROTATE THE TAB PUSHRODS CLOCKWISE UNTIL CONTACT IS MADE. END THE MEASUREMENT. THE MINIMUM TRAVEL (WHILE ROTATING EACH PUSHROD FROM CONTACT TO CONTACT) SHOULD BE 0.040 INCH (1.02 mm).

2412470 S00061530733_V2

Elevator Tab Pushrods Adjustment
Figure 510/27-31-00-990-816 (Sheet 2 of 2)

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1 RIGGING BAR PART NUMBER C27083.

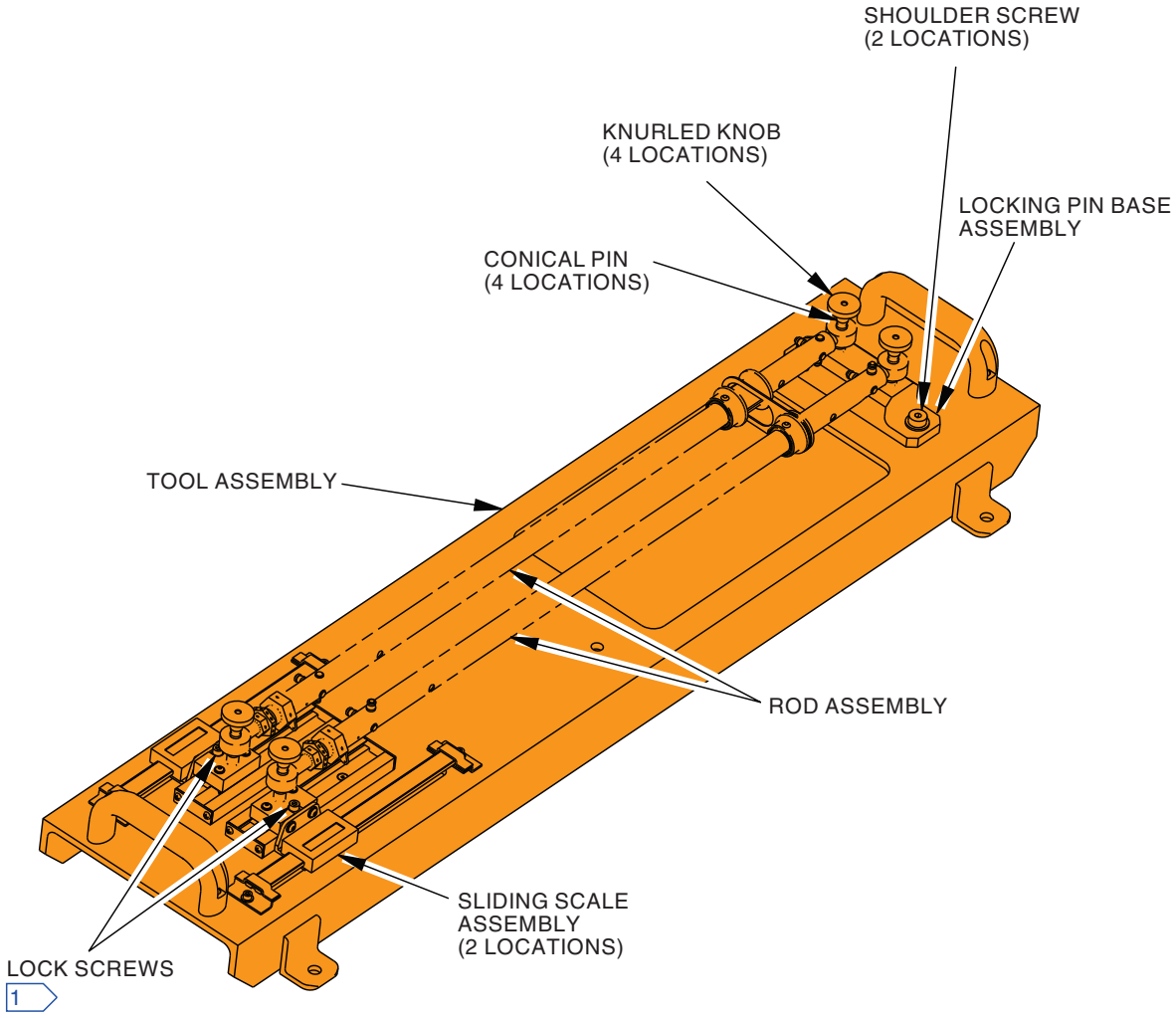
2 PUSH RIGGING BAR FWD UNTIL ELEVATOR STOP INDEXES ON THE AFT EDGE OF ELEVATOR NEAR ELEVATOR REAR SPAR.

3 SLIDE STEP GAGE FWD UNDER ELEVATOR TAB TO OBTAIN READING.

2412471 S00061530734_V3

Elevator Tab Fixture Bar Installation
Figure 511/27-31-00-990-818

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 LOCK SCREW IS USED FOR TOOL TRANSPORT ONLY. REMOVE DURING USE

2412472 S00061530735_V1

Elevator Tab Pushrod Length Adjustment Tool
Figure 512/27-31-00-990-819

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-820-806

12. Input Rod of the Elevator Feel Computer - Adjustment

(Figure 504, Figure 513)

A. General

(1) Use this test to adjust the input rod (stabilizer pushrod) of the elevator feel computer.

B. References

Reference	Title
22-11-31-710-801	Stabilizer Position Sensor Test (P/B 501)
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

D. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Adjustment

SUBTASK 27-31-00-860-066

(1) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-860-065

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-010-018

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-860-068

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677 to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)). (Figure 504)

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677 and not the stabilizer indicator in the flight deck.

G. Input Rod of the Elevator Feel Computer Adjustment

SUBTASK 27-31-00-410-012

(1) Make sure you can install the rig pin E-4 from the rigging pin kit, SPL-1585 in the rig pin holes in the input arm [122] and elevator feel computer [127].

SUBTASK 27-31-00-820-012

(2) If you can not install the rig pin E-4 easily, adjust the input rod [121] of the elevator feel computer [127] (Figure 513):

(a) Remove the nut [125], washer [124], washer [128], and bolt [123] to disconnect the input rod [121] from the input arm [122].

(b) Install the rig pin E-4 in the rig pin holes in the input arm [122] and elevator feel computer [127].

(c) Loosen the checknut [126].

(d) Adjust the input rod [121] until you can install the bolt [123] easily through the input arm [122] and input rod [121].

(e) Install the bolt [123], washer [128], washer [124] and nut [125].

1) Make sure that the recessed side of the washer [128] is oriented towards the head of the bolt [123].

2) Tighten the nut [125] to 63 in-lb (7.1 N·m) to 67 in-lb (7.6 N·m).

(f) Tighten the checknut [126].

1) Tighten the checknut [126] to 63 in-lb (7.1 N·m) to 67 in-lb (7.6 N·m).

(g) Remove the rig pin E-4.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-820-013

(1) Do this task: Stabilizer Position Sensor Test, TASK 22-11-31-710-801.

SUBTASK 27-31-00-410-013

(2) Close this access panel:

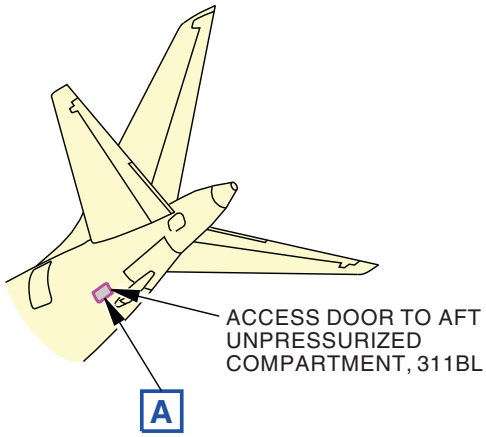
Number	Name/Location
311BL	Stabilizer Trim Access Door

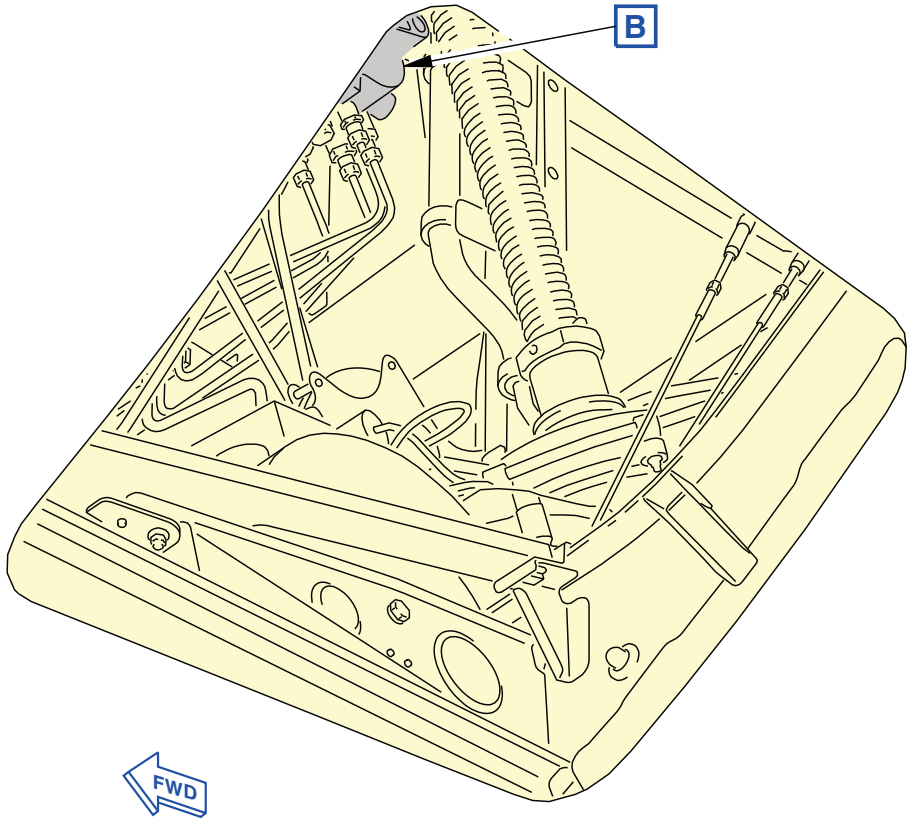
SUBTASK 27-31-00-860-070

(3) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





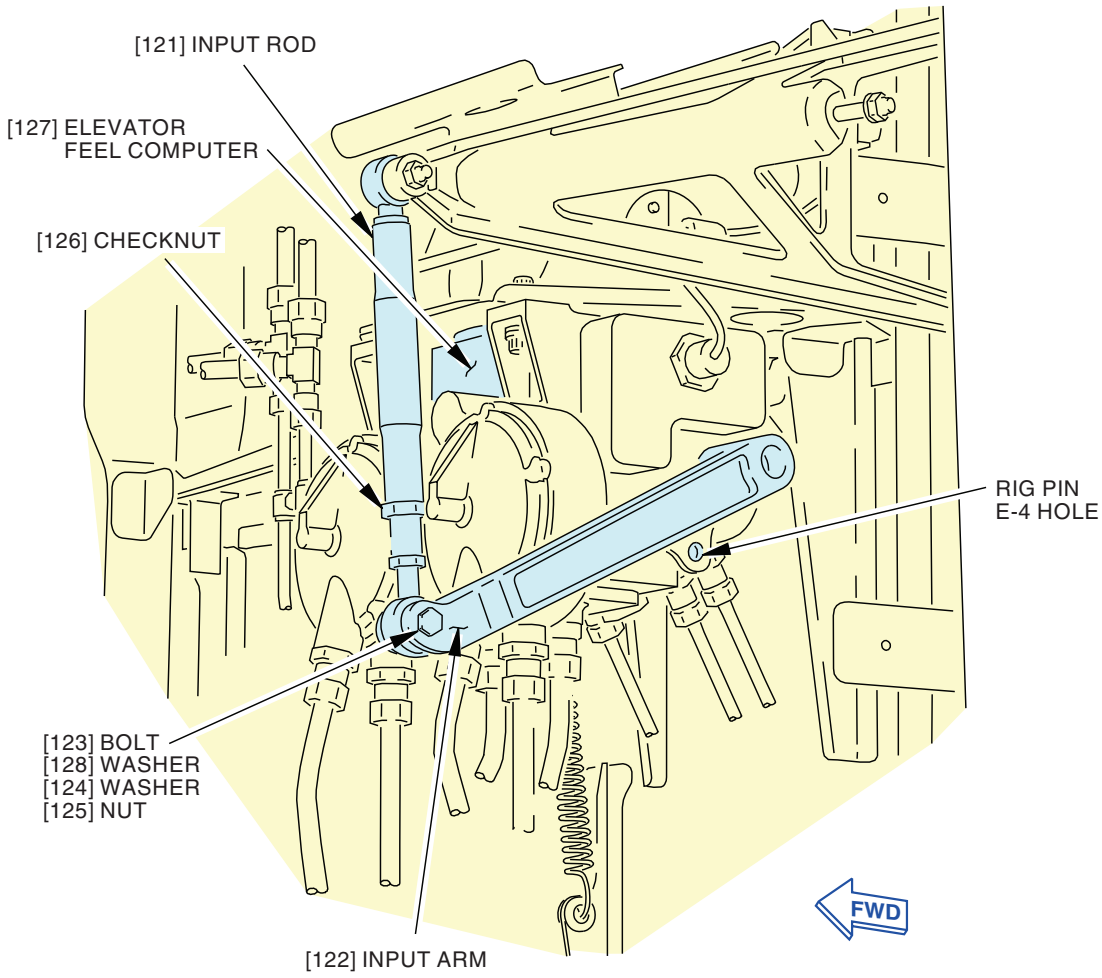
VIEW THROUGH THE ACCESS DOOR TO
AFT UNPRESSURIZED COMPARTMENT, 311BL

A

2412473 S00061530737_V1

Elevator Feel Computer Input Rod Adjustment
Figure 513/27-31-00-990-820 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412474 S00061530738_V2

Elevator Feel Computer Input Rod Adjustment
Figure 513/27-31-00-990-820 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-700-803

13. Elevator Surface Friction and Balance - Test

(Figure 504, Figure 507)

A. General

(1) Use this test to make sure that the elevators are balanced.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
STD-1278	Block - Loading, 1/8 Inch Thick, 3 Inch by 3 Inch Wood or Fiberglass Block with Pad to Prevent Damage to the Skin.

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Test

SUBTASK 27-31-00-860-072

(1) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-860-071

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-010-019

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-860-073

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

SUBTASK 27-31-00-860-146

(5) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-31-00-860-074

(6) Move the control columns in a longitudinal (forward to aft) direction through full travel 20 times.

G. Elevator Surface Friction and Balance Test

SUBTASK 27-31-00-730-002

CAUTION

DO NOT PUT A LOAD OF MORE THAN 65 POUNDS (290 NEWTONS) ON THE ELEVATORS AT ONE TIME. IF THE LOAD IS TOO HEAVY, IT CAN CAUSE DAMAGE TO THE ELEVATORS.

(1) Do the elevator surface friction and balance test:

(a) Hold the captain's control column at the neutral position.

(b) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

(c) Move the elevator with your hand until the right elevator trailing edge aligns with the elevator neutral index mark (Figure 507).

1) Release the elevator trailing edge.

(d) If the right elevator trailing edge moves down from the index mark, then do the elevator balance adjustment (Elevator Balance Procedure/SRM PROCEDURE 51-60-04-0G-0).

(e) If the right elevator trailing edge moves up from the index mark, then do these steps:

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WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Turn on the A and B hydraulic systems to center the elevator, do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.
2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.
3) Hold the captain's control column in the neutral position.
4) Use the block, STD-1278, and force gauge, COM-1557, to measure the minimum force required to move the elevator trailing edge down 0.75 ±0.15 in. (19.05 ±3.81 mm) (1.25 ±0.24 degrees) from the neutral index mark (this force is referred to as the "down force").
   a) Make a record of this down force.
5) Hold the captain' control column in the neutral position.
6) Use the block, STD-1278, and force gauge, COM-1557, to measure the minimum force required to hold the elevator trailing edge at the index mark (which is referred to as the "up force").
   a) Make a record of this up force.
7) Make sure that the difference between the down force and up force is equal to or less than 20 lbf (89 N).
8) If the difference between the down force and up force is greater than 20 lbf (89 N), refer to the IFIM and do the applicable procedure(s).

(f) If the right elevator trailing edge stays at the index mark, then do these steps:

1) Hold the captain's control column in the neutral position.

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

2) Turn on the A and B hydraulic systems to center the elevator, do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.
3) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.
4) Use the force gauge, COM-1557, to measure the force required to move the elevator trailing edge down 0.75 ±0.15 in. (19.05 ±3.81 mm) (1.25 ±0.24 degrees) from the index mark (which is referred to as the "down force").
   a) Make a record of this down force.
5) Use the force gauge, COM-1557, to measure the force required to move the elevator trailing edge up 0.75 ±0.15 in. (19.05 ±3.81 mm) (1.25 ±0.24 degrees) from the index mark (which is referred to as the "up force").
   a) Make a record of this up force.

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6) If the down force is less than the up force, then do a check of the elevator balance adjustment (Elevator Balance Procedure/SRM PROCEDURE 51-60-04-0G-0).

7) Calculate the sum of the down force and up force.

8) Make sure that the sum of the down force and up force is equal to or less than 32 lbf (142 N).

9) If the sum of the down force and up force is greater than 32 lbf (142 N), refer to the IFIM and do the applicable procedure(s).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-075

(1) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-00-410-014

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

TASK 27-31-00-700-804

14. Elevator Balance Panels - Test

(Figure 504)

A. General

(1) Use this test to make sure that the elevator balance panels are installed properly. This task is applicable for the left and right elevator balance panels.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
27-31-41-400-801	Elevator Balance Panel - Installation (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Test

SUBTASK 27-31-00-860-077

(1) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-860-076

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-010-020

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-860-078

(4) Use the stab trim switches on the control wheel to set the stabilizer at approximately 4 units of trim.

SUBTASK 27-31-00-860-079

(5) Move the control columns in a longitudinal direction (forward and aft) through full travel one time.

F. Elevator Balance Panels Test

SUBTASK 27-31-00-200-007

(1) Make sure that the bulb seal gaps in the balance panel bay are 0.02 ±0.01 in. (0.51 ±0.26 mm) (TASK 27-31-41-400-801).

SUBTASK 27-31-00-200-008

(2) Move each elevator (left and right side) by hand through the full travel to make sure that the elevator moves smoothly and freely.

(a) Visually make sure that each of the bulb seals do not bind as the elevator moves through the full travel.

(b) Make sure that the forward hinge seal cloth is not tight at all elevator positions.

SUBTASK 27-31-00-410-015

(3) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-700-001

(4) Do this task: Manual Mode - Test, TASK 27-31-00-700-808.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-080

(1) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

——— END OF TASK ———

TASK 27-31-00-700-805

15. Elevator Travel Limit - Test

(Figure 504, Figure 520)

A. General

(1) Use this test to make sure that the elevator operates correctly through the full travel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-31-00-860-082

(1) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-860-081

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-010-021

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-860-083

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677 to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (995 ±0 mm)) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

G. Elevator Travel Limit Test

SUBTASK 27-31-00-730-003

CAUTION

DO NOT PUT A LOAD OF MORE THAN 65 POUNDS (290 NEWTONS) ON THE ELEVATORS AT ONE TIME. IF THE LOAD IS TOO HEAVY, IT CAN CAUSE DAMAGE TO THE ELEVATORS.

(1) Do the elevator travel limit test with hydraulics off.

(a) Check for binding at the left and right elevators:

1) Use your hand to move the right elevator through the full travel.

a) Make sure that there is no binding through the full travel.

2) Use your hand to move the left elevator through the full travel.

a) Make sure that there is no binding through the full travel.

(b) Move and hold the captain's control column aft until it touches the stops.

1) Make sure that the right elevator moves 12.94 in. (329 mm) - 14.06 in. (357 mm) up from the index mark (Figure 520).

(c) Move and hold the captain's control column forward until it touches the stops.

1) Make sure that the right elevator moves 9.69 in. (246 mm) - 10.56 in. (268 mm) down from the index mark.

(d) Move and hold the captain's control column aft until it touches the stops.

1) Make sure that the left elevator moves 12.94 in. (329 mm) - 14.06 in. (357 mm) up from the index mark (Figure 520).

(e) Move and hold the captain's control column forward until it touches the stops.

1) Make sure that the left elevator moves 9.69 in. (246 mm) - 10.56 in. (268 mm) down from the index mark (Figure 520).

(f) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-00-730-004

(2) Do the elevator travel limit test with hydraulics on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

(b) Move and hold the captain's control column aft until it touches the stops.

1) Make sure that the right elevator moves up 12.94 in. (329 mm) - 14.06 in. (357 mm) from the index mark (Figure 520).

(c) Move and hold the captain's control column forward until it touches the stops.

1) Make sure that the right elevator moves down 9.69 in. (246 mm) - 10.56 in. (268 mm) from the index mark (Figure 520).

(d) Set the FLT CONTROL A switch on the forward overhead panel, P5, to the OFF position.

(e) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

(f) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

(g) Move and hold the captain's control column aft until it touches the stops.

1) Make sure that the right elevator moves up 12.94 in. (329 mm) - 14.06 in. (357 mm) from the index mark (Figure 520).

(h) Move and hold the captain's control column forward until it touches the stops.

1) Make sure that the right elevator moves down 9.69 in. (246 mm) - 10.56 in. (268 mm) from the index mark (Figure 520).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-084

(1) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

SUBTASK 27-31-00-410-016

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

TASK 27-31-00-700-806

16. Elevator Output Torque Tube Buss Linkage Test

(Figure 504)

A. General

(1) Use this test to do a check of the elevator output torque tube buss linkage.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
STD-1278	Block - Loading, 1/8 Inch Thick, 3 Inch by 3 Inch Wood or Fiberglass Block with Pad to Prevent Damage to the Skin.

D. Consumable Materials

Reference	Description	Specification
G50314	Tape - Masking	BAC5034-4 Type VII Class 2

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Prepare for the Test

SUBTASK 27-31-00-860-167

(1) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-860-085

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize the elevator hydraulic systems A and B (TASK 27-31-00-800-801).

SUBTASK 27-31-00-010-022

(3) Open this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-860-087

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

SUBTASK 27-31-00-860-088

(5) Operate the elevator in the full travel 6 times.

SUBTASK 27-31-00-860-089

(6) Make sure that the control column is in the neutral position.

H. Elevator Output Torque Tube Buss Linkage Test

SUBTASK 27-31-00-730-005

CAUTION DO NOT PUT A LOAD OF MORE THAN 65 POUNDS (290 NEWTONS) ON THE ELEVATORS AT ONE TIME. IF THE LOAD IS TOO HEAVY, IT CAN CAUSE DAMAGE TO THE ELEVATORS.

(1) Do a test of the elevator output torque tube:

(a) Set the FLT CONTROL A switch on the forward overhead panel, P5, to the OFF position.

(b) Make sure that the FLT CONTROL B switch on the forward overhead panel, P5, is in the ON position.

(c) Use a block, STD-1278, and force gauge, COM-1557, on the top side of the right elevator trailing edge, to apply a downward force of 17.5 ±2.5 lbf (77.84 ±11.12 N) to the right elevator trailing edge.

1) Slowly release the force.

2) Make a mark of the elevator position on the tailcone after you remove the force.

NOTE: You can use masking tape, G50314, to make the mark.

(d) Use a block, STD-1278, and force gauge, COM-1557, on the bottom side of the right elevator trailing edge, to apply an upward force of 10 ±1 lbf (44 ±5 N) to the right elevator trailing edge.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Hold the elevator in this position with the force you have applied.
2) Make a mark of the elevator position on the tailcone before you remove the force.
   NOTE: You can use masking tape, G50314, to make the mark.
3) Slowly remove the force.
   a) Make sure that the distance between the up and down position of the right elevator is 0.21 in. (5.33 mm) or less.
   b) Remove the mark on the tailcone.

(e) Set the FLT CONTROL A switch on the forward overhead panel, P5, to the ON position.

(f) Set the FLT CONTROL B switch on the forward overhead panel, P5, to the OFF position.

(g) Use a block, STD-1278, and force gauge, COM-1557, on the top side of the left elevator trailing edge, to apply a downward force of 17.5 ±2.5 lbf (77.84 ±11.12 N) to the left elevator trailing edge.
1) Slowly release the force.
2) Make a mark of the elevator position on the tailcone after you remove the force.
   NOTE: You can use masking tape, G50314, to make the mark.

(h) Use a block, STD-1278, and force gauge, COM-1557, on the bottom side of the left elevator trailing edge, to apply an upward force of 10 ±1 lbf (44 ±5 N) to the left elevator trailing edge.
1) Hold the elevator in this position with the force you have applied.
2) Make a mark of the elevator position on the tailcone before you remove the force.
   NOTE: You can use masking tape, G50314, to make the mark.
3) Slowly remove the force.
   a) Make sure that the distance between the up and down position of the left elevator is 0.21 in. (5.33 mm) or less.
   b) Remove the mark on the tailcone.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-090

(1) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

SUBTASK 27-31-00-410-017

(2) Close this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

TASK 27-31-00-700-807

17. Elevator Tab Linkage - Test

(Figure 504, Figure 507, Figure 511, Figure 520)

A. General

(1) Use this test to do a check of the elevator tab control system.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-10345	Rigging Bar - Elevator Tab Part #: C27083-18 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-31-00-860-091

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize elevator hydraulic systems A and B, do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-31-00-860-092

(2) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-760-002

(3) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC
E	6	C01205	SMYD-1 SNSR EXC AC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	4	C01207	SMYD-2 SNSR EXC AC
B	5	C01206	SMYD-2 CMPTR DC
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

SUBTASK 27-31-00-010-023

(4) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-860-093

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(5) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

SUBTASK 27-31-00-860-094

(6) Make sure that the control column is in the neutral position.

G. Elevator Tab Linkage Test

SUBTASK 27-31-00-730-006

CAUTION

DO NOT PUT A LOAD OF MORE THAN 65 POUNDS (290 NEWTONS) ON THE ELEVATORS AT ONE TIME. IF THE LOAD IS TOO HEAVY, IT CAN CAUSE DAMAGE TO THE ELEVATORS.

(1) Do the elevator tab linkage test:

(a) Cycle the control column forward and aft to center the system.

(b) Make sure that the flaps are fully retracted.

1) If necessary, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

(c) Move the left elevator tab up and down by hand to determine if the elevator tab is loose.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the looseness is not greater than 0.03 in. (0.76 mm) total movement at the trailing edge of the left elevator tab.

(d) Move the right elevator tab up and down by hand to determine if the elevator tab is loose.

1) Make sure that the looseness is not greater than 0.03 in. (0.76 mm) total movement at the trailing edge of the right elevator tab.

(e) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

(f) Move the right elevator by hand to align it with the index mark in ±0.06 in. (1.52 mm) (Figure 507).

(g) Install the rigging bar, SPL-10345, below the elevator and elevator tab at the third elevator tab hinge of the right elevator (from inboard) (Figure 511).

1) Make sure that the stand offs for the rigging bar, SPL-10345, rest on the skin fastener heads to prevent skin variation.

2) Make sure that the cross bar on the rigging bar, SPL-10345, touches the lower surface of the elevator trailing edge.

3) Push the rigging bar, SPL-10345, FWD until elevator stop indexes on the AFT edge of elevator near elevator rear spar.

(h) Make sure that the distance measured between the right elevator tab faired/neutral position and rigging bar, SPL-10345, is within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm) (Figure 511).

(i) Move the left elevator by hand to align it with the index mark in ±0.06 in. (1.52 mm).

(j) Install the rigging bar, SPL-10345, below the elevator and elevator tab at the third elevator tab hinge of the left elevator (from inboard) (Figure 511).

1) Make sure that the stand offs for the rigging bar, SPL-10345, rest on the skin fastener heads to prevent skin variation.

2) Make sure that the cross bar on the rigging bar, SPL-10345, touches the lower surface of the elevator trailing edge.

3) Push the rigging bar, SPL-10345, FWD until elevator stop indexes on the AFT edge of elevator near elevator rear spar.

(k) Make sure that the distance measured between the left elevator tab faired/neutral position and rigging bar, SPL-10345, is within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm) (Figure 511).

(l) Remove the rigging bar, SPL-10345.

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(m) Pressurize elevator hydraulic systems A and B, do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

(n) Lightly move the control column forward and aft a small amount to make sure that the system is in the neutral position.

(o) Make sure that the right elevator is aligned with the index mark in ±0.06 in. (1.52 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(p) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the right elevator to the bottom edge of the elevator tab (this is referred to as the "neutral tab position").

1) Make a record of this neutral tab position.

(q) Move and hold the control column full aft.

1) Use a scale to make sure that the trailing edge of the right elevator is in 12.94 in. (329 mm) - 14.06 in. (357 mm) of the index mark (Figure 520).

2) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the right elevator to the bottom edge of the elevator tab (this is referred to as the "down tab position").

a) Make a record of this down tab position.

b) Release the control column.

(r) Move and hold the control column full forward.

1) Use a scale to make sure that the trailing edge of the right elevator is in 9.69 in. (246 mm) - 10.56 in. (268 mm) of the index mark.

2) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the right elevator to the bottom edge of the elevator tab (this is referred to as the "up tab position").

a) Make a record of this up tab position.

b) Release the control column.

(s) Add the neutral tab position to the down tab position.

1) Make sure that the sum is 1.06 in. (26.92 mm) - 1.40 in. (35.56 mm).

(t) Subtract the neutral tab position from up tab position.

1) Make sure that the difference is between 0.80 in. (20.32 mm) and 1.08 in. (27.43 mm).

(u) Lightly move the control column forward and aft a small amount to make sure that the system is in the neutral position.

(v) Make sure that the left elevator is aligned with the index mark in ±0.06 in. (1.52 mm).

(w) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the left elevator to the bottom edge of the elevator tab (this is referred to as the "neutral tab position").

1) Make a record of this neutral tab position.

(x) Move and hold the control column full aft.

1) Use a scale to make sure that the trailing edge of the left elevator is in 12.94 in. (329 mm) - 14.06 in. (357 mm) of the index mark.

2) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the left elevator to the bottom edge of the elevator tab (this is referred to as the "down tab position").

a) Make a record of this down tab position.

b) Release the control column.

(y) Move and hold the control column full forward.

1) Use a scale to make sure that the trailing edge of the left elevator is in 9.69 in. (246 mm) - 10.56 in. (268 mm) of the index mark.

2) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the left elevator to the bottom edge of the elevator tab (this is referred to as the "up tab position").

a) Make a record of this up tab position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) Release the control column.

(z) Add the neutral tab position to the down tab position.

1) Make sure that the sum is 1.06 in. (26.92 mm) - 1.40 in. (35.56 mm).

(aa) Subtract the neutral tab position from the up tab position.

1) Make sure that the difference is 0.80 in. (20.32 mm) - 1.08 in. (27.43 mm).

(ab) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	8	C01464	FLIGHT CONTROL ELEV TAB VLV RIGHT

(ac) Move the flap control lever to the 5-unit position.

(ad) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	8	C01464	FLIGHT CONTROL ELEV TAB VLV RIGHT

1) Make sure that the right tab begins to move in 10 ±2 seconds.

(ae) Lightly move the control column forward and aft a small amount to make sure that the system is in the neutral position.

(af) Make sure that the right elevator is aligned with the index mark in ±0.06 in. (1.52 mm).

1) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the right elevator to the bottom edge of the elevator tab (this is referred to as the "neutral tab position).

a) Make a record of this neutral tab position.

(ag) Move and hold the control column full aft.

1) Use a scale to make sure that the trailing edge of the right elevator is in 12.94 in. (329 mm) - 14.06 in. (357 mm) of the index mark.

2) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the right elevator to the bottom edge of the elevator tab (this is referred to as the "up tab position").

a) Make a record of this up tab position.

b) Release the control column.

(ah) Move and hold the control column full forward.

1) Use a scale to make sure that the trailing edge of the right elevator is in 9.69 in. (246 mm) - 10.56 in. (268 mm) of the index mark.

2) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the right elevator to the bottom edge of the elevator tab (this is referred to as the "down tab position").

a) Make a record of this down tab position.

b) Release the control column.

(ai) Add the neutral tab position to the up tab position.

1) Make sure that the sum is between 0.8 in. (20.32 mm) - 1.2 in. (30.48 mm).

(aj) Subtract the neutral tab position from the down tab position.

1) Make sure that the difference is between 0.6 in. (15.24 mm) - 0.9 in. (22.86 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(ak) Move the left elevator by hand to align it with the index mark in ±0.06 in. (1.52 mm).

1) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the left elevator to the bottom edge of the elevator tab (this is referred to as the "neutral tab position").

a) Make a record of this neutral tab position.

(al) Move and hold the control column full aft.

1) Use a scale to make sure that the left elevator is in 12.94 in. (329 mm) - 14.06 in. (357 mm) of the index mark.

(am) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the left elevator to the bottom edge of the elevator tab (this is referred to as the "up tab position").

1) Make a record of this up tab position.

(an) Move and hold the control column full forward.

1) Use a scale to make sure that the trailing edge of the left elevator is in 9.69 in. (246 mm) - 10.56 in. (268 mm) of the index mark.

2) Use a 6 in. (152.40 mm) scale to measure from the bottom edge of the left elevator to the bottom edge of the elevator tab (this is referred to as the "down tab position").

a) Make a record of this down tab position.

b) Release the control column.

(ao) Add the neutral tab position to the up tab position.

1) Make sure that the sum is between 0.8 in. (20.32 mm) - 1.2 in. (30.48 mm).

(ap) Subtract the neutral tab position from the down tab position.

1) Make sure that the difference is between 0.6 in. (15.24 mm) - 0.9 in. (22.86 mm).

(aq) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-095

(1) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

SUBTASK 27-31-00-410-018

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

TASK 27-31-00-700-808

18. Manual Mode - Test

(Figure 504)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this test to make sure the control columns and control wheels operate freely.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-31-00-860-096

(1) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-040-002

(2) Do the following to isolate the spoilers to keep them from moving when you turn the control wheel:

(a) Turn the A and B Spoiler Shutoff Switches on the P5-3 panel to the OFF position.

SUBTASK 27-31-00-010-024

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

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SUBTASK 27-31-00-860-097

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677 to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)). (Figure 504)

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677 and not the stabilizer indicator in the flight deck.

G. Manual Mode Test

SUBTASK 27-31-00-730-007

(1) Do the manual mode test:

(a) Make sure the control columns and control wheels move freely:

1) Move the control column in a longitudinal direction (forward and aft), and while moving the control column forward and aft, rotate the control wheel between the left and right stops.

a) Repeat this step nine more times.

2) Rotate and hold the control wheel against the left stop and move the control column in a longitudinal direction (forward and aft) between the stops.

3) Rotate and hold the control wheel against the right stop and move the control column in a longitudinal direction (forward and aft) between the stops.

4) Hold the control column full forward and rotate the control wheel left and right between the stops.

5) Hold the control column full aft and rotate the control wheel left and right between the stops.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-098

(1) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-00-040-003

(2) If necessary, turn the A and B Spoiler Shutoff Switches on the P5-3 panel to the ON position.

SUBTASK 27-31-00-410-019

(3) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-700-809

19. Control Column Travel and Centering - Test

(Figure 504, Figure 507, Figure 514, and Figure 520)

A. General

(1) Use this test to make sure that the control columns can move through their full travel. This also includes a centering test of the control columns.

(2) Use this test to verify two separate requirements:

(a) Minimum forward and aft travel.

(b) Total stop to stop travel.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1674	Assembly - Adapter, Control Wheel, Torque and Force Test Part #: C27060-22 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205 Opt Part #: F72867-1 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

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E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-31-00-860-100

(1) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-860-099

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-010-025

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-860-101

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim to 39.17 ±0.01 in. (994.92 ±0.26 mm) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

G. Control Column Travel and Centering Test

SUBTASK 27-31-00-730-008

(1) Do the control column travel test (Figure 514).

(a) Attach the adapter, SPL-1674, and protractor, SPL-1680, to the captain's control column (Figure 514).

(b) Move the captain's control column full aft until it touches the stop and hold it in that position.

NOTE: This measures the minimum aft travel.

1) Make sure that the right elevator trailing edge moves to 11.82 in. (300.23 mm) - 13.83 in. (351.28 mm) up from the index mark (Figure 520).

a) Make sure that the position of the captain's control column is at a minimum of 14 degrees aft of the column neutral rig position (Figure 514).

(c) Move the captain's control column full forward until it touches the stop and hold it in that position.

NOTE: This measures the minimum forward travel.

1) Make sure that the right elevator trailing edge moves to 9.15 in. (232.41 mm) - 10.44 in. (265.18 mm) down from the index mark (Figure 520).

2) Make sure that the position of the captain's control column is at a minimum of 12.8 degrees forward of the column neutral rig position.

a) Release the control column.

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(d) Move the captain's control column full forward and note the degree forward measurement.

NOTE: Enough force is required to stretch the control cables allowing the column to hit the forward stop.

(e) Move the captain's control column full aft and note the degree aft measurement.

NOTE: Enough force is required to stretch the control cables allowing the column to hit the aft stop.

1) Release the control column.

2) Make sure that the total stop to stop travel of the captain's control column is a minimum of 28.4 degrees (Figure 514).

NOTE: This measures the total stop to stop travel.

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(f) Do this task : Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

(g) Move the captain's control column lightly forward and aft to make sure that it is in the neutral position.

1) Make sure that the position of the captain's control column is in 6.1 - 7.7 degrees of the vertical position (90 degrees).

(h) Remove the adapter, SPL-1674, and protractor, SPL-1680, from the captain's control column.

(i) Attach the adapter, SPL-1674, and protractor, SPL-1680, to the first officer's control column (Figure 514).

(j) Move the first officer's control column lightly forward and aft to make sure that it is in the neutral position.

1) Make sure that the position of the first officer's control column is in 6.1 - 7.7 degrees of the vertical position.

(k) Remove the adapter, SPL-1674, and protractor, SPL-1680, from the first officer's control column.

SUBTASK 27-31-00-730-009

(2) Do the elevator centering test.

(a) Set the FLT CONTROL B switch, on the P5 forward overhead panel, to the OFF position.

(b) Make sure that the FLT CONTROL A switch, on the P5 forward overhead panel, is in the ON position.

(c) Make sure that the captain's control column is at the neutral position.

1) Make sure that the right elevator aligns with the index mark in ±0.06 in. (1.52 mm) (Figure 507).

(d) Set the FLT CONTROL B switch to the ON position.

1) Make sure that the right elevator aligns with the index mark in ±0.06 in. (1.52 mm).

(e) Set the FLT CONTROL A switch to the OFF position.

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(f) Make sure that the FLT CONTROL B switch is in the ON position.

1) Make sure that the right elevator aligns with the index mark in ±0.06 in. (1.52 mm).

(g) Set the FLT CONTROL A switch to the ON then to the OFF position.

1) Make sure that there is less than 0.02 in. (0.51 mm) movement at the right elevator trailing edge.

(h) Set the FLT CONTROL B switch to the OFF then to the ON then to the OFF position.

1) Make sure that there is less than 0.02 in. (0.51 mm) movement at the right elevator trailing edge.

(i) Set the FLT CONTROL A and B switches to the ON position.

(j) Make sure that the captain's control column is at the neutral position.

1) Make sure that the right elevator aligns with the index mark in ±0.06 in. (1.52 mm).

(k) Move the captain's control column forward approximately 2 degrees and release.

1) Make sure that the right elevator trailing edge aligns with the index mark in 0.00 ±0.12 in. (0.0 ±3.0 mm).

(l) Move the captain's control column aft approximately 2 degrees and release.

1) Make sure that the right elevator trailing edge aligns with the index mark in 0.00 ±0.12 in. (0.0 ±3.0 mm).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-102

(1) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

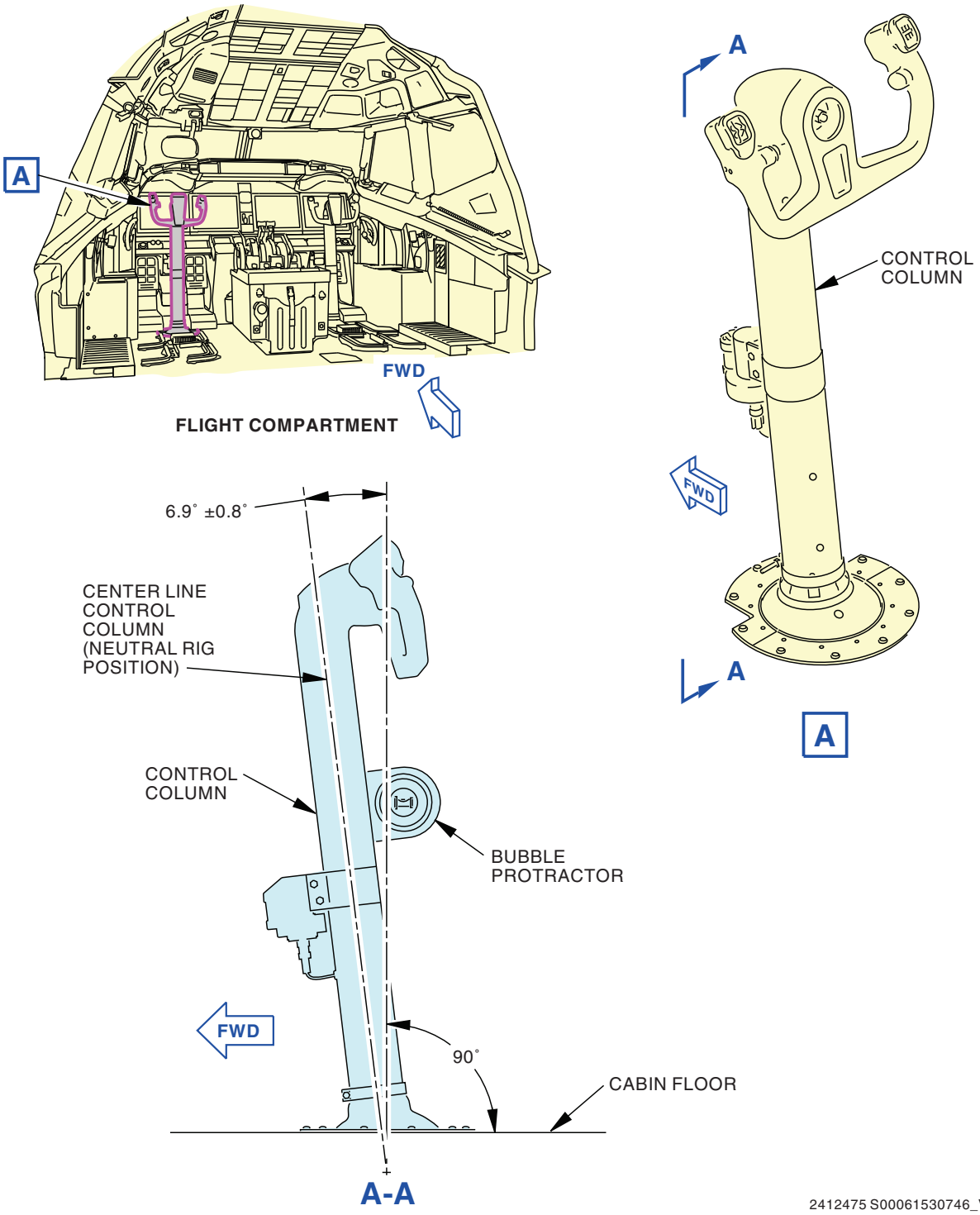
SUBTASK 27-31-00-410-020

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

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Elevator Control Column Protractor Installation
Figure 514/27-31-00-990-821

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-00-700-810

20. Control Column Power Force - Test

A. General

(1) Use this test to make sure that the control columns move with the correct input force.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
SPL-1674	Assembly - Adapter, Control Wheel, Torque and Force Test Part #: C27060-22 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205 Opt Part #: F72867-1 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205

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D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-31-00-860-104

(1) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-860-103

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-00-010-026

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-860-105

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

G. Control Column Power Force Test

SUBTASK 27-31-00-730-010

(1) Do the control column force test (with hydraulics unpressurized):

(a) Move the captain's control column slowly aft and hold at the elevator trailing edge up position of 3.59 in. (91.19 mm) - 3.69 in. (93.73 mm) (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force to hold the control column is between 11 lbf (48.93 N) - 53 lbf (235.76 N) (Figure 514).

2) Center the control column.

(b) Move the captain's control column slowly forward and hold at the elevator trailing edge down position of 3.64 in. (92.46 mm) - 3.74 in. (95.00 mm) (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force to hold the control column is between 19 lbf (84.52 N) - 63 lbf (280.24 N) (Figure 514).

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SUBTASK 27-31-00-730-011

(2) Do the control column force test (with hydraulics pressurized):

(a) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Pressurize elevator hydraulic systems A and B (TASK 27-31-00-800-801).

(c) Move the captain's control column lightly forward and aft to put the system in the neutral position.

1) Make sure that the position of the captain’s control column is in 6.1 - 7.7 degrees of the vertical position (90 degrees) (Figure 516, Figure 514).

2) Attach "DO-NOT-MOVE" tags on the control columns.

(d) Attach approximately 36 in. (914.40 mm) of masking tape to the tailcone, forward of the right elevator index plate (Figure 520).

1) Make sure that there is approximately 18 in. (457.20 mm) of masking tape above the index plate.

2) Make sure that there is approximately 18 in. (457.20 mm) of masking tape below the index plate.

(e) Make a mark of the neutral position of the elevator trailing edge on the masking tape.

1) Use this mark of the neutral position as reference.

(f) Make marks of the right elevator positions on the masking tape (from the two tables below).

(g) Move the captain's control column slowly aft with the force gauge, COM-1557.

1) Hold the captain's control column at these elevator positions (that have been marked on the masking tape of the right elevator) and make sure that the forces to move the control column are in these limits:

NOTE: The "breakout" is when the elevator starts to move.

2) Put a straight edge on the top of the trailing edge surface of the elevator surface to measure the elevator position at the surface of the tailcone.

Table 502/27-31-00-993-811

RIGHT ELEVATOR POSITION (T.E. UP)	REQUIRED COLUMN FORCE (AFT)
BREAKOUT	4.0 lbf (17.79 N) -15.0 lbf (66.72 N)
1.9 in. (48.26 mm) - 2.0 in. (50.80 mm)	13.0 lbf (57.83 N) - 26.0 lbf (115.65 N)
5.05 in. (128.27 mm) - 5.15 in. (130.81 mm)	22.0 lbf (97.86 N) - 33.0 lbf (146.79 N)
8.9 in. (226.06 mm) -9.0 in. (228.60 mm)	29.0 lbf (129.00 N) -42.0 lbf (186.83 N)

(h) Move the captain's control column slowly forward with the force gauge, COM-1557.

1) Hold the captain's control column at these elevator positions and make sure that the forces to move the control column are in these limits:

NOTE: The "breakout" is when the elevator starts to move.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 503/27-31-00-993-803

RIGHT ELEVATOR POSITION (T.E. DOWN)	REQUIRED COLUMN FORCE (FORWARD)
BREAKOUT	4.0 lbf (17.79 N) - 18.0 lbf (80.07 N)
1.9 in. (48.26 mm) - 2.0 in. (50.80 mm)	15.0 lbf (66.72 N) - 28.0 lbf (124.55 N)
5.05 in. (128.27 mm) - 5.15 in. (130.81 mm)	27.5 lbf (122.33 N) - 42.0 lbf (186.83 N)
8.9 in. (226.06 mm) -9.0 in. (228.60 mm)	32.0 lbf (142.34 N) - 47.0 lbf (209.07 N)

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-080-006

(1) Remove the adapter, SPL-1674, and protractor, SPL-1680, from the captain's control column.

SUBTASK 27-31-00-410-021

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

TASK 27-31-00-700-811

21. Neutral Shift - Test

(Figure 504, Figure 507, Figure 520)

A. General

(1) Use this test to make sure that the elevator neutral shift mechanism moves the elevator when the stabilizer moves. When the stabilizer moves, it gives a mechanical input to the elevator PCUs.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left

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(Continued)

Zone	Area
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-31-00-860-107

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize elevator hydraulic systems A and B, do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-31-00-860-108

(2) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-010-027

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-860-109

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677, to set the "B" dimension at 21.32 ±0.10 in. (541.53 ±2.54 mm) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

G. Neutral Shift Test

SUBTASK 27-31-00-730-012

(1) Do the neutral shift test:

(a) Make sure that the right elevator trailing edge moves up 13.9 in. (353.1 mm) - 14.9 in. (378.5 mm) from the index mark (Figure 520).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(b) Use the trammel bar, SPL-1677, to set the "B" dimension at 45.84 ±0.10 in. (1164.34 ±2.54 mm).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

(c) Move the captain's control column aft and forward lightly to make sure that it is in the neutral position.

(d) Make sure that the right elevator trailing edge moves down 3.2 in. (81.3 mm) - 3.8 in. (96.5 mm) from the index mark.

SUBTASK 27-31-00-700-002

(2) Do the following steps to check the elevator system:

NOTE: You need to use the Control Display Unit (CDU) to do the following Built-In-Test Equipment (BITE) test.

(a) Push the INIT REF key on the CDU keyboard.

1) Push the Line Select Key (LSK) that is adjacent to each selection:

a) INDEX

b) MAINT

c) DFCS

d) EXTENDED MAINTENANCE

e) RIGGING

f) ELEVATOR

NOTE: After you make this selection, the CONTINUE will show. Do the instructions that show on the CDU display before you push the LSK that is adjacent to CONTINUE.

g) COMPLETE ELEV RIG

NOTE: This selection contains the ELEV RIG test.

h) During this test, do the instructions that shown on the CDU display.

i) After you completed the instructions that shown on the CDU display, push the LSK that is adjacent to CONTINUE.

2) Do the Elevator Rigging Check:

a) Make sure that the values for the A and B sides are between the limits that shown on the CDU display or in this task.

Table 504/27-31-00-993-804

Test		Limits:
51.01	STAB POS (VAC)	-0.07 to 0.07

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

b) Use the trammel trammel bar, SPL-1677, to measure the B dimension.

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

c) Make sure that the B dimension is 39.17 ±0.01 in. (994.92 ±0.26 mm).

Table 505/27-31-00-993-805

Test		Limits
51.02	MT ACT POS (VAC)	-8.70 to -6.35
Note: No value for B side; N/A		
51.03	MT ACT POS (VAC)	-0.20 TO 0.20
Note: No value for B side; N/A		
51.04	NSS POS (VAC)	-1.77 to -1.67

d) Make sure that the elevator trailing edge aligns with the elevator index on the tail cone at 0.00 ±0.06 in. (0.00 ±1.52 mm).

Table 506/27-31-00-993-806

Test		Limits
51.05	ELEV POS (VAC)	-0.55 to -0.45
51.06	ELEV LVDT (VAC)	-2.05 to -1.95
51.07	ELEV POS (VAC)	-0.55 TO -0.45

e) Make sure that the elevator trailing edge aligns with the elevator index on the tail cone at 0.00 ±0.06 in. (0.00 ±1.52 mm).

Table 507/27-31-00-993-807

Test		Limits
51.08	ELEV POS (VAC)	-0.55 to -0.45
51.09	ELEV POS (VAC)	-0.55 to -0.45
51.10	ELEV POS (VAC)	-0.55 to -0.45
51.11	ELEV POS (VAC)	-0.55 to -0.45

f) Set the B dimension to 28.27 ±0.01 in. (71.81 ±0.03 cm).

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

g) Use the trammel trammel bar, SPL-1677, to measure the B dimension.

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

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h) Make sure that the B dimension is at 28.27 ±0.01 in. (71.81 ±0.03 cm).

Table 508/27-31-00-993-808

Test		Limits
51.12	STAB POS (VAC)	4.47 to 5.29
51.13	ELEV LVDT (VAC)	-0.06 to 0.06
51.14	ELEV POS (VAC)	-0.10 to 0.10
51.15	NSS POS (VAC)	-0.20 to 0.20

3) Do these steps to get out of the BITE test:

NOTE: If there are instructions on the CDU display, it is necessary to do the instructions that show on the CDU display.

a) Push the LSK adjacent to PREV MENU on test 51.16 screen.

b) Push the LSK adjacent to PREV MENU on the ELEVATOR RIGGING screen.

c) Push the LSK adjacent to CONTINUE on the RIGGING INDEX screen.

d) Push the LSK adjacent to EXIT on the RIGGING INDEX screen.

e) Push the LSK adjacent to CONTINUE on the END OF BITE RIGGING screen.

f) Push the LSK adjacent to EXIT until you get out of the BITE test.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-110

(1) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

SUBTASK 27-31-00-410-022

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

TASK 27-31-00-700-812

22. Feel Differential Pressure Light - Test

A. General

(1) Use this test to make sure the FEEL DIFF PRESS light operates correctly.

B. References

Reference	Title
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

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(Continued)

Zone	Area
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Prepare for the Test

SUBTASK 27-31-00-860-111

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT

(1) Pressurize elevator hydraulic systems A and B. To pressurize them, do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-31-00-010-028

(2) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Feel Differential Pressure Light Test

SUBTASK 27-31-00-860-112

(1) Set the FLT CONTROL B switch on the forward overhead panel, P5, to the OFF position.

(a) Make sure the FEEL DIFF PRESS light on the forward overhead panel, P5, illuminates after a minimum of 35 seconds.

NOTE: The time required for the system to bleed down will vary. It may take up to 120 seconds for the FEEL DIFF PRESS light to illuminate.

SUBTASK 27-31-00-860-113

(2) Set the FLT CONTROL A switch to the OFF position.

(a) Make sure the FEEL DIFF PRESS light goes off.

SUBTASK 27-31-00-860-114

(3) Set the FLT CONTROL B switch to the ON position.

(a) Wait for a minimum of 35 seconds.

(b) Make sure the FEEL DIFF PRESS light comes on.

NOTE: The time required for the system to bleed down will vary. It may take up to 120 seconds for the FEEL DIFF PRESS light to illuminate.

SUBTASK 27-31-00-860-115

(4) Set the FLT CONTROL A switch to the ON position.

(a) Make sure the FEEL DIFF PRESS light goes off.

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G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-116

(1) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

SUBTASK 27-31-00-410-023

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

TASK 27-31-00-790-801

23. Pitot System Leak - Test

(Figure 515)

A. General

(1) Use this test to do a leak check of the pitot system.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1914	Test Set - Air Data Model FLMTS (Flight Line Maintenance) Part #: ADSE 650 Supplier: 3BSK6 Part #: ADTS2000 Supplier: 52892 Part #: ADTS3350ER Supplier: 52892 Part #: ADTS405MKII Supplier: U0427 Part #: ADTS530 Supplier: U0427 Part #: ADTS552F Supplier: U0427 Part #: ADTS553F Supplier: U0427 Part #: ADTS554F Supplier: U0427 Part #: D60340MK Supplier: K1474 Part #: DPS1000 Supplier: 21844 Part #: MODEL 6300 Supplier: 0RDZ5 Part #: MPS34C Supplier: 48RQ2 Part #: MPS43B Supplier: A0197 Part #: MPS45 Supplier: 48RQ2 Part #: MPS49 Supplier: 48RQ2 Part #: TES9463 Supplier: 88277 Opt Part #: 01-0987-00 Supplier: 41364 Opt Part #: 18910480000 Supplier: 89944 Opt Part #: 18910920000 Supplier: 89944 Opt Part #: ADTS405F Supplier: U0427 Opt Part #: ADTS505 Supplier: U0427 Opt Part #: D60302 Supplier: K1474 Opt Part #: D60340 Supplier: K1474 Opt Part #: D60383 Supplier: K1474 Opt Part #: DPS350 Supplier: 21844 Opt Part #: DPS450 Supplier: 21844 Opt Part #: DPS500 Supplier: 21844 Opt Part #: MPS31C Supplier: 48RQ2 Opt Part #: MPS43 Supplier: A0197

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(Continued)

Reference	Description
COM-1916	Adapter - Pitot Test (Typically included in Air Data Accessory Kit) Part #: CSA75700HT-3 Supplier: 3BSK6 Part #: P75701-4 Supplier: 38002 Part #: P75701M2-3 Supplier: 38002
COM-16577	Plug - Pitot Part #: BACP20AU6D Supplier: 81205
COM-25177	Accessory Kit - Air Data, 737 (Pitot/Static adapters, hoses, etc.) Part #: ADA737-678 Supplier: 38002 Part #: AK737-900 Supplier: 3BSK6 Part #: PA-0120-A Supplier: 3BSK6
SPL-1742	Regulator - Air Pressure, Elevator Feel Computer Part #: F72928-109 Supplier: 81205 Part #: F72928-110 Supplier: 81205 Part #: F72928-111 Supplier: 81205 Opt Part #: F72928-58 Supplier: 81205 Opt Part #: F72928-62 Supplier: 81205 Opt Part #: F72928-63 Supplier: 81205

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
323	Vertical Fin - Front Spar To Rear Spar

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Prepare for the Pitot System Leak Test

SUBTASK 27-31-00-010-039

(1) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-860-160

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

F. Pitot System Leak Test

SUBTASK 27-31-00-480-008

CAUTION MAKE SURE THAT THE PITOT PROBE HAS NO ADDED WEIGHT ON IT FROM THE TEST HOSE. THE WEIGHT OF THE TEST HOSE CAN BEND OR TWIST THE PITOT PROBE OUT OF TOLERANCE.

(1) At the two elevator pitot probes on the vertical fin, do these steps:

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(a) Install the pitot test adapter, COM-1916, to the pitot probes.

(b) Connect the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to the pitot probes.

1) Do this step for the regulator, SPL-1742:

a) Connect the pressure lines from the regulator, SPL-1742, with the pitot test adapter, COM-1916, to the pitot probes.

2) Do this step for the air data model test set, COM-1914:

a) Use the air data accessory kit, COM-25177, to connect the air data model test set, COM-1914, to the pitot test adapter, COM-1916.

NOTE: The air data accessory kit, COM-25177, contains hoses to connect to the air data model test set, COM-1914, and the pitot test adapter, COM-1916.

SUBTASK 27-31-00-480-007

(2) Inside the aft unpressurized compartment, below the elevator feel computer, do these steps:

(a) Remove the drain plugs from the pitot lines.

(b) If the regulator, SPL-1742, is used for the leak test, do these steps:

1) Connect the pressure gage lines from the regulator, SPL-1742, to the drain holes of the pitot lines.

(c) If the air data model test set, COM-1914, is used for the leak test, do these steps:

1) Install the pitot plugs, COM-16577, on the ends of the pitot lines.

2) Install a streamer on the opening of the aft unpressurized compartment.

a) Make sure that you can easily see the streamer from the ground.

NOTE: This is to remind you to remove the pitot plugs and install the drain plugs again when the test is completed.

SUBTASK 27-31-00-790-001

(3) Do a pitot system leak test:

CAUTION

DO NOT INCREASE THE PRESSURE TO MORE THAN 5 PSIG (34.5 KPA) (437 KNOTS OF COMPUTED AIRSPEED (KCAS)), DURING THE TEST. IF THE PRESSURE IS MORE THAN 5 PSIG (34.5 KPA) (437 KCAS), IT CAN CAUSE DAMAGE TO THE ELEVATOR FEEL COMPUTER.

(a) Operate the air data model test set, COM-1914, or regulator, SPL-1742, to apply pressure 0.90 +0.07 / -0.00 psig (6.21 +0.48 / -0.00 kPa) to simulate 194 to 201 kcas.

(b) Set the shutoff valve in the air data model test set, COM-1914, or regulator, SPL-1742, to the OFF position.

(c) Make sure that the pressure does not decrease more than 0.2 psig (1.4 kPa) (8 kcas) during a 2-minute time.

NOTE: The low pressure pressurization steps above ensure that there is no significant system leak before you apply full pressure in the following steps.

1) Fix any pitot system or pressurization equipment leaks, as necessary, before proceeding.

(d) Operate the air data model test set, COM-1914, or regulator, SPL-1742, to apply pressure 4.8 ±0.1 psig (33.1 ±0.7 kPa) to simulate 425 to 433 kcas.

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(e) Set the shutoff valve in the air data model test set, COM-1914, or regulator, SPL-1742, to the OFF position.

(f) Make sure that the pressure does not decrease more than 0.3 psig (2.1 kPa) (12 kcas) during a 2-minute time.

(g) Operate the air data model test set, COM-1914, or regulator, SPL-1742, to put the system back to ambient pressure (0 psig (0.0 kPa)).

SUBTASK 27-31-00-160-001

(4) Remove the seal from the drain hole in each pitot probe.

SUBTASK 27-31-00-080-007

(5) At the two elevator pitot probes on the vertical fin, do these steps:

(a) Remove the pressure lines from the air data model test set, COM-1914, or regulator, SPL-1742, and pitot test adapter, COM-1916.

(b) Remove the pitot test adapter, COM-1916, from the pitot probes.

SUBTASK 27-31-00-080-008

(6) In the aft unpressurized compartment, below the elevator feel computer, do these steps:

(a) If the regulator, SPL-1742, is used for the leak test, do these steps:

1) Disconnect the pressure gage lines of the regulator, SPL-1742, from the drain holes of the pitot lines.

(b) If the air data model test set, COM-1914, is used for the leak test, do these steps:

1) Remove the pitot plug, COM-16577, that were installed on the ends of the pitot lines.

2) Remove the streamer on the opening of the aft unpressurized compartment.

(c) Install the drain plugs on the pitot lines.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-161

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

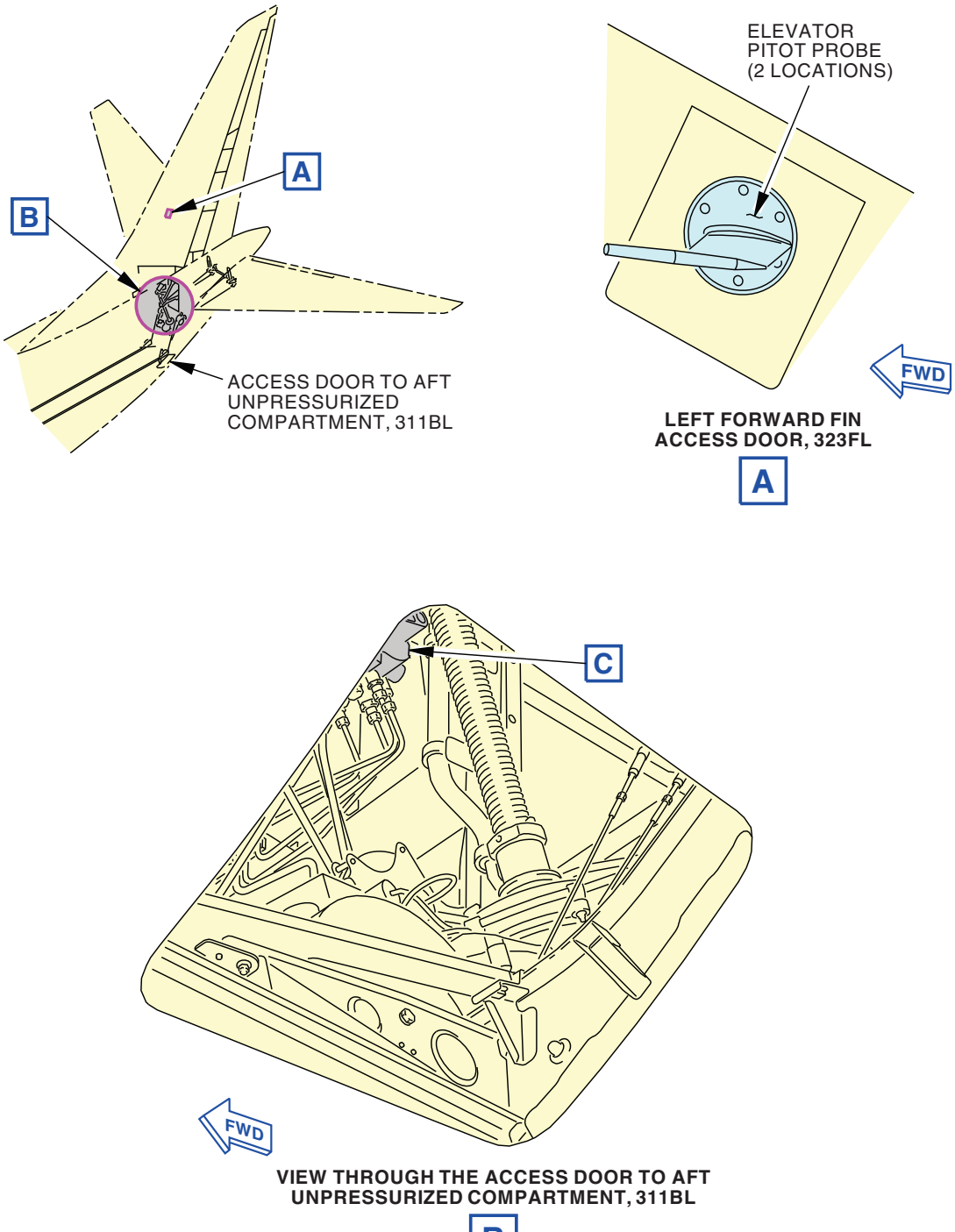
SUBTASK 27-31-00-410-032

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

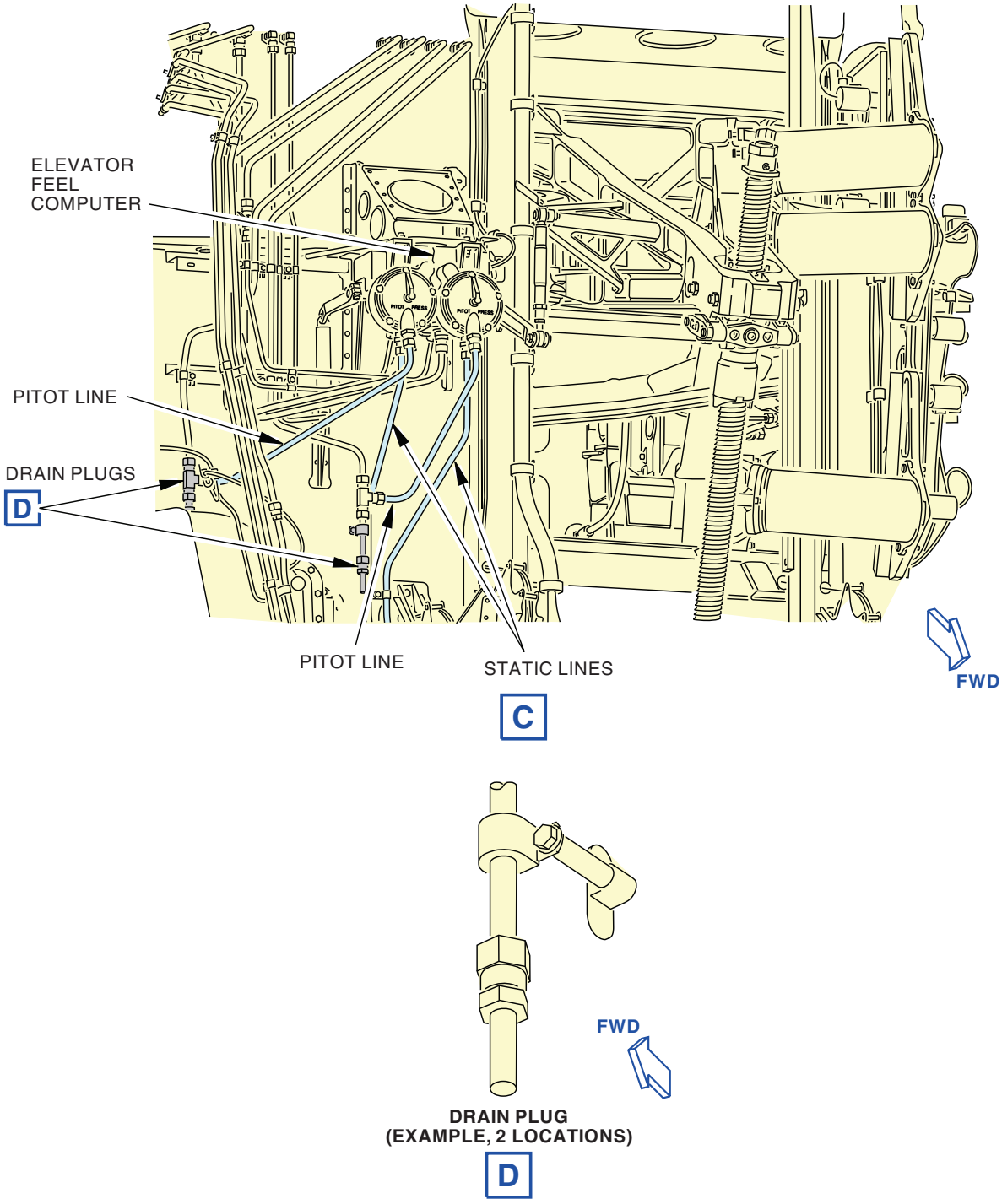
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Elevator Pitot-Static System
Figure 515/27-31-00-990-822 (Sheet 1 of 2)

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Elevator Pitot-Static System
Figure 515/27-31-00-990-822 (Sheet 2 of 2)

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TASK 27-31-00-700-813

24. Feel Force System - Test

(Figure 504, Figure 515, Figure 520)

A. General

(1) Use this test to make sure that the elevator feel computer operates correctly. The elevator feel computer changes the control column forces as the airspeed changes.

B. References

Reference	Title
22-11-00-820-802	Elevator Rigging Test (P/B 501)
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9

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(Continued)

Reference	Description
COM-1914	Test Set - Air Data Model FLMTS (Flight Line Maintenance) Part #: ADSE 650 Supplier: 3BSK6 Part #: ADTS2000 Supplier: 52892 Part #: ADTS3350ER Supplier: 52892 Part #: ADTS405MKII Supplier: U0427 Part #: ADTS530 Supplier: U0427 Part #: ADTS552F Supplier: U0427 Part #: ADTS553F Supplier: U0427 Part #: ADTS554F Supplier: U0427 Part #: D60340MK Supplier: K1474 Part #: DPS1000 Supplier: 21844 Part #: MODEL 6300 Supplier: 0RDZ5 Part #: MPS34C Supplier: 48RQ2 Part #: MPS43B Supplier: A0197 Part #: MPS45 Supplier: 48RQ2 Part #: MPS49 Supplier: 48RQ2 Part #: TES9463 Supplier: 88277 Opt Part #: 01-0987-00 Supplier: 41364 Opt Part #: 18910480000 Supplier: 89944 Opt Part #: 18910920000 Supplier: 89944 Opt Part #: ADTS405F Supplier: U0427 Opt Part #: ADTS505 Supplier: U0427 Opt Part #: D60302 Supplier: K1474 Opt Part #: D60340 Supplier: K1474 Opt Part #: D60383 Supplier: K1474 Opt Part #: DPS350 Supplier: 21844 Opt Part #: DPS450 Supplier: 21844 Opt Part #: DPS500 Supplier: 21844 Opt Part #: MPS31C Supplier: 48RQ2 Opt Part #: MPS43 Supplier: A0197
COM-1916	Adapter - Pitot Test (Typically included in Air Data Accessory Kit) Part #: CSA75700HT-3 Supplier: 3BSK6 Part #: P75701-4 Supplier: 38002 Part #: P75701M2-3 Supplier: 38002
COM-16577	Plug - Pitot Part #: BACP20AU6D Supplier: 81205
COM-25177	Accessory Kit - Air Data, 737 (Pitot/Static adapters, hoses, etc.) Part #: ADA737-678 Supplier: 38002 Part #: AK737-900 Supplier: 3BSK6 Part #: PA-0120-A Supplier: 3BSK6
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-1742	Regulator - Air Pressure, Elevator Feel Computer Part #: F72928-109 Supplier: 81205 Part #: F72928-110 Supplier: 81205 Part #: F72928-111 Supplier: 81205 Opt Part #: F72928-58 Supplier: 81205 Opt Part #: F72928-62 Supplier: 81205 Opt Part #: F72928-63 Supplier: 81205

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D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-31-00-860-153

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

SUBTASK 27-31-00-860-117

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize the elevator hydraulic systems A and B, do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-31-00-860-118

(3) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-00-860-164

(4) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-31-00-480-009

(5) At the two pitot probes on the vertical fin, do these steps:

(a) Seal the drain hole on each pitot probe.

CAUTION MAKE SURE THAT THE PITOT PROBE HAS NO ADDED WEIGHT ON IT FROM THE TEST HOSE. THE WEIGHT OF THE TEST HOSE CAN BEND OR TWIST THE PITOT PROBE OUT OF TOLERANCE.

(b) Connect the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to the pitot probes.

1) Do this step for the regulator, SPL-1742:

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a) Connect the pressure lines from the regulator, SPL-1742, with the pitot test adapter, COM-1916, to the pitot probes.

2) Do this step for the air data model test set, COM-1914:

a) Use the air data accessory kit, COM-25177, to connect the air data model test set, COM-1914, to the pitot test adapter, COM-1916.

NOTE: The air data accessory kit, COM-25177, contains hoses to connect to the air data model test set, COM-1914, and the pitot test adapter, COM-1916.

SUBTASK 27-31-00-010-030

(6) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-480-010

(7) Inside the aft unpressurized compartment, below the elevator feel computer, do these steps:

(a) Remove the restrictor valves from the bottom of the pitot lines.

(b) If the regulator, SPL-1742, or equivalent is used, do these steps:

1) Connect the pressure gage lines from the regulator, SPL-1742, or equivalent, to the drain holes of the pitot lines.

(c) If the air data model test set, COM-1914, is used, do these steps:

1) Install the pitot plugs, COM-16577, over the ends of the pitot lines.

2) Install a highly visible streamer on the opening of the aft unpressurized compartment.

a) Make sure that the streamer can be easily seen from the ground.

NOTE: This is to remind to remove the plugs and install the restrictor valves again when the test is completed.

SUBTASK 27-31-00-860-119

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(8) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

G. Feel Force System Test

SUBTASK 27-31-00-730-013

(1) Do the feel force system test:

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WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Make sure that elevator hydraulic systems A and B is pressurized (Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801).

(b) Make sure that FLT CONTROL A and FLT CONTROL B switches are in the ON position.

CAUTION DO NOT INCREASE THE PRESSURE TO MORE THAN 5 PSIG (437 KCAS) DURING THE TEST. IF THE PRESSURE IS MORE THAN 5 PSIG, IT CAN CAUSE DAMAGE TO THE ELEVATOR FEEL COMPUTER.

(c) Operate the regulator, SPL-1742, or air data model test set, COM-1914, to increase the pressure 0.77 psig (5.3 kPa) - 0.87 psig (6.0 kPa) (185 knots).

(d) Apply masking tape on the tail cone along direction of the travel of the elevator Trailing Edge (TE).

(e) Move the CAPT (Captain)'s control column lightly to put the system in the neutral position, then release.

1) Make a mark of the neutral position of the elevator TE on the masking tape.

(f) Move the CAPT's control column slowly aft until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 17 lbf (76 N) - 28 lbf (125 N).

(g) Move the CAPT's control column slowly aft until the elevator TE is at 4.75 in. (120.65 mm) - 4.85 in. (123.19 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 36 lbf (160 N) - 45 lbf (200 N).

(h) Move the CAPT's control column slowly forward until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 17 lbf (76 N) - 27.5 lbf (122 N).

(i) Move the CAPT's control column slowly forward until the elevator TE is at 4.75 in. (120.65 mm) - 4.85 in. (123.19 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 44 lbf (196 N) - 53 lbf (236 N).

(j) Set the FLT CONTROL B switch, on the P5 forward overhead panel, to the OFF position.

(k) Move the CAPT's control column slowly aft until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 17 lbf (76 N) - 28 lbf (125 N).

(l) Move the CAPT's control column slowly aft until the elevator TE is at 4.75 in. (120.65 mm) - 4.85 in. (123.19 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 36 lbf (160 N) - 45 lbf (200 N).

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(m) Move the CAPT's control column slowly forward until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 17 lbf (76 N) - 27.5 lbf (122 N).

(n) Move the CAPT's control column slowly forward until the elevator TE is at 4.75 in. (120.65 mm) - 4.85 in. (123.19 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 44 lbf (196 N) - 53 lbf (236 N).

(o) Set the FLT CONTROL A switch, on the P5 forward overhead panel, to the OFF position.

(p) Set the FLT CONTROL B switch to the ON position.

(q) Move the CAPT's control column slowly aft until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 17 lbf (76 N) - 28 lbf (125 N).

(r) Move the CAPT's control column slowly aft until the elevator TE is at 4.75 in. (120.65 mm) - 4.85 in. (123.19 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 36 lbf (160 N) - 45 lbf (200 N).

(s) Move the CAPT's control column slowly forward until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 17 lbf (76 N) - 27.5 lbf (122 N).

(t) Move the CAPT's control column slowly forward until the elevator TE is at 4.75 in. (120.65 mm) - 4.85 in. (123.19 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 44 lbf (196 N) - 53 lbf (236 N).

(u) Set the FLT CONTROL A switch to the ON position.

CAUTION DO NOT INCREASE THE PRESSURE TO MORE THAN 5 PSIG (437 KCAS) DURING THE TEST. IF THE PRESSURE IS MORE THAN 5 PSIG, IT CAN CAUSE DAMAGE TO THE ELEVATOR FEEL COMPUTER.

(v) Operate the regulator, SPL-1742, or air data model test set, COM-1914, to increase the pressure 2.42 psig (16.7 kPa) - 2.52 psig (17.4 kPa) (315 knots).

(w) Move the CAPT's control column slowly aft until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold (Figure 520).

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 28 lbf (125 N) - 44 lbf (196 N).

(x) Move the CAPT's control column slowly forward until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold.

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 30 lbf (133 N) - 46 lbf (205 N).

(y) Set the FLT CONTROL B switch to the OFF position.

(z) Move the CAPT's control column slowly aft until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold.

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1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 28 lbf (125 N) - 44 lbf (196 N).

(aa) Move the CAPT's control column slowly forward until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold.

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 30 lbf (133 N) - 46 lbf (205 N).

(ab) Set the FLT CONTROL B switch to the ON position.

(ac) Set the FLT CONTROL A switch, on the P5 forward overhead panel, to the OFF position.

(ad) Move the CAPT's control column slowly aft until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold.

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 28 lbf (125 N) - 44 lbf (196 N).

(ae) Move the CAPT's control column slowly forward until the elevator TE is at 1.15 in. (29.21 mm) - 1.25 in. (31.75 mm) from the neutral position, and hold.

1) Use the force gauge, COM-1557, to make sure that the force required to keep the control column at this position is 30 lbf (133 N) - 46 lbf (205 N).

(af) Move the CAPT's control column slowly to the neutral position.

(ag) Set the FLT CONTROL A switch to the ON position.

(ah) Make sure that the FLT CONTROL B switch is in the ON position.

(ai) Operate the regulator, SPL-1742, or air data model test set, COM-1914, to decrease the pressure to 0.0 psig (0.0 kPa) (0 knots).

SUBTASK 27-31-00-730-014

(2) Do this task: Elevator Rigging Test, TASK 22-11-00-820-802.

SUBTASK 27-31-00-080-009

(3) At the two pitot probes on the vertical fin, do these steps:

(a) Remove the pressure lines from the air data model test set, COM-1914, or regulator, SPL-1742, from the pitot probes.

(b) Remove the pitot test adapter, COM-1916, from the pitot probes.

(c) Remove the seal from the drain hole in each pitot probe.

SUBTASK 27-31-00-430-002

(4) Inside the aft unpressurized compartment, do these steps:

(a) If the regulator, SPL-1742, or equivalent is used for the test, do these steps:

1) Disconnect the pressure gage lines of the regulator, SPL-1742, or equivalent, from the drain holes of the pitot lines.

(b) If the air data model test set, COM-1914, is used for the test, do these steps:

1) Remove the pitot plugs, COM-16577, that were installed over the ends of the pitot lines.

2) Remove the highly visible streamer that is installed on the opening of the aft unpressurized compartment.

(c) Install the restrictor valves on the bottom of the pitot lines.

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-163

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-31-00-410-025

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-860-120

(3) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

SUBTASK 27-31-00-860-154

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

——— END OF TASK ———

TASK 27-31-00-700-814

25. Sensor Null - Test

(Figure 504)

A. General

(1) Use this test to do the elevator rigging test with the DFCS computer BITE. The control display unit (CDU) is necessary for this procedure.

B. References

Reference	Title
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left

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(Continued)

Zone	Area
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-31-00-860-121

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize elevator hydraulic systems A and B. To pressurize them, do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-31-00-010-031

(2) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-00-860-122

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(3) Use the trammel bar, SPL-1677 to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)). (Figure 504)

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677 and not the stabilizer indicator in the flight deck.

G. Sensor Null Test

SUBTASK 27-31-00-740-008

(1) Do the sensor null test:

(a) Do this BITE test on the control display unit (CDU):

1) Push the INIT REF key on the CDU keyboard.

2) Push the Line Select Key (LSK) that is adjacent to these selections:

NOTE: Use the NEXT PAGE or PREV PAGE key to change page if it is necessary. During the BITE test, if the "CONTINUE" shows, then push the LSK that is adjacent to "CONTINUE" after you do the instructions that show on the CDU display.

a) INDEX

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b) MAINT

c) DFCS

d) EXTENDED MAINTENANCE

e) RIGGING

f) ELEVATOR

g) CONTINUE

3) Do the instructions that show on the CDU and push the LSK adjacent to these selections:

a) CONTINUE

b) ELEV RIG

4) Make sure the BITE display values on the A and B sides are in the limit as shown below:

NOTE: During the BITE test, if the "CONTINUE" shows, then push the LSK that is adjacent to "CONTINUE" after you do the instructions that show on the CDU display.

Table 509/27-31-00-993-809

Test 51.01	STAB POS (VAC)	Limits: -0.07 TO 0.07
Use the trammel bar, SPL-1677 to set the "B" dimension as on the CDU. Test 51.02		
Test 51.02	MT ACT POS (VAC)	Limits: -8.70 to -6.35
51.03	MT ACT POS (VAC)	Limits: -0.20 to 0.20
Test 51.04	NSS POS (VAC)	Limits: -1.77 to -1.67
Make sure the elevator trailing edge aligns with the index mark +/- 0.06 inch (1.5 millimeters).		
Test 51.05	ELEV POS (VAC)	Limits: -0.55 to -0.45
Test 51.06	ELEV LVDT (VAC)	Limits: -2.05 to -1.95
Test 51.07	ELEV POS (VAC)	Limits: -0.55 to -0.45
Make sure the elevator trailing edge aligns with the index mark +/- 0.06 inch (1.5 millimeters).		
Test 51.08	ELEV POS (VAC)	Limits: -0.55 to -0.45
Test 51.09	ELEV POS (VAC)	Limits: -0.55 to -0.45
Test 51.10	ELEV POS (VAC)	Limits: -0.55 to -0.45
Test 51.11	ELEV POS (VAC)	Limits: -0.55 to -0.45
Test 51.12	STAB POS (VAC)	Limits: 4.47 to 5.29
Test 51.13	ELEVATOR LVDT	Limits: -0.06 to -0.06
Test 51.14	ELEVATOR POS	Limits: -0.20 to 0.20
Test 51.15	NSS POS	Limits: -0.20 to 0.20

5) Push the INIT/REF key.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-123

(1) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

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SUBTASK 27-31-00-410-026

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

TASK 27-31-00-820-807

26. Elevator Power Control Unit (PCU) Input Rod (Pogo) - Functional Test

(Figure 504, Figure 516)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is applicable for the left and right elevator Power Control Unit (PCU)s.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-1749	Lock Equipment - Aileron/Elevator PCU Input Rod (POGO) Part #: C27066-1 Supplier: 81205

D. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

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F. Prepare for the Functional Test

SUBTASK 27-31-00-860-124

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize the elevator hydraulic systems A and B (TASK 27-31-00-800-801).

SUBTASK 27-31-00-010-032

(2) Open these access doors:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-00-860-125

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(3) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (Figure 504).

NOTE: The “B” dimension must be set with the trammel bar, SPL-1677 and not the stabilizer indicator in the flight deck.

G. Elevator Power Control Unit (PCU) Input Rod (Pogo) Functional Test

SUBTASK 27-31-00-740-009

(1) Do this Built-In-Test Equipment (BITE) test on the Control Display Unit (CDU):

(a) Push the INIT REF key on the CDU keyboard.

(b) Null the mach trim actuator (TASK 27-31-00-700-801).

(c) Push the Line Select Key (LSK) that is adjacent to these selections:

NOTE: Use the NEXT PAGE or PREV PAGE key to change page if it is necessary. During the BITE test, if the CONTINUE shows, then push the LSK that is adjacent to CONTINUE after you do the instructions that show on the CDU display.

1) INDEX
2) MAINT
3) DFCS
4) EXTENDED MAINTENANCE
5) RIGGING
6) ELEVATOR
7) CONTINUE

(d) Do the instructions that show on the CDU and push the LSK adjacent to these selections:

1) CONTINUE

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2) ELEV RIG

(e) Make sure the BITE display values on the A and B sides are in the limit as shown below:

NOTE: During the BITE test, if the "CONTINUE" shows, then push the LSK that is adjacent to "CONTINUE" after you do the instructions that show on the CDU display.

(f) Make sure that the mach trim actuator voltage is 0.00 ± 0.20 Alternating Current Volts (VAC).

(g) Make sure that the flaps are in the full up position.

(h) Make sure that the values for the A side are between the limits that show on the CDU display.

1) Test 51.03 MT ACT POS (VAC) limits: -0.20 to 0.20

(i) Push the INIT/REF key.

(j) Cycle the pilot control column (full forward, full aft) to make sure the elevator has full travel.

(k) Make sure the captain's control column is at the neutral position.

(l) Make sure the right elevator trailing edge position is ±0.06 in. (1.5 mm) (±0.10 degree) with index plate.

NOTE: Use a straight edge on top of the elevator to make a line to the index plate.

SUBTASK 27-31-00-860-126

(2) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

SUBTASK 27-31-00-480-011

(3) Install the lock equipment, SPL-1749, on both elevator PCU pistons.

(a) Make sure that the lock equipment, SPL-1749, is installed correctly to prevent damage to the lockwire.

SUBTASK 27-31-00-860-127

(4) Move the captain and first officer's control column aft until all four elevator PCU pogos extend approximately 0.3 in. (7.6 mm).

NOTE: Total column force will be approximately 140 lbf (623 N).

NOTE: The extension of the PCU pogo is the distance that the PCU pogo clevis moves away from the end of the tube.

SUBTASK 27-31-00-860-128

(5) Remove the force on the control columns slowly.

SUBTASK 27-31-00-080-010

(6) Remove the lock equipment, SPL-1749, from the two elevator PCU pistons.

SUBTASK 27-31-00-860-129

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

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SUBTASK 27-31-00-860-130

(8) Cycle the pilot control column (full forward, full aft) to make sure the elevator has full travel.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-00-860-131

(1) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

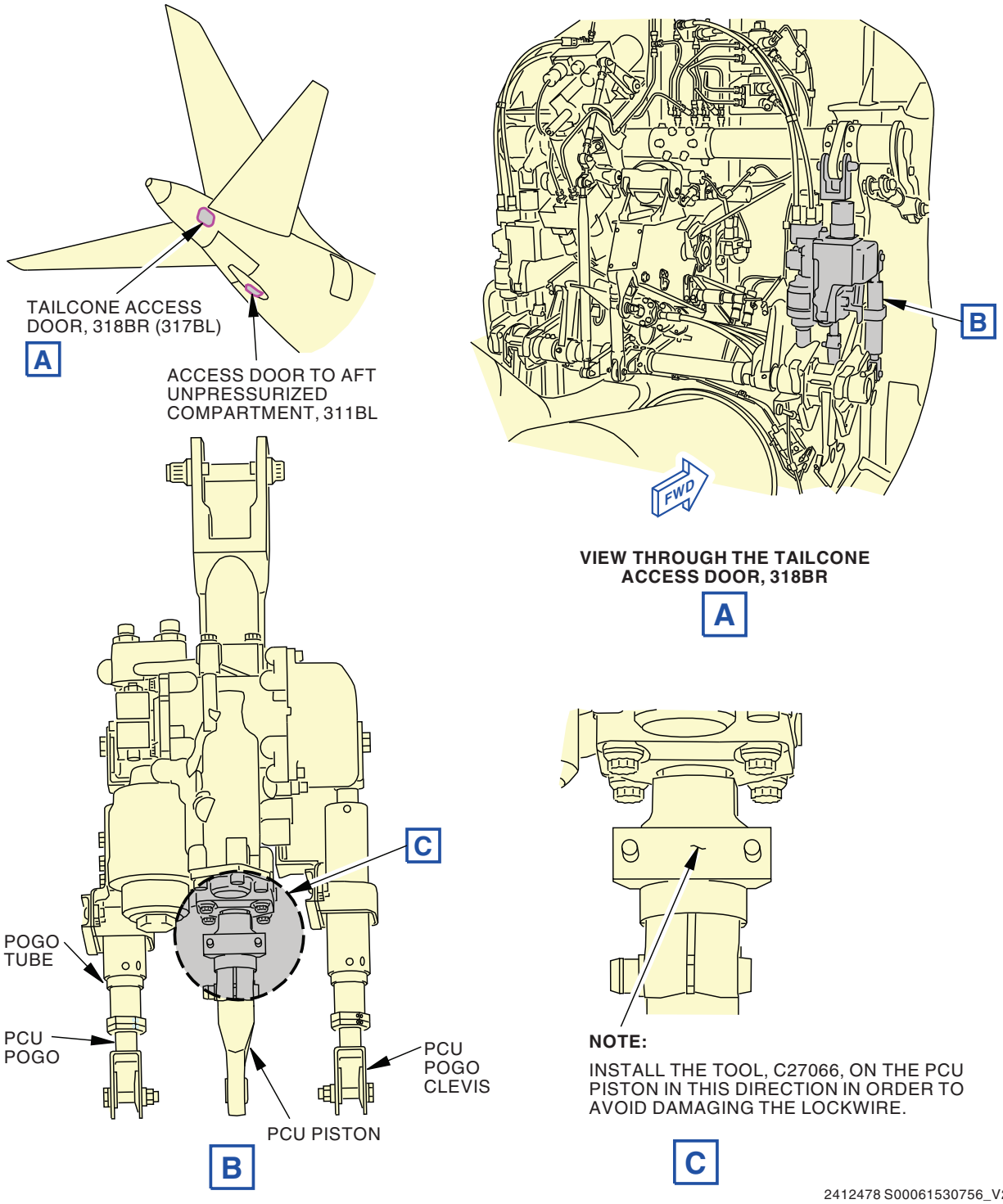
SUBTASK 27-31-00-410-027

(2) Close these access doors:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

——— END OF TASK ———

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Aileron/Elevator Power Control Unit (PCU) Input Rod Functional Test Tool
Figure 516/27-31-00-990-823

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TASK 27-31-00-710-803

27. Elevator Power-Off Flight Test

(Figure 517, Figure 518)

A. General

(1) The flight test is necessary because the maintenance action can affect the flight characteristics of the airplane and incorrect adjustment can cause a pitch and attitude upset in the event of a dual simultaneous hydraulic system loss. Since the magnitude and attitude of this pitch upset cannot be predicted to sufficient accuracy or verified on the ground, a flight test is necessary to determine the unbalanced moment. The unbalanced moment is reflected in terms of the number and direction the pilot must turn the stabilizer trim wheel to return the airplane to level flight after initiating manual reversion. If the number and direction of stabilizer trim wheel turns are not within limits, then the elevator tab pushrods need to be re-adjusted and a subsequent flight test performed. The number and direction of stabilizer trim wheel turns should be recorded during the manual reversion flight test since this information is needed to correctly re-adjust the pushrods.

(2) This flight test is necessary after you have completed one of the following maintenance tasks:

(a) For the elevator:

1) After a replacement of an elevator.

NOTE: Replacement of an elevator with a new elevator, or with one that is not the same as the one that was originally on the airplane, requires a flight test.

NOTE: If the elevator which was removed from the airplane is reinstalled without repair, adjustment, or modification, a flight test is not required.

2) After a replacement or adjustment of an elevator control pushrod.

NOTE: Replacement of the elevator output torque tube crank arm bearing does not require a flight test.

NOTE: Replacement of the elevator mast arm bearing does not require a flight test.

3) After a replacement of an elevator hinge bearing plate (Figure 518).

NOTE: Flight test is not required if all of the following conditions are met:

- The elevator is not removed from the airplane during replacement of the elevator bearing plate(s).

- Adjacent elevator bearing plates are not removed at the same time.

- No more than two elevator bearing plates are removed at the same time on the same side.

NOTE: Flight test is not required if only the bearing is replaced and the original elevator bearing plate is reinstalled in its original location.

4) After a replacement of an elevator hinge fitting plate (Figure 518).

5) After a maintenance where re-balance of the elevator is necessary regardless of procedure used, either on-jig or by calculation (SRM SUBJECT 51-60-04).

NOTE: A flight test is not necessary following repair and/or repainting of the elevator if the re-balance method used does not result in adjustment of the balance weights.

(b) For the elevator tab:

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1) After a replacement of an elevator tab.

NOTE: A flight test is not necessary if the same elevator tab is removed and reinstalled after it was repaired, repainted, and rebalanced and the following conditions are met:

- The elevator tab stays within the balance limits for Center of Gravity (CG) and weight after it was repaired, repainted, and rebalanced (SRM PROCEDURE 51-60-00-0G-0) and no adjustment of the elevator tab adjust weights is necessary following repair and/or repaint.

- Any repair and/or paint application to the elevator tab provides a smooth continuous finish to all edges.

2) After a replacement or adjustment of one or both of the elevator tab pushrods.

NOTE: If only one pushrod is replaced or adjusted and you can verify that the remaining pushrod was not replaced or adjusted since the last flight test, then a flight test is not necessary.

NOTE: If only one pushrod is replaced or adjusted and you cannot verify that the remaining pushrod was not replaced or adjusted since the last flight test, then a flight test is necessary.

NOTE: If the same elevator tab pushrods are removed and re-installed without adjustment in the same left and right locations on the same elevator tab, a flight test is not required.

3) After a replacement of an elevator tab hinge bearing (Figure 519).

NOTE: It is not necessary to do a flight test if you do not remove the elevator tab hinge fitting assemblies from the elevator when you replace the bearing.

4) After replacement of an elevator tab hinge fitting (Figure 519).

5) After a replacement of the inboard elevator tab hinge bushing (Figure 519).

NOTE: It is not necessary to do a flight test if you replace bolts at any elevator tab hinge location, or if you replace bushings at any other tab hinge location.

6) After the replacement of a new or repaired elevator tab control mechanism.

(c) For the stabilizer:

1) After a replacement of the horizontal stabilizer.

NOTE: A power-off flight test is not necessary if the horizontal stabilizer is removed and the same stabilizer, elevator, and elevator tab are re-installed (without elevator or elevator tab rework or adjustment).

(d) For the tailcone:

1) After a replacement of the tailcone.

NOTE: A power-off flight test is not required if the following conditions are met:

- The same tailcone is removed and re-installed (without elevator or elevator tab rework or adjustment).

- The elevator and rudder position is recorded prior to removal (relative to the respective index plates).

- The elevator and rudder position remains the same after re-installation.

(3) Dual Hydraulics Failure: Please read the subsequent instructions before you do the flight test.

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(a) Sudden dual hydraulic failure or selection of both flight control switches to off or standby will put the airplane in manual reversion and cause elevator movement to a deflection that balances the air load. The intent of the airplane design and Aircraft Maintenance Manual procedures is to have the resultant pitch upset be easily manageable, but if an out of tolerance elevator tab rigging exists on the airplane, the pitch upset could be significant. Flight Crews, especially for check flights after elevator tab maintenance, should be prepared in the event of a large upset.

(b) Upset recovery techniques are discussed in the maneuvers section of the Flight Crew Training Maneuvers. These techniques provide good guidance in the case of a large upset. Additional items to consider for a dual hydraulic failure are:

1) Elevator forces will be high and elevator control authority reduced in manual reversion hydraulics should not be restored with large column inputs present.

(c) If the pitch upset is significant, immediately accomplish the Upset Recovery maneuver found in the non-normal maneuvers section in the Quick Reference Handbook. For example, the use of bank angle in the event of large nose up pitch rate, described in this section, may be beneficial.

(d) If performing a maintenance check (where hydraulics are available to be restored), the pilot's should coordinate the restoration of hydraulics to ensure that column input is small at time hydraulics are restored. Restoring hydraulics with large column input will result in a large sudden elevator command that may generate a pitch upset in the opposite direction. The pilot flying should reduce column input at the time hydraulics are restored by using stabilizer trim or by momentarily allowing the airplane to maneuver. At the time hydraulics are restored, the pilot flying should be prepared to apply column inputs to maintain control as normal column feel forces return.

B. References

Reference	Title
27-09-71 P/B 501	FLIGHT CONTROLS TRIM CORRECTION - ADJUSTMENT/TEST

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

D. Prepare for Flight Test

SUBTASK 27-31-00-860-132

(1) Before doing a flight test to verify the elevator tab pushrod adjustment, make sure that you adjust the appropriate system if either of these conditions exist:

(a) Aileron trim requirement is more than 3/4 unit.

NOTE: If the aileron trim requirement is more than 3/4 unit, you need to adjust the aileron trim per the applicable section of PAGEBLOCK 27-09-71/501 before doing an elevator tab adjustment power-off flight test.

(b) Rudder trim requirement is more than 3/4 unit.

NOTE: If the rudder trim requirement is more than 3/4 unit, you need to adjust the rudder trim per the applicable section of PAGEBLOCK 27-09-71/501 before doing an elevator tab adjustment power-off flight test.

SUBTASK 27-31-00-820-014

(2) Adjust the elevator tab pushrods before the flight test, do this task: Elevator Tab Pushrods - Adjustment, TASK 27-31-00-820-805.

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E. Flight Test Procedure

SUBTASK 27-31-00-710-008

(1) Make sure that you satisfy these flight conditions:

(a) Indicated Airspeed: 250 kcas

(b) Altitude: Flight Level 15,000 ft (4572 m).

NOTE: The altitude should be within 500 ft (152 m) of the 15,000 ft (4572 m) level to get good flight results. Altitude variation more than 500 ft (152 m) is large enough to cause misleading test results.

(c) Weight requirement:

1) less than 135,000 lb (61,235 kg)

(d) Percentage of MAC center of gravity

NOTE: See Figure 517.

NOTE: The Center of Gravity (CG) percentage Mean Aerodynamic Chord (MAC) is calculated by dispatch based on fuel loads, cargo loading, and passenger loading. This information is calculated for the pilot to enter into the Flight Management Computer (FMC) to get the recommended stabilizer setting for takeoff.

Because the percent MAC center of gravity (for this test) is being used for a flight test with minimal flight crew, no cargo, and a normal fuel burn is anticipated; recording the CG in percent of MAC at takeoff is sufficient for proper adjustment of the tabs.

(e) N1 equal for both engines

(f) Wing fuel loads equal

(g) Flaps up

SUBTASK 27-31-00-860-133

(2) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand, to the CUTOUT position.

NOTE: A speed trim fail master caution annunciation may occur if thresholds are exceeded.

NOTE: As soon as you complete the power off flight test, you must move the STAB TRIM PRI switch to the NORMAL position.

SUBTASK 27-31-00-710-009

(3) Set the SPOILER A and B switches, on the P5 forward overhead panel, to the OFF position.

NOTE: Hold control wheel firmly when you turn the SPOILER A and B switches OFF to control any upset which might occur in pitch or roll direction. If wheel input required to maintain steady flight increases when SPOILER A and B switches are turned off, this could indicate unusual spoiler movement. If the required wheel input increases more than approximately 2 units, higher than normal wheel forces may occur when the FLT CONTROL A and B are turned to STBY RUD.

SUBTASK 27-31-00-860-134

(4) Set the FLT CONTROL A and B switches, on the P5 forward overhead panel, to the STDBY RUD position.

NOTE: To successfully complete the test, the airplane must be re-trimmed to the same airspeed. To provide a greater precision for the maneuver, someone other than the pilot in command should set the switches.

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SUBTASK 27-31-00-860-135

(5) If the airplane pitches, use the elevator to re-trim the airplane to the same airspeed:

(a) Manually turn the stabilizer trim wheel to reduce column force to zero.

(b) Keep a record of the number of turns required to re-trim the airplane.

NOTE: During the initial rotation of the stabilizer trim wheel, there might be some freeplay. Do not count the turns until you remove the freeplay.

(c) Keep a record of the center of gravity of the airplane.

SUBTASK 27-31-00-860-136

(6) After completing the elevator power-off flight test portion of the flight, return the switches to their normal flight positions as follows:

NOTE: These are the switches that were moved for the power-off flight test

(a) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand, to the NORMAL position.

(b) Set the FLT CONTROL A and B switches, on the P5 forward overhead panel, to the ON position.

(c) Set the SPOILER A and B switches, on the P5 forward overhead panel, to the ON position.

SUBTASK 27-31-00-700-003

(7) Make sure that the number of turns of the stabilizer trim wheel in the airplane Nose Up or Nose Down direction, are within the limits of the corresponding region in the referenced chart (Figure 517).

NOTE: An example on how to use the Elevator Tab Rig Adjustment Chart is provided in TASK 27-31-00-820-808 in the section Elevator Tab Pushrod Adjustment.

SUBTASK 27-31-00-700-004

(8) If the number of turns of the stabilizer trim wheel in the airplane Nose Up or Nose Down direction, are more than the limits, do the steps that follow:

(a) Make sure that the rigging of the elevator control pushrods, input rod of the Power Control Unit (PCU), and output rod of the elevator feel and centering unit is correct.

(b) If the rigging of the elevator control pushrods, input rod of the elevator PCU, and output rod of the elevator feel and centering unit is satisfactory, do this task: Elevator Tab Pushrod - Adjustment after the Elevator Power-Off Flight Test, TASK 27-31-00-820-808.

(c) If the rigging of the elevator control pushrods, input rod of the elevator PCU, or output rod of the elevator feel and centering unit is not correct, do these tasks:

1) Elevator Control Pushrods - Adjustment, TASK 27-31-00-820-802.

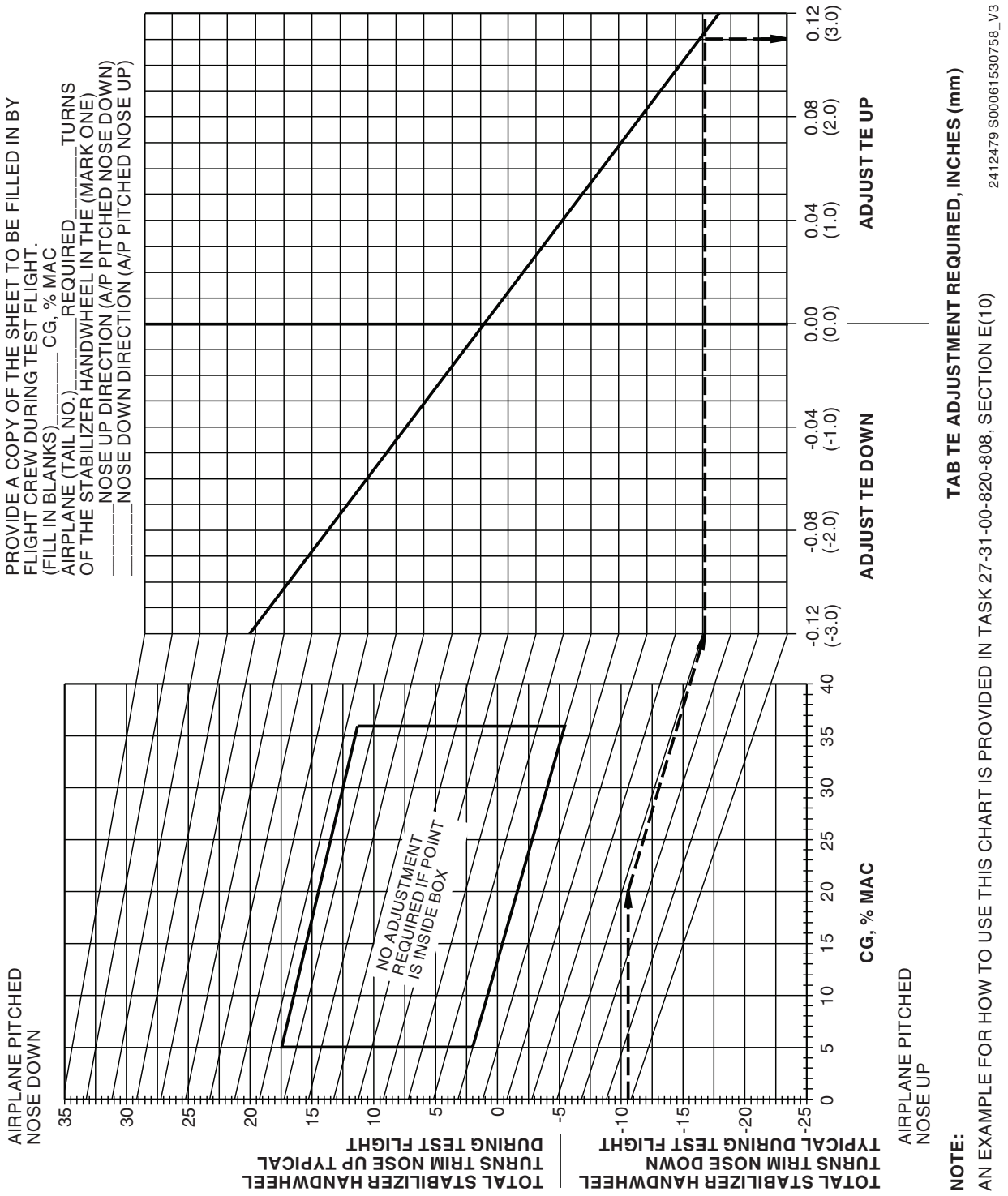
2) Input Rods of the Elevator Power Control Unit - Adjustment, TASK 27-31-00-820-803.

3) Input Rods of the Elevator Neutral Shift Mechanism and Output Rod of the Feel and Centering Unit - Adjustment, TASK 27-31-00-820-804.

4) Elevator Tab Pushrod - Adjustment after the Elevator Power-Off Flight Test, TASK 27-31-00-820-808.

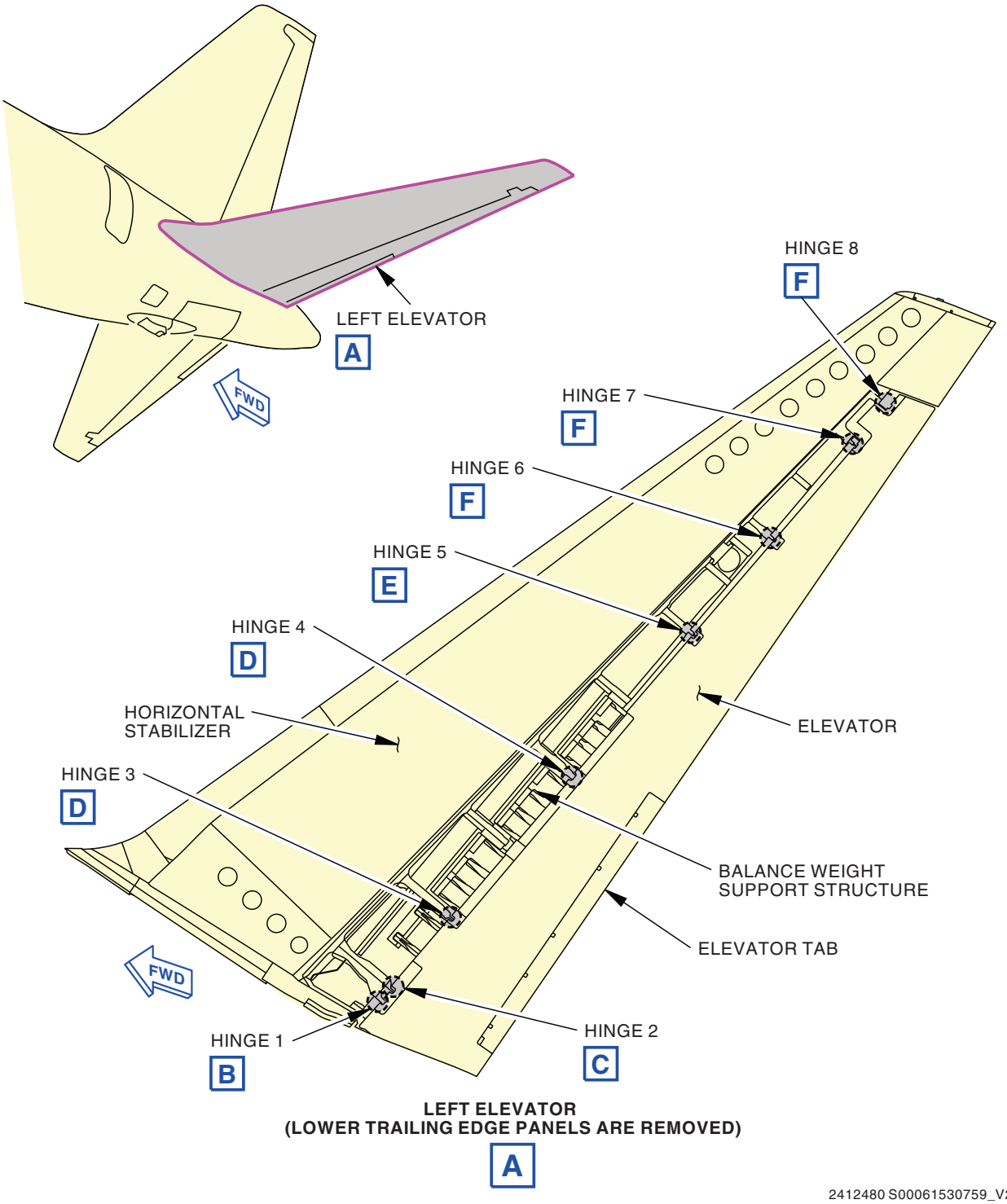
END OF TASK

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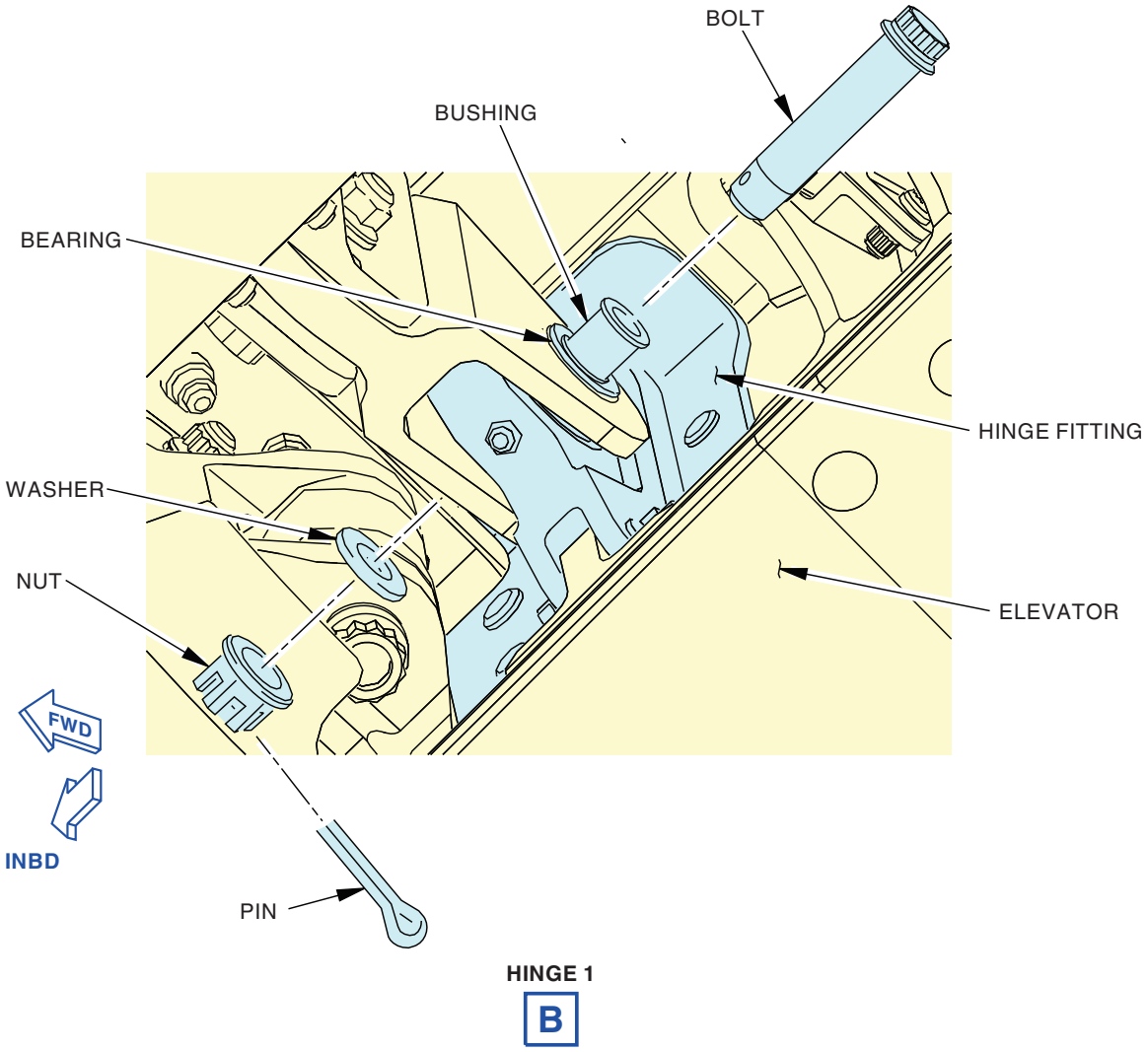
Elevator Tab Rig Adjustment Chart
Figure 517/27-31-00-990-824

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Hinge Assemblies
Figure 518/27-31-00-990-825 (Sheet 1 of 4)

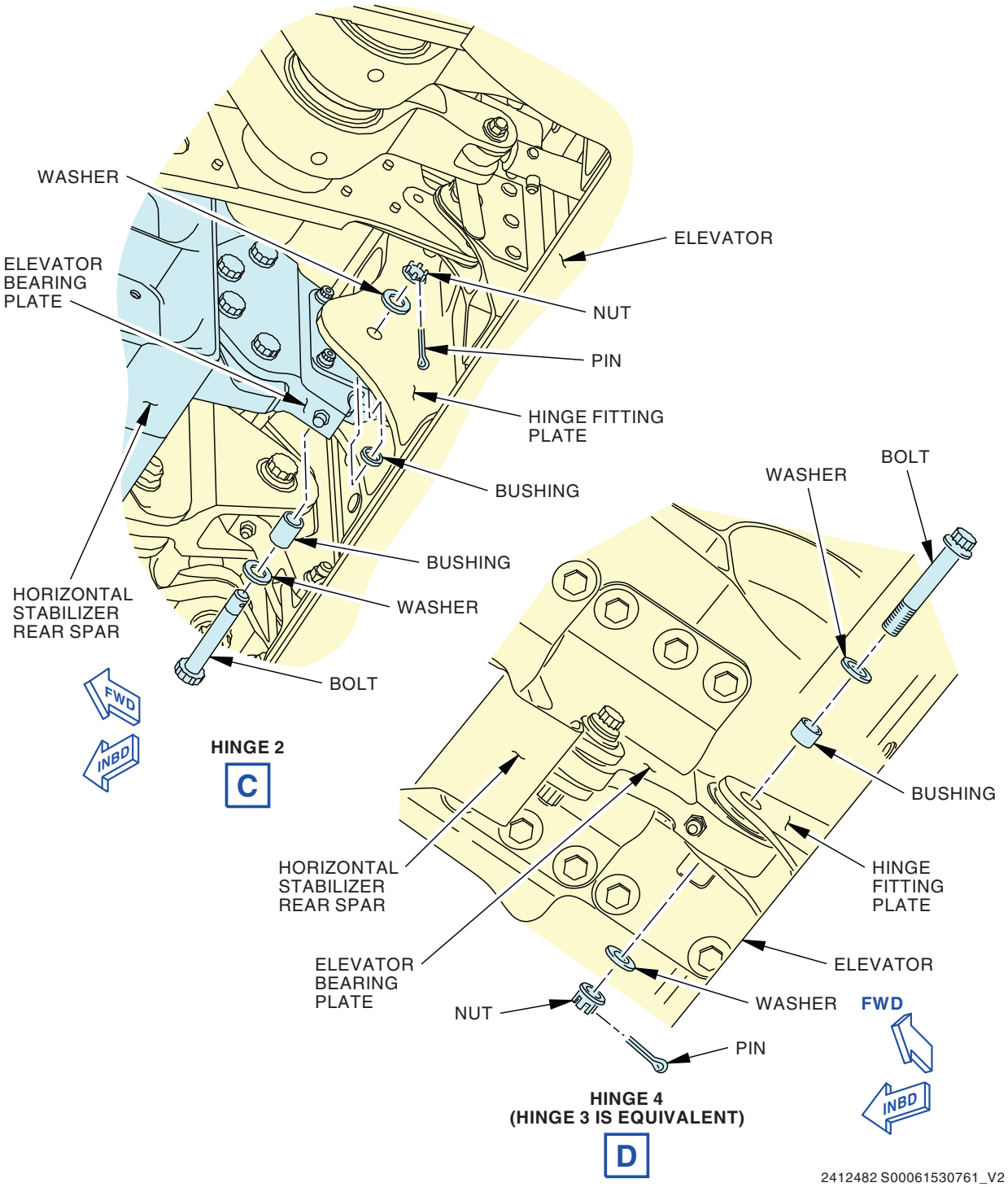
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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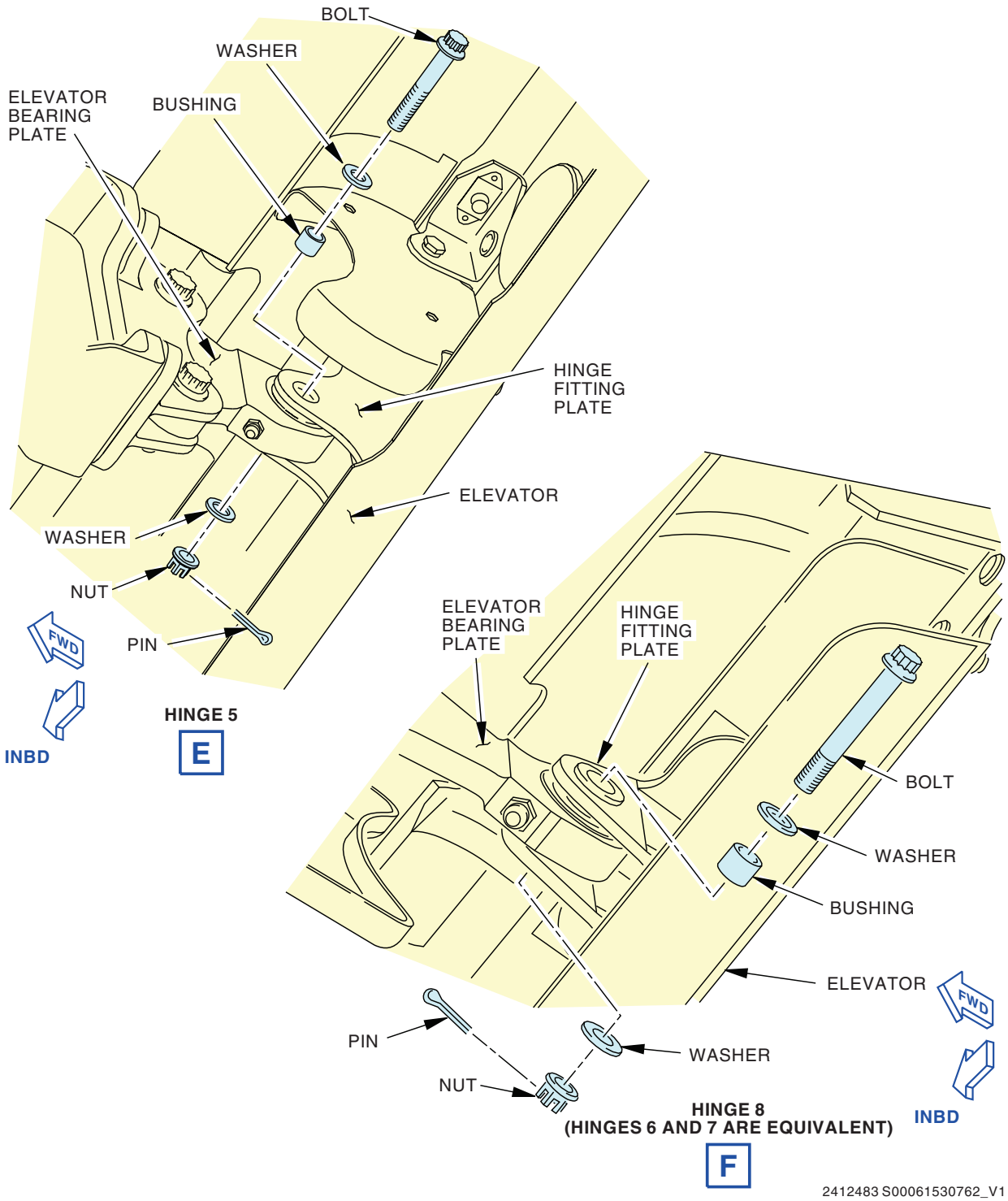
Elevator Hinge Assemblies
Figure 518/27-31-00-990-825 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



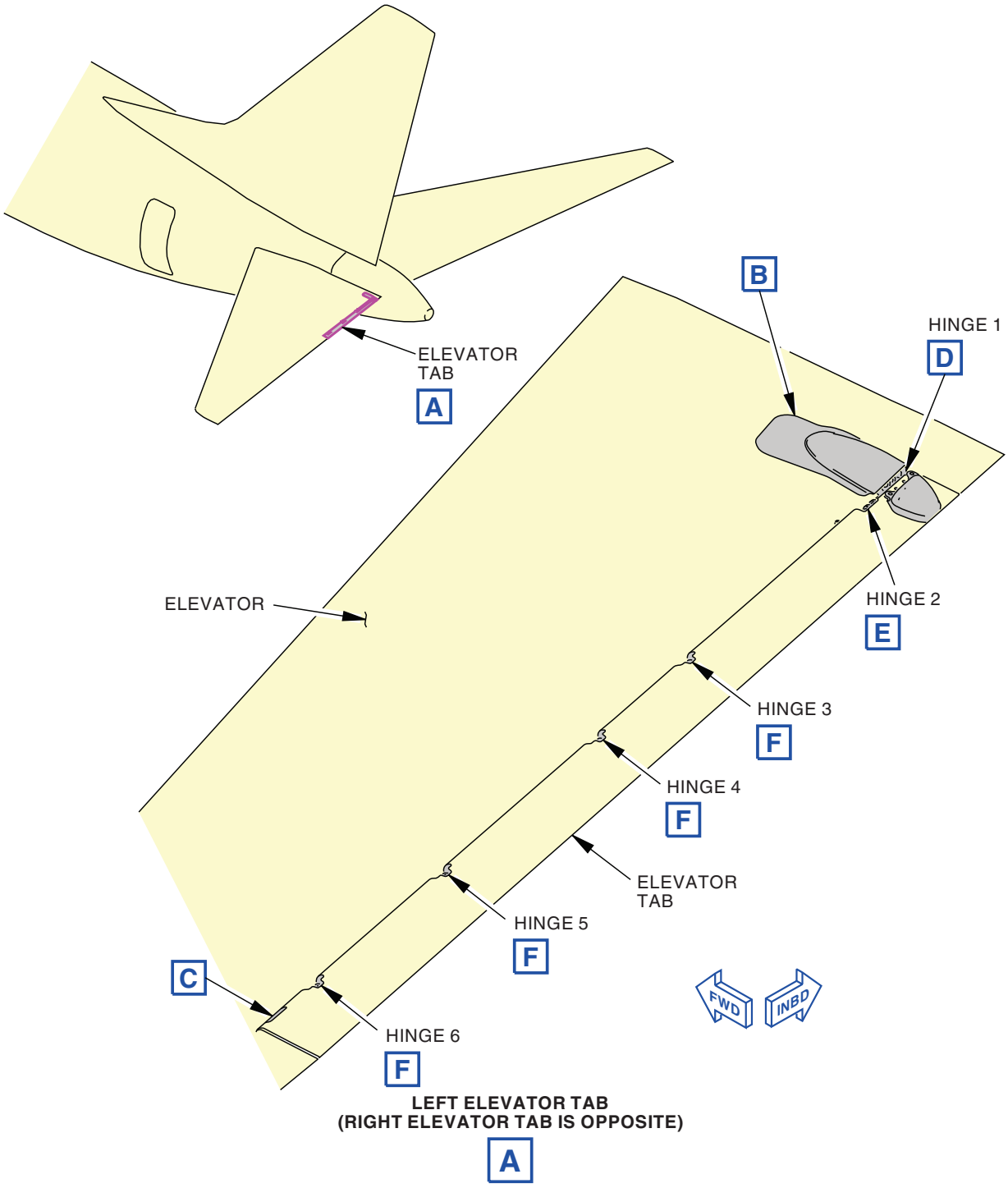
Elevator Hinge Assemblies
Figure 518/27-31-00-990-825 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



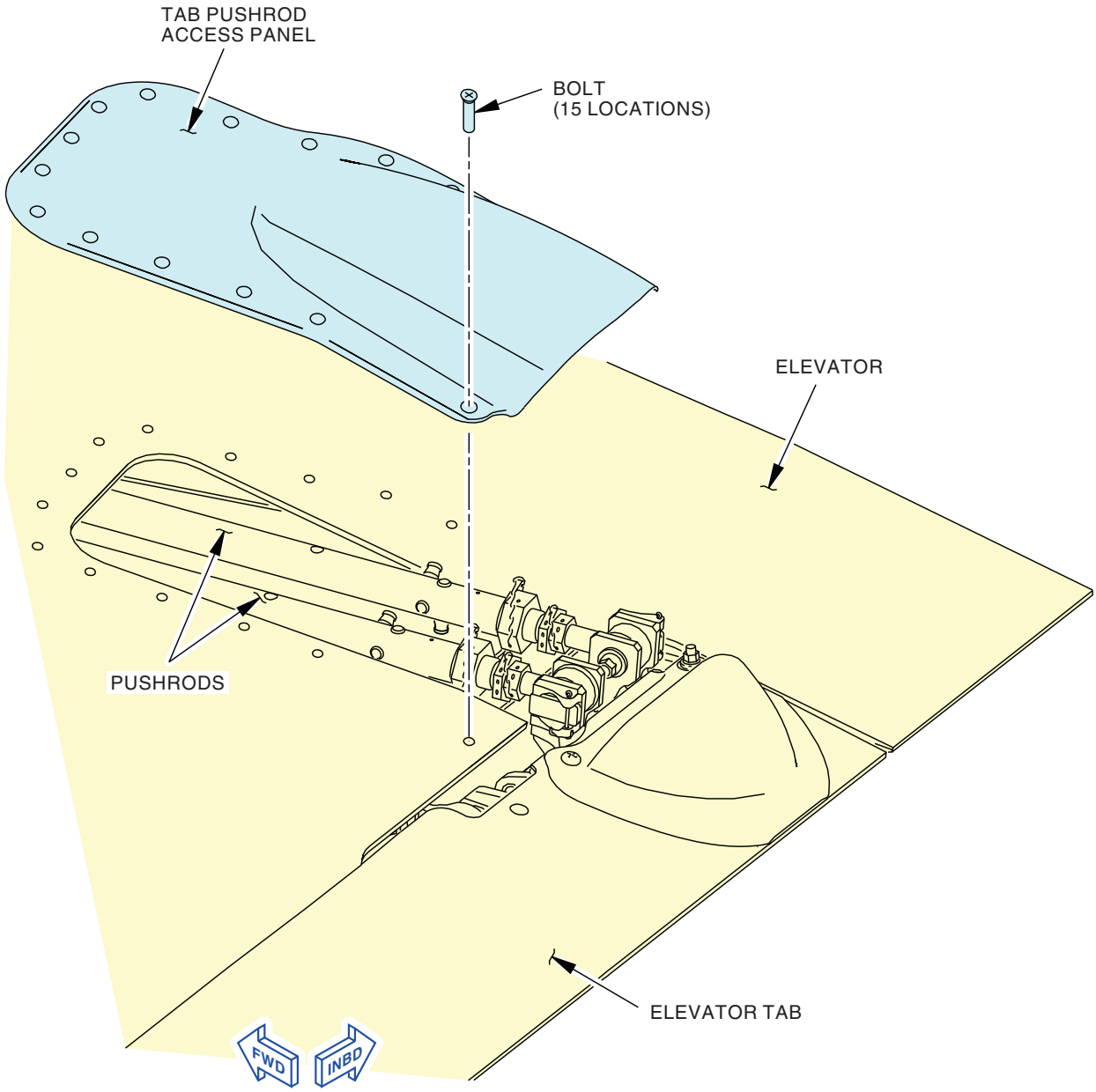
Elevator Hinge Assemblies
Figure 518/27-31-00-990-825 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



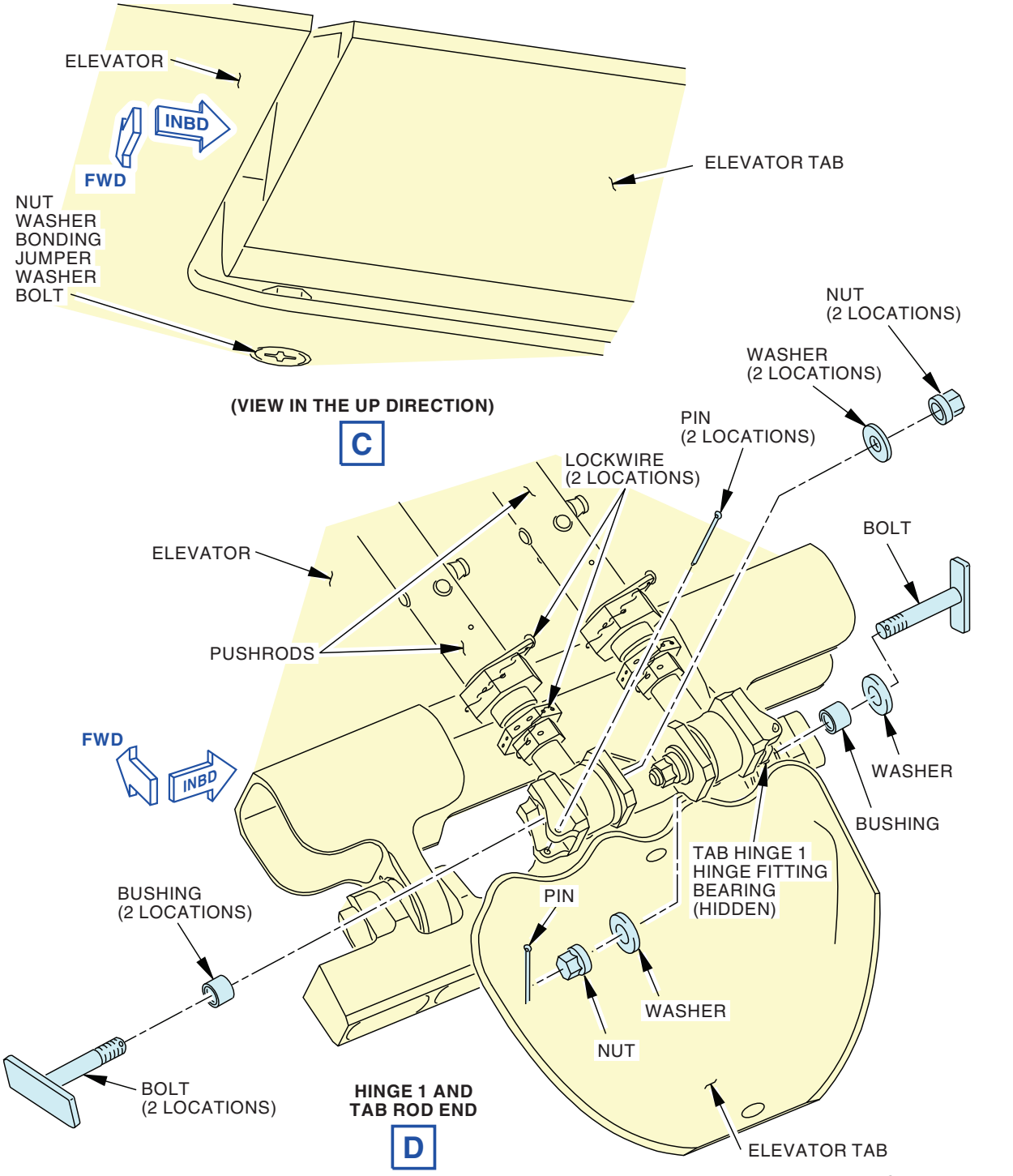
Elevator Tab Hinge Assemblies
Figure 519/27-31-00-990-826 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



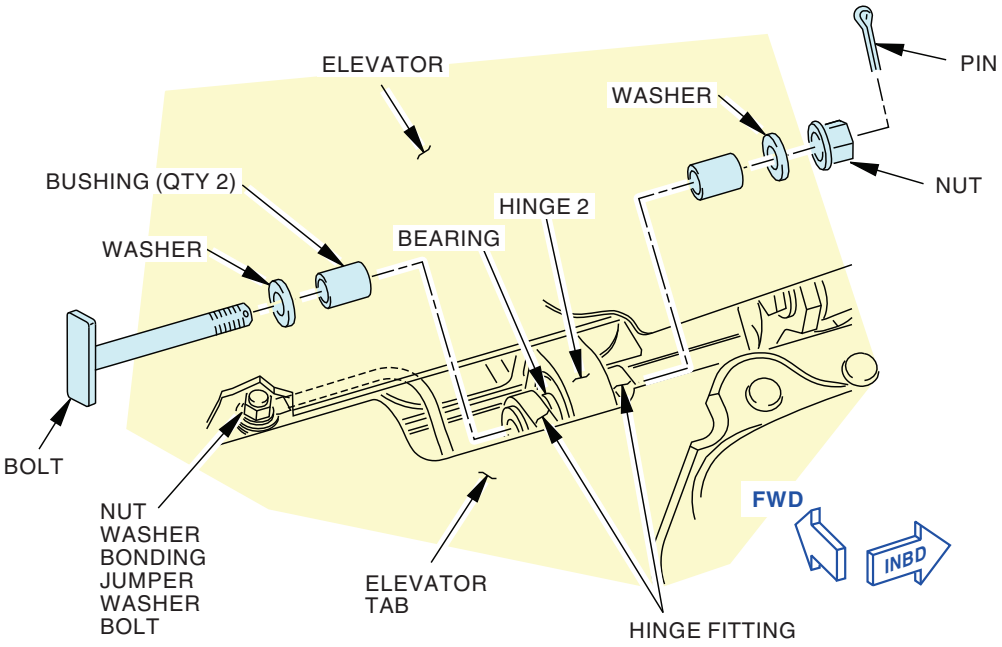
Elevator Tab Hinge Assemblies
Figure 519/27-31-00-990-826 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



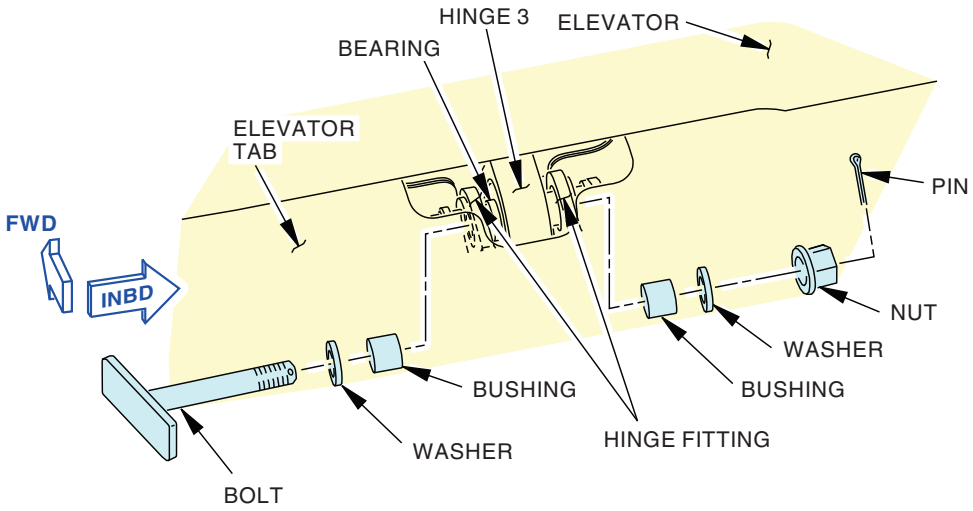
Elevator Tab Hinge Assemblies
Figure 519/27-31-00-990-826 (Sheet 3 of 4)

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HINGE 2

E



HINGE 3
(HINGES 4, 5, AND 6 ARE EQUIVALENT)

F

2412487 S00061530766_V2

Elevator Tab Hinge Assemblies
Figure 519/27-31-00-990-826 (Sheet 4 of 4)

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TASK 27-31-00-820-808

28. Elevator Tab Pushrod - Adjustment after the Elevator Power-Off Flight Test

(Figure 504, Figure 507, Figure 510, Figure 511, Figure 517)

NOTE: For each elevator tab, if both pushrods are adjusted, then the elevator power off flight test is necessary (TASK 27-31-00-710-803). If only one pushrod is adjusted and you can verify that the remaining rod has not been replaced or adjusted since the last flight test, then this flight test is not necessary. If only one pushrod is adjusted and you cannot verify that the remaining pushrod has not been replaced or adjusted since the last flight test, then the elevator power off flight test is necessary (TASK 27-31-00-710-803).

A. General

(1) Use this task to adjust the elevator tab if the number of turns of the stabilizer trim wheel during the elevator power off flight test are more than the allowable limits (Figure 510, Figure 517).

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-10345	Rigging Bar - Elevator Tab Part #: C27083-18 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

E. Elevator Tab Pushrod Adjustment

SUBTASK 27-31-00-970-003

(1) Get the number and direction (airplane nose up or airplane nose down) of stabilizer wheel turns necessary to trim the airplane during the power-off flight test.

SUBTASK 27-31-00-970-004

(2) Get the record of the center of gravity that was made during the power-off flight test.

SUBTASK 27-31-00-010-035

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-00-010-036

(4) For the left elevator tab pushrods, remove this access panel:

Number	Name/Location
334PT	Horizontal Stabilizer, Tab Control Rod Fairing

SUBTASK 27-31-00-010-037

(5) For the right elevator tab pushrods, remove this access panel:

Number	Name/Location
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-00-480-012

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(6) Use the trammel bar, SPL-1677, to set the "B" dimension to 39.17 ±0.01 in. (994.92 ±0.26 mm) (Figure 504).

NOTE: The “B” dimension must be set with the trammel trammel bar, SPL-1677, and not the stabilizer indicator in the flight deck.

SUBTASK 27-31-00-980-003

(7) Push the elevator trailing edge full up by hand.

SUBTASK 27-31-00-820-015

(8) When you move the elevator down from the trailing edge up position, align the elevator with the index mark in ±0.04 in. (1.0 mm), or less (Figure 507).

NOTE: Keep the elevators at this position when you adjust the tab pushrods. If the elevator position is changed, you must put the elevator trailing edge at full up position and do this step again before you continue.

SUBTASK 27-31-00-480-013

(9) Install the rigging bar, SPL-10345 (Figure 511).

(a) Install the rigging bar, SPL-10345, below the elevator and elevator tab at the inboard side of the third elevator tab hinge.

1) Make sure that the stand offs for the rigging bar, SPL-10345, rest on the skin fastener heads to prevent skin variation.

(b) Make sure that the cross bar on the rigging bar, SPL-10345, touches the lower surface of the elevator trailing edge.

(c) Push the rigging bar, SPL-10345, FWD until elevator stop indexes on the AFT edge of elevator near elevator rear spar.

SUBTASK 27-31-00-820-016

(10) Adjust the tab pushrods [110] using the elevator tab rig adjustment chart (Figure 517).

NOTE: If necessary, the example below shows how to use the elevator tab rig adjustment chart.

NOTE: An example of how to use the elevator tab rig adjustment chart is shown with the dashed lines with arrows.

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The example assumes 10.5 airplane nose down handwheel turns are needed to retrim the airplane with a center of gravity of 20 percent MAC on the left chart. Follow the diagonal lines to the left side of the right chart. Make a horizontal line across the chart. Draw a line down from the point where the horizontal line intersects the thick diagonal line on the right chart.

The example shows that you will need to adjust the elevator tab to 0.11 in. (2.8 mm) trailing edge up from the current position to bring the airplane to a neutral elevator tab trim.

NOTE: For coarse adjustment, a half turn of the tab rod end results in approximately 0.10 in. (2.5 mm) at the tab trailing edge (Figure 510).

For finer adjustment, one full turn of the bushing [104] (vernier collar) results in approximately 0.02 in. (0.5 mm) at the elevator tab trailing edge.

The number of threads showing on the pushrod ends should be approximately equal to the number of threads showing on the bushing [104].

The elevator tab should be adjusted to the nearest 0.01 in. (0.3 mm) shown on the scale at the bottom of the right chart of the elevator tab rig adjustment chart.

SUBTASK 27-31-00-820-017

(11) If it is necessary, do this coarse adjustment of the elevator tab pushrods (Figure 510, Figure 517):

(a) Remove the pins [111], nuts [109], washers [108], bolts [106] and bushings [107] from the tab pushrods [110].

(b) Turn the tab pushrod ends until the adjustment dimension is within 0.1 in. (3 mm).

NOTE: One half turn of the tab pushrod end is equal to 0.10 in. (2.5 mm) at the elevator tab trailing edge.

(c) Install the bushings [107], bolts [106], washers [108] and new nuts [109] to connect the tab pushrods [110] to the elevator tab [102].

NOTE: If you loosened or removed the nuts, they need to be replaced and not re-installed. Using new nuts will ensure that they have the required run-on torque of 3.5 in-lb (0.40 N·m).

NOTE: The inboard bolt is installed with the head inboard and outboard bolt is installed with the head outboard.

(d) Install the rigging bar, SPL-10345 (Figure 511).

1) Install the rigging bar, SPL-10345, below the elevator and elevator tab at the third elevator tab hinge (from inboard).

a) Make sure that the stand offs for the rigging bar, SPL-10345, rest on the skin fastener heads to prevent skin variation.

2) Make sure that the cross bar on the rigging bar, SPL-10345, touches the lower surface of the elevator trailing edge.

3) Push the rigging bar, SPL-10345, FWD until elevator stop indexes on the AFT edge of elevator near elevator rear spar.

4) Slide step gage FWD under elevator tab to obtain reading.

(e) Make sure that the adjustment measurement is within 0.1 in. (3 mm).

(f) Tighten the new nuts [109] to 70 in-lb (7.9 N·m) - 80 in-lb (9.0 N·m).

SUBTASK 27-31-00-820-018

(12) Do this fine adjustment of the elevator tab pushrods (Figure 510):

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(a) Turn the bushing [104] to adjust the length of the tab pushrods [110].

NOTE: One half turn of the bushing is equal to 0.01 in. (0.3 mm) at the elevator tab trailing edge.

(b) Loosen the nuts [109] that attach the tab pushrods [110] to the elevator tab [102].

(c) Remove the pins [111] from in front of the head of the bolts [106].

(d) Move the rectangle bolt heads out of the anti-rotation bolt bosses and make sure that you can easily turn the bolts [106] by hand.

1) If it is necessary, adjust the bushing [104] on the tab pushrods [110].

NOTE: One half turn of the bushing/rod end assembly is equal to 0.11 in. (2.8 mm).

CAUTION

TIGHTEN THE NUTS [109] CORRECTLY. IF YOU TIGHTEN THEM INCORRECTLY THEN THE PUSHRODS CAN BECOME DISCONNECTED. THIS CAN CAUSE DAMAGE TO THE ELEVATOR, AND STABILIZER.

(e) Tighten the new nuts [109] to 70 in-lb (7.9 N·m) - 80 in-lb (9.0 N·m) and install pins [111].

NOTE: If you loosened or removed the nut, they need to be replaced and not re-installed. Using new nuts will ensure that they have the required run-on torque of 3.5 in-lb (0.40 N·m).

(f) Tighten the checknuts [103] and checknuts [105] on the tab pushrods [110] to 150 in-lb (16.9 N·m) - 200 in-lb (22.6 N·m).

SUBTASK 27-31-00-220-004

(13) Make sure that the adjustment measurement is the same.

(a) Install the rigging bar, SPL-10345 (Figure 511).

1) Install the rigging bar, SPL-10345, below the elevator and elevator tab at the third elevator tab hinge (from inboard).

a) Make sure that the stand offs for the rigging bar, SPL-10345, rest on the skin fastener heads to prevent skin variation.

2) Make sure that the cross bar on the rigging bar, SPL-10345, touches the lower surface of the elevator trailing edge.

3) Push the rigging bar, SPL-10345, FWD until elevator stop indexes on the AFT edge of elevator near elevator rear spar.

4) Slide step gage FWD under elevator tab to obtain reading.

(b) Make sure that the adjustment measurement is the same.

SUBTASK 27-31-00-220-005

(14) Do a check of the thread engagement of the tab pushrod ends and bushings [104] (Figure 510):

(a) Make sure that the end of the bushing [104] covers the inspection hole on each tab pushrod [110].

(b) Make sure that the inspection hole of each tab pushrod end bearing is not shown.

(c) If the thread engagement is not correct, you must do the coarse adjustment again (Figure 517).

SUBTASK 27-31-00-430-003

(15) Install lockwire on the checknuts [103] and checknuts [105] (Figure 510).

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SUBTASK 27-31-00-410-028

(16) For the right elevator tab pushrods, install this access panel:

Number	Name/Location
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-00-410-029

(17) For the left elevator tab pushrods, install this access panel:

Number	Name/Location
334PT	Horizontal Stabilizer, Tab Control Rod Fairing

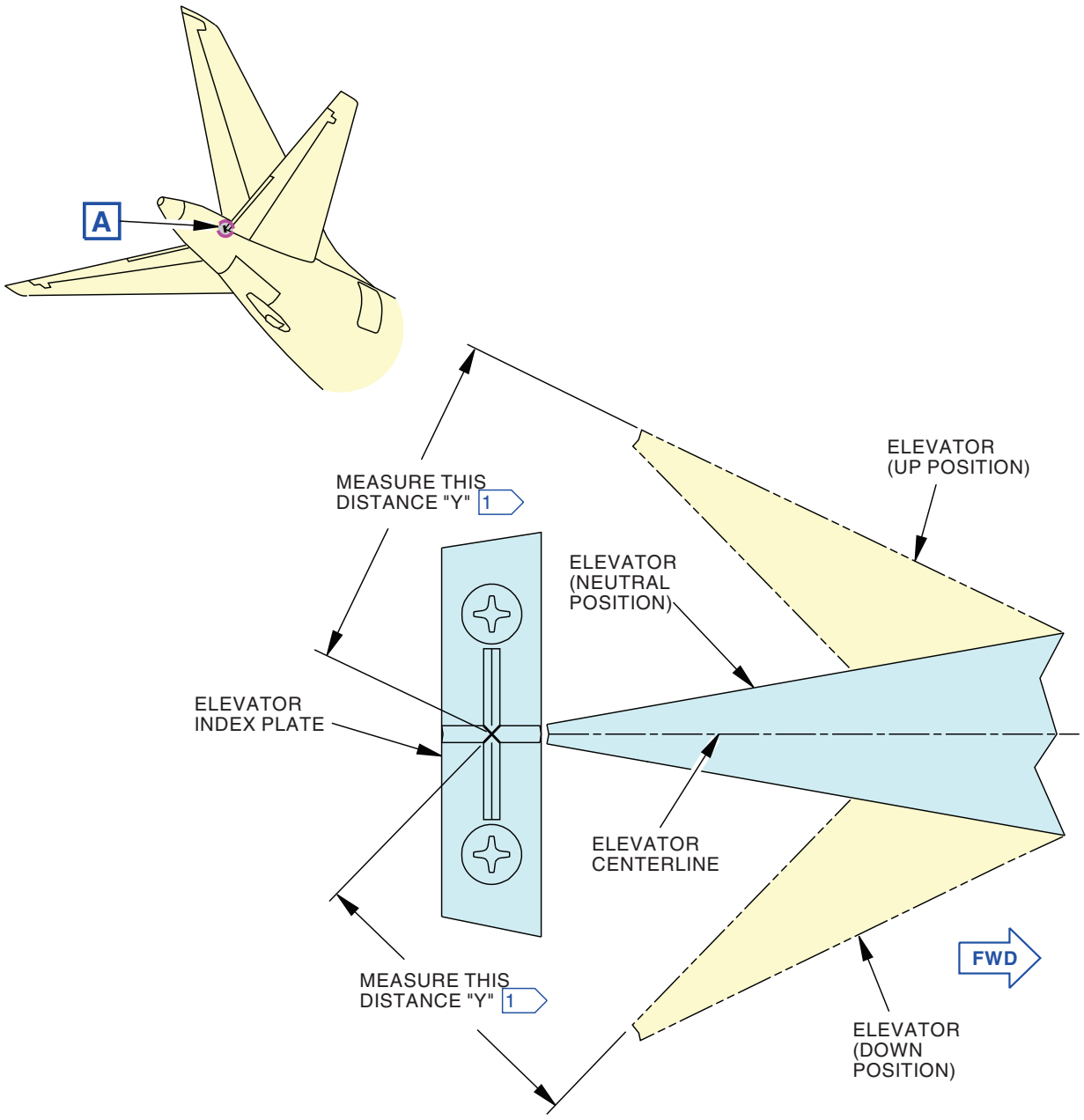
SUBTASK 27-31-00-410-030

(18) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

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1 THE DISTANCE "Y" IS MEASURED FROM THE MIDDLE OF THE ELEVATOR INDEX PLATE TO THE UPPER SURFACE OF THE INBOARD EDGE OF THE ELEVATOR.

MEASUREMENT LOCATION (RIGHT SIDE)

A

2412488 S00061530768_V1

Elevator Full Travel Measurements
Figure 520/27-31-00-990-827

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TASK 27-31-00-710-804

29. Elevator Jam Landing Assist - Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this test to do the operational tests on the Elevator Jam Switch, S1214, and ASSIST ON light, L1383.

B. References

Reference	Title
27-61-45-730-801	Spoiler Control Electronics Self Test (P/B 501)
31-65-00-970-801	System Maintenance Data Pages Function (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 27-31-00-710-010

(1) On the MAX Display System (MDS), go to the 27 FLIGHT CONTROLS display in the MAINT DATA PGS (TASK 31-65-00-970-801).

(a) Make sure that all four GEAR ON GRND values read “NEAR”.

SUBTASK 27-31-00-710-011

(2) On the Onboard Maintenance Function (OMF), go to the EXISTING FAULTS page for Chapter 27, Spoiler Control (TASK 46-13-02-700-805).

(a) Make sure that 27-0266X RADIO ALT DATA INVALID is NOT set.

1) If 27-0266X RADIO ALT DATA INVALID is SET, run the SCE TEST to clear the fault (TASK 27-61-45-730-801).

a) Do the steps in the Prepare for the Test again.

b) Do not proceed until the fault has been cleared.

E. Elevator Jam Landing Assist Test

SUBTASK 27-31-00-860-156

(1) Do these steps to check configuration of ELEVATOR JAM LANDING ASSIST switch:

(a) On the aft P5 panel, lift the ELEVATOR JAM LANDING ASSIST switch guard and attempt to put the switch to the middle detent position.

NOTE: The ELEVATOR JAM SWITCH can have a middle detent position that does not activate the elevator jam landing assist function.

(b) If the switch has a middle detent position, replace this switch with a two position switch PN 64AT34-3 (SWPM 20-83-11).

SUBTASK 27-31-00-720-001

(2) Do these steps to test the ASSIST ON, L1383, light indicator:

(a) On the P5 panel, push and hold the ASSIST ON, L1383, light indicator.

1) Make sure that the light comes on.

(b) Release the ASSIST ON light indicator.

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1) Make sure that the light turns off.

(c) If the light does not come on, refer to the IFIM and do the applicable procedure(s).

(d) Re-test per initial test procedures.

1) Do the test of the ASSIST ON, L1383, light indicator again.

SUBTASK 27-31-00-710-012

(3) Do these steps to test the ELEVATOR JAM LANDING ASSIST switch:

(a) Put the ELEVATOR JAM LANDING ASSIST switch to the ON position (full forward).

(b) Make sure that the ASSIST ON light comes on.

SUBTASK 27-31-00-710-013

(4) While the switch is in the ON position, do these steps:

(a) Open the EXISTING FAULTS for Chapter 27, Spoiler Control, using the OMF (TASK 46-13-02-700-805).

(b) Make sure that the ELEV JAM SW FAULT, 27-0016X does not show.

SUBTASK 27-31-00-860-137

(5) Put the ELEVATOR JAM LANDING ASSIST switch to the OFF position.

(a) Make sure that the ASSIST ON light turns off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the elevator

(2) An installation of the elevator

(3) A removal of the elevator control pushrod

(4) An installation of the elevator control pushrod.

B. When it is necessary to balance the elevator, make sure you do a check of the elevator and the elevator tab part numbers. Refer to Chapter 51 of the Structural Repair Manual when you balance the elevator-elevator tab combination after the structural repairs or after you put on new paint.

C. If you install a tab on the elevator during the elevator installation, make sure the number of the elevator control tab adjust weights agree with the tab data plate.

TASK 27-31-11-000-801

2. Elevator - Removal

(Figure 401, Figure 402, Figure 403, Figure 404, Figure 405, Figure 406)

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-32-200-802	Elevator Tab Freeplay - Check (P/B 601)
53-53-00-000-801	Tailcone Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-2034	Sling Equipment - Elevator Removal/Installation Part #: C27034-54 Supplier: 81205 Opt Part #: C27034-44 Supplier: 81205

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right
334	Left Horizontal Stabilizer - Elevator
335	Left Horizontal Stabilizer - Tip
344	Right Horizontal Stabilizer - Elevator
345	Right Horizontal Stabilizer - Tip

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D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
333CB	Horizontal Stabilizer, Access Panel, Trailing Edge
333DB	Horizontal Stabilizer, Access Panel, Trailing Edge
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334BB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334DB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334EB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334FB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
334HB	Horizontal Stabilizer, Elevator Hinge Cover
334JB	Horizontal Stabilizer, Elevator Hinge Cover
334KB	Horizontal Stabilizer, Elevator Hinge Cover
334LB	Horizontal Stabilizer, Elevator Hinge Cover
334MB	Horizontal Stabilizer, Elevator Hinge Cover
334NB	Horizontal Stabilizer, Elevator Hinge Cover
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
343CB	Horizontal Stabilizer, Access Panel - T.E. Area
343DB	Horizontal Stabilizer, Access Panel - T.E. Area
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344BB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344DB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344EB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344FB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09
344HB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 66.54
344JB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 121.59
344KB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 176.64
344LB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 213.32
344MB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 250.04
344NB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 265.45

E. Prepare for the Removal

SUBTASK 27-31-11-860-001

(1) Make sure that pressure is removed from the elevator hydraulic systems A and B, do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-11-860-003

CAUTION MAKE SURE TO PREVENT DAMAGE TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CHAF OR CAN CAUSE DAMAGE TO THE WIRES OR BUNDLE INSULATION.

(2) Use the trammel bar, SPL-1677, to set “B” dimension at 4 units of trim 39.17 ±0.01 in. (994.92 ±0.26 mm) (Figure 401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-11-820-001

(3) Measure and record the elevator tab position.

NOTE: Measuring and recording the elevator tab position before you remove the tab may improve the manual reversion success rate if the tab needs to be adjusted after installation.

(a) If the same tab will be re-installed, do these steps before removing the tab:

1) Make sure that the elevator tab meets the freeplay limits, do this task: Elevator Tab Freeplay - Check, TASK 27-31-32-200-802.

(b) While you move the elevator trailing edge down, align the projected center of the elevator trailing edge with the index mark within ±0.04 in. (1.02 mm), or less.

1) Measure and record the elevator tab position.

SUBTASK 27-31-11-000-001

(4) Do this task: Tailcone Removal, TASK 53-53-00-000-801.

SUBTASK 27-31-11-010-001

(5) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-11-010-003

(6) For the left elevator, remove these access panels:

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
333CB	Horizontal Stabilizer, Access Panel, Trailing Edge
333DB	Horizontal Stabilizer, Access Panel, Trailing Edge

SUBTASK 27-31-11-010-004

(7) For the right elevator, remove these access panels:

Number	Name/Location
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
343CB	Horizontal Stabilizer, Access Panel - T.E. Area
343DB	Horizontal Stabilizer, Access Panel - T.E. Area

SUBTASK 27-31-11-010-005

(8) For the left elevator, open these access panels:

Number	Name/Location
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334BB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334DB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334EB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334FB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
334HB	Horizontal Stabilizer, Elevator Hinge Cover
334JB	Horizontal Stabilizer, Elevator Hinge Cover
334KB	Horizontal Stabilizer, Elevator Hinge Cover
334LB	Horizontal Stabilizer, Elevator Hinge Cover
334MB	Horizontal Stabilizer, Elevator Hinge Cover
334NB	Horizontal Stabilizer, Elevator Hinge Cover

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SUBTASK 27-31-11-010-006

(9) For the right elevator, open these access panels:

Number	Name/Location
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344BB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344DB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344EB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344FB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09
344HB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 66.54
344JB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 121.59
344KB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 176.64
344LB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 213.32
344MB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 250.04
344NB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 265.45

SUBTASK 27-31-11-860-002

(10) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

SUBTASK 27-31-11-480-005

(11) Install the rig pin E-5 from rigging pin kit, SPL-1585, in the elevator aft quadrant (Figure 402).

F. Elevator Removal

SUBTASK 27-31-11-020-001

(1) Put the elevator tab control mechanism [22] in a neutral position.

SUBTASK 27-31-11-420-010

(2) Attach the elevator tab control mechanism [22] to the elevator with clamps.

NOTE: This step prevents the rods from scraping the elevator surface.

SUBTASK 27-31-11-020-010

(3) Remove the elevator connections to the elevator tab control mechanism [22] (Figure 404).

(a) Remove the screws [17] and washers [25] from the pin retainer [18] and rotate the pin retainer [18] arms away from the bolt [12] heads.

(b) Remove the nut [19], washer [20], bushing [21], and pin assembly [11] that attach the left aft attach lug to the elevator [1].

(c) Remove the nut [19], washer [20], washers [23], and pin assembly [11] that attach the right aft attach lug to the elevator [1].

(d) Remove the nuts [16], washers [15], bolts [12], and spacers [13] that attach the tab pushrods [14] to the elevator tab control mechanism [22].

(e) Do not remove the elevator mast arms [91] from the aft attach lugs.

SUBTASK 27-31-11-480-002

(4) If you do not use the sling equipment, SPL-2034, to remove the elevator, you must properly support the elevator while removing the hinge bolts.

SUBTASK 27-31-11-480-006

(5) Do these steps to install the sling equipment, SPL-2034, to the elevator [1] (Figure 403):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Find the bolts [3] on the elevator [1] to attach the sling equipment, SPL-2034.

NOTE: The bolts are on the upper panel next to elevator hinge 3 and hinge 5 (Figure 403).

(b) Remove the bolts [3] from the elevator [1].

(c) Install the forward sling fittings on the elevator [1].

1) Install the bolts [3] to attach each sling fitting to the elevator [1].

(d) Install the aft sling fitting on the elevator [1].

(e) Attach the straps from the forward sling fittings to the aft sling fitting.

(f) Attach the sling equipment, SPL-2034, to the forward sling fittings and aft sling fitting.

(g) Connect an overhead crane to the primary support sling on the sling equipment, SPL-2034.

(h) Connect a tether line to the aft sling fitting.

(i) Use the sling equipment, SPL-2034, to hold the elevator [1].

SUBTASK 27-31-11-020-002

(6) Disconnect the pushrod [38] of the elevator position transmitter [37] from the elevator [1] (Figure 405).

(a) Disconnect the electrical connector [39] from the elevator position transmitter [37].

(b) Remove the bolt [42], washer [41] and spacer [40] from the pushrod [38].

SUBTASK 27-31-11-020-003

(7) Remove the nut [31], nut [32], washer [33], bushing [34], and bolt assembly [35] to disconnect the elevator control pushrod [36] from the elevator [1].

SUBTASK 27-31-11-020-004

(8) Disconnect the balance panels from the elevator [1] (Figure 405).

(a) The bulb seal of the balance panel can be removed for easier removal of the elevator.

SUBTASK 27-31-11-000-002

(9) If it is necessary, remove the applicable bulb seal (Figure 406).

(a) Remove the bolts [84] and washers [85].

SUBTASK 27-31-11-020-005

(10) Remove the elevator [1] in the following order (Figure 405):

(a) At the elevator hinge 1, remove the pin [50], nut [49], washer [48], and bolt [47].

1) Discard the pin [50].

(b) At the elevator hinge 2, remove the pin [55], nut [54], washer [53], bolt [51], and bushing [52].

1) Discard the pin [55].

(c) At the elevator hinge 4, remove the pin [68], nut [67], washer [66], bolt [63], washer [64], and bushing [65].

1) Discard the pin [68].

(d) At the elevator hinge 5, remove pin [73], nut [74], washer [72], bolt [69], washer [70], and bushing [71].

1) Discard the pin [73].

(e) Remove the nuts [82] and washers [83] to disconnect the bonding jumpers [81] from the horizontal stabilizer [2].

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(f) At the elevator hinges 6, 7, and 8, remove the pins [80], nuts [79], washers [78], bolts [75], washers [76], and bushings [77].

1) If it is necessary, pull on the tether line to remove the bolts [75].

2) Discard the pins [80].

(g) At the elevator hinge 3, remove the lockwire, pin [62], nut [61], washer [60], bushing [59], bolt [56], and washer [57].

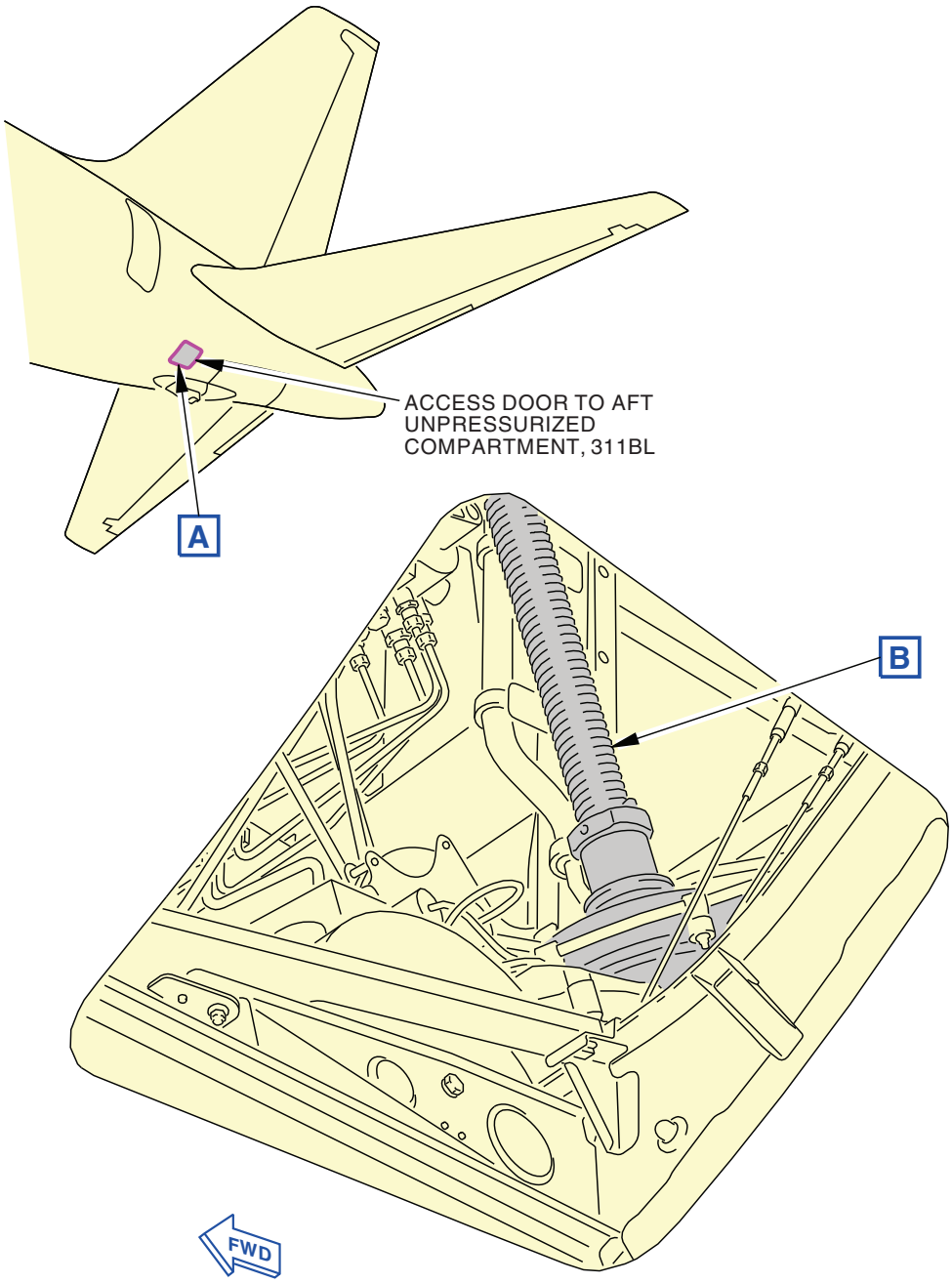
1) Discard the pin [62].

(h) Remove the lockwire from the bushing [58] at the hinge 3 and remove the bushing [58].

(i) Remove the elevator [1] from the airplane.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



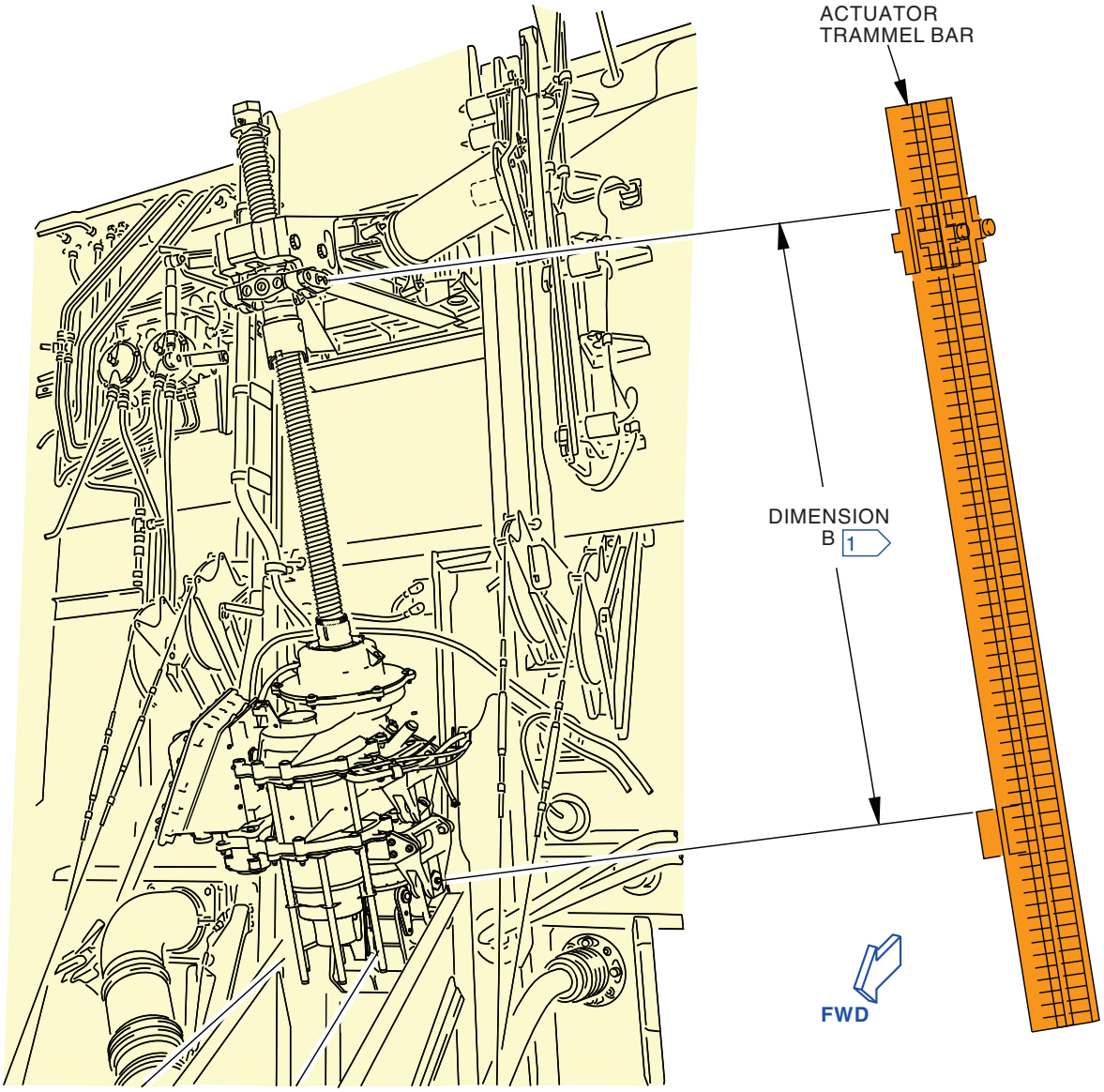
**VIEW THROUGH THE ACCESS DOOR TO
AFT UNPRESSURIZED COMPARTMENT, 311BL**

A

2412489 S00061530772_V1

**Stabilizer Trim Jackscrew Setting
Figure 401/27-31-11-990-801 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACTUATOR

B

NOTE:

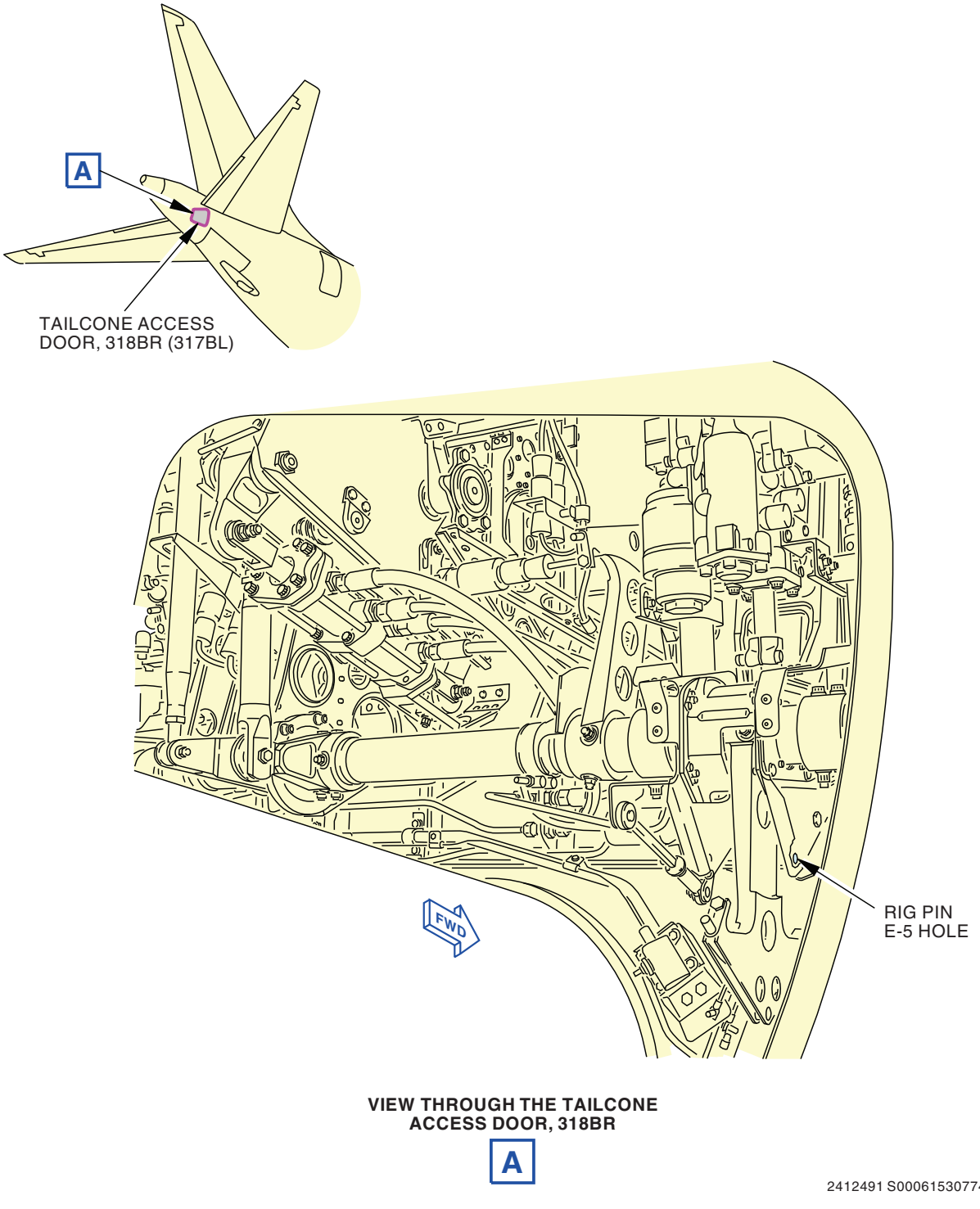
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

2412490 S00061530773_V2

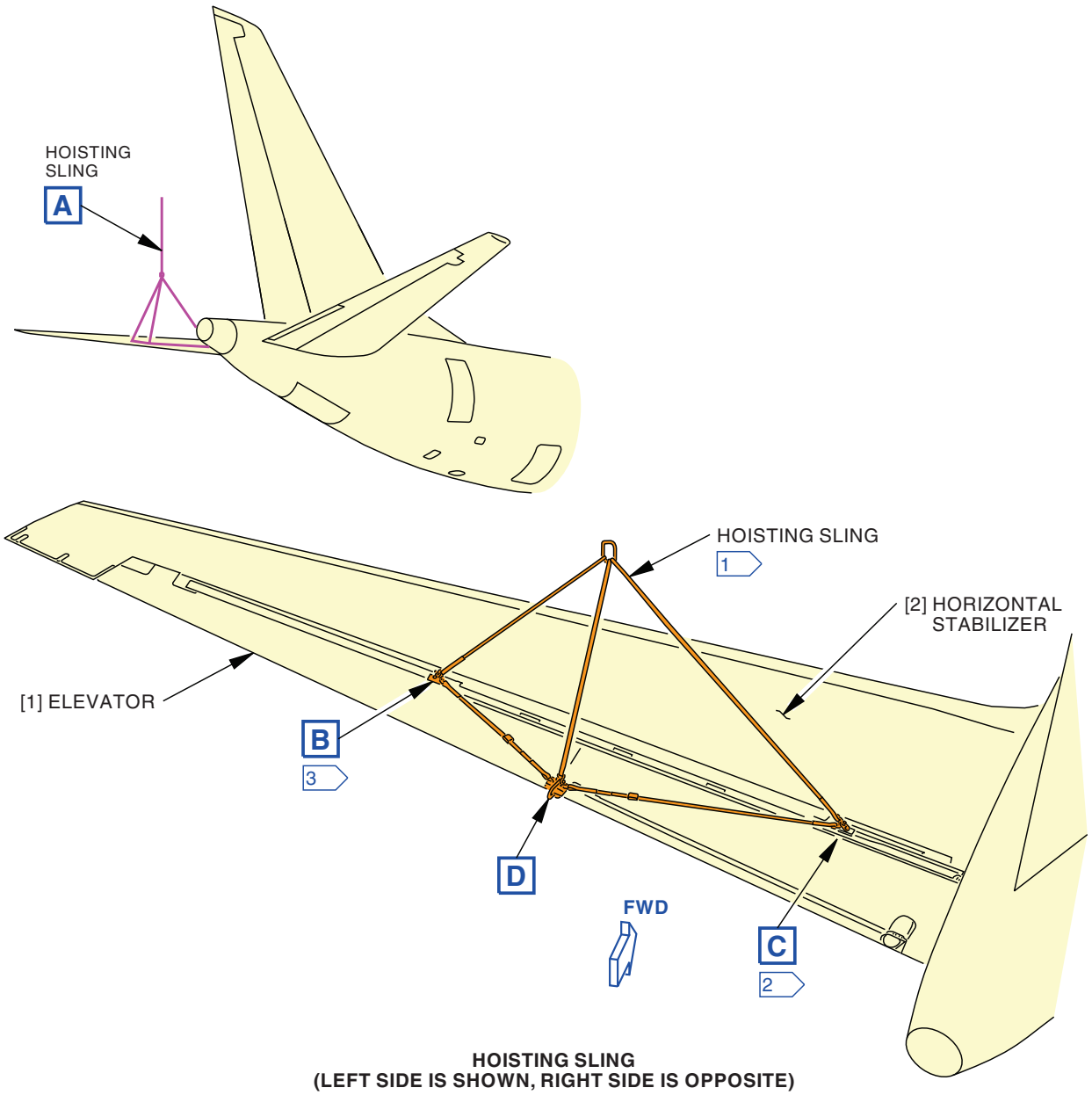
Stabilizer Trim Jackscrew Setting
Figure 401/27-31-11-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Rig Pin Installation
Figure 402/27-31-11-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HOISTING SLING
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A

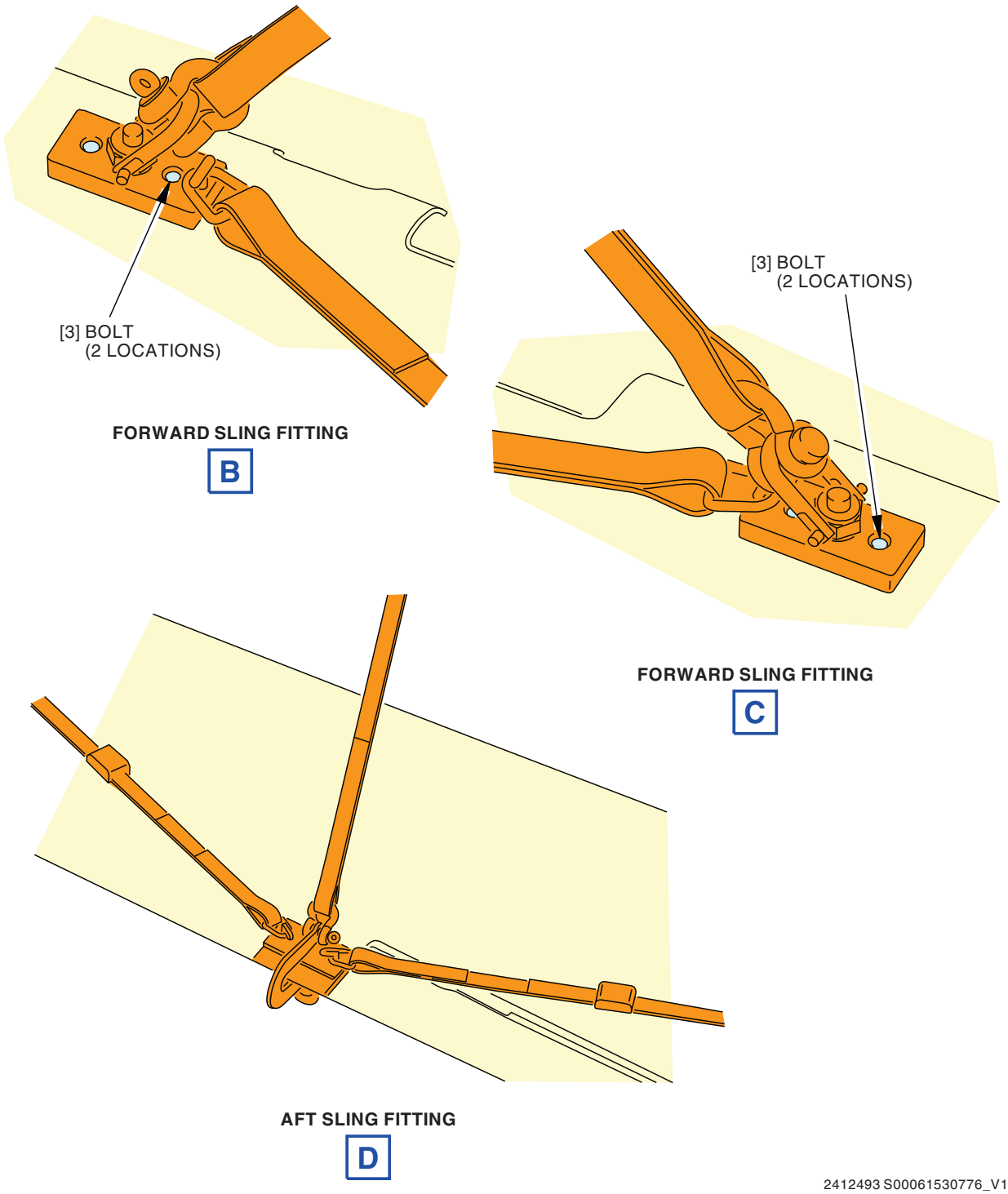
NOTE:

1. THIS SLING CAN BE USED TO FACILITATE THE HANDLING OF THE ELEVATOR.
2. THE INBOARD FORWARD SLING FITTING CAN BE FOUND NEXT TO HINGE 3.
3. THE OUTBOARD FORWARD SLING FITTING CAN BE FOUND NEXT TO HINGE 5.

2412492 S00061530775_V2

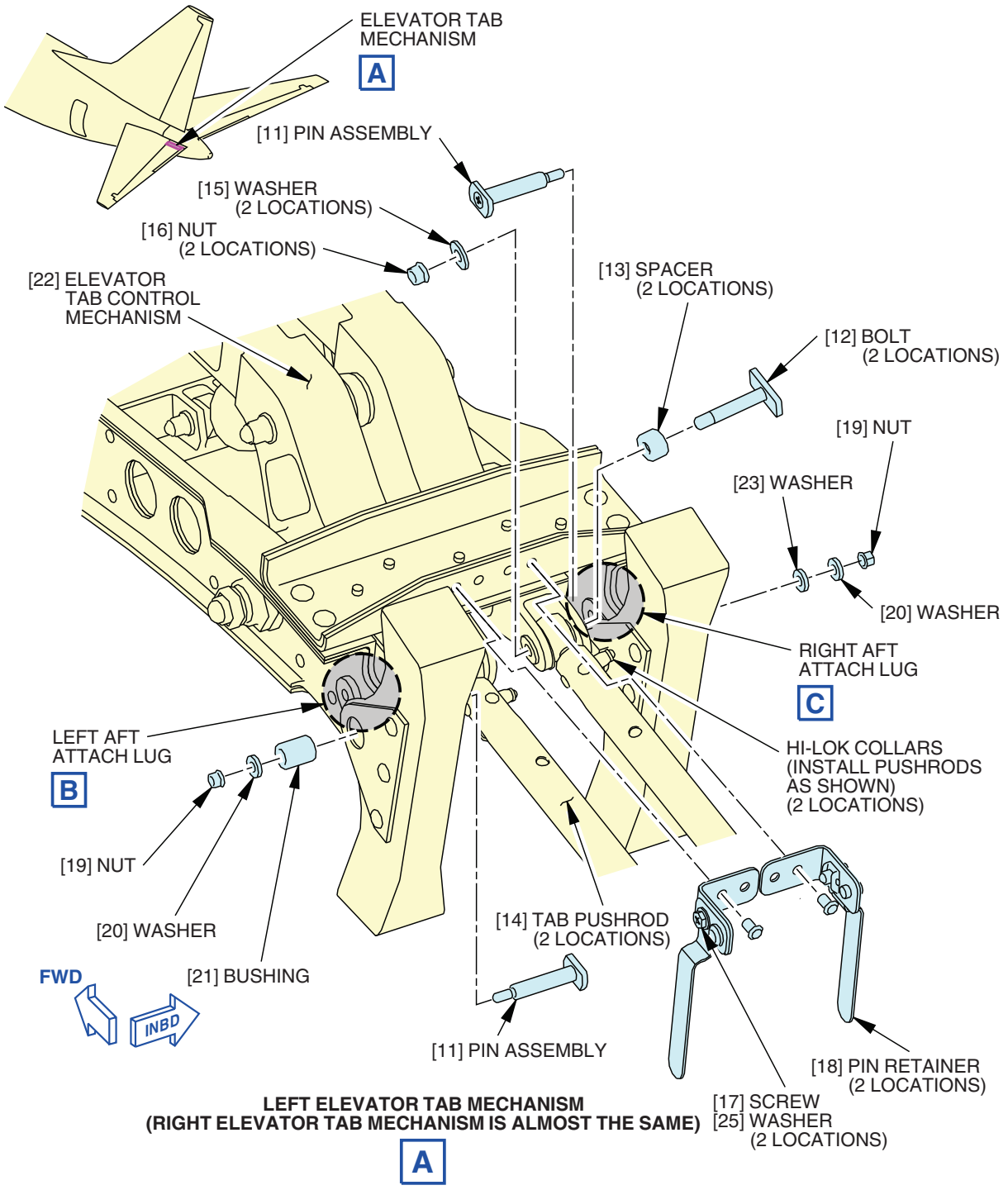
Elevator Hoisting Sling Installation
Figure 403/27-31-11-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



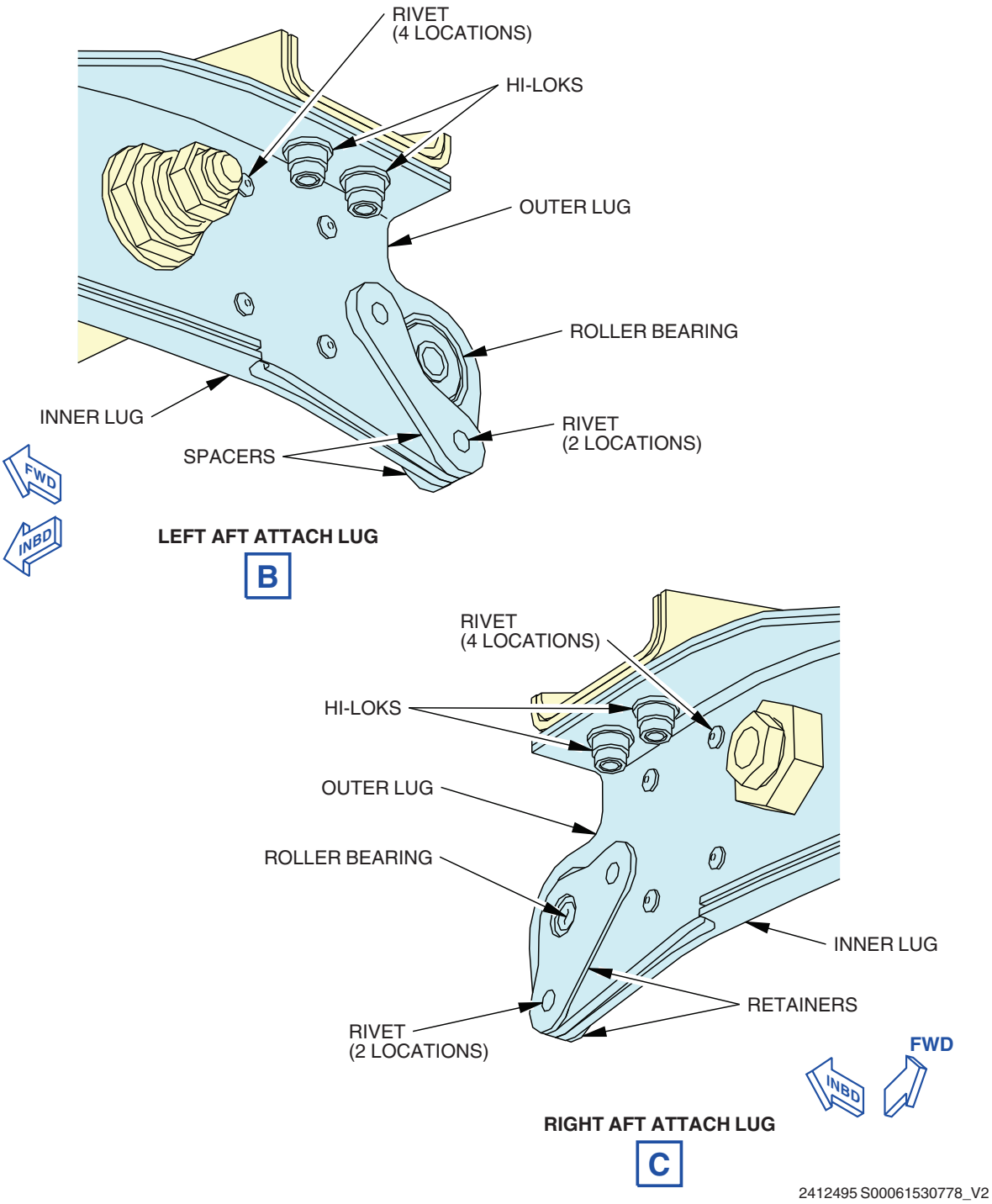
Elevator Hoisting Sling Installation
Figure 403/27-31-11-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



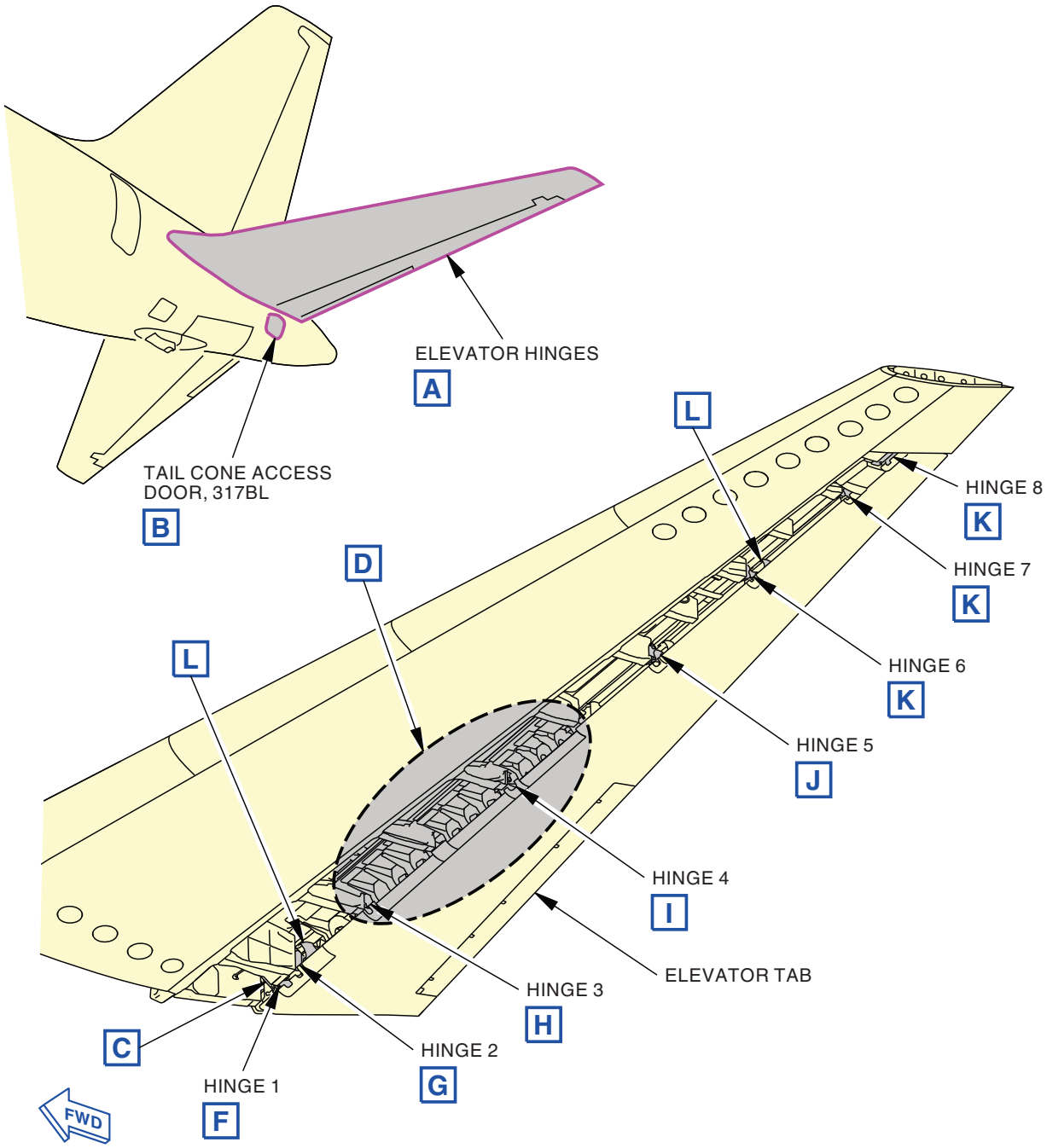
Elevator Tab Mechanism Installation
Figure 404/27-31-11-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Tab Mechanism Installation
Figure 404/27-31-11-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

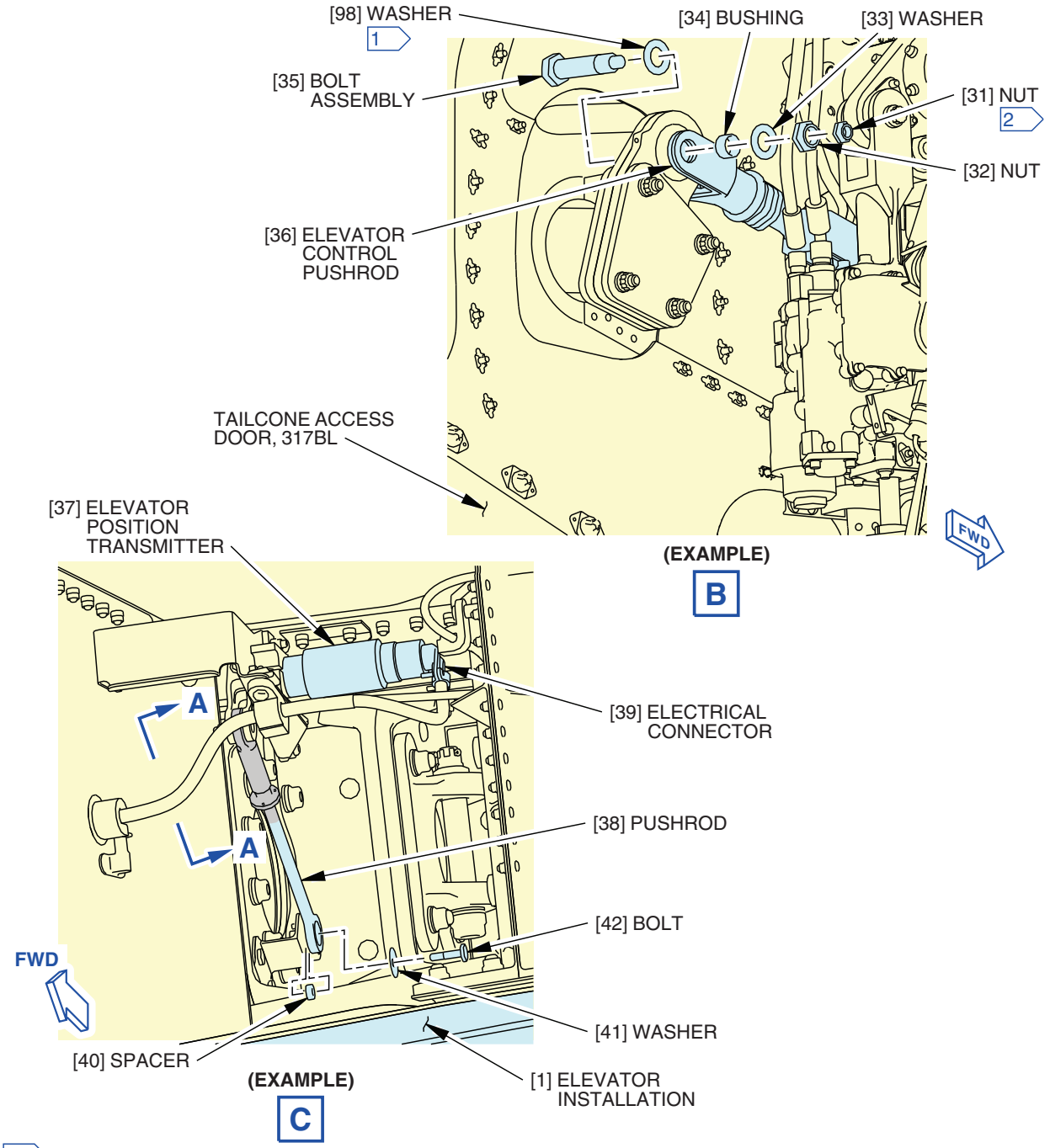


ELEVATOR HINGES
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)
A

2412496 S00061530779_V2

Elevator Installation
Figure 405/27-31-11-990-805 (Sheet 1 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



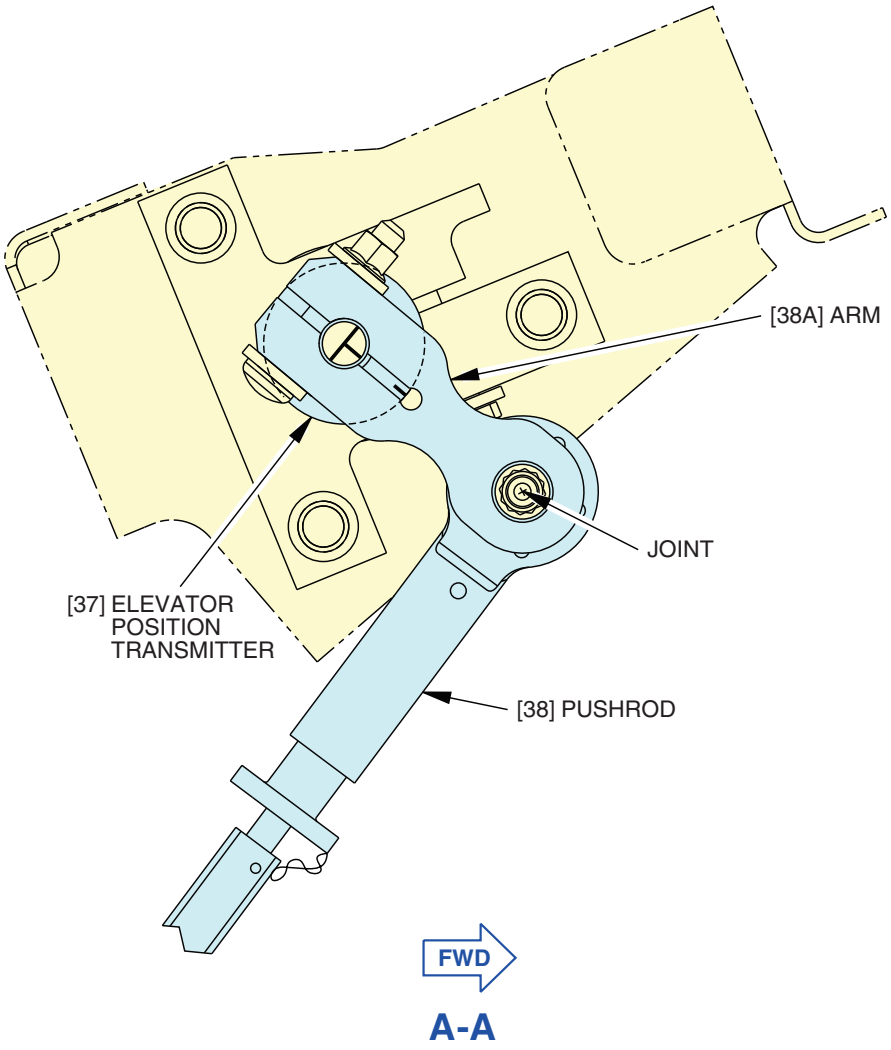
1 IF INSTALLED.

2 DO NOT TIGHTEN NUT [32] AFTER YOU TIGHTEN NUT [31]. THIS CAN CAUSE DAMAGE TO EQUIPMENT. BOLT PROTRUSION THROUGH NUT [31] MAY BE 0.01 INCH MINIMUM, FOR BOLT ASSEMBLY.

2412497 S00061530780_V5

Elevator Installation
Figure 405/27-31-11-990-805 (Sheet 2 of 8)

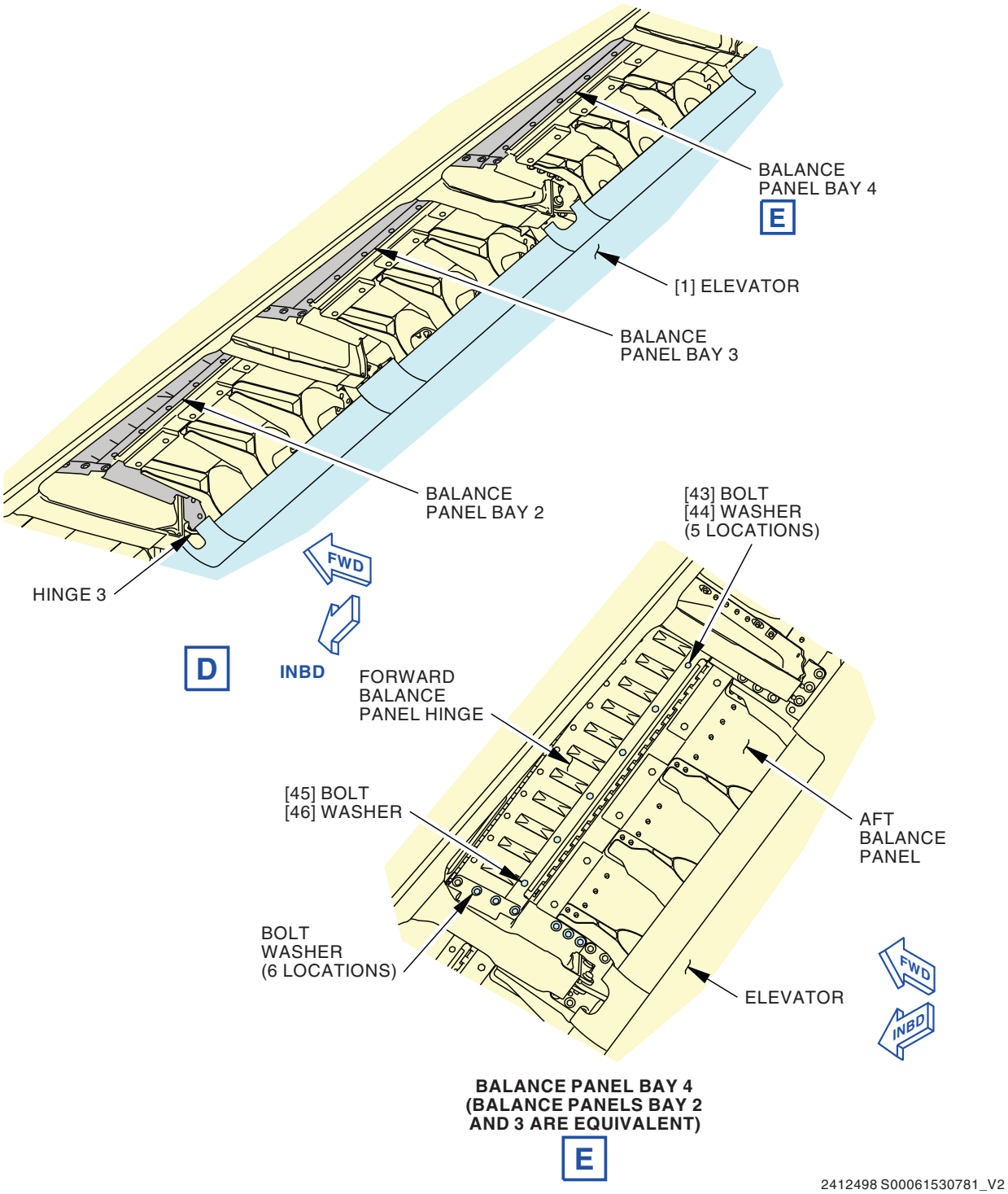
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3089115 S0000840890_V1

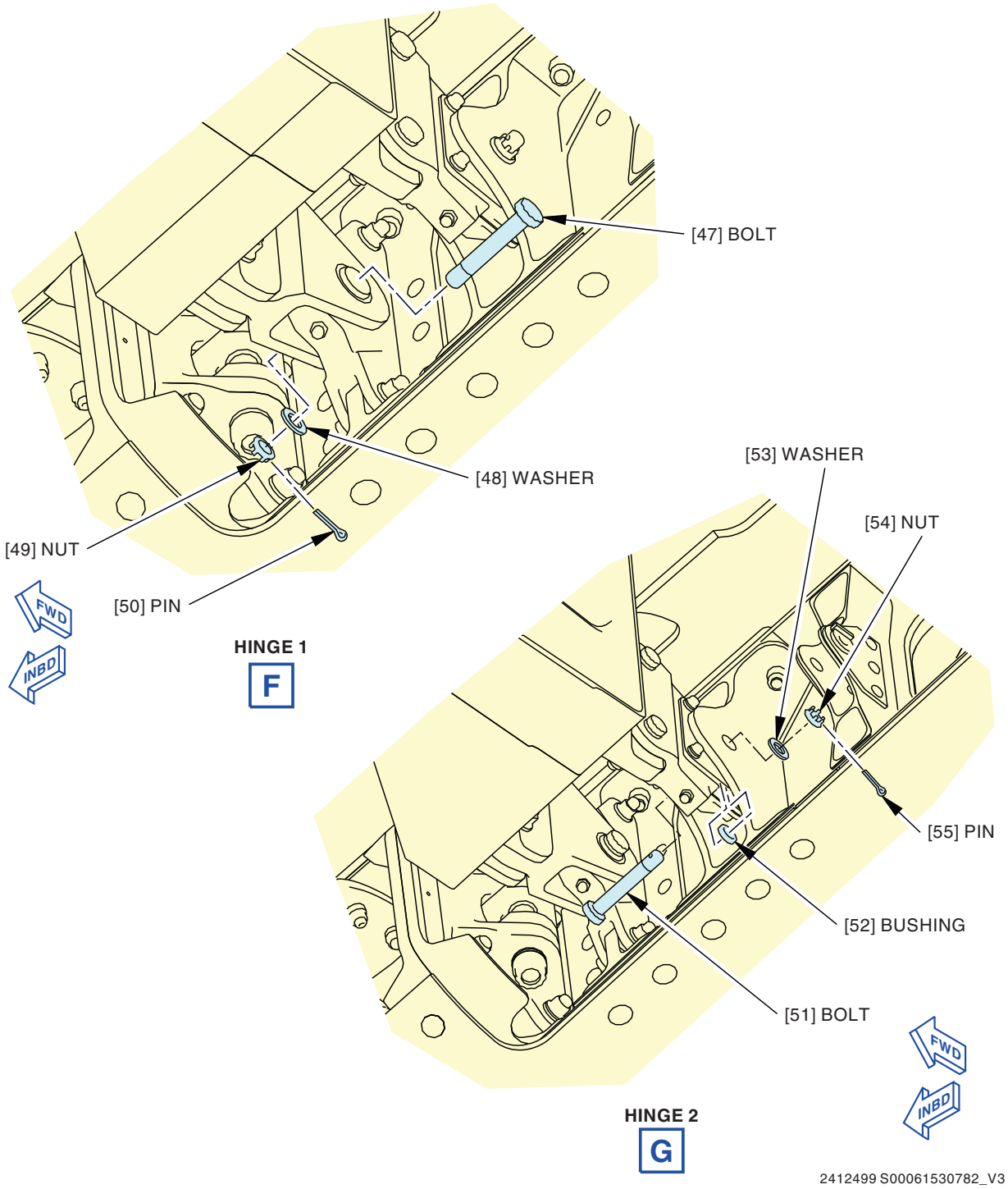
Elevator Installation
Figure 405/27-31-11-990-805 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



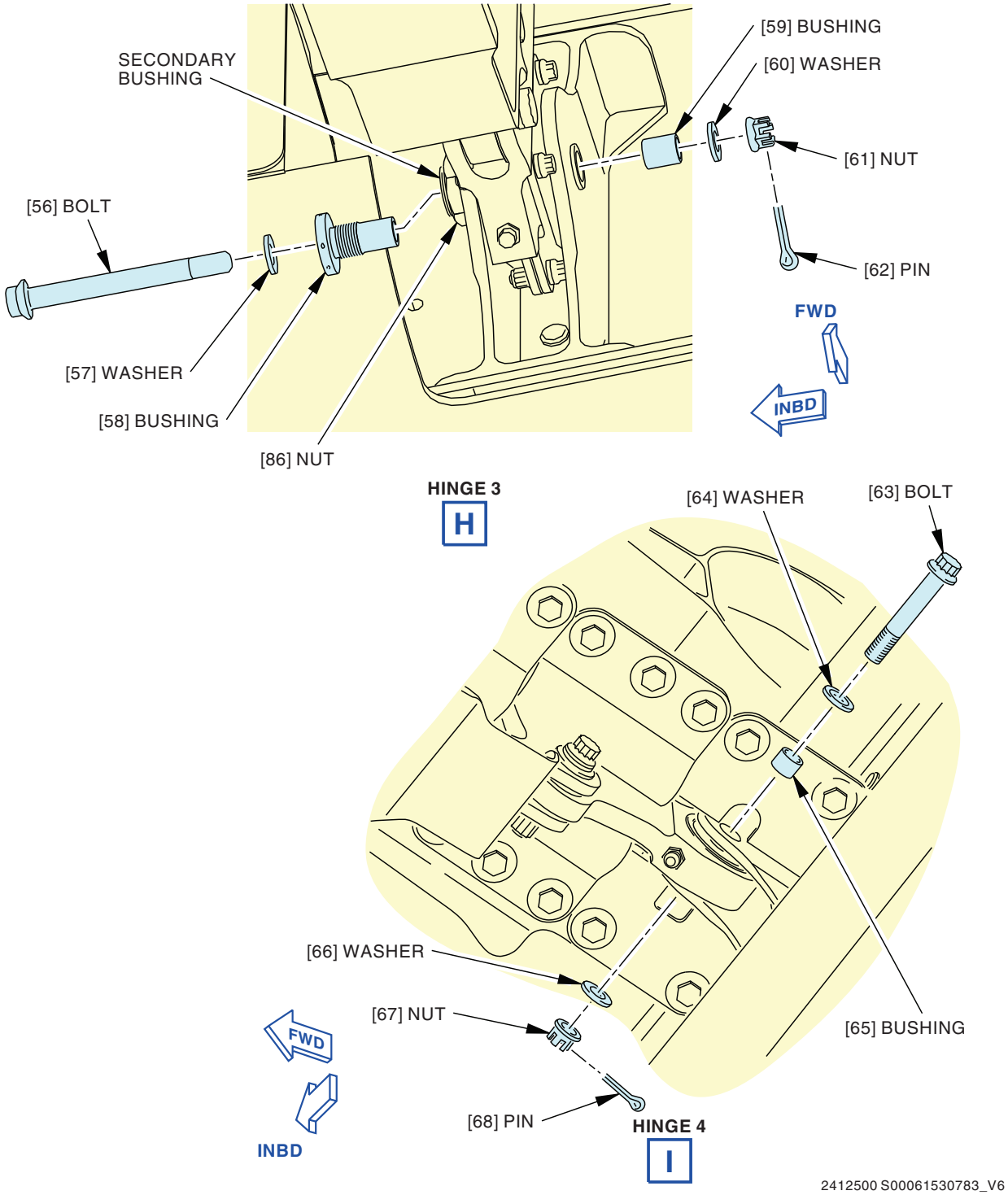
Elevator Installation
Figure 405/27-31-11-990-805 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



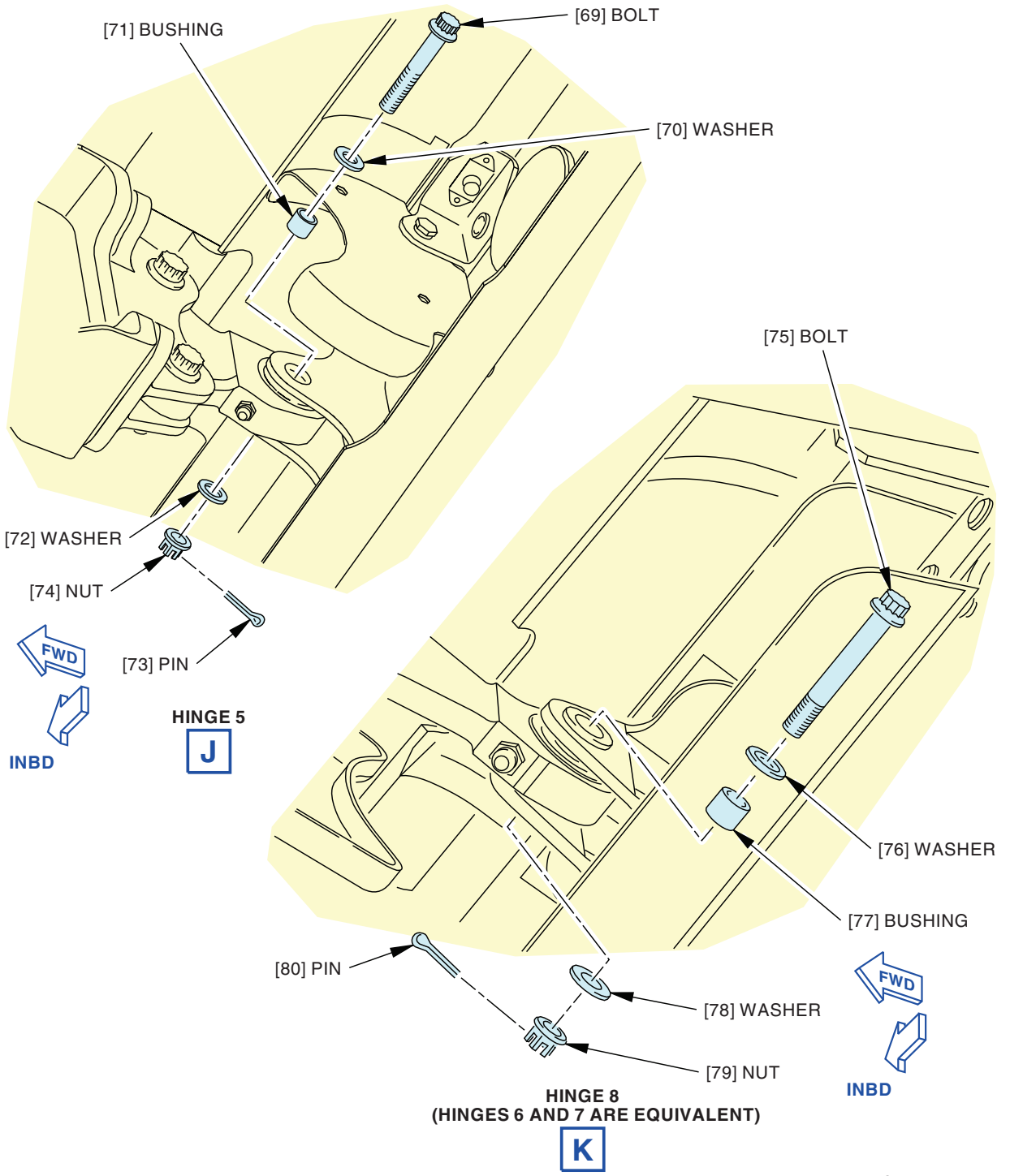
Elevator Installation
Figure 405/27-31-11-990-805 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



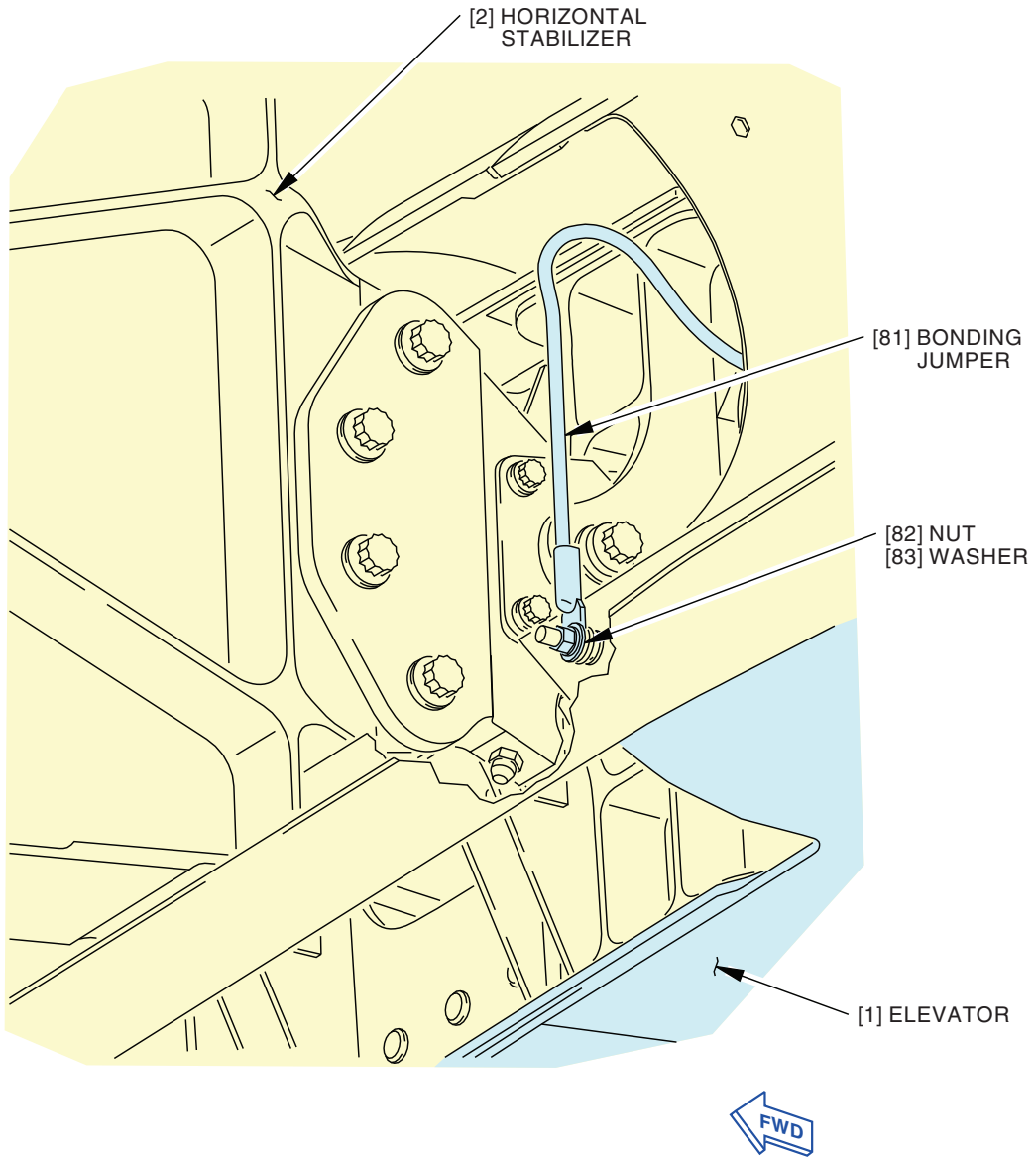
Elevator Installation
Figure 405/27-31-11-990-805 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



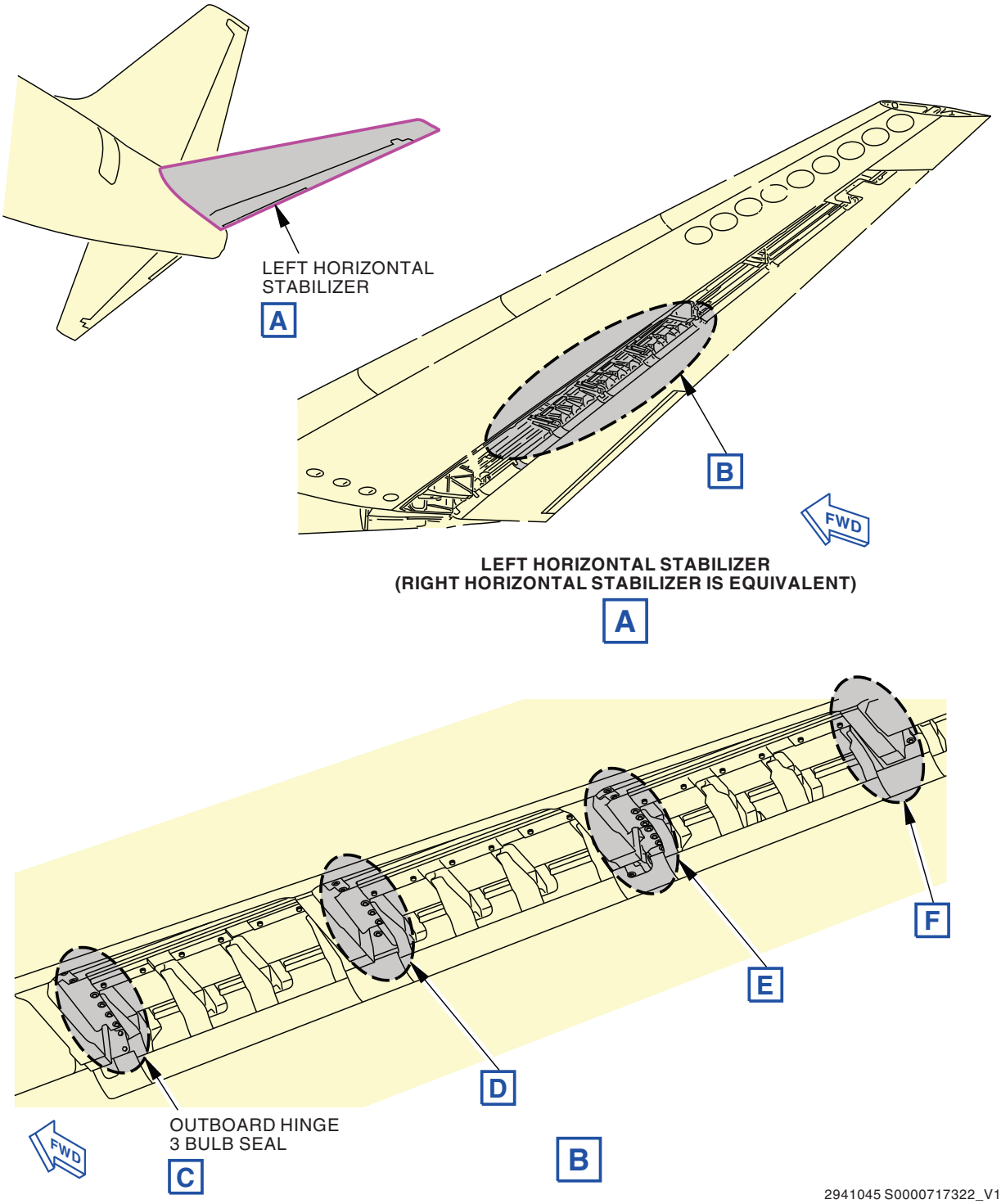
Elevator Installation
Figure 405/27-31-11-990-805 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



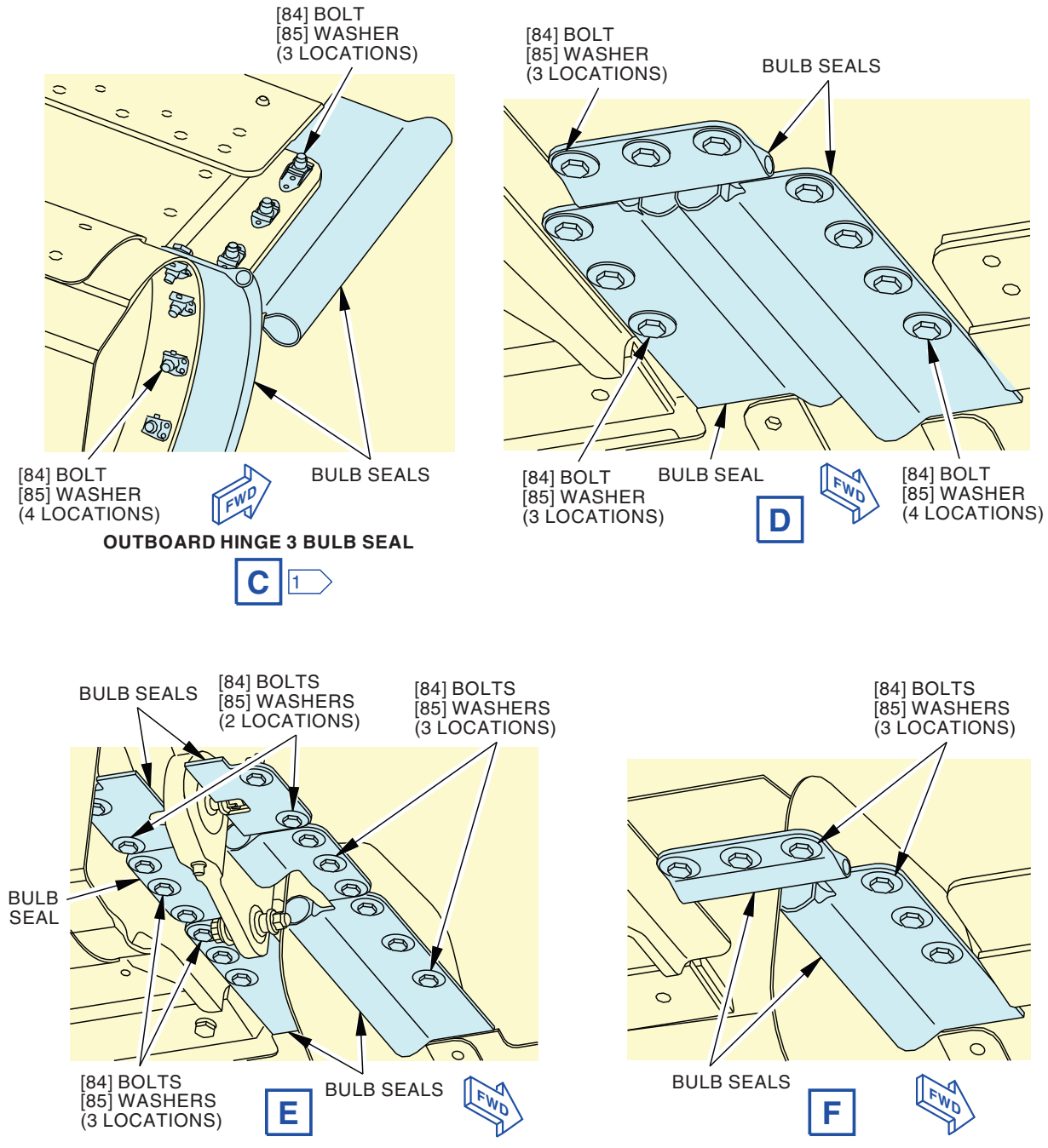
Elevator Installation
Figure 405/27-31-11-990-805 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Bulb Seal Installation
Figure 406/27-31-11-990-810 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

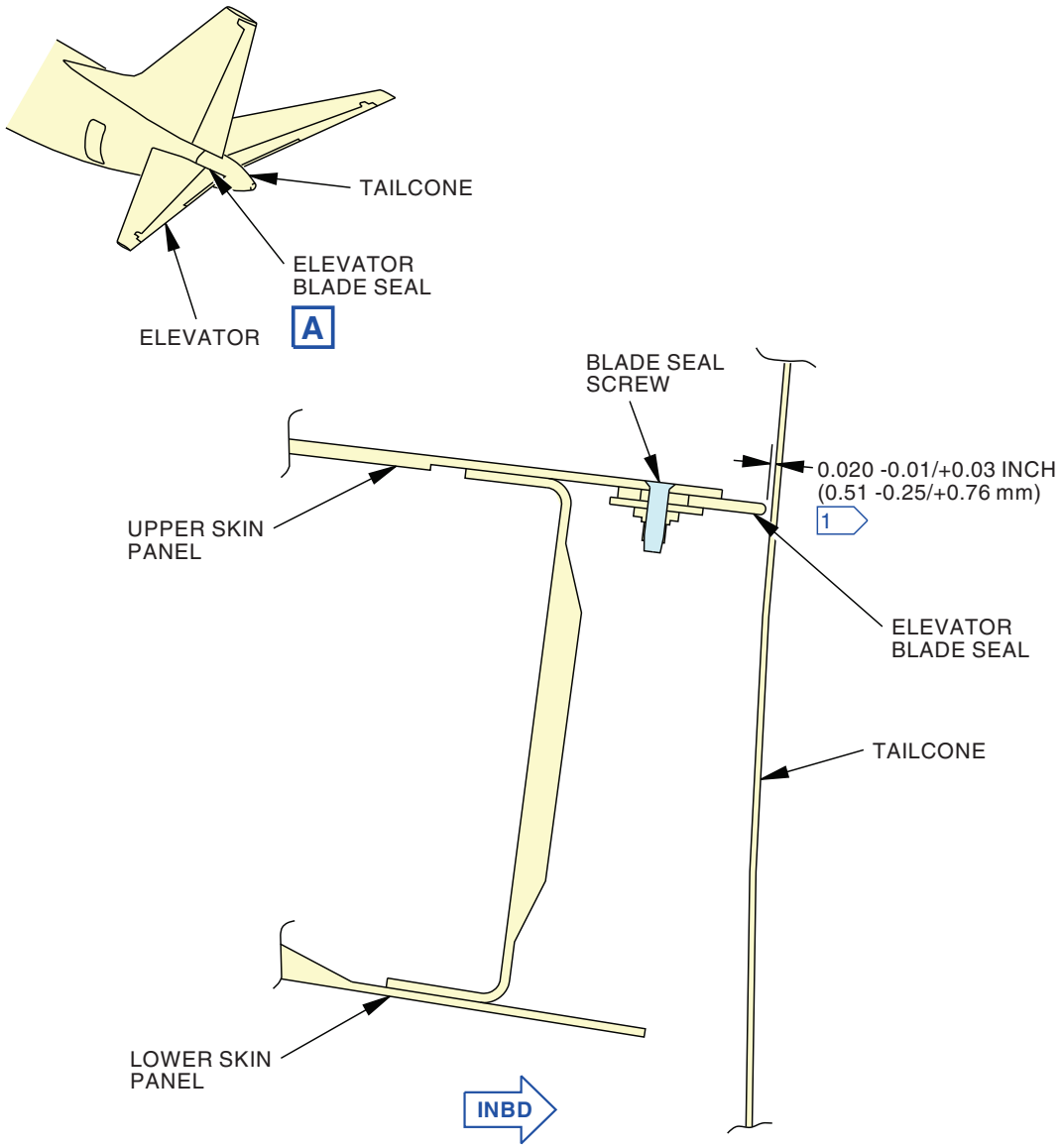


1 YOU NEED TO MOVE THE TRAILING EDGE OF THE FORWARD BALANCE PANEL UP (WHEN IT HAS BEEN DISCONNECTED) AND MOVE THE TRAILING EDGE OF THE ELEVATOR UP TO GET ACCESS TO THE UPPER TWO BOLTS AND WASHERS ON THE OUTBOARD HINGE 3 BULB SEAL.

2941047 S0000717323_V1

Bulb Seal Installation
Figure 406/27-31-11-990-810 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT ELEVATOR BLADE SEAL
(RIGHT ELEVATOR BLADE SEAL IS OPPOSITE)
(VIEW IN THE FORWARD DIRECTION)

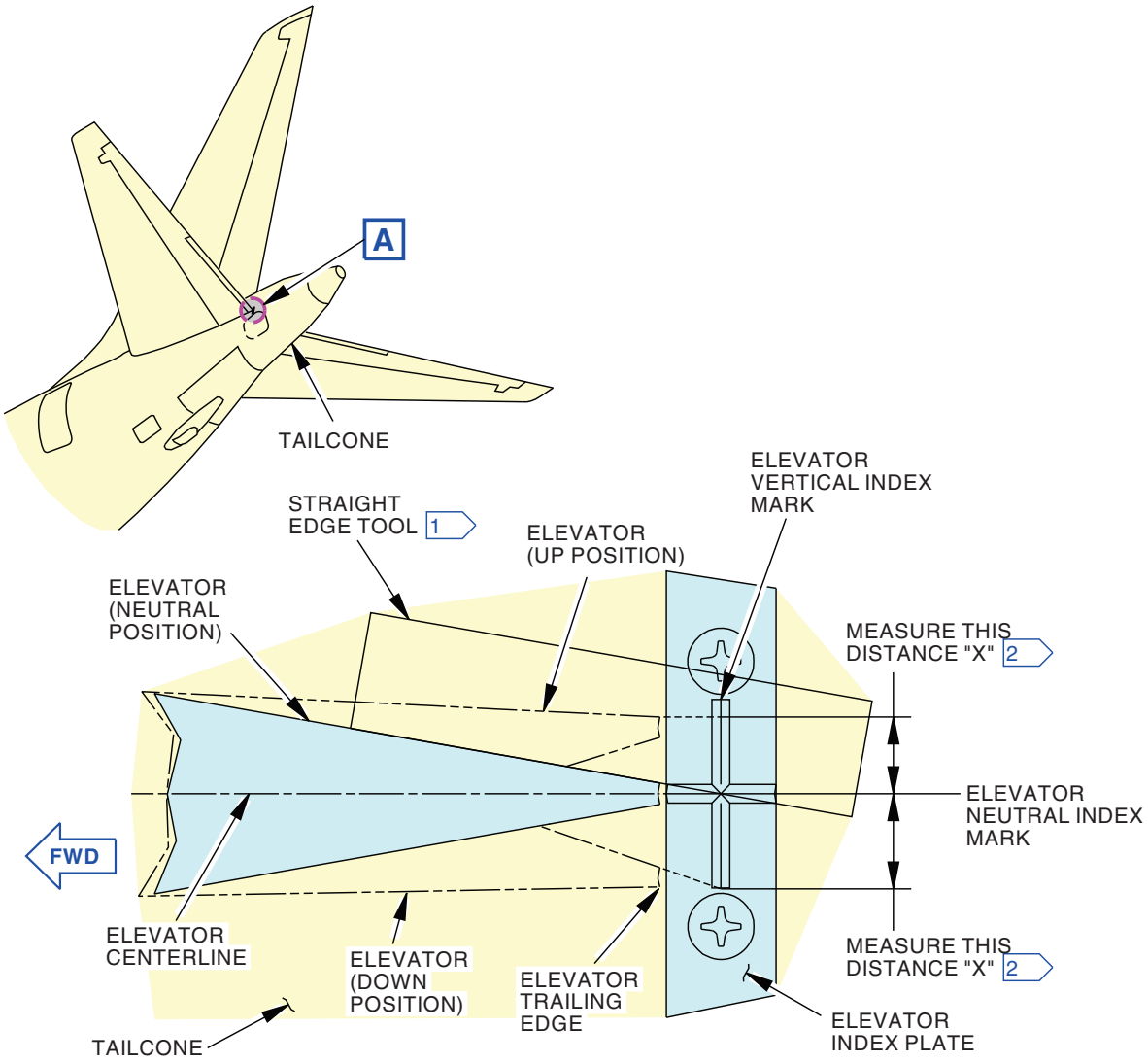
A

1 THE ELEVATOR BLADE SEAL GAP ADJUSTMENT NEEDS TO BE MADE WITHIN THE STABILIZER AND THE ELEVATOR AT THE NEUTRAL POSITIONS

2412503 S00061530787_V1

Elevator Blade Seal Gap Adjustment
Figure 407/27-31-11-990-811

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELEVATOR TAB PUSHROD ADJUSTMENT MEASUREMENTS (LEFT SIDE)

A

1 ELEVATOR IS AT NEUTRAL WHEN THE BOTTOM EDGE OF A STRAIGHT EDGE TOOL PLACED ON THE UPPER ELEVATOR SURFACE PASSES THROUGH THE MIDDLE OF THE ELEVATOR INDEX PLATE, AS SHOWN.

2 THE DISTANCE "X" IS MEASURED USING A STRAIGHT EDGE TOOL ON THE UPPER SURFACE OF THE ELEVATOR TO INTERSECT WITH THE ELEVATOR VERTICAL INDEX MARK AT THE ELEVATOR NEUTRAL INDEX MARK.

2412462 S00061530722_V1

Elevator Tab Adjustment
Figure 408/27-31-11-990-812

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-11-400-801

3. Elevator - Installation

(Figure 401, Figure 402, Figure 403, Figure 404, Figure 405, Figure 406, Figure 407, Figure 408)

A. References

Reference	Title
12-22-31-600-801	Elevator Buss Crank and Master Arm Fitting - Lubrication (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
27-31-00-700-803	Elevator Surface Friction and Balance - Test (P/B 501)
27-31-00-700-805	Elevator Travel Limit - Test (P/B 501)
27-31-00-700-806	Elevator Output Torque Tube Buss Linkage Test (P/B 501)
27-31-00-700-807	Elevator Tab Linkage - Test (P/B 501)
27-31-00-700-808	Manual Mode - Test (P/B 501)
27-31-00-700-809	Control Column Travel and Centering - Test (P/B 501)
27-31-00-700-810	Control Column Power Force - Test (P/B 501)
27-31-00-710-803	Elevator Power-Off Flight Test (P/B 501)
27-31-00-820-802	Elevator Control Pushrods - Adjustment (P/B 501)
27-31-00-820-803	Input Rods of the Elevator Power Control Unit - Adjustment (P/B 501)
27-31-00-820-805	Elevator Tab Pushrods - Adjustment (P/B 501)
27-31-41-400-801	Elevator Balance Panel - Installation (P/B 401)
31-31-51-700-801	Elevator Position Transmitter Installation Test (P/B 401)
53-53-00-400-801	Tailcone Installation (P/B 401)
SWPM 20-20-00, Paragraph 2	CLEANING OF FAY SURFACES
SWPM 20-20-10	Replacement of Ground Studs and Bonding Jumper Installation

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-2034	Sling Equipment - Elevator Removal/Installation Part #: C27034-54 Supplier: 81205 Opt Part #: C27034-44 Supplier: 81205

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C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
C00308	Compound - Corrosion Preventive, Petrolatum Hot Application	MIL-C-11796
D00633	Grease - Aircraft General Purpose	BMS3-33
D50004	Compound - Antiseize	BMS3-28
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Elevator	55-20-00-01-120	PTF ALL
		55-20-00-01-125	PTF ALL
16	Nut	27-31-34-01-085	PTF ALL
50	Pin	55-20-00-01-010	PTF ALL
55	Pin	55-20-00-01-010	PTF ALL
62	Pin	55-20-00-01-010	PTF ALL
68	Pin	55-20-00-01-010	PTF ALL
73	Pin	55-20-00-01-010	PTF ALL
80	Pin	55-20-00-01-010	PTF ALL

E. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right
334	Left Horizontal Stabilizer - Elevator
335	Left Horizontal Stabilizer - Tip
344	Right Horizontal Stabilizer - Elevator
345	Right Horizontal Stabilizer - Tip

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
333CB	Horizontal Stabilizer, Access Panel, Trailing Edge
333DB	Horizontal Stabilizer, Access Panel, Trailing Edge
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334BB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334DB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334EB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator

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(Continued)

Number	Name/Location
334FB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
334HB	Horizontal Stabilizer, Elevator Hinge Cover
334JB	Horizontal Stabilizer, Elevator Hinge Cover
334KB	Horizontal Stabilizer, Elevator Hinge Cover
334LB	Horizontal Stabilizer, Elevator Hinge Cover
334MB	Horizontal Stabilizer, Elevator Hinge Cover
334NB	Horizontal Stabilizer, Elevator Hinge Cover
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
343CB	Horizontal Stabilizer, Access Panel - T.E. Area
343DB	Horizontal Stabilizer, Access Panel - T.E. Area
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344BB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344DB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344EB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344FB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09
344HB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 66.54
344JB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 121.59
344KB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 176.64
344LB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 213.32
344MB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 250.04
344NB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 265.45

G. Elevator Installation

SUBTASK 27-31-11-820-006

WARNING MAKE SURE THAT YOU HAVE THE ELEVATORS BALANCED. IF NOT BALANCED, THE ELEVATORS CAN CAUSE UNWANTED FLUTTER AND INSTABILITY OF THE AIRPLANE. THIS CAN CAUSE AN IN-FLIGHT EMERGENCY. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Balance and stamp the elevator assembly with the new static balance hinge moment if you install a reworked, repaired, repainted, or new elevator assembly (SRM PROCEDURE 51-60-04-0G-0).

SUBTASK 27-31-11-640-001

(2) Lubricate the bolts, bushings with grease, D00633.

SUBTASK 27-31-11-420-001

(3) Install the elevator [1] in the order as shown (Figure 405):

(a) Put the elevator [1] at the stabilizer trailing edge using the sling equipment, SPL-2034.

1) If you do not use the sling equipment, SPL-2034, to install the elevator, you must properly support the elevator while installing the hinge bolts.

(b) At the elevator hinge 3, do these steps:

1) If it is necessary, install the secondary bushing at the hinge 3.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Loosely install the bushing [58].

a) Do not tighten the bushing [58] until it seats against the horizontal stabilizer bearing.

3) Install the washer [57] and bolt [56] with the bolt head inboard.

4) Loosely install the bushing [59], washer [60], and nut [61].

a) Make sure that the clearance between the washer [60] and flanged bushing on the elevator hinge is not less than 0.015 in. (0.381 mm).

(c) At the elevator hinge 1, do these steps:

1) Install the bolt [47] with the bolt head outboard.

2) Loosely install the washer [48] and nut [49].

(d) At the elevator hinge 2, do these steps:

1) Install the bolt [51] and bushing [52] with the bolt head inboard.

2) Loosely install the washer [53] and nut [54].

(e) At the elevator hinges 8, 7, and 6, do the following steps starting at the hinge 8, then hinge 7, and then hinge 6:

1) Install the bushings [77], washers [76], and bolts [75] with the bolt head outboard.

a) Make sure that the clearance between the washer [76] and bushing on the elevator hinge is not less than 0.015 in. (0.381 mm).

2) Loosely install the washers [78] and nuts [79].

(f) At the elevator hinge 4, do these steps:

1) Install the bushing [65], washer [64], and bolt [63] with the bolt heads outboard.

a) Make sure that the clearance between the washer [64] and bushing on the elevator hinge is not less than 0.015 in. (0.381 mm).

2) Loosely install the washer [66] and nut [67].

(g) At the elevator hinge 5, do these steps:

1) Install the bushing [71], washer [70], and bolt [69] with the bolt head outboard.

a) Make sure that the clearance between the washer [70] and bushing on the elevator hinge is not less than 0.015 in. (0.381 mm).

2) Loosely install the washer [72] and nut [74].

(h) Connect the bonding jumpers [81] to the horizontal stabilizer [2]:

1) Clean the fay surfaces of the elevator [1], horizontal stabilizer [2], and bonding jumpers [81] (SWPM 20-20-00, Paragraph 2).

2) Install the washers [83] and nuts [82] to attach the bonding jumpers [81] to the horizontal stabilizer [2].

3) Use the intrinsically safe approved bonding meter, COM-1550, to make sure that there is a maximum of 0.010 ohm (10 milliohms) between the horizontal stabilizer [2] and elevator [1] (SWPM 20-20-10, Paragraph 5).

a) Make sure that the bonding resistance is 0.010 ohm (10 milliohms) or less.

b) Make sure that the measured resistance is less than the maximum bonding resistance as specified in SWPM 20-20-10, Paragraph 5.

4) Seal the bonding jumper ends and fasteners with sealant, A00247.

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SUBTASK 27-31-11-080-001

(4) If the sling equipment, SPL-2034, was used for installation of the elevator, do the following steps to remove it from the elevator [1] (Figure 403):

(a) Remove the bolts [3] to disconnect the forward sling fittings from the elevator [1].

(b) Remove the forward sling fittings from the elevator [1].

(c) Install the bolts [3] on the elevator [1].

(d) Remove the aft sling fitting from the elevator [1].

SUBTASK 27-31-11-080-002

(5) Remove the rig pin E-5 of the rigging pin kit, SPL-1585, from the elevator aft quadrant.

SUBTASK 27-31-11-710-001

(6) Do an initial check of the elevator full travel by turning the elevator [1] full up and full down by hand.

SUBTASK 27-31-11-400-002

(7) If removed, install the applicable bulb seal (Figure 406).

(a) Apply Cor-Ban 27L Compound, G50237, (preferred) or corrosion preventive compound, C00308, (alternate) to all areas of the holes for the bolts [84].

(b) Install the bolts [84] and washers [85].

(c) Tighten the bolts [84] to 25 in-lb (2.8 N·m).

SUBTASK 27-31-11-480-003

(8) Install the rig pin E-5 of the rigging pin kit, SPL-1585, to the elevator aft quadrant.

SUBTASK 27-31-11-420-002

CAUTION

MAKE SURE THAT THE IDLER HINGE MOVES DOWN FROM THE STABILIZER SUPPORT BEAM WHEN YOU ATTACH THE BALANCE PANELS TO THE ELEVATOR. IF YOU DO NOT INSTALL THE BALANCE PANELS CORRECTLY, IT CAN CAUSE DAMAGE TO EQUIPMENT.

(9) Connect the balance panels to the elevator [1] (Figure 405):

(a) Install the washers [46] and bolts [45].

(b) Install the washer [44] and bolt [43].

(c) Check the elevator balance panel seal gap (TASK 27-31-41-400-801).

(d) Make sure that you install the hinge seals correctly before you tighten the bolts [45] and bolt [43].

SUBTASK 27-31-11-020-006

(10) Connect the elevator tab control mechanism [22] to the elevator [1] and tab pushrods [14] (Figure 404):

(a) Apply compound, D50004, to the threads of the pin assemblies [11].

(b) Install the bushing [21], pin assembly [11], washer [20], and nut [19] that attach the left aft attach lug to the elevator [1].

(c) Install the nut [19], washer [20], washer [23], and pin assembly [11] that attach the right aft attach lug to the elevator [1].

(d) Tighten the mechanism nuts [19] to 30 in-lb (3.4 N·m) - 40 in-lb (4.5 N·m).

(e) Apply compound, D50004, to the threads of the bolts [12] and outer surface of the spacers [13].

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(f) Install the spacers [13], bolts [12], washers [15], and new nuts [16] to attach the tab pushrods [14] to the elevator tab control mechanism [22].

1) If you loosened or removed the nuts [16], they must be replaced with new nuts [16].

NOTE: Using new nuts will ensure that they have the required run-on torque.

a) Make sure that the new nuts [16] are installed.

(g) Tighten the nuts [16] to 70 in-lb (7.9 N·m) - 80 in-lb (9.0 N·m).

(h) Install the pin retainers [18] with the washers [25] and screws [17].

(i) Remove the clamps that prevent movement of the elevator tab control mechanism [22].

SUBTASK 27-31-11-420-008

(11) At the elevator hinge 2, do these steps:

(a) Tighten the hinge 2 nut [54] to 95 in-lb (10.7 N·m) - 160 in-lb (18.1 N·m).

(b) Install the new pin [55].

SUBTASK 27-31-11-420-003

(12) At the elevator hinge 4, do these steps:

(a) Tighten the hinge 4 nut [67] to 90 in-lb (10.2 N·m) - 125 in-lb (14.1 N·m).

(b) Install the new pin [68].

SUBTASK 27-31-11-420-011

(13) At the elevator hinge 5, do these steps:

(a) Tighten the hinge 5 nut [74] to 90 in-lb (10.2 N·m) - 125 in-lb (14.1 N·m).

(b) Install the new pin [73].

SUBTASK 27-31-11-420-012

(14) At the elevator hinges 6, 7, and 8, do the following steps starting at the hinge 6, then hinge 7, and then hinge 8:

(a) Tighten the nuts [79] to 50 in-lb (5.6 N·m) - 75 in-lb (8.5 N·m).

(b) Install the new pins [80].

SUBTASK 27-31-11-420-015

(15) At the elevator hinge 1, do these steps:

(a) Tighten the hinge 1 nut [49] to 95 in-lb (10.7 N·m) - 160 in-lb (18.1 N·m).

(b) Install the new pin [50].

SUBTASK 27-31-11-420-004

(16) At the elevator hinge 3, do one of the following steps:

(a) Safety the hinge 3 bushing and nut (Method 1):

1) Tighten the hinge 3 bushing [58] using only hand pressure until it seats against the horizontal stabilizer bearing.

2) Tighten the hinge 3 nut [61] to 90 in-lb (10.2 N·m) - 125 in-lb (14.1 N·m).

3) Install the new pin [62].

NOTE: It is recommended to install the cotter pin part number BACP18BC02A06P at the elevator hinge 3 location. Operators whose expendable/parts AIPC references cotter pin part number BACP18BC02A06P for the elevator hinge 3 location may disregard this note.

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4) Install the MS20995C32 lockwire, G01048, from the bushing [58] to the elevator [1] hinge 3 fitting to the nut [86] to prevent clockwise and counterclockwise rotation (TASK 20-10-44-400-801).

5) Apply Cor-Ban 27L Compound, G50237, to the lockwire hole on the hinge 3 fitting of the elevator [1].

(b) Safety the hinge 3 bushing and nut (Method 2):

1) Tighten the hinge 3 bushing [58] using only hand pressure until it seats against the horizontal stabilizer bearing.

2) Align the lockwire holes of the hinge 3 bushing [58] with the hinge 3 nut [86] by loosening or tightening the bushing [58] no more than a quarter turn.

3) Make sure that the clearance between the washer [60] and flanged bushing on the elevator hinge is not less than 0.015 in. (0.381 mm).

4) Tighten the hinge 3 nut [61] to 90 in-lb (10.2 N·m) - 125 in-lb (14.1 N·m).

5) Install the new pin [62].

NOTE: It is recommended to install the cotter pin part number BACP18BC02A06P at the elevator hinge 3 location. Operators whose expendable/parts AIPC references cotter pin part number BACP18BC02A06P for the elevator hinge 3 location may disregard this note.

6) Install the MS20995C32 lockwire, G01048, from the nut [86] to the elevator [1] hinge 3 fitting (TASK 20-10-44-400-801).

7) Install the MS20995C32 lockwire, G01048, from the bushing [58] to the nut [86] (TASK 20-10-44-400-801).

8) Apply Cor-Ban 27L Compound, G50237, to the lockwire hole on the hinge 3 of the elevator [1].

SUBTASK 27-31-11-710-002

(17) Do a final check of the elevator full travel by turning the elevator [1] full up and full down by hand.

SUBTASK 27-31-11-420-005

(18) Connect the pushrod [38] of the elevator position transmitter [37] to the elevator [1] (Figure 405):

(a) Make sure that the joint between the pushrod [38] and arm [38A] is located forward from the elevator position transmitter [37].

(b) Install the bolt [42], washer [41], and spacer [40] to the pushrod [38].

(c) Connect the electrical connector [39] to the elevator position transmitter [37].

SUBTASK 27-31-11-420-006

(19) Connect the elevator control pushrod [36] to the elevator [1] (Figure 405):

(a) Move the elevator [1] to the neutral position.

(b) Install the bolt assembly [35] (with the pin head outboard), bushing [34], washer [33], and nut [32].

1) Tighten the nut [32] to 600 in-lb (67.8 N·m) - 700 in-lb (79.1 N·m).

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CAUTION

DO NOT TIGHTEN NUT [32] AFTER YOU TIGHTEN NUT [31]. BOLT PROTRUSION THROUGH NUT [31] CAN BE 0.01 INCHES MINIMUM, FOR THE BOLT ASSEMBLY. IF YOU DO NOT OBEY, THIS CAN CAUSE DAMAGE TO EQUIPMENT..

(c) Install the nut [31].

1) Tighten the nut [31] to 170 in-lb (19.2 N·m) - 240 in-lb (27.1 N·m).

SUBTASK 27-31-11-080-003

(20) Remove the rig pin E-5 of the rigging pin kit, SPL-1585, from the elevator aft quadrant.

SUBTASK 27-31-11-400-001

(21) Do this task: Tailcone Installation, TASK 53-53-00-400-801.

SUBTASK 27-31-11-700-001

(22) Check the adjustment of the elevator blade seal gaps:

(a) Make sure that the stabilizer and elevator are at their neutral positions.

(b) Make sure that the elevator blade seal gap is within the tolerance of 0.020 +0.030 / -0.010 in. (0.51 +0.76 / -0.26 mm) (Figure 407).

SUBTASK 27-31-11-640-002

(23) Do this task: Elevator Buss Crank and Master Arm Fitting - Lubrication, TASK 12-22-31-600-801.

SUBTASK 27-31-11-860-004

(24) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

SUBTASK 27-31-11-860-005

(25) Adjust and test the elevator system, do these steps:

NOTE: If you install the same elevator that you removed from the airplane, and no repairs or modifications to the elevator or adjustments to the elevator were made, then it is not necessary to do the adjustment steps of the elevator system below. Continue to the elevator tab pushrod adjustment section of this task.

(a) Do this task: Elevator Control Pushrods - Adjustment, TASK 27-31-00-820-802.

(b) Do this task: Input Rods of the Elevator Power Control Unit - Adjustment, TASK 27-31-00-820-803.

(c) Do the following test for elevator centering:

1) Set the FLT CONTROL B switch, on the P5 forward overhead panel, to the OFF position.

2) Make sure that the FLT CONTROL A switch, on the P5 forward overhead panel, is in the ON position.

3) Make sure that the captain's control column is at the neutral position.

a) Make sure that the right elevator aligns with the index mark in ± 0.06 in. (1.5 mm) (Figure 408).

4) Set the FLT CONTROL B switch to the ON position.

a) Make sure that the right elevator aligns with the index mark in ± 0.06 in. (1.5 mm).

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5) Set the FLT CONTROL A switch to the OFF position.

6) Make sure that the FLT CONTROL B switch is in the ON position.

7) Set the FLT CONTROL A switch to the ON then to the OFF position.

   a) Make sure that there is less than 0.02 in. (0.5 mm) movement at the right elevator trailing edge.

8) Set the FLT CONTROL B switch to the OFF then to the ON then to the OFF position.

   a) Make sure that there is less than 0.02 in. (0.5 mm) movement at the right elevator trailing edge.

9) Set the FLT CONTROL A and B switches to the ON position.

10) Make sure that the captain's control column is at the neutral position.

    a) Make sure that the right elevator aligns with the index mark in ± 0.06 in. (1.5 mm).

11) Move the captain's control column forward approximately 2 degrees and release.

    a) Make sure that the right elevator trailing edge aligns with the index mark in 0.00 ±0.12 in. (0.0 ±3.0 mm).

12) Move the captain's control column aft approximately 2 degrees and release.

    a) Make sure that the right elevator trailing edge aligns with the index mark in 0.00 ±0.12 in. (0.0 ±3.0 mm).

SUBTASK 27-31-11-820-002

(26) Adjust the elevator tab pushrods, do these steps:

NOTE: If the elevator tabs, tab control mechanism, or tab rods were not replaced, repaired or modified, and no adjustments to the tab pushrods were made, then it is not necessary to do the adjustment of the elevator tab pushrods in the following step. Do the elevator position transmitter installation test.

(a) Do this task: Elevator Tab Pushrods - Adjustment, TASK 27-31-00-820-805.

NOTE: If the elevator tab position measurements were recorded in the elevator removal steps, then adjust the elevator and tab control system to the position measured prior to removal.

(b) Do this task: Elevator Position Transmitter Installation Test, TASK 31-31-51-700-801.

(c) Do this task: Manual Mode - Test, TASK 27-31-00-700-808.

(d) Do this task: Elevator Surface Friction and Balance - Test, TASK 27-31-00-700-803.

(e) For the left elevator, install these access panel:

(TASK 27-31-41-400-801)

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
333CB	Horizontal Stabilizer, Access Panel, Trailing Edge
333DB	Horizontal Stabilizer, Access Panel, Trailing Edge

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

MAKE SURE THAT THE CUTOUT IN THE HINGE COVER IS FACING THE FORWARD DIRECTION. IF THE HINGE COVER INSTALLATION IS INCORRECT, DAMAGE TO THE HINGE COVER CAN OCCUR.

(f) For the left elevator, install these access panels:

Number	Name/Location
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334BB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334DB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334EB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334FB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
334HB	Horizontal Stabilizer, Elevator Hinge Cover
334JB	Horizontal Stabilizer, Elevator Hinge Cover
334KB	Horizontal Stabilizer, Elevator Hinge Cover
334LB	Horizontal Stabilizer, Elevator Hinge Cover
334MB	Horizontal Stabilizer, Elevator Hinge Cover
334NB	Horizontal Stabilizer, Elevator Hinge Cover

(g) For the right elevator, install these access panels:

(TASK 27-31-41-400-801)

Number	Name/Location
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
343CB	Horizontal Stabilizer, Access Panel - T.E. Area
343DB	Horizontal Stabilizer, Access Panel - T.E. Area

CAUTION

MAKE SURE THAT THE CUTOUT IN THE HINGE COVER IS FACING THE FORWARD DIRECTION. IF THE HINGE COVER INSTALLATION IS INCORRECT, DAMAGE TO THE HINGE COVER CAN OCCUR.

(h) For the right elevator, install these access panels:

Number	Name/Location
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344BB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344DB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344EB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344FB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09
344HB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 66.54
344JB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 121.59
344KB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 176.64
344LB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 213.32
344MB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 250.04
344NB	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 265.45

(i) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

(j) Do this task: Elevator Travel Limit - Test, TASK 27-31-00-700-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(k) Do this task: Elevator Output Torque Tube Buss Linkage Test, TASK 27-31-00-700-806.

(l) Do this task: Elevator Tab Linkage - Test, TASK 27-31-00-700-807.

(m) Do this task: Control Column Travel and Centering - Test, TASK 27-31-00-700-809.

(n) Do this task: Control Column Power Force - Test, TASK 27-31-00-700-810.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-11-410-002

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-11-710-003

(2) Do this task: Elevator Power-Off Flight Test, TASK 27-31-00-710-803.

NOTE: Depending on which systems were disturbed, you may not have to do a power-off flight test. To see if you need to do a power-off flight test, check the requirements in the "General" section at the beginning of the power-off flight test task.

——— END OF TASK ———

TASK 27-31-11-000-802

4. Elevator Control Pushrod - Removal

(Figure 403, Figure 409 and Figure 410)

A. General

(1) This task is for the removal of an elevator control pushrod.

B. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-2034	Sling Equipment - Elevator Removal/Installation Part #: C27034-54 Supplier: 81205 Opt Part #: C27034-44 Supplier: 81205

D. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

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(Continued)

Zone	Area
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09

F. Prepare for the Removal

SUBTASK 27-31-11-970-001

(1) Measure and record the elevator tab trailing edge position.

SUBTASK 27-31-11-860-006

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-11-010-013

(3) Gain access to the bolt assembly [35] as follows:

(a) For the left horizontal stabilizer, open these access panels:

Number	Name/Location
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover

(b) For the right horizontal stabilizer, open these access panels:

Number	Name/Location
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09

(c) Remove the bolts [100], washers [101], and the front sliding seal rail [102].

(d) Remove the bolts [103], washers [104], and the aft sliding seal rail [105].

(e) Remove the bolts [106], washers [107], and nuts [108] that hold the sliding seal [109] to the bracket [110].

(f) Remove the sliding seal [109] to gain access to the bolt assembly [35].

SUBTASK 27-31-11-010-007

(4) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

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SUBTASK 27-31-11-860-007

(5) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

SUBTASK 27-31-11-860-008

CAUTION MAKE SURE TO PREVENT DAMAGE TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CHAF OR CAN CAUSE DAMAGE TO THE WIRES OR BUNDLE INSULATION.

(6) Use the trammel bar, SPL-1677, to set “B” dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)). (Figure 401)

SUBTASK 27-31-11-820-003

(7) Install the rig pin E-5 from rigging pin kit, SPL-1585, in the elevator aft quadrant (Figure 402).

SUBTASK 27-31-11-480-004

(8) If the sling equipment, SPL-2034, is used to support the elevator assembly while the elevator control pushrod [36] is removed, do the following (Figure 403):

(a) Find the bolts [3] on the elevator [1] to attach the sling equipment, SPL-2034 (Figure 405).

NOTE: The bolts are on the upper panel next to elevator hinge 3 and hinge 5.

(b) Remove the bolts [3] from the elevator [1].

(c) Install the forward sling fittings on the elevator [1].

1) Install the fasteners to attach each sling fitting to the elevator [1].

(d) Install the aft sling fitting on the elevator [1].

(e) Attach the straps from the forward sling fittings to the aft sling fitting.

(f) Attach the sling equipment, SPL-2034, to the forward sling fittings and aft sling fitting.

(g) Connect an overhead crane to the primary support sling on the sling equipment, SPL-2034.

(h) Connect a tether line to the aft sling fitting.

(i) Use the sling equipment, SPL-2034, to hold the elevator [1].

G. Pushrod Removal

SUBTASK 27-31-11-020-007

(1) Remove the nut [31], nut [32], washer [33], bushing [34], and bolt assembly [35] to disconnect the elevator control pushrod [36] from the elevator mast arm [91].

SUBTASK 27-31-11-020-008

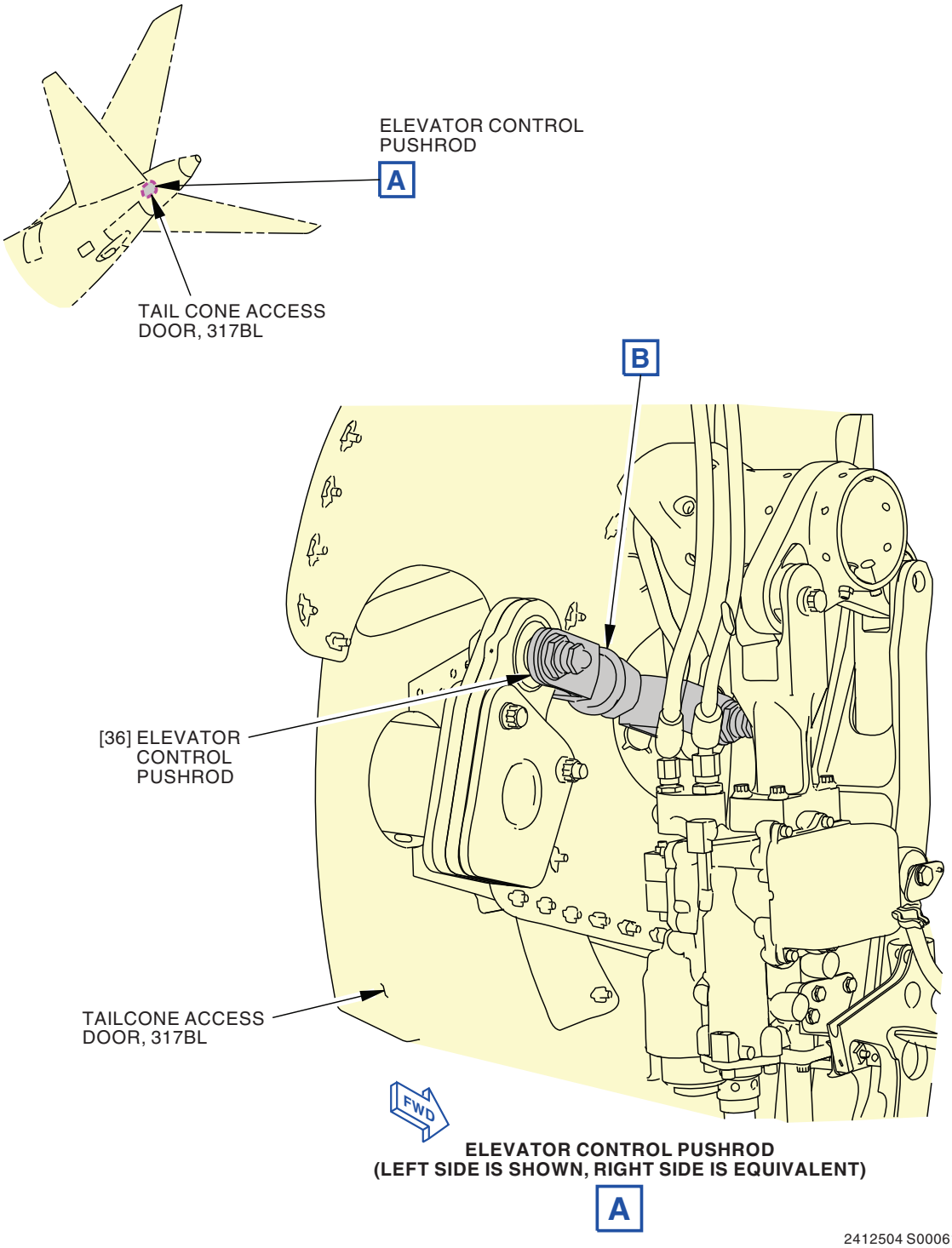
(2) Remove the nut [92], nut [93], washer [94], bushing [95], and bolt assembly [96] to disconnect the elevator control pushrod [36] from the elevator buss crank [97].

SUBTASK 27-31-11-020-009

(3) Remove the elevator control pushrod [36].

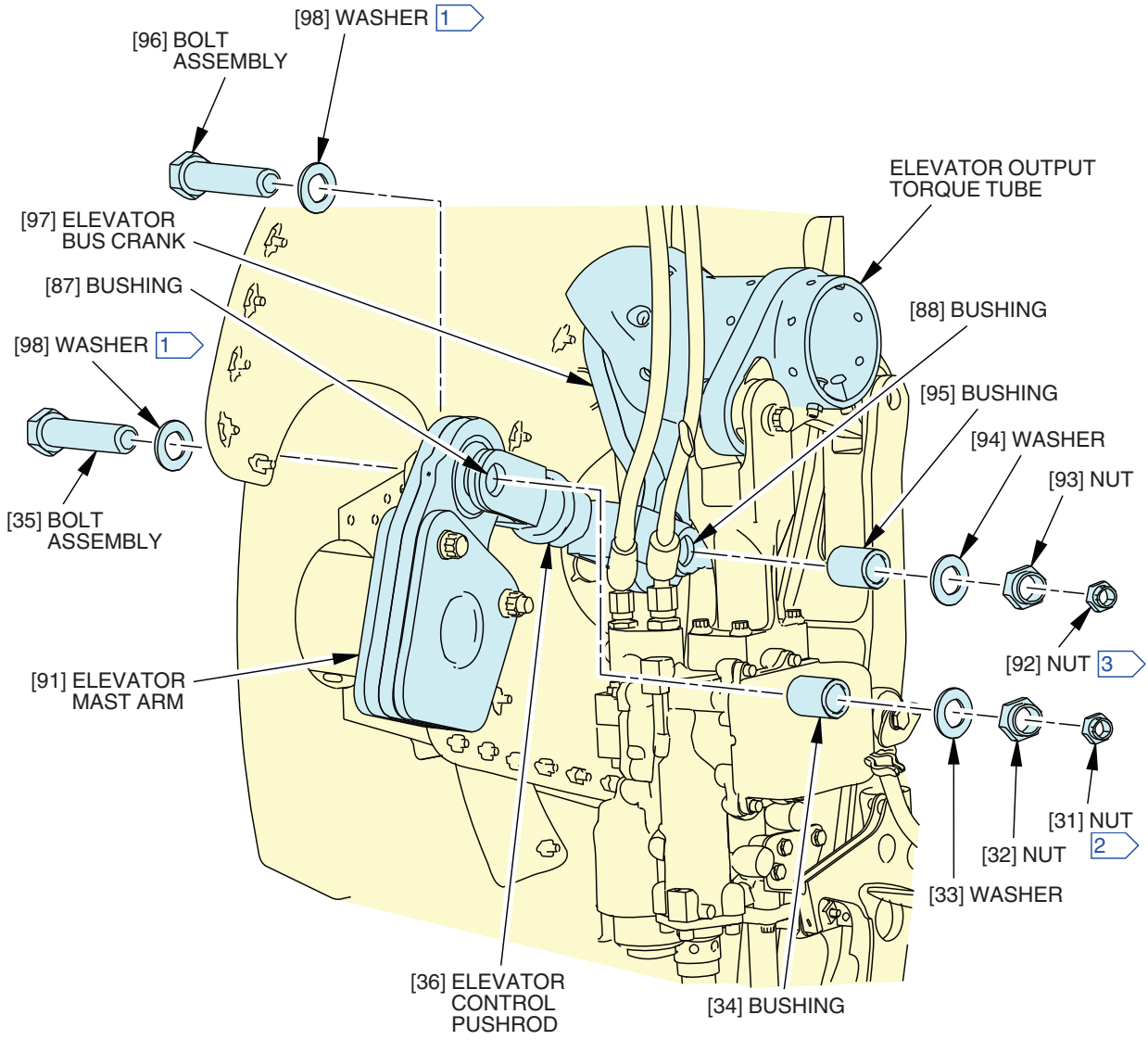
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Control Pushrod Installation
Figure 409/27-31-11-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

B

1 IF INSTALLED.

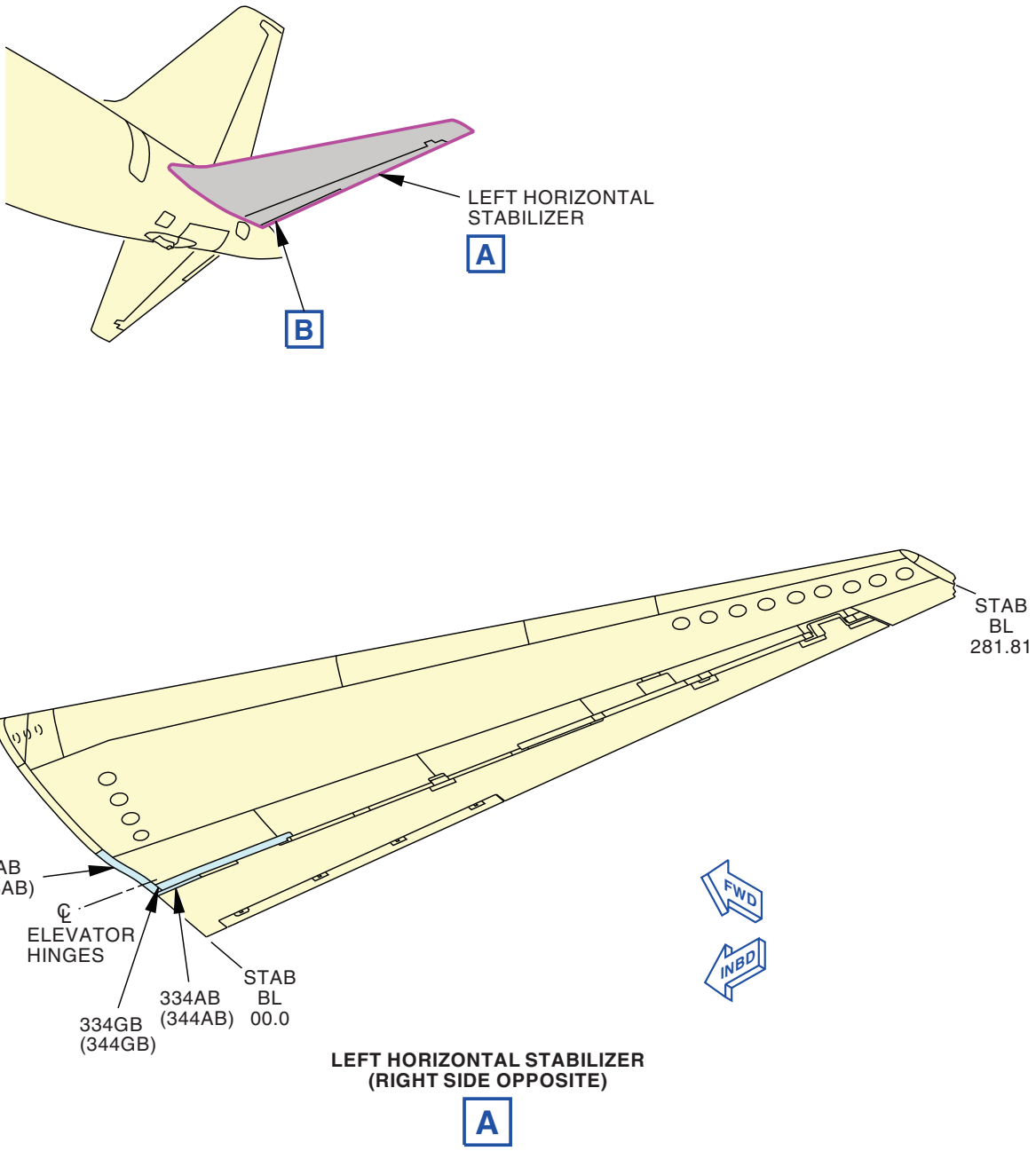
2 DO NOT TIGHTEN NUT [32] AFTER YOU TIGHTEN NUT [31]. THIS CAN CAUSE DAMAGE TO EQUIPMENT. BOLT PROTRUSION THROUGH NUT [31] MAY BE 0.01 INCH MINIMUM, FOR BOLT ASSEMBLY.

3 DO NOT TIGHTEN NUT [93] AFTER YOU TIGHTEN NUT [92]. THIS CAN CAUSE DAMAGE TO EQUIPMENT. BOLT PROTRUSION THROUGH NUT [92] MAY BE 0.01 INCH MINIMUM, FOR BOLT ASSEMBLY.

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Elevator Control Pushrod Installation
Figure 409/27-31-11-990-807 (Sheet 2 of 2)

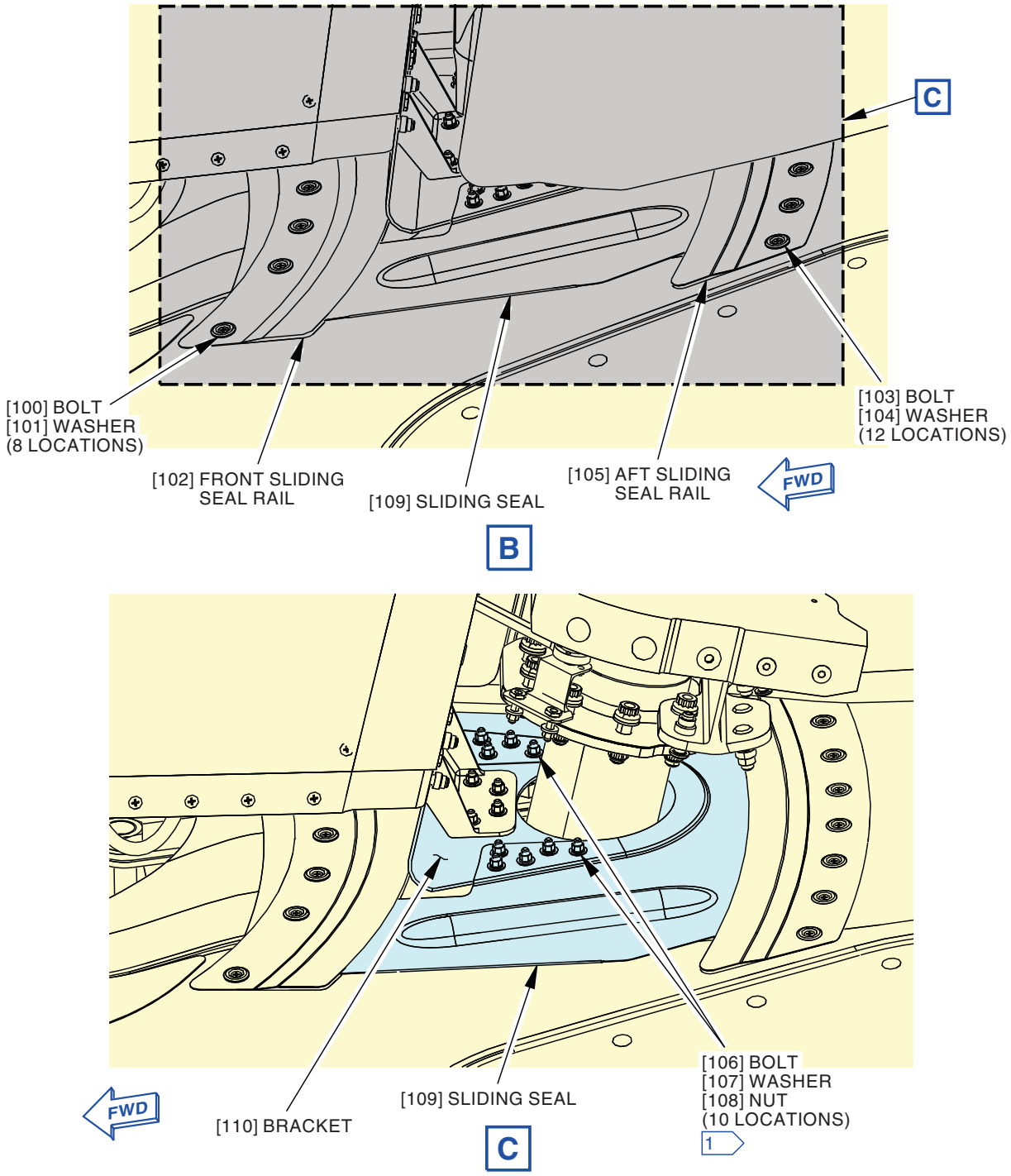
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Sliding Seal Installation
Figure 410/27-31-11-990-809 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Sliding Seal Installation
Figure 410/27-31-11-990-809 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-11-400-802

5. Elevator Control Pushrod - Installation

(Figure 403, Figure 409 and Figure 410)

A. General

(1) This task gives the steps on how to install an elevator control pushrod.

(2) The elevator power-off flight test procedure is required after replacement of an elevator control pushrod.

B. References

Reference	Title
20-10-02-900-801	Bearings and Bushings - Removal/Installation (P/B 401)
27-31-00-700-803	Elevator Surface Friction and Balance - Test (P/B 501)
27-31-00-700-805	Elevator Travel Limit - Test (P/B 501)
27-31-00-700-806	Elevator Output Torque Tube Buss Linkage Test (P/B 501)
27-31-00-700-807	Elevator Tab Linkage - Test (P/B 501)
27-31-00-700-808	Manual Mode - Test (P/B 501)
27-31-00-700-809	Control Column Travel and Centering - Test (P/B 501)
27-31-00-700-810	Control Column Power Force - Test (P/B 501)
27-31-00-710-803	Elevator Power-Off Flight Test (P/B 501)
27-31-00-820-802	Elevator Control Pushrods - Adjustment (P/B 501)
27-31-00-820-803	Input Rods of the Elevator Power Control Unit - Adjustment (P/B 501)
27-31-00-820-805	Elevator Tab Pushrods - Adjustment (P/B 501)
31-31-51-700-801	Elevator Position Transmitter Installation Test (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2034	Sling Equipment - Elevator Removal/Installation Part #: C27034-54 Supplier: 81205 Opt Part #: C27034-44 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A50473	Sealant - Brush (BMS5-95 Type I Class A-2 (See BAC5000 8.5.3 a. for preparation instructions))	BAC5000 Section 8.5.3 a.
D00633	Grease - Aircraft General Purpose	BMS3-33

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
36	Elevator control pushrod	27-31-00-32-040	PTF ALL

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F. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

G. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09

H. Elevator Control Pushrod Installation

SUBTASK 27-31-11-640-003

(1) Lubricate the bolts and bushings with grease, D00633.

SUBTASK 27-31-11-420-007

(2) Install the bolt assembly [96]:

(a) Make sure that the elevator [1] is in the neutral position.

(b) Connect the end of the elevator control pushrod [36] labeled FWD to the elevator buss crank [97] (Figure 409)

1) Make sure that the bushing [88] is installed on the elevator control pushrod [36].

a) If it is necessary, install the bushing [88] with sealant, A00247 (TASK 20-10-02-900-801).

2) Install the bolt assembly [96] (with the bolt head outboard), bushing [95], washer [94], and nut [93].

3) Tighten the nut [93] to 600 in-lb (67.8 N·m) – 700 in-lb (79.1 N·m).

CAUTION DO NOT TIGHTEN NUT [93] AFTER YOU TIGHTEN NUT [92]. BOLT PROTRUSION THROUGH NUT [92] CAN BE 0.01 INCHES MINIMUM, FOR THE BOLT ASSEMBLY. IF YOU DO NOT OBEY, THIS CAN CAUSE DAMAGE TO EQUIPMENT.

4) Install the nut [92].

5) Tighten the nut [92] to 170 in-lb (19.2 N·m) – 240 in-lb (27.1 N·m).

(c) Connect the elevator control pushrod [36] to the elevator mast arm [91]:

1) Make sure that the bushing [87] is installed on the elevator control pushrod [36].

a) If it is necessary, install the bushing [87] with sealant, A00247 (TASK 20-10-02-900-801).

2) Install the bolt assembly [35] (with the bolt head outboard), bushing [34], washer [33], and nut [32].

3) Tighten the nut [32] to 600 in-lb (67.8 N·m) – 700 in-lb (79.1 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

DO NOT TIGHTEN NUT [32] AFTER YOU TIGHTEN NUT [31]. BOLT PROTRUSION THROUGH NUT [31] CAN BE 0.01 INCHES MINIMUM, FOR THE BOLT ASSEMBLY. IF YOU DO NOT OBEY, THIS CAN CAUSE DAMAGE TO EQUIPMENT..

4) Install the nut [31].

5) Tighten the nut [31] to 170 in-lb (19.2 N·m) – 240 in-lb (27.1 N·m).

SUBTASK 27-31-11-080-004

(3) If you used the sling equipment, SPL-2034, to support the elevator assembly, do these steps to remove the sling equipment, SPL-2034, from the elevator [1]:

(a) Remove the fasteners to disconnect the forward sling fittings from the elevator [1].

(b) Remove the forward sling fittings from the elevator [1].

(c) Install the bolts [3] on the elevator [1].

(d) Remove the aft sling fitting from the elevator [1].

SUBTASK 27-31-11-860-009

(4) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	8	C00544	FLIGHT RECORDER POSITION SENSOR

SUBTASK 27-31-11-820-004

(5) Do this task: Elevator Control Pushrods - Adjustment, TASK 27-31-00-820-802.

SUBTASK 27-31-11-700-002

(6) Do this task: Input Rods of the Elevator Power Control Unit - Adjustment, TASK 27-31-00-820-803.

SUBTASK 27-31-11-700-003

(7) Do this task: Elevator Tab Pushrods - Adjustment, TASK 27-31-00-820-805.

SUBTASK 27-31-11-820-005

(8) Do this task: Elevator Position Transmitter Installation Test, TASK 31-31-51-700-801.

SUBTASK 27-31-11-710-004

(9) Do this task: Manual Mode - Test, TASK 27-31-00-700-808.

NOTE: Keep the hydraulic system pressurized after the test.

SUBTASK 27-31-11-790-001

(10) Do a check of the hydraulic connection on the elevator tab mechanism actuator for hydraulic fluid leaks.

SUBTASK 27-31-11-710-005

(11) Do this task: Elevator Surface Friction and Balance - Test, TASK 27-31-00-700-803.

SUBTASK 27-31-11-710-006

(12) Do this task: Elevator Travel Limit - Test, TASK 27-31-00-700-805.

SUBTASK 27-31-11-710-007

(13) Do this task: Elevator Output Torque Tube Buss Linkage Test, TASK 27-31-00-700-806.

SUBTASK 27-31-11-710-008

(14) Do this task: Elevator Tab Linkage - Test, TASK 27-31-00-700-807.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-11-710-009

(15) Do this task: Control Column Travel and Centering - Test, TASK 27-31-00-700-809.

SUBTASK 27-31-11-700-004

(16) Do this task: Control Column Power Force - Test, TASK 27-31-00-700-810.

SUBTASK 27-31-11-700-005

(17) Do this task: Elevator Power-Off Flight Test, TASK 27-31-00-710-803.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-11-410-008

(1) Install the sliding seal [109] as follows:

(a) Move the sliding seal [109] back to its usual position.

(b) Install the bolts [106], washers [107], and nuts [108] that hold the sliding seal [109] to the bracket [110].

1) Install the bolts [106] with the bolt heads facing inboard.

2) Fay seal entire mating surface with polysulfide sealant, A50473.

(c) Install the bolts [103], washers [104] and the aft sliding seal rail [105].

1) Install the bolts [103] with polysulfide sealant, A50473.

(d) Install the bolts [100], washer [101] and the front sliding seal rail [102].

1) Install the bolts [100] with polysulfide sealant, A50473.

CAUTION

MAKE SURE THAT THE CUTOUT IN THE HINGE COVER IS FACING THE FORWARD DIRECTION. IF THE HINGE COVER INSTALLATION IS INCORRECT, DAMAGE TO THE HINGE COVER CAN OCCUR.

(e) For the left horizontal stabilizer, close these access panels:

Number	Name/Location
333AB	Horizontal Stabilizer, Gap Cover, Horizontal Stabilizer to Body
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover

CAUTION

MAKE SURE THAT THE CUTOUT IN THE HINGE COVER IS FACING THE FORWARD DIRECTION. IF THE HINGE COVER INSTALLATION IS INCORRECT, DAMAGE TO THE HINGE COVER CAN OCCUR.

(f) For the right horizontal stabilizer, close these access panels:

Number	Name/Location
343AB	Horizontal Stabilizer, Gap Cover - H. Stab. to Body
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09

SUBTASK 27-31-11-410-004

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

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SUBTASK 27-31-11-710-010

(3) Do this task: Elevator Power-Off Flight Test, TASK 27-31-00-710-803.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR POWER CONTROL UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of an elevator power control unit (PCU)

(2) An installation of an elevator PCU.

B. This procedure is applicable for the left and right elevator PCUs.

TASK 27-31-14-000-801

2. Elevator Power Control Unit - Removal

(Figure 401, Figure 402, Figure 403)

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-1683	Fixture - Elevator Rigging and Restraining Part #: F80063-501 Supplier: 81205

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

E. Prepare for the Removal

SUBTASK 27-31-14-010-001

(1) Open these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-14-860-001

CAUTION

MAKE SURE THAT YOU PROTECT WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR. THE TOOL HAS SHARP EDGES THAT CAN CAUSE CHAFING OR DAMAGE TO THE WIRES.

(2) Use the trammel bar, SPL-1677, to set the B dimension at 39.16 in. (994.66 mm) - 39.18 in. (995.17 mm) (Figure 401).

SUBTASK 27-31-14-860-002

(3) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

F. Elevator PCU Removal

SUBTASK 27-31-14-020-001

(1) Remove the nuts [22], washers [21], washer [20], and bolts [19] to disconnect the four input rods [23] from the four input cranks [18] (two on each elevator Power Control Unit (PCU)) (Figure 402).

(a) Move the control column to move the quadrant out of the way when there are clearance problems with the lower input rod attach bolt.

SUBTASK 27-31-14-020-002

(2) Move the four input cranks [18] forward.

SUBTASK 27-31-14-480-001

(3) Install the rig pin E-5 from the rigging pin kit, SPL-1585, in the elevator aft control quadrant (Figure 402).

NOTE: To easily get access to the lower connection, set the input cranks on the rig pin E-5.

(a) If it is necessary, disconnect the vernier control rod [30] and vernier control rod [31] so that the torque tube can be rotated to gain access to the PCU attach bolt, do these steps (View D, Figure 402).

1) Remove the nut [39], washer [38], washer [37], and bolt [36] to disconnect the vernier control rod [30].

2) Remove the nut [35], washer [34], washer [33], and bolt [32] to disconnect the vernier control rod [31].

SUBTASK 27-31-14-020-003

(4) Remove the elevator PCU [6]:

(a) Remove the nut [17], washer [16], nut [15], washer [14], bolt [12], and bushing [13] that attach the elevator PCU [6] to the mounting brackets.

1) If it is necessary, move the control quadrant to rotate the input cranks out of the way of the PCU attachment bolt head.

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WARNING RELEASE THE HYDRAULIC CONNECTION SLOWLY AND LET THE PRESSURE DECREASE BEFORE YOU FULLY REMOVE THE HYDRAULIC LINES. IF YOU DISCONNECT THE HYDRAULIC LINE WHEN IT HAS PRESSURE, A SPRAY OF HYDRAULIC FLUID CAN CAUSE INJURY TO PERSONS.

(b) Disconnect the hydraulic line [3] an hydraulic line [24] from the applicable elevator PCU [6].

1) Install caps and plugs on the hydraulic lines and ports.

(c) Remove the nut [11], washer [10], washer [9], and bolt [8] that attach the elevator PCU [6] to the output torque tube [7].

(d) Remove the elevator PCU [6] from the airplane.

SUBTASK 27-31-14-480-002

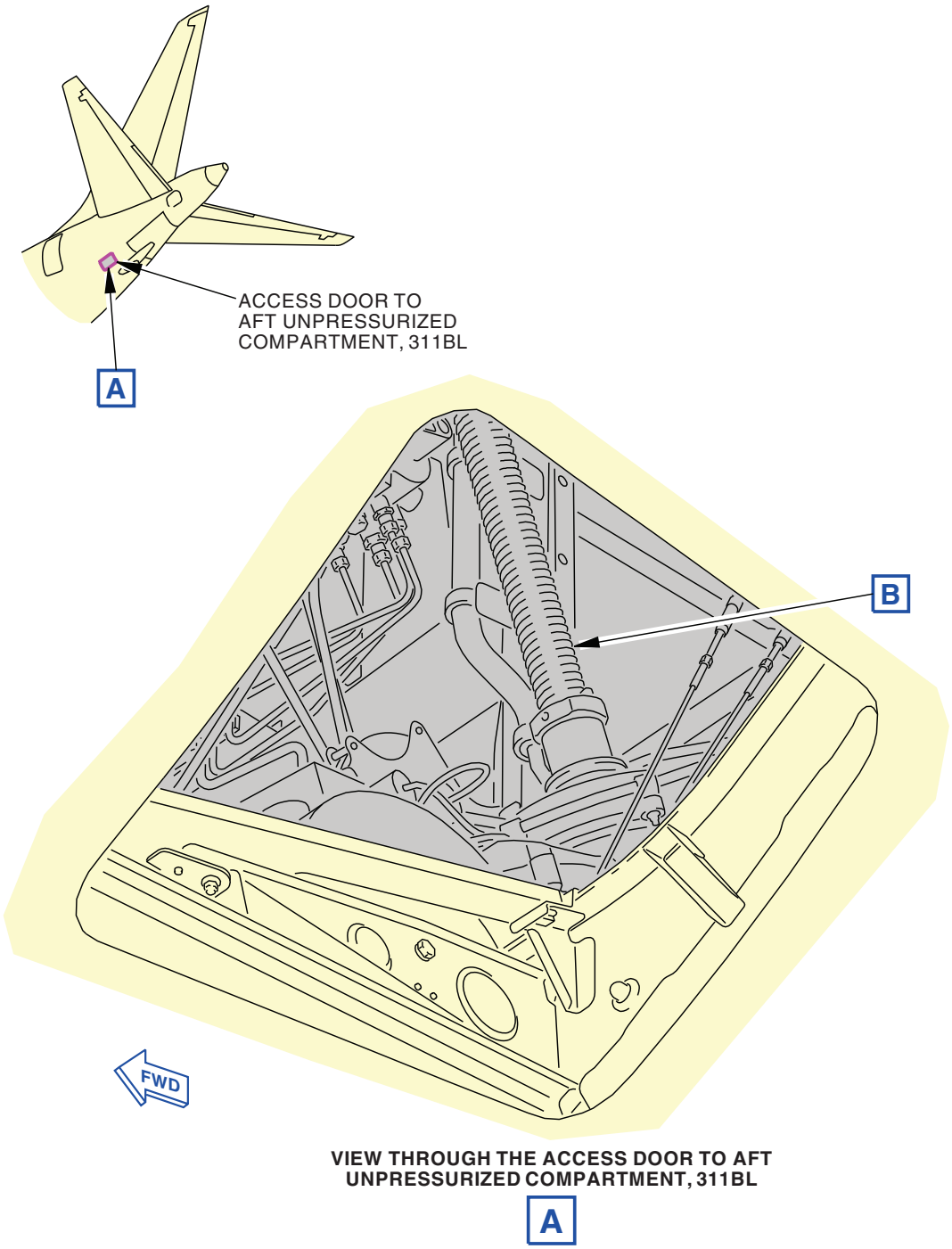
(5) If you remove the two elevator PCUs at the same time, then install the fixture, SPL-1683, in the position of the right elevator PCU [6] (Figure 403).

(a) Put the fixture, SPL-1683, between the output torque tube and input crank.

(b) Install the pins of fixture, SPL-1683, to hold the fixture, SPL-1683, in position.

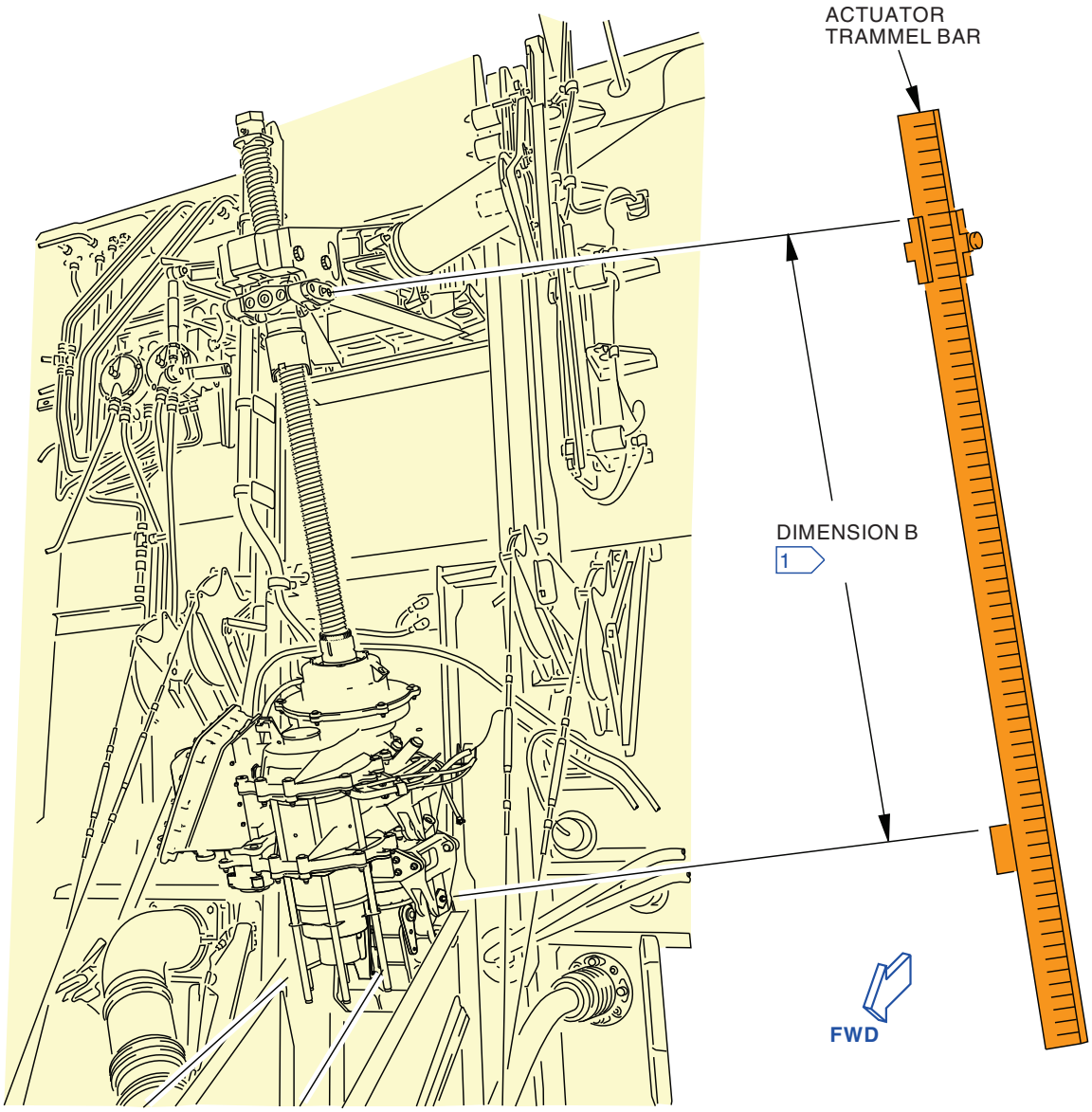
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Jackscrew Adjustment
Figure 401/27-31-14-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACTUATOR

B

NOTE:

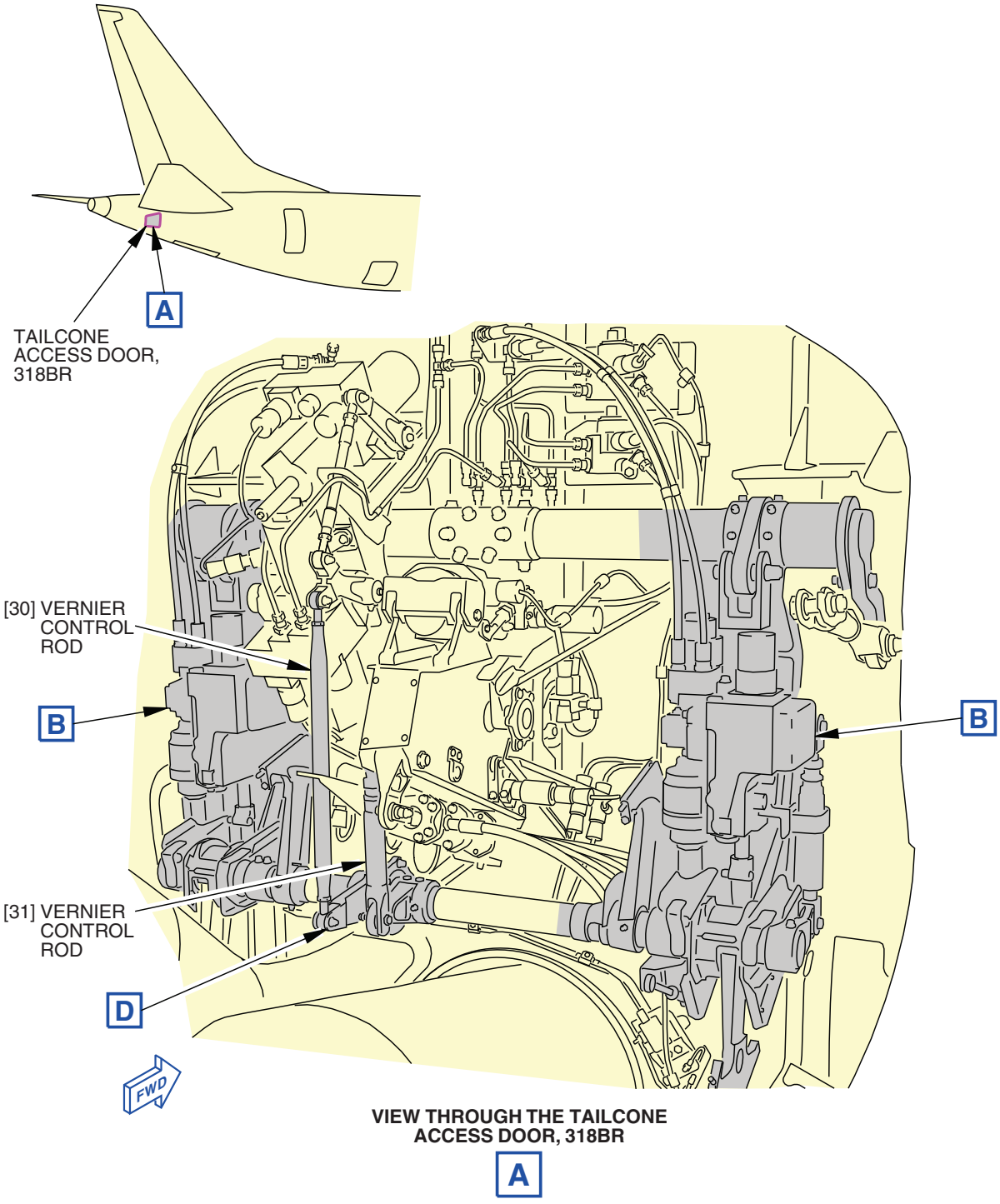
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

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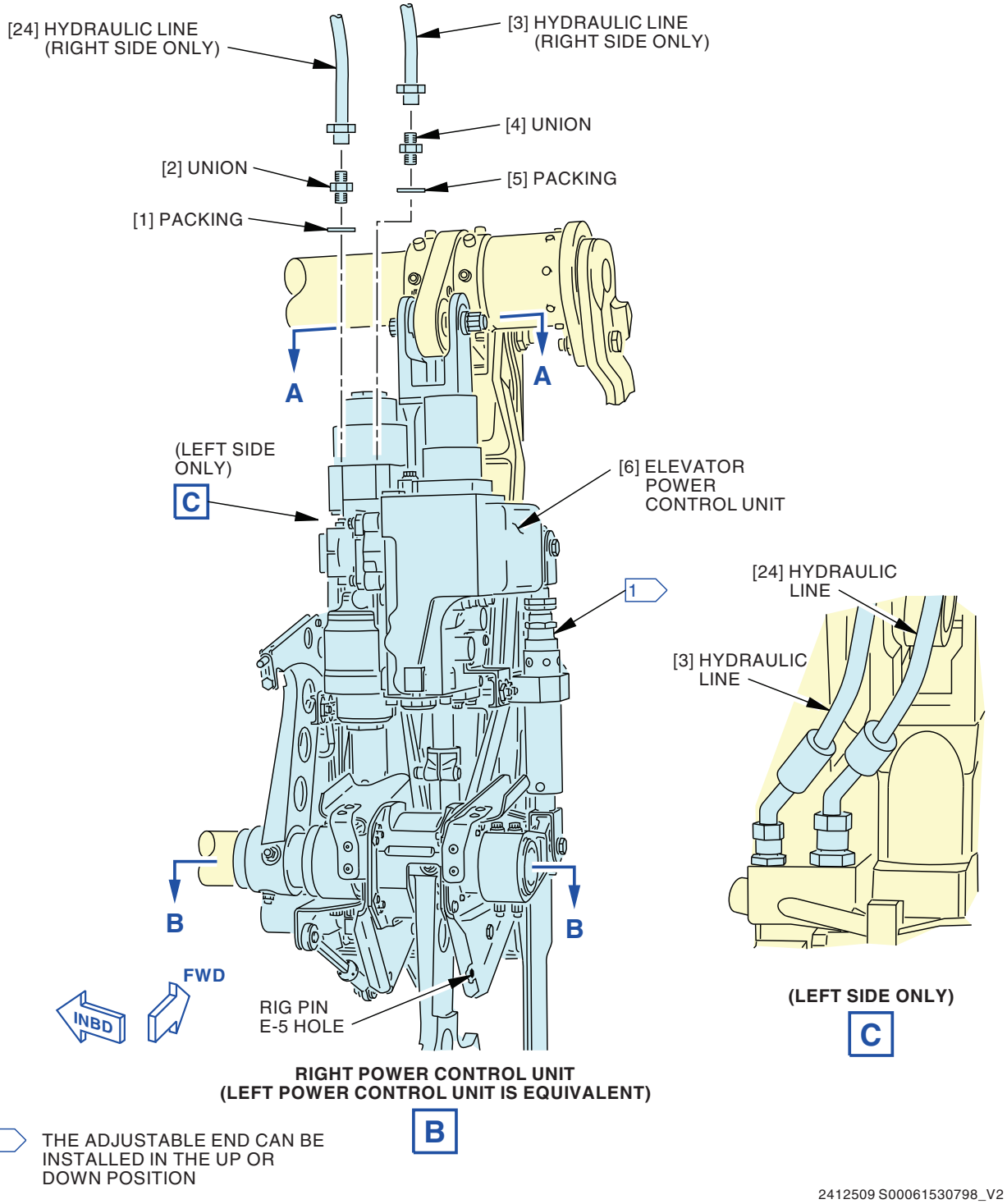
Stabilizer Trim Jackscrew Adjustment
Figure 401/27-31-14-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



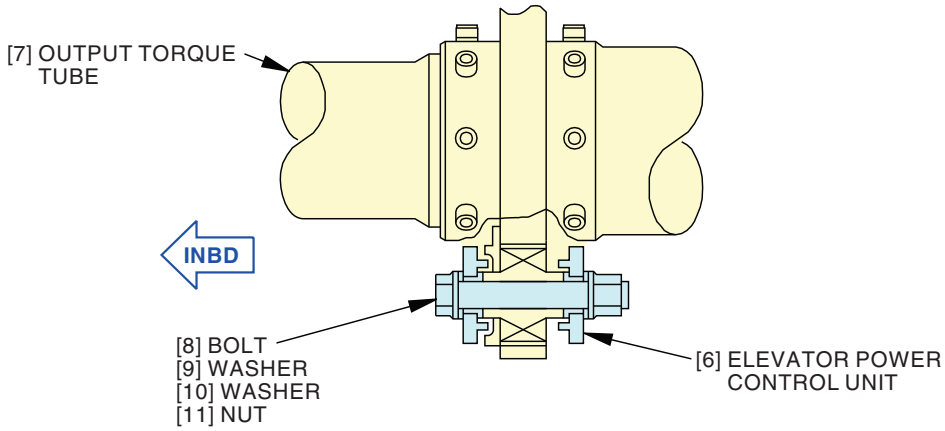
Elevator Power Control Unit Installation
Figure 402/27-31-14-990-802 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

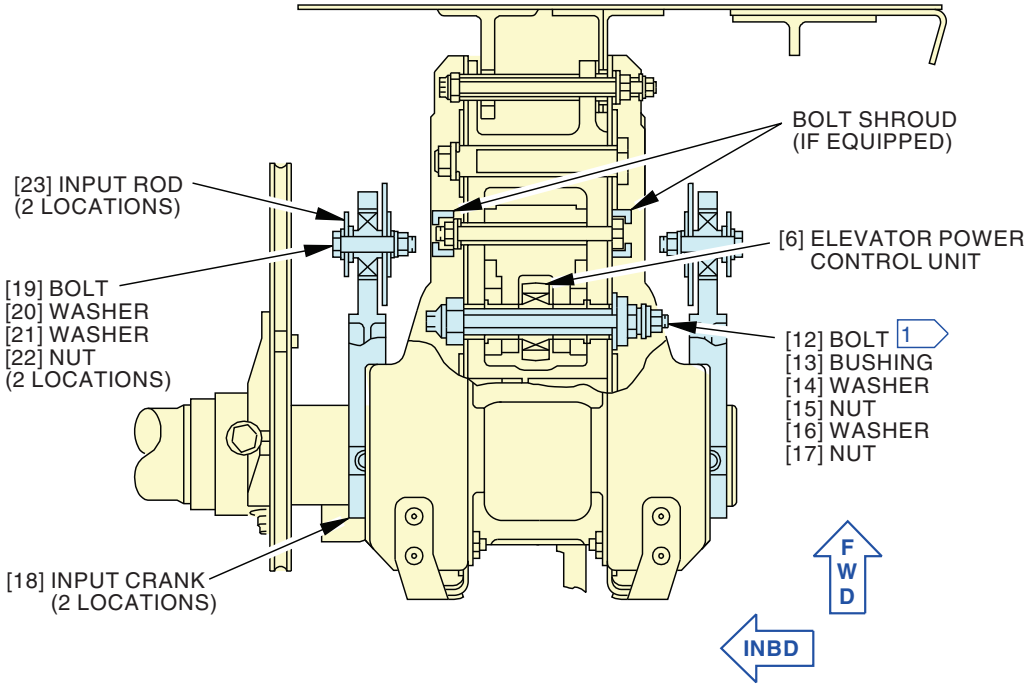


Elevator Power Control Unit Installation
Figure 402/27-31-14-990-802 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



(RIGHT SIDE SHOWN)

B-B

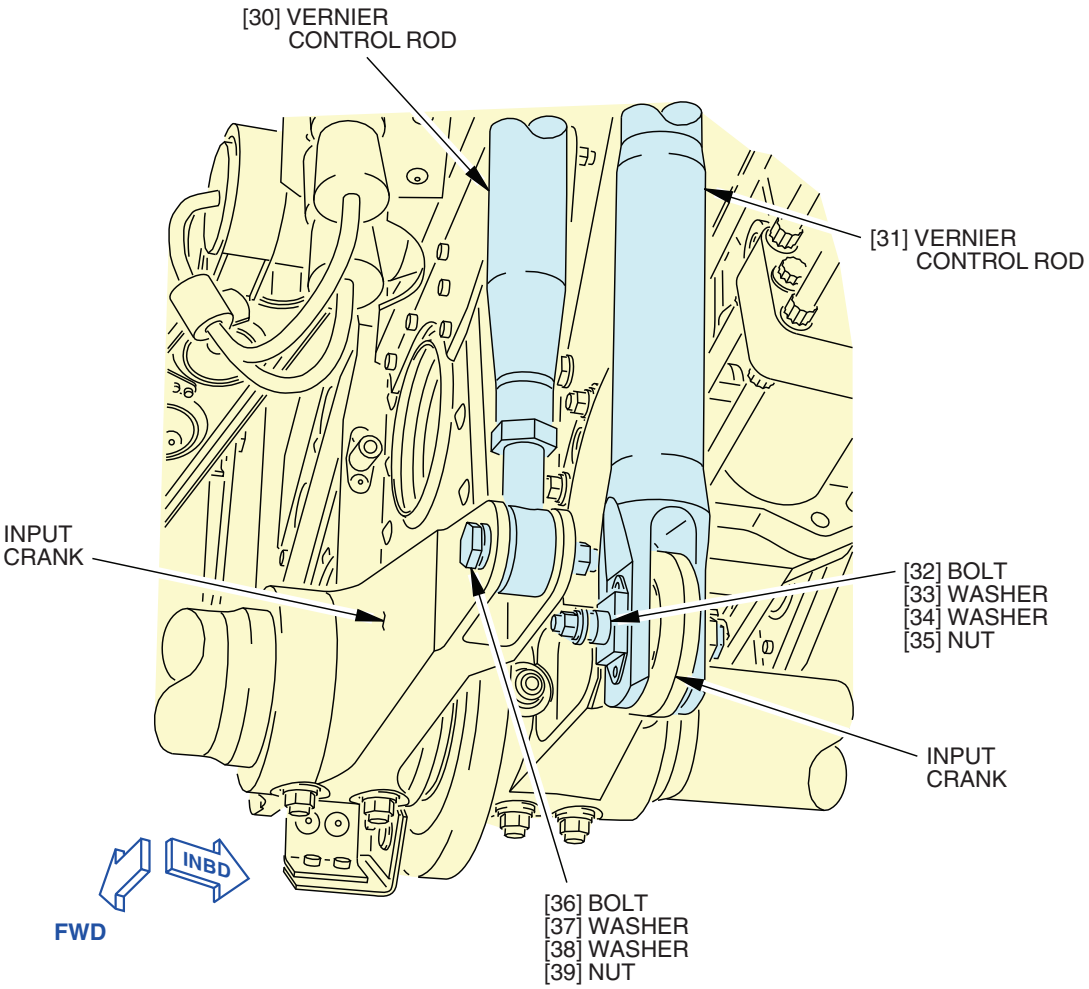
NOTE:

1 INSTALL THE BOLT WITH THE BOLT HEAD INBOARD FOR THE LEFT ELEVATOR PCU AND THE RIGHT ELEVATOR PCU (SHOWN).

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Elevator Power Control Unit Installation
Figure 402/27-31-14-990-802 (Sheet 3 of 4)

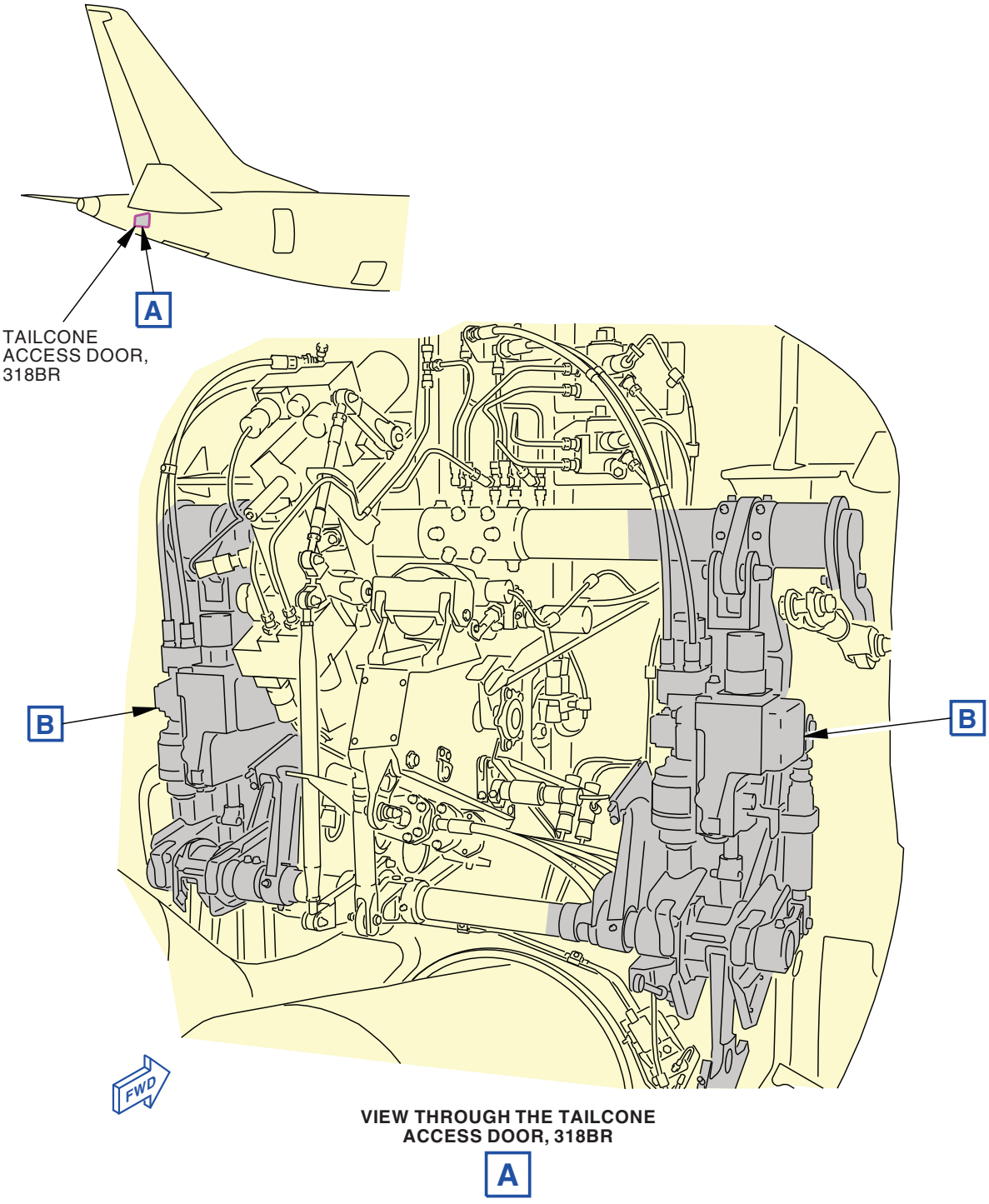
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3073434 S0000832755_V1

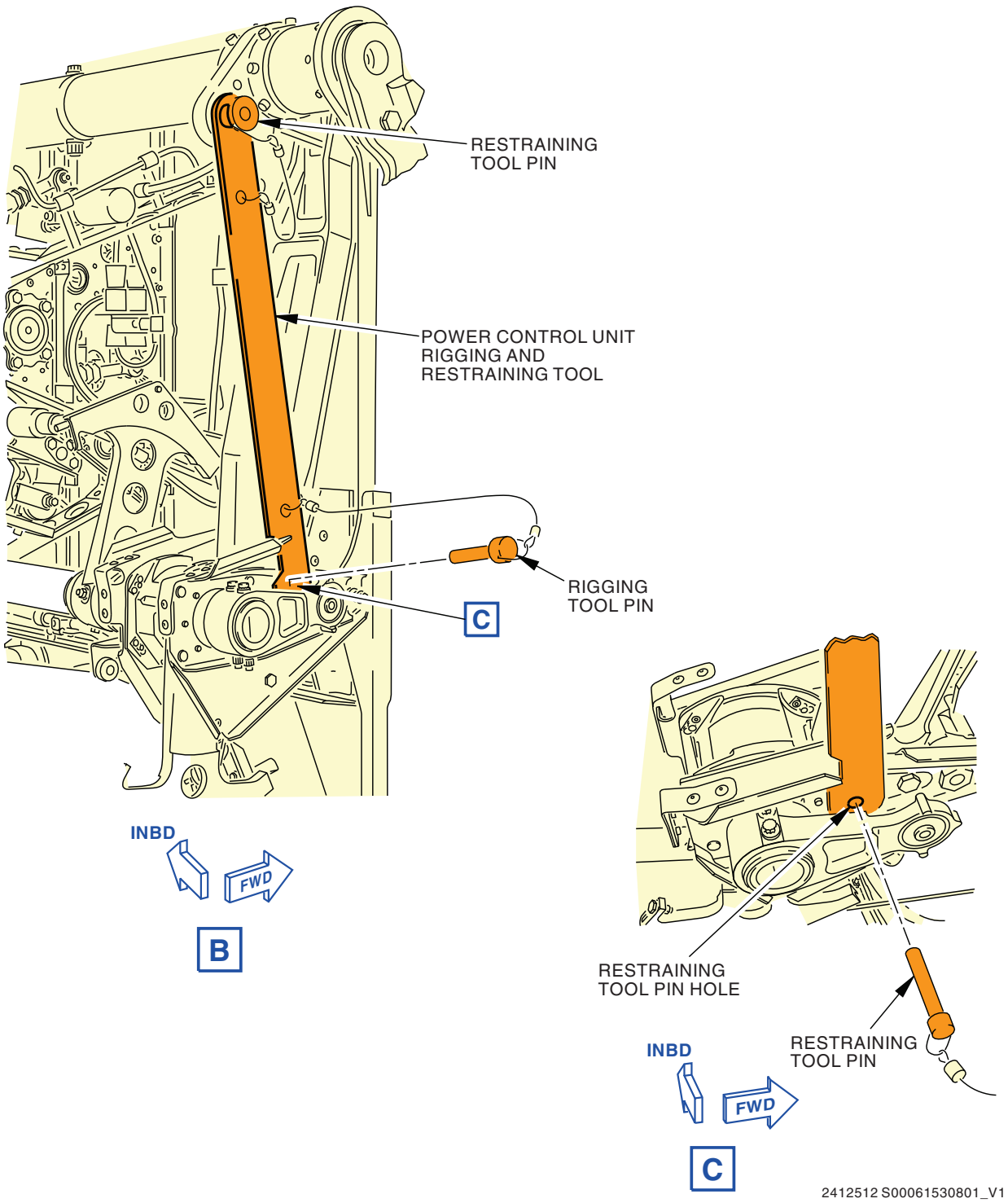
Elevator Power Control Unit Installation
Figure 402/27-31-14-990-802 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Power Control Unit Rigging and Restraining Tool Installation
Figure 403/27-31-14-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Power Control Unit Rigging and Restraining Tool Installation
Figure 403/27-31-14-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-14-400-801

3. Elevator Power Control Unit Installation

(Figure 401, Figure 402, Figure 403)

A. References

Reference	Title
27-31-00-700-803	Elevator Surface Friction and Balance - Test (P/B 501)
27-31-00-700-805	Elevator Travel Limit - Test (P/B 501)
27-31-00-700-808	Manual Mode - Test (P/B 501)
27-31-00-700-809	Control Column Travel and Centering - Test (P/B 501)
27-31-00-700-810	Control Column Power Force - Test (P/B 501)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-820-803	Input Rods of the Elevator Power Control Unit - Adjustment (P/B 501)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1683	Fixture - Elevator Rigging and Restraining Part #: F80063-501 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Packing	27-31-14-02-500	PTF ALL
5	Packing	27-31-14-02-505	PTF ALL
6	Elevator PCU	27-31-14-03A-285	PTF ALL

E. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
315	APU Compartment - Left
316	APU Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

G. Prepare for the Installation

SUBTASK 27-31-14-860-003

(1) Make sure that the pressure to the elevator hydraulic systems A and B is removed (TASK 27-31-00-800-802).

SUBTASK 27-31-14-080-001

(2) Remove the fixture, SPL-1683, if it is applicable.

H. Elevator PCU Installation

SUBTASK 27-31-14-420-001

(1) If the packings and unions are not installed in the elevator PCU [6], then do these steps:

(a) Lubricate the new packing [1] and new packing [5] with hydraulic fluid, D00153.

(b) Install the packing [1], union [2], packing [5], and union [4] in the elevator PCU [6].

SUBTASK 27-31-14-420-002

(2) Install the elevator PCU [6]:

(a) Apply grease, D00633, to the bolt [8] and bolt [12].

(b) Put the elevator PCU [6] in its position.

(c) Install the bolt [8] with the bolt head inboard, washer [9], washer [10], and nut [11] to attach the elevator PCU [6] to the output torque tube [7].

1) Tighten the nut [11] to 558 in-lb (63.0 N·m) - 592 in-lb (66.9 N·m).

(d) Install the bushing [13], bolt [12], washer [14], and nut [15] to attach the elevator PCU [6] to the mounting bracket.

1) For the right elevator Power Control Unit (PCU), install the bolt [12] with the bolt head inboard.

2) For the left elevator PCU, install the bolt [12] with the bolt head inboard.

3) Tighten the nut [15] to 550 in-lb (62.1 N·m) - 600 in-lb (67.8 N·m).

CAUTION

DO NOT TIGHTEN THE NUT [15] AGAIN AFTER YOU TIGHTEN THE NUT [17]. YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(e) Install the washer [16] and nut [17].

1) Tighten the nut [17] to 225 in-lb (25.4 N·m) - 250 in-lb (28.2 N·m).

SUBTASK 27-31-14-080-002

(3) Remove the rig pin E-5 of the rigging pin kit, SPL-1585, from the elevator aft control quadrant.

(a) If it is necessary, connect the vernier control rod [30] and vernier control rod [31] if they were disconnected to give access to the PCU attach bolt, do these steps:

1) Install the nut [39], washer [38] (under nut [39]), washer [37] (under the bolt [36]), and bolt [36] to connect the vernier control rod [30].

a) Tighten the nut [39] to 76 in-lb (8.6 N·m) - 80 in-lb (9.0 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

DO NOT TRY TO TIGHTEN THE BOLT AGAIN AFTER YOU TIGHTEN THE NUT. YOU CAN CAUSE DAMAGE TO EQUIPMENT.

2) Install the nut [35], washer [34] (under the nut [35]), washer [33] (under the bolt [32]), and bolt [32] to connect the vernier control rod [31].
   a) Tighten the outer bolt of the bolt [32] to 75 in-lb (8.5 N·m) - 85 in-lb (9.6 N·m).
   b) Tighten the nut [35] to 25 in-lb (2.8 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 27-31-14-860-004

(4) Move the elevator PCU [6] to the fully retracted position.

SUBTASK 27-31-14-420-003

(5) Connect the four input rods [23] to the four input cranks [18] (two on each PCU):

NOTE: Move the control column to move the quadrant out of the way when there are clearance problems with the lower input rod attach bolt.

(a) Apply grease, D00633, to the bolts [19].

(b) Install the bolts [19], washers [20], washers [21], and nuts [22].
   1) For the outboard bolt, install the bolt [19] with the bolt head outboard.
   2) For the inboard bolt, install the bolt [19] with the bolt head inboard.

SUBTASK 27-31-14-420-004

(6) Remove the caps and plugs from the hydraulic lines and ports.

(a) Remove hydraulic fluid that falls on the floor.

SUBTASK 27-31-14-420-005

(7) For the right elevator PCU [6], do these steps:

(a) Connect the hydraulic line [3] and hydraulic line [24] to the right elevator PCU [6].

(b) Tighten the B-nut on the RETURN port hydraulic line [3] to 161 in-lb (18.2 N·m) - 179 in-lb (20.2 N·m).

(c) Tighten the B-nut on the PRESSURE port hydraulic line [24] to 104 in-lb (11.8 N·m) - 116 in-lb (13.1 N·m).

SUBTASK 27-31-14-420-006

(8) For the left elevator PCU [6], do these steps:

(a) Connect the hydraulic line [3] and hydraulic line [24] to the right elevator PCU [6].

(b) Tighten the B-nut on the RETURN port hydraulic line [3] to 161 in-lb (18.2 N·m) - 179 in-lb (20.2 N·m).

(c) Tighten the B-nut on the PRESSURE port hydraulic line [24] to 104 in-lb (11.8 N·m) - 116 in-lb (13.1 N·m).

SUBTASK 27-31-14-860-005

(9) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-14-790-001

(10) Examine the hydraulic connections of the elevator PCU [6] for leaks.

SUBTASK 27-31-14-820-001

(11) Do this task: Input Rods of the Elevator Power Control Unit - Adjustment, TASK 27-31-00-820-803.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-14-820-002

(12) Do this task: Elevator Surface Friction and Balance - Test, TASK 27-31-00-700-803.

SUBTASK 27-31-14-820-003

(13) Do this task: Elevator Travel Limit - Test, TASK 27-31-00-700-805.

SUBTASK 27-31-14-820-004

(14) Do this task: Manual Mode - Test, TASK 27-31-00-700-808.

SUBTASK 27-31-14-820-005

(15) Do this task: Control Column Travel and Centering - Test, TASK 27-31-00-700-809.

SUBTASK 27-31-14-820-006

(16) Do this task: Control Column Power Force - Test, TASK 27-31-00-700-810.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-14-410-001

(1) Close these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR POWER CONTROL UNIT - INSPECTION/CHECK

1. General

A. This procedure has these inspection tasks:

(1) Elevator Power Control Unit (PCU) Visual Inspection

(2) Elevator PCU Inspection.

(a) This task supplies the data for a dimensional inspection of the elevator PCU.

(b) This task does not remove or install the elevator PCU, refer to the Elevator Power Control Unit - Removal/Installation procedures to remove and install the elevator PCU.

TASK 27-31-14-210-801

2. Elevator Power Control Unit Visual Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-41-00-800-801	Horizontal Stabilizer Trim Control System - Deactivation (P/B 201)
27-41-00-800-802	Horizontal Stabilizer Trim Control System - Activation (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

C. Access Panels

Number	Name/Location
318BR	Tailcone Access Door

D. Elevator Power Control Unit Visual Inspection

SUBTASK 27-31-14-860-008

(1) Set the horizontal stabilizer to the full up (airplane nose down) position.

SUBTASK 27-31-14-040-001

(2) Do this task: Horizontal Stabilizer Trim Control System - Deactivation, TASK 27-41-00-800-801.

SUBTASK 27-31-14-010-004

(3) Remove this access panel to get access to the elevator Power Control Units (PCUs):

Number	Name/Location
318BR	Tailcone Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-14-860-006

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SPOILERS AND THE THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) To pressurize hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-31-14-210-001

(5) Do a general visual inspection of the elevator PCU.

SUBTASK 27-31-14-210-002

(6) Do a general visual inspection to make sure that there are no hydraulic leaks from the elevator PCU.

SUBTASK 27-31-14-860-007

(7) To remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-31-14-410-004

(8) Install this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-14-440-001

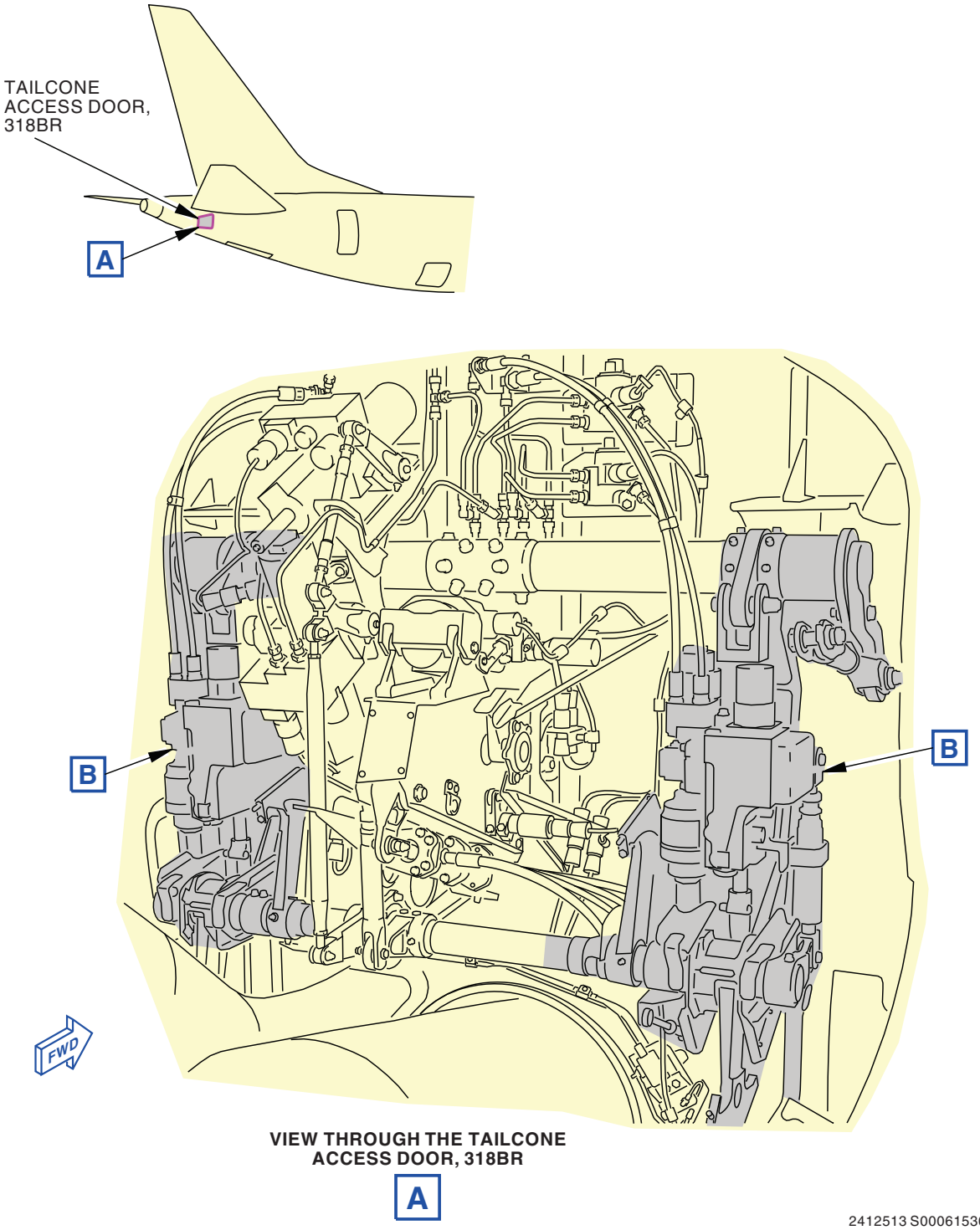
(9) Do this task: Horizontal Stabilizer Trim Control System - Activation, TASK 27-41-00-800-802.

SUBTASK 27-31-14-860-009

(10) Set the horizontal stabilizer to the NORMAL position.

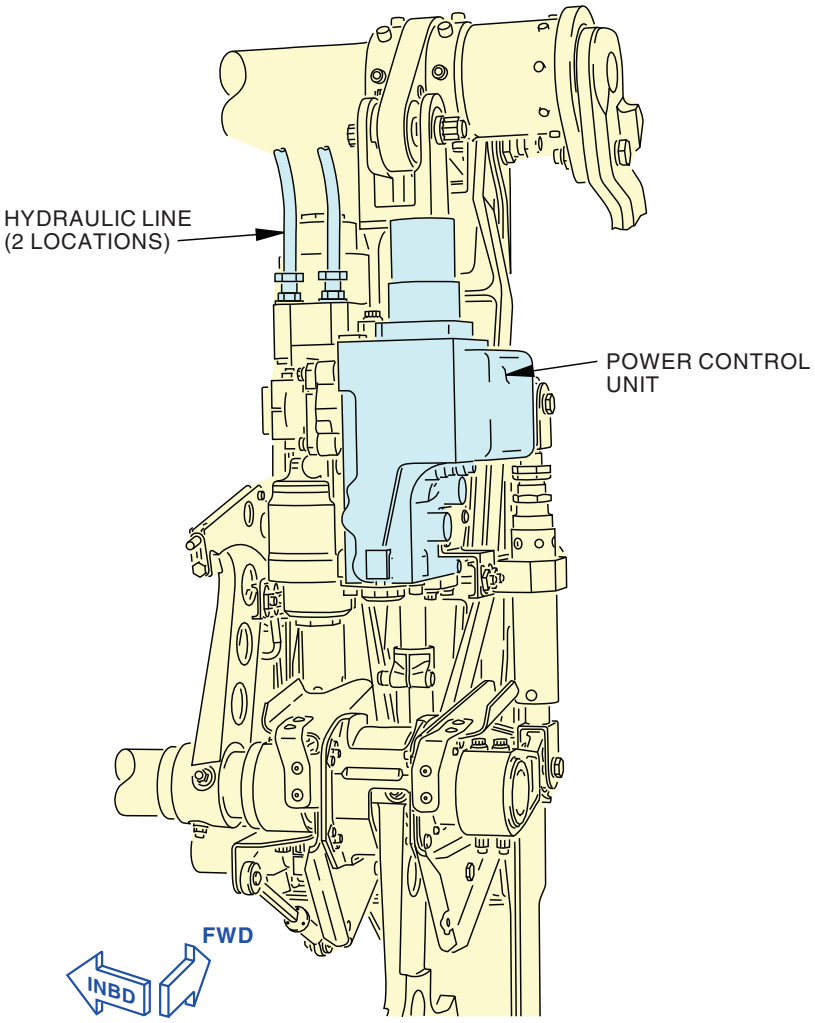
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Power Control Unit Location
Figure 601/27-31-14-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RIGHT POWER CONTROL UNIT
(LEFT POWER CONTROL UNIT IS OPPOSITE)

B

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Elevator Power Control Unit Location
Figure 601/27-31-14-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-14-200-801

3. Elevator Power Control Unit - Inspection

(Figure 602)

A. General

(1) This procedure gives wear limits and repair data for the elevator power control unit (PCU).

B. References

Reference	Title
27-31-14-000-801	Elevator Power Control Unit - Removal (P/B 401)
27-31-14-400-801	Elevator Power Control Unit Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

E. Elevator Power Control Unit Inspection

SUBTASK 27-31-14-020-004

(1) Remove the elevator power control unit (PCU). To remove it, do this task: Elevator Power Control Unit - Removal, TASK 27-31-14-000-801.

SUBTASK 27-31-14-020-005

(2) Use these tools to check for wear on the parts of the elevator PCU connections:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-31-14-020-006

(3) Examine the parts of the elevator PCU connections for worn areas:

(a) Remove the parts to be checked.

(b) Measure the parts as necessary with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096, or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions that follow.

(d) If the parts are not in the tolerance, replace or repair the part as required.

Table 601/27-31-14-993-801

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 BEARING	I.D.	0.4995 in 12.687 mm	0.5000 in 12.700 mm	0.5030 in 12.776 mm	0.0035 in

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-31-14-993-801 (Continued)

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 BOLT	O.D.	0.4985 in 12.662 mm	0.4995 in 12.687 mm	0.4981 in 12.652 mm	0.089 mm
#2 PCU END FITTING BUSHING	I.D.	0.4995 in 12.687 mm	0.5000 in 12.700 mm	0.5030 in 12.776 mm	0.0035 in 0.089 mm
#2 BOLT	O.D.	0.4985 in 12.662 mm	0.4995 in 12.687 mm	0.4981 in 12.652 mm	
#3 BUSHING	I.D.	0.6245 in 15.862 mm	0.6250 in 15.875 mm	0.6275 in 15.938 mm	0.0033 in 0.084 mm
#3 BOLT ASSEMBLY	O.D.	0.6237 in 15.842 mm	0.6242 in 15.855 mm	0.6212 in 15.778 mm	
#4 BUSHING	I.D.	0.6245 in 15.862 mm	0.6250 in 15.875 mm	0.6265 in 15.913 mm	0.0033 in 0.084 mm
#4 BOLT ASSEMBLY	O.D.	0.6237 in 15.842 mm	0.6242 in 15.855 mm	0.6212 in 15.778 mm	
#5 BEARING	I.D.	0.6245 in 15.862 mm	0.6250 in 15.875 mm	0.6275 in 15.938 mm	0.0033 in 0.084 mm
#5 BOLT ASSEMBLY	O.D.	0.6237 in 15.842 mm	0.6242 in 15.855 mm	0.6212 in 15.778 mm	
#6 SUPPORT LINK	I.D.	0.7500 in 19.050 mm	0.7507 in 19.068 mm	0.7522 in 19.106 mm	0.0022 in 0.056 mm
#6 BUSHING	O.D.	0.7495 in 19.037 mm	0.7500 in 19.050 mm	0.7491 in 19.027 mm	
#7 BUSHING	I.D.	0.6245 in 15.862 mm	0.6250 in 15.875 mm	0.6265 in 15.913 mm	0.0023 in 0.058 mm
#7 BOLT	O.D.	0.6237 in 15.842 mm	0.6242 in 15.855 mm	0.6212 in 15.778 mm	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-31-14-993-801 (Continued)

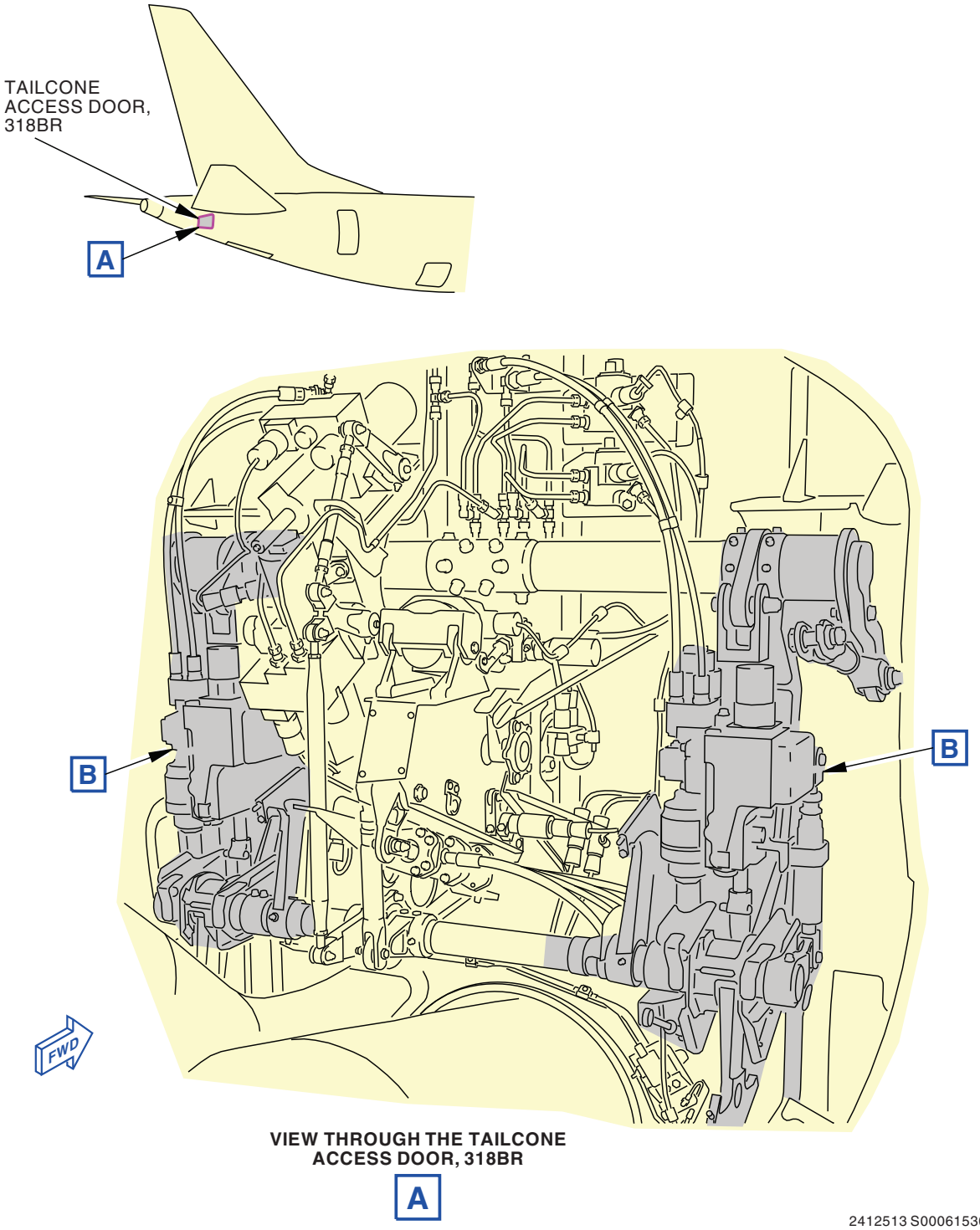
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#8 BEARING	I.D.	0.3122 in 7.930 mm	0.3125 in 7.937 mm	0.3155 in 8.014 mm	0.0035 in 0.089 mm
#8 BOLT	O.D.	0.3110 in 7.899 mm	0.3120 in 7.925 mm	0.3106 in 7.889 mm	
#9 INPUT ROD	I.D.	0.3120 in 7.925 mm	0.3130 in 7.950 mm	0.3160 in 8.026 mm	0.0040 in 0.102 mm
#9 BOLT	O.D.	0.3110 in 7.899 mm	0.3120 in 7.925 mm	0.3106 in 7.889 mm	

SUBTASK 27-31-14-020-007

(4) Install the elevator PCU. To install it, do this task: Elevator Power Control Unit Installation, TASK 27-31-14-400-801.

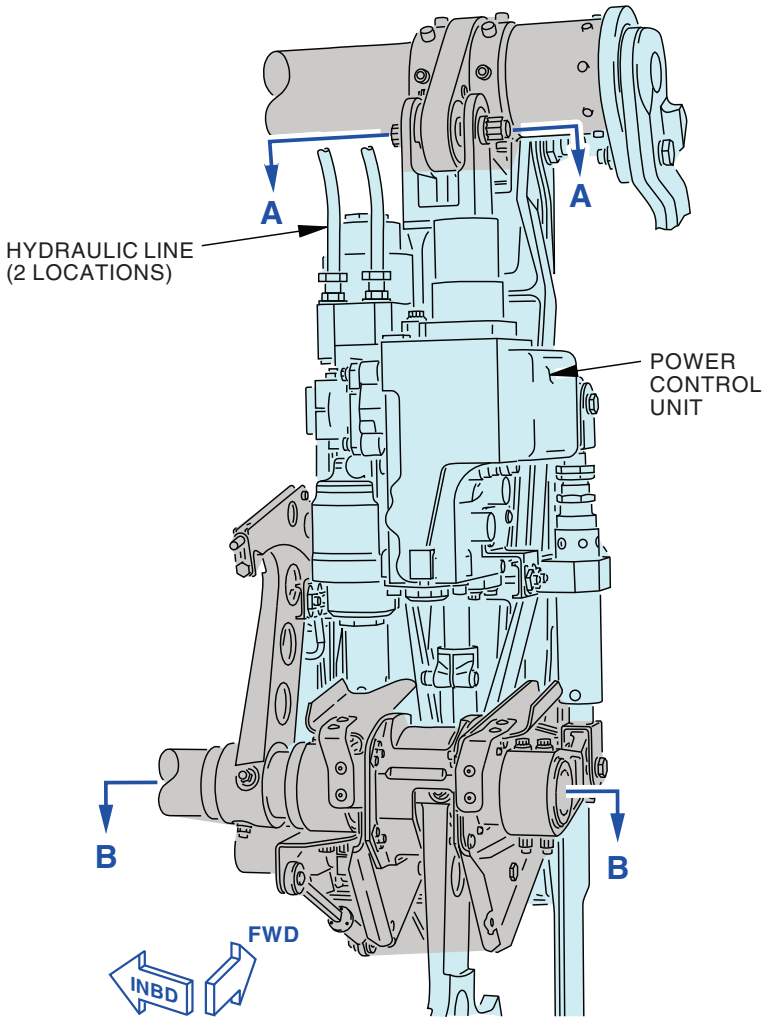
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Power Control Unit Wear Limits
Figure 602/27-31-14-990-805 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



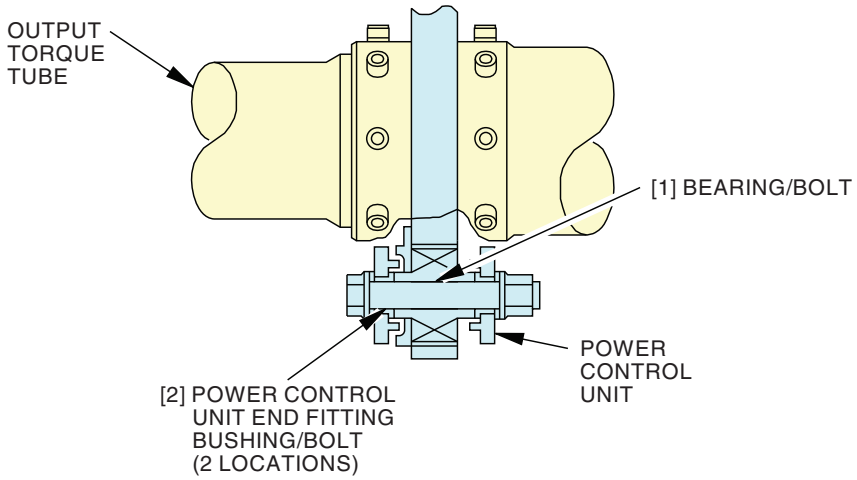
RIGHT POWER CONTROL UNIT
(LEFT POWER CONTROL UNIT IS OPPOSITE)

B

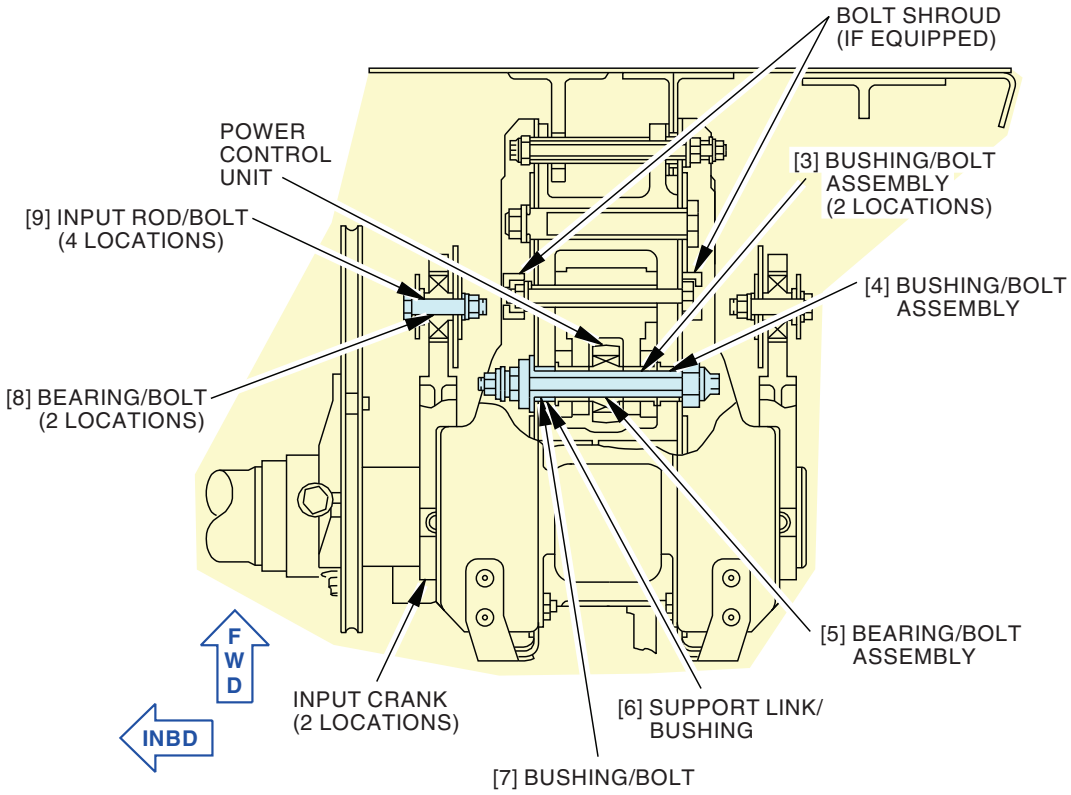
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Elevator Power Control Unit Wear Limits
Figure 602/27-31-14-990-805 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



B-B

2412516 S00061530809_V1

Elevator Power Control Unit Wear Limits
Figure 602/27-31-14-990-805 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR PITOT-STATIC SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Elevator Pitot-Static System Flushing

(2) Elevator Pitot-Static System Drainage Inspection

(3) Elevator Pitot Probe Inspection

(4) Elevator Pitot System Leak Test

(5) Elevator Pitot System Pressurization.

TASK 27-31-17-170-801

2. Elevator Pitot-Static System Flushing

(Figure 201)

A. General

(1) Air Pressure in PSIG/kPa will simulate knots of computed airspeed (kcas).

B. References

Reference	Title
27-31-94-000-801	Feel System Pitot Probe Removal (P/B 401)
27-31-94-400-801	Feel System Pitot Probe Installation (P/B 401)
34-11-01-100-801	Pitot Probe Cleaning (P/B 701)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1916	Adapter - Pitot Test (Typically included in Air Data Accessory Kit) Part #: CSA75700HT-3 Supplier: 3BSK6 Part #: P75701-4 Supplier: 38002 Part #: P75701M2-3 Supplier: 38002
COM-16577	Plug - Pitot Part #: BACP20AU6D Supplier: 81205
STD-1154	Container - 5 Gallon (19 Liter)
STD-1280	Source - Air, Regulated, Dry Filtered, 0-30 PSIG

D. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B50126	Solvent - Degreasing, General Purpose	MIL-PRF-680

E. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
323	Vertical Fin - Front Spar To Rear Spar

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F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Prepare for the Flushing

SUBTASK 27-31-17-010-001

(1) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-17-860-001

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

H. Elevator Pitot-Static System Flushing

SUBTASK 27-31-17-170-001

(1) Flush the pitot system:

(a) Inspect the pitot probes, do this task: Elevator Pitot Probe - Inspection, TASK 27-31-17-200-802.

1) If debris is visually detected or visual inspection can not be completed, do these steps:

a) Remove the elevator pitot probe, do this task: Feel System Pitot Probe Removal, TASK 27-31-94-000-801.

b) Clean the elevator pitot probe, refer to task: Pitot Probe Cleaning, TASK 34-11-01-100-801.

c) Install the elevator pitot probe, do this task: Feel System Pitot Probe Installation, TASK 27-31-94-400-801.

CAUTION

MAKE SURE THAT YOU DISCONNECT THE FEEL COMPUTER LINES BEFORE YOU FLUSH THE LINES WITH A DRY FILTERED AIR SOURCE OF 20-30 PSIG, STD-1280. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Disconnect the left and right pitot lines [4] from the elevator feel computer [5].

(c) Install plugs on the elevator feel computer [5] ports.

(d) Remove the left and right restrictor valves [3] from the pitot lines [4].

(e) Clean the restrictor valves [3] with degreasing solvent, B50126, or solvent, B00062 (TASK 27-31-17-200-801).

(f) Put a 5 gallon (19 liter) container, STD-1154, below the pitot lines to catch any debris.

(g) Install the pitot plug, COM-16577, on the pitot line [4] for the left elevator feel computer [5].

(h) Install the pitot test adapter, COM-1916, on the left pitot probe.

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(i) From the left pitot probe, flush the left pitot line [4] with 20 psig (137.9 kPa) - 30 psig (206.8 kPa) of air constantly for 3 minutes and no unwanted material comes out from the tubes using a 0-30 psig dry filtered regulated air source, STD-1280.

1) Make sure that the pitot line [4] is not blocked and air flows out of the other open end.

2) If air does not flow out of the other end from the left pitot system, do the following steps:

a) Remove the elevator pitot probe, do this task: Feel System Pitot Probe Removal, TASK 27-31-94-000-801.

b) Clean the elevator pitot probe, refer to task: Pitot Probe Cleaning, TASK 34-11-01-100-801.

c) Install the elevator pitot probe, do this task: Feel System Pitot Probe Installation, TASK 27-31-94-400-801.

(j) Remove the pitot plug, COM-16577, from the pitot line [4] for the left elevator feel computer [5].

(k) Install the pitot plug, COM-16577, on the pitot line [4] for the left restrictor valve [3].

(l) From the left pitot probe, flush the left pitot line [4] with 20 psig (137.9 kPa) - 30 psig (206.8 kPa) of air constantly for 3 minutes and no unwanted material comes out from the tubes using a 0-30 psig dry filtered regulated air source, STD-1280.

1) Make sure that the pitot line [4] is not blocked and air flows out of the other open end.

(m) Remove the pitot plug, COM-16577, from the pitot line [4] for the left restrictor valve [3].

(n) Remove the pitot test adapter, COM-1916, from the left pitot probe.

(o) Install the pitot test adapter, COM-1916, on the right pitot probe.

(p) Install the pitot plug, COM-16577, on the pitot line [4] for the right elevator feel computer [5].

(q) From the right pitot probe, flush the right pitot line [4] with 20 psig (137.9 kPa) - 30 psig (206.8 kPa) of air constantly for 3 minutes and no unwanted material comes out from the tubes using a 0-30 psig dry filtered regulated air source, STD-1280.

1) Make sure that the pitot line [4] is not blocked and air flows out of the other open end.

2) If air does not flow out of the other end from the right pitot system, do the following steps:

a) Remove the elevator pitot probe, do this task: Feel System Pitot Probe Removal, TASK 27-31-94-000-801.

b) Clean the elevator pitot probe, refer to task: Pitot Probe Cleaning, TASK 34-11-01-100-801.

c) Install the elevator pitot probe, do this task: Feel System Pitot Probe Installation, TASK 27-31-94-400-801.

(r) Remove the pitot plug, COM-16577, from the pitot line [4] for the right elevator feel computer [5].

(s) Install the pitot plug, COM-16577, on the pitot line [4] for the restrictor valve [3].

(t) From the right pitot probe, flush the right pitot line [4] with 20 psig (137.9 kPa) - 30 psig (206.8 kPa) of air constantly for 3 minutes and no unwanted material comes out from the tubes using a 0-30 psig dry filtered regulated air source, STD-1280.

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1) Make sure that the pitot line [4] is not blocked and air flows out of the other open end.

(u) Remove the pitot test adapter, COM-1916, from the right pitot probe.

(v) Remove the pitot plug, COM-16577, from the pitot line [4] for the right restrictor valve [3].

(w) Check the 5 gallon (19 liter) container, STD-1154, for unwanted material from the pitot lines.

1) If unwanted material is found in the container, examine the pitot lines.

(x) Remove the plugs from the elevator feel computer [5] ports.

(y) Connect the pitot lines [4] to the elevator feel computer [5].

1) Tighten the pitot lines [4] to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

(z) Install the restrictor valves [3] on the pitot lines [4].

1) Tighten the restrictor valves [3] to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

SUBTASK 27-31-17-170-002

(2) Flush the static system:

(a) Disconnect the static lines [2] from the elevator feel computer [5].

(b) Install plugs on the elevator feel computer [5] ports.

(c) Flush the static lines [2] from each end with 20 psig (137.9 kPa) - 30 psig (206.8 kPa) of air for 3 minutes and no unwanted material comes out from the tubes using a 0-30 psig dry filtered regulated air source, STD-1280.

1) Make sure that the static lines [2] are not blocked and air flows out of the other open end.

(d) Remove the plugs from the elevator feel computer [5] ports.

(e) Connect the static lines [2] to the elevator feel computer [5].

1) Tighten the B-nuts on the static lines [2] to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

SUBTASK 27-31-17-210-001

(3) Make sure that there is no unwanted material in the ports of the pitot-static system.

SUBTASK 27-31-17-210-002

(4) Do this task: Elevator Pitot-Static System Drainage - Inspection, TASK 27-31-17-200-801.

SUBTASK 27-31-17-790-001

(5) Do this task: Elevator Pitot System Leak - Test, TASK 27-31-17-790-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-17-860-002

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-31-17-410-001

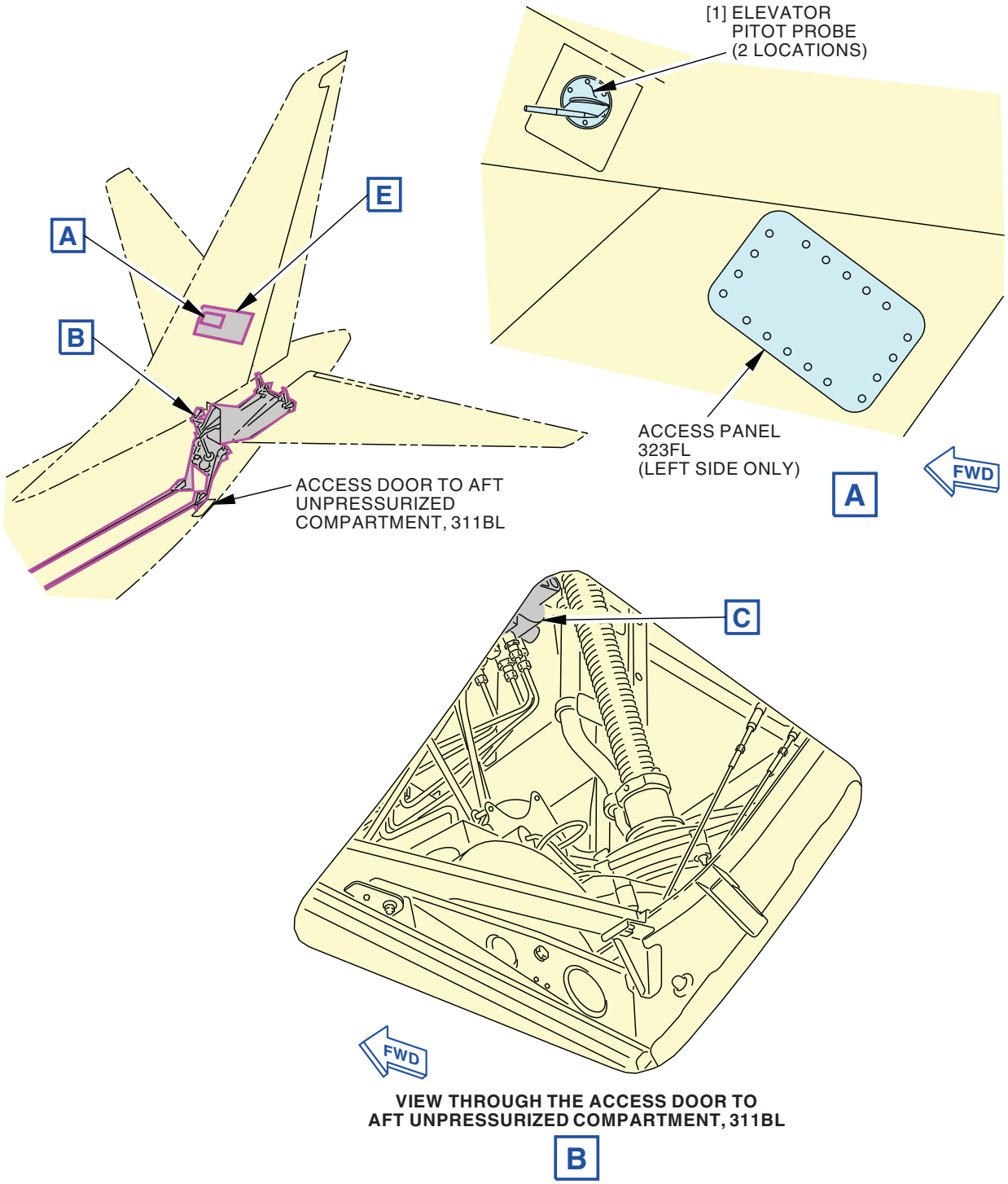
(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

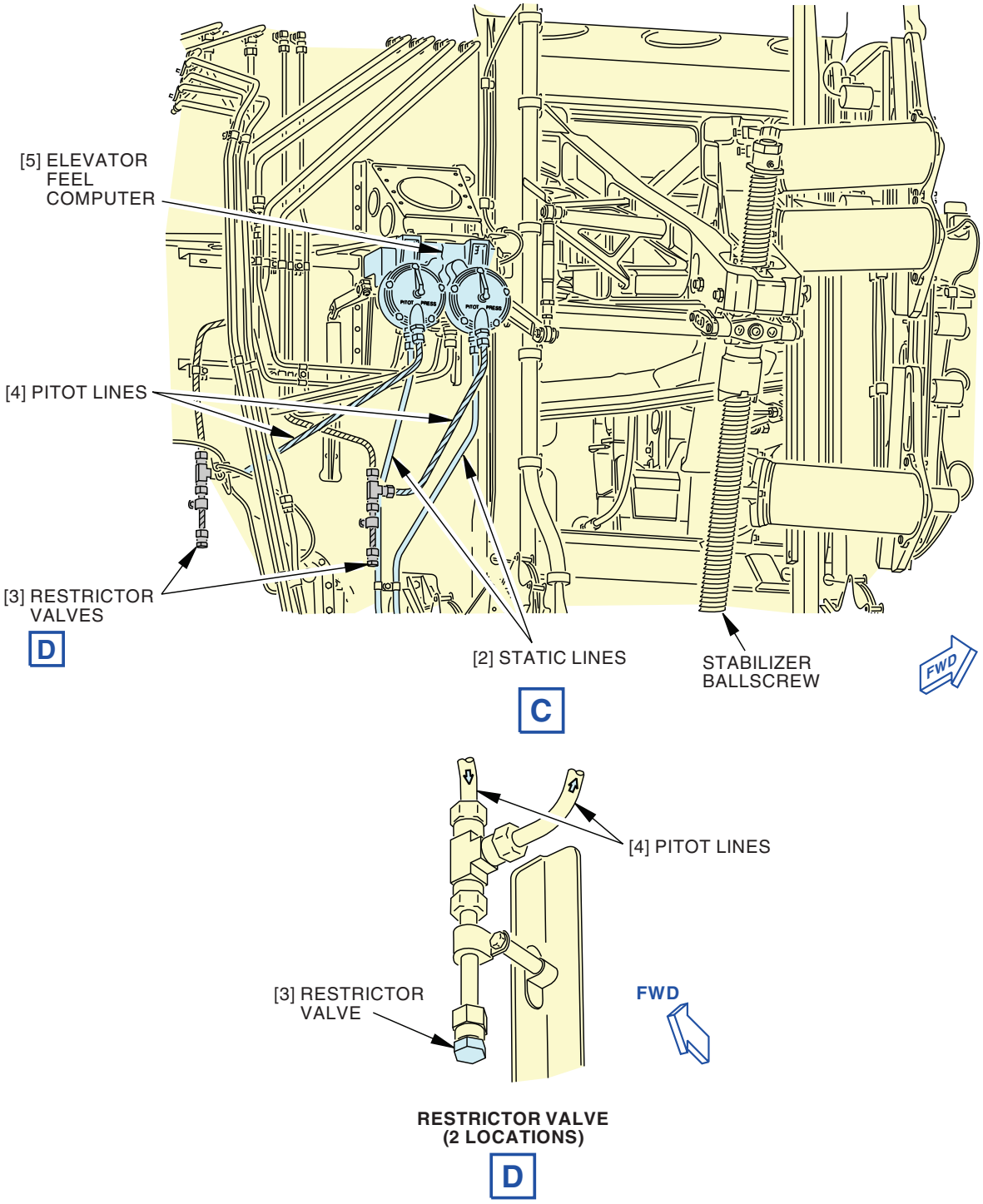
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



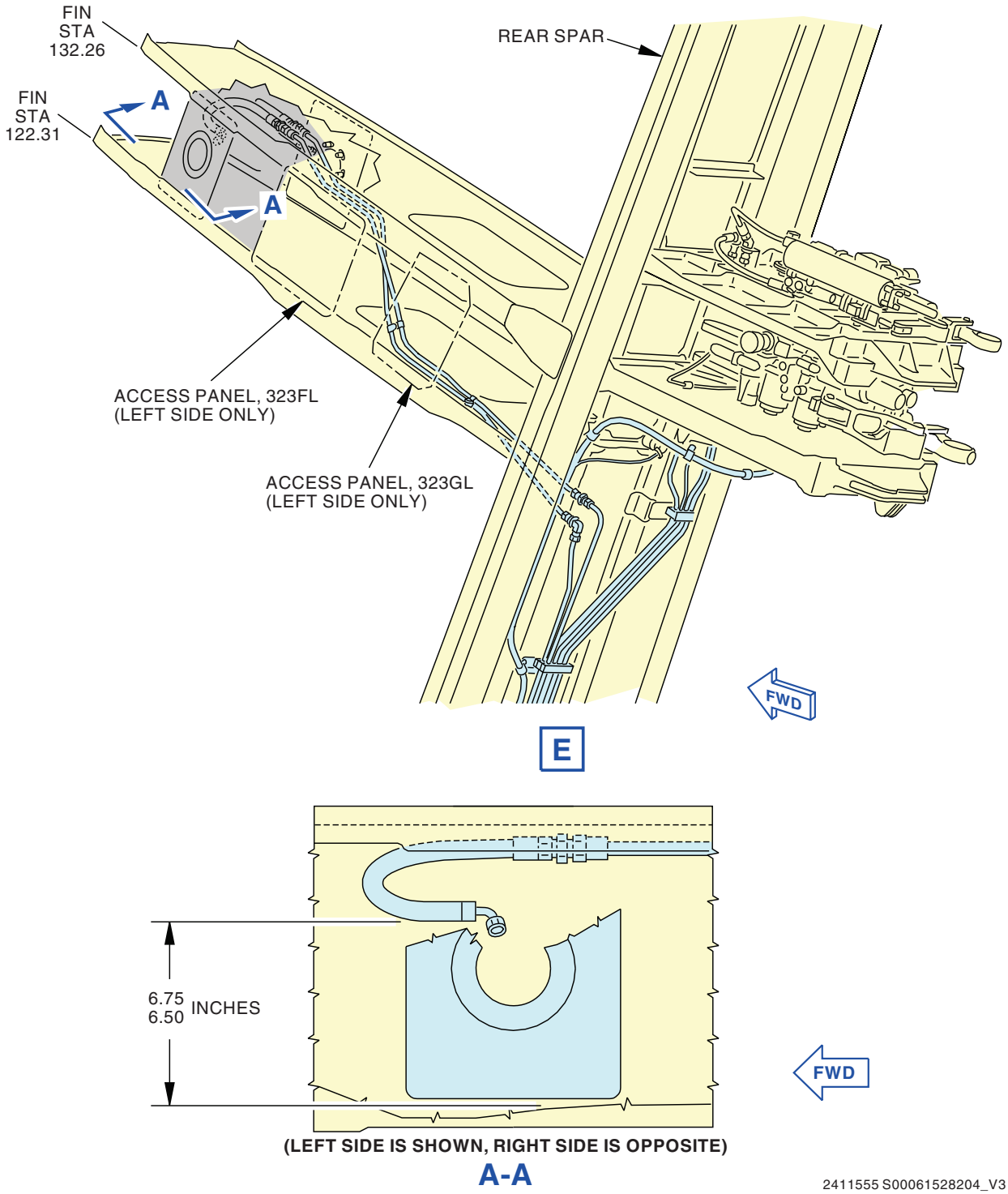
Elevator Pitot-Static System Installation
Figure 201/27-31-17-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Pitot-Static System Installation
Figure 201/27-31-17-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Pitot-Static System Installation
Figure 201/27-31-17-990-801 (Sheet 3 of 3)

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TASK 27-31-17-200-801

3. Elevator Pitot-Static System Drainage - Inspection

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
23-11-00-040-801	High Frequency Communication System - Deactivation (P/B 201)
23-11-00-440-801	High Frequency Communication System - Activation (P/B 201)
27-41-00-800-801	Horizontal Stabilizer Trim Control System - Deactivation (P/B 201)
27-41-00-800-802	Horizontal Stabilizer Trim Control System - Activation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00065	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002 (Supersedes O-A-396)
B00068	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002, MIL-E-51454 Type II
B00130	Alcohol - Isopropyl	TT-I-735

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
323	Vertical Fin - Front Spar To Rear Spar

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
323FL	Vertical Fin, Access

E. Prepare for the Inspection

SUBTASK 27-31-17-010-003

(1) Open these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
323FL	Vertical Fin, Access

SUBTASK 27-31-17-860-003

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-17-860-009

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(3) Do this task: High Frequency Communication System - Deactivation, TASK 23-11-00-040-801.

SUBTASK 27-31-17-860-010

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Do this task: Horizontal Stabilizer Trim Control System - Deactivation, TASK 27-41-00-800-801.

F. Elevator Pitot-Static System Drainage Inspection

SUBTASK 27-31-17-170-003

(1) Drain the pitot system:

(a) Remove the restrictor valves [3] from the pitot lines [4].

(b) Make sure that the pitot lines [4] drain correctly.

(c) Make sure that the hoses between the pitot-static probes and the rigid metal tubing loop up and do not trap water.

(d) If necessary, flush the pitot system (Elevator Pitot-Static System Flushing, TASK 27-31-17-170-801).

(e) Clean the restrictor valves [3] with alcohol, B00130, alcohol, B00065, or alcohol, B00068.

(f) Install the restrictor valves [3].

1) Tighten the restrictor valves [3] to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

SUBTASK 27-31-17-170-004

(2) Examine the static system:

(a) Disconnect the static lines [2] from elevator feel computer [5].

(b) Make sure that the static lines [2] drain correctly.

1) If it is necessary, flush the static system (Elevator Pitot-Static System Flushing, TASK 27-31-17-170-801).

(c) Connect the static lines [2] to the elevator feel computer [5].

1) Tighten the B-nuts on the static lines [2] to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-17-860-004

(1) Do this task: High Frequency Communication System - Activation, TASK 23-11-00-440-801.

SUBTASK 27-31-17-860-011

(2) Do this task: Horizontal Stabilizer Trim Control System - Activation, TASK 27-41-00-800-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-17-860-016

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-31-17-410-003

(4) Close these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
323FL	Vertical Fin, Access

SUBTASK 27-31-17-390-002

(5) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

——— END OF TASK ———

TASK 27-31-17-200-802

4. Elevator Pitot Probe - Inspection

(Figure 201)

A. General

(1) The tasks that follow give more data on how to examine the elevator pitot probes [1].

(a) See Pitot Probe Special Detailed Inspection, TASK 34-11-01-200-802.

(b) See Pitot Probe Detailed Inspection, TASK 34-11-01-200-801.

B. References

Reference	Title
27-31-94-000-801	Feel System Pitot Probe Removal (P/B 401)
27-31-94-400-801	Feel System Pitot Probe Installation (P/B 401)
34-11-01-200-801	Pitot Probe Detailed Inspection (P/B 601)
34-11-01-200-802	Pitot Probe Special Detailed Inspection (P/B 601)

C. Location Zones

Zone	Area
323	Vertical Fin - Front Spar To Rear Spar

D. Elevator Pitot Probe Inspection

SUBTASK 27-31-17-170-005

(1) Examine the elevator pitot probe [1] for damage.

PTF ALL; AIRPLANES WITHOUT PITOT PROBE PART NUMBER 0851HT-2

(a) Make sure that the opening sharpness is not more than 0.025 in. (0.635 mm).

PTF ALL; AIRPLANES WITH PITOT PROBE PART NUMBER 0851HT-2

(b) Make sure that the opening sharpness is not more than 0.015 in. (0.381 mm).

PTF ALL

(c) Make sure that the elevator pitot probe [1] is not bent or the electrical heater is damaged.

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(d) If it is necessary, replace the elevator pitot probe [1], do these tasks:

- Feel System Pitot Probe Removal, TASK 27-31-94-000-801
- Feel System Pitot Probe Installation, TASK 27-31-94-400-801.

——— END OF TASK ———

TASK 27-31-17-790-801

5. Elevator Pitot System Leak - Test

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Air Pressure in PSIG/kPa will simulate knots of computed airspeed (kcas).

B. References

Reference	Title
23-11-00-040-801	High Frequency Communication System - Deactivation (P/B 201)
23-11-00-440-801	High Frequency Communication System - Activation (P/B 201)
27-41-00-800-801	Horizontal Stabilizer Trim Control System - Deactivation (P/B 201)
27-41-00-800-802	Horizontal Stabilizer Trim Control System - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-1914	Test Set - Air Data Model FLMTS (Flight Line Maintenance) Part #: ADSE 650 Supplier: 3BSK6 Part #: ADTS2000 Supplier: 52892 Part #: ADTS3350ER Supplier: 52892 Part #: ADTS405MKII Supplier: U0427 Part #: ADTS530 Supplier: U0427 Part #: ADTS552F Supplier: U0427 Part #: ADTS553F Supplier: U0427 Part #: ADTS554F Supplier: U0427 Part #: D60340MK Supplier: K1474 Part #: DPS1000 Supplier: 21844 Part #: MODEL 6300 Supplier: 0RDZ5 Part #: MPS34C Supplier: 48RQ2 Part #: MPS43B Supplier: A0197 Part #: MPS45 Supplier: 48RQ2 Part #: MPS49 Supplier: 48RQ2 Part #: TES9463 Supplier: 88277 Opt Part #: 01-0987-00 Supplier: 41364 Opt Part #: 18910480000 Supplier: 89944 Opt Part #: 18910920000 Supplier: 89944 Opt Part #: ADTS405F Supplier: U0427 Opt Part #: ADTS505 Supplier: U0427 Opt Part #: D60302 Supplier: K1474 Opt Part #: D60340 Supplier: K1474 Opt Part #: D60383 Supplier: K1474 Opt Part #: DPS350 Supplier: 21844 Opt Part #: DPS450 Supplier: 21844 Opt Part #: DPS500 Supplier: 21844 Opt Part #: MPS31C Supplier: 48RQ2 Opt Part #: MPS43 Supplier: A0197
COM-1916	Adapter - Pitot Test (Typically included in Air Data Accessory Kit) Part #: CSA75700HT-3 Supplier: 3BSK6 Part #: P75701-4 Supplier: 38002 Part #: P75701M2-3 Supplier: 38002
COM-16577	Plug - Pitot Part #: BACP20AU6D Supplier: 81205
COM-25177	Accessory Kit - Air Data, 737 (Pitot/Static adapters, hoses, etc.) Part #: ADA737-678 Supplier: 38002 Part #: AK737-900 Supplier: 3BSK6 Part #: PA-0120-A Supplier: 3BSK6
SPL-1742	Regulator - Air Pressure, Elevator Feel Computer Part #: F72928-109 Supplier: 81205 Part #: F72928-110 Supplier: 81205 Part #: F72928-111 Supplier: 81205 Opt Part #: F72928-58 Supplier: 81205 Opt Part #: F72928-62 Supplier: 81205 Opt Part #: F72928-63 Supplier: 81205

D. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

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(Continued)

Zone	Area
323	Vertical Fin - Front Spar To Rear Spar

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Test

SUBTASK 27-31-17-010-005

(1) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-17-860-005

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-31-17-860-012

WARNING MAKE SURE THAT PERSONNEL STAY A MINIMUM OF 10 FT (3 M) AWAY FROM THE VERTICAL STABILIZER WHEN THE HF SYSTEM TRANSMITS. RF ENERGY FROM THE HF ANTENNA CAN CAUSE INJURIES TO PERSONNEL.

(3) Do this task: High Frequency Communication System - Deactivation, TASK 23-11-00-040-801.

SUBTASK 27-31-17-860-013

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Do this task: Horizontal Stabilizer Trim Control System - Deactivation, TASK 27-41-00-800-801.

G. Elevator Pitot System Leak Test

SUBTASK 27-31-17-390-001

WARNING WHEN THE PITOT PROBES HAVE COVERS ON THEM, MAKE SURE THAT A PERSON ON THE GROUND CAN SEE THE COVERS. ALSO MAKE SURE YOU ATTACH A TAG TO THE LEFT CONTROL WHEEL IN THE FLIGHT COMPARTMENT AS A REMINDER THAT THE PITOT PROBES HAVE COVERS ON THEM. IF THE COVERS ARE NOT REMOVED FROM THE PITOT PROBES, INCORRECT AIRSPEED-SENSING AND ALTITUDE-SENSING SIGNALS CAN OCCUR. THIS CAN CAUSE DANGEROUS FLIGHT CONDITIONS.

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(WARNING PRECEDES)

CAUTION MAKE SURE THAT THE PITOT PROBE HAS NO ADDED WEIGHT ON IT FROM THE TEST HOSE. THE WEIGHT OF THE TEST HOSE CAN BEND OR TWIST THE PITOT PROBE OUT OF TOLERANCE.

(1) Connect the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to the elevator pitot probes [1].

(a) Do this step for the regulator, SPL-1742:

1) Connect the pressure lines from the regulator, SPL-1742, with the pitot test adapter, COM-1916, to the elevator pitot probes [1].

(b) Do this step for the air data model test set, COM-1914:

1) Use the air data accessory kit, COM-25177, to connect the air data model test set, COM-1914, to the pitot test adapter, COM-1916.

NOTE: The air data accessory kit, COM-25177, contains hoses to connect to the air data model test set, COM-1914, and the pitot test adapter, COM-1916.

SUBTASK 27-31-17-020-001

(2) In the aft unpressurized compartment, below the elevator feel computer [5], do these steps:

(a) Remove the restrictor valves [3] from the bottom of the pitot lines [4] (Figure 201).

(b) If you are using the regulator, SPL-1742, or equivalent, for the leak test, do these steps:

NOTE: The DPS 400 Test Set can be used as an alternative tool to the DPS 500.

1) Connect the pressure gage lines from the regulator, SPL-1742, or equivalent, to the drain holes of the pitot lines [4].

(c) If you are using the air data model test set, COM-1914, for the leak test, do these steps:

1) Install the pitot plugs, COM-16577, on the ends of the pitot lines [4].

2) Install a streamer on the opening of the aft unpressurized compartment.

a) Make sure that you can easily see the streamer from the ground.

NOTE: This is to remind you to remove the plugs and install the restrictor valves again when the test is completed.

SUBTASK 27-31-17-790-002

(3) Do the pitot system leak test:

CAUTION DO NOT INCREASE THE PRESSURE TO MORE THAN 5 PSIG (34.5 KPA) (437 KNOTS OF COMPUTED AIRSPEED (KCAS)), DURING THE TEST. IF THE PRESSURE IS MORE THAN 5 PSIG (34.5 KPA) (437 KCAS), IT CAN CAUSE DAMAGE TO THE ELEVATOR FEEL COMPUTER.

(a) Operate the air data model test set, COM-1914, or regulator, SPL-1742, or equivalent, to apply pressure of 0.90 +0.07 / -0.00 psig (6.21 +0.48 / -0.00 kPa) to simulate 194 to 201 kcas.

(b) Set the shutoff valve in the air data model test set, COM-1914, or regulator, SPL-1742, or equivalent to the OFF position.

(c) Make sure that the pressure does not decrease more than 0.2 psig (1.4 kPa) (8 kcas) during a 2-minute time.

NOTE: The low pressure pressurization steps above make sure that there is no system leak before you apply full pressure in the following steps.

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1) If it is necessary, repair the pitot system or pressurization equipment leak.

(d) Operate the air data model test set, COM-1914, or regulator, SPL-1742, or equivalent, to apply pressure of 4.8 ±0.1 psig (33.1 ±0.7 kPa) to simulate 425 to 433 kcas.

(e) Set the shutoff valve in the air data model test set, COM-1914, or regulator, SPL-1742, or equivalent to the OFF position.

(f) Make sure that the pressure does not decrease more than 0.3 psig (2.1 kPa) (12 kcas) during a 2-minute time.

(g) Operate the air data model test set, COM-1914, or regulator, SPL-1742, to put the system back to ambient pressure (0 psig (0.0 kPa)).

SUBTASK 27-31-17-160-001

(4) At the elevator pitot probes [1] on the vertical fin, do these steps:

(a) Remove the pressure lines with the pitot test adapter, COM-1916, from the air data model test set, COM-1914, or regulator, SPL-1742, or equivalent.

(b) Remove the pitot test adapter, COM-1916, from the elevator pitot probes [1].

SUBTASK 27-31-17-080-001

(5) In the aft unpressurized compartment, below the elevator feel computer [5], do these steps:

(a) If you used the regulator, SPL-1742, or equivalent, for the leak test, do these steps:

1) Disconnect the pressure gage lines of the regulator, SPL-1742, or equivalent from the drain holes of the pitot lines [4].

(b) If you used the air data model test set, COM-1914, for the leak test, do these steps:

1) Remove the pitot plugs, COM-16577, that were installed on the ends of the pitot lines [4].

2) Remove the streamer on the opening of the aft unpressurized compartment.

(c) Install the restrictor valves [3] on the bottom of the pitot lines [4].

1) Tighten the restrictor valves [3] to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-17-860-014

(1) Do this task: High Frequency Communication System - Activation, TASK 23-11-00-440-801.

SUBTASK 27-31-17-860-015

(2) Do this task: Horizontal Stabilizer Trim Control System - Activation, TASK 27-41-00-800-802.

SUBTASK 27-31-17-860-006

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-31-17-410-005

(4) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

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TASK 27-31-17-780-801

6. Elevator Pitot System - Pressurization

(Figure 201)

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1914	Test Set - Air Data Model FLMTS (Flight Line Maintenance) Part #: ADSE 650 Supplier: 3BSK6 Part #: ADTS2000 Supplier: 52892 Part #: ADTS3350ER Supplier: 52892 Part #: ADTS405MKII Supplier: U0427 Part #: ADTS530 Supplier: U0427 Part #: ADTS552F Supplier: U0427 Part #: ADTS553F Supplier: U0427 Part #: ADTS554F Supplier: U0427 Part #: D60340MK Supplier: K1474 Part #: DPS1000 Supplier: 21844 Part #: MODEL 6300 Supplier: 0RDZ5 Part #: MPS34C Supplier: 48RQ2 Part #: MPS43B Supplier: A0197 Part #: MPS45 Supplier: 48RQ2 Part #: MPS49 Supplier: 48RQ2 Part #: TES9463 Supplier: 88277 Opt Part #: 01-0987-00 Supplier: 41364 Opt Part #: 18910480000 Supplier: 89944 Opt Part #: 18910920000 Supplier: 89944 Opt Part #: ADTS405F Supplier: U0427 Opt Part #: ADTS505 Supplier: U0427 Opt Part #: D60302 Supplier: K1474 Opt Part #: D60340 Supplier: K1474 Opt Part #: D60383 Supplier: K1474 Opt Part #: DPS350 Supplier: 21844 Opt Part #: DPS450 Supplier: 21844 Opt Part #: DPS500 Supplier: 21844 Opt Part #: MPS31C Supplier: 48RQ2 Opt Part #: MPS43 Supplier: A0197
COM-1916	Adapter - Pitot Test (Typically included in Air Data Accessory Kit) Part #: CSA75700HT-3 Supplier: 3BSK6 Part #: P75701-4 Supplier: 38002 Part #: P75701M2-3 Supplier: 38002
COM-25177	Accessory Kit - Air Data, 737 (Pitot/Static adapters, hoses, etc.) Part #: ADA737-678 Supplier: 38002 Part #: AK737-900 Supplier: 3BSK6 Part #: PA-0120-A Supplier: 3BSK6

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(Continued)

Reference	Description
SPL-1742	Regulator - Air Pressure, Elevator Feel Computer Part #: F72928-109 Supplier: 81205 Part #: F72928-110 Supplier: 81205 Part #: F72928-111 Supplier: 81205 Opt Part #: F72928-58 Supplier: 81205 Opt Part #: F72928-62 Supplier: 81205 Opt Part #: F72928-63 Supplier: 81205

B. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

C. Prepare for the Pressurization

SUBTASK 27-31-17-010-006

(1) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-17-860-007

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

D. Elevator Pitot System Pressurization

SUBTASK 27-31-17-480-001

(1) Prepare the elevator pitot system for pressurization:

NOTE: The next two steps are not necessary if you are using the air data model test set, COM-1914.

(a) Remove the restrictor valves [3] from the pitot lines [4].

(b) Connect the pressure gage lines from the regulator, SPL-1742, or equivalent to the drain holes of the pitot line [4].

NOTE: The DPS 400 Test Set can be used as an alternative tool to the DPS 500.

WARNING

WHEN THE PITOT PROBES HAVE COVERS ON THEM, MAKE SURE THAT A PERSON ON THE GROUND CAN SEE THE COVERS. ALSO MAKE SURE YOU ATTACH A TAG TO THE LEFT CONTROL WHEEL IN THE FLIGHT COMPARTMENT AS A REMINDER THAT THE PITOT PROBES HAVE COVERS ON THEM. IF THE COVERS ARE NOT REMOVED FROM THE PITOT PROBES, INCORRECT AIRSPEED-SENSING AND ALTITUDE-SENSING SIGNALS CAN OCCUR. THIS CAN CAUSE DANGEROUS FLIGHT CONDITIONS.

(c) Seal the drain hole in each elevator pitot probe [1].

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CAUTION

MAKE SURE THAT THE WEIGHT OF THE TEST HOSE IS NOT ON THE PITOT PROBE. THE WEIGHT OF THE HOSE CAN BEND OR TWIST THE PROBE UNTIL IT IS NOT IN TOLERANCE.

(d) Connect the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to the two elevator pitot probes [1].

1) Do this step for the regulator, SPL-1742:

a) Connect the pressure lines from the regulator, SPL-1742, with the pitot test adapter, COM-1916, to the elevator pitot probes [1].

2) Do this step for the air data model test set, COM-1914:

a) Use the air data accessory kit, COM-25177, to connect the air data model test set, COM-1914, to the pitot test adapter, COM-1916.

NOTE: The air data accessory kit, COM-25177, contains hoses to connect to the air data model test set, COM-1914, and the pitot test adapter, COM-1916.

SUBTASK 27-31-17-480-002

CAUTION

DO NOT LET THE PRESSURE BECOME MORE THAN 5 PSIG (34.5 KPA). DAMAGE TO THE COMPONENT COULD OCCUR.

(2) Pressurize the elevator pitot system:

(a) Operate the air data model test set, COM-1914, or regulator, SPL-1742, to pressurize the system, as necessary, to get the system test data.

(b) Slowly operate the air data model test set, COM-1914, or regulator, SPL-1742, to remove pressure from the system until it is at zero.

SUBTASK 27-31-17-010-007

(3) Remove the pressurization equipment from the airplane:

(a) Remove the pressure gage lines from the air data model test set, COM-1914, or regulator, SPL-1742, or equivalent to the drain holes.

(b) Install the restrictor valves [3] in each of the pitot lines [4].

1) Tighten the restrictor valves [3] to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

(c) Remove the pressure lines from the air data model test set, COM-1914, or regulator, SPL-1742, or equivalent from the elevator pitot probes [1].

(d) Remove the pitot test adapter, COM-1916, from the elevator pitot probes [1].

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-17-860-008

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

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SUBTASK 27-31-17-410-006

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

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ELEVATOR FEEL SHIFT MODULE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of an elevator feel shift module.

(2) An installation of an elevator feel shift module.

TASK 27-31-20-000-801

2. Elevator Feel Shift Module Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

C. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

D. Prepare for the Removal

SUBTASK 27-31-20-010-001

(1) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-20-860-001

(2) Put the SYSTEM A FLIGHT CONTROL SHUTOFF and the SYSTEM B FLIGHT CONTROL SHUTOFF switches on the P5 forward overhead panel to the OFF position.

SUBTASK 27-31-20-860-002

(3) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	4	C01392	STICK SHAKER LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	6	C01393	STICK SHAKER RIGHT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	2	C01064	AFCS SYS B MACH TRIM DC

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(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	3	C01046	AFCS SYS B FCC DC
B	4	C00716	AFCS SYS B ENGAGE INTLK
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
C	1	C01037	AFCS SYS B MACH TRIM AC
C	2	C01042	AFCS SYS B SNSR EXC AC
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2

SUBTASK 27-31-20-860-003

(4) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

E. Elevator Feel Shift Module Removal

(Figure 402)

SUBTASK 27-31-20-020-001

WARNING RELEASE THE CONNECTION SLOWLY AND LET THE PRESSURE DECREASE BEFORE YOU COMPLETELY REMOVE THE HYDRAULIC LINE. IF YOU DISCONNECT THE HYDRAULIC LINE WHEN IT HAS PRESSURE, A SPRAY OF HYDRAULIC FLUID CAN CAUSE INJURY TO PERSONS.

(1) Disconnect the hydraulic lines [4] and hydraulic line [6] from the elevator feel shift module [1].

(a) Install the caps and plugs on the open ports and hydraulic lines.

(b) Discard the packings [7].

SUBTASK 27-31-20-480-001

(2) Disconnect the electrical connector [5] from the elevator feel shift module [1].

SUBTASK 27-31-20-020-002

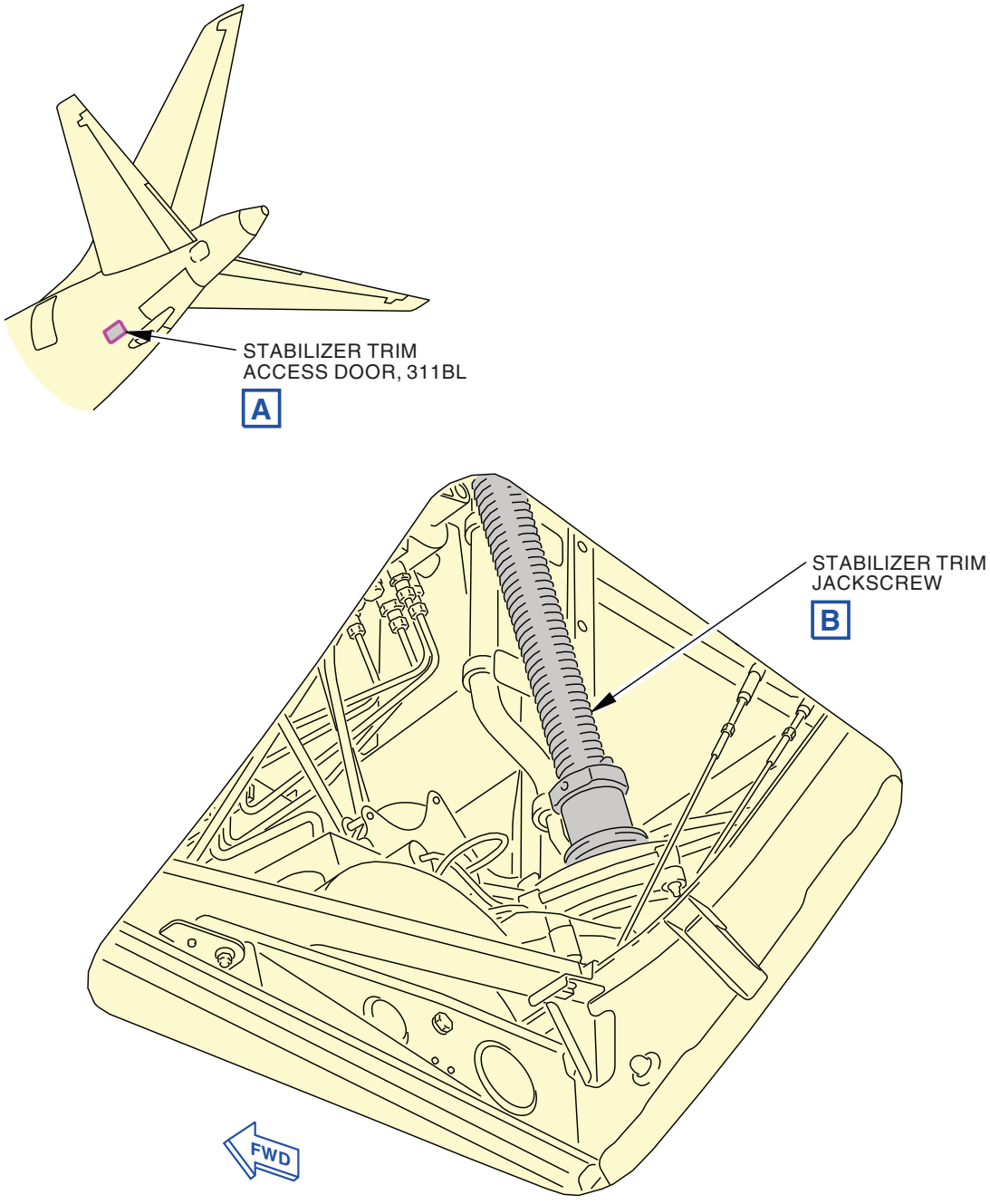
(3) Hold the elevator feel shift module [1] and remove the bolts [2] and washers [3] that connect the elevator feel shift module [1] to the bracket.

SUBTASK 27-31-20-480-002

(4) Remove the elevator feel shift module [1].

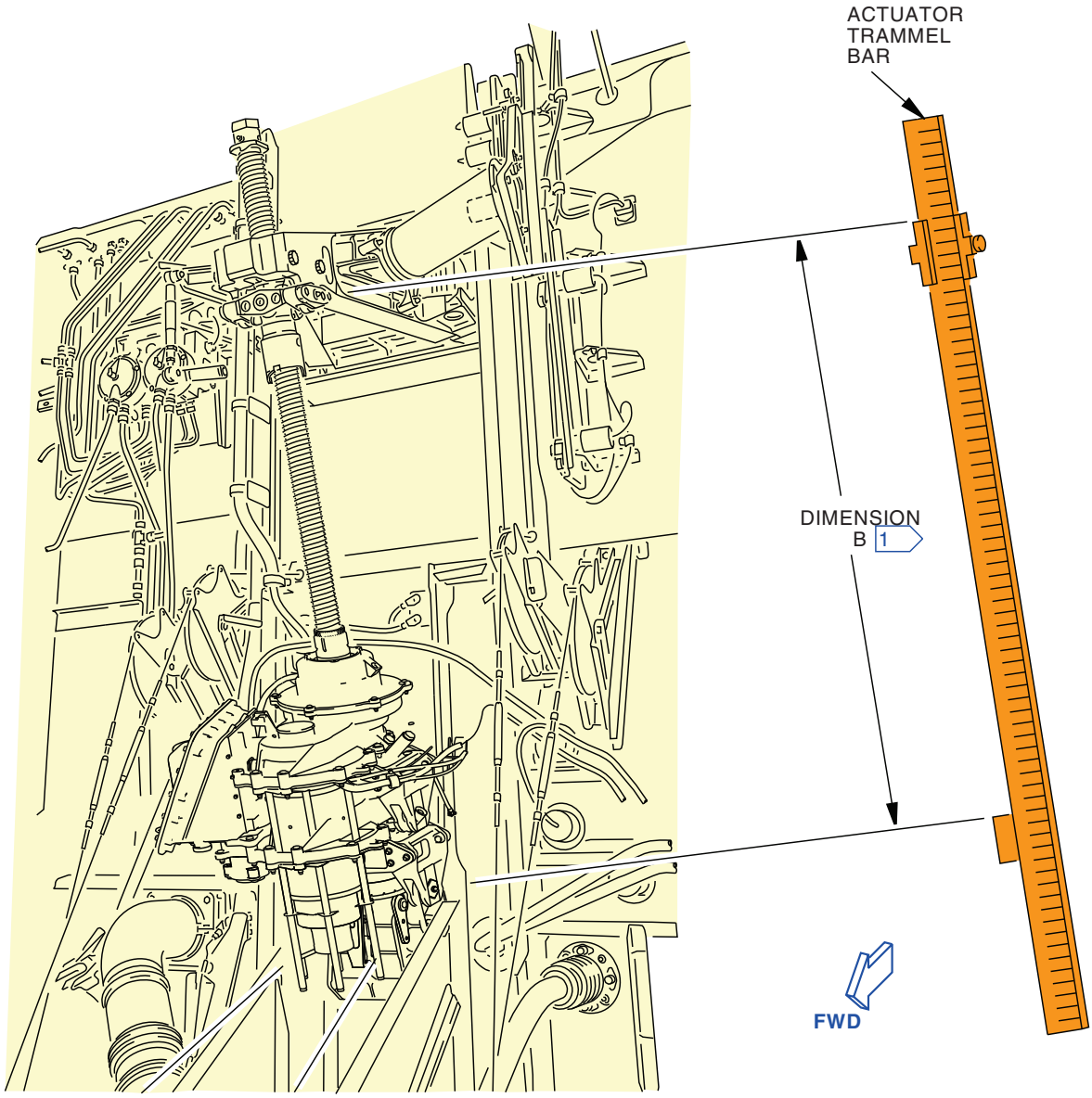
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Trim Jackscrew Adjustment
Figure 401/27-31-20-990-801 (Sheet 1 of 2)

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STABILIZER TRIM ACTUATOR

B

NOTE:

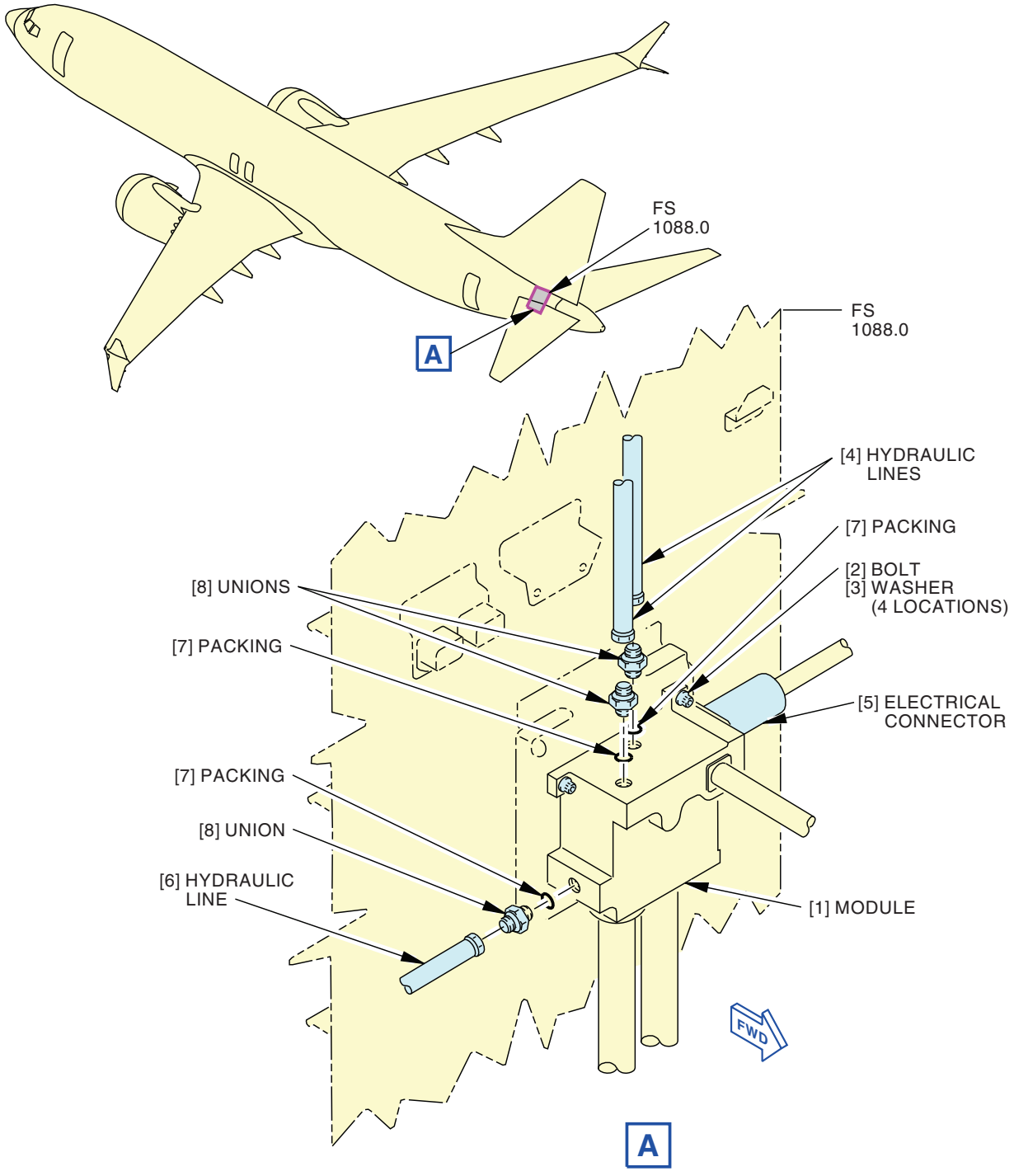
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

2412518 S00061530821_V1

Stabilizer Trim Jackscrew Adjustment
Figure 401/27-31-20-990-801 (Sheet 2 of 2)

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Elevator Feel Shift Module - Installation
Figure 402/27-31-20-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-20-400-801

3. Elevator Feel Shift Module - Installation

(Figure 401, Figure 402)

A. References

Reference	Title
27-31-00-700-813	Feel Force System - Test (P/B 501)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-20-740-801	Elevator Feel Shift Module - Test (P/B 501)
51-22-00-390-806	Fillet Seal Application (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Elevator feel shift module	27-31-20-01-015	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

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G. Prepare for the Installation

SUBTASK 27-31-20-860-004

(1) Make sure that the pressure to the elevator hydraulic systems A and B is removed (TASK 27-31-00-800-802).

H. Elevator Feel Shift Module Installation

SUBTASK 27-31-20-420-001

(1) Install the elevator feel shift module [1]:

(a) Install the washers [3] and bolts [2] to attach the elevator feel shift module [1] to the bracket.

1) With the electrical bonding intrinsically safe approved bonding meter, COM-1550, make sure that the electrical resistance between the elevator feel shift module [1] and the bracket, that it is attached to, is 0.005 ohms or less.

2) With the electrical bonding intrinsically safe approved bonding meter, COM-1550, make sure that the electrical resistance between the elevator feel shift module [1] and the airplane structure is 0.005 ohms or less.

3) If the electrical resistance at the two locations is more than 0.005 ohms, then remove the elevator feel shift module [1] and prepare the electrical faying surfaces.

(b) Apply a fillet seal of sealant, A00247 along the joint edges (TASK 51-22-00-390-806).

SUBTASK 27-31-20-420-002

(2) Connect electrical electrical connector [5] to the elevator feel shift module [1].

SUBTASK 27-31-20-420-003

(3) Remove the caps and plugs from the open ports, hydraulic lines [4] and hydraulic line [6].

SUBTASK 27-31-20-420-006

(4) If the packings [7] and unions [8] are not installed on the elevator feel shift module [1], then do these steps:

(a) Lightly apply hydraulic fluid, D00153 to the new packings [7].

(b) Install the packings [7] and unions [8] on the elevator feel shift module [1].

SUBTASK 27-31-20-420-004

(5) Connect the hydraulic lines [4] and hydraulic line [6] to the elevator feel shift module [1].

NOTE: Apply a layer of hydraulic fluid to the hydraulic fittings before you install the hydraulic lines.

SUBTASK 27-31-20-420-007

(6) Tighten the B-nuts on the hydraulic lines [4] to 270 ±14 in-lb (31 ±2 N·m).

SUBTASK 27-31-20-420-008

(7) Tighten the B-nut on the hydraulic line [6] to 170 ±9 in-lb (19 ±1 N·m).

SUBTASK 27-31-20-720-001

(8) Do this task: Elevator Feel Shift Module - Test, TASK 27-31-20-740-801.

SUBTASK 27-31-20-720-002

(9) Do this task: Feel Force System - Test, TASK 27-31-00-700-813.

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I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-20-860-005

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	4	C01392	STICK SHAKER LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	6	C01393	STICK SHAKER RIGHT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	2	C01064	AFCS SYS B MACH TRIM DC
B	3	C01046	AFCS SYS B FCC DC
B	4	C00716	AFCS SYS B ENGAGE INTLK
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
C	1	C01037	AFCS SYS B MACH TRIM AC
C	2	C01042	AFCS SYS B SNSR EXC AC
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2

SUBTASK 27-31-20-860-006

(2) Put the SYSTEM A FLIGHT CONTROL SHUTOFF and the SYSTEM B FLIGHT CONTROL SHUTOFF switches on the P5 forward overhead panel to the ON position.

SUBTASK 27-31-20-860-007

(3) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand to the NORMAL position.

SUBTASK 27-31-20-410-001

(4) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR FEEL SHIFT MODULE - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) A test of the elevator feel shift module

TASK 27-31-20-740-801

2. Elevator Feel Shift Module - Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Make sure that the two Stall Management Yaw Damper (SMYD) computers are installed for this test.

NOTE: The Stall Management Yaw Damper Computer P/N 285A1010-4 or subsequent is necessary for this test.

B. References

Reference	Title
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Test

SUBTASK 27-31-20-760-005

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	8	C01464	FLIGHT CONTROL ELEV TAB VLV RIGHT
C	10	C01676	ELEVATOR FEEL LIMIT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	13	C00311	INDICATOR MASTER DIM BUS 1
E	14	C00316	INDICATOR MASTER DIM SECT 4

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SUBTASK 27-31-20-420-005

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-31-20-410-002

(3) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

F. Elevator Feel Shift Module Test

SUBTASK 27-31-20-700-001

(1) Do a test of the Elevator Feel Shift Module (EFSM) with the SMYD number 1 energized:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	5	C01206	SMYD-2 CMPTR DC

(b) Push the ON/OFF button on the BITE panel on the front of SMYD number 1.

1) Make sure that EXISTING FAULTS? shows on the display.

(c) Push the down arrow button until OTHER FUNCTIONS? shows on the display.

(d) Push the YES button on the BITE panel on the front of SMYD number 1.

1) Make sure that SYSTEM CONFIG? shows on the display.

(e) Push the down arrow button until SET OUTPUTS? shows on the display.

(f) Push the YES button on the BITE panel on the front of SMYD number 1.

1) Make sure that SS CMD? shows on the display.

(g) Push the YES button on the BITE panel on the front of SMYD number 1.

NOTE: SS ACTIVATE will show on the display, and then REMOVE CMD? will show on the display.

1) Make sure that the Captain's control column shakes.

2) After a minimum of 60 seconds, make sure that the FEEL DIFF PRESS light on the P5 forward overhead panel does not come on.

3) Make sure that CMD REMOVED shows on the display after the Captain's control column stops shaking.

(h) Push the MENU button on the BITE panel on the front of SMYD number 1.

1) Make sure that SS CMD? shows on the display.

(i) Push the NO button on the BITE panel on the front of SMYD number 1.

1) Make sure that SS CMD & EFS CMD? shows on the display.

(j) Push the YES button on the BITE panel on the front of SMYD number 1.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that SS & EFS ACTIVATE shows on the display before REMOVE CMD? shows on the display.
2) Make sure that the Captain's control column shakes.
3) Make sure that these lights come on and stay on after a minimum of 20 to 60 seconds:
   a) Left and right MASTER CAUTION lights on the P7 glareshield panel
   b) FEEL DIFF PRESS light on the P5 forward overhead panel.
4) Make sure that CMD REMOVED shows on the display after the Captain's control column stops shaking.

   NOTE: The FEEL DIFF PRESS light must go off.

(k) Push the master caution PUSH TO RESET on the P7 glareshield panel.
1) Make sure that these lights go off:
   a) Left MASTER CAUTION light on the P7 glareshield panel
   b) Right MASTER CAUTION light on the P7 glareshield panel.

(l) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	5	C01206	SMYD-2 CMPTR DC

(m) Push the ON/OFF button on the BITE panel on the front of SMYD number 1.

SUBTASK 27-31-20-700-002

(2) Do a test of the Elevator Feel Shift Module (EFSM) with the SMYD number 2 energized:

(a) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC

(b) Push the ON/OFF button on the BITE panel on the front of SMYD number 2.
1) Make sure that EXISTING FAULTS? shows on the display.

(c) Push the down arrow button until OTHER FUNCTIONS? shows on the display.

(d) Push the YES button on the BITE panel on the front of SMYD number 2.
1) Make sure that SYSTEM CONFIG? shows on the display.

(e) Push the down arrow button until SET OUTPUTS? shows on the display.

(f) Push the YES button on the BITE panel on the front of SMYD number 2.
1) Make sure that SS CMD? shows on the display.

(g) Push the YES button on the BITE panel on the front of SMYD number 2.

NOTE: SS ACTIVATE will show on the display, and then REMOVE CMD? will show on the display.

1) Make sure that the First Officer's control column shakes.
2) After a minimum of 60 seconds, make sure that the FEEL DIFF PRESS light on the P5 forward overhead does not come on.
3) Make sure that CMD REMOVED shows on the display after the First Officer's control column stops shaking.

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(h) Push the MENU button on the BITE panel on the front of SMYD number 2.

1) Make sure that SS CMD? shows on the display.

(i) Push the NO button on the BITE panel on the front of SMYD number 2.

1) Make sure that SS CMD & EFS CMD? shows on the display.

(j) Push the YES button on the BITE panel on the front of SMYD number 2.

1) Make sure that SS & EFS ACTIVATE shows on the display before REMOVE CMD? shows on the display.

2) Make sure that the First Officer's control column shakes.

3) Make sure that these lights come on and stay on after a minimum of 20 to 60 seconds:

a) Left and right MASTER CAUTION lights on the P7 glareshield panel

b) FEEL DIFF PRESS light on the P5-3 forward overhead panel.

4) Make sure that CMD REMOVED shows on the display after the First Officer's control column stops shaking.

NOTE: The FEEL DIFF PRESS light must go off.

(k) Push the master caution PUSH TO RESET on the P7 glareshield panel.

1) Make sure that these lights go off:

a) Left MASTER CAUTION light on the P7 glareshield panel

b) Right MASTER CAUTION light on the P7 glareshield panel.

(l) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC

(m) Push the ON/OFF button on the BITE panel on the front of SMYD number 2.

SUBTASK 27-31-20-700-003

(3) Do a test of the Elevator Feel Shift Module (EFSM) with the SMYD number 1 and the SMYD number 2 energized:

(a) Do the steps to SMYD number 1 and SMYD number 2 at the same time.

(b) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	5	C01206	SMYD-2 CMPTR DC

(c) Push the ON/OFF button on the BITE panel that is on the front of each SMYD.

1) Make sure that EXISTING FAULTS? shows on each of the two displays.

(d) Push the down arrow button on each SMYD until OTHER FUNCTIONS? shows on each of the two displays.

(e) Push the YES button on the BITE panel that is on the front of each SMYD.

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1) Make sure that SYSTEM CONFIG? shows on each of the two displays.

(f) Push the down arrow button on each SMYD until SET OUTPUTS? shows on each of the two displays.

(g) Push the YES button on the BITE panel that is on the front of each SMYD.

1) Make sure that SS CMD? shows on each of the two displays.

(h) Push the YES button on the BITE panel that is on the front of each SMYD.

NOTE: SS ACTIVATE will show on the two displays, and then REMOVE CMD? will show on the two displays.

1) Make sure that the Captain's and First Officer's control columns shake.

2) After a minimum of 60 seconds, make sure that the FEEL DIFF PRESS light on the P5 forward overhead panel does not come on.

3) Make sure that CMD REMOVED shows on each of the two displays after the two control columns stop shaking.

(i) Push the MENU button on the BITE panel that is on the front of each SMYD.

1) Make sure that SS CMD? shows on each of the two displays.

(j) Push the NO button on the BITE panel that is on the front of each SMYD.

1) Make sure that SS CMD & EFS CMD? shows on each of the two displays.

(k) Push the YES button on the BITE panel that is on the front of each SMYD.

1) Make sure that SS & EFS ACTIVATE shows on each of the two displays before REMOVE CMD? shows on each of the two displays.

2) Make sure that the Captain's and First Officer's control columns shake.

3) Make sure that these lights come on and stay on after a minimum of 20 to 60 seconds:

a) Left and right MASTER CAUTION lights on the P7 glareshield panel

b) FEEL DIFF PRESS light on the P5-3 forward overhead panel.

4) Make sure that CMD REMOVED shows on each of the two displays after the two control columns stop shaking.

NOTE: The FEEL DIFF PRESS light should go off.

(l) Push the master caution PUSH TO RESET on the P7 glareshield panel.

1) Make sure that these lights go off:

a) Left MASTER CAUTION light on the P7 glareshield panel

b) Right MASTER CAUTION light on the P7 glareshield panel.

(m) Push the ON/OFF button on the BITE panel that is on the front of each SMYD.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-20-860-008

(1) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

SUBTASK 27-31-20-410-003

(2) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

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——— END OF TASK ———

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ELEVATOR TAB - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the elevator tab

(2) An installation of the elevator tab.

B. This procedure is applicable to the left and right elevator tabs.

TASK 27-31-31-000-801

2. Elevator Tab Removal

(Figure 401, Figure 402, Figure 403)

A. References

Reference	Title
27-31-00-000-801	Elevator and Tab Control System - Deactivation (P/B 201)
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-990-810	Figure: Stabilizer Trim Jackscrew Setting (P/B 501)
27-31-00-990-818	Figure: Elevator Tab Fixture Bar Installation (P/B 501)
27-31-32-200-802	Elevator Tab Freeplay - Check (P/B 601)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-10345	Rigging Bar - Elevator Tab Part #: C27083-18 Supplier: 81205

C. Location Zones

Zone	Area
316	APU Compartment - Right
334	Left Horizontal Stabilizer - Elevator
335	Left Horizontal Stabilizer - Tip
344	Right Horizontal Stabilizer - Elevator
345	Right Horizontal Stabilizer - Tip

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

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E. Prepare for the Removal

SUBTASK 27-31-31-860-006

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-31-31-040-001

(2) Do this task: Elevator and Tab Control System - Deactivation, TASK 27-31-00-000-801.

SUBTASK 27-31-31-820-001

(3) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-31-010-001

(4) Remove these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-31-010-002

(5) For the left elevator tab pushrods, remove this access panel:

Number	Name/Location
334PT	Horizontal Stabilizer, Tab Control Rod Fairing

SUBTASK 27-31-31-010-003

(6) For the right elevator tab pushrods, remove this access panel:

Number	Name/Location
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-31-480-001

(7) Use the trammel bar, SPL-1677, to set the “B” dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (Figure 27-31-00-990-810).

(a) Make sure that the wire bundles are protected when the trammel bar, SPL-1677, is used.

SUBTASK 27-31-31-220-001

(8) Measure and record the elevator tab [22] position (Figure 27-31-00-990-818):

NOTE: Measuring and recording the elevator tab position before the elevator tab removal may improve the manual reversion success rate, if the elevator tab needs to be adjusted after installation. The elevator tab position can be measured with the rigging bar, SPL-10345, installed on the third elevator tab hinge (from inboard).

(a) If the same elevator tab [22] will be re-installed, do these steps before the elevator tab [22] removal:

1) Make sure that the elevator tab [22] meets the freeplay limits.

a) To do a check of the freeplay limits, do this task: Elevator Tab Freeplay - Check, TASK 27-31-32-200-802.

(b) Push the elevator trailing edge up by hand (Figure 403).

(c) While the elevator trailing edge is moved down, align the projected center of the elevator trailing edge with the index mark within ± 0.04 in. (1.0 mm), or less.

1) Measure and record the elevator tab [22] position.

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2) Make sure that the distance between the left elevator tab [22] and rigging bar, SPL-10345, is within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm).

3) Make sure that the distance between the right elevator tab [22] and rigging bar, SPL-10345, is within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm).

SUBTASK 27-31-31-480-002

(9) Install the rig pin E-5 from the rigging pin kit, SPL-1585, in the aft control quadrant (Figure 401).

F. Elevator Tab Removal

SUBTASK 27-31-31-020-001

(1) Remove the bolts [24] from the access panel [23] to get access to the elevator tab pushrods [25].

SUBTASK 27-31-31-020-002

(2) Remove the elevator tab [22]:

(a) Remove the pins [48], nuts [30], washers [31], bushings [32], and bolts [33] from the elevator tab [22] and elevator tab pushrods [25].

1) Discard the pins [48].

(b) Hold the elevator tab [22] while the elevator tab [22] is disconnected from the elevator [21].

1) For the hinge 1 (inboard hinge), remove the pin [44], nut [45], washers [43], bushing [42], washer [41], and bolt [40].

a) Discard the pin [44].

2) For the hinge 2 (inboard hinge), remove pin [44], nut [45], washers [47], bushings [42], washer [41], and bolt [46].

3) For the hinges 3, 4, 5, and 6 (outboard hinges), remove the pins [44], nuts [45], washers [47], bushings [42], washers [41], and bolts [46].

a) Discard the pins [44].

CAUTION DO NOT LET THE BONDING JUMPERS HOLD THE WEIGHT OF THE ELEVATOR TAB ASSEMBLY. IF YOU DO, YOU CAN CAUSE DAMAGE TO THE BONDING JUMPERS.

4) Remove the nut [26], washers [27], and bolt [29] to disconnect the bonding jumper [28] from the elevator [21].

5) Remove the nut [34], washers [35], and bolt [37] to disconnect the bonding jumper [36] from the elevator [21].

(c) Remove the elevator tab [22].

SUBTASK 27-31-31-020-004

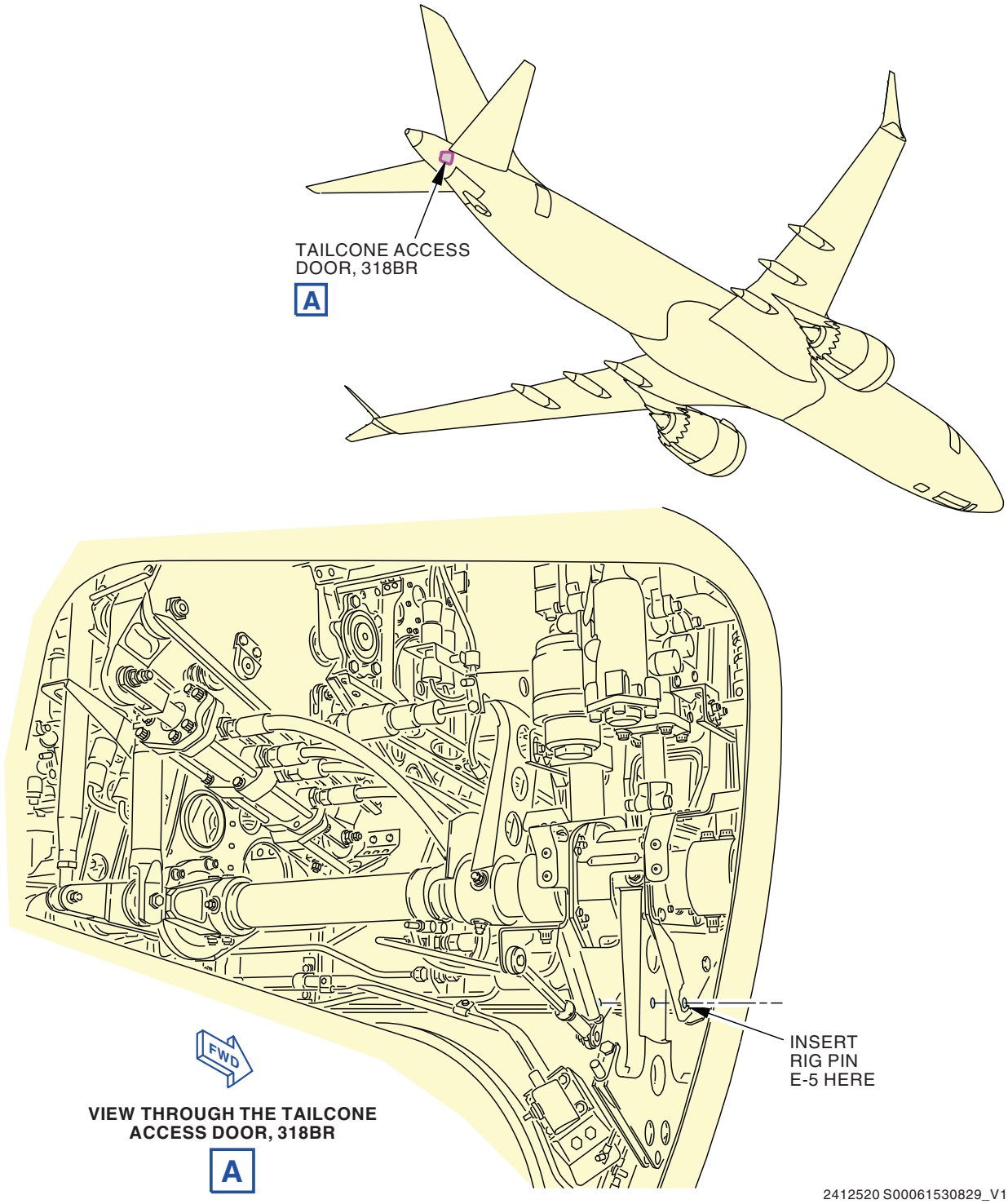
(3) If it is necessary to remove the bonding jumper [28] or bonding jumper [36] from the elevator tab [22], do these steps:

(a) Remove the nut [49], washers [50], bolt [51] to disconnect the bonding jumper [36] from the elevator tab [22].

(b) Remove the nut [52], washers [53], and bolt [54] to disconnect the bonding jumper [28] from the elevator tab [22].

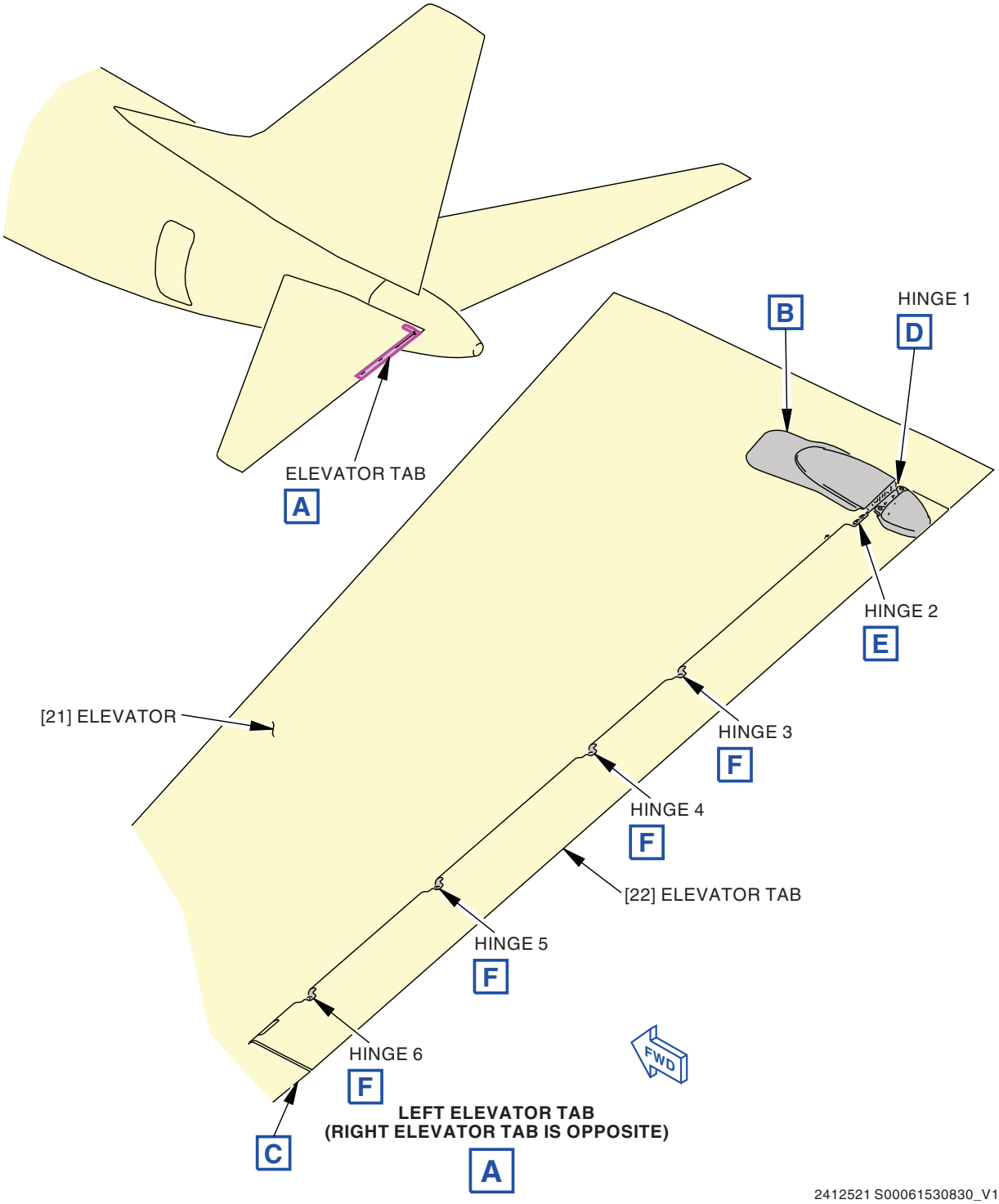
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



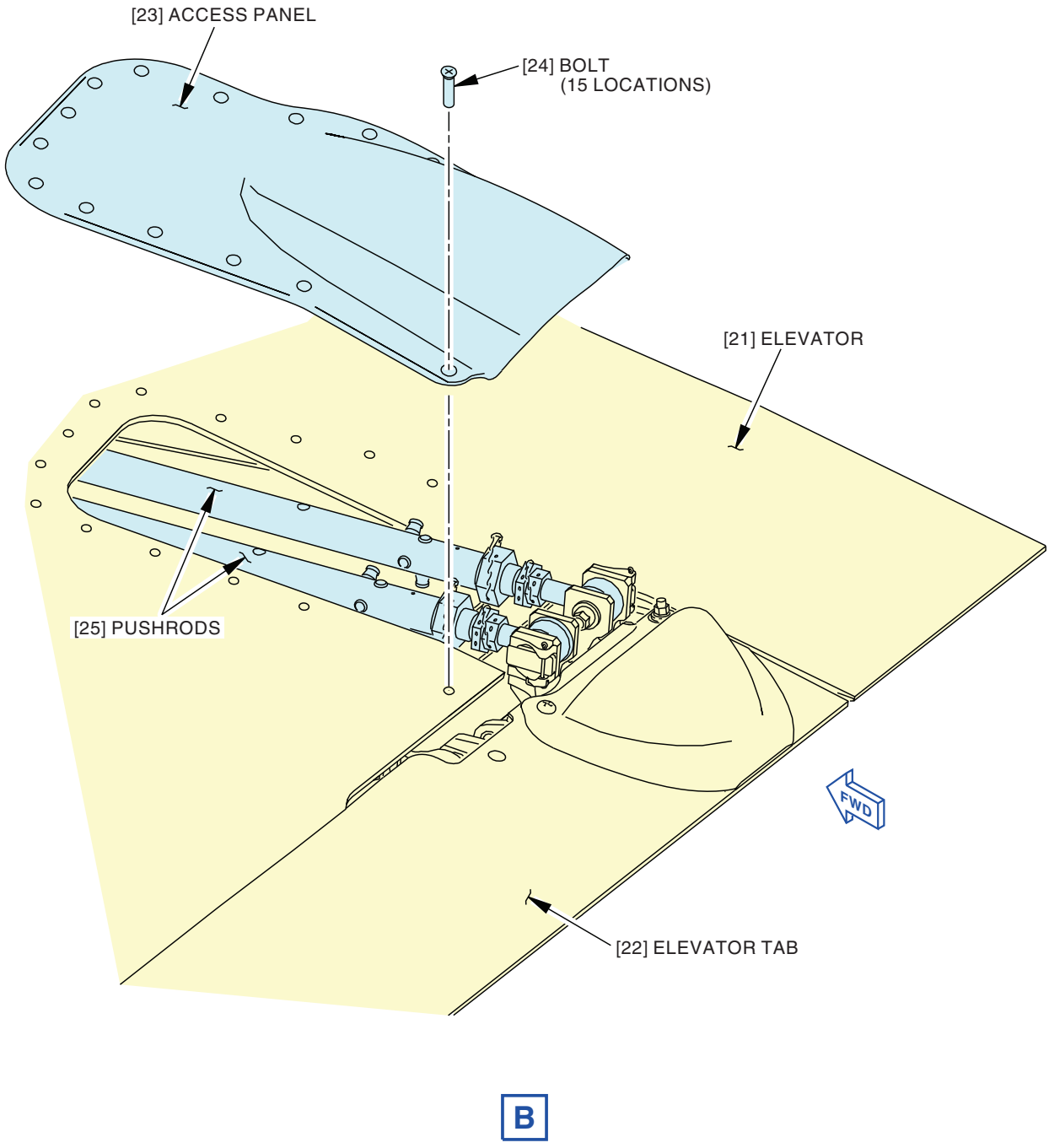
Elevator Rig Pin Installation
Figure 401/27-31-31-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



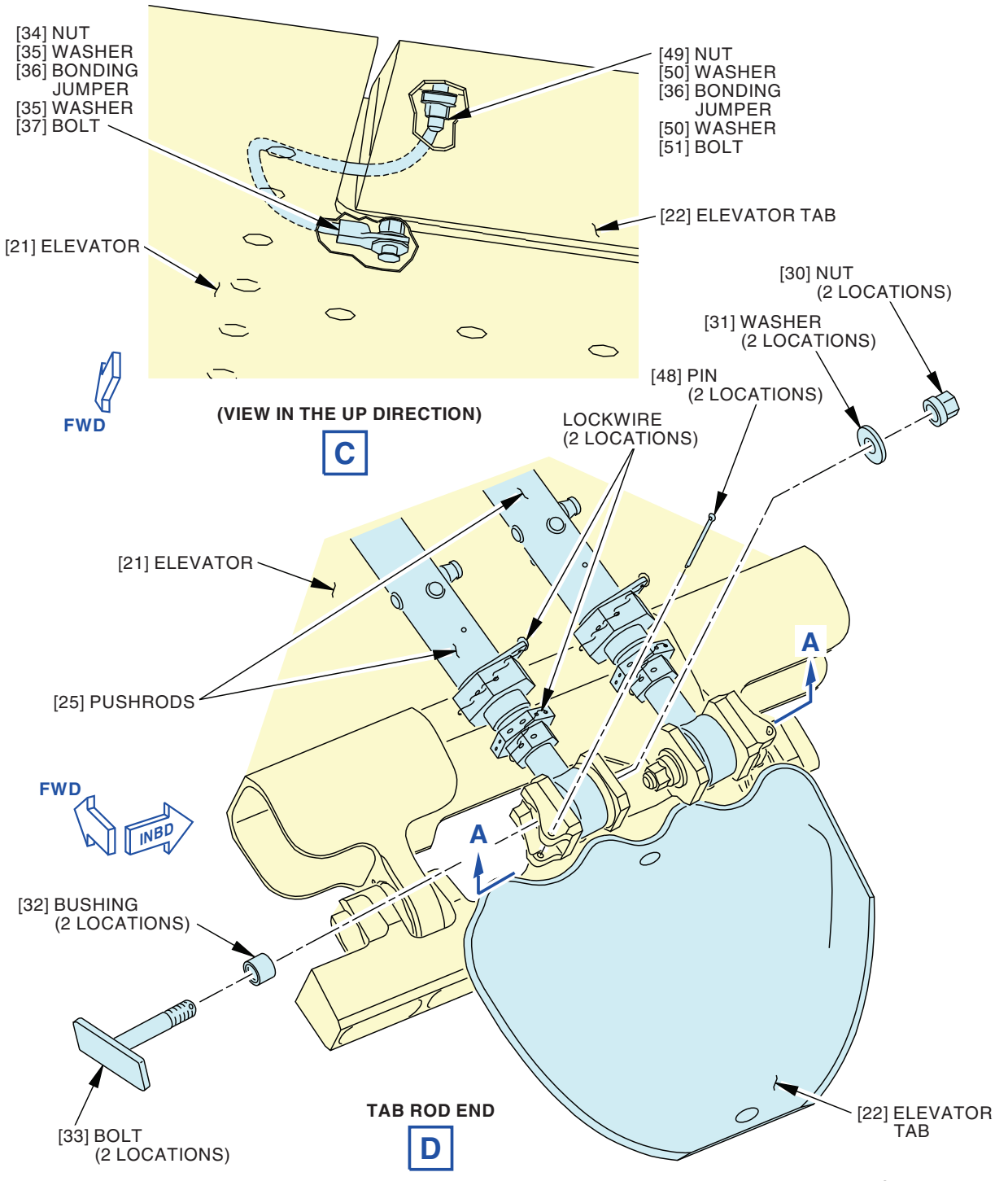
Elevator Tab Installation
Figure 402/27-31-31-990-802 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



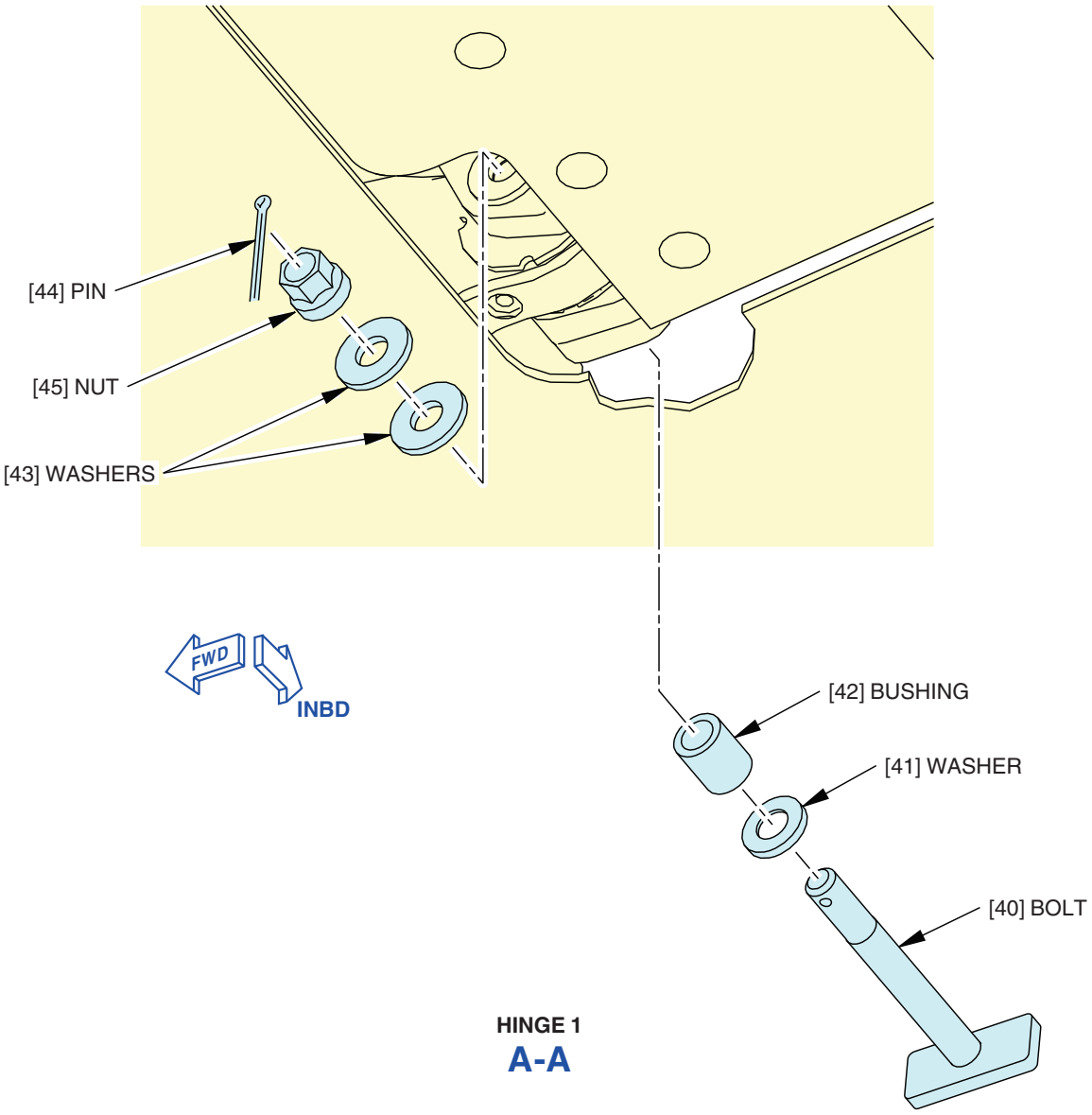
Elevator Tab Installation
Figure 402/27-31-31-990-802 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



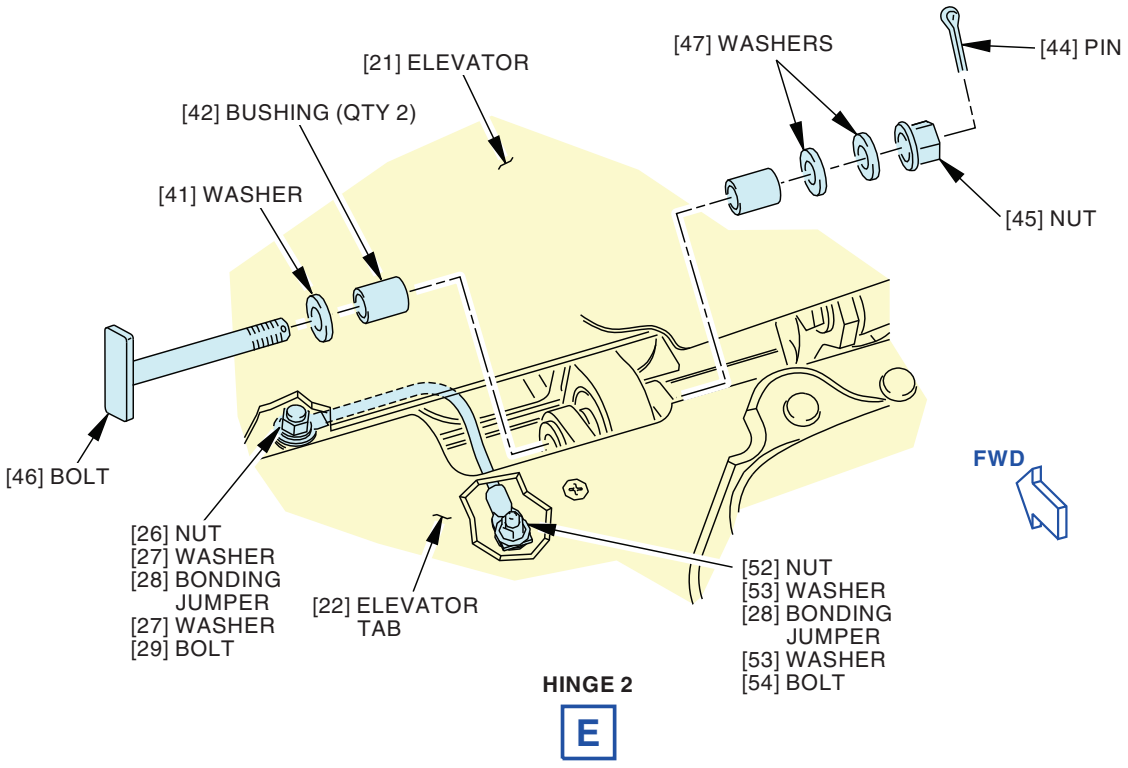
Elevator Tab Installation
Figure 402/27-31-31-990-802 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



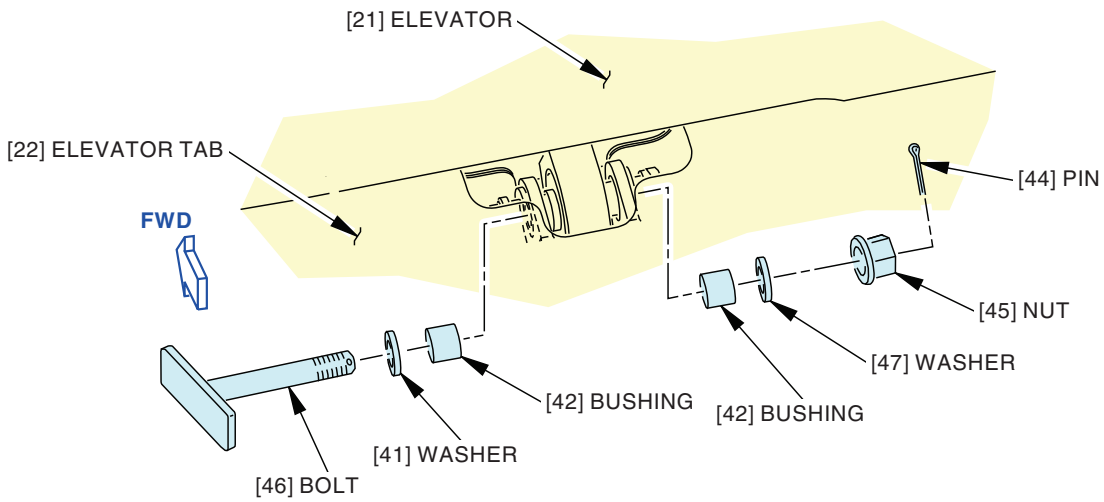
Elevator Tab Installation
Figure 402/27-31-31-990-802 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HINGE 2

E



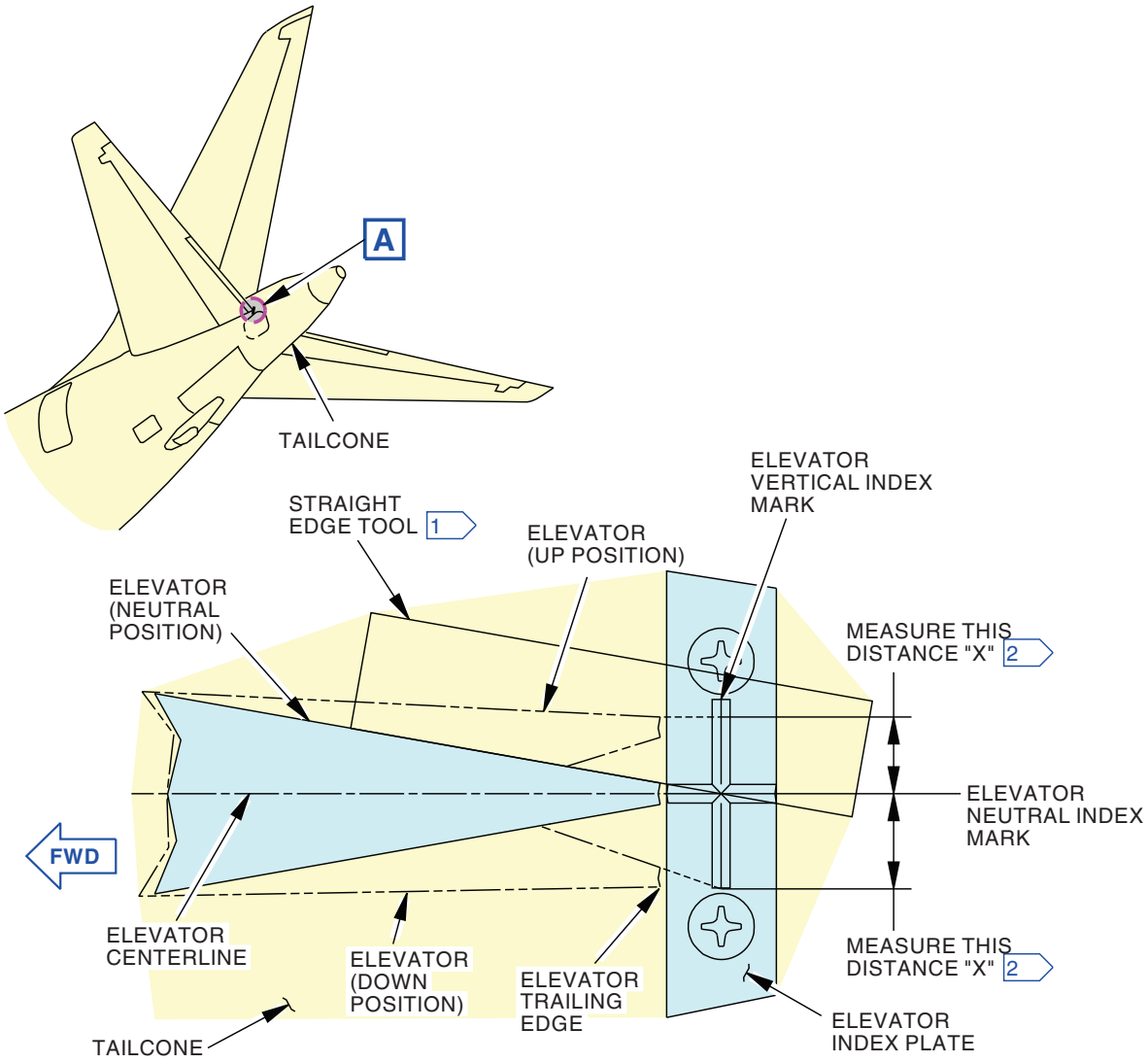
HINGE 3
(HINGES 4, 5, AND 6 ARE EQUIVALENT)

F

2412524 S00061530833_V2

Elevator Tab Installation
Figure 402/27-31-31-990-802 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELEVATOR TAB PUSHROD ADJUSTMENT MEASUREMENTS (LEFT SIDE)

A

1 ELEVATOR IS AT NEUTRAL WHEN THE BOTTOM EDGE OF A STRAIGHT EDGE TOOL PLACED ON THE UPPER ELEVATOR SURFACE PASSES THROUGH THE MIDDLE OF THE ELEVATOR INDEX PLATE, AS SHOWN.

2 THE DISTANCE "X" IS MEASURED USING A STRAIGHT EDGE TOOL ON THE UPPER SURFACE OF THE ELEVATOR TO INTERSECT WITH THE ELEVATOR VERTICAL INDEX MARK AT THE ELEVATOR NEUTRAL INDEX MARK.

2412462 S00061530722_V1

Elevator Tab Adjustment
Figure 403/27-31-31-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-31-400-801

3. Elevator Tab Installation

(Figure 401, Figure 402)

A. References

Reference	Title
12-22-31-640-802	Elevator Tab Hinge Lubrication (P/B 301)
27-31-00-400-801	Elevator and Tab Control System - Activation (P/B 201)
27-31-00-700-803	Elevator Surface Friction and Balance - Test (P/B 501)
27-31-00-700-807	Elevator Tab Linkage - Test (P/B 501)
27-31-00-710-803	Elevator Power-Off Flight Test (P/B 501)
27-31-00-820-805	Elevator Tab Pushrods - Adjustment (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-10 Paragraph 4.C	Category 1 Unsealed and Category 6 Fillet Sealed Pre-Installed Ground Stud Assembly

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A00767	Sealant - Fuel Tank	BMS5-45
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
D50004	Compound - Antiseize	BMS3-28
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

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D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
22	Elevator tab	27-31-31-01-070	PTF ALL
30	Nut	27-31-34-01-085	PTF ALL
		27-31-34-02-085	PTF ALL
44	Pin	27-31-31-01-025	PTF ALL
45	Nut	27-31-31-01-055	PTF ALL
48	Pin	27-31-34-01-035	PTF ALL
		27-31-34-02-035	PTF ALL

E. Location Zones

Zone	Area
316	APU Compartment - Right
334	Left Horizontal Stabilizer - Elevator
335	Left Horizontal Stabilizer - Tip
344	Right Horizontal Stabilizer - Elevator
345	Right Horizontal Stabilizer - Tip

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

G. Prepare for the Installation

SUBTASK 27-31-31-860-003

WARNING

MAKE SURE THAT YOU HAVE THE ELEVATOR TABS BALANCED. IF NOT BALANCED, THE ELEVATOR TABS CAN CAUSE UNWANTED FLUTTER AND INSTABILITY OF THE AIRPLANE. THIS CAN CAUSE AN IN-FLIGHT EMERGENCY. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Examine the data plate on the elevator tab [22] for the necessary number of the tab adjust weights (SRM SUBJECT 51-60-06).

SUBTASK 27-31-31-860-004

(2) Make sure that the elevator [21] has the correct number of the tab adjust weights.

(a) Add or remove weights on the elevator balance panels as applicable.

H. Elevator Tab Installation

SUBTASK 27-31-31-020-003

(1) Install the elevator tab [22] (Figure 402):

(a) If the bonding jumper [36] was removed from the elevator tab [22], do these steps:

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the elevator tab [22], nut [49], washers [50], bonding jumper [36], and bolt [51].

2) Install the bolt [51], washers [50], bonding jumper [36], and nut [49].

3) Tighten the nut [49] to 33 in-lb (3.7 N·m) - 40 in-lb (4.5 N·m).

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4) Measure the electrical resistance between the bonding jumper [36] and bolt [51] (SWPM 20-20-00).
   a) Use a non-intrinsically safe bonding meter, COM-614.
   b) Make sure that the electrical bonding resistance is 0.0015 ohm (1.5 milliohm) or less.
5) Apply a fillet seal with sealant, A00247, or sealant, A00767, to cover the nut [49], washers [50], and bonding jumper [36] (SWPM 20-20-10 Paragraph 4.C).
   a) Do not apply sealant, A00247, or sealant, A00767, to the head side of the bolt.

(b) If the bonding jumper [28] was removed from the elevator tab [22], do these steps:
1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the elevator tab [22], nut [52], washers [53], bonding jumper [28], and bolt [54].
2) Install the bolt [54], washers [53], bonding jumper [28], and nut [52].
3) Tighten the nut [52] to 26 in-lb (2.9 N·m) - 30 in-lb (3.4 N·m).
4) Measure the electrical resistance between the bonding jumper [28] and bolt [54] (SWPM 20-20-00).
   a) Use a non-intrinsically safe bonding meter, COM-614.
   b) Make sure that the electrical bonding resistance is 0.0015 ohm (1.5 milliohm) or less.
5) Apply a fillet seal with sealant, A00247, or sealant, A00767, to cover the nut [52], washers [53], and bonding jumper [28] (SWPM 20-20-10 Paragraph 4.C).
   a) Do not apply sealant, A00247, or sealant, A00767, to the head side of the bolt.

(c) Put the elevator tab [22] near its position behind the elevator [21].

CAUTION

DO NOT LET THE BONDING JUMPERS HOLD THE WEIGHT OF THE ELEVATOR TAB ASSEMBLY. IF YOU DO, YOU CAN CAUSE DAMAGE TO THE BONDING JUMPERS.

(d) Connect the bonding jumper [28] to the elevator [21]:
1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the elevator [21], nut [26], washers [27], bonding jumper [28], and bolt [29].
2) Install the bolt [29], washers [27], and nut [26] to attach the bonding jumper [28] to the elevator [21].
   a) Install the head of the bolt [29] on outside surface.
   b) Tighten the nut [26] to a torque tool setpoint of 30 in-lb (3.4 N·m).
3) Measure the electrical resistance between the bonding jumper [28] and bolt [29] (SWPM 20-20-00).
   a) Use a non-intrinsically safe bonding meter, COM-614.
   b) Make sure that the electrical bonding resistance is 0.015 ohm (15.0 milliohm) or less.
4) Apply a fillet seal with sealant, A00247, to the cover the nut [26], washers [27], bonding jumper [28], and end of the bolt [29] (SWPM 20-20-10 Paragraph 4.C).
   a) Do not apply sealant, A00247, to the head side of the bolt.

(e) Connect the bonding jumper [36] to the elevator [21]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the elevator [21], nut [34], washers [35], bonding jumper [36], and bolt [37].
2) Install the bolt [37], washers [35], and nut [34] to attach the bonding jumper [36] to the elevator [21].
   a) Install the head of the bolt [37] on outside surface.
   b) Tighten the nut [34] to a torque tool setpoint of 30 in-lb (3.4 N·m).
3) Measure the electrical resistance between terminal of the bonding jumper [36] and bolt [37] (SWPM 20-20-00).
   a) Use a non-intrinsically safe bonding meter, COM-614.
   b) Make sure that the electrical bonding resistance is 0.015 ohm (15.0 milliohm) or less.
4) Apply a fillet seal with sealant, A00247, to the nut [34], washers [35], bonding jumper [28], and end of the bolt [37] (SWPM 20-20-10 Paragraph 4.C).
   a) Do not apply sealant, A00247, to the head side of the bolt.

SUBTASK 27-31-31-420-001

(2) Connect the elevator tab [22] to the elevator [21]:

(a) If the nuts [30] or nuts [45] were loosened or removed, they need to be replaced with new nuts.

NOTE: Use of new nuts [30] or new nuts [45] will make sure that they have the necessary run-on torque.

(b) Apply compound, D50004, to the threads of the bolts [40] and bolts [46].
   1) Do not apply compound, D50004, to the unthreaded part of the bolts.

(c) For the hinges 3, 4, 5 and 6 (outboard hinges), install the bolts [46], washers [41], bushings [42], washers [47], and new nuts [45].
   1) Tighten the nuts [45] to 95 ±5 in-lb (10.7 ±0.6 N·m).
   2) Install the new pin [44] in the end of the bolts [46].

(d) For hinge 2 (inboard hinge), install the bolt [46], washer [41], bushings [42], washers [47], and new nut [45].
   1) Tighten the new nuts [45] to 95 ±5 in-lb (10.7 ±0.6 N·m).
   2) Install new pin [44] in the end of the bolt [46].
      a) If it is necessary, you can remove the washer [47] under the nut [45] to install pin [44].

(e) For hinge 1 (inboard hinge), install the bolt [40], washer [41], bushing [42], washers [43], and new nut [45].
   1) Tighten the new nuts [45] to 95 ±5 in-lb (10.7 ±0.6 N·m).
   2) Install new pin [44] in the end of the bolt [40].
      a) If it is necessary, you can remove the washer [43] under the nut [45] to install the pin [44].

(f) Apply a coating of compound, D50004, on the outside diameter of bolts [33].

(g) Install the bolts [33], bushings [32], washers [31], and new nuts [30] to connect the tab pushrods [25] to the elevator tab [22].

NOTE: The inboard bolt is installed with the head inboard and the outboard bolt is installed with the head outboard.

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CAUTION TIGHTEN THE NEW NUTS CORRECTLY. IF YOU TIGHTEN THEM INCORRECTLY, THE PUSHRODS CAN BECOME DISCONNECTED. THIS CAN CAUSE DAMAGE TO THE ELEVATOR AND STABILIZER.

1) Tighten the new nuts [30] to 75 ±5 in-lb (8.5 ±0.6 N·m).
2) Install the new pins [48] in the anti-rotation boss at the head of the bolts [33].

SUBTASK 27-31-31-420-002

(3) Install the bolts [24] to connect the access panel [23] to the elevator [21].

SUBTASK 27-31-31-610-001

(4) Do this task: Elevator Tab Hinge Lubrication, TASK 12-22-31-640-802.

SUBTASK 27-31-31-080-001

(5) Remove rig pin E-5 of the rigging pin kit, SPL-1585, from the aft control quadrant.

SUBTASK 27-31-31-860-007

(6) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-31-31-860-008

(7) Do this task: Elevator and Tab Control System - Activation, TASK 27-31-00-400-801.

SUBTASK 27-31-31-820-002

(8) Adjust the tab pushrods until you get the same elevator tab [22] that was measured before the elevator tab [22] removal (TASK 27-31-00-820-805).

NOTE: If the elevator tab is in the same position as before the removal, then it is not necessary to do the adjustment of the elevator tab pushrods.

I. Elevator Tab Installation Test

SUBTASK 27-31-31-710-001

(1) Do this task: Elevator Surface Friction and Balance - Test, TASK 27-31-00-700-803.

SUBTASK 27-31-31-710-002

(2) Do this task: Elevator Tab Linkage - Test, TASK 27-31-00-700-807.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-31-410-001

(1) Install these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-31-410-002

(2) For the left elevator tab pushrods, install this access panel:

Number	Name/Location
334PT	Horizontal Stabilizer, Tab Control Rod Fairing

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SUBTASK 27-31-31-410-003

(3) For the right elevator tab pushrods, install this access panel:

Number	Name/Location
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-31-390-001

(4) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

SUBTASK 27-31-31-710-003

(5) If it is necessary, do this task: Elevator Power-Off Flight Test, TASK 27-31-00-710-803.

——— END OF TASK ———

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ELEVATOR AND ELEVATOR TAB FREEPLAY - INSPECTION/CHECK

1. General

A. This procedure contains tasks to inspect the following for freeplay:

(1) The elevator output torque tube buss linkage.

(2) The elevator tab.

TASK 27-31-32-200-801

2. Elevator Output Torque Tube Buss Linkage Freeplay - Check

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this test to do a check of the elevator output torque tube buss linkage freeplay.

B. References

Reference	Title
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)
27-51-00-860-801	Trailing Edge Flap System Operation With Primary Control (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
STD-1278	Block - Loading, 1/8 Inch Thick, 3 Inch by 3 Inch Wood or Fiberglass Block with Pad to Prevent Damage to the Skin.

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D. Consumable Materials

Reference	Description	Specification
G50393	Tape - Adhesive, Label	BAC5307 Type III Polyester (Mylar)

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Prepare for the Check

SUBTASK 27-31-32-860-001

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Pressurize the elevator hydraulic systems A and B (TASK 27-31-00-800-801).

SUBTASK 27-31-32-860-002

(2) Make sure that the flaps are in the full up position (TASK 27-51-00-860-801).

SUBTASK 27-31-32-010-001

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-32-860-003

CAUTION MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677, to set the "B" dimension at 39.17 ±0.01 in. (994.92 ±0.26 mm) or use the trim system to move the stabilizer to 4 units of trim (Figure 601).

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SUBTASK 27-31-32-860-004

(5) Operate the elevator in the full travel 6 times.

SUBTASK 27-31-32-860-005

(6) Make sure that the control column is in the neutral position.

H. Elevator Output Torque Tube Buss Linkage Freeplay Check

SUBTASK 27-31-32-220-001

CAUTION DO NOT PUT A LOAD OF MORE THAN 65 POUNDS (290 NEWTONS) ON THE ELEVATORS AT ONE TIME. IF THE LOAD IS TOO HEAVY, IT CAN CAUSE DAMAGE TO THE ELEVATORS.

(1) Set the FLT CONTROL A switch, on the P5 forward overhead panel, to the OFF position.

SUBTASK 27-31-32-860-013

(2) Make sure that the FLT CONTROL B switch, on the P5 forward overhead panel, is in the ON position.

SUBTASK 27-31-32-420-001

(3) Put a piece of adhesive tape, G50393, over the right elevator index plate [1].

SUBTASK 27-31-32-220-007

(4) Use a block, STD-1278, and force gauge, COM-1557, to apply a force of 15 ±1 lbf (66.7 ±4.4 N) to the right elevator trailing edge in the down direction and release slowly.

(a) Apply the force approximately 0.5 in. (12.7 mm) forward of the trailing edge and 1 in. (25 mm) outboard of the inboard edge of the elevator (next to the tailcone).

(b) Slowly remove the force from the elevator.

(c) Make a mark of the right elevator position on the tape.

SUBTASK 27-31-32-220-006

(5) Use a block, STD-1278, and force gauge, COM-1557, to apply a force of 10 ±1 lbf (44.5 ±4.4 N) to right elevator trailing edge in the up direction.

(a) Apply the force approximately 0.5 in. (12.7 mm) forward of the trailing edge and 1 in. (25 mm) outboard of the inboard edge of the elevator (next to the tailcone).

(b) Hold the force of 10 ±1 lbf (44.5 ±4.4 N) on the elevator.

(c) Make a mark of the right elevator position on the tape.

(d) Slowly remove the force from the elevator.

SUBTASK 27-31-32-220-005

(6) Make sure that the total up and down freeplay of the right elevator is 0.21 in. (5.33 mm) or less.

(a) If the total up and down freeplay is more than 0.21 in. (5.33 mm), do a check of the torque tube and the torque tube crank arm bearing for excessive wear.

SUBTASK 27-31-32-020-001

(7) Remove the tape.

SUBTASK 27-31-32-860-014

(8) Set the FLT CONTROL A switch, on the P5 forward overhead panel, to the ON position.

SUBTASK 27-31-32-860-015

(9) Set the FLT CONTROL B switch, on the P5 forward overhead panel, to the OFF position.

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SUBTASK 27-31-32-420-002

(10) Put a piece of tape over the left elevator index plate [1].

SUBTASK 27-31-32-220-008

(11) Use a block, STD-1278, and force gauge, COM-1557, to apply a force of 15 ±1 lbf (66.7 ±4.4 N) to left elevator trailing edge in the down direction and release slowly.

(a) Apply the force approximately 0.5 in. (12.7 mm) forward of the trailing edge and 1 in. (25 mm) outboard of the inboard edge of the elevator (next to the tailcone).

(b) Slowly remove the force from the elevator.

(c) Make a mark of the left elevator position on the tape.

SUBTASK 27-31-32-220-009

(12) Use a block, STD-1278, and force gauge, COM-1557, to apply a force of10 ±1 lbf (44.5 ±4.4 N) to left elevator trailing edge in the up direction.

(a) Apply the force approximately 0.5 in. (12.7 mm) forward of the elevator trailing edge and 1 in. (25 mm) outboard of the inboard edge of the elevator (next to the tailcone).

(b) Hold the force of 10 ±1 lbf (44.5 ±4.4 N) on the elevator.

(c) Make a mark of the left elevator position on the tape.

(d) Slowly remove the force from the elevator.

SUBTASK 27-31-32-220-010

(13) Make sure that the total up and down freeplay of the left elevator is 0.21 in. (5.33 mm) or less.

(a) If the total up and down freeplay is more than 0.21 in. (5.33 mm), do a check of the torque tube and the torque tube crank arm bearing for excessive wear.

SUBTASK 27-31-32-020-002

(14) Remove the tape.

SUBTASK 27-31-32-860-016

(15) Set the FLT CONTROL B switch, on the P5 forward overhead panel, to the ON position.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-32-860-006

(1) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

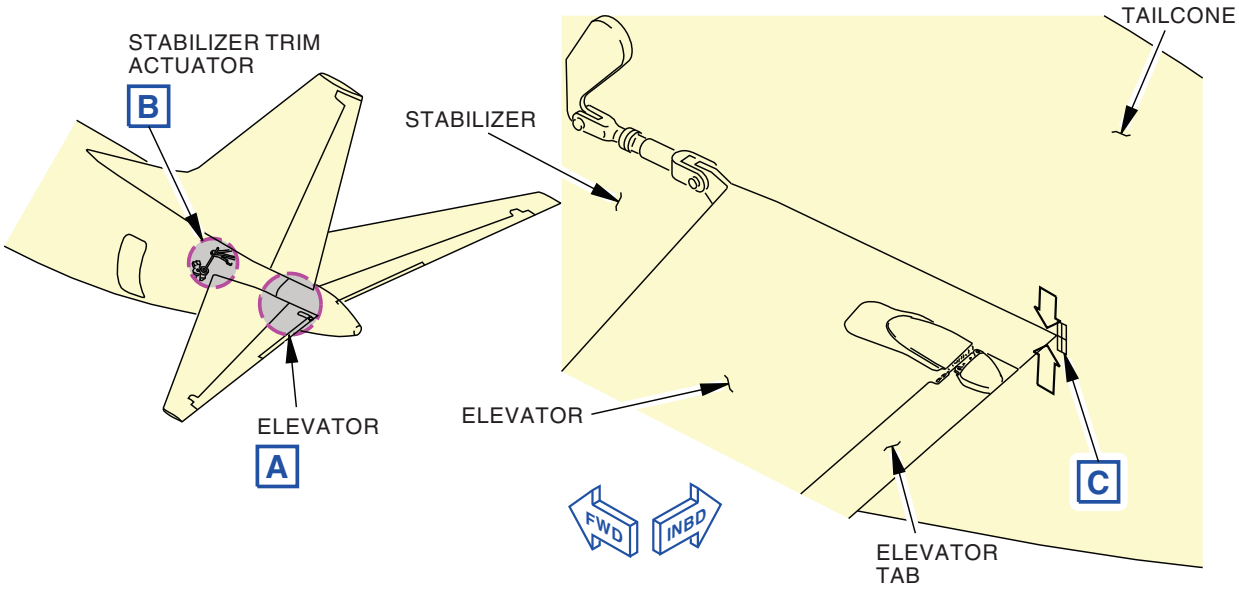
SUBTASK 27-31-32-410-001

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

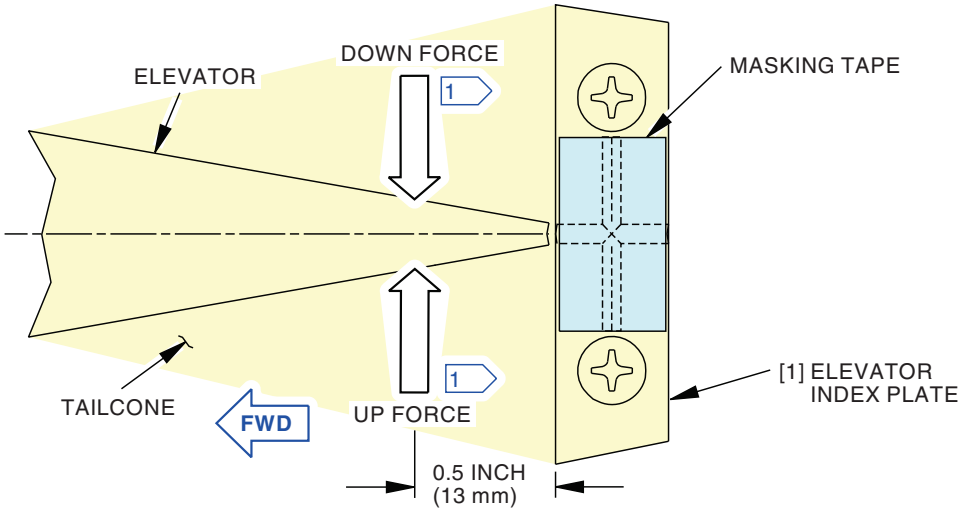
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT ELEVATOR
(RIGHT ELEVATOR IS OPPOSITE)

A



BUSS LINKAGE FREEPLAY

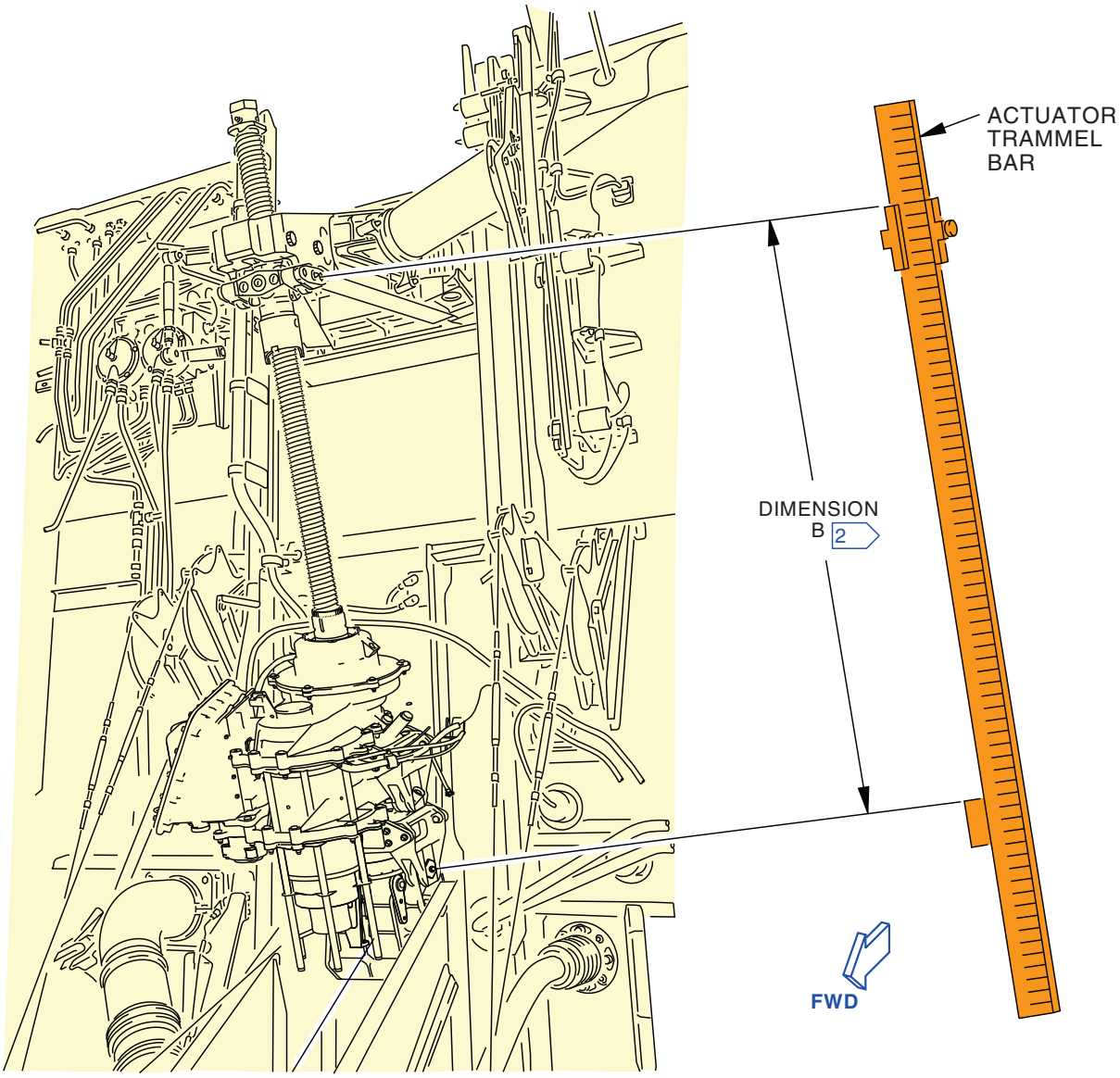
C

1 APPLY UP FORCE AND DOWN FORCE APPROXIMATELY 0.5 INCH (13 mm) FORWARD OF THE TRAILING EDGE AND 1 INCH (25 mm) OUTBOARD OF THE INBOARD EDGE OF THE ELEVATOR (NEXT TO THE TAILCONE).

2412526 S00061530839_V1

Elevator Output Torque Tube Buss Linkage Freeplay Inspection
Figure 601/27-31-32-990-801 (Sheet 1 of 2)

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STABILIZER TRIM ACTUATOR

B

NOTE:

THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES (4.0 PILOT UNITS).

1 THE DIMENSIONS B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS (CENTER OF GREASE FITTINGS).

2412525 S00061530838_V1

Elevator Output Torque Tube Buss Linkage Freeplay Inspection
Figure 601/27-31-32-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

27-CMR-02

TASK 27-31-32-200-802

3. Elevator Tab Freeplay - Check

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this test to do a check of the elevator tab trailing edge freeplay, the elevator tab hinge line axial freeplay, and the elevator tab hinge line radial freeplay.

(2) This check is applicable to the left and right side elevator tabs.

B. References

Reference	Title
12-22-31-640-802	Elevator Tab Hinge Lubrication (P/B 301)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
27-51-00-860-801	Trailing Edge Flap System Operation With Primary Control (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
STD-1107	Gauge - Feeler, 0.0 - 0.5 Inch, Readable to 1/1000th
STD-1238	Indicator - Dial
STD-1279	Mount - Device, Mounting, Holds the Dial Indicator
STD-1303	Caliper - Dial

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Check

SUBTASK 27-31-32-860-007

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES. MAKE SURE THAT THE INBOARD FAN DUCT COWL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED. THE FLIGHT CONTROL SURFACES CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize the elevator hydraulic systems A and B (TASK 27-31-00-800-801).

SUBTASK 27-31-32-860-008

(2) Make sure that the flaps are in the full up position (TASK 27-51-00-860-801).

SUBTASK 27-31-32-010-002

(3) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-32-860-009

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(4) Use the trammel bar, SPL-1677, to set the "B" dimension at 39.17 ±0.01 in. (994.92 ±0.26 mm) or use the trim system to move the stabilizer to 4 units of trim (Figure 602).

SUBTASK 27-31-32-860-010

(5) Operate the elevator in the full travel 6 times.

SUBTASK 27-31-32-860-011

(6) Make sure that the control column is in the neutral position.

G. Elevator Tab Trailing Edge Freeplay Check

SUBTASK 27-31-32-220-002

(1) Do a check of the elevator tab trailing edge freeplay (Figure 602):

(a) Remove pressure from the elevator hydraulic systems A and B (TASK 27-31-00-800-802).

(b) Make sure that the flaps are retracted to the full up position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Make sure that the stabilizer is set to 4.0 ± 0.5 units of trim.

CAUTION DO NOT USE MORE THAN HAND PRESSURE WHEN YOU INSTALL THE WOOD BLOCK. IF YOU USE MORE THAN HAND PRESSURE, YOU CAN CAUSE DAMAGE TO THE ELEVATOR OR TAIL CONE.

(d) Install a wood block between the inboard edge of the elevator and the tail cone to keep the elevator from moving (Figure 602).

1) Make sure that you cover the wood block with duct tape or equivalent so that you do not damage the paint of the elevator or tail cone.

WARNING DO NOT MOVE THE CONTROL COLUMNS WHILE THE MOUNTING PLATE IS IN ITS POSITION. MOVEMENT OF THE CONTROL COLUMNS CAN CAUSE INJURY TO PERSONNEL OR DAMAGE TO THE ELEVATOR.

CAUTION MAKE SURE THAT YOU INSTALL THE MOUNTING PLATE CORRECTLY. THE MOUNTING PLATE CAN CAUSE DAMAGE TO THE ELEVATOR IF YOU DO NOT INSTALL IT CORRECTLY.

(e) Attach a mounting plate to the elevator trailing edge forward of the elevator tab hinge 2.

1) If double-back tape is used to attach the mounting plate, make sure that it is not made of compressible foam.

2) Attach the mount, STD-1279, and the dial indicator, STD-1238, plunger on the top of the elevator tab, approximately 0.1 in. (3 mm) forward of the trailing edge of the elevator tab and within 0.5 in. (13 mm) of the inboard side of the hinge 2 centerline.

3) Adjust the dial indicator, STD-1238, to zero.

(f) Use a block and a force gauge, COM-1557, to push down on the top of the elevator tab with a force of 10.0 ±0.2 lbf (44.5 ±0.9 N).

1) Apply the force on the elevator tab at the trailing edge, as close as possible to the tab hinge 2 centerline.

2) Make a record of the travel shown on the dial indicator, STD-1238.

3) Slowly remove the force.

(g) Do not adjust the dial indicator, STD-1238.

(h) Push up on the bottom surface of the elevator tab with a force of 10.0 ±0.2 lbf (44.5 ±0.9 N), directly below the force applied before.

1) Make a record of the travel shown on the dial indicator, STD-1238.

2) Slowly remove the force.

(i) Make sure that the sum of the two travel records is 0.03 in. (0.76 mm) or less.

1) If the sum of the two travel records is more than 0.03 in. (0.76 mm), do a check of the hinges for excessive wear and replace if it is necessary.

(j) Remove the mount, STD-1279, and the dial indicator, STD-1238.

(k) Remove the mounting plate.

(l) If no other checks are necessary, remove the wood block from between the inboard edge of the elevator and the tail cone.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Elevator Tab Hinge Axial Freeplay Check

SUBTASK 27-31-32-220-003

(1) Do a check of the elevator tab hinge axial freeplay (Figure 602):

(a) Remove pressure from the elevator hydraulic systems A and B (TASK 27-31-00-800-802).

(b) Make sure that the flaps are retracted to the full up position.

(c) Make sure that the stabilizer is set at 4.0 ± 0.5 units of trim.

CAUTION

DO NOT USE MORE THAN HAND PRESSURE WHEN YOU INSTALL THE WOOD BLOCK. IF YOU USE MORE THAN HAND PRESSURE, YOU CAN CAUSE DAMAGE TO THE ELEVATOR OR TAIL CONE.

(d) Install a wood block between the inboard edge of the elevator and the tail cone to keep the elevator from moving (Figure 602).

1) Make sure that you cover the wood block with duct tape or equivalent so that you do not damage the paint of the elevator or tail cone.

(e) Do these steps to make sure that the axial play (inboard - outboard direction) of the tab is 0.036 in. (0.91 mm) or less.

1) Hold the elevator tab at the trailing edge and move it outboard with hand pressure.

2) Use a 0.0 - 0.5 Inch feeler gauge, STD-1107, or a dial caliper, STD-1303, to measure the clearance between the elevator and the outboard end of the elevator tab.

a) If it is necessary, insert a flat metal filler block between the elevator and the elevator tab to help in this measurement.

b) Make a record of the clearance dimension.

3) Hold the elevator tab at the trailing edge and move it inboard with hand pressure.

4) Measure the clearance between the elevator and the outboard end of the elevator tab.

a) Make a record of the clearance dimension.

5) Make sure that the difference between the two measured clearances, or the axial freeplay, is less than 0.036 in. (0.91 mm).

(f) If no other checks are necessary, remove the wood block from between the inboard edge of the elevator and the tail cone.

I. Elevator Tab Hinge Line Freeplay Check

SUBTASK 27-31-32-220-004

(1) Do a check of the elevator tab hinge line freeplay (Figure 602):

(a) If it is necessary, remove pressure from the elevator hydraulic systems A and B (TASK 27-31-00-800-802).

(b) Make sure that the stabilizer is set at 4.0 ± 0.5 units of trim.

(c) Make sure that the flaps are retracted to the full up position.

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CAUTION DO NOT USE MORE THAN HAND PRESSURE WHEN YOU INSTALL THE WOOD BLOCK. IF YOU USE MORE THAN HAND PRESSURE, YOU CAN CAUSE DAMAGE TO THE ELEVATOR OR TAIL CONE.

(d) Install a wood block between the inboard edge of the elevator and the tail cone to keep the elevator from moving (Figure 602).

1) Make sure that you cover the wood block with duct tape or equivalent so that you do not damage the paint of the elevator or tail cone.

CAUTION MAKE SURE THAT YOU INSTALL THE MOUNTING PLATE CORRECTLY. THE MOUNTING PLATE CAN CAUSE DAMAGE TO THE ELEVATOR IF YOU DO NOT INSTALL IT CORRECTLY.

(e) Attach a mounting plate to the elevator trailing edge, forward of elevator tab hinge 3.

1) If double-back tape is used to attach the mounting plate, make sure that it is not made of compressible foam.

(f) Attach the mount, STD-1279, and the dial indicator, STD-1238, plunger on the elevator tab leading edge directly over the top of the tab hinge centerline (Figure 602).

(g) Adjust the dial indicator, STD-1238, to zero once there is contact with the elevator tab leading edge.

(h) Use a block and a force scale to apply a 10.0 ±0.5 lbf (44.5 ±2.2 N) in the up direction.

NOTE: The force is to be applied in line with the plunger on the bottom side of the tab leading edge.

(i) Make a record of the travel shown on the dial indicator, STD-1238.

(j) Slowly remove the force.

(k) Do not adjust the dial indicator, STD-1238.

(l) Use a block and a force scale to apply 10.0 ±0.5 lbf (44.5 ±2.2 N) in the down direction.

NOTE: The force is to be applied to the top side of the tab leading edge.

(m) Make a record of the travel shown on the dial indicator, STD-1238.

(n) Slowly remove the force.

(o) Make sure that the sum of the two travel records is 0.03 in. (0.76 mm) or less.

1) If the sum of the two travel records is more than 0.03 in. (0.76 mm), do a check of the hinges for excessive wear and replace if it is necessary.

(p) Move the mount, STD-1279, the dial indicator, STD-1238, and the mounting plate to the next hinge centerline.

(q) Repeat the above steps for the other three outboard tab hinges (hinges 4, 5, and 6).

(r) Remove the mount, STD-1279, and the dial indicator, STD-1238.

(s) Remove the mounting plate.

(t) Remove the wood block from between the inboard edge of the elevator and the tail cone.

SUBTASK 27-31-32-640-001

(2) Lubricate the elevator tab hinges (TASK 12-22-31-640-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

J. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-32-860-012

(1) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

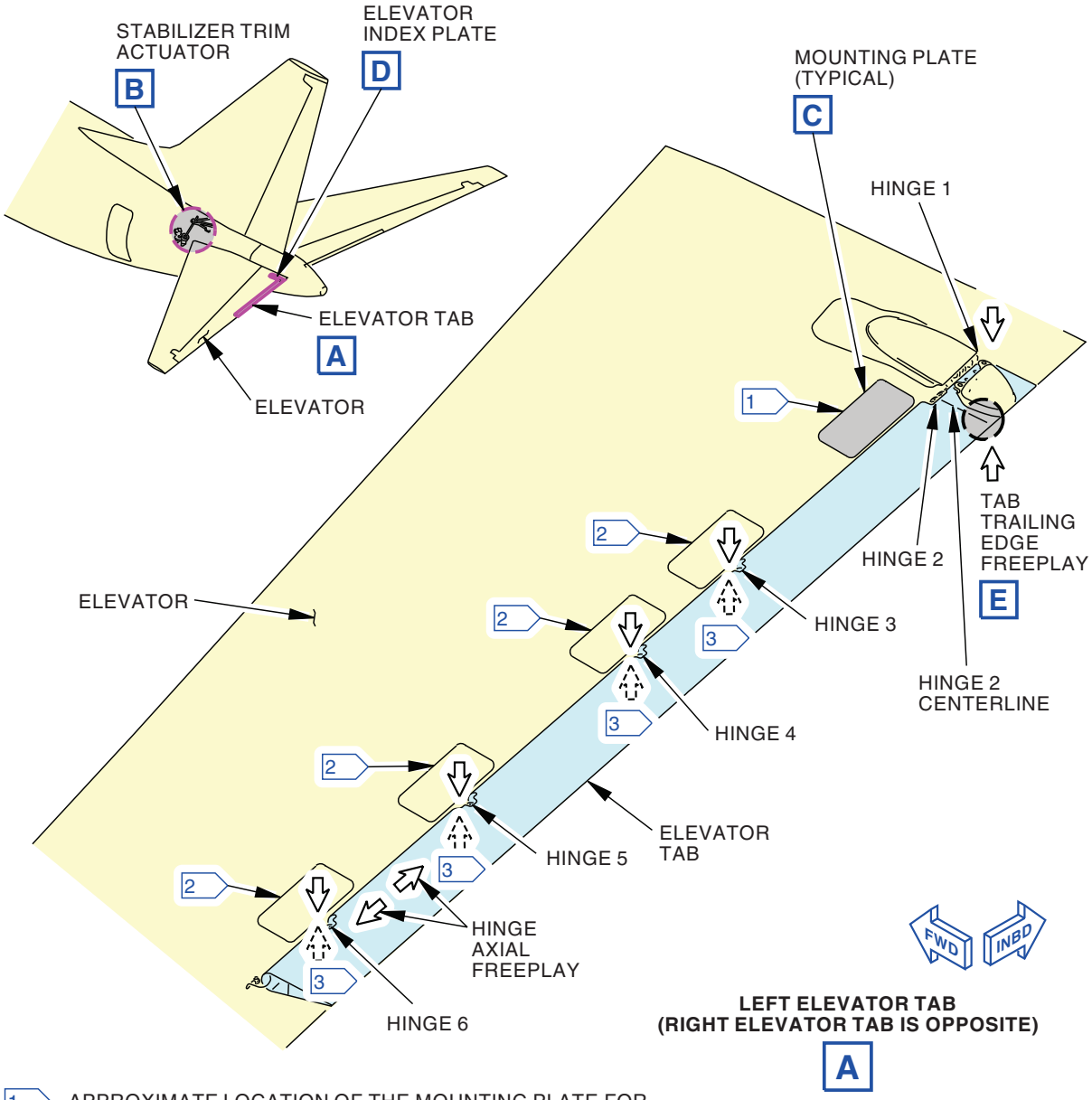
SUBTASK 27-31-32-410-002

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 APPROXIMATE LOCATION OF THE MOUNTING PLATE FOR THE ELEVATOR TAB TRAILING EDGE FREEPLAY.

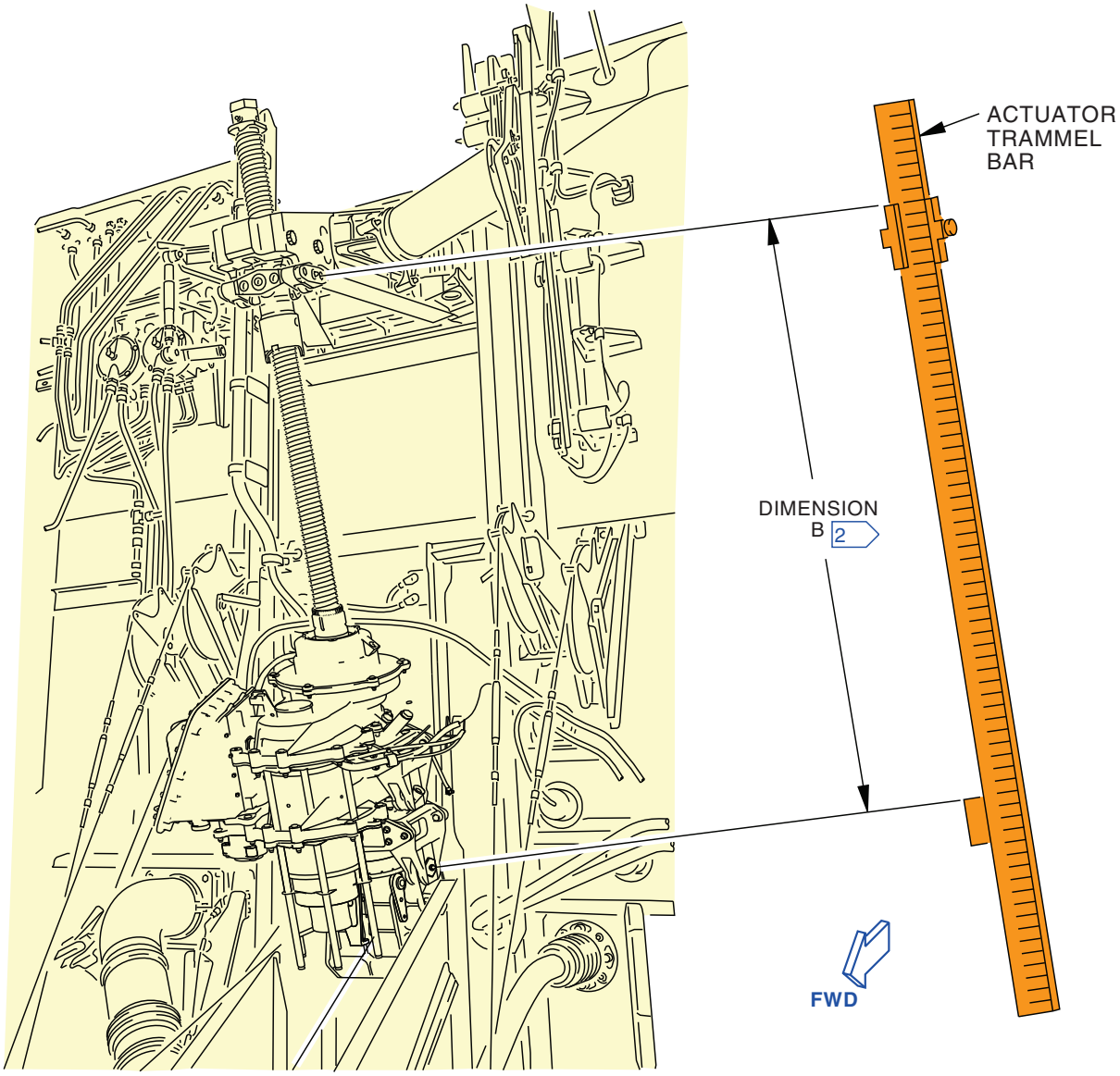
2 APPROXIMATE LOCATION OF THE MOUNTING PLATE FOR THE ELEVATOR TAB HINGE LINE FREEPLAY.

3 APPLY FORCE FOR TAB HINGE LINE FREEPLAY AT HINGES 3, 4, 5 AND 6, AS CLOSE TO THE HINGE LUG CENTERLINE, AS POSSIBLE.

2412527 S00061530841_V1

Elevator Tab Freeplay Check
Figure 602/27-31-32-990-802 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACTUATOR

B

NOTE:

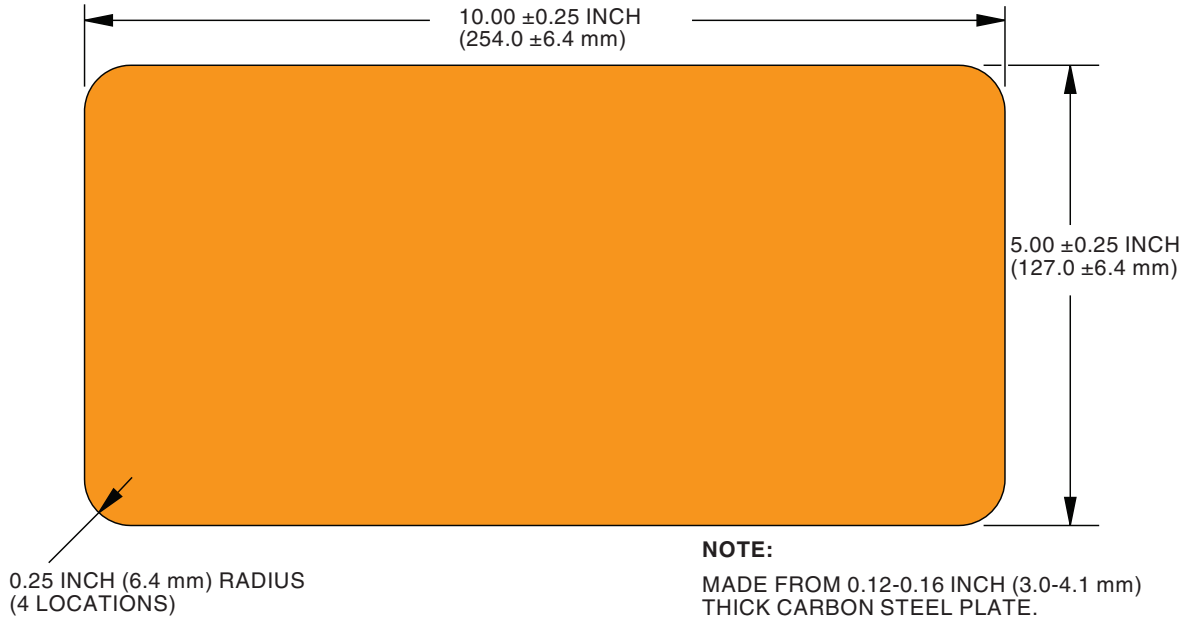
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES (4.0 PILOT UNITS).

4 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS (CENTER OF GREASE FITTINGS).

2412528 S00061530842_V1

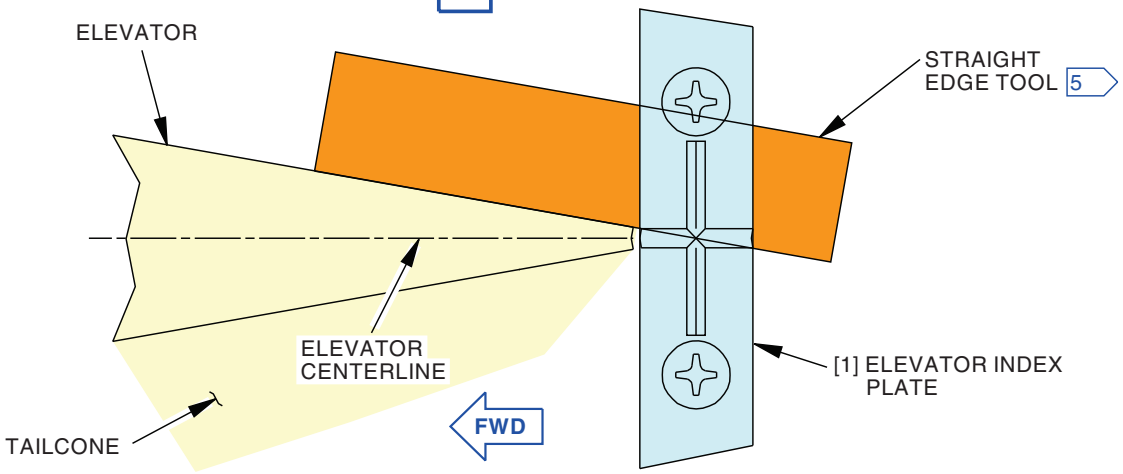
Elevator Tab Freeplay Check
Figure 602/27-31-32-990-802 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MOUNTING PLATE (TYPICAL)

C



ELEVATOR INDEX PLATE

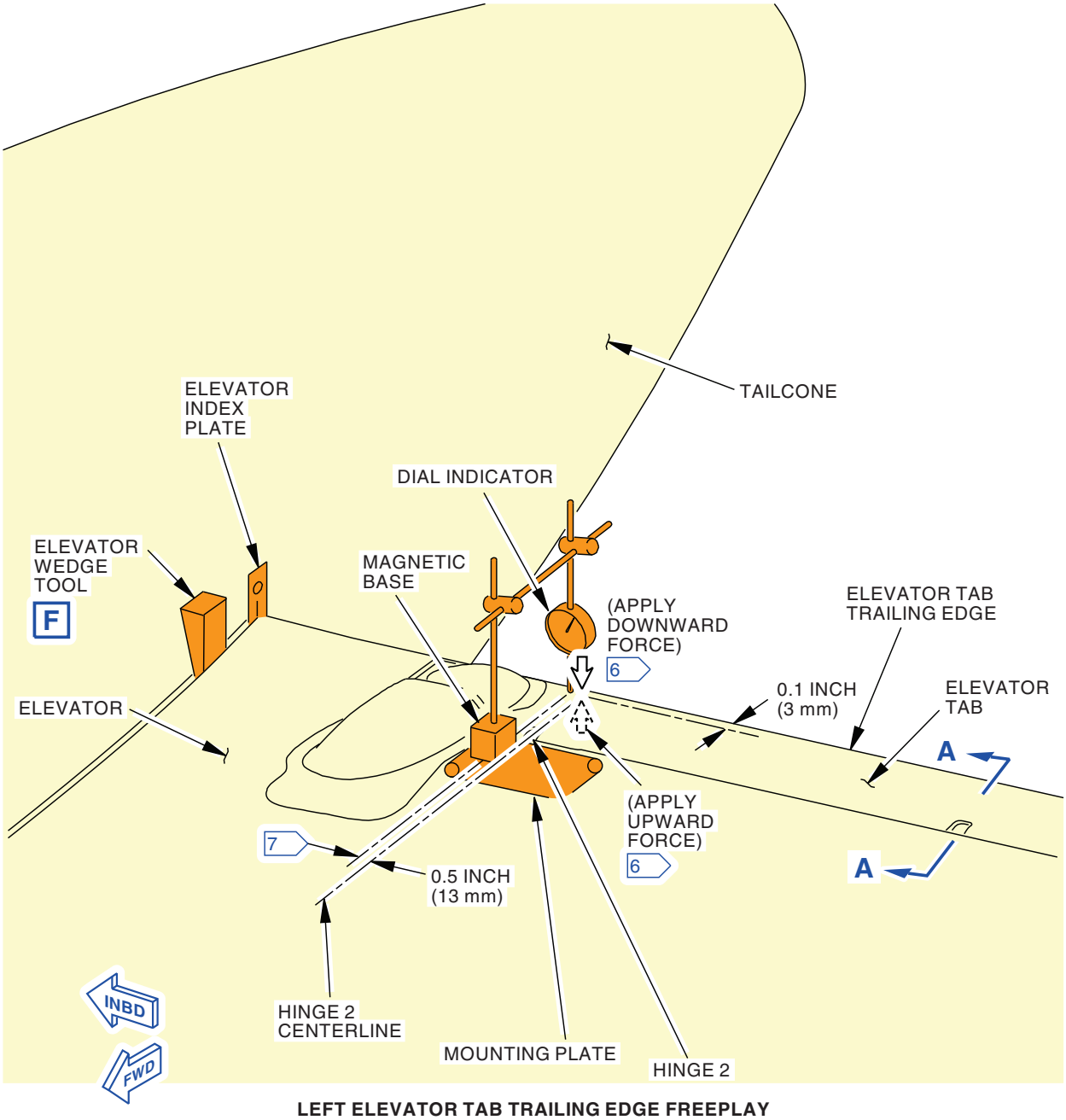
A

5 ELEVATOR IS AT THE NEUTRAL POSITION WHEN THE BOTTOM EDGE OF A STRAIGHT EDGE TOOL PLACED ON THE UPPER ELEVATOR SURFACE PASSES THROUGH THE MIDDLE OF THE ELEVATOR INDEX PLATE, AS SHOWN.

2412529 S00061530843_V1

Elevator Tab Freeplay Check
Figure 602/27-31-32-990-802 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT ELEVATOR TAB TRAILING EDGE FREEPLAY
(RIGHT ELEVATOR TAB IS OPPOSITE)

E

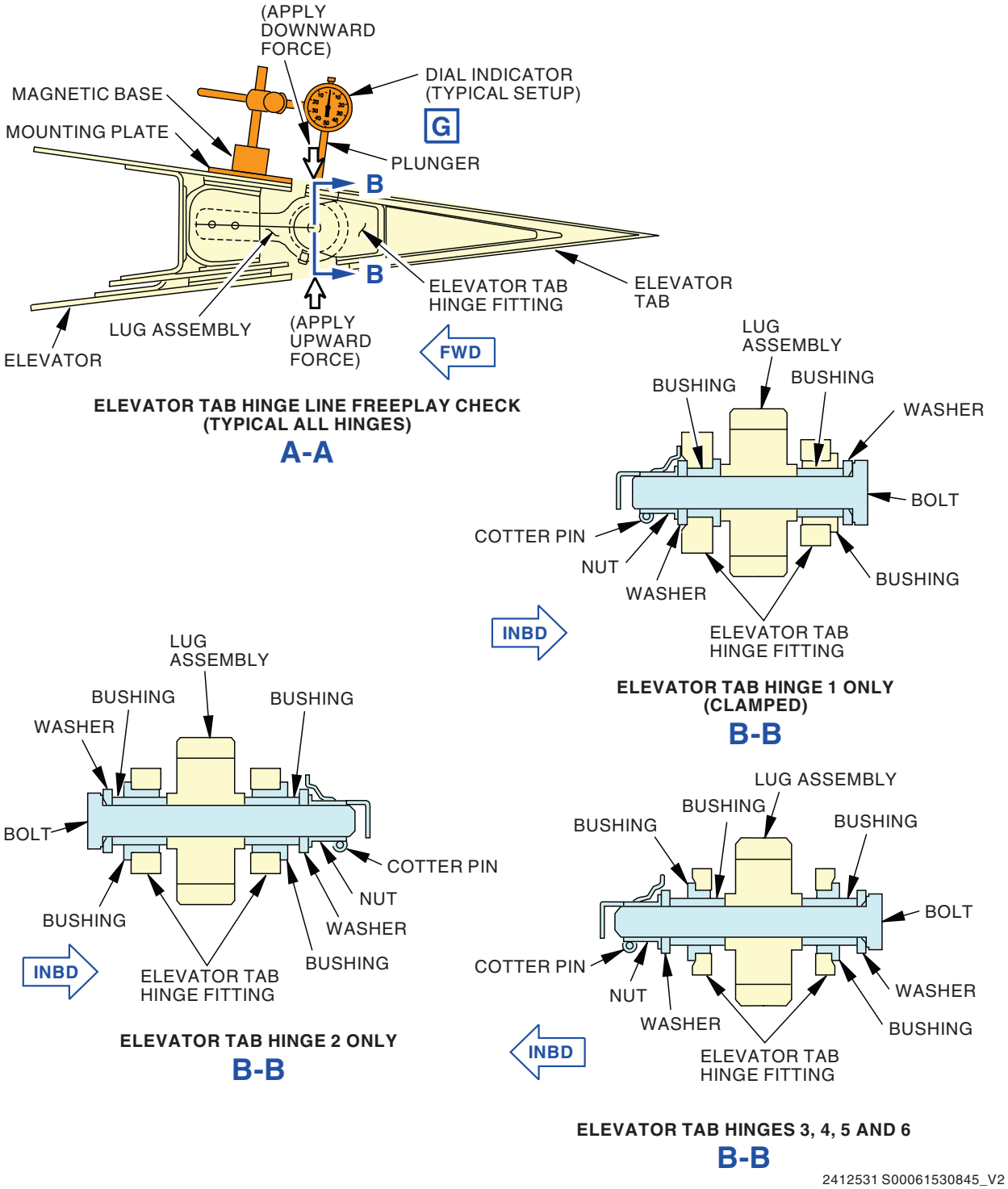
6 APPLY FORCE TO TAB WITHIN 0.1 INCH (3 mm) OF THE TRAILING EDGE OF THE TAB AND AS CLOSE AS POSSIBLE TO THE HINGE 2 CENTERLINE.

7 DIAL INDICATOR PLUNGER CENTERLINE WITHIN 0.5 INCH (13 mm) OF THE HINGE 2 CENTERLINE.

2412530 S00061530844_V1

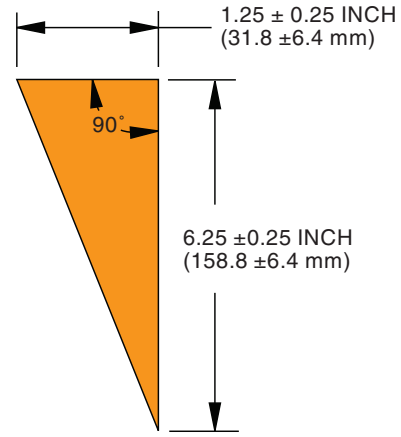
Elevator Tab Freeplay Check
Figure 602/27-31-32-990-802 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Tab Freeplay Check
Figure 602/27-31-32-990-802 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



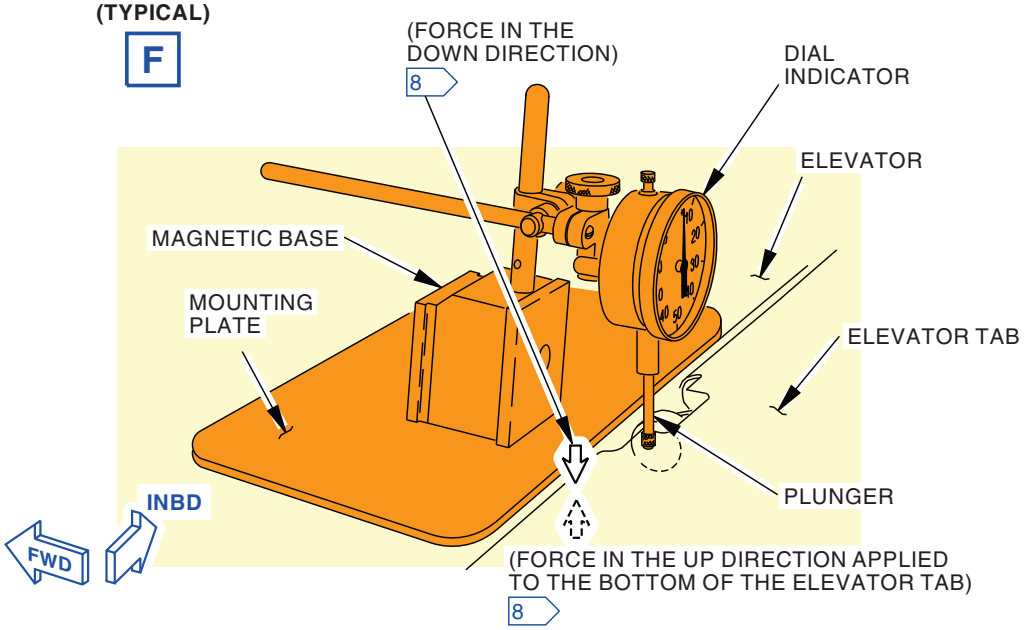
NOTE:

USE WOOD TO MAKE THE WEDGE TO THE APPROXIMATE DIMENSIONS SHOWN. COVER THE WEDGE WITH DUCT TAPE OR EQUIVALENT TO KEEP IT FROM DAMAGING THE PAINT.

APPLY WEDGE WITH HAND PRESSURE ONLY. IF YOU FORCE THE WEDGE, DAMAGE TO THE AIRPLANE STRUCTURE MAY OCCUR.

ELEVATOR WEDGE TOOL (TYPICAL)

F



ELEVATOR TAB HINGE LINE FREEPLAY CHECK (HINGES 3, 4, 5 AND 6) (FORCE IN THE UP AND DOWN DIRECTION)

G

8 APPLY THE UP AND DOWN FORCES AS CLOSE AS POSSIBLE TO THE AXIS OF THE DIAL INDICATOR PLUNGER.

2412532 S00061530846_V1

Elevator Tab Freeplay Check
Figure 602/27-31-32-990-802 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR TAB CONTROL MECHANISM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the elevator tab control mechanism

(2) An installation of the elevator tab control mechanism

(3) A removal of the tab pushrod

(4) An installation of the tab pushrod

(5) A removal of the elevator tab lock actuator

(6) An installation of the elevator tab lock actuator

(7) A removal of the elevator tab control mechanism springs

(8) An installation of the elevator tab control mechanism springs.

B. The elevator tab control mechanism is referred to as the "tab mechanism" in this procedure.

C. The tab mechanism includes the crank assembly and two tab pushrods.

D. This procedure is applicable to the left and right tab mechanisms.

TASK 27-31-34-000-801

2. Elevator Tab Control Mechanism - Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-820-801	Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment (P/B 501)
27-31-00-820-802	Elevator Control Pushrods - Adjustment (P/B 501)
27-31-00-820-805	Elevator Tab Pushrods - Adjustment (P/B 501)
27-31-41-000-801	Elevator Balance Panel - Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-10345	Rigging Bar - Elevator Tab Part #: C27083-18 Supplier: 81205
SPL-12558	Removal/Installation Equipment - Tab Mechanism Assembly, Elevator Control Part #: C27085-1 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
318	Tail Cone Compartment - Right
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
334	Left Horizontal Stabilizer - Elevator
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge
344	Right Horizontal Stabilizer - Elevator

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

E. Prepare for the Removal

SUBTASK 27-31-34-860-001

(1) Remove pressure from the elevator hydraulic systems A and B, do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-34-730-001

(2) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-34-410-001

(3) Open this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-34-410-022

(4) Open the applicable access panels:

Number	Name/Location
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-34-860-002

CAUTION MAKE SURE THAT YOU ARE CAREFUL WITH THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE SHARP EDGES ON THE TRAMMEL BAR, SPL-1677 CAN CAUSE CHAFING OR DAMAGE TO THE WIRES.

(5) Use the trammel trammel bar, SPL-1677, to set the “B” dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (TASK 27-31-00-820-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-34-700-001

(6) Measure and record the elevator tab [22] position in relation to the elevator [21].

NOTE: Measure and record the elevator tab position before you remove the tab to improve the manual reversion success rate, if the tab needs to be adjusted after installation. The elevator tab position can be measured with the rigging bar, SPL-10345. It is installed on the third elevator tab hinge (from inboard).

(a) Manually push the elevator trailing edge up.

(b) While you move the elevator trailing edge down, align the projected center of the elevator trailing edge with the index mark within 0.00 ±0.04 in. (0.0 ±1.0 mm) (TASK 27-31-00-820-802).

1) Install the rigging bar, SPL-10345 (TASK 27-31-00-820-805).

a) Install the rigging bar, SPL-10345, below the elevator and elevator tab at the third elevator tab hinge (from inboard).

<1> Make sure that the stand offs for the rigging bar, SPL-10345, rest on the skin fastener heads to prevent skin variation.

b) Make sure that the cross bar on the rigging bar, SPL-10345, touches the lower surface of the elevator trailing edge.

c) Push the rigging bar, SPL-10345, FWD until elevator stop indexes on the AFT edge of elevator near elevator rear spar.

d) Slide step gage FWD under elevator tab to obtain reading.

2) Measure and record the elevator tab [22] position.

3) Make sure that the distance between the left elevator tab [22] and rigging bar, SPL-10345, is within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm).

4) Make sure that the distance between the right elevator tab [22] and rigging bar, SPL-10345, is within 0.20 in. (5.08 mm) - 0.50 in. (12.70 mm).

SUBTASK 27-31-34-010-001

(7) Open this access door:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-34-010-020

(8) Open the applicable access panels:

(TASK 27-31-41-000-801)

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

SUBTASK 27-31-34-010-002

(9) Open the applicable access panels:

Number	Name/Location
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Elevator Tab Control Mechanism Removal

SUBTASK 27-31-34-020-001

(1) Disconnect the hydraulic line [31] from the elevator tab control mechanism [32].

SUBTASK 27-31-34-020-002

(2) Disconnect the electrical line clamp [30] from the elevator tab control mechanism [32].

SUBTASK 27-31-34-020-003

(3) Remove the elevator tab control mechanism [32], do these steps (Figure 402):

(a) Remove the screws [48] and washers [55] from the pin retainers [49].

1) Rotate the pin retainer [49] arms up.

(b) Remove the nut [52], washer [43], bushing [95], and pin assembly [40] that attach the left aft attach lug to the elevator [21].

(c) Remove the nuts [47], washers [46], spacers [54], and bolts [41] that attach the tab pushrods [29] to the elevator tab control mechanism [32].

(d) Remove the nut [52], washer [43], washer [94], and pin assembly [40] that attach the right aft attach lug to the elevator [21].

PTF ALL PRE SB 737-27-1327

(e) Remove the nut [37], washer [36], nut [35], washer [34], bushing [60], bushing [38], bolt assembly [39], and pin [61] that attach the elevator tab control mechanism [32] to the horizontal stabilizer [33].

NOTE: For ease of removal of bushing [38] use the removal/installation equipment, SPL-12558.

(f) Discard the pin [61].

PTF ALL POST SB 737-27-1327

(g) Remove the nut [37], washer [36], nut [35], washer [34], bushing [60], bushing [38], bolt assembly [39], and safety cable that attach the elevator tab control mechanism [32] to the horizontal stabilizer [33].

NOTE: For ease of removal of bushing [38] use the removal/installation equipment, SPL-12558.

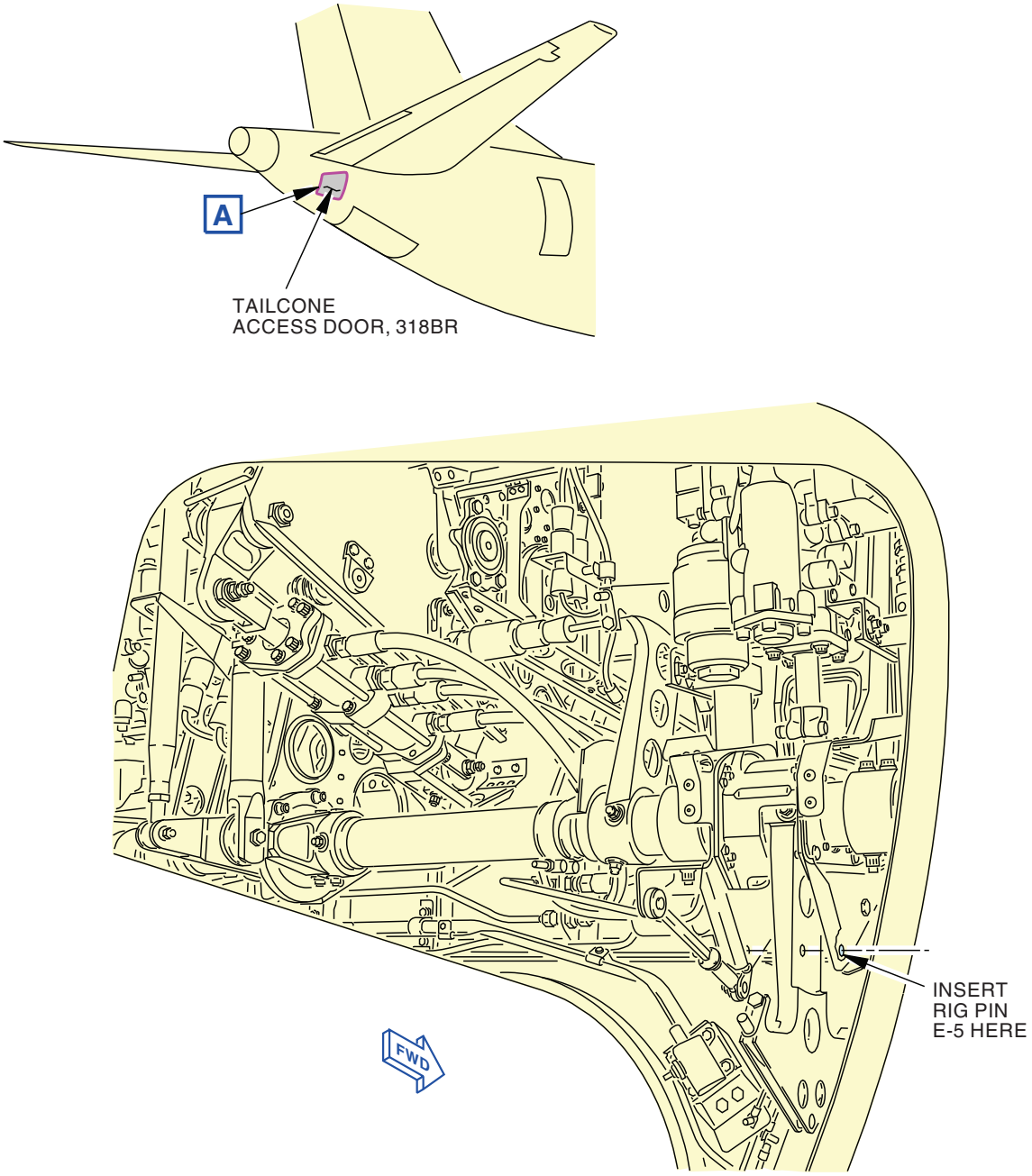
(h) Discard the safety cable.

PTF ALL

(i) Remove the elevator tab control mechanism [32] from the airplane.

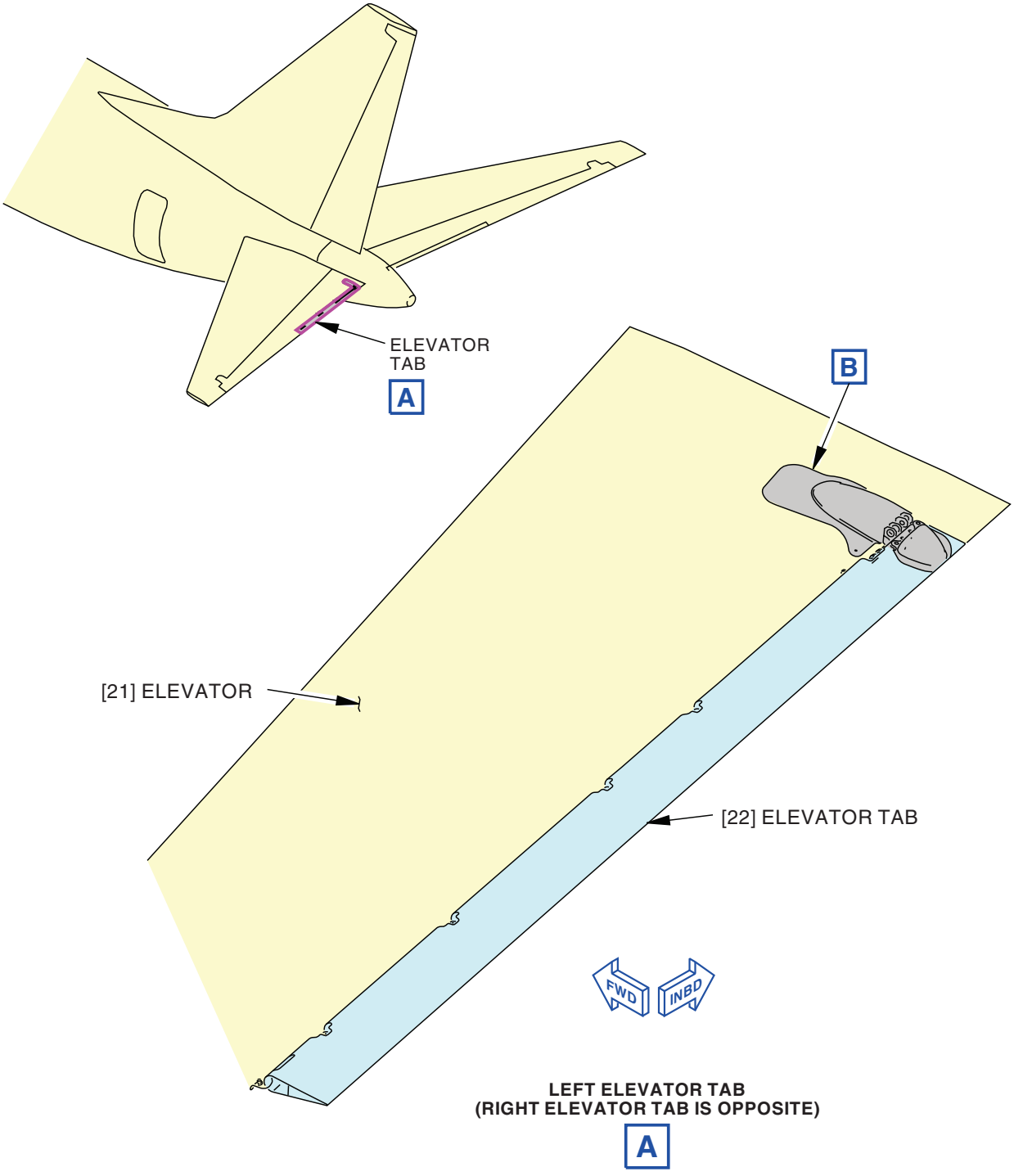
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



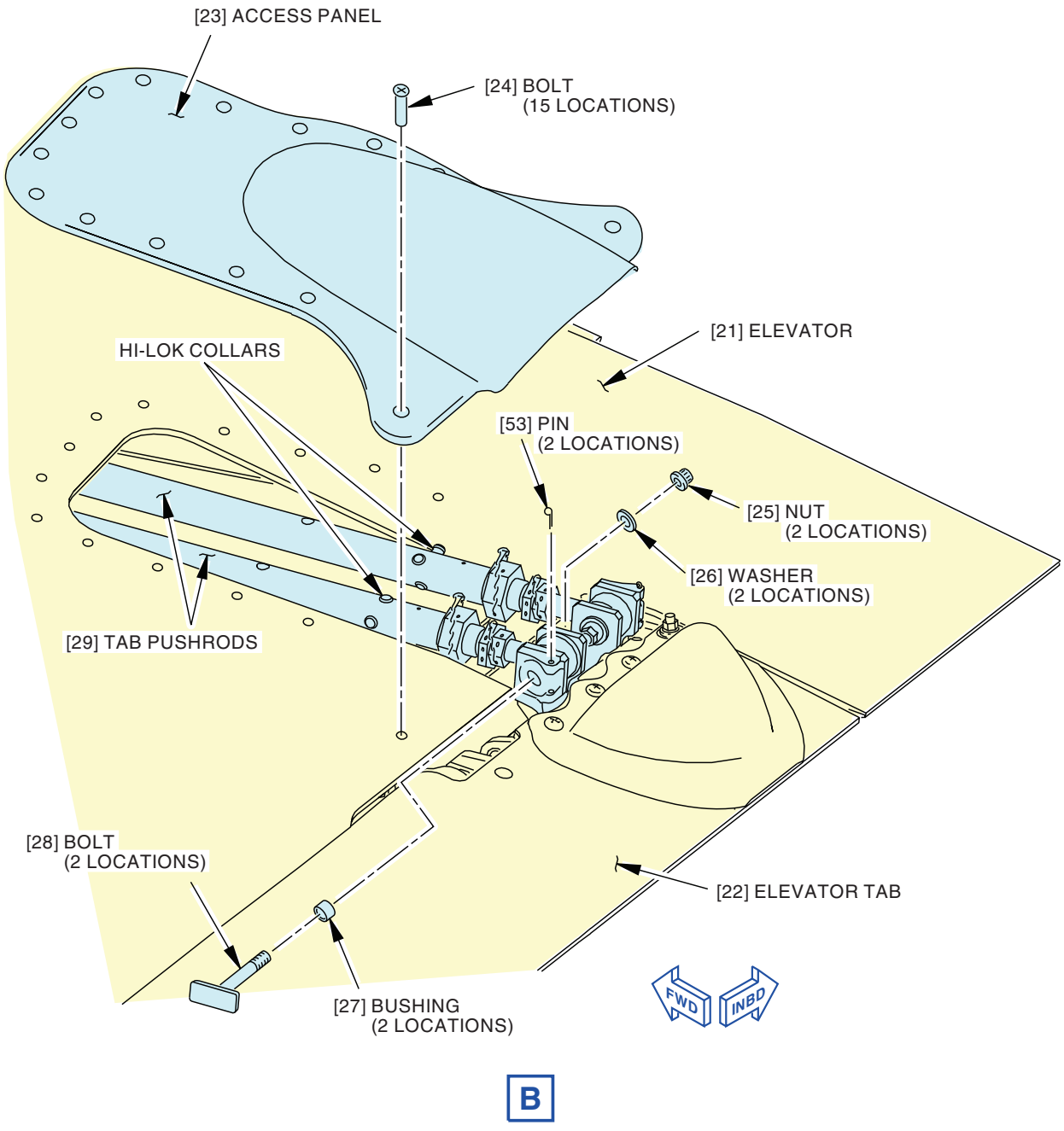
Elevator Aft Quadrant Rig Pin Installation
Figure 401/27-31-34-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



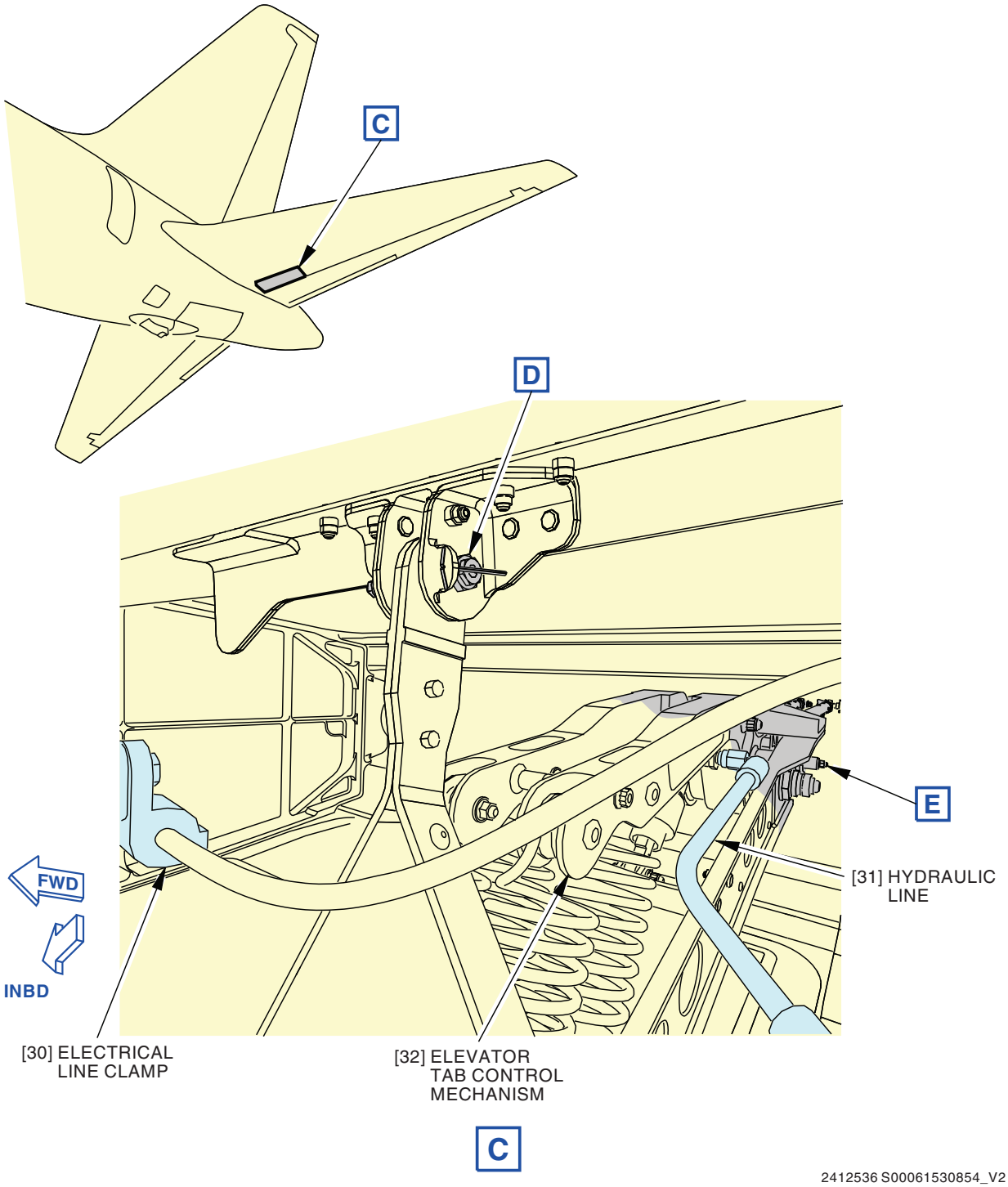
Elevator Tab Control Mechanism Installation
Figure 402/27-31-34-990-802 (Sheet 1 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



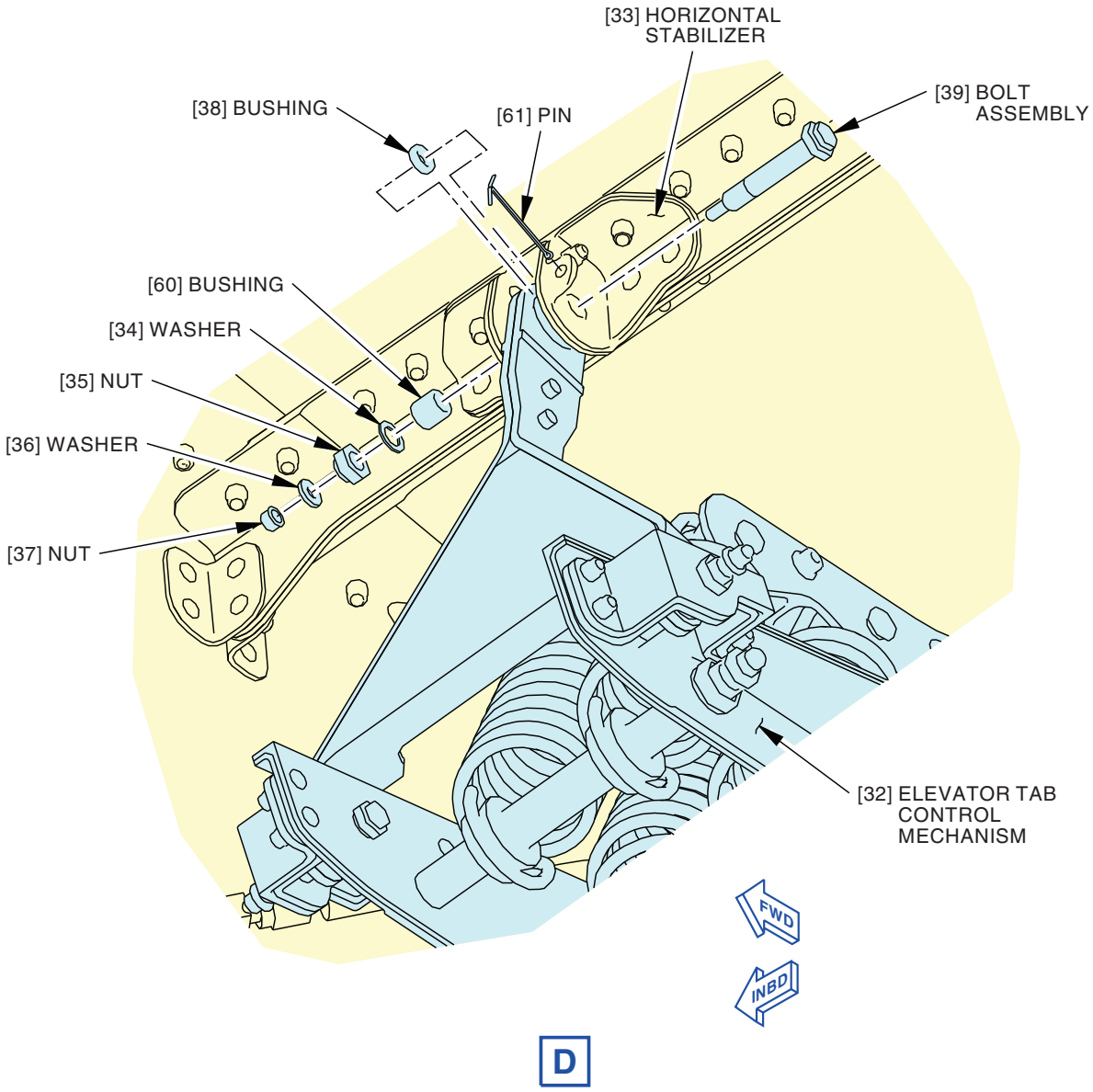
Elevator Tab Control Mechanism Installation
Figure 402/27-31-34-990-802 (Sheet 2 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Tab Control Mechanism Installation
Figure 402/27-31-34-990-802 (Sheet 3 of 9)

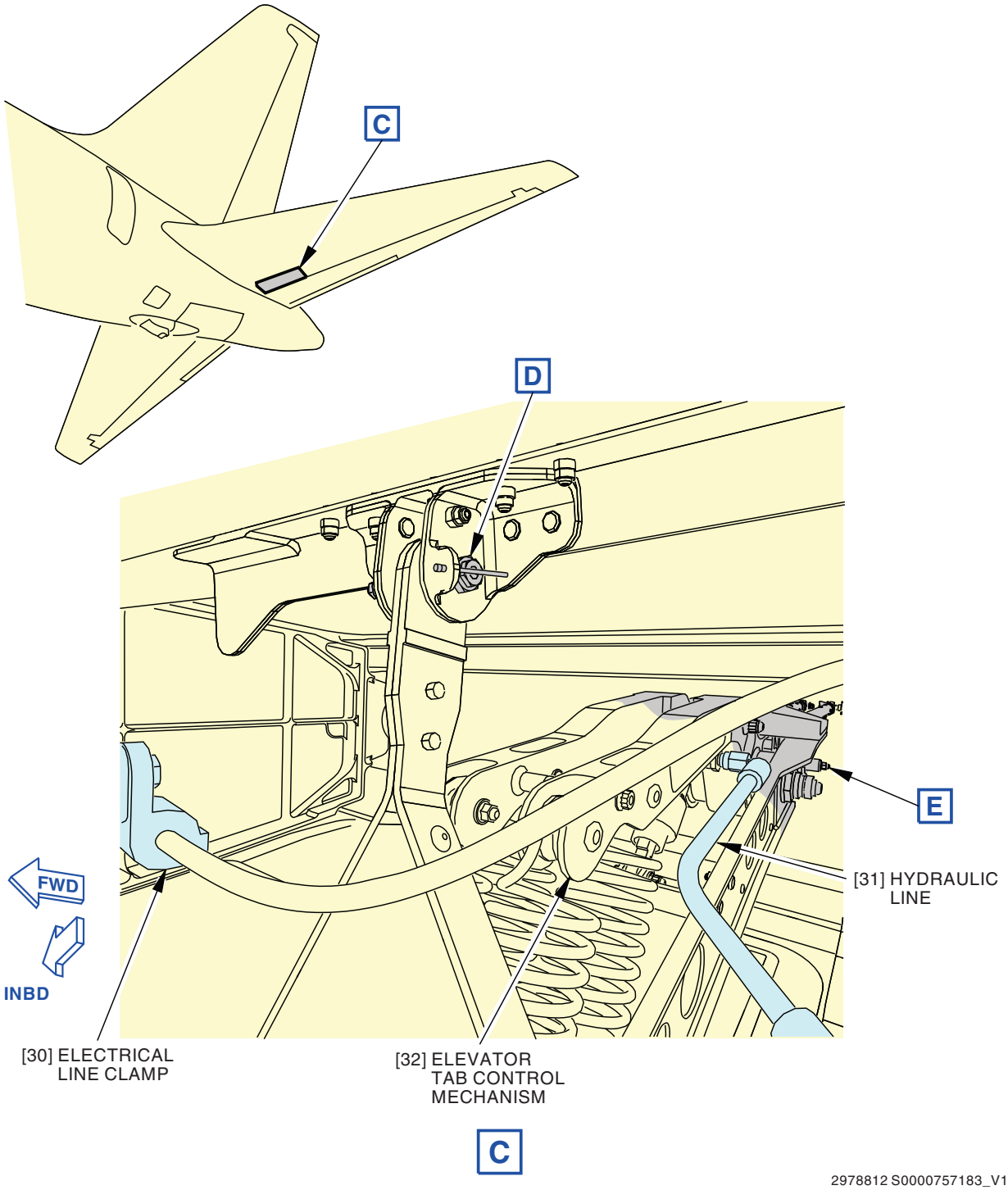
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412537 S00061530855_V2

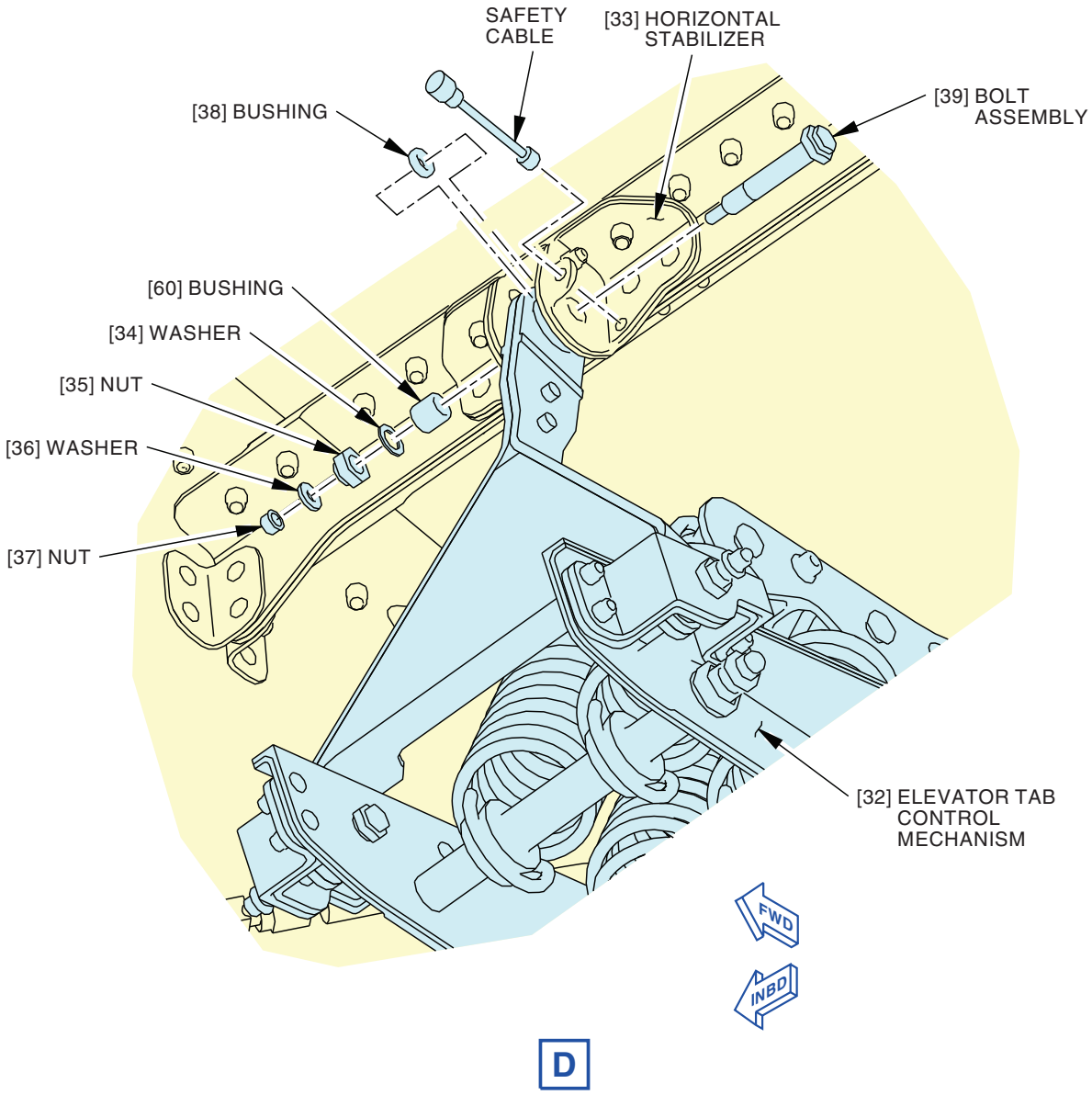
Elevator Tab Control Mechanism Installation
Figure 402/27-31-34-990-802 (Sheet 4 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Tab Control Mechanism Installation
Figure 402/27-31-34-990-802 (Sheet 5 of 9)

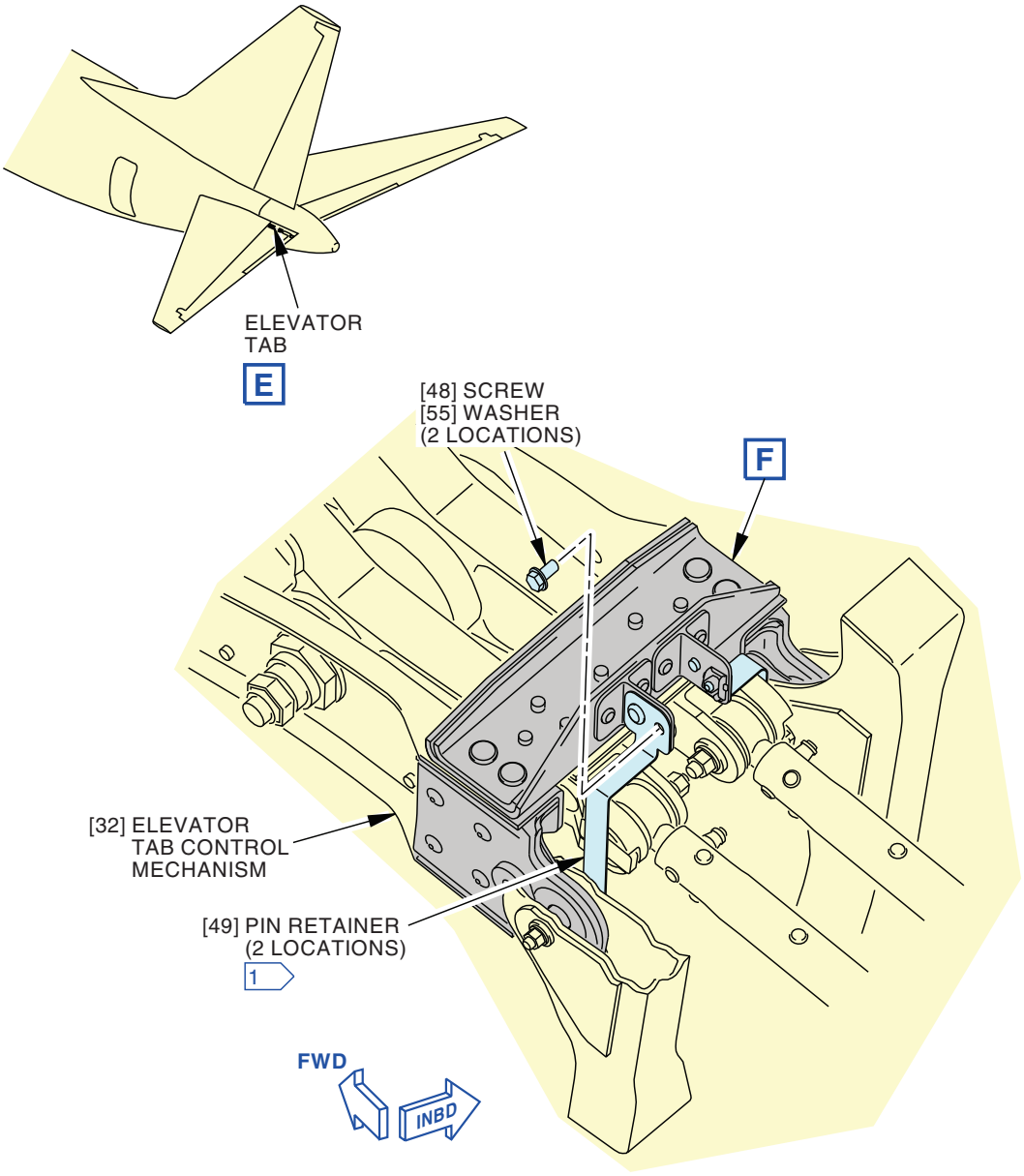
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2978841 S0000757184_V1

Elevator Tab Control Mechanism Installation
Figure 402/27-31-34-990-802 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

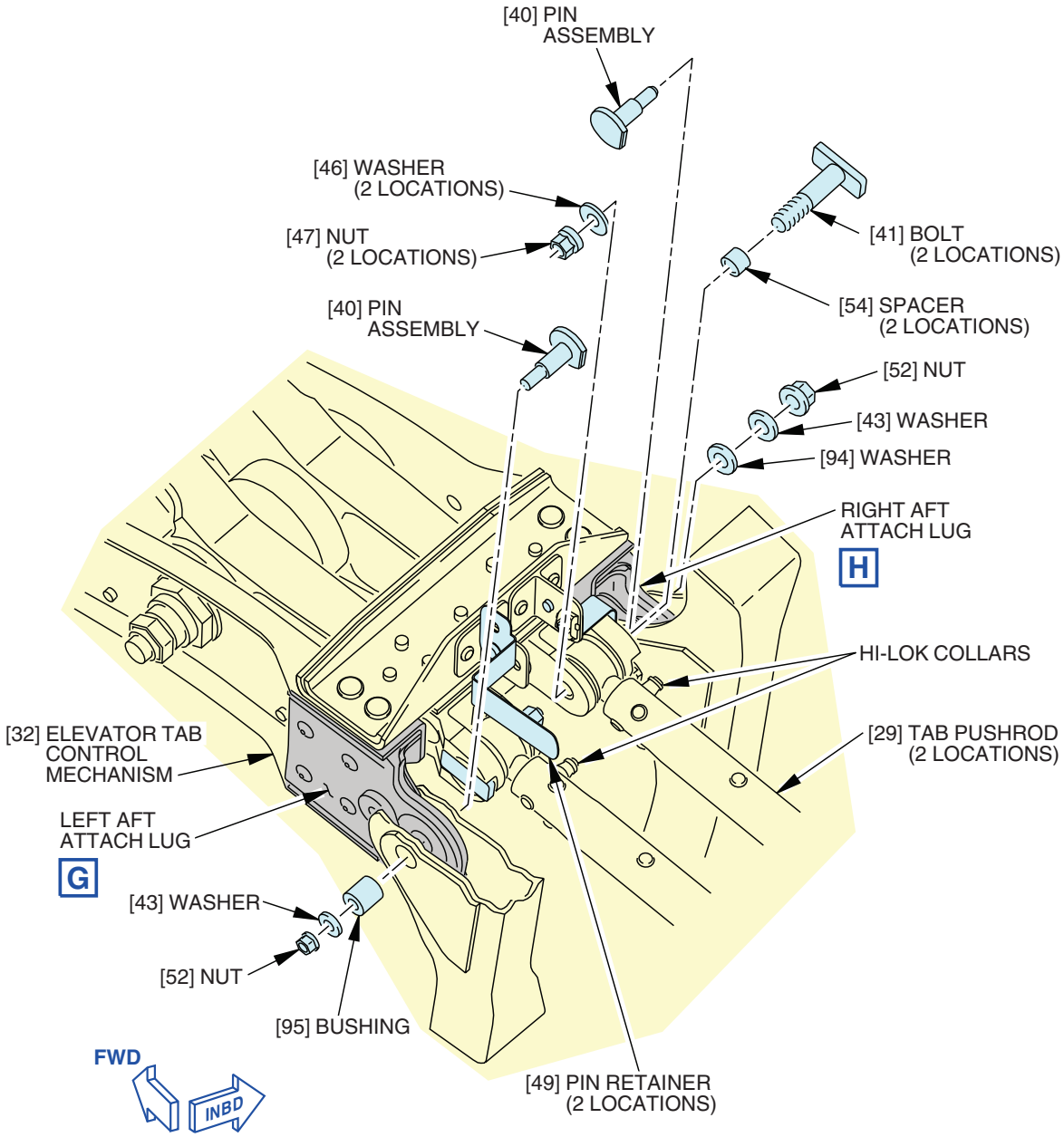


1 SCREWS AND WASHERS CAN BE INSTALLED IN DIFFERENT PLACES DEPENDING ON THE PIN RETAINER CONFIGURATIONS.

3016390 S0000789554_V1

Elevator Tab Control Mechanism Installation
Figure 402/27-31-34-990-802 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



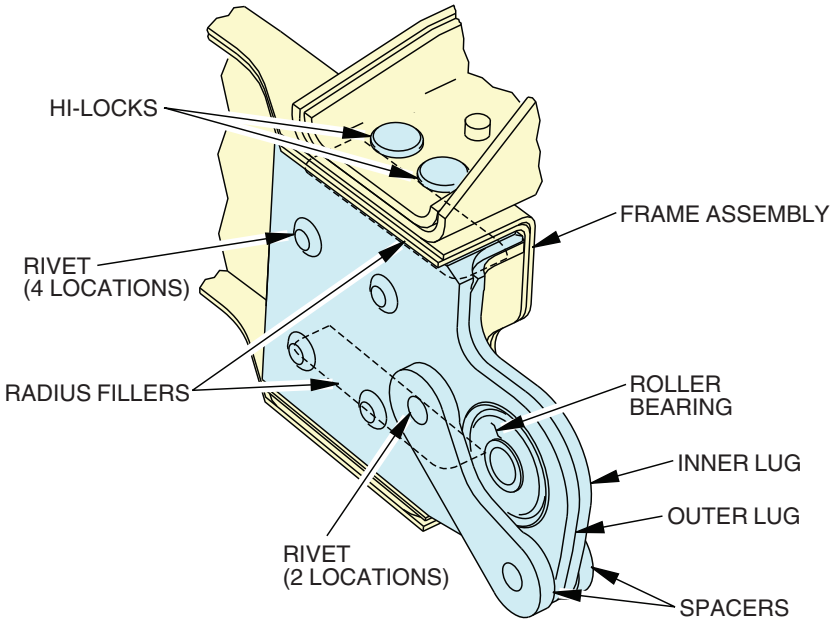
LEFT ELEVATOR TAB
(RIGHT ELEVATOR TAB IS ALMOST THE SAME)
(EXAMPLE)

F

2412539 S00061530857_V4

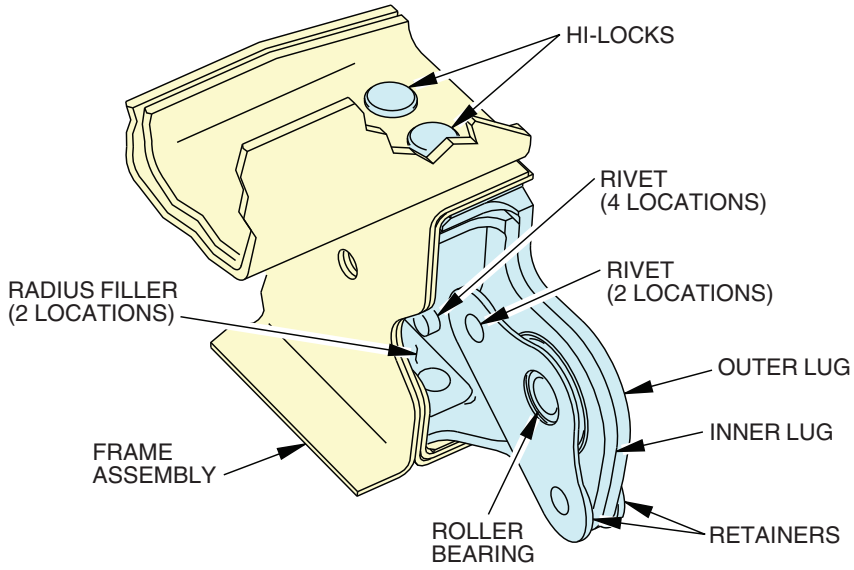
Elevator Tab Control Mechanism Installation
Figure 402/27-31-34-990-802 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT AFT ATTACH LUG

G



RIGHT AFT ATTACH LUG

H

2412541 S00061530859_V3

Elevator Tab Control Mechanism Installation
Figure 402/27-31-34-990-802 (Sheet 9 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-34-400-801

3. Elevator Tab Control Mechanism - Installation

(Figure 401, Figure 402)

A. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
27-31-00-220-801	Elevator Tab and Tab Mechanism - Detailed Visual Inspection (P/B 201)
27-31-00-700-805	Elevator Travel Limit - Test (P/B 501)
27-31-00-700-807	Elevator Tab Linkage - Test (P/B 501)
27-31-00-700-810	Control Column Power Force - Test (P/B 501)
27-31-00-710-803	Elevator Power-Off Flight Test (P/B 501)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-820-805	Elevator Tab Pushrods - Adjustment (P/B 501)
27-31-41-400-801	Elevator Balance Panel - Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-7502	Screwdriver

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
D50004	Compound - Antiseize	BMS3-28
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G51674	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 12 Inches Long	BACC13AT3K12

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
32	Elevator tab control mechanism	27-31-34-01-170	PTF ALL
		27-31-34-01-175	PTF ALL
		27-31-34-01-180	PTF ALL
		27-31-34-02-170	PTF ALL
		27-31-34-02-175	PTF ALL
		27-31-34-02-180	PTF ALL
47	Nut	27-31-34-01-085	PTF ALL
		27-31-34-02-085	PTF ALL
61	Pin	27-31-34-01-040	PTF ALL
		27-31-34-02-040	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
334	Left Horizontal Stabilizer - Elevator
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge
344	Right Horizontal Stabilizer - Elevator

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GB	Horizontal Stabilizer, Elevator Hinge Cover
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

G. Elevator Tab Control Mechanism Installation

SUBTASK 27-31-34-860-003

(1) Make sure that the hydraulic power is removed from the elevator hydraulic systems A and B (TASK 27-31-00-800-802).

SUBTASK 27-31-34-420-001

(2) Install the elevator tab control mechanism [32], do these steps (Figure 402):

(a) Install the retention clips on the aft attach lugs.

1) Make sure that the retention clips lock into their positions.

(b) Put the elevator tab control mechanism [32] in its position.

(c) Connect the tab pushrods [29] to the elevator tab control mechanism [32], do these steps:

1) Apply compound, D50004, to the threads of the bolts [41] and outer surface of the spacers [54].

2) Install the bolts [41], washers [46], spacers [54], and new nuts [47] to connect the tab pushrods [29] to the elevator tab control mechanism [32].

NOTE: The inboard bolt is installed with the head inboard and the outboard bolt is installed with the head outboard.

a) Tighten the nuts [47] to 70 in-lb (7.9 N·m) - 80 in-lb (9.0 N·m).

NOTE: If you loosened or removed the nuts, they need to be replaced and not re-installed. Using new nuts will ensure that they have the required run-on torque of 3.5 in-lb (0.4 N·m).

(d) Apply compound, D50004, to the threads of the pin assembly [40].

(e) Install the pin assembly [40] with the head at the inside of the elevator tab control mechanism [32], bushing [95], washer [43], and nut [52] to connect the left aft attach lug to the elevator [21].

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1) Use a Phillips screwdriver, STD-7502, set-up to hold the heads of the pin assembly [40] while tightening the nut [52].

2) Tighten the nut [52] to 30 in-lb (3.4 N·m) - 40 in-lb (4.5 N·m).

(f) Apply compound, D50004, to the threads of the pin assembly [40].

(g) Install the pin assembly [40] with the head at the inside of the elevator tab control mechanism [32], washer [94], washer [43], and nut [52] to connect the right aft attach lug to the elevator [21].

1) Tighten the nut [52] to 30 in-lb (3.4 N·m) - 40 in-lb (4.5 N·m).

(h) Turn the pin retainer [49] arms down and install the screws [48] and washers [55] to attach the pin retainers [49] to the elevator tab control mechanism [32].

(i) Apply compound, D50004, to the threads of the bolt assemblies [39] and the outer surface of the bushings [60].

(j) Install the bushing [38], bolt assembly [39], bushing [60], washer [34], and nut [35] to connect the elevator tab control mechanism [32] to the horizontal stabilizer [33].

NOTE: The bolts are installed with the heads at the outside of the tab mechanism.

1) Tighten the nut [35] to 50 in-lb (5.6 N·m) - 60 in-lb (6.8 N·m).

CAUTION

DO NOT TIGHTEN THE NUTS [35] AGAIN AFTER YOU TIGHTEN THE NUT [37]. YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(k) Install the washer [36] and nut [37].

1) Tighten the nut [37] to 12 in-lb (1.4 N·m) - 15 in-lb (1.7 N·m).

PTF ALL PRE SB 737-27-1327

(l) Install a new pin [61].

PTF ALL POST SB 737-27-1327

(m) Install the safety cable kit, G51674.

1) Apply corrosion inhibiting material, G50136, to both sides of the clevis assembly of the horizontal stabilizer [33] and to the cable holes.

NOTE: It is not required to completely fill the inside of the cable holes with corrosion inhibiting material, G50136.

2) Install the safety cable through the cable holes (TASK 20-10-44-400-801).

PTF ALL

(n) Connect the hydraulic line [31] to the elevator tab control mechanism [32].

(o) Tighten the B-nut on the hydraulic line [31] to 140 ±7 in-lb (16 ±1 N·m).

(p) Connect the electrical line and the electrical line clamp [30] to the elevator tab control mechanism [32].

SUBTASK 27-31-34-211-001

(3) Do a detailed inspection of the inboard and outboard aft attach lugs of the elevator tab control mechanism after installation (TASK 27-31-00-220-801).

SUBTASK 27-31-34-700-002

(4) Adjust the tab rods to the position that you measured before removing the mechanism, do this task: Elevator Tab Pushrods - Adjustment, TASK 27-31-00-820-805.

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SUBTASK 27-31-34-730-002

(5) Do this task: Elevator Travel Limit - Test, TASK 27-31-00-700-805.

SUBTASK 27-31-34-730-003

(6) Do this task: Elevator Tab Linkage - Test, TASK 27-31-00-700-807.

SUBTASK 27-31-34-730-004

(7) Do this task: Control Column Power Force - Test, TASK 27-31-00-700-810.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-34-410-002

(1) Close these access doors:

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-34-410-020

(2) Close the applicable access panels:

(TASK 27-31-41-400-801)

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

SUBTASK 27-31-34-410-003

(3) Close the applicable access panels:

Number	Name/Location
334GB	Horizontal Stabilizer, Elevator Hinge Cover
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
344GB	Horizontal Stabilizer, Hinge Cover, Elevator Station 24.09
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-34-410-021

CAUTION

TORQUE FASTENERS AFTER SEAL ADJUSTMENT AS HIGH AS POSSIBLE WITHIN A 10-15 IN-LB RANGE. TOO MUCH TORQUE CAN CAUSE DAMAGE TO THE SEAL PANELS.

(4) Close the applicable access panels:

Number	Name/Location
334AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344AB	Horizontal Stabilizer, Seal, Trailing Edge to Elevator

SUBTASK 27-31-34-390-001

(5) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

SUBTASK 27-31-34-730-005

(6) Do this task: Elevator Power-Off Flight Test, TASK 27-31-00-710-803.

END OF TASK

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TASK 27-31-34-000-802

4. Elevator Tab Pushrod - Removal

(Figure 401, Figure 402)

A. General

(1) This task is applicable for the two left elevator tab and two right elevator tab pushrods.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-820-801	Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment (P/B 501)
27-31-00-820-802	Elevator Control Pushrods - Adjustment (P/B 501)
27-31-00-820-805	Elevator Tab Pushrods - Adjustment (P/B 501)
27-31-41-000-801	Elevator Balance Panel - Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-10345	Rigging Bar - Elevator Tab Part #: C27083-18 Supplier: 81205
SPL-11021	Fixture Eqpt - Pushrod Length Adjustment, Elevator Tab & Aileron Part #: C27082-1 Supplier: 81205
SPL-12558	Removal/Installation Equipment - Tab Mechanism Assembly, Elevator Control Part #: C27085-1 Supplier: 81205

D. Location Zones

Zone	Area
318	Tail Cone Compartment - Right
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
334	Left Horizontal Stabilizer - Elevator
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
334AT	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GT	Horizontal Stabilizer, Elevator Hinge Cover

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(Continued)

Number	Name/Location
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
344AT	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GT	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 39.02
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

F. Prepare for the Removal

SUBTASK 27-31-34-860-004

(1) Remove pressure from the elevator hydraulic systems A and B, do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-34-010-004

(2) Open these access doors:

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-34-010-005

(3) Open the applicable access panels:

(TASK 27-31-41-000-801)

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

SUBTASK 27-31-34-010-007

(4) Open the applicable access panels:

Number	Name/Location
334AT	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GT	Horizontal Stabilizer, Elevator Hinge Cover
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
344AT	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GT	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 39.02
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-34-860-005

(5) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-34-860-006

CAUTION MAKE SURE THAT YOU ARE CAREFUL WITH THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE SHARP EDGES ON THE TRAMMEL BAR, SPL-1677 CAN CAUSE CHAFING OR DAMAGE TO THE WIRES.

(6) Use the trammel bar, SPL-1677, to set the "B" dimension at 4 units of trim (39.17 ±0.01 in. (994.92 ±0.26 mm)) (TASK 27-31-00-820-801).

SUBTASK 27-31-34-860-007

(7) Manually push the elevator trailing edge full up.

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SUBTASK 27-31-34-860-008

(8) While you move the elevator trailing edge down, align the center of the elevator trailing edge with the index mark to 0.00 ±0.04 in. (0.0 ±1.0 mm), or less (TASK 27-31-00-820-802).

NOTE: Keep the elevators at this position while you measure and adjust the tab pushrods. If the elevator position is changed, you must put the elevator trailing edge at the full up position and do this step again before you continue.

SUBTASK 27-31-34-480-001

(9) Install the rigging bar, SPL-10345 (TASK 27-31-00-820-805).

(a) Install the rigging bar, SPL-10345, below the elevator and elevator tab at the third elevator tab hinge (from inboard).

1) Make sure that the stand offs for the rigging bar, SPL-10345, rest on the skin fastener heads to prevent skin variation.

(b) Make sure that the cross bar on the rigging bar, SPL-10345, touches the lower surface of the elevator trailing edge.

(c) Push the rigging bar, SPL-10345, FWD until elevator stop indexes on the AFT edge of elevator near elevator rear spar.

(d) Slide step gage FWD under elevator tab to obtain reading.

(e) Record the elevator tab dimension measured from the elevator tab faired/neutral position to the rigged up position.

NOTE: Adjusting the elevator tab back to this recorded position may reduce the need for a second flight test.

G. Tab Pushrod Removal

SUBTASK 27-31-34-020-004

(1) Remove the bolts [24] that attach the access panel [23] to the elevator [21].

SUBTASK 27-31-34-020-005

(2) Remove the tab pushrod [29], do these steps:

(a) Remove the screws [48] and washers [55] from the pin retainer [49].

(b) Remove the pin [53], nut [25], washer [26], bushing [27], and bolt [28] from the tab pushrod [29] and elevator tab [22].

1) Discard the pin [53].

(c) Remove the upper access panels to disconnect the elevator tab control mechanism [32] from the elevator [21], do these steps:

1) Remove the nut [52], washer [43], bushing [95], and pin assembly [40] that attach the left aft attach lug to the elevator [21].

NOTE: For ease of removal of pin assembly use removal/installation equipment, SPL-12558.

2) Remove the nut [52], washer [43], washer [94], and pin assembly [40] that attach the right aft attach lug to the elevator [21].

NOTE: For ease of removal of pin assembly use removal/installation equipment, SPL-12558.

(d) Remove the nut [47], washer [46], spacer [54], and bolt [41] from the elevator tab control mechanism [32] and tab pushrod [29].

(e) Disconnect the tab pushrod [29] from the elevator tab control mechanism [32].

(f) Remove the tab pushrod [29] from the airplane.

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SUBTASK 27-31-34-420-002

(3) Use the fixture, SPL-11021, to assist in measuring and replacing the new or replacement tab pushrods [29] to match the removed old tab pushrods [29] (TASK 27-31-00-820-805).

NOTE: Measuring the new or replacement tab pushrod using fixture, SPL-11021 is optional. The tool is used to adjust the new or replacement tab rods off the airplane to the same length as the removed tab rods. This adjustment will reduce the chance that a second flight test will be required.

NOTE: Use the previously recorded elevator tab dimension obtained using (TASK 27-31-00-820-805).

(a) Make sure that no changes to the recorded elevator tab dimensions are made after removal.

(b) Install the old removed tab pushrods [29] onto the fixture, SPL-11021.

(c) Set the digital scales to zero.

(d) Remove the old tab pushrods [29].

(e) Install the new or replacement tab pushrods [29] onto the fixture, SPL-11021.

NOTE: The inboard and outboard pushrod must be identified prior to adjustment. The slot in the retainer that attaches the pushrod is on the inboard for the RH elevator and outboard for the LH elevator.

(f) Record readings from the scale.

(g) Adjust the new or replacement tab pushrods [29] by coarse or fine method until scale is back to 0.0 ± 0.0005 inches (TASK 27-31-00-820-805).

NOTE: The fine adjustments may be made with the pushrod installed on the tool. The coarse adjustments must be made with the pushrods removed from the tool.

(h) Tighten the checknut to prevent movement of the rod end prior to installation.

NOTE: Do not lockwire the checknuts until final adjustment is done on the airplane.

(i) Remove the new or replacement tab pushrods [29] from fixture, SPL-11021.

(j) Using the information from the previously recorded old elevator tab dimension, tag the new tab pushrods [29] with the same information.

(k) Install the new or replacement tab pushrods [29] onto the fixture, SPL-11021.

(l) Make sure that the scales reads 0.0 ± 0.0005 inches.

(m) Remove the tab pushrods [29].

1) Make sure that no adjustment is made to tab pushrods [29] when removing from fixture, SPL-11021.

(n) Tighten the checknuts on the tab pushrods [29] (TASK 27-31-00-820-805).

(o) Install the new or replacement tab pushrods [29] onto the airplane (TASK 27-31-34-400-802).

(p) Make sure that the tab position is the same as the tab dimension first obtained from the old tab pushrods rigging bar, SPL-10345, readings within ± 0.01 inches.

NOTE: If required, adjust the tab pushrods on the airplane.

END OF TASK

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TASK 27-31-34-400-802

5. Elevator Tab Pushrod - Installation

(Figure 401, Figure 403)

A. References

Reference	Title
27-31-00-700-805	Elevator Travel Limit - Test (P/B 501)
27-31-00-700-810	Control Column Power Force - Test (P/B 501)
27-31-00-710-803	Elevator Power-Off Flight Test (P/B 501)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-820-805	Elevator Tab Pushrods - Adjustment (P/B 501)
27-31-41-400-801	Elevator Balance Panel - Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-7502	Screwdriver

C. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
D50004	Compound - Antiseize	BMS3-28

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
25	Nut	27-31-34-01-085	PTF ALL
		27-31-34-02-085	PTF ALL
29	Tab pushrod	27-31-34-01-120	PTF ALL
		27-31-34-02-120	PTF ALL
47	Nut	27-31-34-01-085	PTF ALL
		27-31-34-02-085	PTF ALL
53	Pin	27-31-34-01-035	PTF ALL
		27-31-34-02-035	PTF ALL

E. Location Zones

Zone	Area
318	Tail Cone Compartment - Right
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
334	Left Horizontal Stabilizer - Elevator
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge
344	Right Horizontal Stabilizer - Elevator

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
334AT	Horizontal Stabilizer, Seal, Trailing Edge to Elevator

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(Continued)

Number	Name/Location
334GT	Horizontal Stabilizer, Elevator Hinge Cover
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
344AT	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GT	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 39.02
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

G. Tab Pushrod Installation

SUBTASK 27-31-34-860-009

(1) Make sure that the hydraulic power is removed from the elevator hydraulic systems A and B (TASK 27-31-00-800-802).

SUBTASK 27-31-34-420-003

(2) Install the tab pushrods [29], do these steps (Figure 402):

(a) Put the tab pushrods [29] in its position.

1) Make sure that the orientation of the tab pushrods [29] is correct.

a) For the left tab pushrod [29], the HI-LOK collars must be in the 3 o'clock and 6 o'clock positions at each end. For the right tab pushrod [29], the HI-LOK collars must be in the 2 o'clock and 5 o'clock positions at each end. This orientation is applicable to the left and right elevator tab mechanisms.

2) If you loosened or removed the nuts [47] or nuts [25], replace them with new nuts [47] or nuts [25].

NOTE: New nuts will make sure that they have the required run-on torque.

(b) Install the elevator tab control mechanism [32] to the elevator [21], do these steps:

1) Apply compound, D50004, to the threads of the pin assembly [40].

2) Install the pin assembly [40] with the head at the inside of the elevator tab control mechanism [32], bushing [95], washer [43], and nut [52] to connect the left aft attach lug to the to elevator [21].

a) Use a Phillips screwdriver, STD-7502, set-up to hold the heads of the pin assembly [40] while tightening the nut [52].

b) Tighten the nut [52] to 30 in-lb (3.4 N·m) - 40 in-lb (4.5 N·m).

3) Apply compound, D50004, to the threads of the pin assembly [40].

4) Install the pin assembly [40] with the head at the inside of the elevator tab control mechanism [32], washer [94], washer [43], and nut [52] to connect the right aft attach lug to the elevator [21].

a) Tighten the nut [52] to 30 in-lb (3.4 N·m) - 40 in-lb (4.5 N·m).

(c) Apply compound, D50004, to the threads of the bolts [41] and outer surface of the spacers [54].

(d) Install the bolts [41], spacers [54], washers [46], and new nuts [47] to connect the tab pushrods [29] to the elevator tab control mechanism [32].

NOTE: The inboard bolt is installed with the head inboard and the outboard bolt is installed with the head outboard.

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CAUTION

IF THE NEW NUTS [47] ARE NOT TIGHTENED CORRECTLY, THEN THE PUSHRODS CAN BECOME DISCONNECTED AND CAUSE DAMAGE TO THE ELEVATOR AND STABILIZER.

(e) Tighten the nuts [47] to 70 in-lb (7.9 N·m) - 80 in-lb (9.0 N·m).

(f) Install the screws [48] and washers [55] to attach the pin retainers [49] to the elevator tab control mechanism [32].

(g) Apply compound, D50004, to the threads of the bolts [28] and outer surface of the bushings [27].

(h) Install the bolts [28], bushings [27], washers [26], and new nuts [25] to connect the tab pushrods [29] to the elevator tab [22].

NOTE: The inboard bolt is installed with the head inboard and the outboard bolt is installed with the head outboard.

CAUTION

IF THE NEW NUTS [25] ARE NOT TIGHTENED CORRECTLY, THEN THE PUSHRODS CAN BECOME DISCONNECTED AND CAUSE DAMAGE TO THE ELEVATOR AND STABILIZER.

(i) Tighten the nuts [25] to 70 in-lb (7.9 N·m) - 80 in-lb (9.0 N·m).

1) Install a new pins [53] in the anti-rotation boss at the head of the bolt [28].

SUBTASK 27-31-34-820-001

(3) Adjust the tab pushrods [29] until you get the same tab position that you measured before you removed the elevator tab pushrods [29], do this task: Elevator Tab Pushrods - Adjustment, TASK 27-31-00-820-805.

SUBTASK 27-31-34-420-004

(4) Install the bolts [24] to attach the access panel [23] to the elevator [21].

SUBTASK 27-31-34-730-006

(5) Do this task: Elevator Travel Limit - Test, TASK 27-31-00-700-805.

SUBTASK 27-31-34-730-007

(6) Do this task: Control Column Power Force - Test, TASK 27-31-00-700-810.

SUBTASK 27-31-34-730-008

(7) Do this task: Elevator Power-Off Flight Test, TASK 27-31-00-710-803.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-34-410-008

(1) Close the applicable access panels:

Number	Name/Location
334AT	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
334GT	Horizontal Stabilizer, Elevator Hinge Cover
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
344AT	Horizontal Stabilizer, Seal, Trailing Edge to Elevator
344GT	Horizontal Stabilizer, Elevator Hinge Cover, Elevator Sta 39.02
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

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SUBTASK 27-31-34-410-011

(2) Close the applicable access panels:

(TASK 27-31-41-400-801)

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

SUBTASK 27-31-34-410-013

(3) Close these access doors:

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-34-390-002

(4) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

SUBTASK 27-31-34-730-009

(5) Do this task: Elevator Power-Off Flight Test, TASK 27-31-00-710-803.

——— END OF TASK ———

TASK 27-31-34-000-803

6. Elevator Tab Lock Actuator - Removal

(Figure 401, Figure 402, Figure 403)

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-820-805	Elevator Tab Pushrods - Adjustment (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
318	Tail Cone Compartment - Right
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
334	Left Horizontal Stabilizer - Elevator
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge
344	Right Horizontal Stabilizer - Elevator

D. Access Panels

Number	Name/Location
318BR	Tailcone Access Door

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(Continued)

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

E. Prepare for the Removal

SUBTASK 27-31-34-860-010

(1) Remove pressure from the elevator hydraulic systems A and B, do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-34-010-010

(2) Open this access door:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-34-010-011

(3) Open the applicable access panels:

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

SUBTASK 27-31-34-480-002

(4) Install the rig pin E-5 from rigging pin kit, SPL-1585, in the aft control quadrant (Figure 401).

SUBTASK 27-31-34-220-001

(5) If the adjustment of the elevator tab pushrods is required, do these steps:

(a) Measure the elevator tab position.

(b) Keep a record of the elevator tab position.

(c) Do this task: Elevator Tab Pushrods - Adjustment, TASK 27-31-00-820-805.

F. Tab Lock Actuator Removal

SUBTASK 27-31-34-020-006

(1) Disconnect the hydraulic line [31] from the actuator [78] and install caps and plugs on the hydraulic line and port (Figure 402).

(a) If necessary, remove the union [96] and packing [97].

1) Discard the old packing [97].

NOTE: The spring must be at the neutral position when the hydraulic line is disconnected.

SUBTASK 27-31-34-020-007

(2) Remove the actuator [78]:

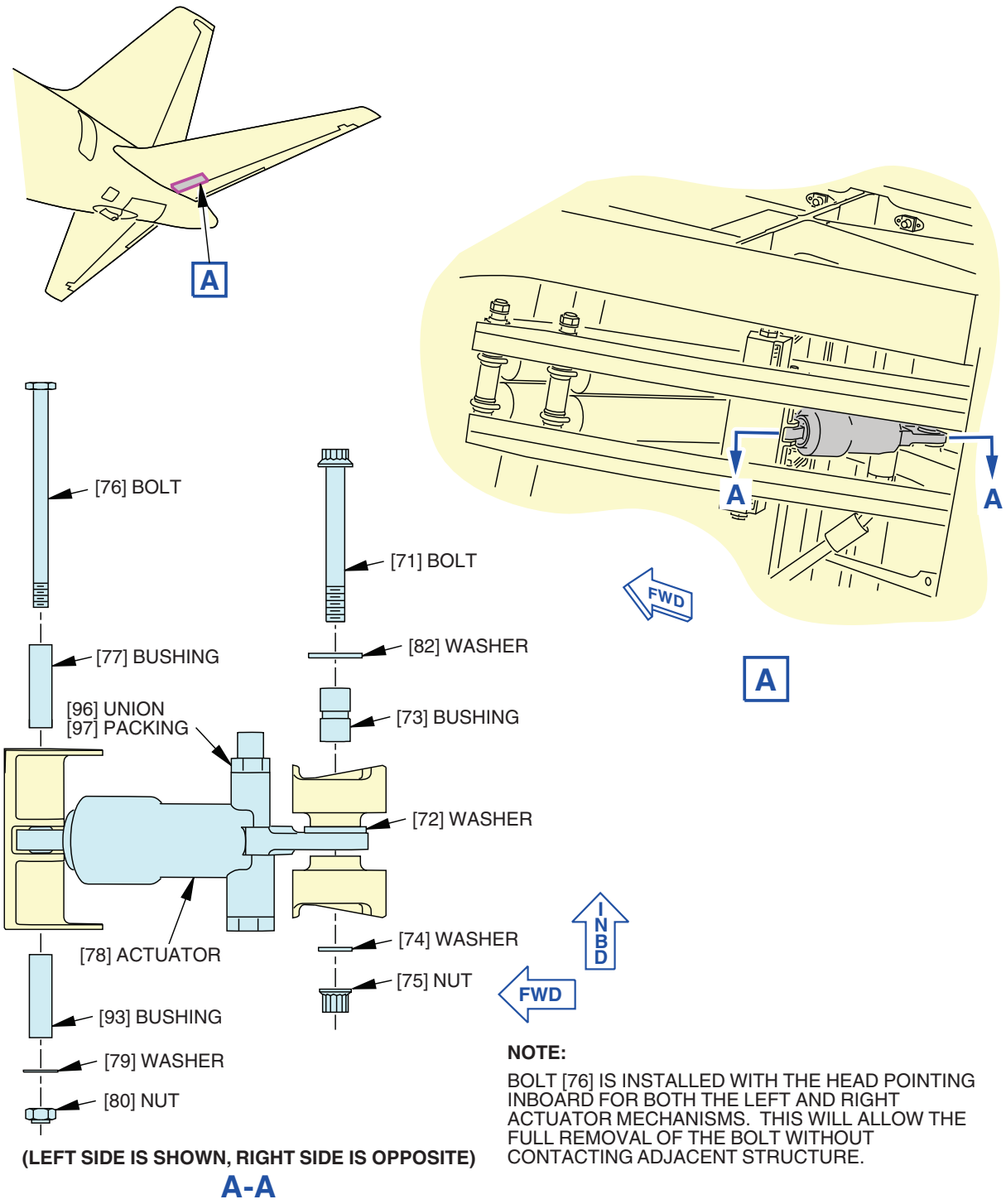
(a) Remove the nut [80], washer [79], bushing [93], bushing [77], and bolt [76] from the actuator [78].

(b) Remove the nut [75], washer [74], washer [72], bushing [73], washer [82], and bolt [71] from the actuator [78].

(c) Remove the actuator [78] from the airplane.

END OF TASK

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(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A-A

NOTE:

BOLT [76] IS INSTALLED WITH THE HEAD POINTING INBOARD FOR BOTH THE LEFT AND RIGHT ACTUATOR MECHANISMS. THIS WILL ALLOW THE FULL REMOVAL OF THE BOLT WITHOUT CONTACTING ADJACENT STRUCTURE.

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Elevator Tab Lock Actuator Installation
Figure 403/27-31-34-990-803

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TASK 27-31-34-400-803

7. Elevator Tab Lock Actuator - Installation

(Figure 401, Figure 402, Figure 403)

A. References

Reference	Title
27-31-00-700-802	Elevator Tab Control System - Operational Test (P/B 501)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D50122	Fluid - Hydraulic	BMS3-11 Type IV or Type V

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
78	Actuator	27-31-34-03-065	PTF ALL
97	Packing	27-31-34-03-075	PTF ALL

E. Location Zones

Zone	Area
318	Tail Cone Compartment - Right
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
334	Left Horizontal Stabilizer - Elevator
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge
344	Right Horizontal Stabilizer - Elevator

F. Access Panels

Number	Name/Location
318BR	Tailcone Access Door
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

G. Tab Mechanism Installation

SUBTASK 27-31-34-860-011

(1) Make sure that the hydraulic power is removed from the elevator hydraulic systems A and B (Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802).

SUBTASK 27-31-34-420-005

(2) Install the actuator [78]:

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(a) Install the bolt [71], washer [82], bushing [73], washer [72], washer [74], and nut [75].

(b) Install the bolt [76], bushing [77], bushing [93], washer [79], and nut [80].

NOTE: The bolt [76] is installed with the head pointing inboard on both the left and right actuator mechanisms. This will allow the full removal of the bolt from the actuator mechanism without it contacting the adjacent structure.

SUBTASK 27-31-34-420-006

(3) Connect the hydraulic line:

(a) If removed, install the union [96].

1) Apply hydraulic fluid, D50122, or MCS 352B fluid, D00054, to the new packing [97] and threads of the union [96].

2) Tighten the union [96] to 110 ±6 in-lb (12 ±1 N·m).

(b) Remove the caps and plugs from the hydraulic and electrical lines.

(c) Connect hydraulic line [31] from the elevator tab control mechanism [32].

(d) Tighten the B-nut on the hydraulic line [31] to 140 ±7 in-lb (16 ±1 N·m).

SUBTASK 27-31-34-080-001

(4) Remove rig pin E-5 of the rigging pin kit, SPL-1585, from the aft control quadrant (Figure 401).

SUBTASK 27-31-34-730-010

(5) Do this task: Elevator Tab Control System - Operational Test, TASK 27-31-00-700-802.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-34-410-014

(1) Close the applicable access panels:

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

SUBTASK 27-31-34-410-016

(2) Close this access door:

Number	Name/Location
318BR	Tailcone Access Door

——— END OF TASK ———

TASK 27-31-34-000-804

8. Elevator Tab Control Mechanism Springs - Removal

Figure 404

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-41-000-801	Elevator Balance Panel - Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-6048	Wrench - Socket
SPL-13742	Removal/Installation Equipment - Spring, Tab Mechanism Assembly, Elevator Control Part #: C27086-27 Supplier: 81205

C. Location Zones

Zone	Area
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
334	Left Horizontal Stabilizer - Elevator
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge
344	Right Horizontal Stabilizer - Elevator

D. Access Panels

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

E. Prepare for the Removal

SUBTASK 27-31-34-860-012

(1) Remove pressure from the elevator hydraulic systems A and B. To remove it, do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-34-010-013

(2) Open the applicable access panels:

(TASK 27-31-41-000-801)

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

F. Elevator Tab Control Mechanism Spring Removal

SUBTASK 27-31-34-020-008

(1) Do these steps to remove one or both of the aft springs [83]: (Figure 404)

(a) Install the hook assembly and the tool handle from the removal/installation equipment, SPL-13742 on the short fitting assembly from the removal/installation equipment, SPL-13742 in the correct position for the aft springs [83] to be removed.

1) If both aft springs [83] are installed, attach the hook assembly on the center of the aft bushing [85].

2) If one of the aft springs [83] is missing or broken, attach the hook assembly on the aft bushing [85] around the remaining aft spring [83].

(b) Install the removal/installation equipment, SPL-13742 on the elevator control tab mechanism assembly.

(c) Put a socket wrench, COM-6048 or equivalent on the bolt [87].

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WARNING DO NOT TIGHTEN THE TOOL HANDLE TOOL MUCH. THIS CAN CAUSE DAMAGE TO THE TOOL, AIRPLANE STRUCTURE OR INJURY TO PERSONNEL.

(d) At the same time, slowly turn the tool handle and the socket wrench, COM-6048 until the bolt [87] starts to come loose.

1) Make sure that you do not tighten the tool handle too much, it should take approximately 1/4 turn to loosen the bolt [87].

(e) After the bolt [87] is loose, remove the nut [92], washer [91], bolt [87], nut [90], washer [89], and bolt [88].

(f) Loosen the tool handle.

(g) Remove one or both of the aft springs [83] as necessary.

SUBTASK 27-31-34-020-009

(2) Do these steps to remove one or both of the forward springs [84]: (Figure 404)

(a) Install the hook assembly and the tool handle from the removal/installation equipment, SPL-13742 on the long fitting assembly from the removal/installation equipment, SPL-13742 in the correct position for the forward springs [84] to be removed.

1) If both forward springs [84] are installed, attach the hook assembly on the center of the foward bushing [86].

2) If one of the forward springs [84] is missing or broken, attach the hook assembly on the foward bushing [86] around the remaining forward spring [84].

(b) Install the removal/installation equipment, SPL-13742 on the elevator control tab mechanism assembly.

(c) Put a socket wrench, COM-6048 or equivalent on the bolt [87].

WARNING DO NOT TIGHTEN THE TOOL HANDLE TOOL MUCH. THIS CAN CAUSE DAMAGE TO THE TOOL, AIRPLANE STRUCTURE OR INJURY TO PERSONNEL.

(d) At the same time, slowly turn the tool handle and the socket wrench, COM-6048 until the bolt [87] starts to come loose.

1) Make sure you do not tighten the tool handle too much, it should take approximately 1/4 turn to loosen the bolt [87].

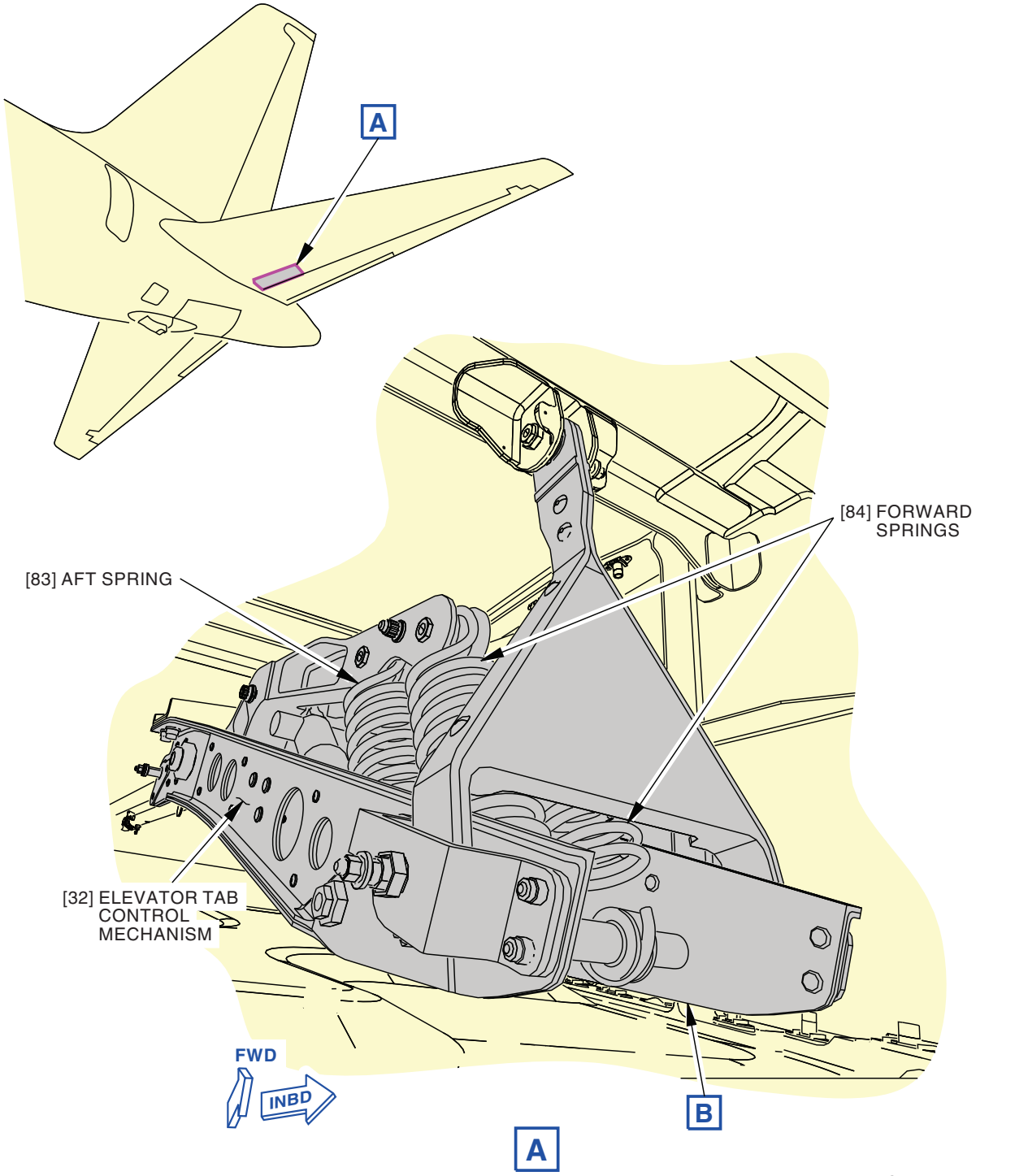
(e) After the bolt [87] is loose, remove the nut [92], washer [91], bolt [87], nut [90], washer [89], and bolt [88].

(f) Loosen the tool handle.

(g) Remove one or both of the forward springs [84] as necessary.

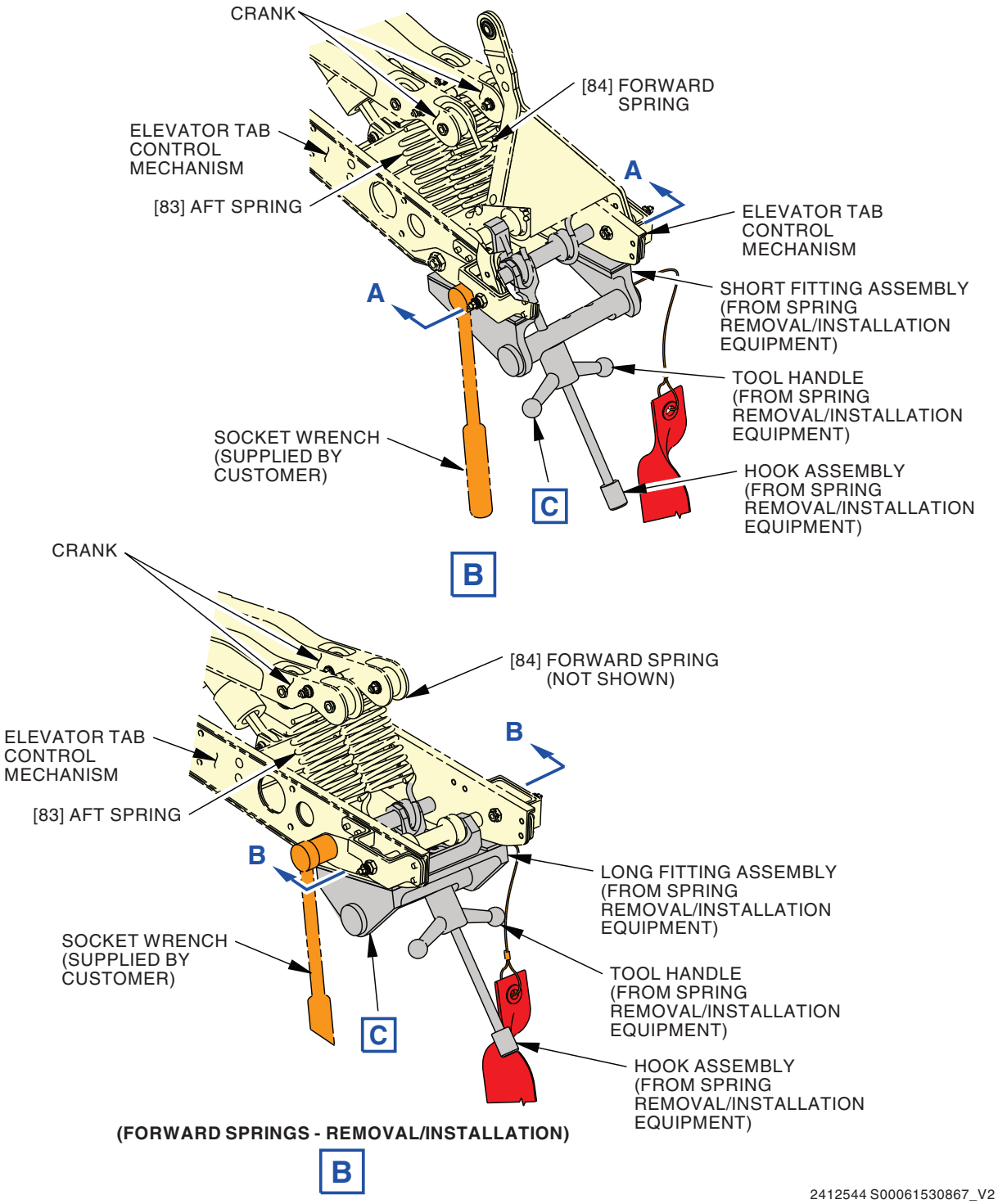
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



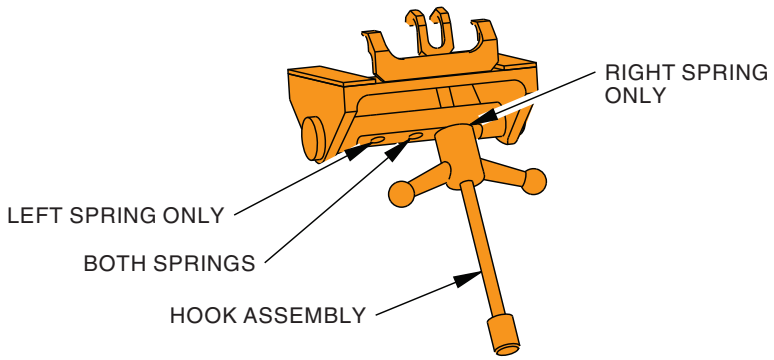
Elevator Tab Control Mechanism Spring - Removal/Installation
Figure 404/27-31-34-990-804 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



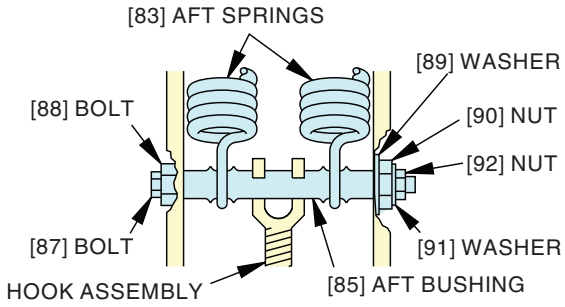
Elevator Tab Control Mechanism Spring - Removal/Installation
Figure 404/27-31-34-990-804 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



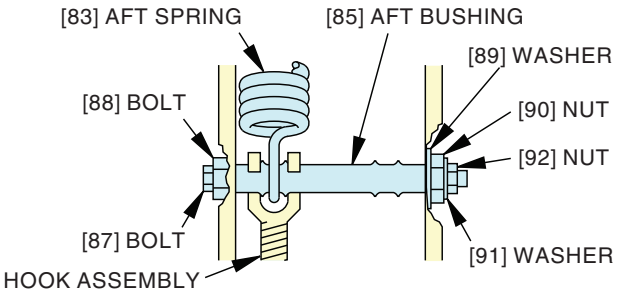
(SHORT FITTING ASSEMBLY IS SHOWN, LONG FITTING ASSEMBLY IS ALMOST THE SAME)

C 1



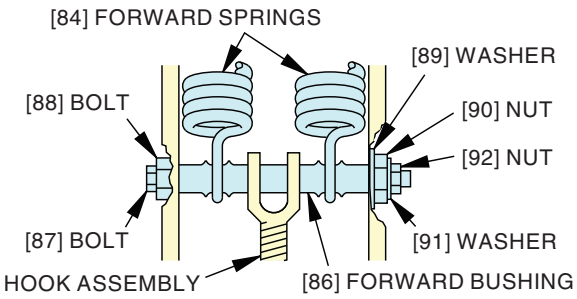
(ATTACH AS SHOWN IF BOTH SPRINGS ARE INSTALLED)

A-A



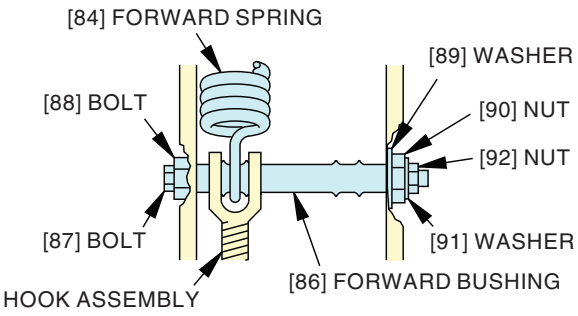
(ATTACH AS SHOWN IF THERE IS A BROKEN OR MISSING SPRING)

A-A



(ATTACH AS SHOWN IF BOTH SPRINGS ARE INSTALLED)

B-B



(ATTACH AS SHOWN IF THERE IS A BROKEN OR MISSING SPRING)

B-B

NOTE:

1 INSTALL HOOK ASSEMBLY AT LOCATION SHOWN DEPENDING ON THE INSTALLATION OR REMOVAL OF ONE OR BOTH SPRINGS.

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Elevator Tab Control Mechanism Spring - Removal/Installation
Figure 404/27-31-34-990-804 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-34-400-804

9. Elevator Tab Control Mechanism Springs - Installation

Figure 404

A. References

Reference	Title
27-31-00-700-802	Elevator Tab Control System - Operational Test (P/B 501)
27-31-00-710-801	Elevator and Elevator Trim Control System - Operational Test (P/B 501)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-41-000-801	Elevator Balance Panel - Removal (P/B 401)
27-31-41-400-801	Elevator Balance Panel - Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-13742	Removal/Installation Equipment - Spring, Tab Mechanism Assembly, Elevator Control Part #: C27086-27 Supplier: 81205

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
83	Aft spring	27-31-34-03-390	PTF ALL
84	Forward spring	27-31-34-03-390	PTF ALL

D. Location Zones

Zone	Area
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
334	Left Horizontal Stabilizer - Elevator
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge
344	Right Horizontal Stabilizer - Elevator

E. Access Panels

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

F. Prepare for the Installation

SUBTASK 27-31-34-860-013

(1) Make sure the pressure is removed from the elevator hydraulic systems A and B. To remove it, do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-34-010-015

(2) Open the applicable access panels:

(TASK 27-31-41-000-801)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

G. Elevator Tab Control Mechanism Spring Installation

SUBTASK 27-31-34-420-007

(1) Do these steps to install the aft springs [83]: (Figure 404)

NOTE: The direction of the coil wind (twist) and the number of coils differ by the manufacture. It is acceptable to install the appropriate springs from different suppliers on the same elevator tab control mechanism.

(a) Install the hook assembly and the tool handle from the removal/installation equipment, SPL-13742 on the short fitting assembly from the removal/installation equipment, SPL-13742.

(b) Install the removal/installation equipment, SPL-13742 on the elevator control tab mechanism assembly.

(c) Install one end of the aft springs [83] in the correct holes in the crank.

(d) Put the other end of the aft springs [83] in the correct groove on the aft bushing [85].

(e) Attach the hook assembly from the removal/installation equipment, SPL-13742 to the aft bushing [85].

(f) Slowly turn the tool handle until the bolt [88] can be installed through the aft bushing [85].

(g) Install the bolt [88], bolt [87], washer [89], nut [90], washer [91], and nut [92].

1) Tighten the nut [90] to 115 ±15 in-lb (13 ±2 N·m).

2) Tighten the nut [92] to 90 ±15 in-lb (10 ±2 N·m).

(h) Remove the removal/installation equipment, SPL-13742.

SUBTASK 27-31-34-420-008

(2) Do these steps to install the forward springs [84]: (Figure 404)

NOTE: The direction of the coil wind (twist) and the number of coils differ by the manufacture. It is acceptable to install the appropriate springs from different suppliers on the same elevator tab control mechanism.

(a) Install the hook assembly and the tool handle from the removal/installation equipment, SPL-13742 on the long fitting assembly from the removal/installation equipment, SPL-13742.

(b) Install the removal/installation equipment, SPL-13742 on the elevator control tab mechanism assembly.

(c) Install one end of the forward springs [84] in the correct holes in the crank.

(d) Put the other end of the forward springs [84] in the correct groove on the foward bushing [86].

(e) Attach the hook assembly from the removal/installation equipment, SPL-13742 to the foward bushing [86].

(f) Slowly turn the tool handle until the bolt [88] can be installed through the foward bushing [86].

(g) Install the bolt [88], bolt [87], washer [89], nut [90], washer [91], and nut [92].

(h) Remove the removal/installation equipment, SPL-13742.

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SUBTASK 27-31-34-730-011

(3) Do this task: Elevator and Elevator Trim Control System - Operational Test, TASK 27-31-00-710-801.

SUBTASK 27-31-34-730-012

(4) Do this task: Elevator Tab Control System - Operational Test, TASK 27-31-00-700-802.

H. Put the Airplane Back to its Usual Condition

SUBTASK 27-31-34-410-017

(1) Close the applicable access panels:

(TASK 27-31-41-400-801)

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR TAB CONTROL MECHANISM - INSPECTION/CHECK

1. General

A. This procedure has this inspection task:

(1) Elevator Balance Weight Installation and Elevator Tab Control Mechanism General Visual Inspection

TASK 27-31-34-210-801

2. Elevator Balance Weight Installation and Elevator Tab Control Mechanism - General Visual Inspection

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
55-10-11-000-801	Balance Bay Panels - Removal (P/B 401)
55-10-11-400-801	Balance Bay Panels - Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II

C. Location Zones

Zone	Area
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

D. Access Panels

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
333CB	Horizontal Stabilizer, Access Panel, Trailing Edge
333DB	Horizontal Stabilizer, Access Panel, Trailing Edge
334PT	Horizontal Stabilizer, Tab Control Rod Fairing
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
343CB	Horizontal Stabilizer, Access Panel - T.E. Area
343DB	Horizontal Stabilizer, Access Panel - T.E. Area
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

E. Elevator Balance Weight Installation and Elevator Tab Control Mechanism General Visual Inspection

SUBTASK 27-31-34-010-022

(1) Remove these access panels to get access to the applicable balance bays. Do this task: Balance Bay Panels - Removal, TASK 55-10-11-000-801.

(a) For the left elevator, remove these access panels:

Number	Name/Location
333CB	Horizontal Stabilizer, Access Panel, Trailing Edge
333DB	Horizontal Stabilizer, Access Panel, Trailing Edge

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) For the right elevator, remove these access panels:

Number	Name/Location
343CB	Horizontal Stabilizer, Access Panel - T.E. Area
343DB	Horizontal Stabilizer, Access Panel - T.E. Area

SUBTASK 27-31-34-010-023

(2) Remove these access panels to get access to the applicable elevator tab control mechanism.

(a) For the left elevator, remove these access panels:

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
334PT	Horizontal Stabilizer, Tab Control Rod Fairing

(b) For the right elevator, remove these access panels:

Number	Name/Location
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-34-210-001

(3) Do a general visual inspection of the left or right elevator balance weight installations and the left or right elevator tab control mechanisms (Figure 601 or Figure 602).

SUBTASK 27-31-34-410-023

(4) Install these access panels to remove access to the applicable balance bays. Do this task: Balance Bay Panels - Installation, TASK 55-10-11-400-801.

(a) For the left elevator, install these access panels:

Number	Name/Location
333CB	Horizontal Stabilizer, Access Panel, Trailing Edge
333DB	Horizontal Stabilizer, Access Panel, Trailing Edge

(b) For the right elevator, install these access panels:

Number	Name/Location
343CB	Horizontal Stabilizer, Access Panel - T.E. Area
343DB	Horizontal Stabilizer, Access Panel - T.E. Area

SUBTASK 27-31-34-410-024

(5) Install these access panels to remove access to the applicable elevator tab control mechanism.

(a) For the left elevator, install these access panels:

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
334PT	Horizontal Stabilizer, Tab Control Rod Fairing

(b) For the right elevator, install these access panels:

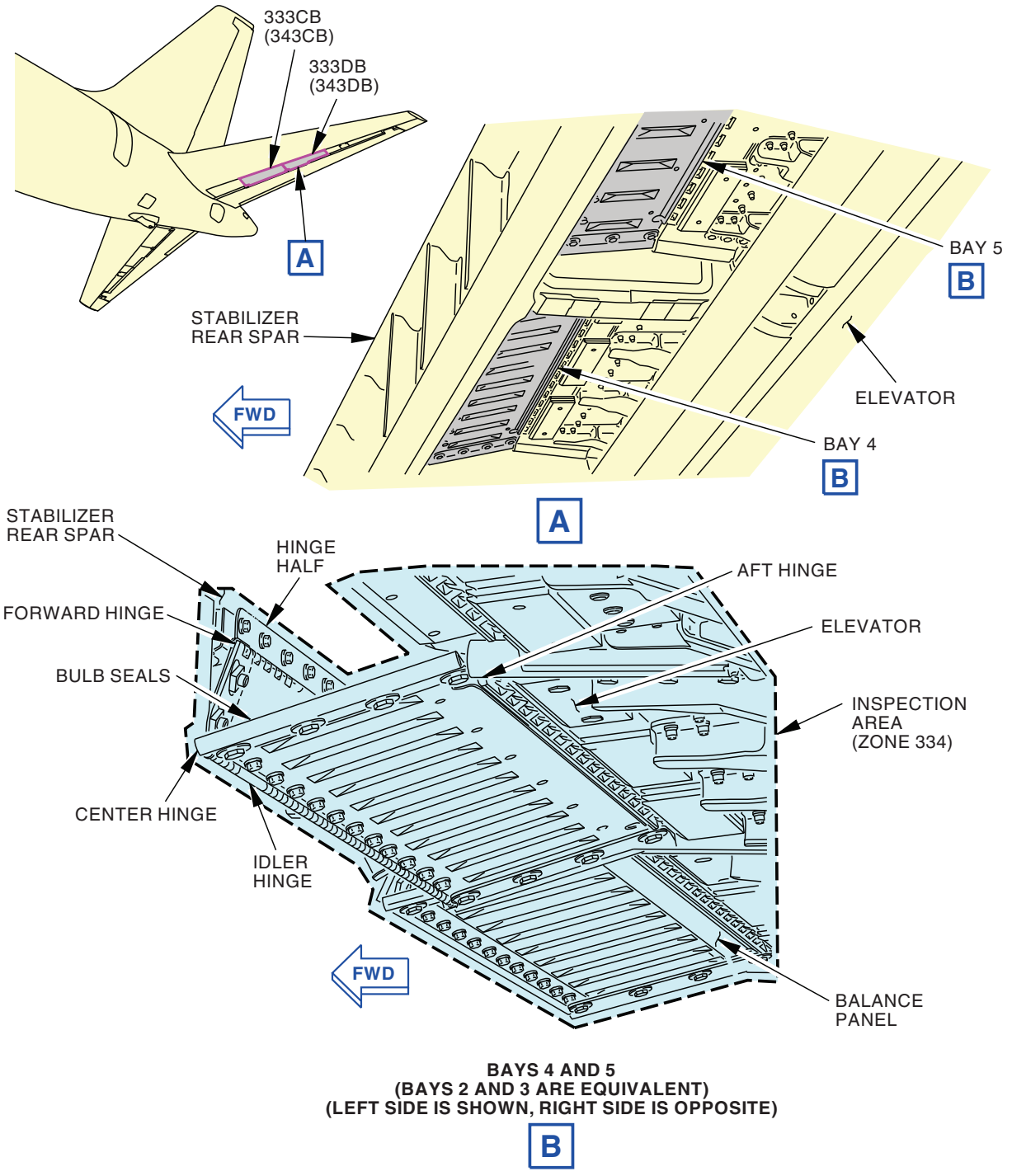
Number	Name/Location
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
344PT	Horizontal Stabilizer, Tab Control Rod Fairing, Elevator Sta 34.0

SUBTASK 27-31-34-390-003

(6) Apply an aerodynamic smoother sealant, A02315, for each applicable panel (SRM SUBJECT 51-10-01).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



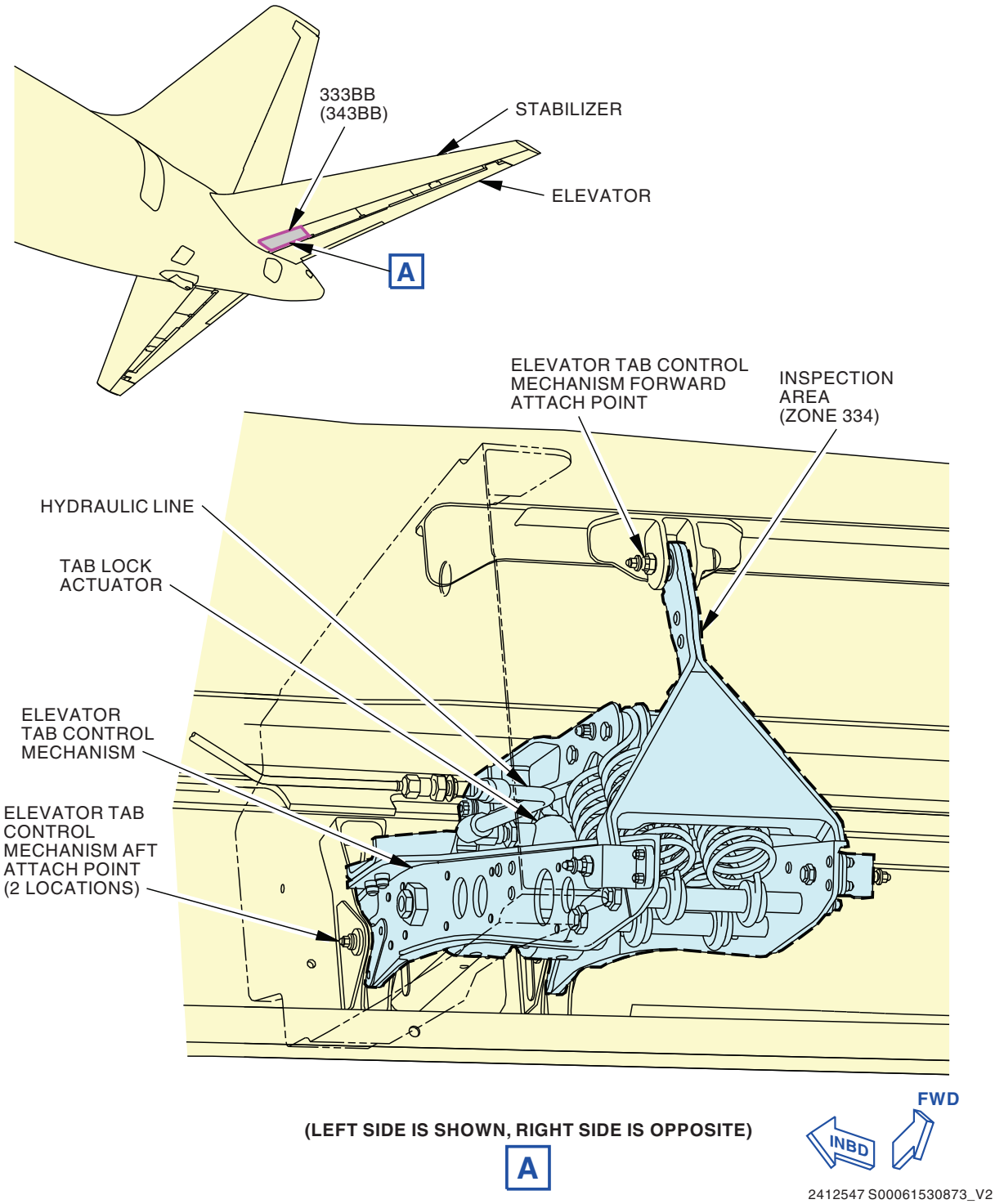
BAYS 4 AND 5
(BAYS 2 AND 3 ARE EQUIVALENT)
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

B

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Elevator Balance Weight Installation - General Visual Inspection
Figure 601/27-31-34-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Tab Control Mechanism - General Visual Inspection
Figure 602/27-31-34-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR TAB CONTROL VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the elevator tab control valve

(2) An installation of the elevator tab control valve

B. The elevator tab control valve is referred to as the "control valve" in this procedure.

C. This procedure is applicable to the control valves for the left and right elevator tab control mechanisms. The upper control valve is for the left elevator tab control mechanism and the lower control valve is for the right elevator tab control mechanism.

TASK 27-31-35-000-801

2. Elevator Tab Control Valve - Removal

(Figure 401)

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Location Zones

Zone	Area
318	Tail Cone Compartment - Right

C. Access Panels

Number	Name/Location
318BR	Tailcone Access Door

D. Prepare for the Removal

SUBTASK 27-31-35-860-001

(1) Remove pressure from the elevator hydraulic systems A and B (TASK 27-31-00-800-802).

SUBTASK 27-31-35-860-005

(2) For the left elevator tab control valve, open these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	4	C00456	AFCS SYS A MACH TRIM AC
C	5	C01041	AFCS SYS A SNSR EXC AC
D	1	C01049	AFCS SYS A WARN LIGHT (BAT)
D	2	C01045	AFCS SYS A FCC DC
D	3	C01048	AFCS SYS A ENGAGE INTLK
D	4	C00457	AFCS SYS A MACH TRIM DC
D	5	C01044	AFCS MCP DC 1
E	1	C00721	AUTOTHROTTLE DC 1

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	1	C00374	AFCS SYS B WARN LIGHT (BAT)
B	2	C01064	AFCS SYS B MACH TRIM DC

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(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	3	C01046	AFCS SYS B FCC DC
B	4	C00716	AFCS SYS B ENGAGE INTLK
C	1	C01037	AFCS SYS B MACH TRIM AC
C	2	C01042	AFCS SYS B SNSR EXC AC
C	3	C01047	AFCS MCP DC 2

SUBTASK 27-31-35-860-004

(3) For the right elevator tab control valve, open this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	8	C01464	FLIGHT CONTROL ELEV TAB VLV RIGHT

SUBTASK 27-31-35-010-001

(4) To get access to the control valve, open this access panel:

Number	Name/Location
318BR	Tailcone Access Door

E. Control Valve Removal

SUBTASK 27-31-35-020-001

(1) Disconnect hydraulic lines [10] from the control valve [11].

(a) Discard the packings [9].

(b) Install the caps and plugs on the control valve [11] ports and hydraulic lines [10].

SUBTASK 27-31-35-020-002

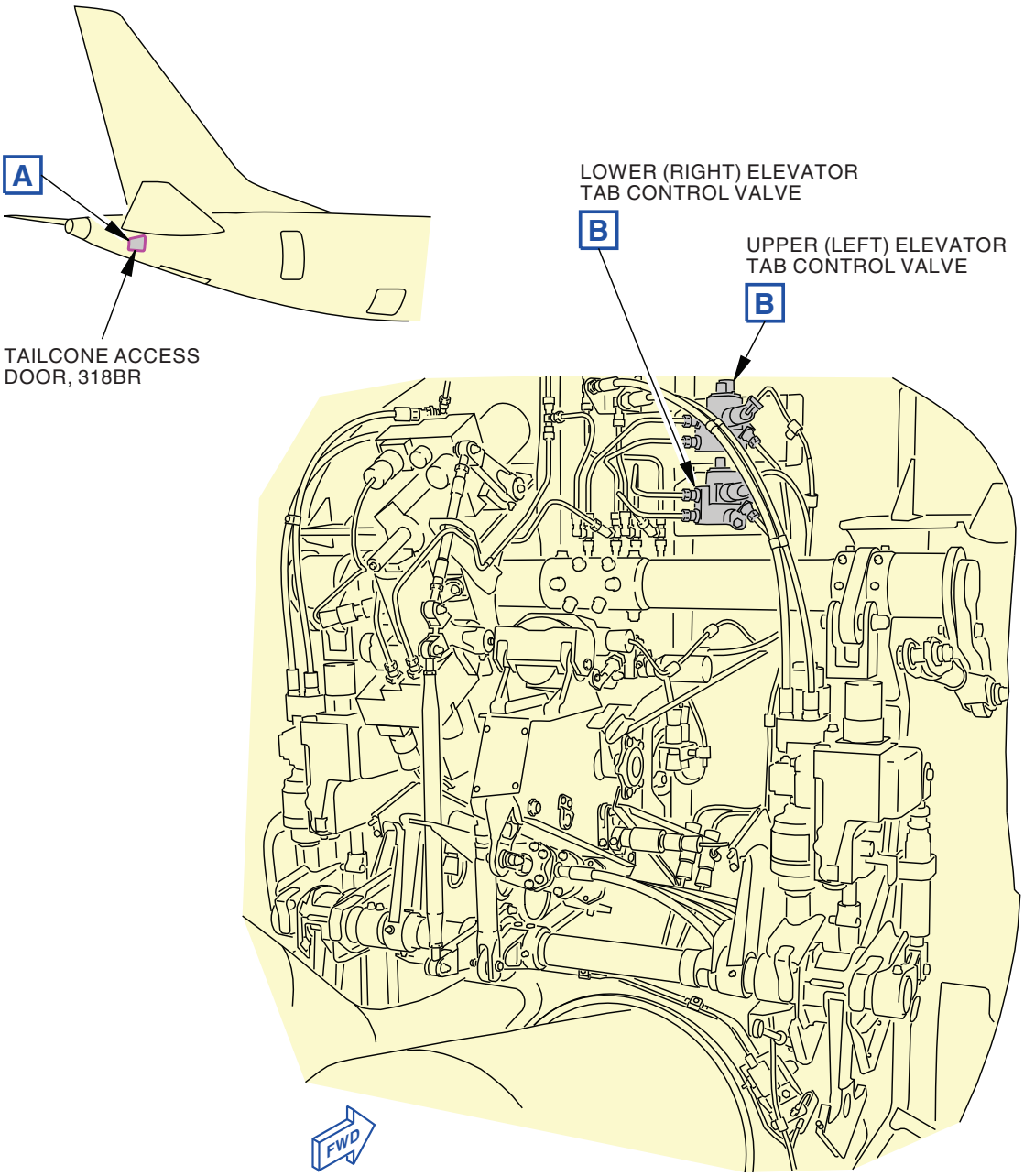
(2) Disconnect the electrical line [5] from the control valve [11].

SUBTASK 27-31-35-020-003

(3) Remove the bolts [6] and washers [7] that attach the control valve [11] to the airplane structure.

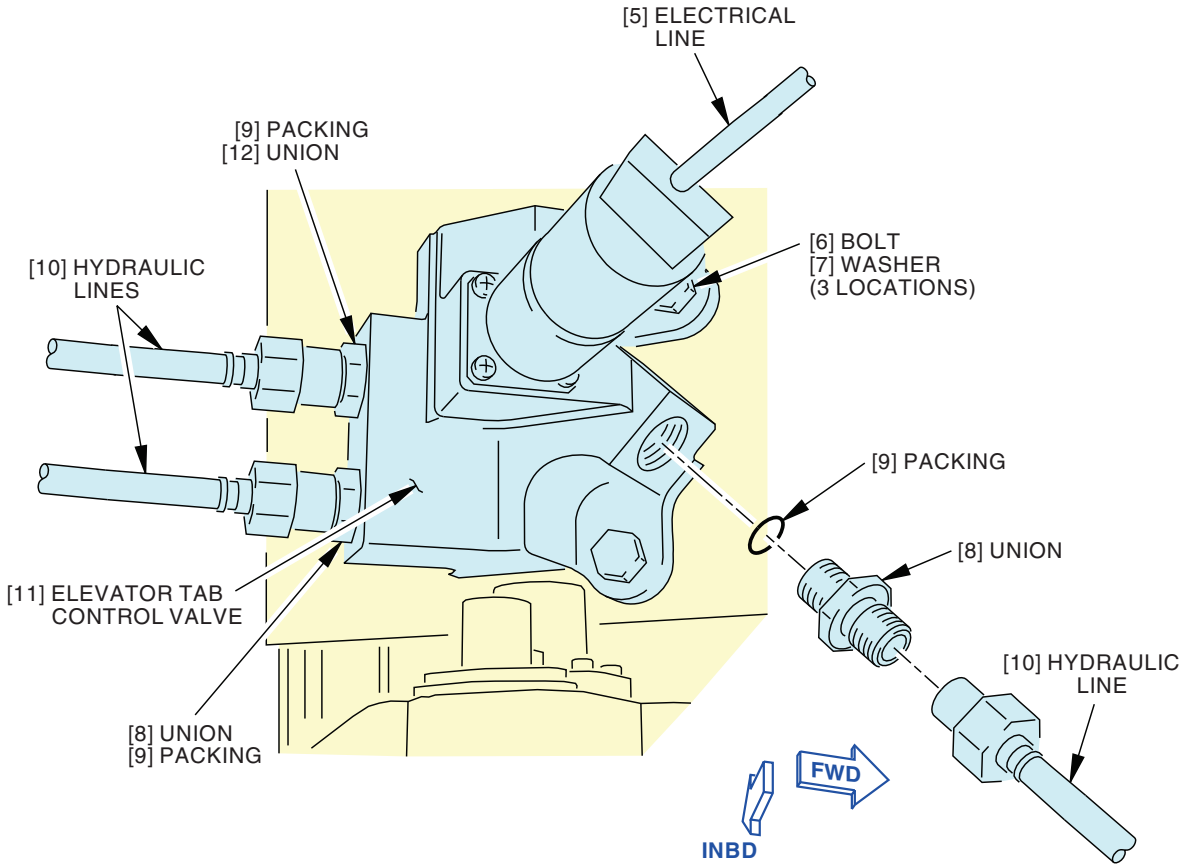
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Tab Control Valve Installation
Figure 401/27-31-35-990-801 (Sheet 1 of 2)

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ELEVATOR TAB CONTROL VALVE
(UPPER (OR LEFT TAB) IS SHOWN,
LOWER (OR RIGHT TAB) IS EQUIVALENT)

B

2412549 S00061530878_V2

Elevator Tab Control Valve Installation
Figure 401/27-31-35-990-801 (Sheet 2 of 2)

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TASK 27-31-35-400-801

3. Elevator Tab Control Valve - Installation

(Figure 401)

A. References

Reference	Title
27-31-00-700-802	Elevator Tab Control System - Operational Test (P/B 501)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Packing	27-31-35-05-060	PTF ALL
11	Control valve	27-31-35-01-015	PTF ALL
		27-31-35-05-010	PTF ALL

D. Location Zones

Zone	Area
318	Tail Cone Compartment - Right

E. Access Panels

Number	Name/Location
318BR	Tailcone Access Door

F. Control Valve Installation

SUBTASK 27-31-35-860-002

(1) Make sure hydraulic power is removed from the elevator hydraulic systems A and B (TASK 27-31-00-800-801).

SUBTASK 27-31-35-420-001

(2) If the unions [8] and packings [9] are not installed in the control valve [11], then do these steps:

(a) Lubricate the unions [8], union [12] and new packings [9] with hydraulic fluid, D00153.

(b) Install the new packings [9], unions [8], and union [12] to the control valve [11].

SUBTASK 27-31-35-420-002

(3) Install the control valve [11] to the airplane structure.

(a) Install the bolts [6] and washers [7] in the control valve [11].

(b) Remove the caps and plugs from the hydraulic lines [10] and electrical line [5].

(c) Connect the hydraulic lines [10] to the control valve [11].

(d) Tighten the B-nuts on the hydraulic lines [10] for unions [8] 140 ±14 in-lb (16 ±2 N·m).

(e) Tighten the B-nut on the hydraulic line [10] for union [12] 110 ±6 in-lb (12 ±1 N·m).

(f) Connect the electrical line [5] to the control valve [11].

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SUBTASK 27-31-35-860-003

(4) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-35-730-001

(5) Do this task: Elevator Tab Control System - Operational Test, TASK 27-31-00-700-802.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-35-410-001

(1) Close this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-35-860-006

(2) For the left elevator tab control valve, close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	4	C00456	AFCS SYS A MACH TRIM AC
C	5	C01041	AFCS SYS A SNSR EXC AC
D	1	C01049	AFCS SYS A WARN LIGHT (BAT)
D	2	C01045	AFCS SYS A FCC DC
D	3	C01048	AFCS SYS A ENGAGE INTLK
D	4	C00457	AFCS SYS A MACH TRIM DC
D	5	C01044	AFCS MCP DC 1
E	1	C00721	AUTOTHROTTLE DC 1

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	1	C00374	AFCS SYS B WARN LIGHT (BAT)
B	2	C01064	AFCS SYS B MACH TRIM DC
B	3	C01046	AFCS SYS B FCC DC
B	4	C00716	AFCS SYS B ENGAGE INTLK
C	1	C01037	AFCS SYS B MACH TRIM AC
C	2	C01042	AFCS SYS B SNSR EXC AC
C	3	C01047	AFCS MCP DC 2

SUBTASK 27-31-35-860-007

(3) For the right elevator tab control valve, close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	8	C01464	FLIGHT CONTROL ELEV TAB VLV RIGHT

——— END OF TASK ———

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ELEVATOR FEEL COMPUTER - MAINTENANCE PRACTICES

1. General

A. This procedure has two tasks:

(1) The removal of the Elevator Feel Computer Solenoid Valve.

(2) The installation of the Elevator Feel Computer Solenoid Valve.

TASK 27-31-37-000-802

2. Elevator Feel Computer Solenoid Valve - Removal

(Figure 201)

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1673	Pin - Safety, Elevator Feel Computer Part #: F80153-8 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
G50225	Lockwire - MS20995C20, Corrosion Resistant Steel - 0.020 Inch (0.508 mm) Diameter	NASM20995

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Prepare for the Removal

SUBTASK 27-31-37-010-002

(1) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-37-860-011

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-37-010-003

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	2	C01064	AFCS SYS B MACH TRIM DC
B	3	C01046	AFCS SYS B FCC DC
B	4	C00716	AFCS SYS B ENGAGE INTLK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
C	1	C01037	AFCS SYS B MACH TRIM AC
C	2	C01042	AFCS SYS B SNSR EXC AC
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2

SUBTASK 27-31-37-820-001

CAUTION YOU MUST INSTALL THE RIG PIN IN THE RIG PIN FITTING OF THE ELEVATOR-FEEL-COMPUTER. IF YOU DO NOT ATTACH THE ELEVATOR-FEEL-COMPUTER TO THE INPUT ARM, IT CAN CAUSE DAMAGE WHEN YOU REMOVE IT.

(4) Install the pin, SPL-1673, through the input arm [3] and into the rig pin fitting [9] on the elevator feel computer [1].

F. Removal

SUBTASK 27-31-37-020-003

(1) Remove MS20995C20 lockwire, G50225, from the four bolts [4].

NOTE: Use care to prevent scoring or damage to finished surfaces.

SUBTASK 27-31-37-020-004

(2) Remove the bolts [4] and washers [5] from the solenoid valve [2].

SUBTASK 27-31-37-020-005

(3) Remove the solenoid valve [2] from the plug assembly [6] and manifold [10].

NOTE: Use care to prevent damage to the electrical connector.

SUBTASK 27-31-37-020-008

(4) Remove and discard the cup seal [7].

SUBTASK 27-31-37-020-006

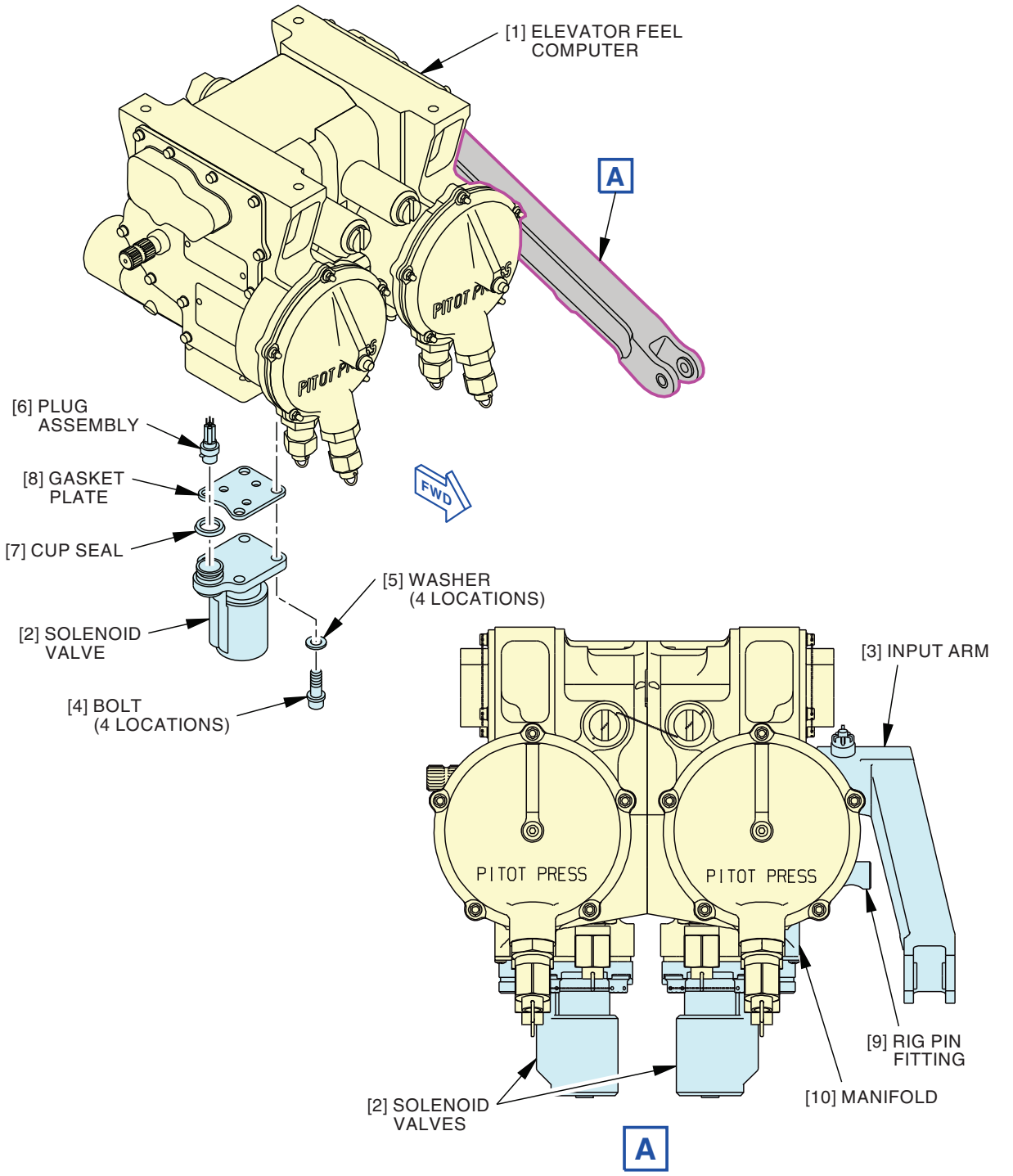
(5) Remove and discard the gasket plate [8].

SUBTASK 27-31-37-020-007

(6) Repeat the above steps for the opposite system.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Feel Computer Solenoid Valve Installation
Figure 201/27-31-37-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-37-400-802

3. Elevator Feel Computer Solenoid Valve - Installation

(Figure 201)

A. References

Reference	Title
27-31-37-200-801	Elevator Feel Computer Leakage Check (P/B 601)
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
29-11-00-860-803	Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP) (P/B 201)
29-11-00-860-804	Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP) (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1914	Test Set - Air Data Model FLMTS (Flight Line Maintenance) Part #: ADSE 650 Supplier: 3BSK6 Part #: ADTS2000 Supplier: 52892 Part #: ADTS3350ER Supplier: 52892 Part #: ADTS405MKII Supplier: U0427 Part #: ADTS530 Supplier: U0427 Part #: ADTS552F Supplier: U0427 Part #: ADTS553F Supplier: U0427 Part #: ADTS554F Supplier: U0427 Part #: D60340MK Supplier: K1474 Part #: DPS1000 Supplier: 21844 Part #: MODEL 6300 Supplier: 0RDZ5 Part #: MPS34C Supplier: 48RQ2 Part #: MPS43B Supplier: A0197 Part #: MPS45 Supplier: 48RQ2 Part #: MPS49 Supplier: 48RQ2 Part #: TES9463 Supplier: 88277 Opt Part #: 01-0987-00 Supplier: 41364 Opt Part #: 18910480000 Supplier: 89944 Opt Part #: 18910920000 Supplier: 89944 Opt Part #: ADTS405F Supplier: U0427 Opt Part #: ADTS505 Supplier: U0427 Opt Part #: D60302 Supplier: K1474 Opt Part #: D60340 Supplier: K1474 Opt Part #: D60383 Supplier: K1474 Opt Part #: DPS350 Supplier: 21844 Opt Part #: DPS450 Supplier: 21844 Opt Part #: DPS500 Supplier: 21844 Opt Part #: MPS31C Supplier: 48RQ2 Opt Part #: MPS43 Supplier: A0197

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(Continued)

Reference	Description
COM-1916	Adapter - Pitot Test (Typically included in Air Data Accessory Kit) Part #: CSA75700HT-3 Supplier: 3BSK6 Part #: P75701-4 Supplier: 38002 Part #: P75701M2-3 Supplier: 38002
COM-25177	Accessory Kit - Air Data, 737 (Pitot/Static adapters, hoses, etc.) Part #: ADA737-678 Supplier: 38002 Part #: AK737-900 Supplier: 3BSK6 Part #: PA-0120-A Supplier: 3BSK6
SPL-1673	Pin - Safety, Elevator Feel Computer Part #: F80153-8 Supplier: 81205
SPL-1742	Regulator - Air Pressure, Elevator Feel Computer Part #: F72928-109 Supplier: 81205 Part #: F72928-110 Supplier: 81205 Part #: F72928-111 Supplier: 81205 Opt Part #: F72928-58 Supplier: 81205 Opt Part #: F72928-62 Supplier: 81205 Opt Part #: F72928-63 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50225	Lockwire - MS20995C20, Corrosion Resistant Steel - 0.020 Inch (0.508 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Solenoid valve	27-31-37-01-370	PTF ALL
7	Cup seal	27-31-37-01-390	PTF ALL
8	Gasket plate	27-31-37-01-385	PTF ALL

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Elevator Feel Computer Solenoid Valve Installation

SUBTASK 27-31-37-420-003

(1) Install the new gasket plate [8] on the manifold [10].

SUBTASK 27-31-37-420-004

(2) Install the new cup seal [7] on the solenoid valve [2].

SUBTASK 27-31-37-420-005

(3) Attach the solenoid valve [2] to the plug assembly [6].

SUBTASK 27-31-37-420-006

(4) Install the solenoid valve [2] to the manifold [10] with the washers [5] and bolts [4].

(a) Tighten the bolts [4] to 50 in-lb (6 N·m) - 70 in-lb (8 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-37-420-007

(5) Safety wire the bolts [4] with MS20995C20 lockwire, G50225.

SUBTASK 27-31-37-420-008

(6) Make sure that there is no leakage of hydraulic fluid, D00153, at the hydraulic components.

SUBTASK 27-31-37-420-009

(7) Repeat the above steps for the opposite system.

SUBTASK 27-31-37-000-001

(8) Remove the pin, SPL-1673, from the rig pin fitting [9] of the elevator feel computer [1].

SUBTASK 27-31-37-870-002

(9) Bleed the elevator control system.

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	2	C01064	AFCS SYS B MACH TRIM DC
B	3	C01046	AFCS SYS B FCC DC
B	4	C00716	AFCS SYS B ENGAGE INTLK
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
C	1	C01037	AFCS SYS B MACH TRIM AC
C	2	C01042	AFCS SYS B SNSR EXC AC
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2

(b) Move the stabilizer to the maximum leading edge up (APL NOSE DOWN) position (the mechanical limits).

NOTE: This can be done with the control wheel STAB TRIM switches or the stabilizer trim control wheel.

1) Make sure that the ballnut is against the jackscrew upstop.

(c) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

CAUTION

MAKE SURE THAT THE PITOT PROBE HAS NO ADDED WEIGHT ON IT FROM THE TEST HOSE. THE WEIGHT OF THE TEST HOSE CAN BEND OR TWIST THE PITOT PROBE OUT OF TOLERANCE.

(d) Connect the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to the two pitot probes on the vertical fin.

1) Do this step for the regulator, SPL-1742:

a) Connect the pressure lines from the regulator, SPL-1742, with the pitot test adapter, COM-1916, to the pitot probes.

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2) Do this step for the air data model test set, COM-1914:

a) Use the air data accessory kit, COM-25177, to connect the air data model test set, COM-1914, to the pitot test adapter, COM-1916.

NOTE: The air data accessory kit, COM-25177, contains hoses to connect to the air data model test set, COM-1914, and the pitot test adapter, COM-1916.

(e) Use the regulator, SPL-1742, or air data model test set, COM-1914, to raise the airspeed to 2.46 psig (169.61 mBar) (300 knots).

NOTE: When pressure to the pitot system is applied or removed, the rate must not be more than 2.46 psig (169.61 mBar) (300 knots) for each minute.

WARNING

KEEP PERSONS AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES. THE AILERONS, ELEVATORS, RUDDER, FLAPS, SLATS, SPOILERS, STABILIZER AND NOSE GEAR CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(f) Pressurize Hydraulic System A and B, do one of these steps:

1) If a Portable Hydraulic Cart is used, do this task: Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, TASK 29-11-00-860-802.

NOTE: Two carts are recommended.

a) If only one cart is available, bleed hydraulic A and B, as follows:

<1> Pressurize hydraulic system A with a cart and hydraulic system B with the Electric Motor-Driven Pump (EMDP).

<2> Pressurize hydraulic system B with a cart and hydraulic system A with the EMDP.

2) If a Portable Hydraulic Cart is not used, do these tasks:

NOTE: Both the EMDP and Engine Driven Pump (EDP) should be run at the same time.

- Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), TASK 29-11-00-860-803
- Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), TASK 29-11-00-860-804.

(g) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

(h) Wait for 10 seconds.

(i) Do these steps to cycle the power between Hydraulic Systems A and B:

1) Set the flight controls hydraulic power switches as follows:

a) Set the FLT CONTROL B switch to OFF position.

b) Set the FLT CONTROL A switch to A ON position.

c) Wait for 2 seconds.

d) Set the FLT CONTROL A switch to OFF position.

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e) Set the FLT CONTROL B switch to B ON position.

f) Wait for 2 seconds.

2) Repeat the cycle 10 times.

(j) Slowly, do control column sweeps with pitot pressure applied.

(k) Do these steps to cycle pitot pressure:

1) Use the regulator, SPL-1742, or air data model test set, COM-1914, connected at the elevator pitot probes on the vertical stabilizer, to lower the airspeed to 0 psig (0.00 mBar) (0 knots).

2) Use the regulator, SPL-1742, or air data model test set, COM-1914, connected at the elevator pitot probes on the vertical stabilizer, to increase the airspeed to 2.46 psig (169.61 mBar) (300 knots).

3) Use the regulator, SPL-1742, or air data model test set, COM-1914, connected at the elevator pitot probes on the vertical stabilizer, to lower the airspeed to 0 psig (0.00 mBar) (0 knots).

(l) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

(m) Wait for 10 seconds.

(n) Remove the pressure from the Hydraulic Systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-31-37-710-001

(10) Do this task: Elevator Feel Computer Leakage Check, TASK 27-31-37-200-801.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-37-860-019

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-31-37-020-009

(2) Do these steps at the two pitot probes on the vertical fin:

(a) Remove the pressure lines from the air data model test set, COM-1914, or regulator, SPL-1742, or equivalent from the pitot probes.

(b) Remove the pitot test adapter, COM-1916, from the pitot probes.

SUBTASK 27-31-37-410-003

(3) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR FEEL COMPUTER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the elevator feel computer

(2) An installation of the elevator feel computer.

TASK 27-31-37-000-801

2. Elevator Feel Computer Removal

(Figure 401)

A. References

Reference	Title
20-10-51-000-801	Flareless Tubing Assembly Removal (P/B 401)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1673	Pin - Safety, Elevator Feel Computer Part #: F80153-8 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Prepare for the Removal

SUBTASK 27-31-37-010-001

(1) Open this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-37-860-001

(2) Do this task: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-37-860-002

(3) Do these steps to set the B Dimension (Figure 401):

NOTE: The B Dimension is measured between the center of the upper and lower Gimbal pins on the stabilizer trim jackscrew. This requires two persons: one person will be in the flight compartment, the second person will be in the tail of the airplane with a headset.

(a) Turn the stabilizer trim wheel on the control stand to set the B Dimension.

(b) Use the trammel bar, SPL-1677, to measure the B Dimension at 39.17 ±0.01 in. (994.92 ±0.26 mm) (Figure 401).

CAUTION

MAKE SURE TO PREVENT DAMAGE TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CHAFE OR CAN CAUSE DAMAGE THE WIRES OR BUNDLE INSULATION.

1) Make sure that the trammel bar, SPL-1677, does not cause damage to the wire bundles.

SUBTASK 27-31-37-860-003

(4) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-37-860-004

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	2	C01064	AFCS SYS B MACH TRIM DC
B	3	C01046	AFCS SYS B FCC DC
B	4	C00716	AFCS SYS B ENGAGE INTLK
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
C	1	C01037	AFCS SYS B MACH TRIM AC
C	2	C01042	AFCS SYS B SNSR EXC AC
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-31-37-480-001

CAUTION

YOU MUST INSTALL THE SAFETY PIN IN THE RIG PIN HOLE OF THE ELEVATOR FEEL COMPUTER. IF YOU DO NOT ATTACH THE ELEVATOR FEEL COMPUTER TO THE INPUT ARM, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(6) Install the pin, SPL-1673, in the input arm [8] and elevator feel computer assembly [1].

F. Elevator Feel Computer Removal

SUBTASK 27-31-37-020-001

CAUTION

DO NOT LIFT THE ELEVATOR FEEL COMPUTER BY THE INPUT ARM, OR APPLY A TORQUE OF MORE THAN 300 POUND-INCHES (33.9 NEWTON-METERS) TO THE INPUT ARM. THE ELEVATOR FEEL COMPUTER IS VERY SENSITIVE AND IT CAN BE DAMAGED.

(1) Remove the elevator feel computer assembly [1]:

(a) Disconnect the spring [16] from the crank [25].

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(b) Remove the nut [26], washer [31], washer [27], and bolt [28] to disconnect the crank [25] from the elevator feel computer assembly [1].

(c) Remove the nut [23], washer [22], and bolt [21] to disconnect the stabilizer pushrod [2] from the input arm [8].

(d) Disconnect the electrical connectors [7] from the elevator feel computer assembly [1].

WARNING

RELEASE THE CONNECTION SLOWLY AND LET THE PRESSURE DECREASE BEFORE YOU COMPLETELY REMOVE THE HYDRAULIC LINE. IF YOU DISCONNECT THE HYDRAULIC LINE WHEN IT HAS PRESSURE, A SPRAY OF HYDRAULIC FLUID CAN CAUSE INJURY TO PERSONS.

(e) Disconnect the return hydraulic lines [29], pressure hydraulic lines [30], pitot lines, and static lines from the elevator feel computer assembly [1].

1) Install tags on lines and port fittings to clearly show the correct connection location (TASK 20-10-51-000-801).

(f) Install caps to the hydraulic lines, pitot lines, and static lines.

(g) Install plugs to all ports.

(h) Remove the nuts [6], washers [5], washers [4], and bolts [3] that attach the elevator feel computer assembly [1] to the airplane structure.

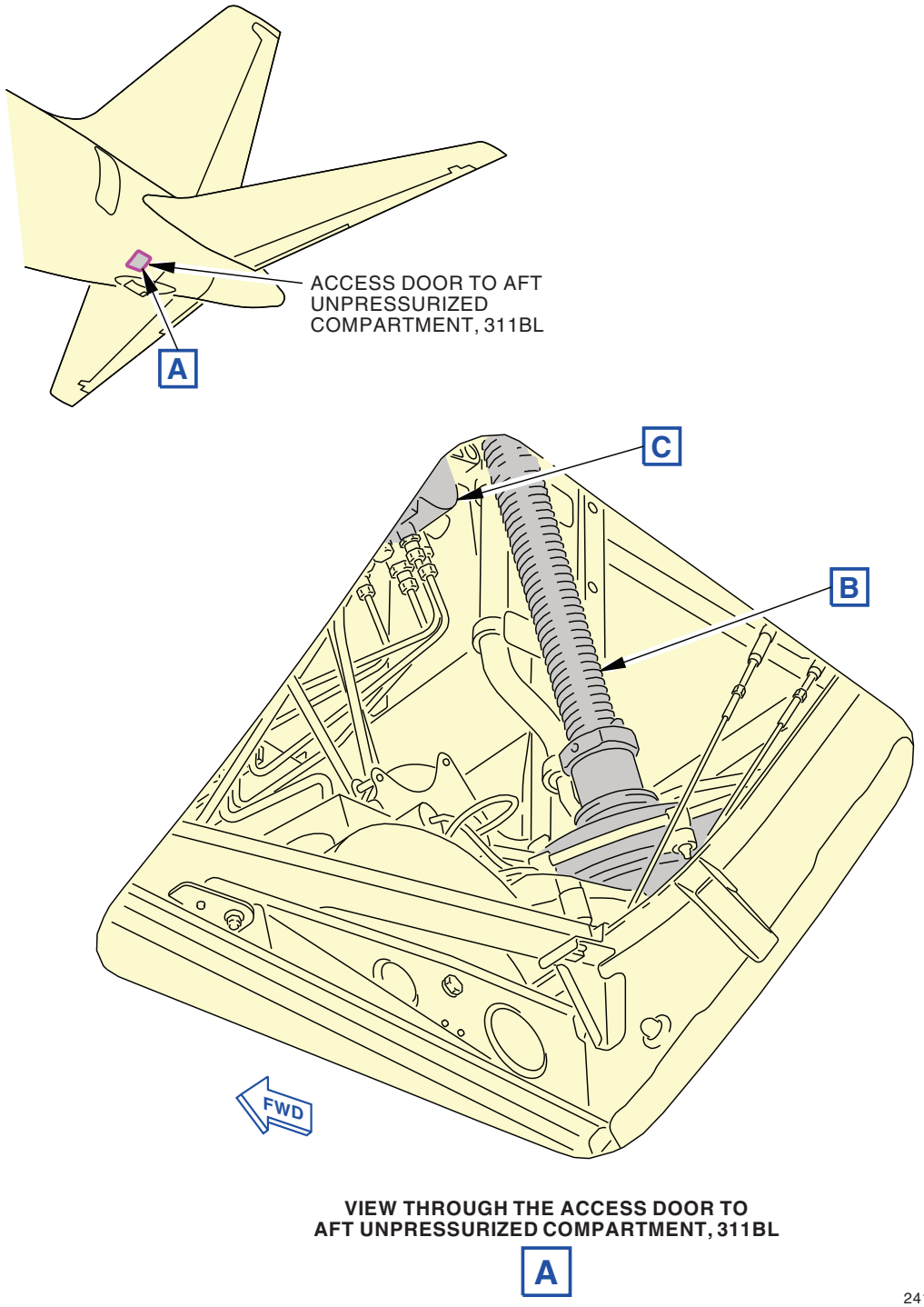
(i) Remove the elevator feel computer assembly [1] from the airplane.

SUBTASK 27-31-37-540-001

(2) Discard the packings [9], packings [12], packings [14], packings [17], and packings [19].

——— END OF TASK ———

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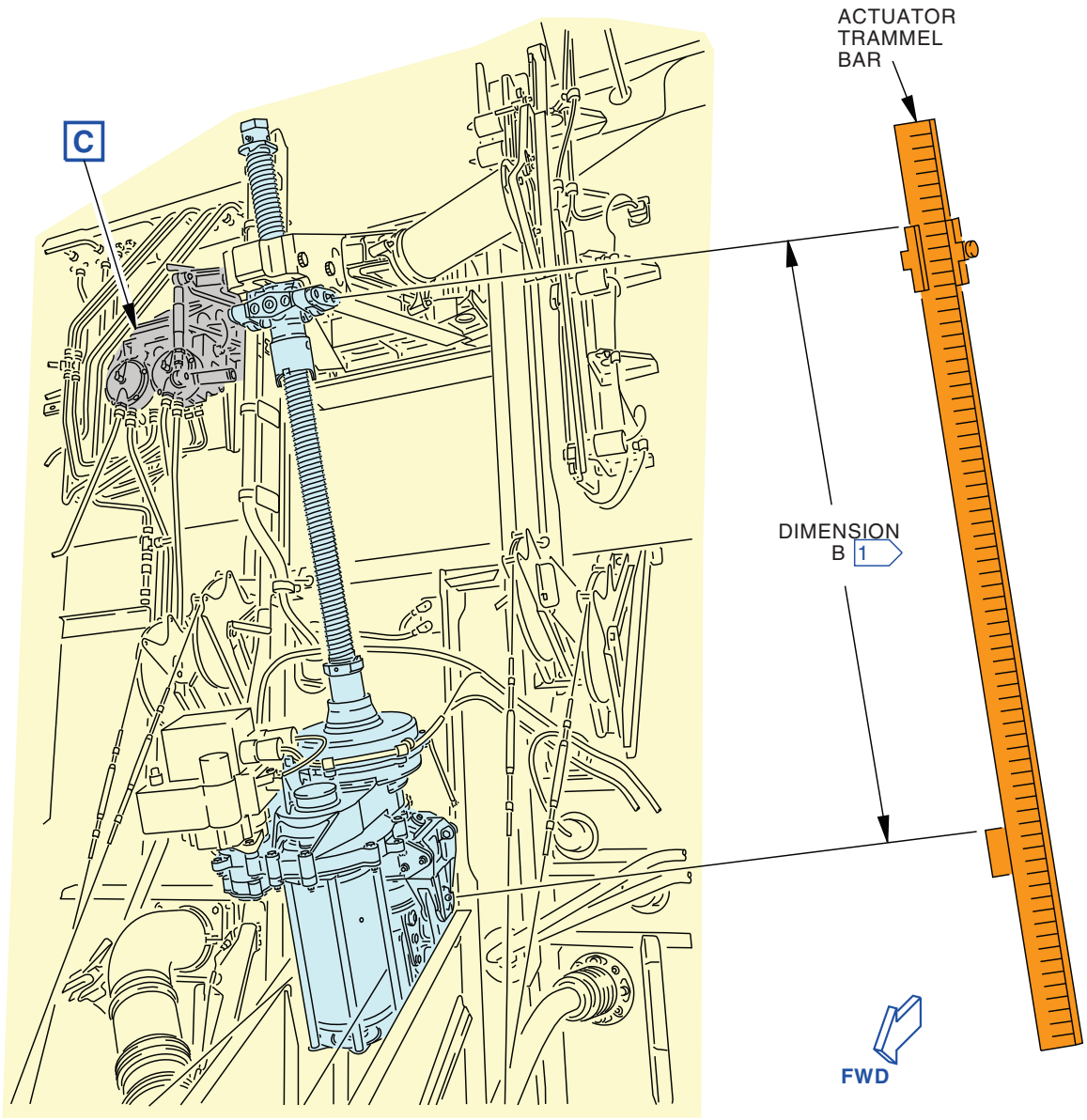
**VIEW THROUGH THE ACCESS DOOR TO
AFT UNPRESSURIZED COMPARTMENT, 311BL**

A

2412550 S00061530883_V1

**Elevator Feel Computer Installation
Figure 401/27-31-37-990-801 (Sheet 1 of 4)**

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STABILIZER TRIM ACTUATOR

B

NOTE:

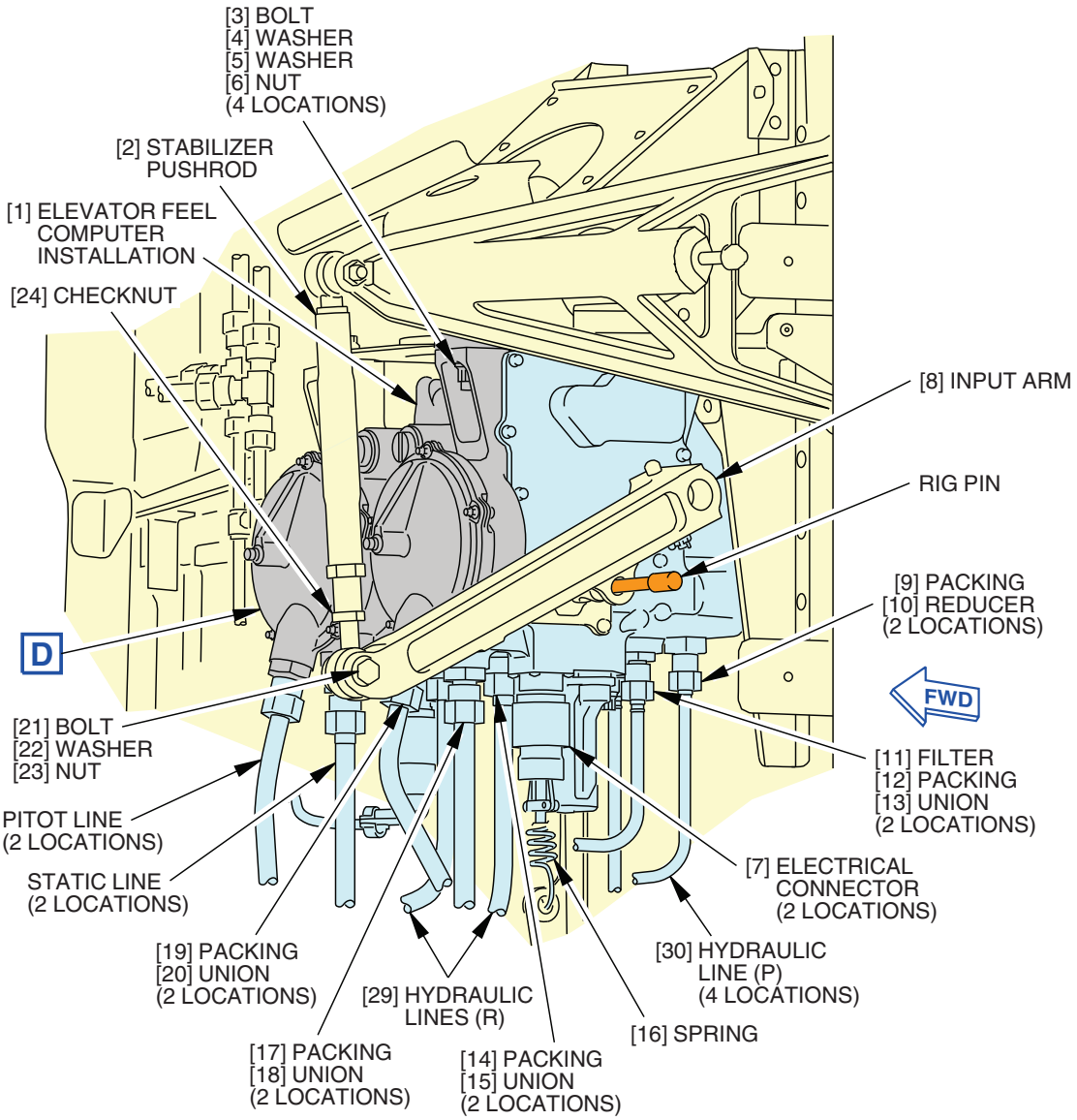
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

2412551 S00061530884_V1

Elevator Feel Computer Installation
Figure 401/27-31-37-990-801 (Sheet 2 of 4)

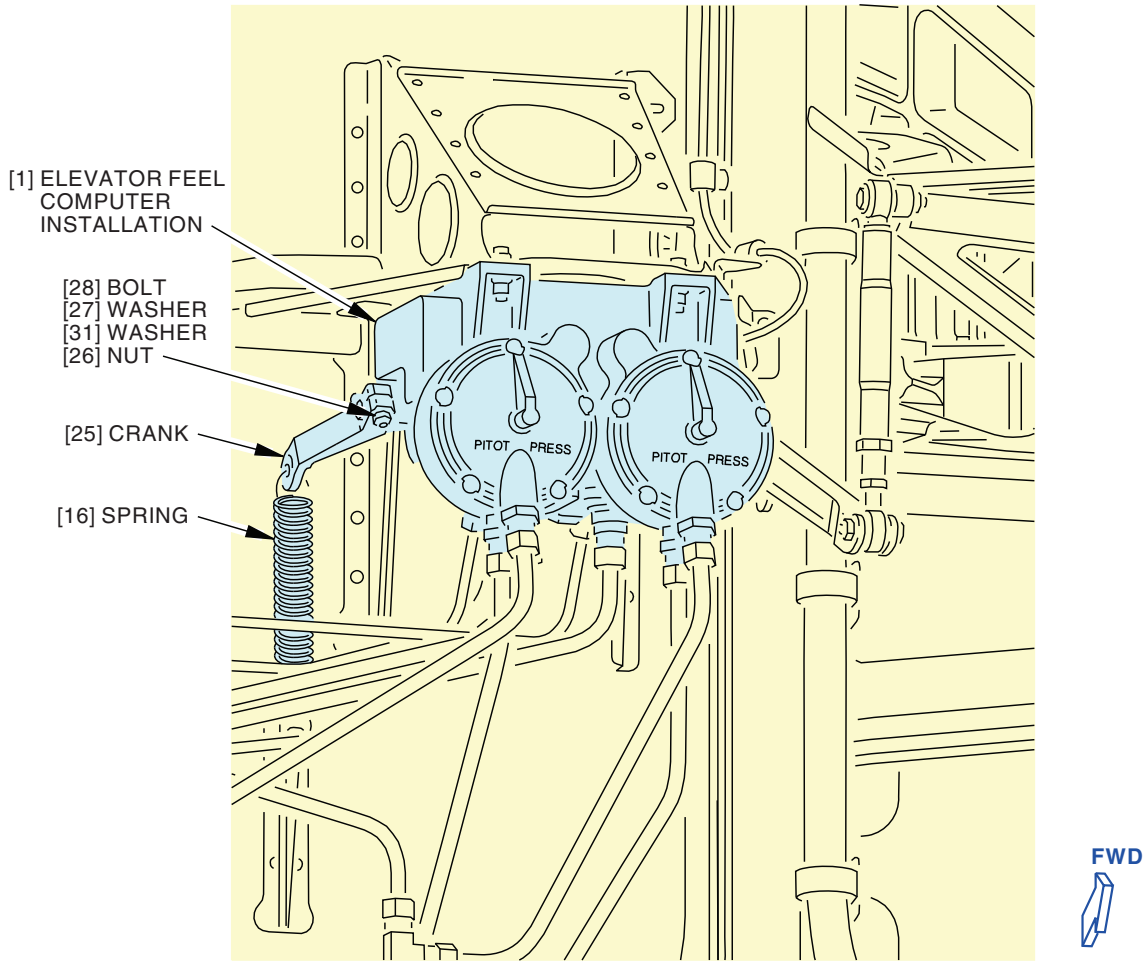
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412552 S00061530885_V4

Elevator Feel Computer Installation
Figure 401/27-31-37-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELEVATOR FEEL COMPUTER (EXAMPLE)

D

2571383 S0000616626_V2

Elevator Feel Computer Installation
Figure 401/27-31-37-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-37-400-801

3. Elevator Feel Computer Installation

(Figure 401)

A. References

Reference	Title
20-10-51-400-801	Flareless Tubing Assembly Installation (P/B 401)
27-31-00-700-812	Feel Differential Pressure Light - Test (P/B 501)
27-31-00-700-813	Feel Force System - Test (P/B 501)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
27-31-17-790-801	Elevator Pitot System Leak - Test (P/B 201)
27-31-37-200-801	Elevator Feel Computer Leakage Check (P/B 601)
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
29-11-00-860-803	Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP) (P/B 201)
29-11-00-860-804	Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP) (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-1914	Test Set - Air Data Model FLMTS (Flight Line Maintenance) Part #: ADSE 650 Supplier: 3BSK6 Part #: ADTS2000 Supplier: 52892 Part #: ADTS3350ER Supplier: 52892 Part #: ADTS405MKII Supplier: U0427 Part #: ADTS530 Supplier: U0427 Part #: ADTS552F Supplier: U0427 Part #: ADTS553F Supplier: U0427 Part #: ADTS554F Supplier: U0427 Part #: D60340MK Supplier: K1474 Part #: DPS1000 Supplier: 21844 Part #: MODEL 6300 Supplier: 0RDZ5 Part #: MPS34C Supplier: 48RQ2 Part #: MPS43B Supplier: A0197 Part #: MPS45 Supplier: 48RQ2 Part #: MPS49 Supplier: 48RQ2 Part #: TES9463 Supplier: 88277 Opt Part #: 01-0987-00 Supplier: 41364 Opt Part #: 18910480000 Supplier: 89944 Opt Part #: 18910920000 Supplier: 89944 Opt Part #: ADTS405F Supplier: U0427 Opt Part #: ADTS505 Supplier: U0427 Opt Part #: D60302 Supplier: K1474 Opt Part #: D60340 Supplier: K1474 Opt Part #: D60383 Supplier: K1474 Opt Part #: DPS350 Supplier: 21844 Opt Part #: DPS450 Supplier: 21844 Opt Part #: DPS500 Supplier: 21844 Opt Part #: MPS31C Supplier: 48RQ2 Opt Part #: MPS43 Supplier: A0197
COM-1916	Adapter - Pitot Test (Typically included in Air Data Accessory Kit) Part #: CSA75700HT-3 Supplier: 3BSK6 Part #: P75701-4 Supplier: 38002 Part #: P75701M2-3 Supplier: 38002
COM-25177	Accessory Kit - Air Data, 737 (Pitot/Static adapters, hoses, etc.) Part #: ADA737-678 Supplier: 38002 Part #: AK737-900 Supplier: 3BSK6 Part #: PA-0120-A Supplier: 3BSK6
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1673	Pin - Safety, Elevator Feel Computer Part #: F80153-8 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

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(Continued)

Reference	Description
SPL-1742	Regulator - Air Pressure, Elevator Feel Computer Part #: F72928-109 Supplier: 81205 Part #: F72928-110 Supplier: 81205 Part #: F72928-111 Supplier: 81205 Opt Part #: F72928-58 Supplier: 81205 Opt Part #: F72928-62 Supplier: 81205 Opt Part #: F72928-63 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Elevator feel computer assembly	27-31-37-01-265	PTF ALL
9	Packing	27-31-37-01-345	PTF ALL
12	Packing	27-31-37-01-340	PTF ALL
14	Packing	27-31-37-01-335	PTF ALL
17	Packing	27-31-37-01-345	PTF ALL
19	Packing	27-31-37-01-345	PTF ALL

E. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Elevator Feel Computer Installation

SUBTASK 27-31-37-020-010

(1) For a new elevator feel computer assembly [1], do these steps:

(a) Remove the transport/storage strings from the input arm [8] and elevator feel computer assembly [1].

CAUTION YOU MUST INSTALL THE SAFETY PIN IN THE RIG PIN HOLE OF THE ELEVATOR FEEL COMPUTER. IF YOU DO NOT ATTACH THE ELEVATOR FEEL COMPUTER TO THE INPUT ARM, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(b) Install the pin, SPL-1673, in the input arm [8] and elevator feel computer assembly [1].

SUBTASK 27-31-37-860-005

(2) Make sure that pressure is removed from the elevator hydraulic systems A and B.

(a) If it is necessary, do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-37-860-006

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(3) Use the trammel bar, SPL-1677, to make sure that the B Dimension is at 39.17 ±0.01 in. (994.92 ±0.26 mm) (Figure 401).

SUBTASK 27-31-37-420-001

(4) If the packings [9], packings [12], packings [14], packings [17], packings [19], filters [11], unions [13], unions [15], unions [18], unions [20], and reducers [10] are not installed in the elevator feel computer assembly [1], do these steps:

(a) Lightly lubricate the new packings [9], new packings [12], new packings [14], new packings [17], and new packings [19] with hydraulic fluid, D00153.

(b) Install these items in the elevator feel computer assembly [1]:

1) The packings [9] and reducers [10] to the fittings for the computer pressure lines

2) The filters [11], packings [12], and unions [13] to the fittings for the hydraulic lines

3) The packings [14] and unions [15] to the fittings for the return lines

4) The packings [17] and unions [18] to the fittings for the static lines

5) The packings [19] and unions [20] to the fittings for the pitot lines.

SUBTASK 27-31-37-080-001

(5) Remove the pin, SPL-1673, from the input arm [8] and elevator feel computer assembly [1].

SUBTASK 27-31-37-480-002

(6) Install the rig pin E-4 from the test set rigging pin kit, SPL-1585, in the input arm [8] and elevator feel computer assembly [1].

SUBTASK 27-31-37-420-002

(7) Install the elevator feel computer assembly [1].

(a) Put the elevator feel computer assembly [1] in its position.

(b) Install the bolts [3], washers [4] (under the bolt), washers [5] (under the nut), and nuts [6] to attach the elevator feel computer assembly [1] to the airplane structure.

(c) Connect the stabilizer pushrod [2] to the input arm [8].

1) If the bolt [21] cannot be installed, do these steps to adjust the length of the stabilizer pushrod [2]:

a) Loosen the checknut [24] on the stabilizer pushrod [2].

b) Turn the rod end fitting of the stabilizer pushrod [2] until the bolt [21] can be installed easily by hand.

c) Make sure that the rod end extends more than half of the distance across the inspection hole in the stabilizer pushrod [2].

d) Tighten the checknut [24].

2) Install the washer [22] and nut [23] to the bolt [21].

(d) Connect the crank [25] to the elevator feel computer assembly [1].

(e) Install the bolt [28], washer [27], washer [31], and nut [26].

(f) Connect the spring [16] to the crank [25].

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CAUTION MAKE SURE THAT YOU CONNECT THE HYDRAULIC LINES TO THE CORRECT PORTS. IF YOU INSTALL THE HYDRAULIC LINES TO INCORRECT PORTS, YOU CAN CAUSE DAMAGE TO THE ELEVATOR FEEL COMPUTER.

(g) Connect the hydraulic lines to the elevator feel computer assembly [1] (TASK 20-10-51-400-801).

1) Remove the caps and plugs.

2) Connect the return hydraulic lines [29] to the elevator feel computer assembly [1].

a) Tighten the B-nuts on the return hydraulic lines [29] to 170 ±9 in-lb (19 ±1 N·m).

3) Connect the pressure hydraulic lines [30] to the elevator feel computer assembly [1].

a) Tighten the B-nuts on the pressure hydraulic lines [30] to 140 ±7 in-lb (16 ±1 N·m).

(h) Remove the caps and plugs from pitot lines and static lines.

(i) Connect the pitot lines and static lines to the fittings on the elevator feel computer assembly [1].

1) Tighten the pitot lines to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

(j) Connect the electrical connectors [7] to the elevator feel computer assembly [1].

SUBTASK 27-31-37-080-002

(8) Remove the rig pin E-4 of the rigging pin kit, SPL-1585, from the input arm [8] and elevator feel computer assembly [1].

SUBTASK 27-31-37-860-007

(9) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-37-860-021

(10) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	2	C01064	AFCS SYS B MACH TRIM DC
B	4	C00716	AFCS SYS B ENGAGE INTLK
C	1	C01037	AFCS SYS B MACH TRIM AC
C	2	C01042	AFCS SYS B SNSR EXC AC
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-31-37-860-022

(11) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	3	C01046	AFCS SYS B FCC DC

SUBTASK 27-31-37-870-001

(12) Bleed the elevator control system.

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(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

WARNING

MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER AND CONTROL CABLES IN THE FORWARD EQUIPMENT COMPARTMENT OR AREA AFT OF THE BULKHEAD. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Move the stabilizer to the maximum leading edge up (APL NOSE DOWN) position (the mechanical limits).

NOTE: This can be done with the control wheel STAB TRIM switches or the stabilizer trim control wheel.

1) Make sure that the ballnut is against the jackscrew upstop.

(c) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

CAUTION

MAKE SURE THAT THE PITOT PROBE HAS NO ADDED WEIGHT ON IT FROM THE TEST HOSE. THE WEIGHT OF THE TEST HOSE CAN BEND OR TWIST THE PITOT PROBE OUT OF TOLERANCE.

(d) Connect the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to the two pitot probes on the vertical fin.

1) Do this step for the regulator, SPL-1742:

a) Connect the pressure lines from the regulator, SPL-1742, with the pitot test adapter, COM-1916, to the pitot probes.

2) Do this step for the air data model test set, COM-1914:

a) Use the air data accessory kit, COM-25177, to connect the air data model test set, COM-1914, to the pitot test adapter, COM-1916.

NOTE: The air data accessory kit, COM-25177, contains hoses to connect to the air data model test set, COM-1914, and the pitot test adapter, COM-1916.

(e) Remove the restrictor valves from the bottom of the pitot lines.

(f) If the regulator, SPL-1742, or equivalent is used, do this step:

1) Connect the pressure gage lines from the regulator, SPL-1742, or equivalent, to the drain holes of the pitot lines.

(g) If the air data model test set, COM-1914, is used, do these steps:

1) Install plugs over the ends of the pitot lines.

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2) Install a highly visible streamer on the opening of the aft unpressurized compartment.

a) Make sure that the streamer can be easily seen from the ground.

NOTE: This is to remind to remove the plugs and install the restrictor valves again when the test is completed.

(h) Use the regulator, SPL-1742, or air data model test set, COM-1914, to raise the airspeed to 2.46 psig (169.61 mBar) (300 knots).

NOTE: When pressure to the pitot system is applied or removed, the rate must not be more than 2.46 psig (169.61 mBar) (300 knots) for each minute.

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(i) Pressurize Hydraulic System A and B, do one of these steps:

1) If a Portable Hydraulic Cart is used, do this task: Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, TASK 29-11-00-860-802.

NOTE: Two carts are recommended.

a) If only one cart is available, bleed hydraulic A and B, as follows:

<1> Pressurize hydraulic system A with a cart and hydraulic system B with the Electric Motor-Driven Pump (EMDP).

<2> Pressurize hydraulic system B with a cart and hydraulic system A with the EMDP.

2) If a Portable Hydraulic Cart is not used, do these tasks:

NOTE: Both the EMDP and Engine Driven Pump (EDP) should be run at the same time.

- Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), TASK 29-11-00-860-803
- Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), TASK 29-11-00-860-804.

(j) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

(k) Wait for 10 seconds.

(l) Do these steps to cycle the power between Hydraulic Systems A and B:

1) Set the flight controls hydraulic power switches as follows:

- A ON, B OFF - Hold for 2 seconds
- B ON, A OFF - Hold for 2 seconds.

2) Repeat the cycle 10 times.

(m) Slowly, do control column sweeps with pitot pressure applied.

(n) Do these steps to cycle pitot pressure:

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1) Use the regulator, SPL-1742, or air data model test set, COM-1914, connected at the elevator pitot probes on the vertical stabilizer, to lower the airspeed to 0 psig (0.00 mBar) (0 knots).

2) Use the regulator, SPL-1742, or air data model test set, COM-1914, connected at the elevator pitot probes on the vertical stabilizer, to increase the airspeed to 2.46 psig (169.61 mBar) (300 knots).

3) Use the regulator, SPL-1742, or air data model test set, COM-1914, connected at the elevator pitot probes on the vertical stabilizer, to lower the airspeed to 0 psig (0.00 mBar) (0 knots).

(o) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

(p) Wait for 10 seconds.

SUBTASK 27-31-37-860-008

(13) Remove the pressure from the Hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-31-37-210-001

(14) Make sure that there is no hydraulic leakage from the elevator feel computer (TASK 27-31-37-200-801).

SUBTASK 27-31-37-860-009

WARNING

MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER AND CONTROL CABLES IN THE FORWARD EQUIPMENT COMPARTMENT OR AREA AFT OF THE BULKHEAD. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(15) Move the horizontal stabilizer to the neutral stabilizer position (zero degrees or 4 units of trim).

NOTE: This can be done with the control wheel STAB TRIM switches or the stabilizer trim wheel.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-37-860-023

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	2	C01064	AFCS SYS B MACH TRIM DC
B	3	C01046	AFCS SYS B FCC DC
B	4	C00716	AFCS SYS B ENGAGE INTLK
C	1	C01037	AFCS SYS B MACH TRIM AC
C	2	C01042	AFCS SYS B SNSR EXC AC
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-37-860-018

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-31-37-020-002

(3) Do these steps at the two pitot probes on the vertical fin:

(a) Remove the pressure lines from the air data model test set, COM-1914, or regulator, SPL-1742, or equivalent from the pitot probes.

(b) Remove the pitot test adapter, COM-1916, from the pitot probes.

SUBTASK 27-31-37-420-010

(4) Do these steps:

(a) If the regulator, SPL-1742, or equivalent is used, do this step:

1) Disconnect the pressure gage lines of the regulator, SPL-1742, or equivalent, to the drain holes of the pitot lines.

(b) If the air data model test set, COM-1914, is used, do these steps:

1) Remove the plugs over the ends of the pitot lines.

2) Remove the highly visible streamer on the opening of the aft unpressurized compartment.

(c) Install the restrictor valves on the bottom of the pitot lines.

SUBTASK 27-31-37-730-001

(5) Do this task: Elevator Pitot System Leak - Test, TASK 27-31-17-790-801.

SUBTASK 27-31-37-730-002

(6) Do this task: Feel Differential Pressure Light - Test, TASK 27-31-00-700-812.

SUBTASK 27-31-37-730-003

(7) Do this task: Feel Force System - Test, TASK 27-31-00-700-813.

SUBTASK 27-31-37-410-001

(8) Close this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

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ELEVATOR FEEL COMPUTER - INSPECTION/CHECK

1. General

A. This procedure contains this task:

(1) Elevator Feel Computer Leakage Check

(2) Elevator Feel Computer Solenoid Valve Inspection

TASK 27-31-37-200-801

2. Elevator Feel Computer Leakage Check

(Figure 27-31-37-990-801)

A. General

(1) Use this task to do a check for leakage at the elevator feel computer.

B. References

Reference	Title
20-10-51-400-801	Flareless Tubing Assembly Installation (P/B 401)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-37 P/B 401	ELEVATOR FEEL COMPUTER - REMOVAL/INSTALLATION
27-31-37-990-801	Figure: Elevator Feel Computer Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
G01043	Cloth - Lint-free	
G50641	Swab - ITW Texwipe TX762 Absorbond Swab With Long Handle	

D. Elevator Feel Computer Leakage Check

SUBTASK 27-31-37-200-001

(1) Do these steps to do a check for leakage:

(a) Clean any residual hydraulic fluid from the elevator feel computer with a lint-free cloth, G01043.

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Pressurize the elevator hydraulic system A and B for one minute and make sure that there is no leakage (TASK 27-31-00-800-801).

(c) Move the Captain's or First Officer's yoke forward then aft then center (neutral).

(d) Visually examine the external surface of the elevator feel computer static tubes for signs of hydraulic fluid.

(e) Visually examine the supply, return and feel pressure tubing B-nuts connection to the elevator feel computer.

(f) Remove pressure from elevator hydraulic system A and B (TASK 27-31-00-800-802).

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(g) If there is no sign of hydraulic fluid at these locations, then continue.

(h) If there is a sign of hydraulic fluid at these locations, the B-nuts may not be correctly installed or may have become loose.

1) Make sure that the B-nut is seated and tightened (TASK 20-10-51-400-801).

(i) Use a lint-free cloth, G01043, to clean off the ends of the static lines of hydraulic fluid.

(j) From the open end of the static lines, carefully put a Texwipe TX762 swab, G50641 (Q-tip), approximately 1 in. (25 mm) or more into the static lines.

1) Wipe the inside surface in the static lines.

2) Remove the Texwipe TX762 swab, G50641 (Q-tip), and visually examine it for signs of hydraulic fluid.

a) If the leakage rate exceeds 1 drop per hour, replace the elevator feel computer. Do this task: ELEVATOR FEEL COMPUTER - REMOVAL/INSTALLATION, PAGEBLOCK 27-31-37/401.

SUBTASK 27-31-37-840-001

(2) Put the airplane back to its usual condition.

——— END OF TASK ———

TASK 27-31-37-200-802

3. Elevator Feel Computer Solenoid Valve Inspection

A. General

(1) This procedure contains steps to inspect the elevator feel computer solenoid valve.

(2) The elevator feel computer solenoid valve is referred to as the solenoid valve in this procedure.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
27-31-37 P/B 201	ELEVATOR FEEL COMPUTER - MAINTENANCE PRACTICES

C. Tools/Equipment

Reference	Description
STD-6151	Gaussmeter

D. Procedure

SUBTASK 27-31-37-860-014

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 27-31-37-860-012

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

SUBTASK 27-31-37-240-001

(3) Put the gaussmeter, STD-6151, below the A-side solenoid valve to measure its magnetic field.

(a) Make sure that there is no magnetic field.

NOTE: The gauss meter must show 0.

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SUBTASK 27-31-37-240-002

(4) Put the gaussmeter, STD-6151, below the B-side solenoid valve to measure its magnetic field.

(a) Make sure that there is no magnetic field.

NOTE: The gauss meter must show 0.

SUBTASK 27-31-37-860-013

(5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

SUBTASK 27-31-37-240-003

(6) Put the gaussmeter, STD-6151, below the A-side solenoid valve to measure its magnetic field.

(a) Record the value of the A-side solenoid valve magnitude.

SUBTASK 27-31-37-240-004

(7) Put the gaussmeter, STD-6151, below the B-side solenoid valve to measure its magnetic field.

(a) Record the value of the B-side solenoid valve magnitude.

SUBTASK 27-31-37-240-005

(8) Compare the A-side and B-side solenoid valve magnitude values.

(a) If one of the solenoid valves shows a much less magnitude or if it shows 0, then replace this solenoid valve (PAGEBLOCK 27-31-37/201).

——— END OF TASK ———

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ELEVATOR BALANCE PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the elevator balance panel

(2) An installation of the elevator balance panel.

B. The elevator balance panel is referred to as the "balance panel" in this procedure.

TASK 27-31-41-000-801

2. Elevator Balance Panel - Removal

(Figure 401, Figure 402)

A. General

(1) This task includes the steps to remove elevator balance panel.

B. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
55-10-11-000-801	Balance Bay Panels - Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
316	APU Compartment - Right
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge

E. Access Panels

Number	Name/Location
318BR	Tailcone Access Door
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
333CB	Horizontal Stabilizer, Access Panel, Trailing Edge
333DB	Horizontal Stabilizer, Access Panel, Trailing Edge
333EB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
343CB	Horizontal Stabilizer, Access Panel - T.E. Area
343DB	Horizontal Stabilizer, Access Panel - T.E. Area
343EB	Horizontal Stabilizer, Access Panel - T.E. Area

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F. Prepare for the Removal

SUBTASK 27-31-41-860-001

(1) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-41-010-001

(2) Remove this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-41-010-002

(3) For the left elevator, open these access panels:

(TASK 55-10-11-000-801)

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
333CB	Horizontal Stabilizer, Access Panel, Trailing Edge
333DB	Horizontal Stabilizer, Access Panel, Trailing Edge
333EB	Horizontal Stabilizer, Access Panel, Trailing Edge

SUBTASK 27-31-41-010-003

(4) For the right elevator, open these access panels:

(TASK 55-10-11-000-801)

Number	Name/Location
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
343CB	Horizontal Stabilizer, Access Panel - T.E. Area
343DB	Horizontal Stabilizer, Access Panel - T.E. Area
343EB	Horizontal Stabilizer, Access Panel - T.E. Area

SUBTASK 27-31-41-480-001

(5) Install the rig pin E-5 from the rigging pin kit, SPL-1585, in the aft control quadrant (Figure 401).

G. Elevator Balance Panel Removal

SUBTASK 27-31-41-020-001

CAUTION

DO NOT MIX THE ELEVATOR BALANCE PANELS. IF YOU MIX THE ELEVATOR BALANCE PANELS, DAMAGE TO THE AIRCRAFT CAN OCCUR.

(1) Remove the balance panel [10] (bays 2, 3, and 4).

(a) Remove the bolts [15], bolts [22], and washers [16] that attach the bulb seals [17] to the balance panel [10].

(b) Remove the washers [14] and bolts [13] that attach the balance panel [10] to the idler hinge [18].

(c) Remove the washers [11] and bolts [12] that attach the balance panel [10] to the elevator [6].

(d) Remove the washer [20] and bolt [21] that attach the balance panel [10] to the elevator [6].

(e) Remove the balance panel [10] from the airplane.

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SUBTASK 27-31-41-020-002

(2) Remove the idler hinge [18] (bays 2, 3, and 4).

(a) Remove the washers [8] and bolts [7] that attach the flat seals [5] to the idler hinge [18].

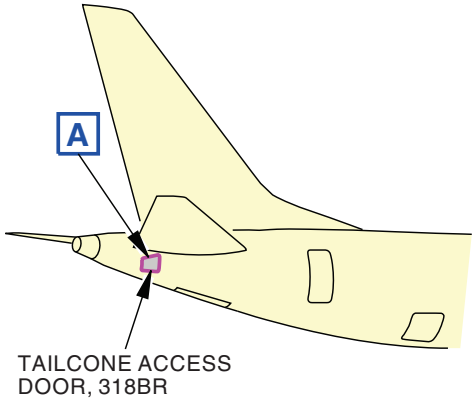
(b) Remove the flat seals [5] from the airplane.

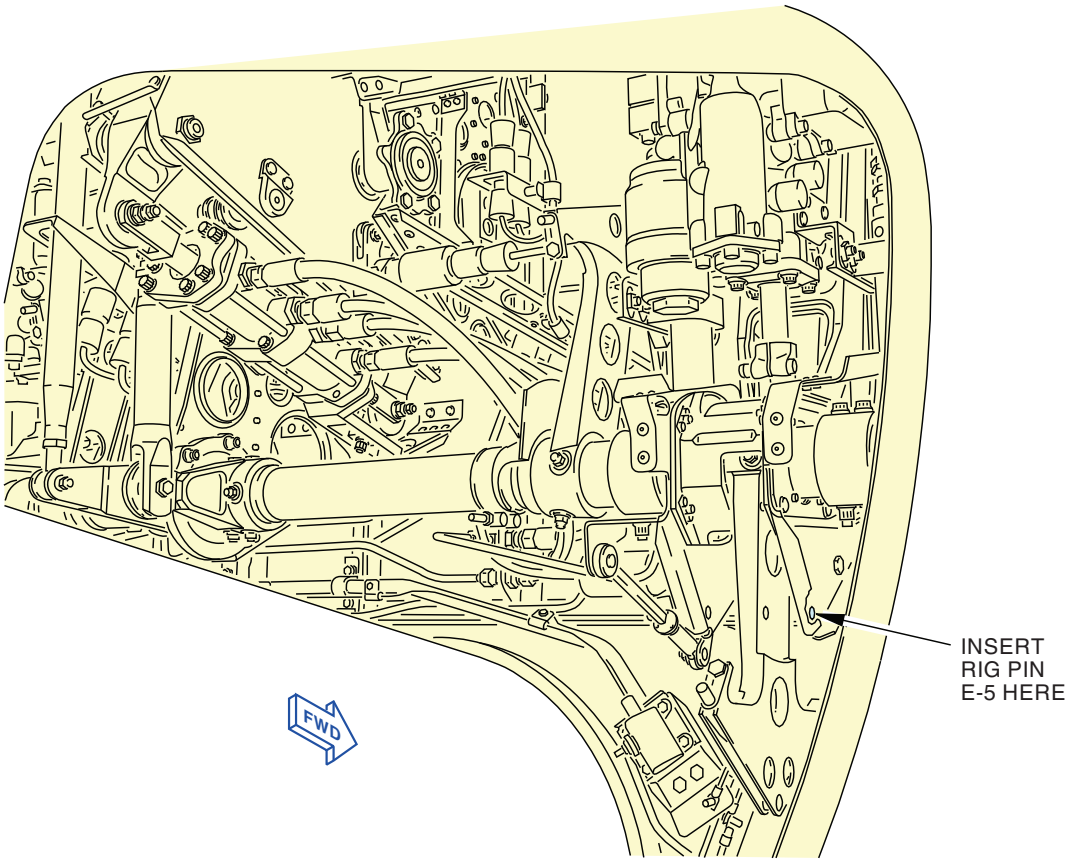
(c) Remove the washers [4] and bolts [3] that attach the hinge half [2] to the balance panel support [1].

(d) Remove the idler hinge [18] and seal cloth [19] from the airplane.

——— END OF TASK ———

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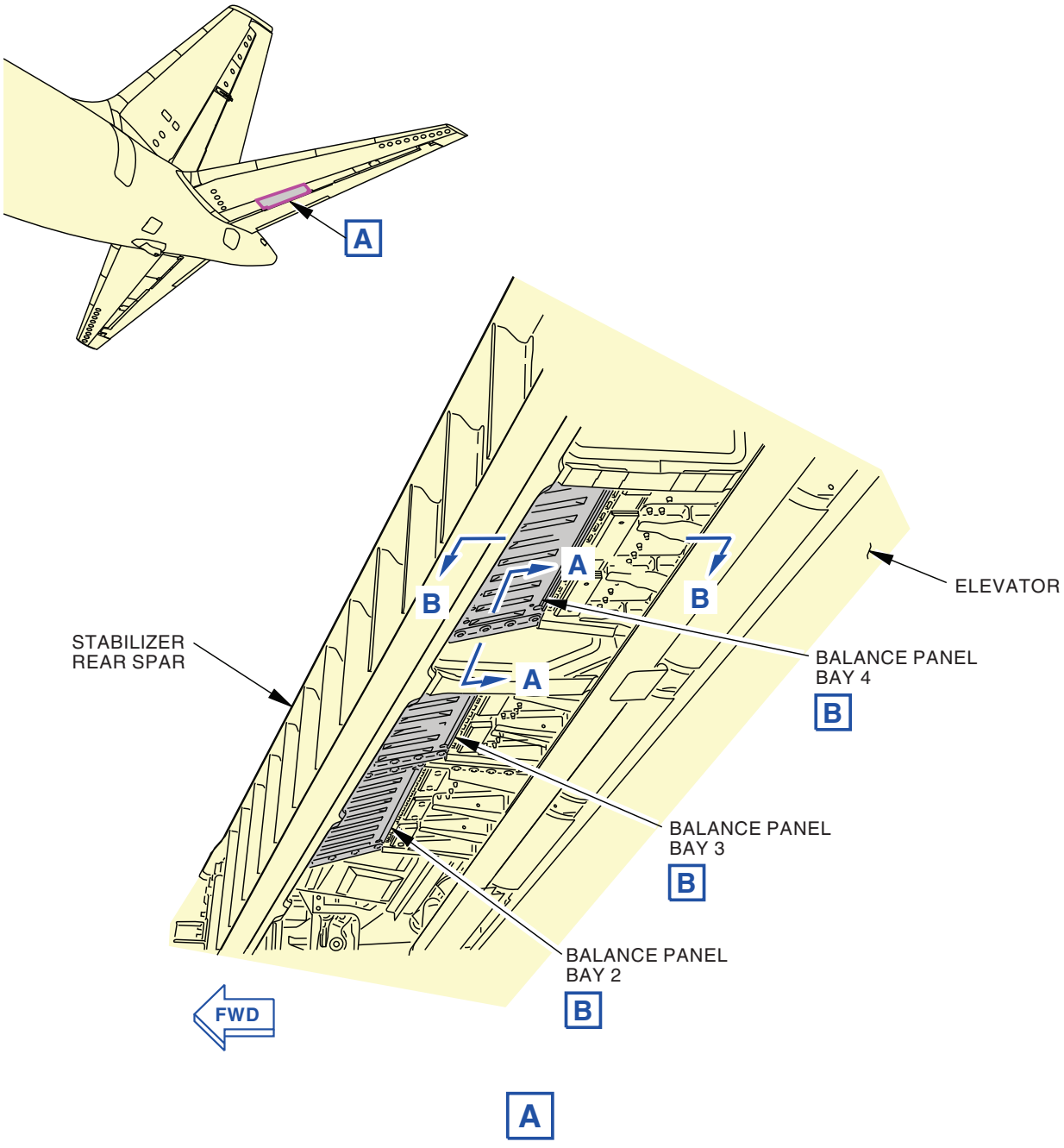
VIEW THROUGH THE TAILCONE ACCESS DOOR, 318BR

A

2412553 S00061530892_V1

Elevator Rig Pin Installation
Figure 401/27-31-41-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

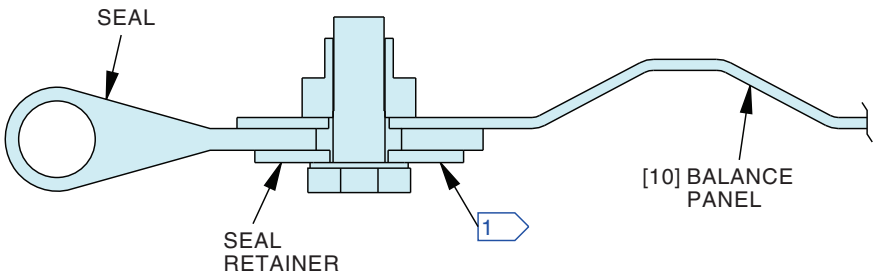


NOTE:
LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE.

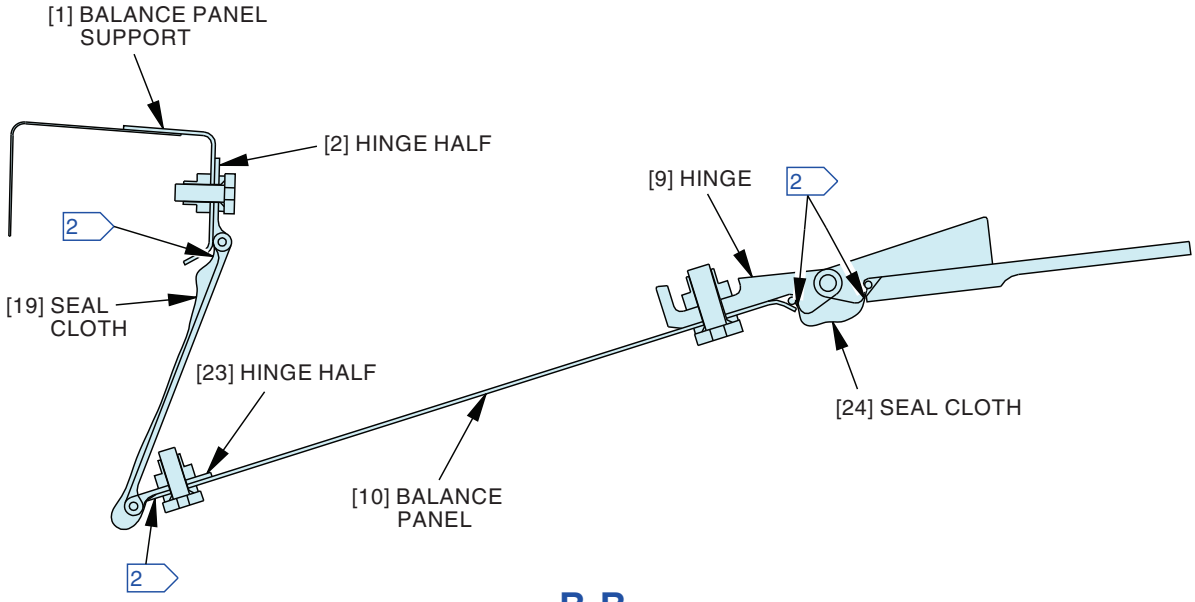
2412554 S00061530893_V2

Elevator Balance Panel Installation
Figure 402/27-31-41-990-802 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



B-B

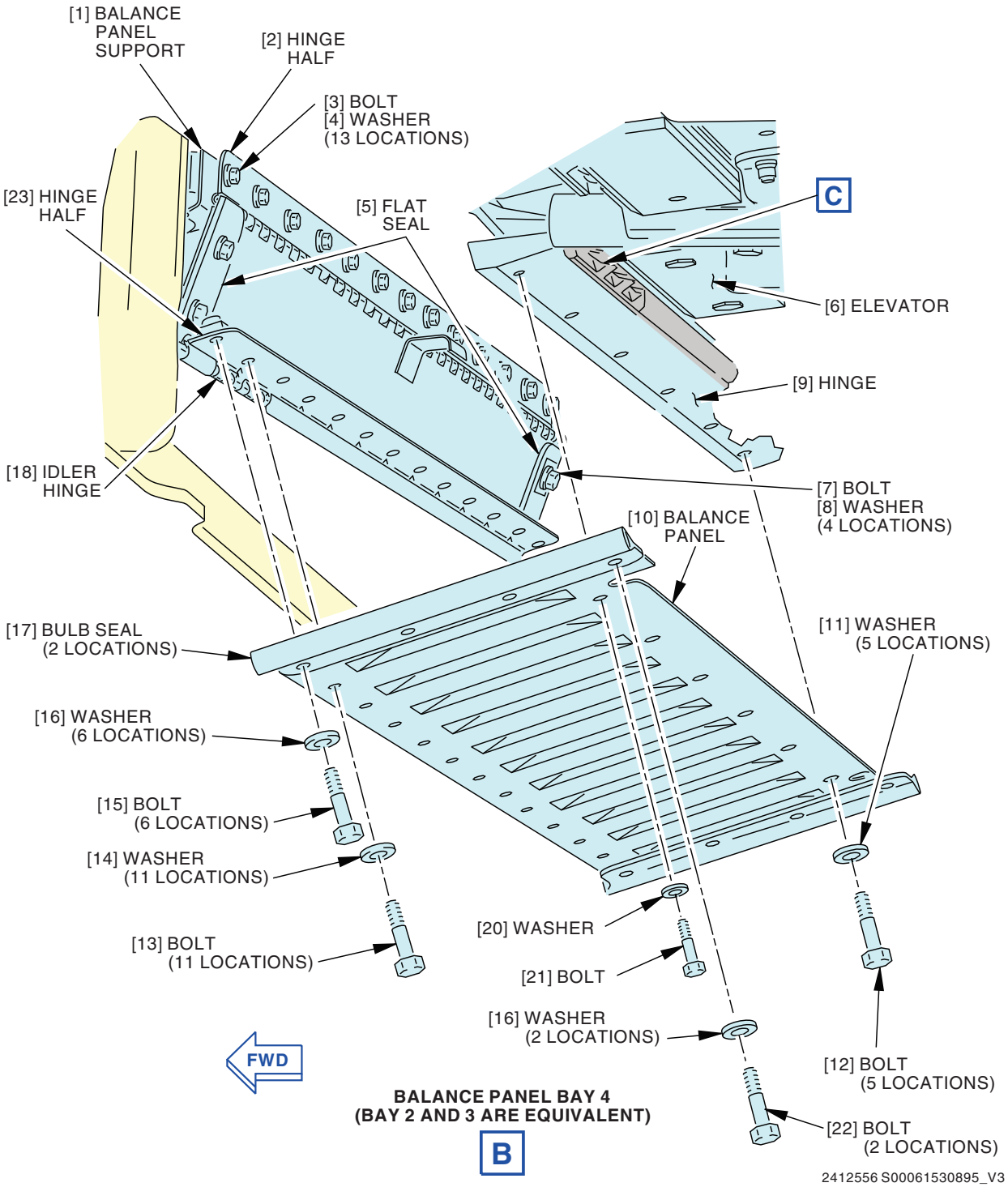
1 APPLY BMS 3-23 BY BRUSH ONLY TO ALL EDGES OF THE SEAL RETAINER AND INTERFACE BETWEEN THE SEAL RETAINER AND THE SEAL. THIS ONLY NEEDS TO BE APPLIED WHEN NEW COMPONENTS ARE INSTALLED.

2 APPLY BMS 3-23 BY BRUSH ONLY TO ALL SURFACES WHERE THE BALANCE PANEL SURFACES CONTACT THE SEAL CLOTH. THIS ONLY NEEDS TO BE APPLIED WHEN NEW COMPONENTS ARE BEING INSTALLED. IF YOU INSTALL THE SAME PART THAT WAS REMOVED, YOU DO NOT NEED TO APPLY THE BMS 3-23.

2412555 S00061530894_V2

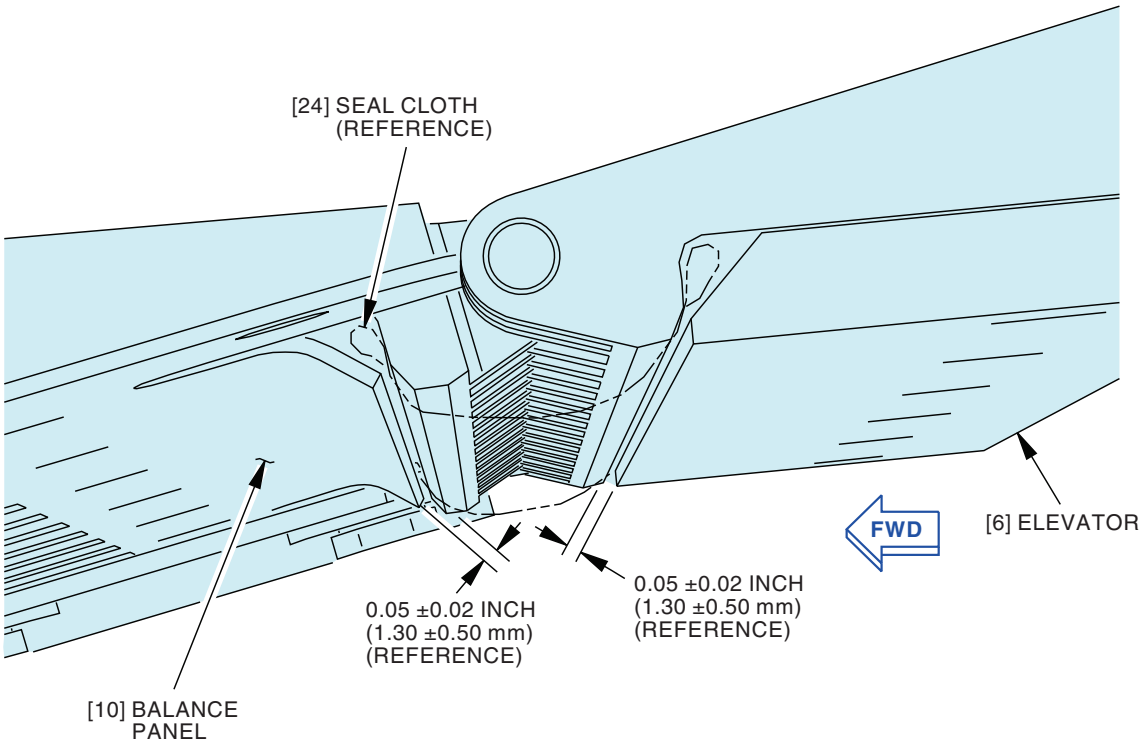
Elevator Balance Panel Installation
Figure 402/27-31-41-990-802 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Balance Panel Installation
Figure 402/27-31-41-990-802 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



C

2412557 S00061530896_V2

Elevator Balance Panel Installation
Figure 402/27-31-41-990-802 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-41-400-801

3. Elevator Balance Panel - Installation

(Figure 401, Figure 402)

A. General

(1) This task includes the steps to install the elevator balance panel.

B. References

Reference	Title
20-50-11-910-801	Standard Torque Values (P/B 201)
27-31-00-700-804	Elevator Balance Panels - Test (P/B 501)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
55-10-11-400-801	Balance Bay Panels - Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
G00009	Compound - Organic Corrosion Inhibiting	BMS3-23

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
10	Balance panel	27-31-41-01-010	PTF ALL
		27-31-41-01-015	PTF ALL
		27-31-41-01-170	PTF ALL
		27-31-41-01-175	PTF ALL
		27-31-41-01-330	PTF ALL
		27-31-41-01-335	PTF ALL
18	Idler hinge	27-31-41-01-140	PTF ALL
		27-31-41-01-145	PTF ALL
		27-31-41-01-300	PTF ALL
		27-31-41-01-305	PTF ALL
		27-31-41-01-460	PTF ALL
		27-31-41-01-465	PTF ALL

F. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
316	APU Compartment - Right
333	Left Horizontal Stabilizer - Rear Spar to Trailing Edge

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(Continued)

Zone	Area
343	Right Horizontal Stabilizer - Rear Spar to Trailing Edge

G. Access Panels

Number	Name/Location
318BR	Tailcone Access Door
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
333CB	Horizontal Stabilizer, Access Panel, Trailing Edge
333DB	Horizontal Stabilizer, Access Panel, Trailing Edge
333EB	Horizontal Stabilizer, Access Panel, Trailing Edge
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
343CB	Horizontal Stabilizer, Access Panel - T.E. Area
343DB	Horizontal Stabilizer, Access Panel - T.E. Area
343EB	Horizontal Stabilizer, Access Panel - T.E. Area

H. Elevator Balance Panel Installation

SUBTASK 27-31-41-860-002

(1) Make sure that you remove the pressure from the elevator hydraulic system A and B.

(a) If it is necessary, do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-41-420-001

(2) Install the idler hinge [18] (bays 2, 3, and 4).

(a) If you install new balance panel assemblies or seals, apply corrosion inhibiting compound, G00009, as necessary (Figure 402).

NOTE: If you use the same parts that you removed, it is not necessary to apply the corrosion inhibiting compound, G00009.

(b) Position one loose end of the seal cloth [19] between the hinge half [2] and balance panel support [1].

(c) Install the bolts [3] and washers [4] that attach the hinge half [2] to the balance panel support [1].

(d) Tighten the bolt [3] to 23 in-lb (3 N·m) - 27 in-lb (3 N·m).

(e) Install the bolts [7] and washers [8] to attach the flat seals [5] to the idler hinge [18].

1) Do not tighten the bolts [7] until the clearance between the flat seals [5] and the stabilizer trailing edge rib is adjusted.

(f) Make sure that the clearance between the flat seals [5] and stabilizer trailing edge rib is 0.01 +0.06 / -0.00 in. (0.25 +1.53 / -0.00 mm).

(g) Tighten the bolts [7] (TASK 20-50-11-910-801).

SUBTASK 27-31-41-420-002

CAUTION MAKE SURE THAT YOU INSTALL THE CORRECT BALANCE PANEL IN THE CORRECT BAY. BALANCE PANELS IN BAYS 2, 3, AND 4 ARE NOT INTERCHANGEABLE. IF YOU DO NOT FOLLOW THIS CAUTION, DAMAGE TO THE AIRCRAFT CAN OCCUR.

(3) Install the balance panel [10].

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(a) If you install new balance panel assemblies or seals, apply corrosion inhibiting compound, G00009, as necessary (Figure 402).

NOTE: If you use the same parts that you removed, it is not necessary to apply corrosion inhibiting compound, G00009.

(b) Put and hold the balance panel [10] in its position.

(c) Position the loose end of the seal cloth [24] between the balance panel [10] and hinge [9].

(d) Install the bolt [21] and washer [20] to attach the balance panel [10] and hinge [9] to the elevator [6].

(e) Install the bolts [12] and washers [11] to attach the balance panel [10] and hinge [9] to the elevator [6].

1) Do not tighten the bolts [12] and bolt [21] until the gap between the balance panel [10] and hinge [9] is adjusted.

2) Carefully install the seal cloth [24] between the hinge [9] and balance panel [10].

3) Make sure that the gap between the balance panel [10] and hinge [9] for the seal cloth [24] does not allow either of the following conditions:

a) Pinching of the seal material.

b) Movement of the seal inboard or outboard.

NOTE: The nominal design dimension for the gap between the balance panel and hinge for the seal cloth that meets this condition is 0.05 ±0.02 in. (1.3 ±0.5 mm).

(f) Tighten the bolts [12] to 68 in-lb (8 N·m) - 82 in-lb (9 N·m).

(g) Tighten the bolt [21] to 23 in-lb (3 N·m) - 27 in-lb (3 N·m).

(h) Make sure that the idler hinge [18] is turned down from the balance panel support [1].

(i) Position the loose end of the seal cloth [19] between the hinge half [23] and balance panel [10].

(j) Install the bolts [13] and washers [14] to attach the balance panel [10] to the hinge half [23].

1) Do not tighten the bolts [13] until the elevator has moved through the full range of travel.

(k) Manually move the elevator through the full range of travel.

1) Make sure that the seal cloth [19] is not tight for all elevator positions.

(l) Tighten the bolts [13] to 23 in-lb (3 N·m) - 27 in-lb (3 N·m).

(m) Install the bolts [15], bolts [22], and washers [16] that attach the bulb seals [17] to the balance panel [10].

1) Do not tighten the bolts [15] and bolts [22] until the clearance between the flat seals [5] and stabilizer trailing edge rib is adjusted.

CAUTION

MAKE SURE THAT YOU ADJUST THE BULB SEALS CORRECTLY. IF YOU DO NOT ADJUST THE BULB SEALS CORRECTLY, DAMAGE TO THE TRAILING EDGE RIBS CAN OCCUR.

(n) Make sure that the clearance between the bulb seals [17] and stabilizer rib is 0.02 ±0.01 in. (0.51 ±0.26 mm).

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(o) Make sure that the clearance between the bulb seals [17] and aft end of the flat seals [5] is 0.02 in. (0.51 mm).

1) If it is necessary, loosen the bolts [7] and adjust the flat seals [5] to get the necessary clearance.

(p) Tighten the bolts [15] and bolts [22] to 23 in-lb (3 N·m) - 27 in-lb (3 N·m).

SUBTASK 27-31-41-080-001

(4) Remove the rig pin E-5 of the rigging pin kit, SPL-1585, from the aft control quadrant (Figure 401).

SUBTASK 27-31-41-710-001

(5) Do this task: Elevator Balance Panels - Test, TASK 27-31-00-700-804.

SUBTASK 27-31-41-010-004

(6) Install this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-41-410-001

(7) For the left elevator, close these access panels:

(TASK 55-10-11-400-801)

Number	Name/Location
333BB	Horizontal Stabilizer, Access Panel, Trailing Edge
333CB	Horizontal Stabilizer, Access Panel, Trailing Edge
333DB	Horizontal Stabilizer, Access Panel, Trailing Edge
333EB	Horizontal Stabilizer, Access Panel, Trailing Edge

SUBTASK 27-31-41-410-002

(8) For the right elevator, close these access panels:

(TASK 55-10-11-400-801)

Number	Name/Location
343BB	Horizontal Stabilizer, Access Panel - T.E. Area
343CB	Horizontal Stabilizer, Access Panel - T.E. Area
343DB	Horizontal Stabilizer, Access Panel - T.E. Area
343EB	Horizontal Stabilizer, Access Panel - T.E. Area

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR BALANCE PANEL - INSPECTION/CHECK

1. General

A. This task supplies the data for a dimensional inspection of the elevator balance panel.

B. This task does not remove or install the elevator balance panel. Refer to the Elevator Balance Panel - Removal/Installation for procedures to remove and install the elevator balance panel.

TASK 27-31-41-200-801

2. Elevator Balance Panel - Inspection

(Figure 601)

A. General

(1) This task gives the wear limits and the repair data for the elevator balance panel.

(2) This task also examines the elevator balance panel for loose or missing attached fasteners.

B. References

Reference	Title
27-31-41-000-801	Elevator Balance Panel - Removal (P/B 401)
27-31-41-400-801	Elevator Balance Panel - Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
334	Left Horizontal Stabilizer - Elevator
344	Right Horizontal Stabilizer - Elevator

E. Procedure

SUBTASK 27-31-41-211-001

(1) Examine the elevator balance panel for missing or loose fasteners.

(a) If a fastener is loose or missing, then replace it with a new fastener.

SUBTASK 27-31-41-020-003

(2) Remove the elevator balance panel. Do this task: Elevator Balance Panel - Removal, TASK 27-31-41-000-801.

SUBTASK 27-31-41-020-004

(3) Use these tools to do a check for worn areas on the parts of the elevator balance panel connections:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-31-41-020-005

(4) Examine the parts of the elevator balance panel connections for worn areas. Do these steps:

(a) Remove the parts that you will examine.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096, or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

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(c) Compare the dimensions that you measured with the permitted wear dimensions that are shown in Table 601.

(d) If a part is out of tolerance, then replace or repair the part.

Table 601/27-31-41-993-802

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS inches (mm)		WEAR LIMITS inches (mm)	
		MINIMUM	MAXIMUM	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 FORWARD HINGE HALF	I.D.	0.0930 (2.362)	0.0980 (2.489)	0.1040 (2.642)	0.0175 (0.445) *[1]
#1 FORWARD HINGE PIN	O.D.	0.0880 (2.235)	0.0900 (2.286)	0.0865 (2.197)	
#2 AFT HINGE HALF	I.D.	0.1830 (4.648)	0.1880 (4.775)	0.1940 (4.928)	0.0175 (0.445) *[1]
#2 AFT HINGE PIN	O.D.	0.1780 (4.521)	0.1800 (4.572)	0.1765 (4.483)	

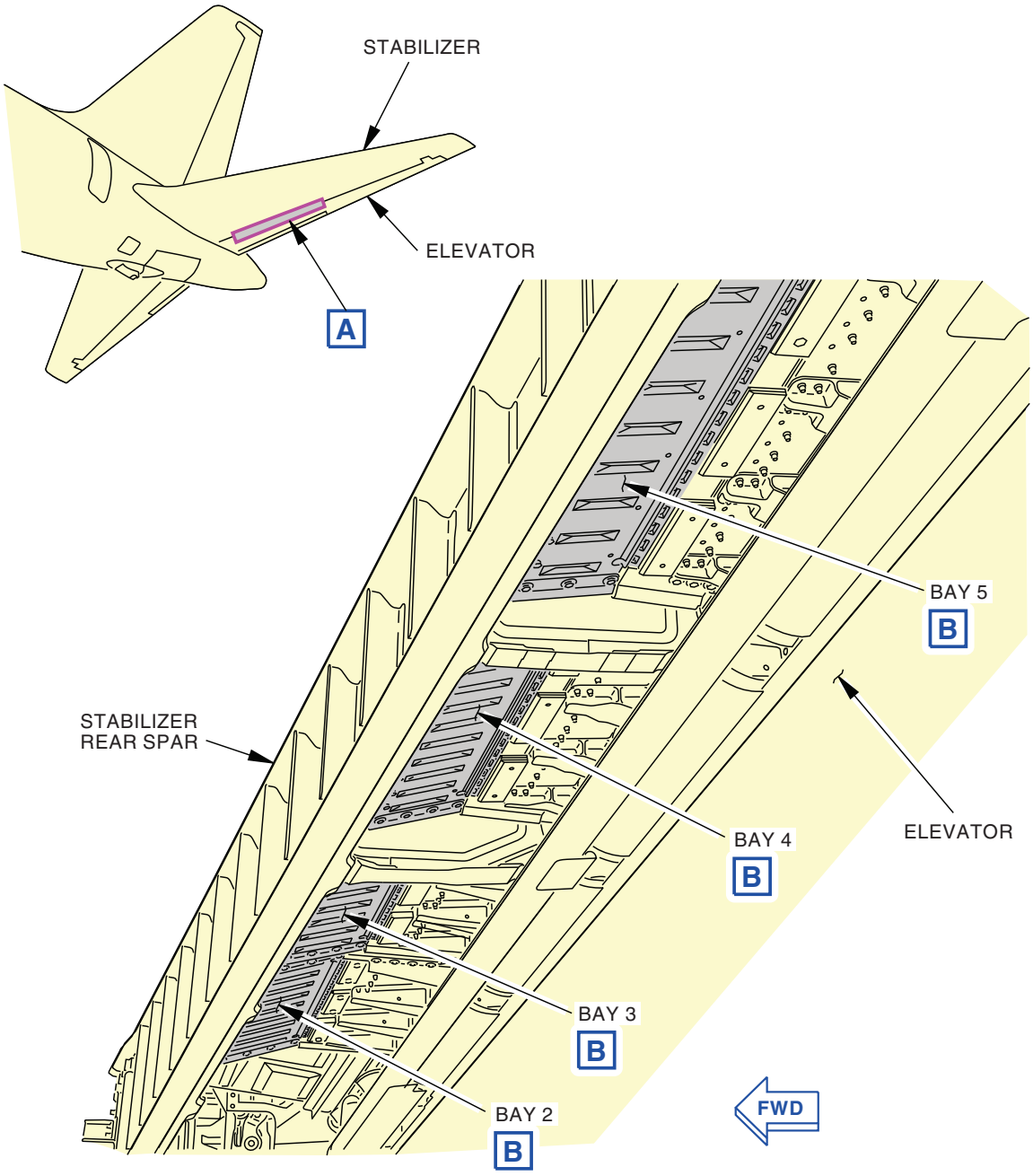
*[1] TOTAL PLAY IN THE JOINT = 0.0350 INCH (0.889 MILLIMETER)

SUBTASK 27-31-41-020-006

(5) Install the elevator balance panel. Do this task: Elevator Balance Panel - Installation, TASK 27-31-41-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



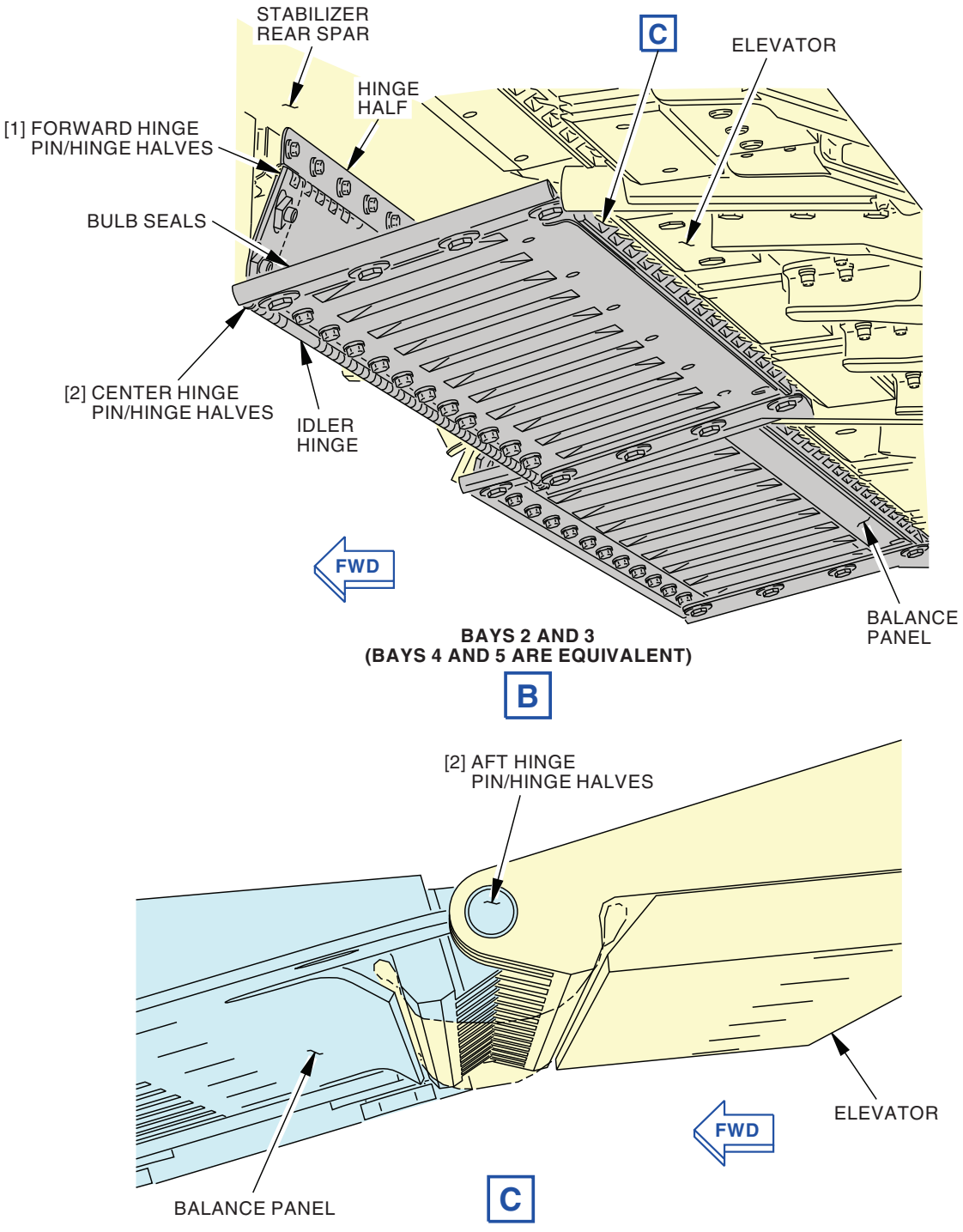
NOTE:

LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT.

2412558 S00061530900_V1

Elevator Balance Panel Wear Limits
Figure 601/27-31-41-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Balance Panel Wear Limits
Figure 601/27-31-41-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CONTROL COLUMN - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of a control column

(2) An installation of a control column.

B. This procedure is applicable for the captain's and first officer's control column.

TASK 27-31-51-000-801

2. Control Column - Removal

(Figure 401)

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door

E. Prepare for the Removal

SUBTASK 27-31-51-860-001

(1) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-51-860-002

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	7	C01208	STALL WARN ASYM MODE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT

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(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-31-51-010-001

(3) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-31-51-480-001

(4) Install the rig pin E-1 from the rigging pin kit, SPL-1585 in the left forward quadrant.

F. Control Column Removal

SUBTASK 27-31-51-020-001

(1) Disconnect the electrical wire [11] for the control column shaker [13] from the terminal strip in the lower nose compartment.

SUBTASK 27-31-51-020-002

(2) Disconnect the electrical wire [5] to the control column [1] from the terminal strip in the lower nose compartment.

SUBTASK 27-31-51-020-003

(3) Remove the ring assemblies and dust seals:

(a) Remove the screws [15] and washers [16] from the forward ring assembly [14] and aft ring assembly [17].

(b) Remove the forward ring assembly [14] and aft ring assembly [17].

(c) Remove the aft dust seal [18] and forward dust seal [19].

SUBTASK 27-31-51-020-004

(4) Remove the control column [1]:

CAUTION

HOLD THE CONTROL COLUMN WHEN YOU REMOVE THE COUPLINGS. THE CONTROL COLUMN CAN FALL AND CAUSE DAMAGE TO EQUIPMENT.

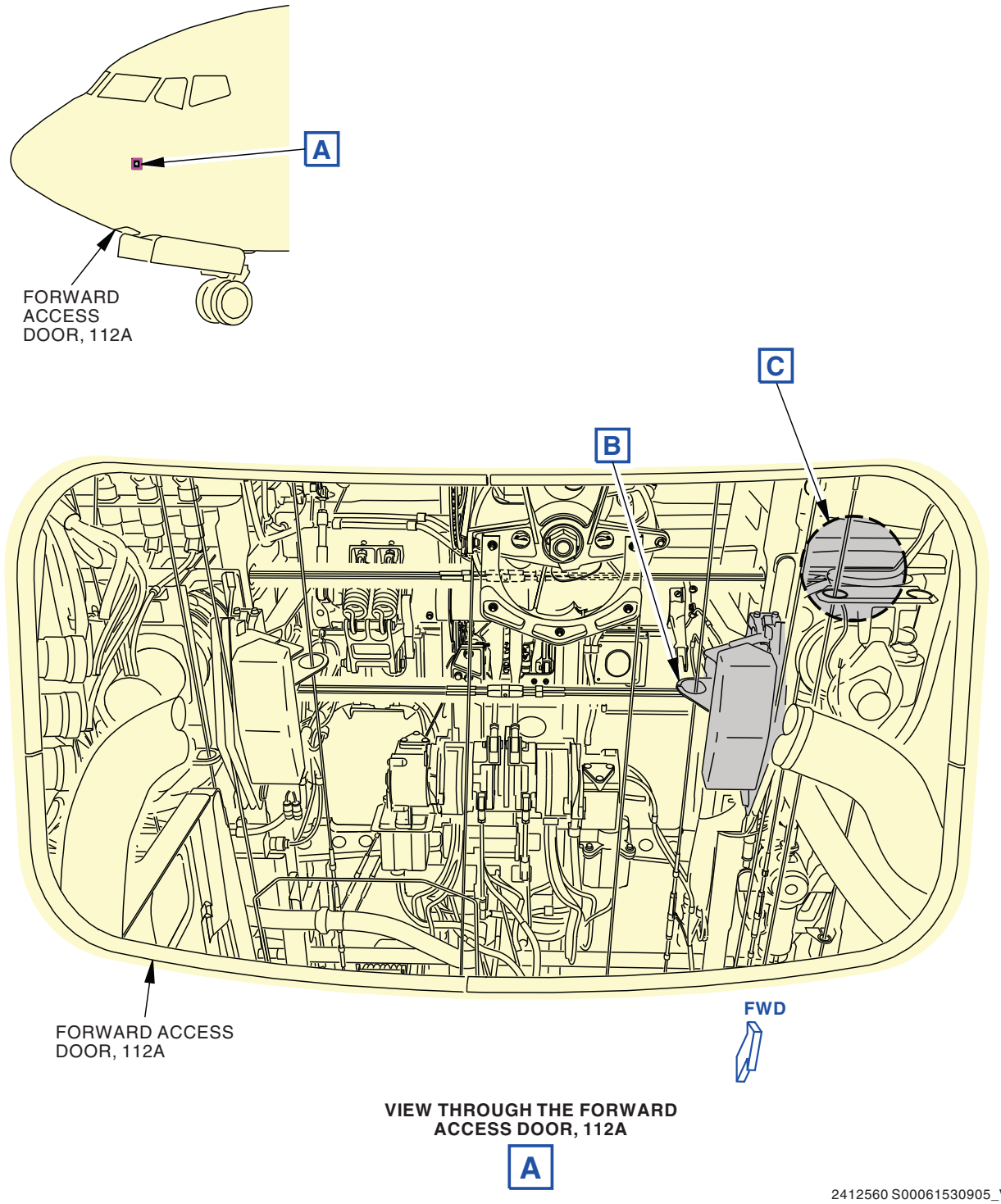
(a) Hold the control column while you remove the nuts [7], washers [8], and bolts [12] from the outboard couplings [9].

(b) Remove the outboard couplings [9].

(c) Lift and remove the control column [1] from the forward quadrant assembly.

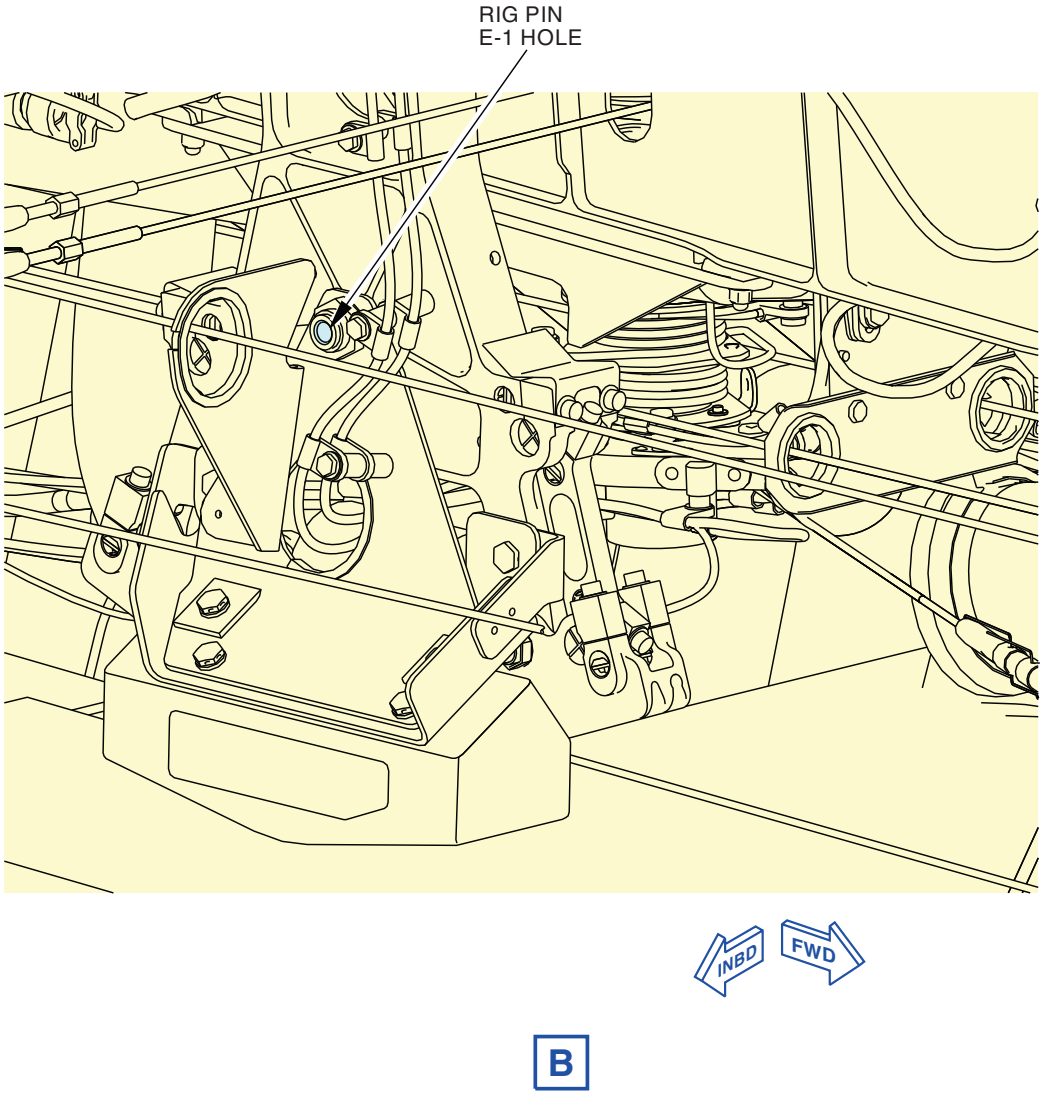
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Control Column Installation
Figure 401/27-31-51-990-801 (Sheet 1 of 7)

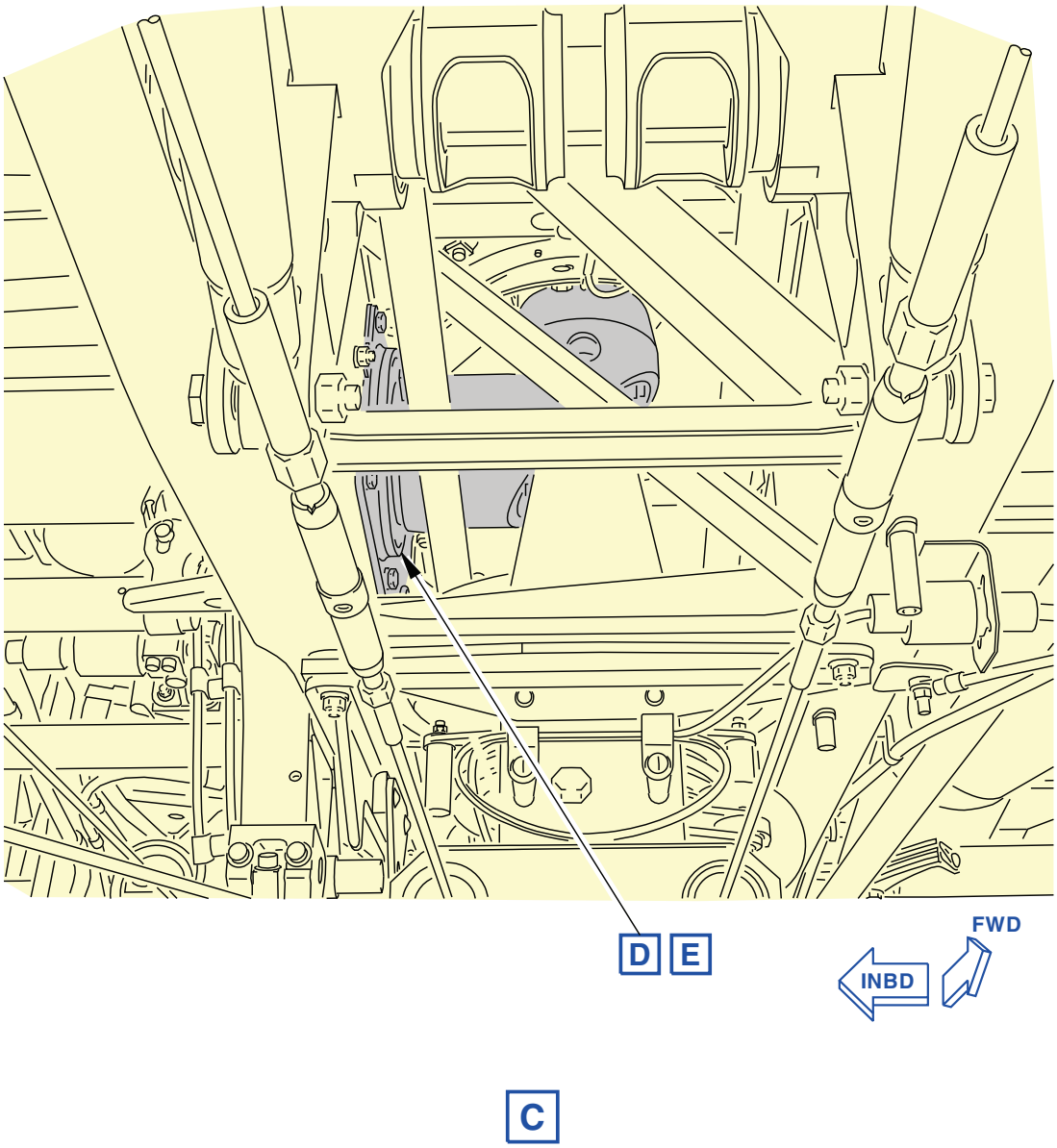
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412561 S00061530906_V2

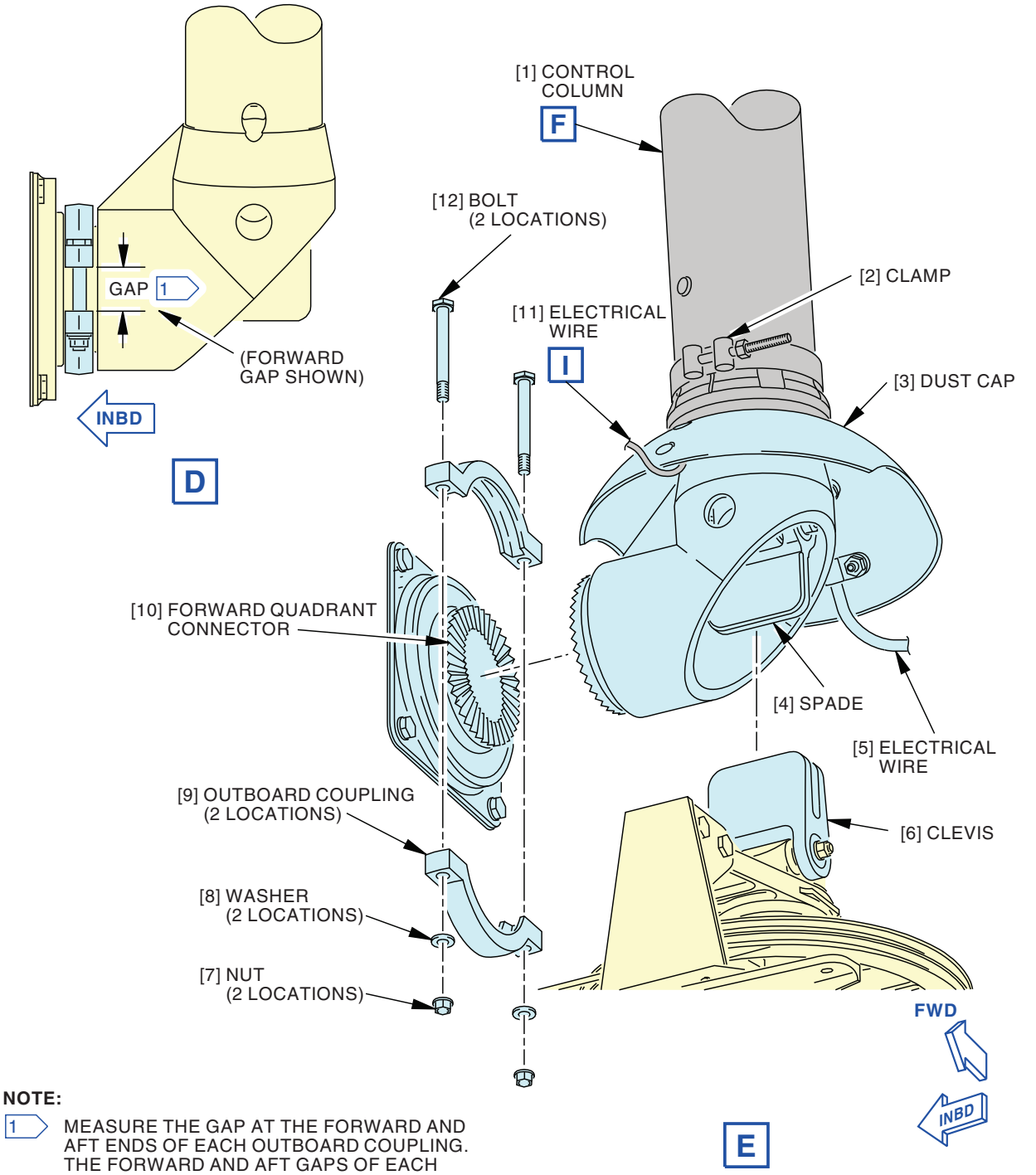
Control Column Installation
Figure 401/27-31-51-990-801 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Control Column Installation
Figure 401/27-31-51-990-801 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



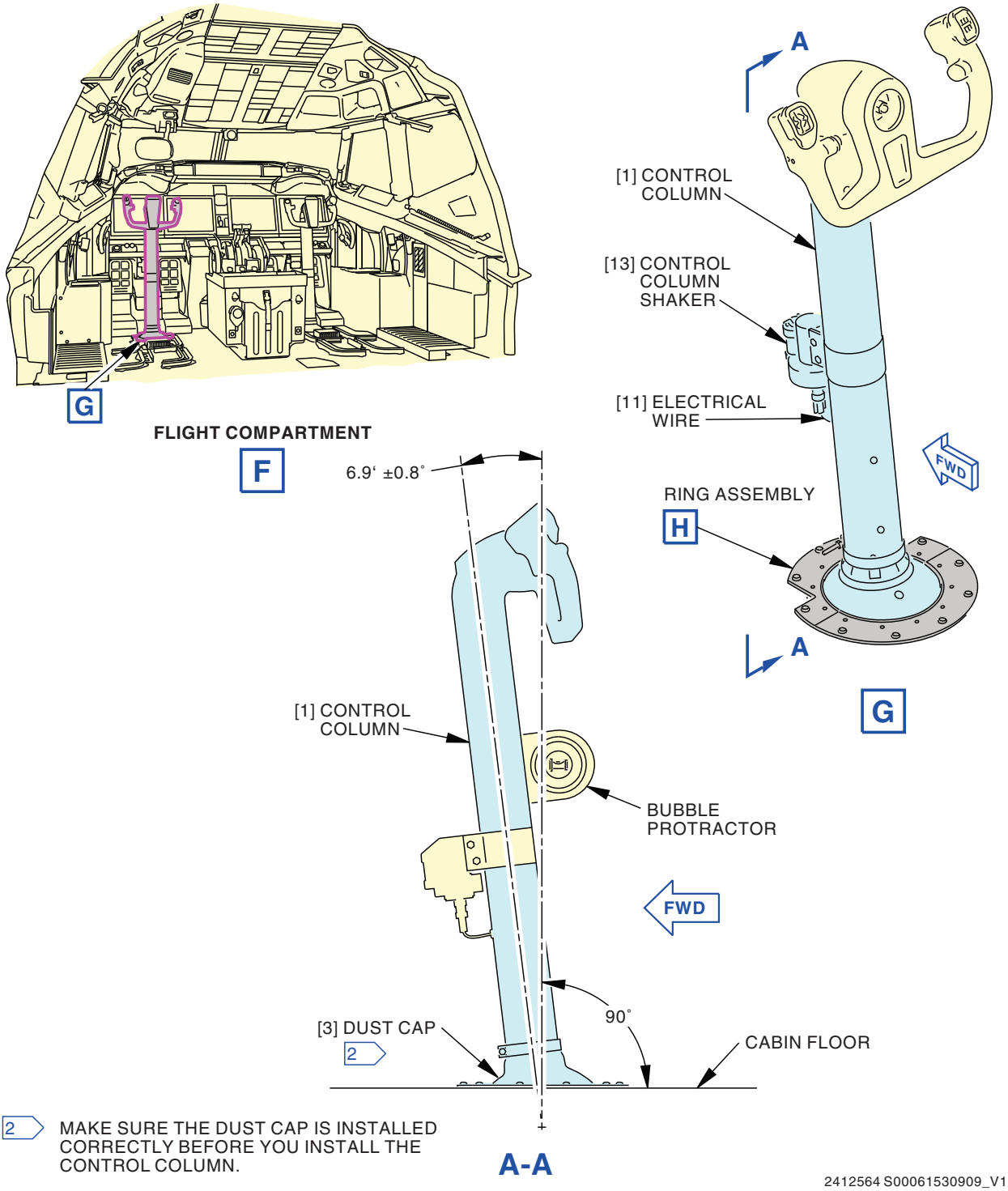
NOTE:

1 MEASURE THE GAP AT THE FORWARD AND AFT ENDS OF EACH OUTBOARD COUPLING. THE FORWARD AND AFT GAPS OF EACH OUTBOARD COUPLING SHOULD BE WITHIN 0.03 INCH (0.8 mm) OF EACH OTHER.

2412563 S00061530908_V2

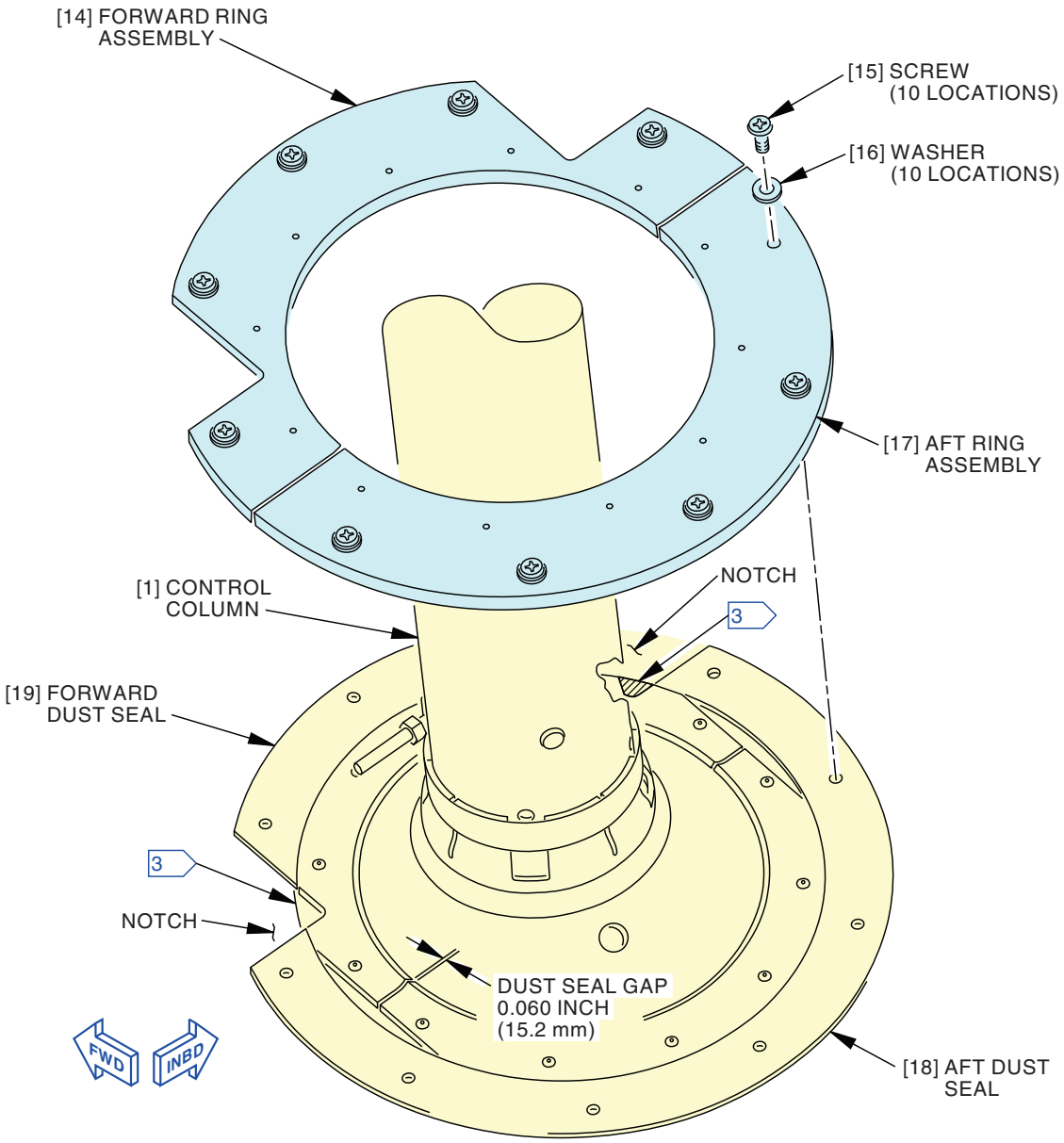
Control Column Installation
Figure 401/27-31-51-990-801 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



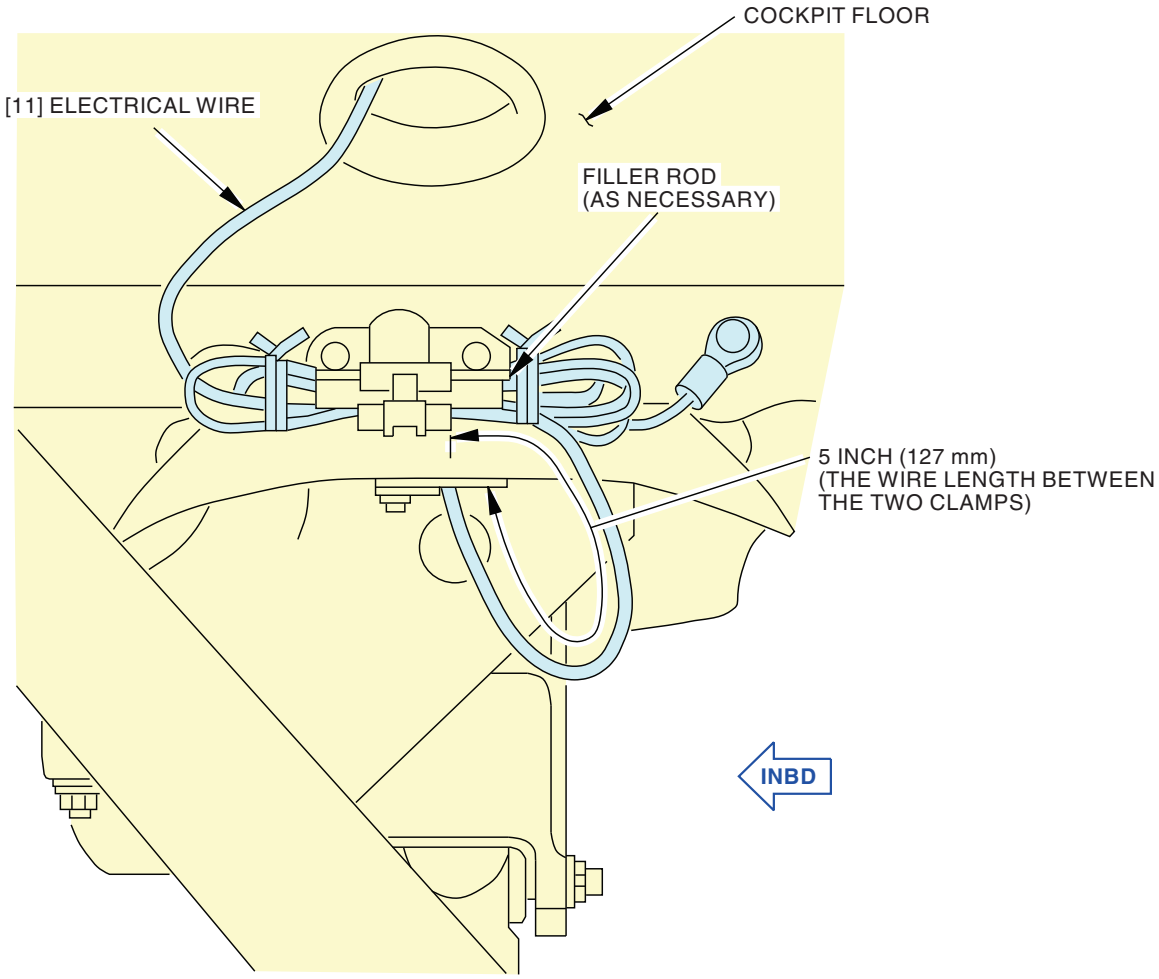
Control Column Installation
Figure 401/27-31-51-990-801 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Control Column Installation
Figure 401/27-31-51-990-801 (Sheet 6 of 7)

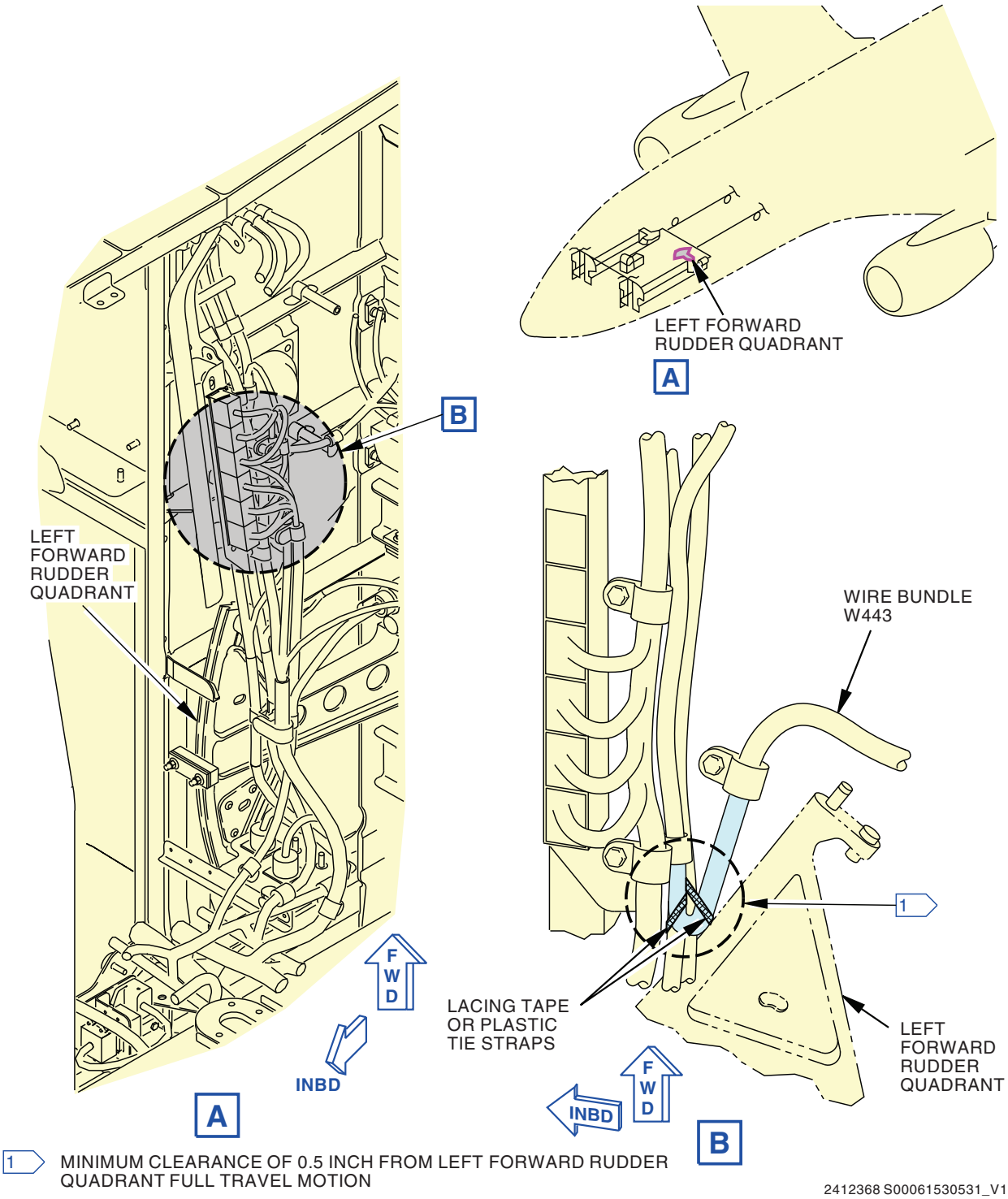
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Control Column Installation
Figure 401/27-31-51-990-801 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Wire Bundle W443 Clearance
Figure 402/27-31-51-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-51-400-801

3. Control Column - Installation

(Figure 401)

A. References

Reference	Title
23-51-04-400-801	Push To Talk (PTT) Control Wheel Switch Installation (P/B 401)
27-31-00-700-805	Elevator Travel Limit - Test (P/B 501)
27-31-00-700-808	Manual Mode - Test (P/B 501)
27-31-00-700-809	Control Column Travel and Centering - Test (P/B 501)
SOPM 20-30-03	General Cleaning Procedures

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1691	Protractor - Digital, Vertical or Horizontal Part #: 21465A82 Supplier: 3A054 Part #: 31-038-3 (Model Pro 360) Supplier: 1X822 Part #: 31-040-9 (Model Pro 3600) Supplier: 1X822 Part #: 950-317 (Model Pro 360) Supplier: 01EP7 Part #: 950-318 (Model Pro 3600) Supplier: 01EP7 Part #: KS5549 Supplier: 75245 Part #: KS6005 Supplier: 75245

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D00113	Lubricant - Solid Film, Liquid Dispersed	BMS3-8
D50108	Lubricant - Solid Film, Air Cured, Corrosion Inhibiting	MIL-L-23398
G50063	Lubricant - PTFE Release Agent - Miller-Stephenson MS-122XD (Replaces MS-122DF)	

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Control Column Installation

SUBTASK 27-31-51-420-001

(1) Put the control column [1] in its position.

(a) Clean the spade [4] and clevis [6] (SOPM 20-30-03).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Apply MS-122XD release agent, G50063, or solid film lubricant, D00113, or lubricant, D50108, to the spade [4] and clevis [6].

(c) Set the control wheel to the neutral position to engage the spade [4] with the clevis [6].

SUBTASK 27-31-51-480-002

(2) Use the protractor, COM-1691, to set the control column to 6.9 +/- 0.8 degrees forward of a vertical line from the floor (Figure 401).

SUBTASK 27-31-51-420-002

(3) Install the control column [1]:

(a) Engage the face splines on the control column [1] with the splines on the forward quadrant connector [10].

(b) Install the outboard couplings [9]:

1) Install the bolts [12], washers [8], and nuts [7] on the outboard couplings [9].

a) Make sure that you use the correct bolts.

(c) Do these steps to make the clearances equal at each end of each outboard coupling:

1) Tighten the nuts [7] to 50 in-lb (5.6 N·m) - 80 in-lb (9.0 N·m).

2) Make sure that the gaps of each end of each outboard coupling are equal and less than 0.03 in. (0.76 mm).

3) Make sure that treads are equal within 1 tread each end of each outboard coupling.

(d) Make sure that the control column angle is 6.9 +/- 0.8 degrees forward of the vertical line from the floor.

(e) Connect the electrical wire [5] and electrical wire [11] to the terminal strip in the lower nose compartment.

1) Make sure that there is a 5.0 in. (12.7 cm) length of the electrical wire [11] between the two clamps.

SUBTASK 27-31-51-420-003

(4) Install the ring assemblies and dust seals:

(a) Install the forward dust seal [19], aft dust seal [18], forward ring assembly [14], and aft ring assembly [17] with the screws [15] and washers [16].

CAUTION

DO NOT HOLD THE DUST CAP TOO TIGHTLY AGAINST THE WIRE BUNDLE. IT CAN CAUSE DAMAGE TO THE ELECTRICAL WIRES.

(b) If it is necessary, adjust the dust cap [3] to set the edge of the dust seals lightly up against the dust cap [3].

1) Loosen the clamp [2] to adjust the dust cap [3].

(c) Make sure that the gap between the two dust seals is 0.060 in. (1.52 mm).

(d) Fill the hole between the notch in the dust seals and the cutout in the floor with sealant, A00247.

SUBTASK 27-31-51-080-001

(5) Remove the rig pin E-1 from the left forward quadrant.

(a) Move the control column [1] full forward then full aft.

1) Make sure that the electrical wire [5] and electrical wire [11] are free of any interference.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-51-410-001

(6) Close this access panel:

Number	Name/Location
112A	Forward Access Door

G. Control Column Installation Test

SUBTASK 27-31-51-860-003

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	7	C01208	STALL WARN ASYM MODE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-31-51-710-001

(2) Do a check of the control column:

(a) Move the control column full forward then full aft.

1) Make sure that the control column operates correctly and smoothly.

(b) Turn the control wheel through full travel.

1) Make sure that the control wheel operates correctly and smoothly.

SUBTASK 27-31-51-710-002

(3) Do this task: Control Column Travel and Centering - Test, TASK 27-31-00-700-809.

SUBTASK 27-31-51-710-003

(4) Do this task: Manual Mode - Test, TASK 27-31-00-700-808.

SUBTASK 27-31-51-710-004

(5) Do this task: Elevator Travel Limit - Test, TASK 27-31-00-700-805.

SUBTASK 27-31-51-710-005

(6) Do the operational test of the PTT switch in this task: Push To Talk (PTT) Control Wheel Switch Installation, TASK 23-51-04-400-801.

SUBTASK 27-31-51-710-006

(7) Do a test of the stabilizer trim control switch on the control wheel:

(a) Make sure that the STAB TRIM PRI switch, S272, on the aft of the control stand is in the NORMAL position.

(b) Move the STAB TRIM switches to APL NOSE UP and then to APL NOSE DOWN.

1) Make sure that the stabilizer moves correctly.

SUBTASK 27-31-51-740-001

(8) Do a test of the autopilot disengage switch on the control wheel:

(a) Push the CMD A or CMD B autopilot engage switch on the mode control panel to engage the autopilot.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the CMD A or CMD B autopilot engage switch light comes on.

(b) Push the captain or first officer autopilot disengage switch on the control wheel to disengage the autopilot.

1) Make sure that the A/P warning lights flash.

2) Make sure that the A/P disengage warning sound comes on.

3) Push the captain or first officer autopilot disengage switch again.

a) Make sure that the A/P warning lights go off.

b) Make sure that there is no A/P disengage warning sound.

SUBTASK 27-31-51-750-001

(9) Make sure that the wire bundle W443 has a minimum clearance of 0.5 in. (12.700 mm) from left forward rudder quadrant full travel motion, while operating the rudder pedals (Figure 402).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CONTROL COLUMN SUPPORT BEARING - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of a control column support bearing

(2) An installation of a control column support bearing.

B. The control column support bearing is referred to as "support bearing" in this procedure.

C. the support bearing includes the forward quadrant connector, bearing housing, shim, and bearing.

D. This procedure is applicable for the support bearings on the captain's and first officer's control column.

TASK 27-31-54-000-801

2. Control Column Support Bearing Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-51-000-801	Control Column - Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1747	Support Set - Forward Elevator Control Quadrant Part #: C27036-1 Supplier: 81205

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right
318	Tail Cone Compartment - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door
318BR	Tailcone Access Door

E. Prepare for the Removal

SUBTASK 27-31-54-860-001

(1) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-54-010-001

(2) Get access to the elevator forward quadrant.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this access door:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-31-54-010-002

(3) Get access to the elevator aft quadrant.

(a) Remove this access door:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-54-480-001

(4) Install the rig pin E-5 from the rigging pin kit, SPL-1585, in the elevator aft quadrant (Figure 401).

SUBTASK 27-31-54-480-002

(5) Install the rig pin E-1 from the rigging pin kit, SPL-1585, in the left forward quadrant (Figure 402).

SUBTASK 27-31-54-020-001

(6) Remove the applicable control column (TASK 27-31-51-000-801).

F. Control Column Support Bearing Removal

(Figure 402)

SUBTASK 27-31-54-020-002

(1) Turn the cable turnbuckles [1] to release the tension in the elevator control cables.

NOTE: Do not disconnect the cable turnbuckles.

SUBTASK 27-31-54-480-003

CAUTION

MAKE SURE THAT YOU INSTALL A SUPPORT FOR THE FORWARD QUADRANT ASSEMBLY BEFORE YOU REMOVE THE COUPLINGS. KEEP THE SUPPORT UNTIL YOU REMOVE THE COUPLINGS. IF YOU DO NOT HOLD THE FORWARD QUADRANT ASSEMBLY CORRECTLY, YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Install the support set, SPL-1747, to support the forward quadrant assembly.

SUBTASK 27-31-54-020-003

(3) Remove the nuts [5], washers [4], and bolts [3] from the inboard couplings [6].

SUBTASK 27-31-54-020-004

(4) Remove the forward quadrant connector [11].

SUBTASK 27-31-54-020-005

(5) Remove the bearing housing [15]:

(a) Remove the nuts [7], washers [9], upper bolts [10] from the bearing housing [15].

(b) Remove the nuts [17], washers [16], washers [14], and lower bolts [13] from the bearing housing [15].

(c) Remove the bearing housing [15] and shim [8] from the airplane.

SUBTASK 27-31-54-020-006

(6) Remove the bearing [12] from the bearing housing [15]:

(a) Turn the bearing [12] at the right angles to the bearing housing [15].

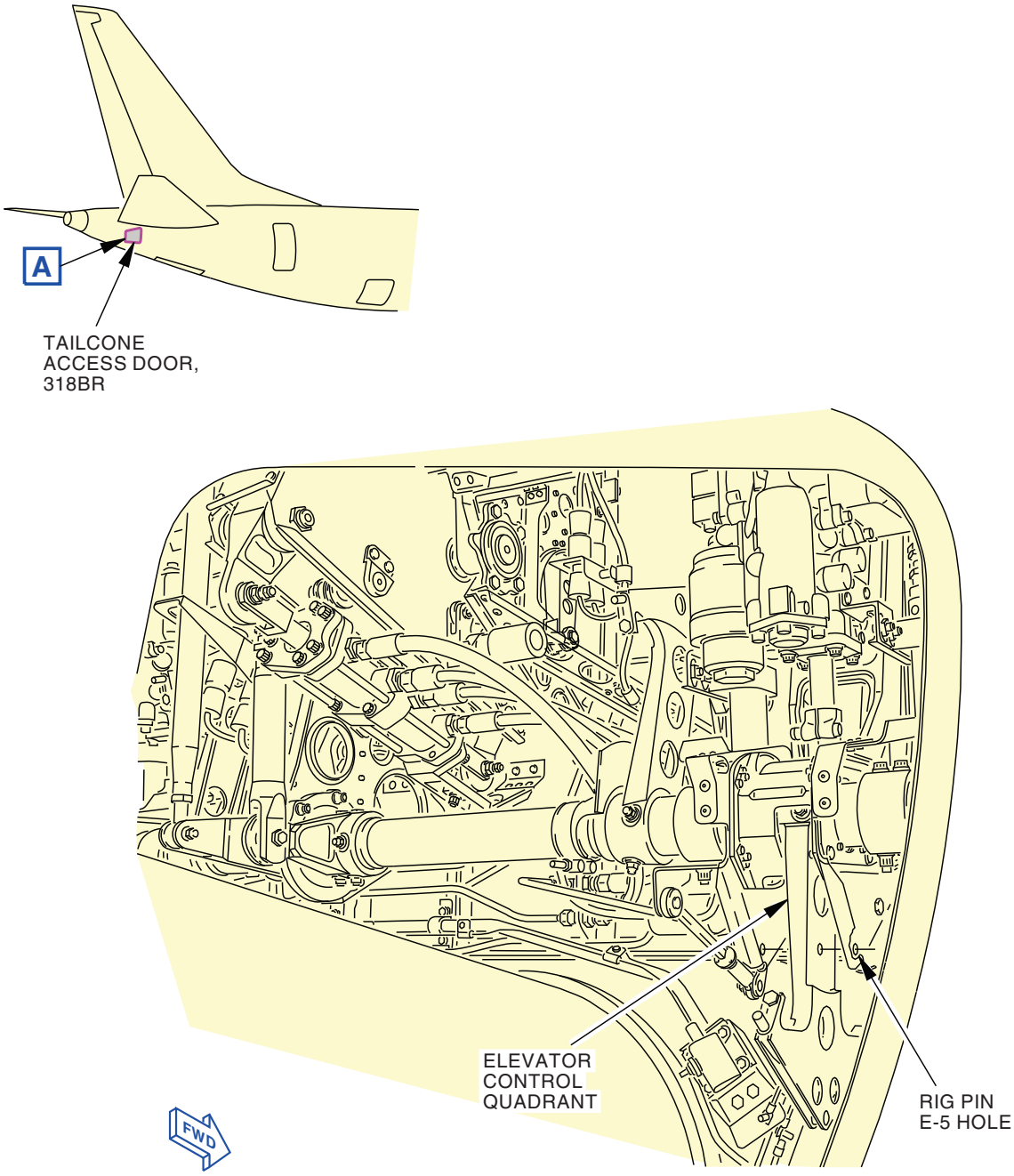
(b) Align the bearing [12] with the slots in the bearing housing [15].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Remove the bearing [12] from the bearing housing [15].

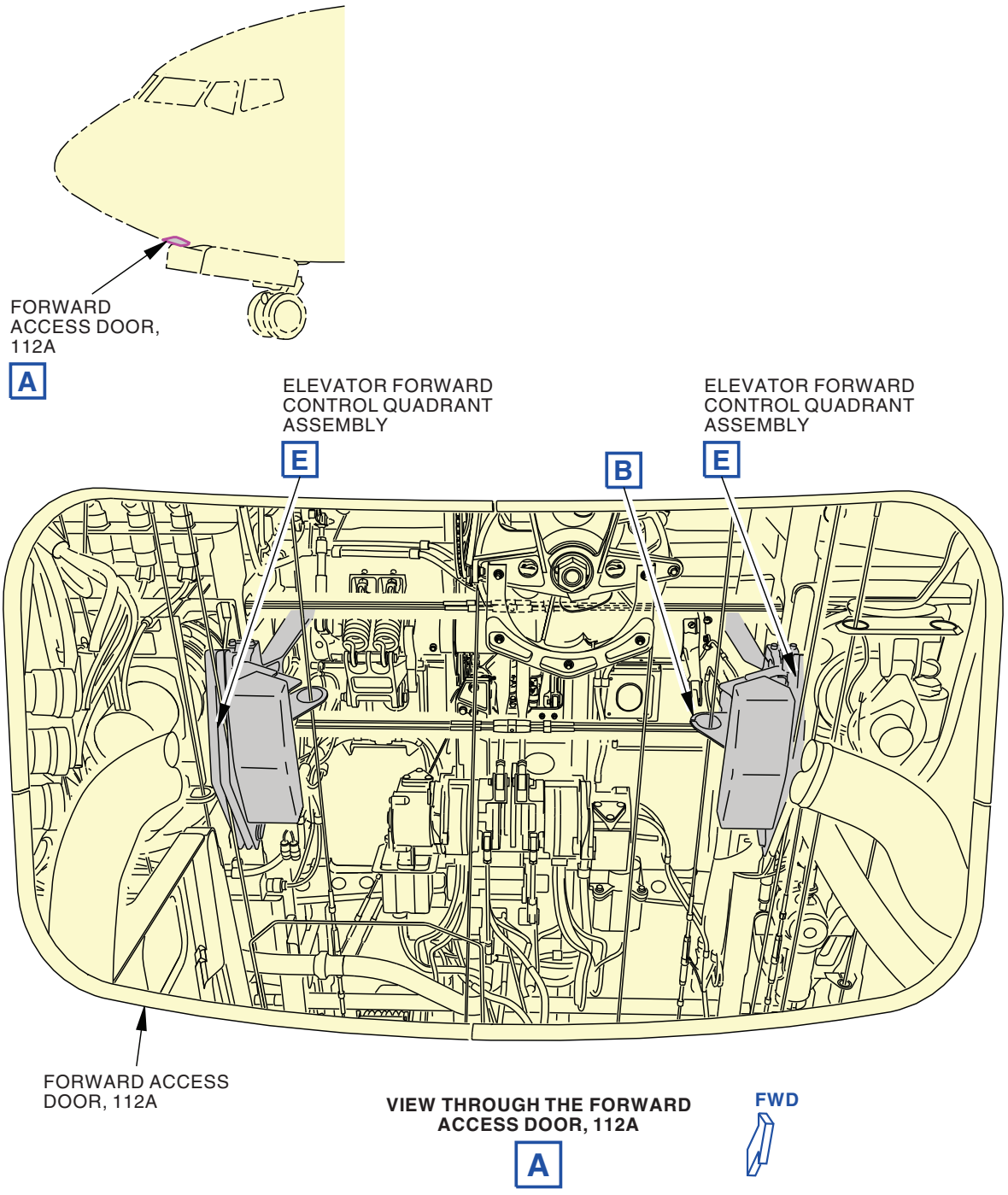
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Rig Pin Installation
Figure 401/27-31-54-990-801

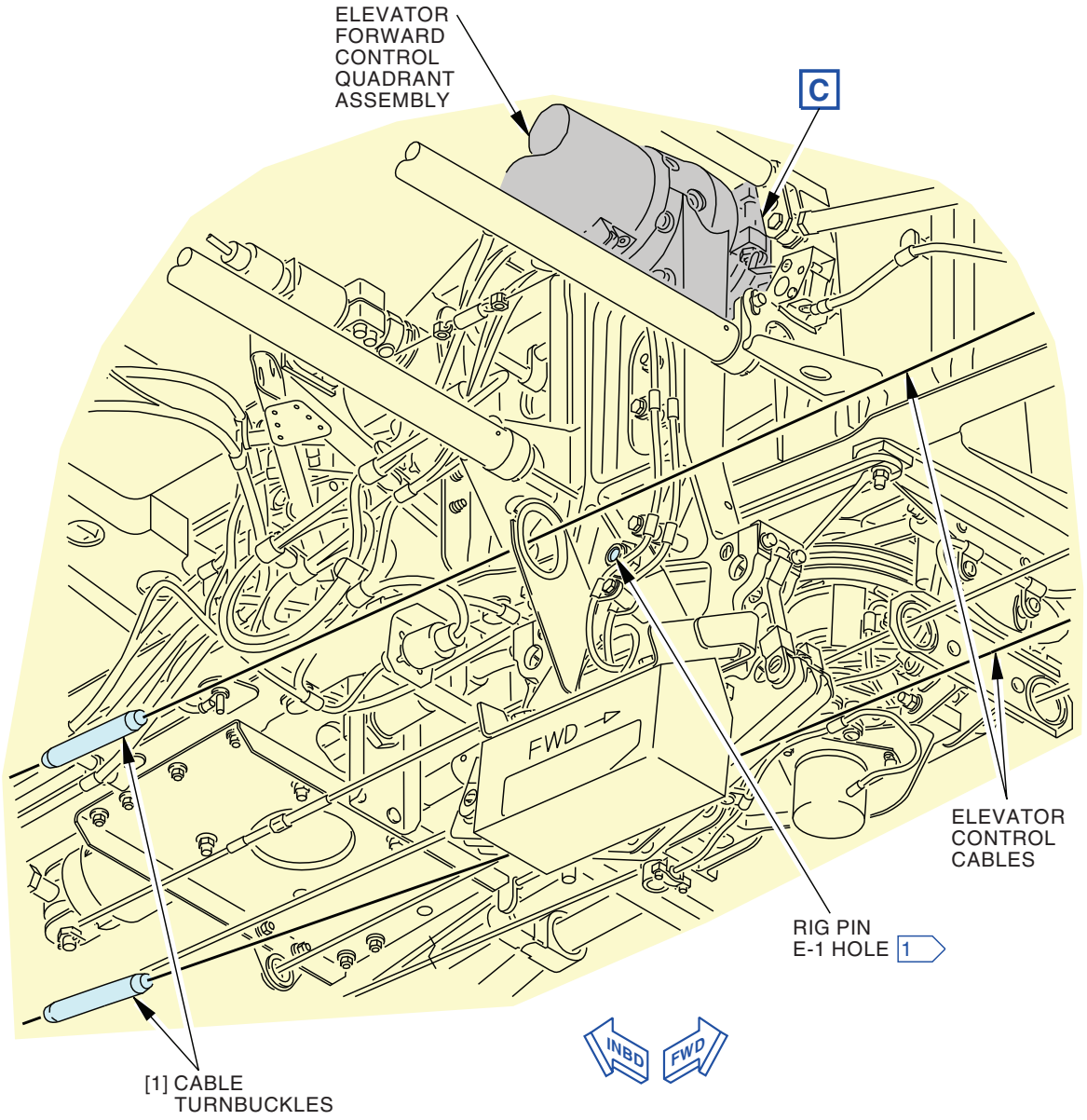
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Control Column Support Bearing Installation
Figure 402/27-31-54-990-802 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

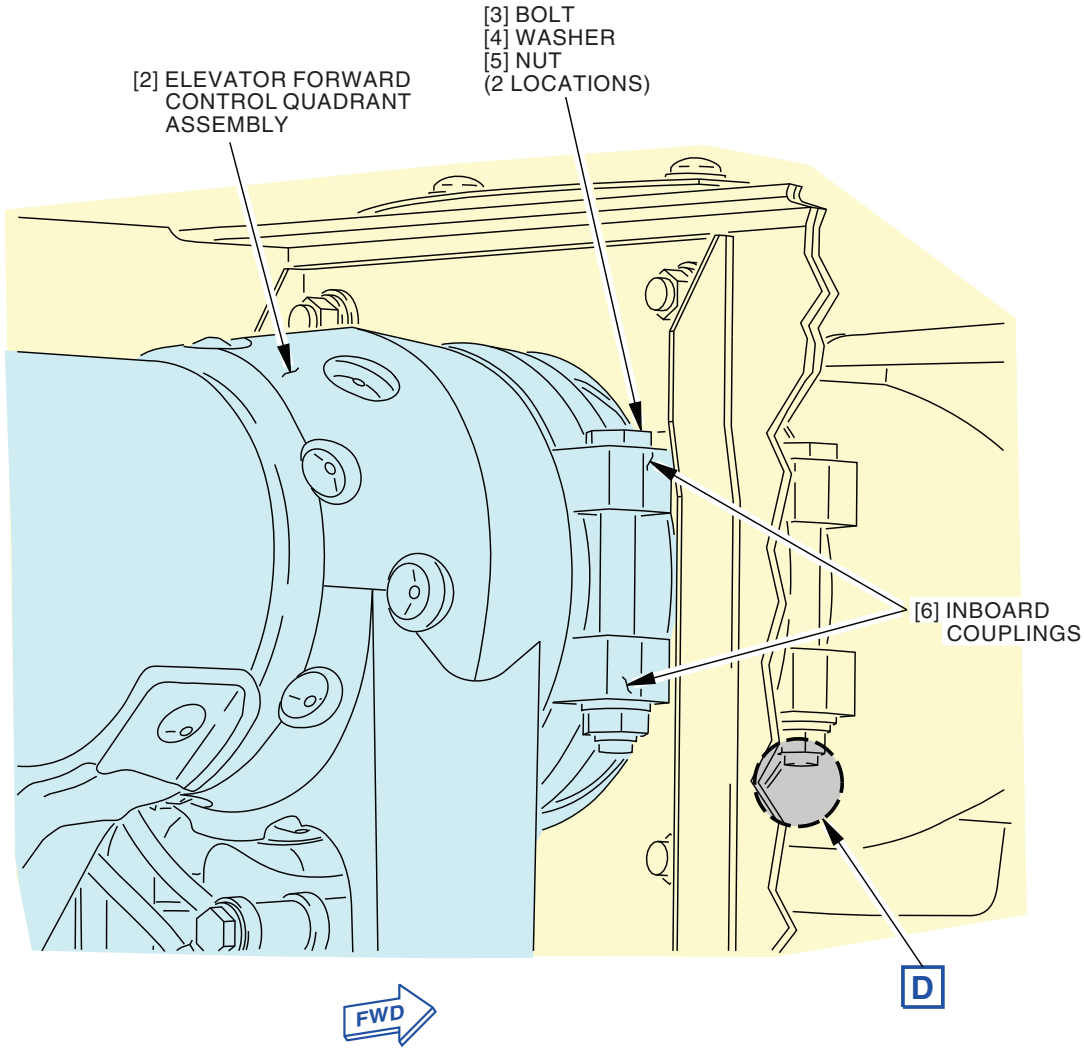


1 LEFT FORWARD CONTROL QUADRANT ONLY

2412569 S00061530918_V1

Control Column Support Bearing Installation
Figure 402/27-31-54-990-802 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



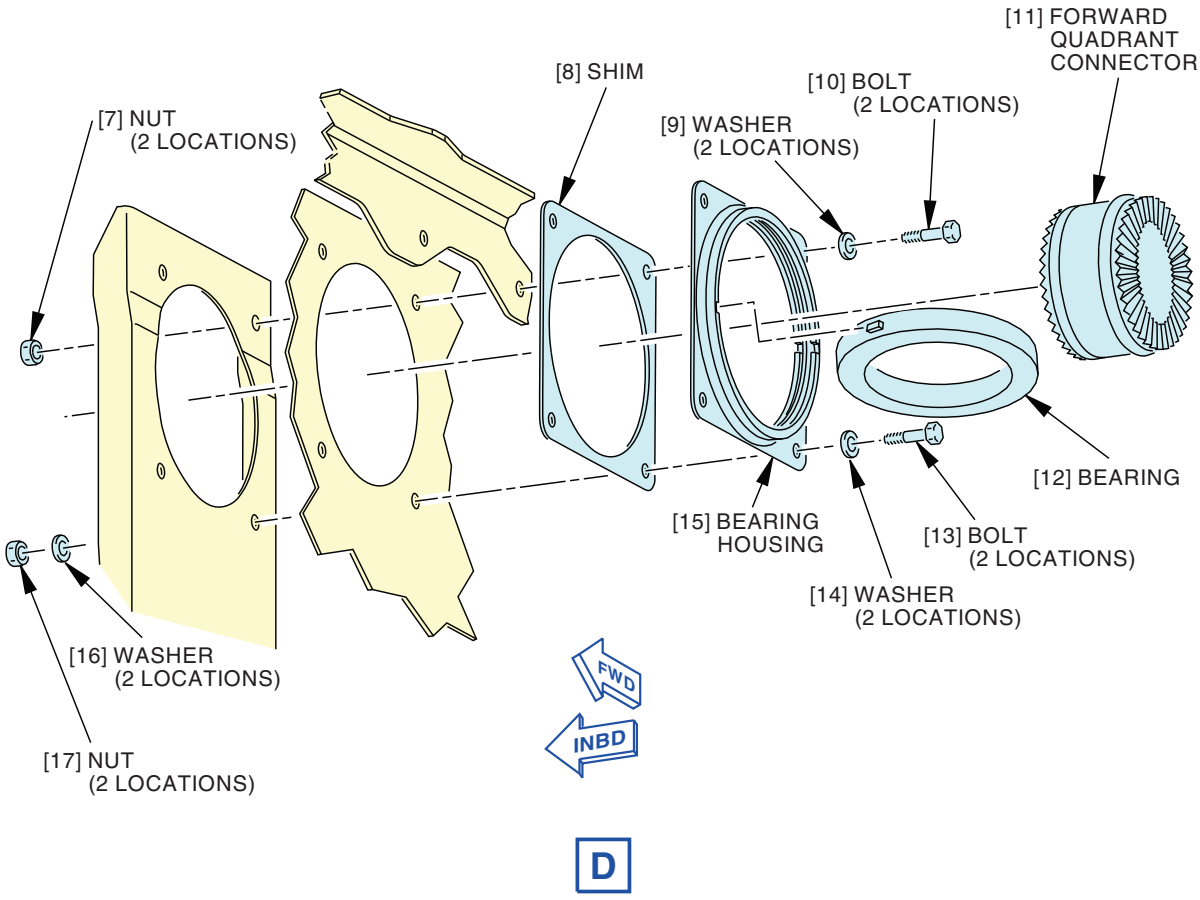
ELEVATOR FORWARD CONTROL QUADRANT ASSEMBLY

C

2412570 S00061530919_V1

Control Column Support Bearing Installation
Figure 402/27-31-54-990-802 (Sheet 3 of 5)

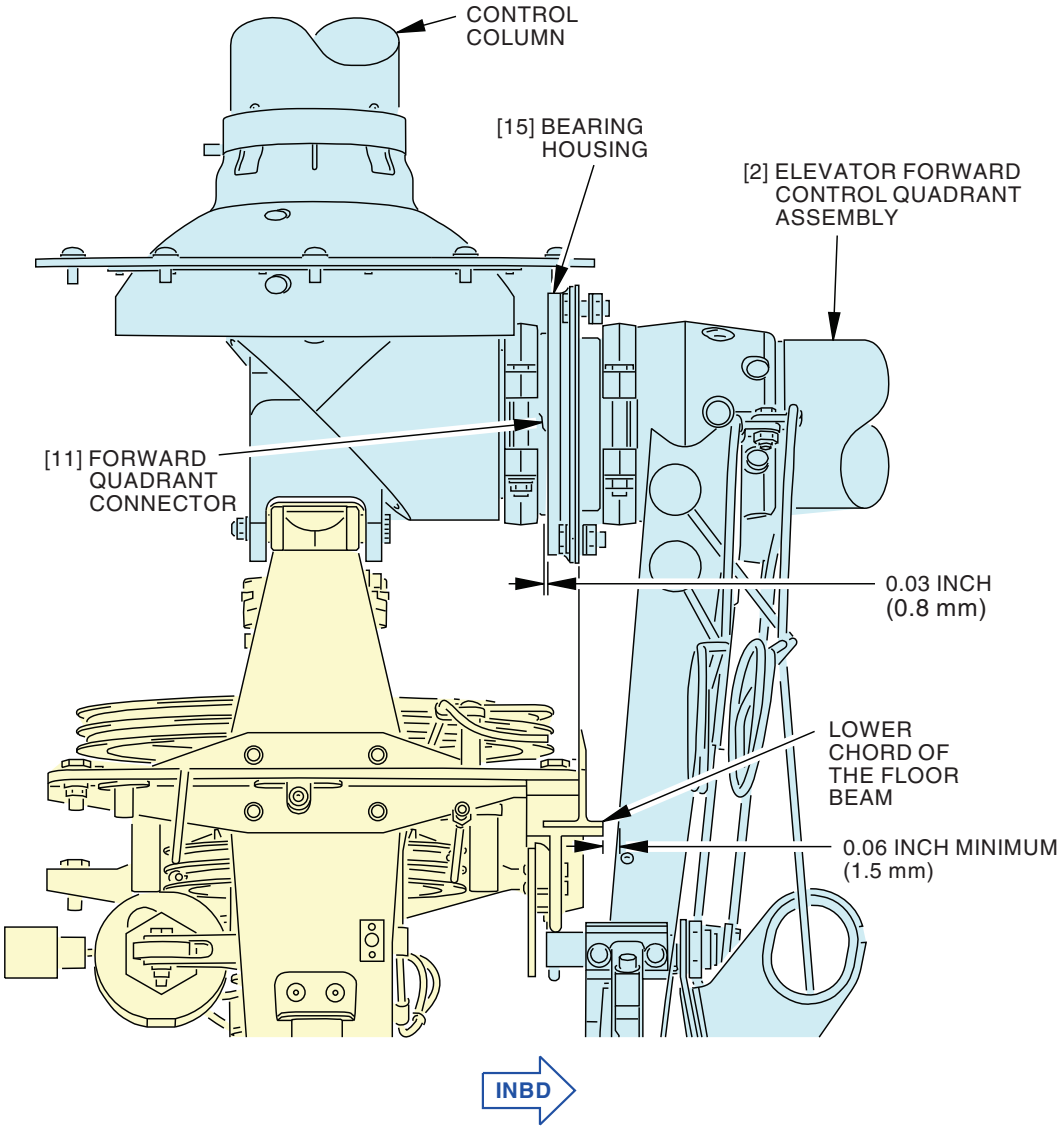
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412571 S00061530920_V1

Control Column Support Bearing Installation
Figure 402/27-31-54-990-802 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELEVATOR FORWARD CONTROL QUADRANT ASSEMBLY
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

E

2412572 S00061530921_V1

Control Column Support Bearing Installation
Figure 402/27-31-54-990-802 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-54-400-801

3. Control Column Support Bearing Installation

(Figure 401, Figure 402)

A. References

Reference	Title
27-31-00-820-801	Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment (P/B 501)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
27-31-51-400-801	Control Column - Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1747	Support Set - Forward Elevator Control Quadrant Part #: C27036-1 Supplier: 81205

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
12	Bearing	27-31-54-01-035	PTF ALL

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right
318	Tail Cone Compartment - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door
318BR	Tailcone Access Door

F. Control Column Support Bearing Installation

SUBTASK 27-31-54-420-001

(1) Install the bearing [12] in the bearing housing [15]:

(a) Put the bearing [12] at the right angle of the bearing housing [15].

(b) Move the bearing [12] in the bearing housing [15] slots.

1) Turn the bearing [12] to the correct position.

SUBTASK 27-31-54-420-002

(2) Install the bearing housing [15]:

(a) Put the bearing housing [15] and the shim [8] against the structure.

(b) Install the upper bolts [10], washers [9], and nuts [7] that attach the bearing housing [15] to the structure.

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(c) Install the lower bolts [13], washers [14], washers [16], and nuts [17] on the bearing housing [15].

SUBTASK 27-31-54-420-003

(3) Install the elevator forward control quadrant assembly [2] to the support bearing:

(a) Install the forward quadrant connector [11] in the bearing housing [15].

(b) Align the spines of the elevator elevator forward control quadrant assembly [2] with the forward quadrant connector [11].

(c) Install inboard couplings [6] with the bolts [3], washers [4], and nuts [5].

(d) Do these steps to make the clearances equal at each end of each inboard coupling [6]:

1) Tighten the bolts [3] to 50 in-lb (5.6 N·m).

2) Tighten the bolt [3] and then the other to 50 in-lb (5.6 N·m) - 70 in-lb (7.9 N·m) in approximately 10 in-lb (1.1 N·m) increments until you get the clearances equal to ± 0.03 in. (0.76 mm) of each other.

(e) Adjust the shim [8] below the bearing housing [15]:

1) Make sure the clearance between the forward quadrant connector [11] and inner race of the bearing [12] is 0.01 in. (0.25 mm) - 0.03 in. (0.76 mm).

2) Make sure the clearance between lower chord of the floor beam and each side of the elevator forward control quadrant assembly [2] is 0.06 in. (1.5 mm) minimum through the full travel.

3) If it is necessary, add or remove shims [8] under the bearing housing [15].

SUBTASK 27-31-54-480-004

(4) Remove the support set, SPL-1747, from the elevator forward control quadrant assembly [2].

SUBTASK 27-31-54-420-004

(5) Install the applicable control column (TASK 27-31-51-400-801).

SUBTASK 27-31-54-080-001

(6) Remove the rig pin E-1 of rigging pin kit, SPL-1585, from the left forward quadrant.

SUBTASK 27-31-54-080-002

(7) Remove the rig pin E-5 of the rigging pin kit, SPL-1585, from the aft control quadrant.

SUBTASK 27-31-54-710-001

(8) Do a check of the control column:

(a) Move the control column full forward then full aft.

1) Make sure the control column operates correctly and smoothly.

SUBTASK 27-31-54-820-001

(9) Adjust the elevator control cables EA and EB (TASK 27-31-00-820-801).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-54-410-001

(1) Install this access door:

Number	Name/Location
318BR	Tailcone Access Door

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SUBTASK 27-31-54-410-002

(2) Close this access door:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-31-54-860-002

(3) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

——— END OF TASK ———

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ELEVATOR FORWARD CONTROL QUADRANT ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the elevator forward control quadrant assembly

(2) An installation of the elevator forward control quadrant assembly.

B. The elevator forward control quadrant assembly is referred to as "forward quadrant assembly" in this procedure.

TASK 27-31-61-000-801

2. Elevator Forward Control Quadrant Assembly Removal

(Figure 401, Figure 402)

A. References

Reference	Title
22-11-92-000-801	Pitch Force Transducer Removal (P/B 401)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1569	Clamp - Control Cable Part #: A20005-9 Supplier: 81205
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right
318	Tail Cone Compartment - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door
318BR	Tailcone Access Door

E. Prepare for the Removal

SUBTASK 27-31-61-860-001

(1) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-61-010-001

(2) To get access to the elevator forward quadrant, do this task:

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Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-31-61-010-002

(3) To get access to the elevator aft quadrant, do this task:

Remove this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-61-480-001

(4) Install the rig pin E-5 from the test set rigging pin kit, SPL-1585 in the elevator aft quadrant (Figure 401).

SUBTASK 27-31-61-480-002

(5) Install the rig pin E-1 from the rigging pin kit, SPL-1585 in the left side of forward quadrant assembly (Figure 402).

SUBTASK 27-31-61-020-001

(6) Remove the pitch control wheel steering force transducers from the forward quadrant assembly. To remove them, do this task: Pitch Force Transducer Removal, TASK 22-11-92-000-801.

F. Forward Quadrant Assembly Removal

(Figure 402)

SUBTASK 27-31-61-020-002

(1) Turn the cable turnbuckles to release the tension in the EA cables [2] and EB cables [3].

NOTE: Do not disconnect the cable turnbuckles.

(a) If it is necessary, install control cable clamp, SPL-1569 on the EA cables [2] and EB cables [3].

SUBTASK 27-31-61-020-003

(2) Remove the cables EA cables [2] and EB cables [3] from the forward quadrant assembly [1]:

(a) Remove the screws [10] and retainers [4] from the EA cables [2] and EB cables [3].

(b) Put identification tags on the cables.

(c) Remove the bolts [6], washers [7], and nuts [8] that attach the fairleads [9] to the forward quadrant assembly [1].

SUBTASK 27-31-61-020-004

(3) Remove the bolts [11] and washers [12] that attach the mass balance weight [5] to the forward quadrant assembly [1].

SUBTASK 27-31-61-020-005

(4) Remove the ring assemblies and dust seals from the control column [15]:

(a) Remove the screws [22] and washers [23] from the forward ring assembly [21] and aft ring assembly [24].

(b) Remove the forward ring assembly [21] and aft ring assembly [24].

(c) Remove the aft dust seals [25] and forward dust seals [26].

SUBTASK 27-31-61-080-001

(5) Remove the rig pin E-1 from the left side of the forward quadrant assembly [1].

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SUBTASK 27-31-61-020-006

CAUTION

HOLD THE CONTROL COLUMNS AND THE FORWARD QUADRANT ASSEMBLY WHEN YOU REMOVE THEM. THEY CAN FALL AND CAUSE DAMAGE TO EQUIPMENT.

(6) Remove the forward quadrant assembly [1]:

(a) Remove the nuts [18], washers [17], and bolts [13] from the inboard couplings [16].

(b) Remove the inboard couplings [16].

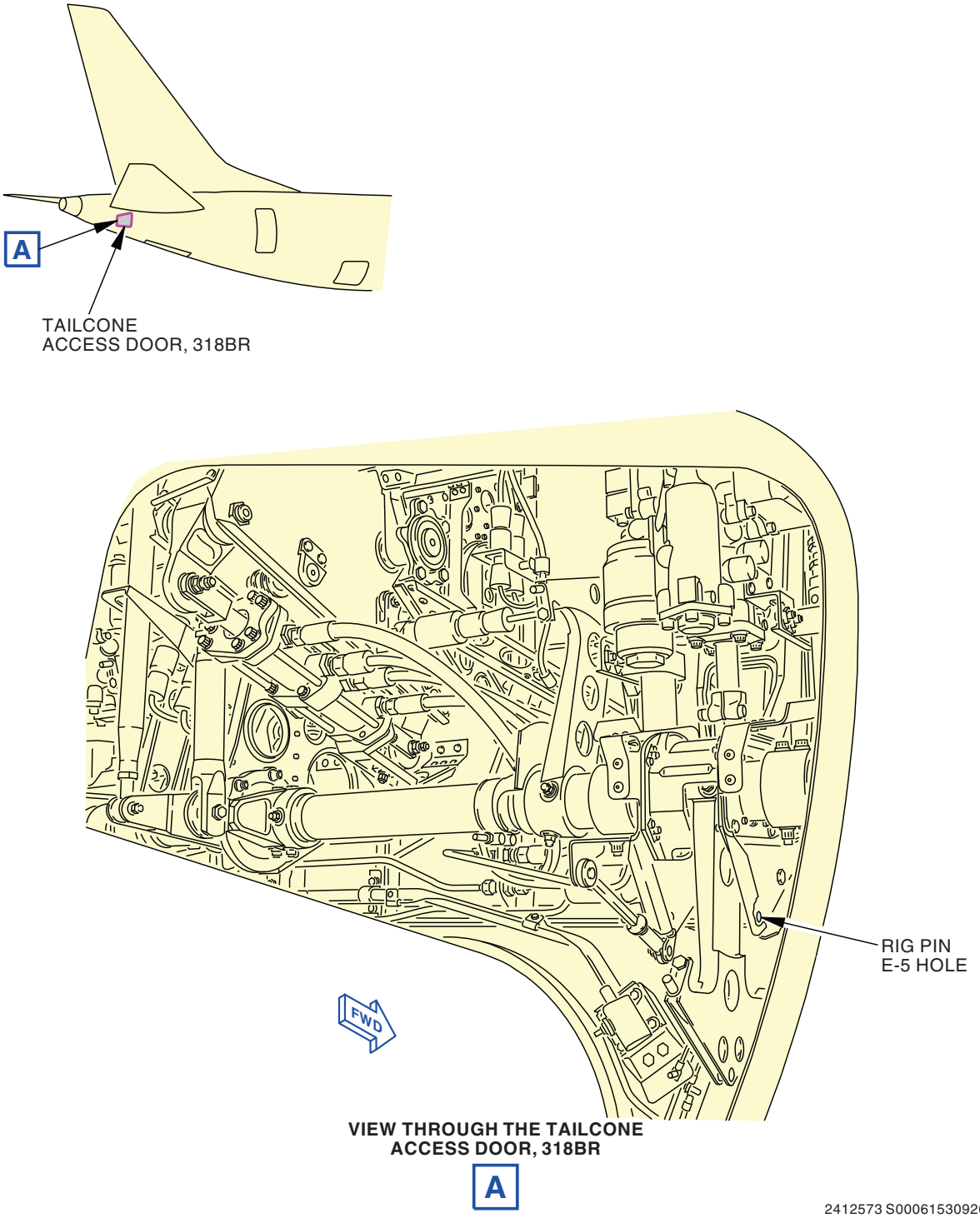
(c) Move the control columns [15] outboard until you can disengage the forward quadrant connector [14] from the forward quadrant assembly [1].

NOTE: Use a marker or a pen to make marks on the forward quadrant connector [14] and the forward quadrant assembly [1] to align the control column during the forward quadrant assembly [1] installation.

(d) Remove the forward quadrant assembly [1].

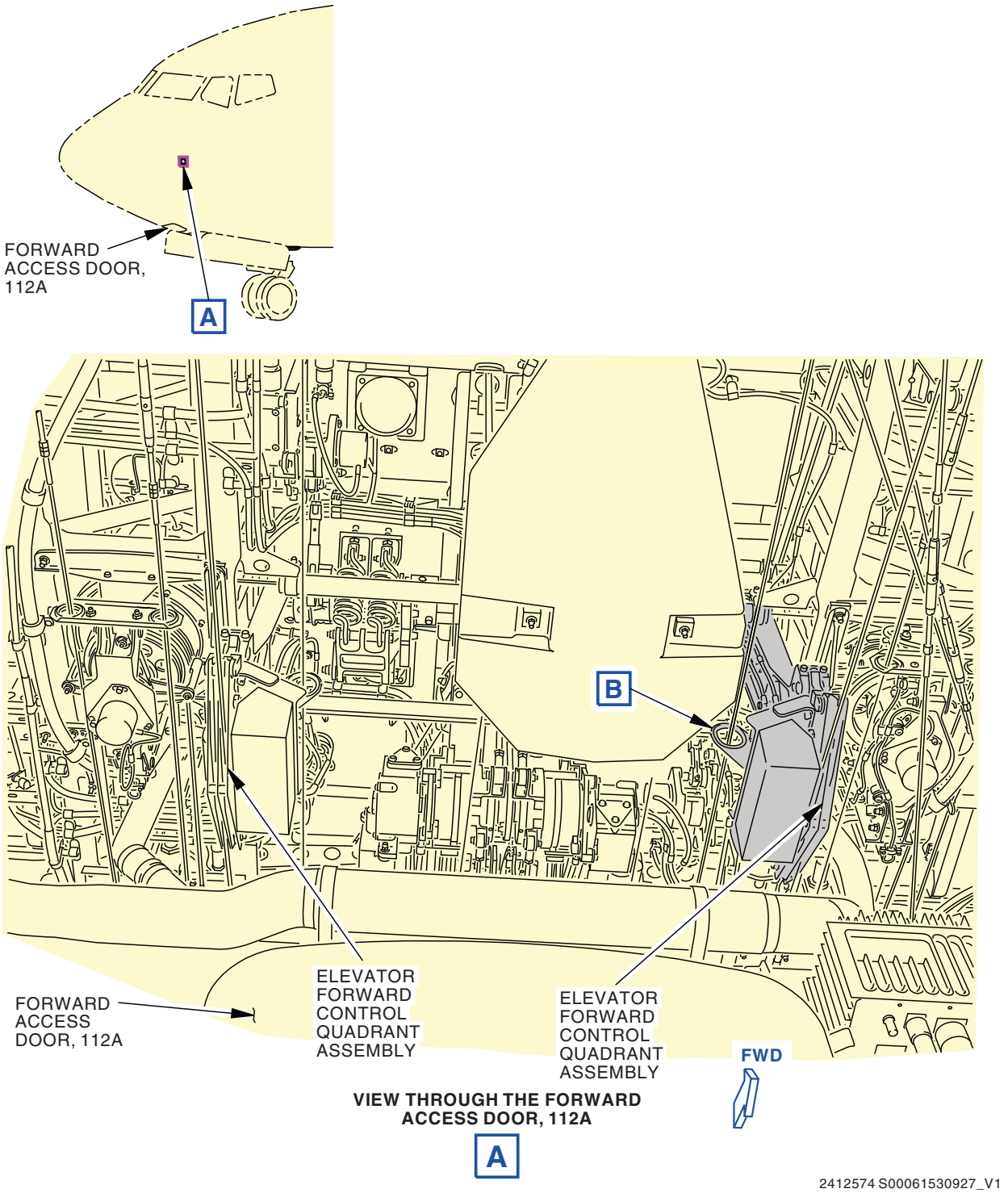
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



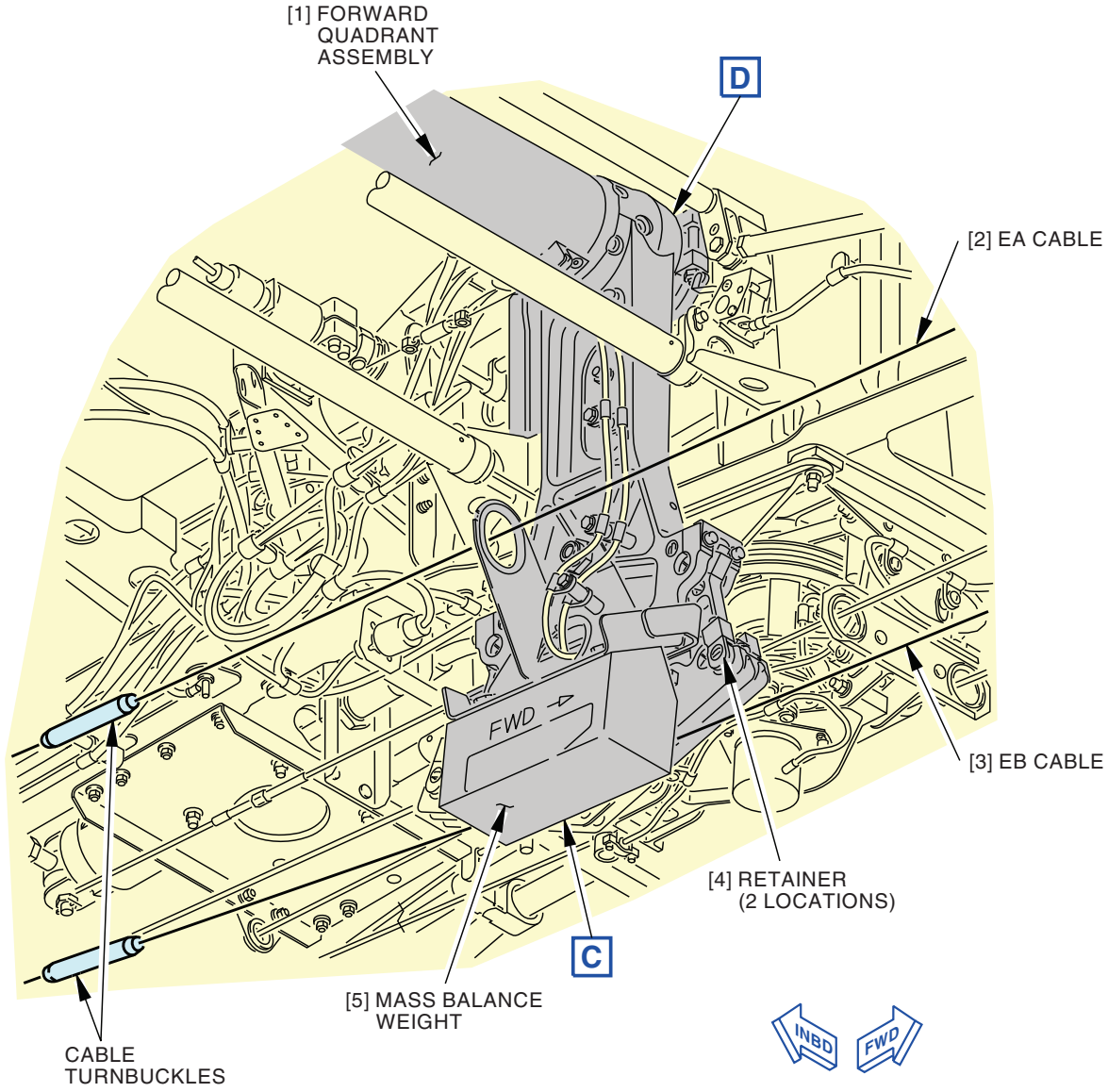
Elevator Rig Pin Installation
Figure 401/27-31-61-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Forward Control Quadrant Assembly Installation
Figure 402/27-31-61-990-802 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



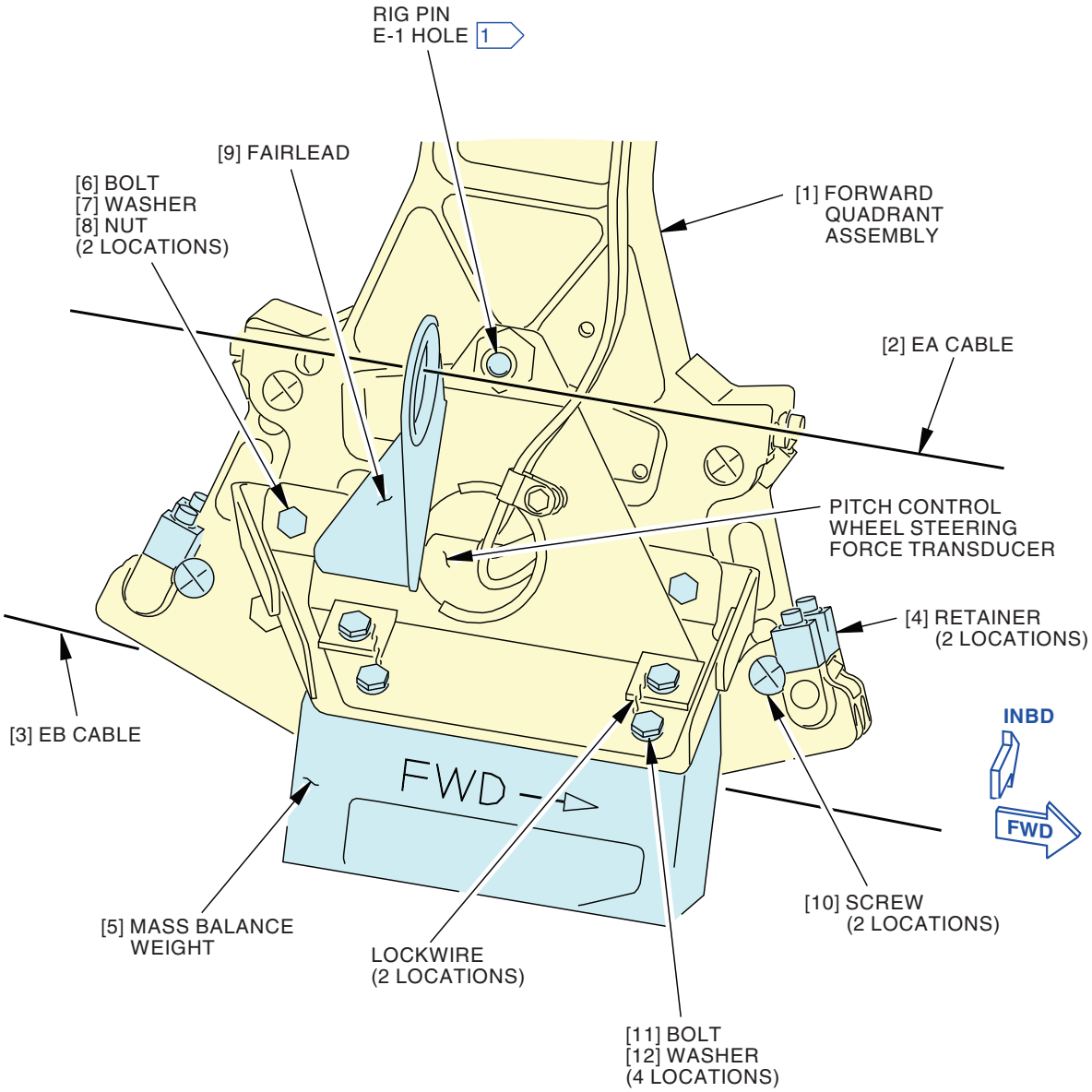
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

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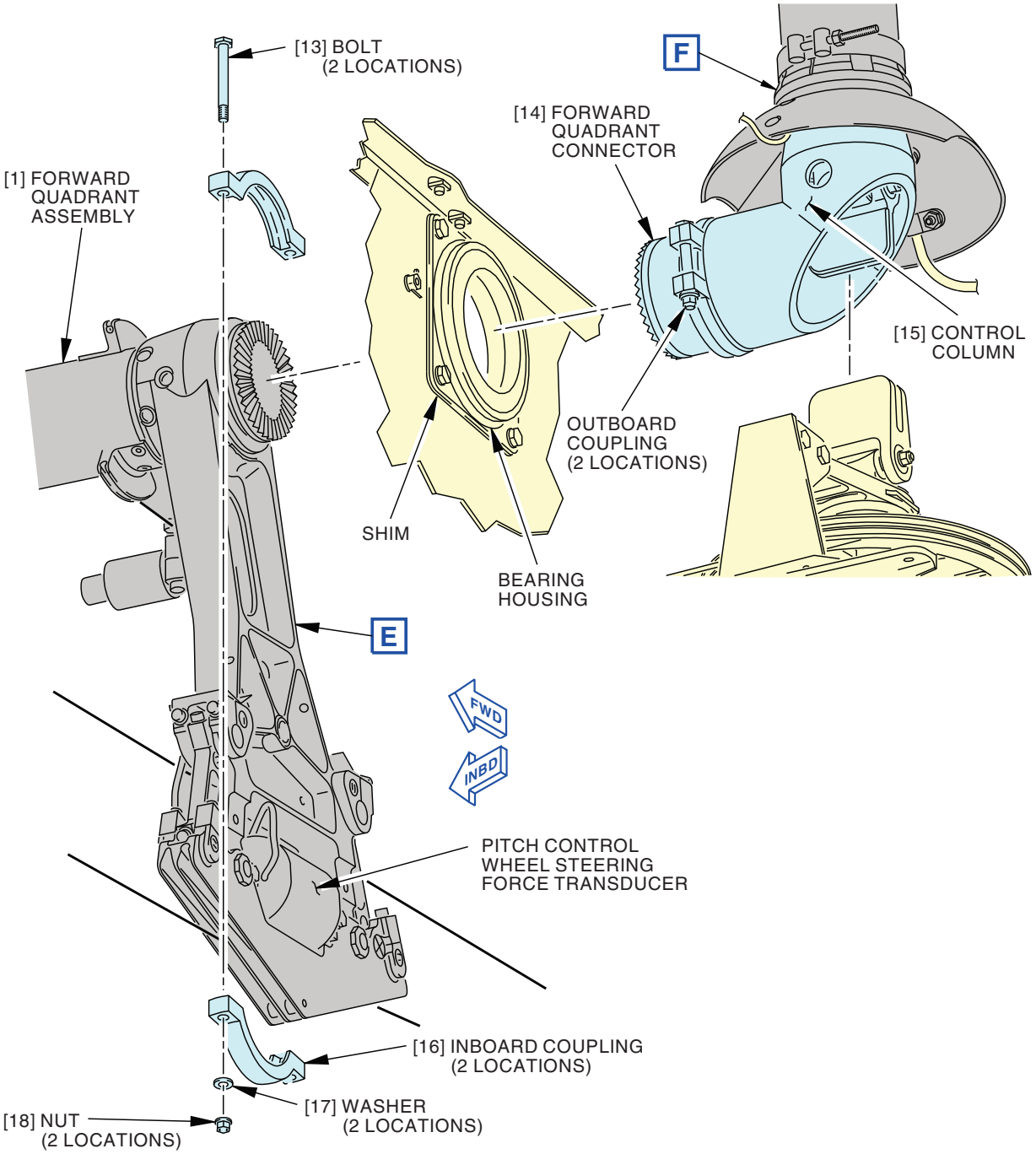
Elevator Forward Control Quadrant Assembly Installation
Figure 402/27-31-61-990-802 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Forward Control Quadrant Assembly Installation
Figure 402/27-31-61-990-802 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



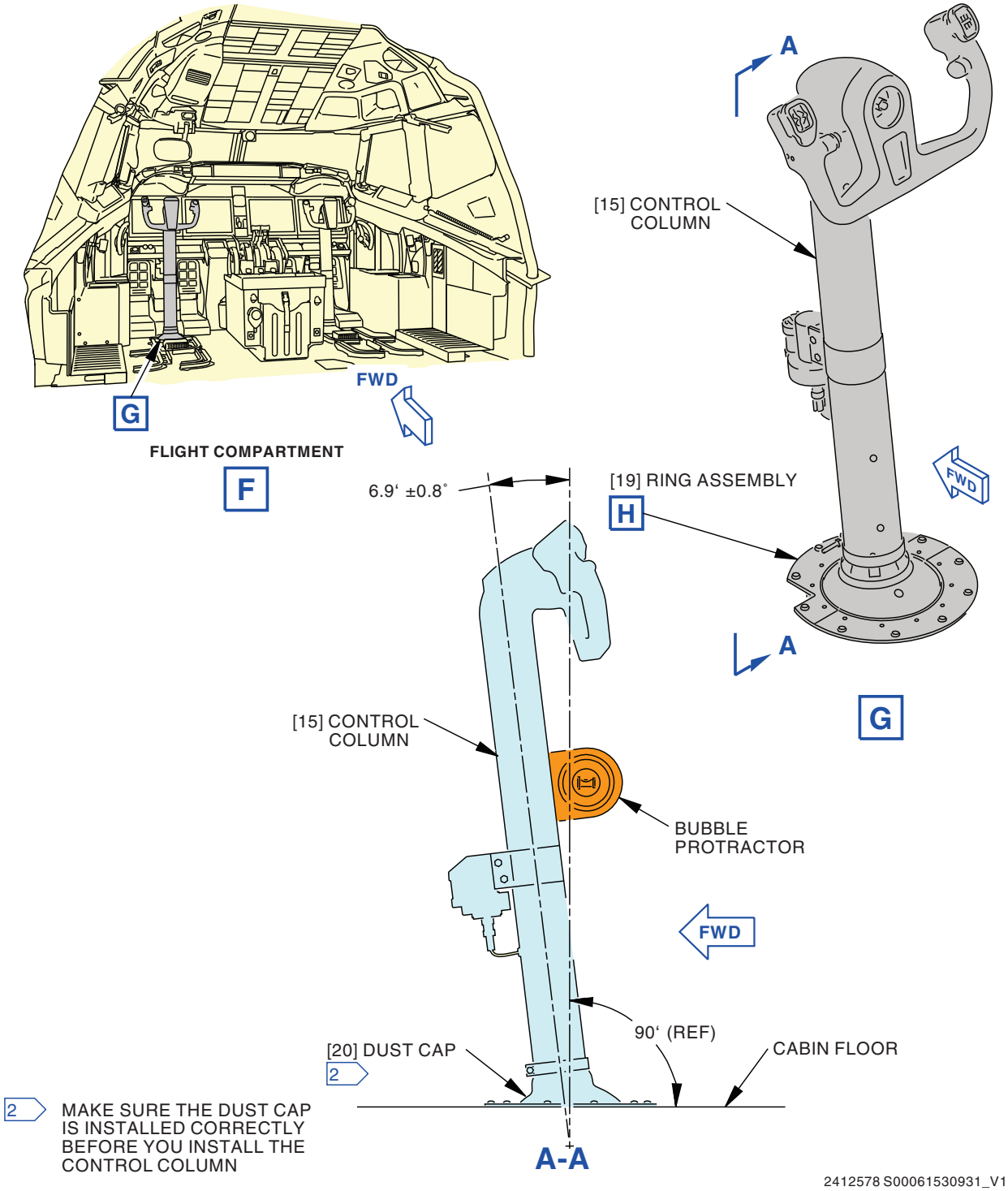
ELEVATOR FORWARD CONTROL QUADRANT ASSEMBLY

D

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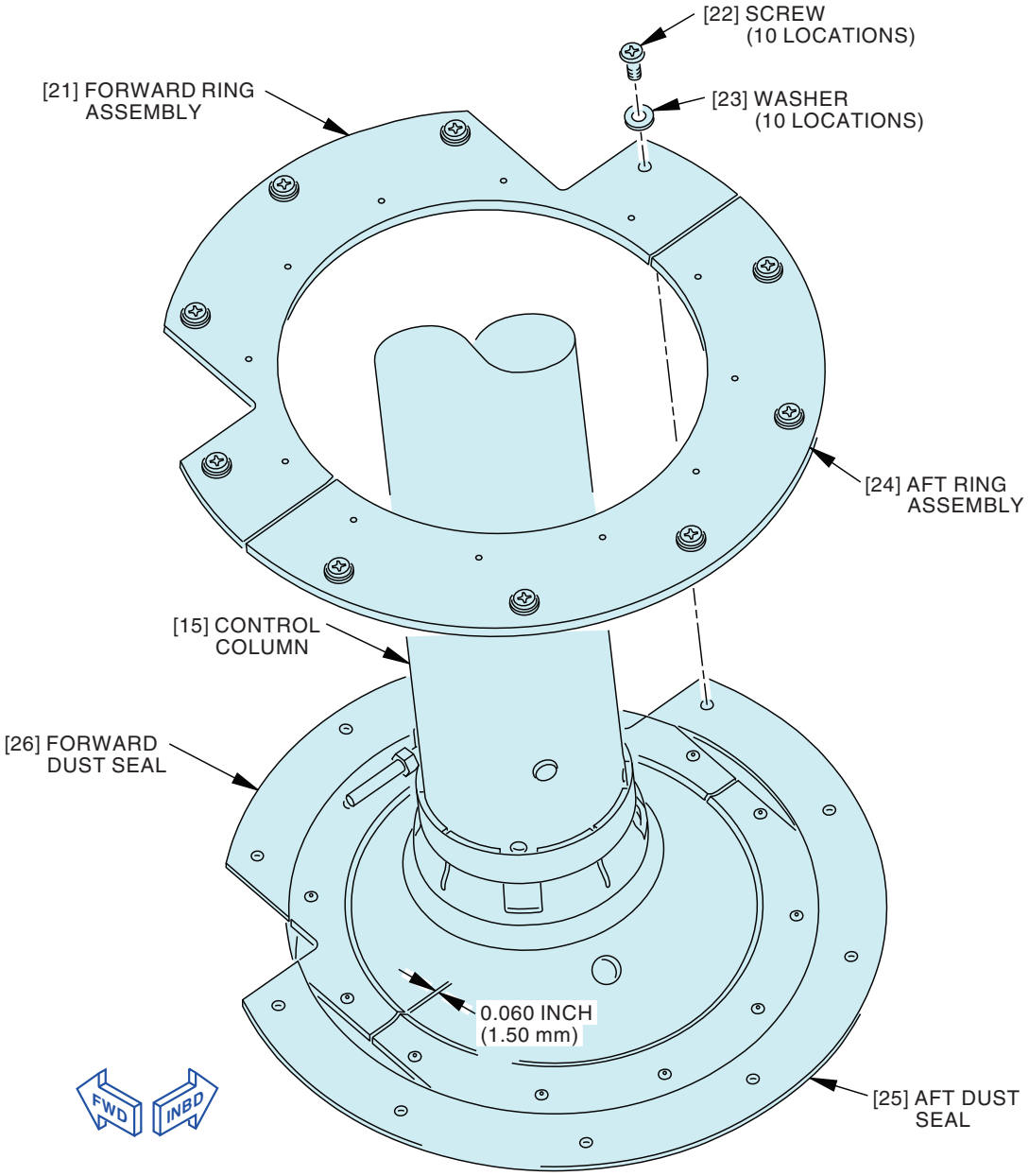
Elevator Forward Control Quadrant Assembly Installation
Figure 402/27-31-61-990-802 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



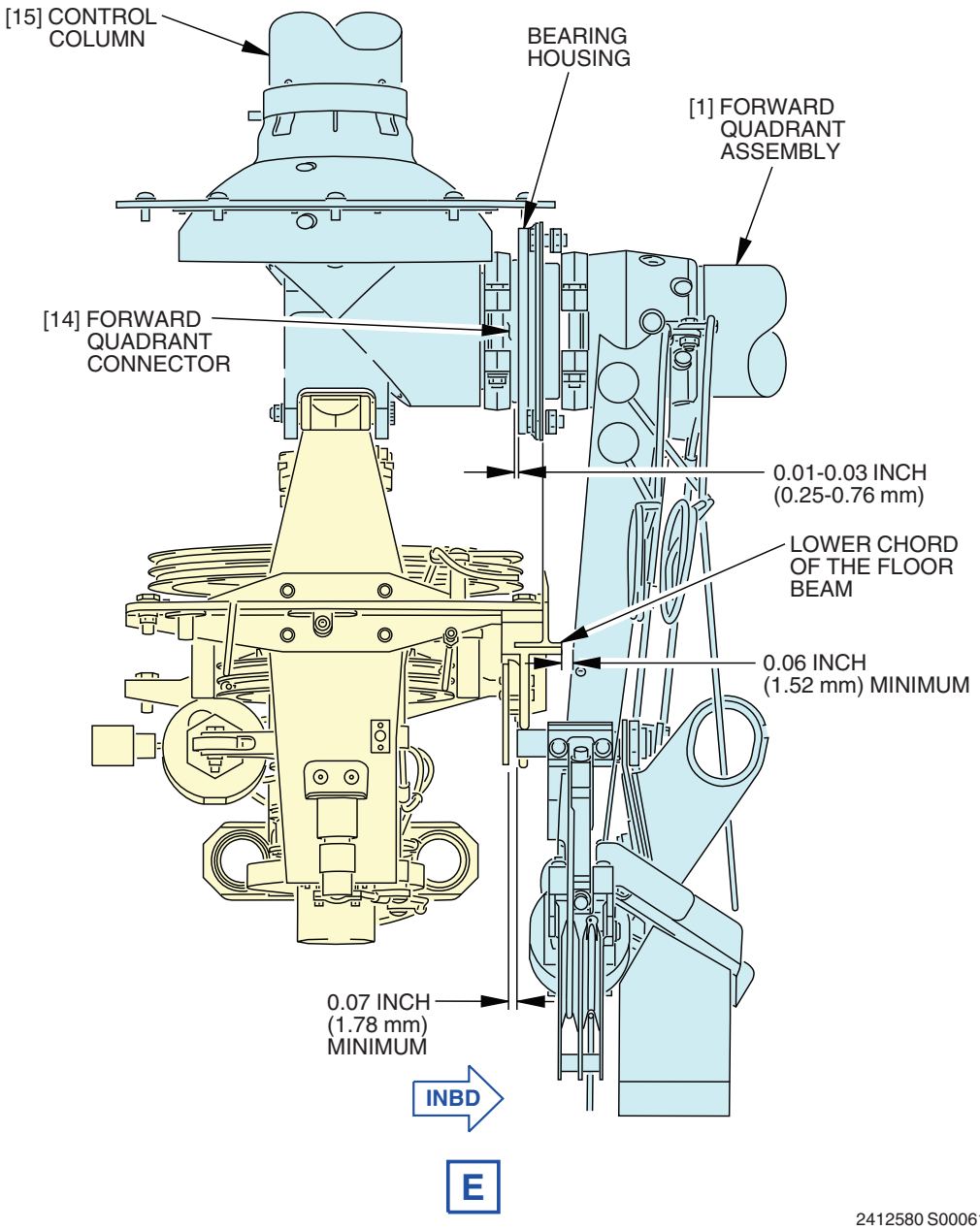
Elevator Forward Control Quadrant Assembly Installation
Figure 402/27-31-61-990-802 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Forward Control Quadrant Assembly Installation
Figure 402/27-31-61-990-802 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Forward Control Quadrant Assembly Installation
Figure 402/27-31-61-990-802 (Sheet 7 of 7)

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TASK 27-31-61-400-801

3. Elevator Forward Control Quadrant Assembly Installation

(Figure 401, Figure 402)

A. References

Reference	Title
22-11-92-400-801	Pitch Force Transducer Installation (P/B 401)
27-31-00-700-808	Manual Mode - Test (P/B 501)
27-31-00-820-801	Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment (P/B 501)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1569	Clamp - Control Cable Part #: A20005-9 Supplier: 81205
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00225	Compound - Two Part Hole Filling, Polysulfide	BMS5-16
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right
318	Tail Cone Compartment - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door
318BR	Tailcone Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Forward Control Quadrant Installation

(Figure 402)

SUBTASK 27-31-61-480-003

(1) Use the protractor, SPL-1680, to set the control column [15] to 6.9 ± 0.8 degrees forward of a vertical line from the floor (Figure 402, View A-A).

SUBTASK 27-31-61-420-001

(2) Install the mass balance weight [5] to the forward quadrant assembly [1]:

(a) Put the mass balance weight [5] in its position.

(b) Install the bolts [11] and washers [12]].

(c) Install the MS20995NC32 lockwire, G01912.

SUBTASK 27-31-61-480-004

(3) Install the rig pin E-1 in the left side of the forward quadrant assembly [1].

SUBTASK 27-31-61-420-002

(4) Install the forward quadrant assembly [1]:

(a) Set the control wheels to the neutral position.

(b) Put the forward quadrant assembly [1] in its position.

(c) Move the control columns inboard and at the same time align the spines of the elevator forward quadrant assembly [1] with the forward quadrant connector [14].

(d) Install inboard couplings [16] with the bolts [13], washers [17], and nuts [18].

NOTE: Make sure that you use the correct bolts.

(e) Do these steps to make the clearances equal at each end of each inboard coupling:

1) Tighten the bolts [13] to 50 in-lb (5.6 N·m) of torque.

2) Tighten the bolt [13] and then the other to 50 in-lb (5.6 N·m)-70 in-lb (7.9 N·m) in approximately 10 in-lb (1.1 N·m) increments until you get the clearances equal to ± 0.03 in. (0.76 mm) of each other.

SUBTASK 27-31-61-820-001

(5) Adjust the shim below the bearing housings:

(a) Make sure the clearance between lower chord of the floor beam and each side of the forward quadrant assembly [1] is 0.06 in. (1.5 mm) minimum through the full travel.

(b) Make sure the total clearance (sum of the clearances of the two side of forward quadrant assembly) between the bearing and forward quadrant connector is 0.01 in. (0.25 mm) - 0.03 in. (0.76 mm).

(c) Make sure the total side play of the control column assembly is 0.01 in. (0.25 mm) - 0.03 in. (0.76 mm).

(d) Make sure the outboard end of the stop on the left side of the forward quadrant assembly [1] is clear of the inboard face of the bracket by a minimum of 0.07 in. (1.78 mm).

(e) Make sure that the difference in left and right shim thickness does not exceed 0.010 in. (0.254 mm).

(f) Make sure that the total end play of the forward quadrant assembly [1] is 0.01 in. (0.25 mm) - 0.03 in. (0.76 mm).

(g) If it is necessary, add or remove shim(s) below the bearing housing.

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SUBTASK 27-31-61-420-003

(6) Install the ring assemblies and dust seals:

(a) Install the forward dust seals [26], aft dust seals [25], forward ring assemblies [21], and aft ring assemblies [24] with the screws [22] and washers [23].

CAUTION

DO NOT HOLD THE DUST CAP TOO TIGHTLY AGAINST THE WIRE BUNDLE. IT CAN CAUSE DAMAGE TO THE ELECTRICAL WIRES.

(b) If it is necessary, adjust the dust cap [20] to set the edge of the dust seals lightly against the dust caps [20].

1) Loosen the clamps to adjust the dust caps [20].

(c) Make sure the gap between the two dust seals is 0.6 in. (15.2 mm).

(d) Fill the gap between the two dust seals with compound, A00225.

(e) Fill the hole between the notch in the dust seals and the cutout in the floor with compound, A00225.

SUBTASK 27-31-61-080-002

(7) Remove the rig pin E-1 from the rigging pin kit, SPL-1585, from the left side of the forward quadrant assembly [1].

SUBTASK 27-31-61-710-001

(8) Do a check of the control column [15]:

(a) Move the control column [15] full forward then full aft.

1) Make sure the control column [15] operates correctly and smoothly.

SUBTASK 27-31-61-420-004

(9) Connect the EA cables [2] and EB cables [3] to the forward quadrant assembly [1]:

(a) Install the bolts [6], washers [7], and nuts [8] that attach the fairleads [9] to the forward quadrant assembly [1].

(b) Remove the identification tags from the EA cable [2] and EB cable [3].

(c) Install the EA cables [2] and EB cables [3] to the forward quadrant assembly [1].

(d) Install the retainers [4] and screws [10] to the EA cables [2] and EB cables [3].

(e) Remove the control cable clamp, SPL-1569, if it is installed.

SUBTASK 27-31-61-080-003

(10) Remove the rig pin E-5 from the rigging pin kit, SPL-1585, from the aft control quadrant (Figure 401).

SUBTASK 27-31-61-420-005

(11) Install the pitch control wheel steering force transducers to the forward quadrant assembly (TASK 22-11-92-400-801).

SUBTASK 27-31-61-860-002

(12) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-61-820-002

(13) Adjust the elevator control cables EA and EB (TASK 27-31-00-820-801).

SUBTASK 27-31-61-710-002

(14) Do this task: Manual Mode - Test, TASK 27-31-00-700-808.

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G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-61-410-001

(1) Install this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-61-410-002

(2) Close this access panel:

Number	Name/Location
112A	Forward Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR FORWARD CONTROL QUADRANT ASSEMBLY - INSPECTION/CHECK

1. General

A. This procedure has this inspection task:

(1) Elevator Mechanical Control Path.

TASK 27-31-61-210-801

2. Elevator Mechanical Control Path - Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

B. Access Panels

Number	Name/Location
112A	Forward Access Door
318BR	Tailcone Access Door

C. Elevator Mechanical Control Path Inspection

SUBTASK 27-31-61-010-003

(1) To get access to the elevator mechanical control path, do this task:

Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-31-61-210-001

(2) Do a general visual inspection of the forward elevator mechanical control path.

SUBTASK 27-31-61-410-003

(3) Close this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-31-61-010-004

(4) To get access to the elevator mechanical control path, do this task:

Remove this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-61-210-002

(5) Do a general visual inspection of the aft elevator mechanical control path.

SUBTASK 27-31-61-410-004

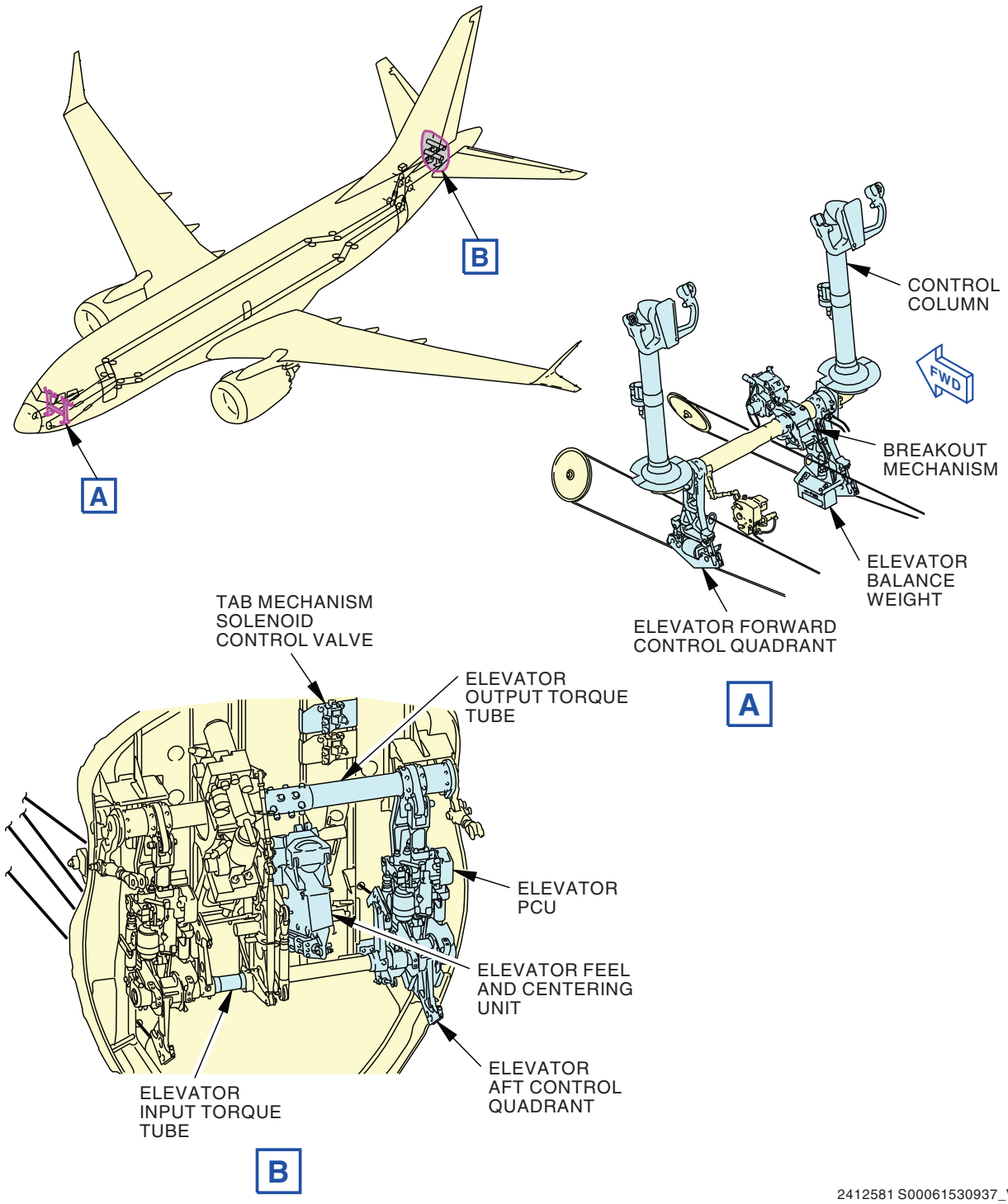
(6) Install this access panel:

Number	Name/Location
318BR	Tailcone Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Elevator Mechanical Control Path Inspection
Figure 601/27-31-61-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR FEEL AND CENTERING UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the elevator feel and centering unit

(2) An installation of elevator feel and centering unit.

(3) A removal of the elevator feel and centering unit spring.

(4) An installation of the elevator feel and centering unit spring.

TASK 27-31-64-000-801

2. Elevator Feel and Centering Unit Removal

(Figure 401)

A. References

Reference	Title
22-11-24-000-801	Mach Trim Actuator Removal (P/B 401)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-67-000-801	Elevator Feel Actuator Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Elevator feel and centering unit	27-31-64-01-180	PTF ALL

D. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

E. Access Panels

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

F. Prepare for the Removal

SUBTASK 27-31-64-010-001

(1) Remove these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

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SUBTASK 27-31-64-860-001

(2) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-64-480-001

(3) Install rig pin E-5 from the rigging pin kit, SPL-1585, in the elevator aft control quadrant.

SUBTASK 27-31-64-480-002

(4) Attach DO NOT OPERATE tags, STD-858, on the control columns.

SUBTASK 27-31-64-020-001

(5) Do this task: Mach Trim Actuator Removal, TASK 22-11-24-000-801.

SUBTASK 27-31-64-020-002

(6) Do this task: Elevator Feel Actuator Removal, TASK 27-31-67-000-801.

G. Elevator Feel and Centering Unit Removal

SUBTASK 27-31-64-020-003

(1) Remove the elevator feel and centering unit [1]:

(a) Remove the nut [7], washer [6], nut [18], washer [5], outer bolt [3], and bolt [4] to disconnect the output rod [8] from the crank [2].

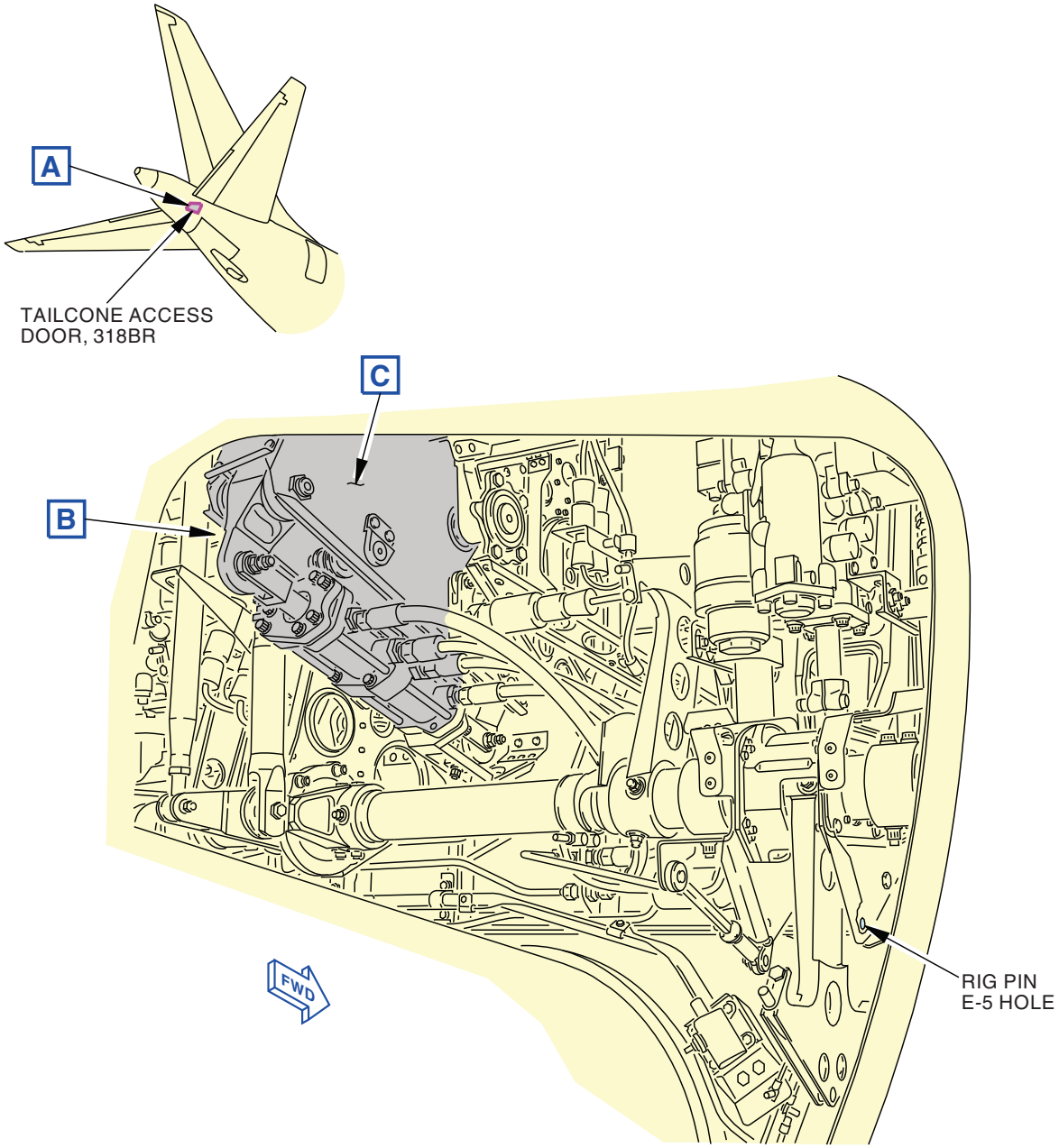
(b) Remove the nut [12], washer [11], washer [10], and bolt [9] to disconnect the micro-adjustment rod [13] from the elevator feel and centering unit [1].

(c) Hold the elevator feel and centering unit [1] and remove the bolts [16] and washers [17] from the bearing assemblies [14].

(d) Remove the elevator feel and centering unit [1] from the support brackets [15].

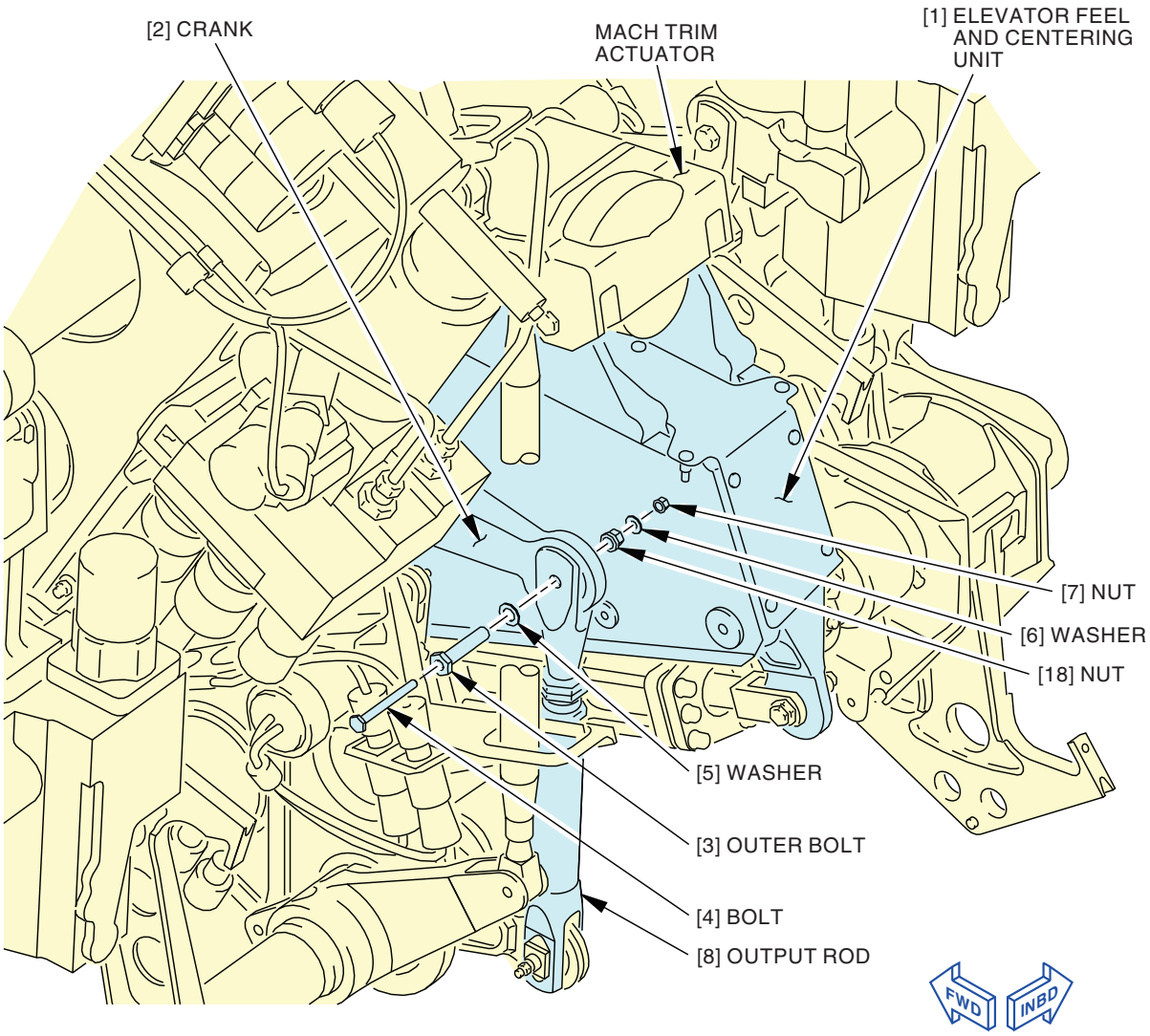
——— END OF TASK ———

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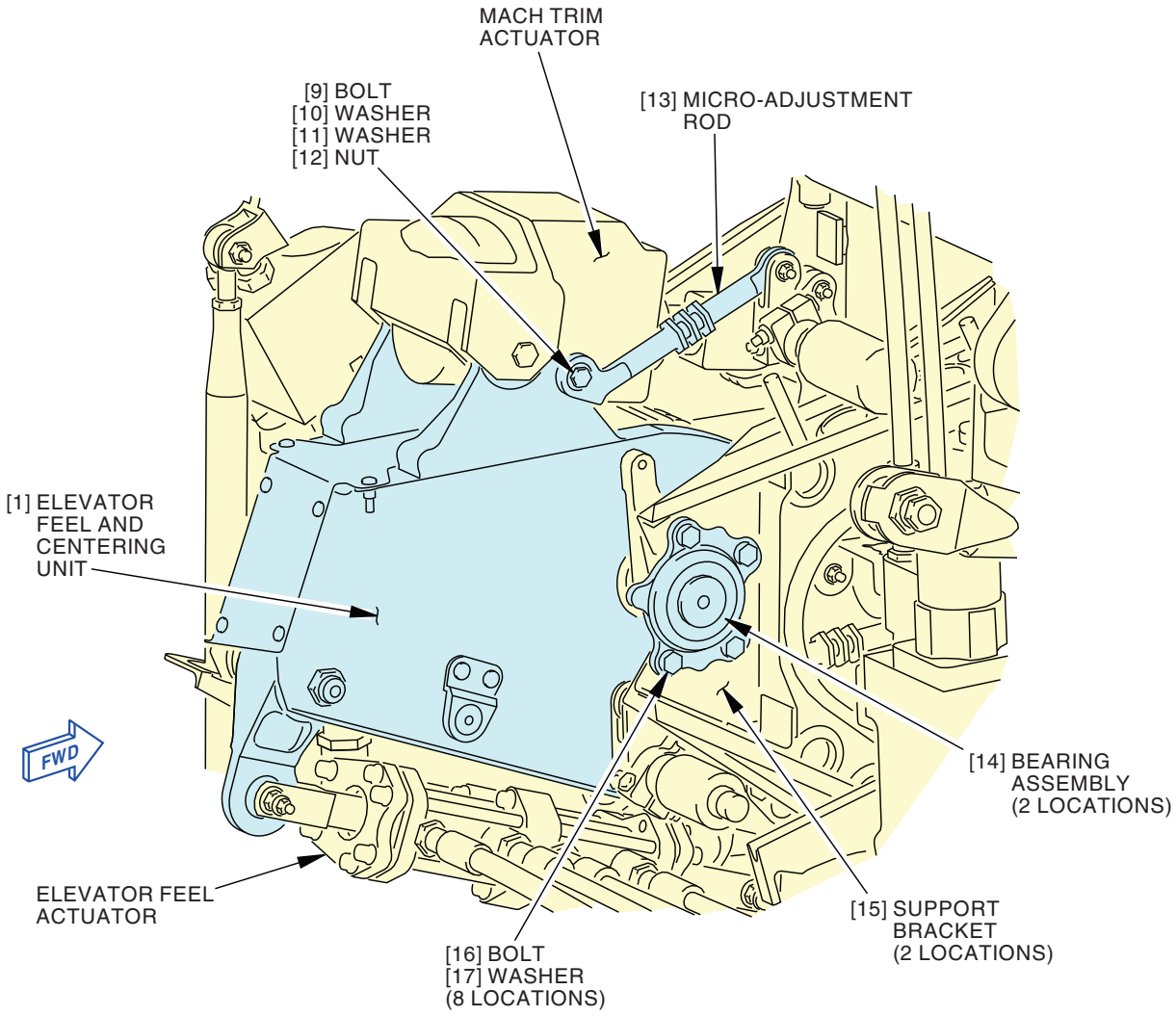
Elevator Feel and Centering Unit Installation
Figure 401/27-31-64-990-801 (Sheet 1 of 3)

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Elevator Feel and Centering Unit Installation
Figure 401/27-31-64-990-801 (Sheet 2 of 3)

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C

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Elevator Feel and Centering Unit Installation
Figure 401/27-31-64-990-801 (Sheet 3 of 3)

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TASK 27-31-64-400-801

3. Elevator Feel and Centering Unit Installation

(Figure 401)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
22-11-14-820-801	Elevator Neutral Shift Sensor Adjustment and Test (P/B 501)
22-11-24-400-801	Mach Trim Actuator Installation (P/B 401)
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-700-813	Feel Force System - Test (P/B 501)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
27-31-67-400-801	Elevator Feel Actuator Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Elevator feel and centering unit	27-31-64-01-180	PTF ALL

D. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

E. Access Panels

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

F. Elevator Feel and Centering Unit Installation

SUBTASK 27-31-64-420-001

(1) Install the elevator feel and centering unit [1]:

(a) Put the elevator feel and centering unit [1] in its position.

(b) Align the bolt installation holes in the bearing assemblies [14] with support brackets [15].

(c) Install the bolts [16] and washers [17] to attach the bearing assemblies [14] to the support brackets [15].

(d) Install the bolt [9], washer [10], washer [11], and nut [12] to connect the micro-adjustment rod [13] to the elevator feel and centering unit [1].

SUBTASK 27-31-64-420-002

(2) Do this task: Elevator Feel Actuator Installation, TASK 27-31-67-400-801.

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SUBTASK 27-31-64-420-003

(3) Do this task: Mach Trim Actuator Installation, TASK 22-11-24-400-801.

SUBTASK 27-31-64-860-002

(4) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-64-820-001

(5) Adjust the output rod [8]:

(a) Align the bolt hole of the output rod [8] and bolt hole of the crank [2].

(b) If the alignment difference is more that 0.06 in. (1.52 mm), then do these steps:

1) Turn the sleeve on the output rod [8] until you can install the bolt [4] easily.

NOTE: Always turn the clevis and the sleeve in the same direction and number of threads are equal in the two parts.

SUBTASK 27-31-64-420-004

(6) Install the bolt [4], outer bolt [3], washer [5] (under the outer bolt [3]), nut [18], washer [6] and nut [7] to connect the output rod [8] to the crank [2].

(a) Tighten the outer bolt [3] to 75 in-lb (8.5 N·m) - 85 in-lb (9.6 N·m).

CAUTION

DO NOT TIGHTEN THE OUTER BOLT AGAIN AFTER YOU TIGHTEN THE LAST NUT. YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(b) Tighten the nut [7] to 25 in-lb (2.8 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 27-31-64-080-001

(7) Remove the rig pin E-5 of the rigging pin kit, SPL-1585, from the elevator aft control quadrant.

SUBTASK 27-31-64-820-002

(8) Do this task: Elevator Neutral Shift Sensor Adjustment and Test, TASK 22-11-14-820-801.

SUBTASK 27-31-64-860-003

(9) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-64-080-002

(10) Remove the DO NOT OPERATE tags, STD-858, from the control columns.

SUBTASK 27-31-64-710-001

(11) Move the control column full forward and full aft.

(a) Make sure the control column moves freely and smoothly.

SUBTASK 27-31-64-710-002

(12) Do this task: Feel Force System - Test, TASK 27-31-00-700-813.

SUBTASK 27-31-64-790-001

(13) Make sure that there is no hydraulic leakage at the elevator feel actuator.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-64-410-001

(1) Install these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

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SUBTASK 27-31-64-610-001

(2) If it is necessary, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

——— END OF TASK ———

TASK 27-31-64-000-802

4. Elevator Feel and Centering Unit Spring Removal

(Figure 402)

A. References

Reference	Title
27-31-00-000-801	Elevator and Tab Control System - Deactivation (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

D. Access Panels

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

E. Prepare for the Removal

SUBTASK 27-31-64-010-002

(1) Open these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-64-040-001

(2) Do this task: Elevator and Tab Control System - Deactivation, TASK 27-31-00-000-801.

SUBTASK 27-31-64-860-004

(3) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-64-400-005

(4) Install the rig pin E-5 from the rigging pin kit, SPL-1585 in the rig pin E-5 hole.

SUBTASK 27-31-64-040-002

(5) Attach a DO NOT OPERATE tag, STD-858 to the control columns.

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F. Elevator Feel and Centering Unit Spring Removal

SUBTASK 27-31-64-020-009

(1) Disconnect the output rod [8] from the crank [2].

(a) Remove the nut [7], washer [6], nut [18], washer [5], outer bolt [3], and bolt [4] to disconnect the output rod [8] from the crank [2].

(b) Move the output rod [8] out of the way.

SUBTASK 27-31-64-000-002

(2) Remove the elevator feel actuator [20] from the elevator feel and centering unit [1].

NOTE: Do not disconnect the hydraulic lines from the elevator feel actuator [20].

(a) Remove the nut [37], washer [38], nut [39], washer [40], and bolt assembly [41] from the aft connection of the elevator feel actuator [20].

(b) Remove the nut [27], washer [26], nut [25], washer [24], antirotation fitting [23], spacer [22], and bolt assembly [21] from the elevator feel actuator [20].

(c) Remove the elevator feel actuator [20].

(d) Set the elevator feel actuator [20] carefully on a piece of soft fabric to prevent damage of the elevator feel actuator [20] or adjacent components.

1) Be careful not to strain the hydraulic lines.

SUBTASK 27-31-64-000-003

(3) Remove the elevator feel and centering cover [19] from the elevator feel and centering unit [1].

NOTE: Removal of the cover is done only to increase visibility of the springs and is not mandatory for this procedure.

SUBTASK 27-31-64-000-004

(4) Remove the inner spring [43] and outer spring [42]:

(a) Remove the bolts [34] to loosen the retainer [47].

(b) Turn the spring post adjustment wrench flats [33] to loosen the springs.

(c) Remove the aft hook of the outer spring [42] from the spring post assembly [44].

1) Use a non-metallic tie-wrap or strap to pull the spring in the aft direction until the outer spring [42] hook is completely removed from the eccentric groove of the spring post assembly [44].

NOTE: Use a protective thin plastic to prevent rubbing the housing surface of the springs.

(d) Remove the aft hook of the inner spring [43] from the spring spring post assembly [44].

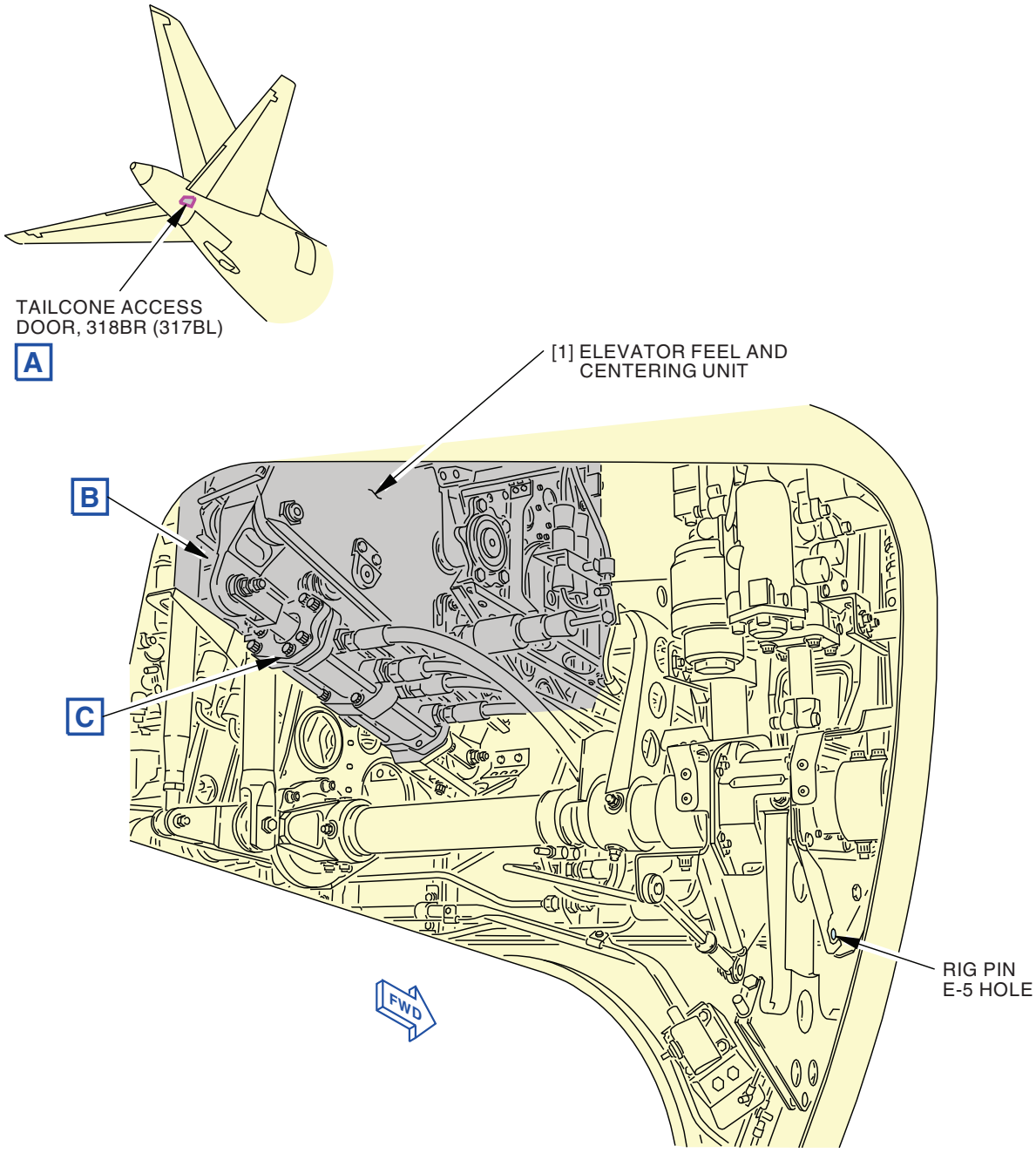
1) Use a non-metallic tie-wrap or strap to pull the spring in the aft direction until the inner spring [43] hook is completely removed from the eccentric groove of the spring post assembly [44].

NOTE: Use a protective thin plastic to prevent rubbing the housing surface of the springs.

(e) Remove the forward hooks of the outer spring [42] and inner spring [43] from the cam follower assembly [45].

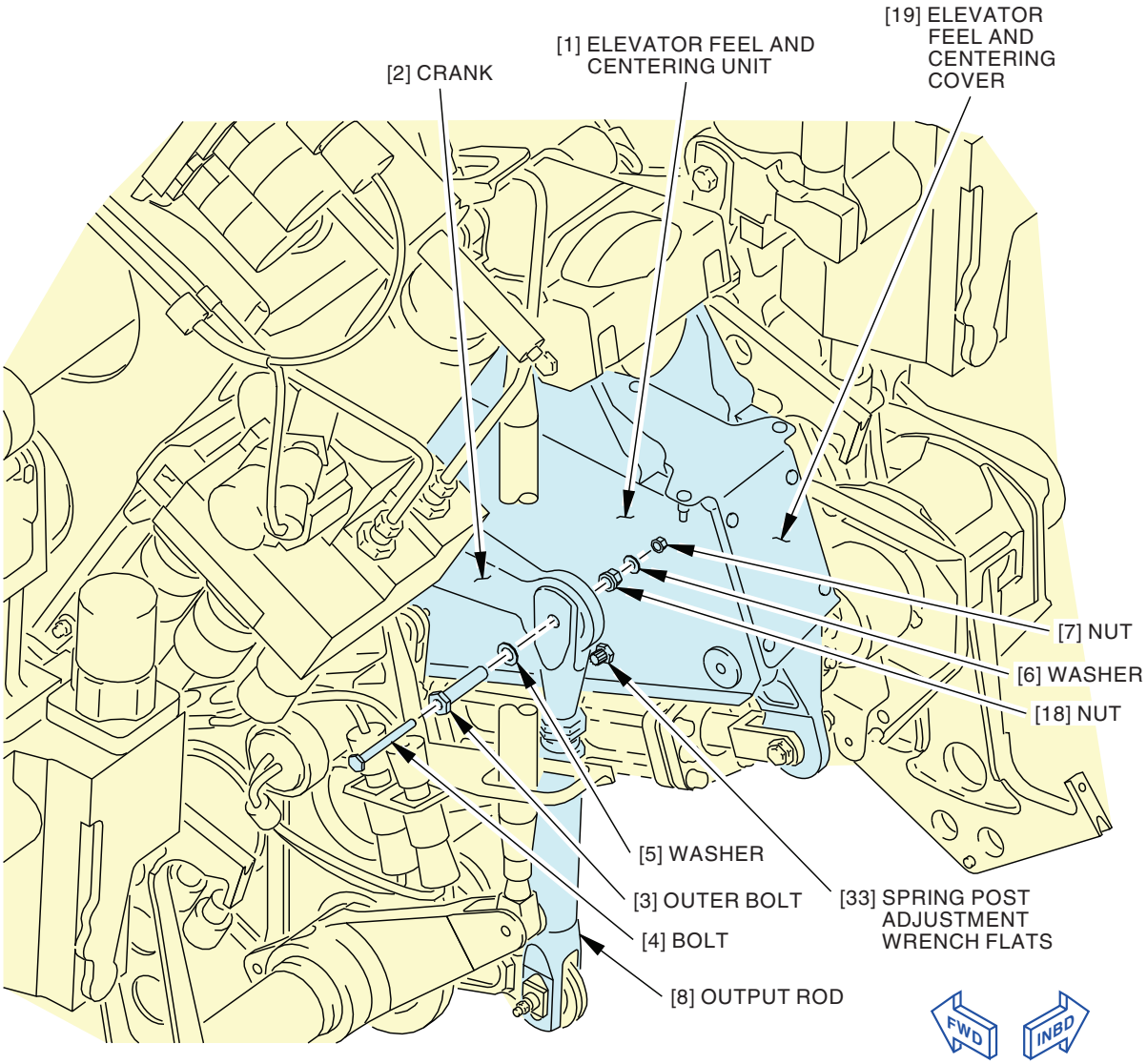
END OF TASK

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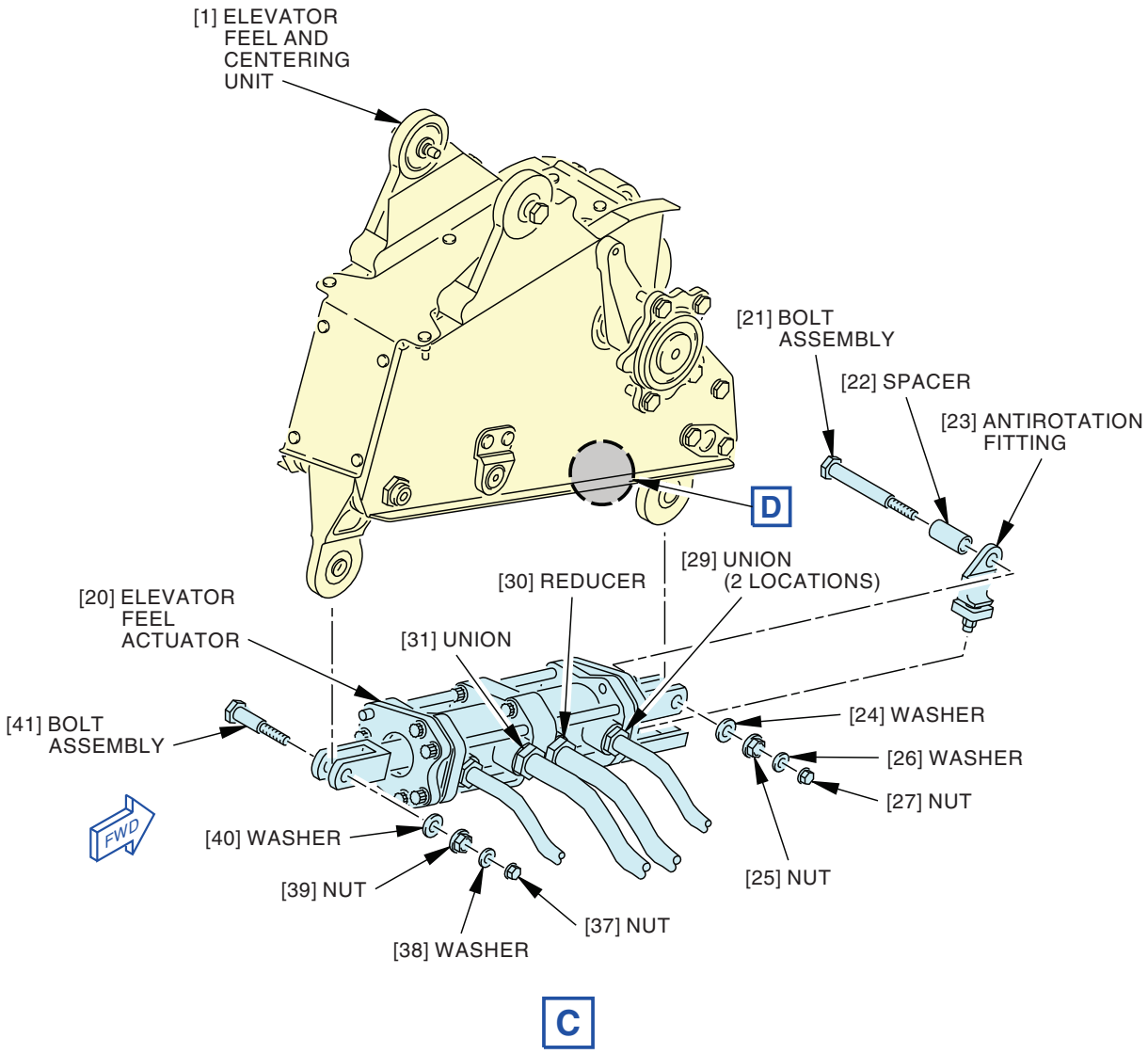
Elevator Feel and Centering Unit Spring Installation
Figure 402/27-31-64-990-803 (Sheet 1 of 4)

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Elevator Feel and Centering Unit Spring Installation
Figure 402/27-31-64-990-803 (Sheet 2 of 4)

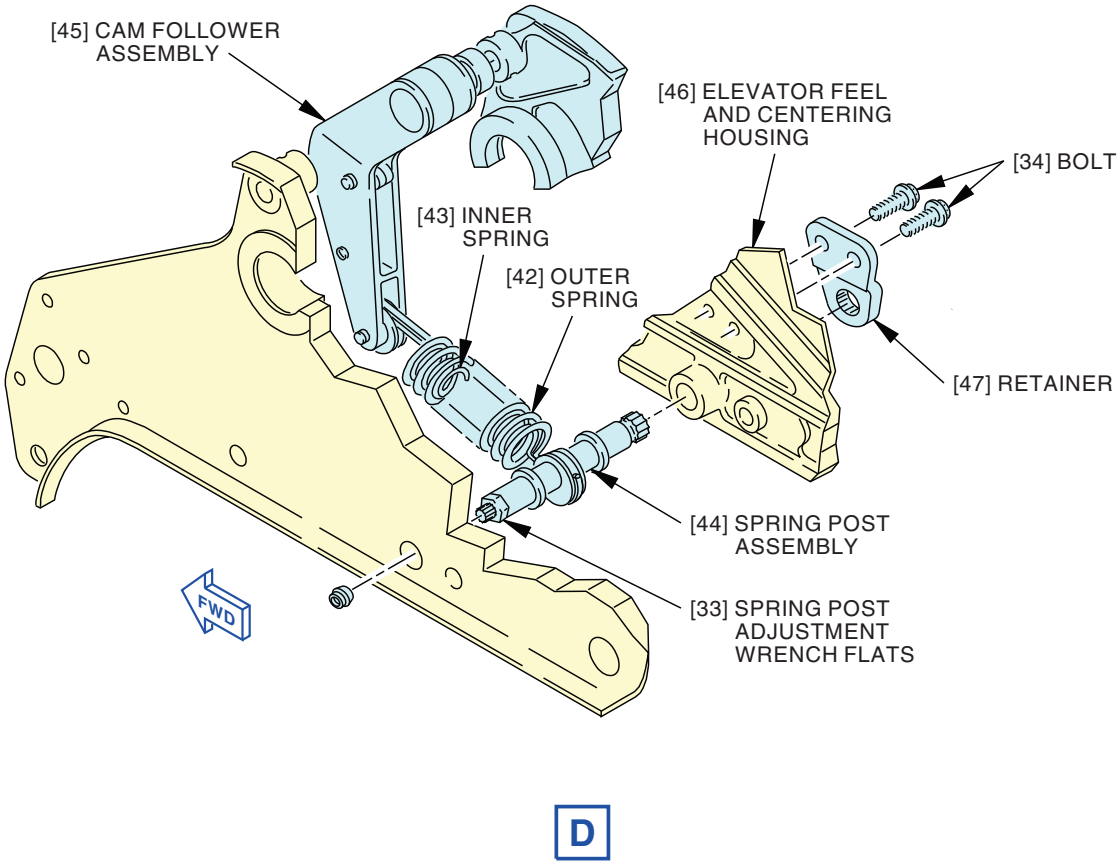
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Elevator Feel and Centering Unit Spring Installation
Figure 402/27-31-64-990-803 (Sheet 3 of 4)

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Elevator Feel and Centering Unit Spring Installation
Figure 402/27-31-64-990-803 (Sheet 4 of 4)

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TASK 27-31-64-400-802

5. Elevator Feel and Centering Unit Spring Installation

(Figure 402)

A. References

Reference	Title
27-31-00-400-801	Elevator and Tab Control System - Activation (P/B 201)
27-31-00-700-810	Control Column Power Force - Test (P/B 501)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1452	Gauge - Mechanical Force, 0-100 lbs. (0.5 lbs. Resolution) Part #: LG-100 Supplier: 92456
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-858	Tag - DO NOT OPERATE
STD-11969	Dial Indicator - 0.001 inch graduations

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
42	Outer spring	27-31-64-02-400	PTF ALL
43	Inner spring	27-31-64-02-405	PTF ALL

D. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

E. Access Panels

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

F. Elevator Feel and Centering Unit Spring Installation

SUBTASK 27-31-64-400-001

(1) Replace the inner spring [43] and outer spring [42]:

(a) Install the forward spring hooks on the cam follower assembly [45].

1) Make sure that the hooks are put in the eccentric groove correctly.

(b) Use a non-metallic tie-wrap or strap to install the aft spring hooks.

1) Install the inner spring [43] on the spring post assembly [44] by pulling the inner spring coil aft and bending it slightly.

a) Make sure that the hook is put in the eccentric groove correctly.

2) Install the outer spring [42] on the spring post assembly [44] by pulling the outer spring coil aft and bending it slightly.

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a) Make sure that the hook is put in the eccentric groove correctly.

3) Make sure that the forward spring hooks stay on the cam follower assembly [45].

(c) Install the bolts [34] and retainer [47].

(d) Do a detailed visual inspection of the inner spring [43] and outer spring [42] to make sure the spring hooks are in the eccentric groove correctly.

(e) Turn the spring post adjustment wrench flat [33] to tighten the springs.

SUBTASK 27-31-64-700-001

(2) Do a breakout test of the crank [2].

(a) Use a non-metallic tie-wrap or strap to cycle the crank [2] a minimum of 3 times:

1) Use the 0-100 lbs. (0.5 lbs. resolution) mechanical force gauge, COM-1452 to pull down on a non-metallic tie-wrap or strap attached to the crank and note the force when the dial indicator, STD-11969 shows 0.025 in. (0.64 mm).

a) Force must measure 13.0 lbf (57.8 N).

2) Relax the tension on the strap and note the force when the crank starts to move upward.

a) Force must measure 12.0 lbf (53.4 N).

3) Use the 0-100 lbs. (0.5 lbs. resolution) mechanical force gauge, COM-1452 to pull up on a non-metallic tie-wrap or strap attached to the crank and note the force when the dial indicator, STD-11969 shows 0.025 in. (0.64 mm).

a) Force must measure 11.5 lbf (51.2 N).

4) Relax the tension on the strap and note the force when the crank starts to move downward.

a) Force must measure 10.5 lbf (46.7 N).

(b) If the force values exceed these limits, loosen the retainer [47] and rotate the spring post adjustment wrench flat [33] if necessary.

1) Repeat the breakout test until the force values are in the specified limits.

SUBTASK 27-31-64-400-002

(3) Install the elevator feel actuator [20].

(a) Put the elevator feel actuator [20] in its position and hold it in place.

(b) Install the bolt assembly [41], washer [40], and nut [39] to attach the aft connection of the washer [40] to the elevator feel and centering unit [1].

1) Tighten the nut [39] to 85 in-lb (9.6 N·m) - 100 in-lb (11.3 N·m).

(c) Install the washer [38] and nut [37].

CAUTION

DO NOT TIGHTEN THE FIRST NUT AGAIN AFTER YOU TIGHTEN THE LAST NUT. YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

1) Tighten the nut [27] to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

(d) Install the bolt assembly [21], nut [12], antirotation fitting [23], washer [24], and the nut [25] to attach the forward connection of the elevator feel actuator [20] to the elevator feel and centering unit [1].

1) Tighten the nut [25] to 85 in-lb (9.6 N·m) - 100 in-lb (11.3 N·m).

(e) Install the washer [26] and nut [27].

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CAUTION

DO NOT TIGHTEN THE FIRST NUT AGAIN AFTER YOU TIGHTEN THE LAST NUT. YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

1) Tighten the nut [27] to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 27-31-64-400-003

(4) Install the elevator feel and centering cover [19] if removed.

SUBTASK 27-31-64-400-004

(5) Install the elevator feel and centering unit output rod [8].

(a) Install the bolt [4], outer bolt [3], washer [5] (under the outer bolt [3]), and nut [18] to connect the output rod [8] to the crank [2].

1) Tighten the outer bolt [3] to 75 in-lb (8.5 N·m) - 85 in-lb (9.6 N·m).

(b) Install washer [6] and nut [7].

CAUTION

DO NOT TIGHTEN THE OUTER BOLT AGAIN AFTER YOU TIGHTEN THE LAST NUT. YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

1) Tighten the nut [7] to 25 in-lb (2.8 N·m) - 35 in-lb (4.0 N·m).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-64-000-001

(1) Remove the rig pin E-5 of the rigging pin kit, SPL-1585 from the rig pin E-5 hole.

SUBTASK 27-31-64-410-002

(2) Close these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-64-440-001

(3) Do this task: Elevator and Tab Control System - Activation, TASK 27-31-00-400-801.

SUBTASK 27-31-64-860-005

(4) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-64-440-002

(5) Remove the DO NOT OPERATE tag, STD-858 from the control columns.

SUBTASK 27-31-64-700-002

(6) Do this task: Control Column Power Force - Test, TASK 27-31-00-700-810.

——— END OF TASK ———

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ELEVATOR FEEL AND CENTERING UNIT - INSPECTION/CHECK

1. General

A. This procedure contains this task:

(1) Elevator Feel and Centering Unit Inspection

(a) This task supplies the data for a dimensional inspection of the elevator feel and centering unit.

(b) This task does not remove or install the elevator feel and centering unit. Refer to the Elevator Feel and Centering Unit - Removal/Installation for procedures to remove and install the elevator feel and centering unit.

TASK 27-31-64-200-801

2. Elevator Feel and Centering Unit Inspection

(Figure 601)

A. General

(1) This procedure gives the wear limits and the repair data for the elevator feel and centering unit.

B. References

Reference	Title
27-31-64-000-801	Elevator Feel and Centering Unit Removal (P/B 401)
27-31-64-400-801	Elevator Feel and Centering Unit Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

E. Procedure

SUBTASK 27-31-64-020-004

(1) Remove the elevator feel and centering unit. Do this task: Elevator Feel and Centering Unit Removal, TASK 27-31-64-000-801.

SUBTASK 27-31-64-020-005

(2) Use these tools to do a check for worn areas on the parts of the elevator feel and centering unit connections:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-31-64-020-006

(3) Examine the parts of the elevator feel and centering unit connections for worn areas. Do these steps:

(a) Remove the parts that you will examine.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096, or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

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(c) Compare the dimensions that you measured with the permitted wear dimensions that are shown in Table 601.

(d) If a part is out of tolerance, then replace or repair the part.

Table 601/27-31-64-993-801 Wear Limits

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 BEARING	I.D.	0.3120 in 7.925 mm	0.3125 in 7.938 mm	0.3155 in 8.014 mm	0.0035 in 0.089 mm
#1 BOLT	O.D.	0.3110 in 7.899 mm	0.3120 in 7.925 mm	0.3106 in 7.889 mm	
#2 OUTPUT ROD	I.D.	0.3120 in 7.925 mm	0.3130 in 7.950 mm	0.3160 in 8.026 mm	0.0040 in 0.102 mm
#2 BOLT	O.D.	0.3110 in 7.899 mm	0.3120 in 7.925 mm	0.3106 in 7.889 mm	

SUBTASK 27-31-64-020-007

(4) Examine the spring of the elevator feel and centering unit. Refer to Figure 601 (Sheet 3).

(a) If the spring is broken, then replace the elevator feel and centering unit. Do these tasks:

1) Elevator Feel and Centering Unit Removal, TASK 27-31-64-000-801

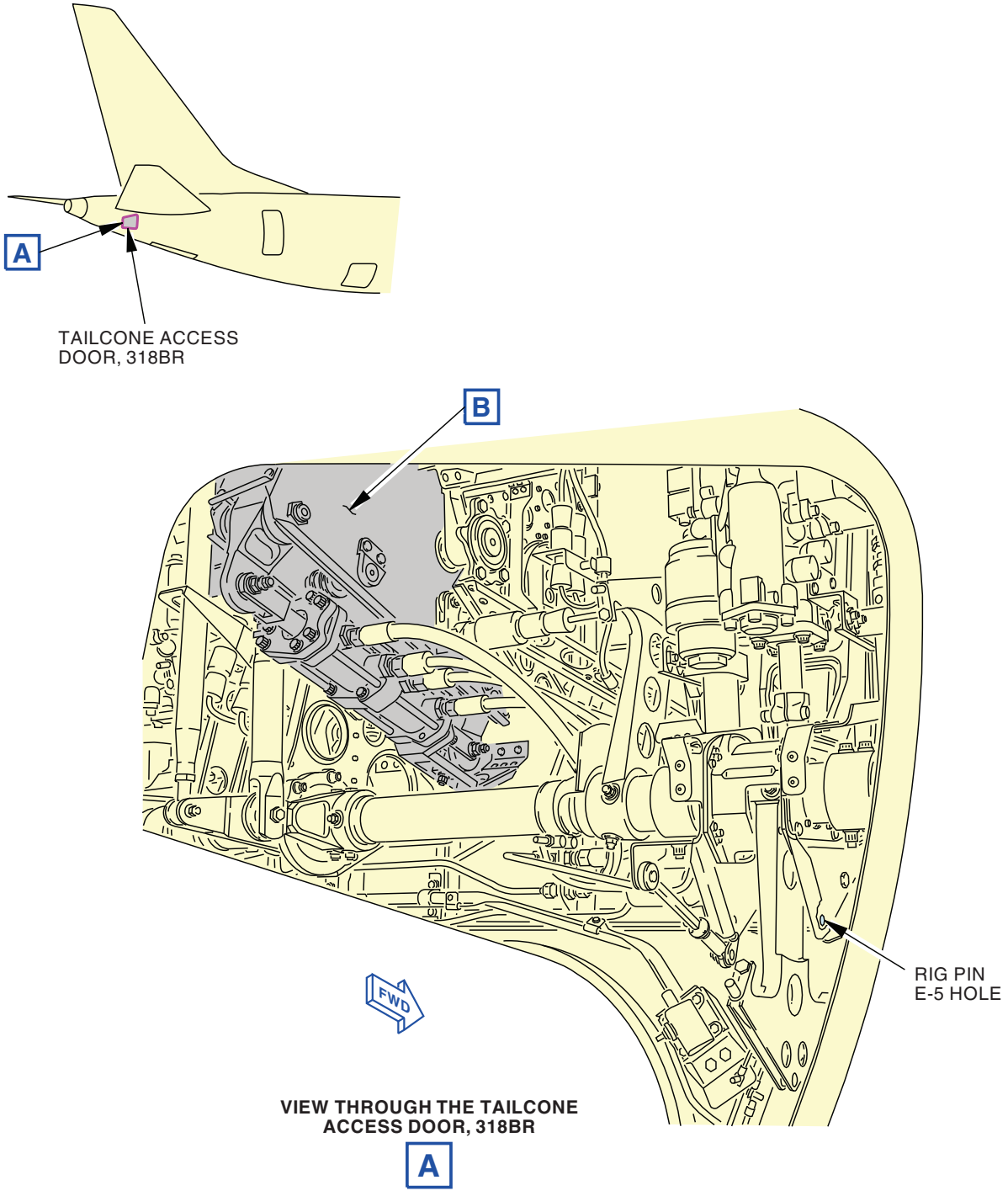
2) Elevator Feel and Centering Unit Installation, TASK 27-31-64-400-801

SUBTASK 27-31-64-020-008

(5) Install the elevator feel and centering unit. Do this task: Elevator Feel and Centering Unit Installation, TASK 27-31-64-400-801.

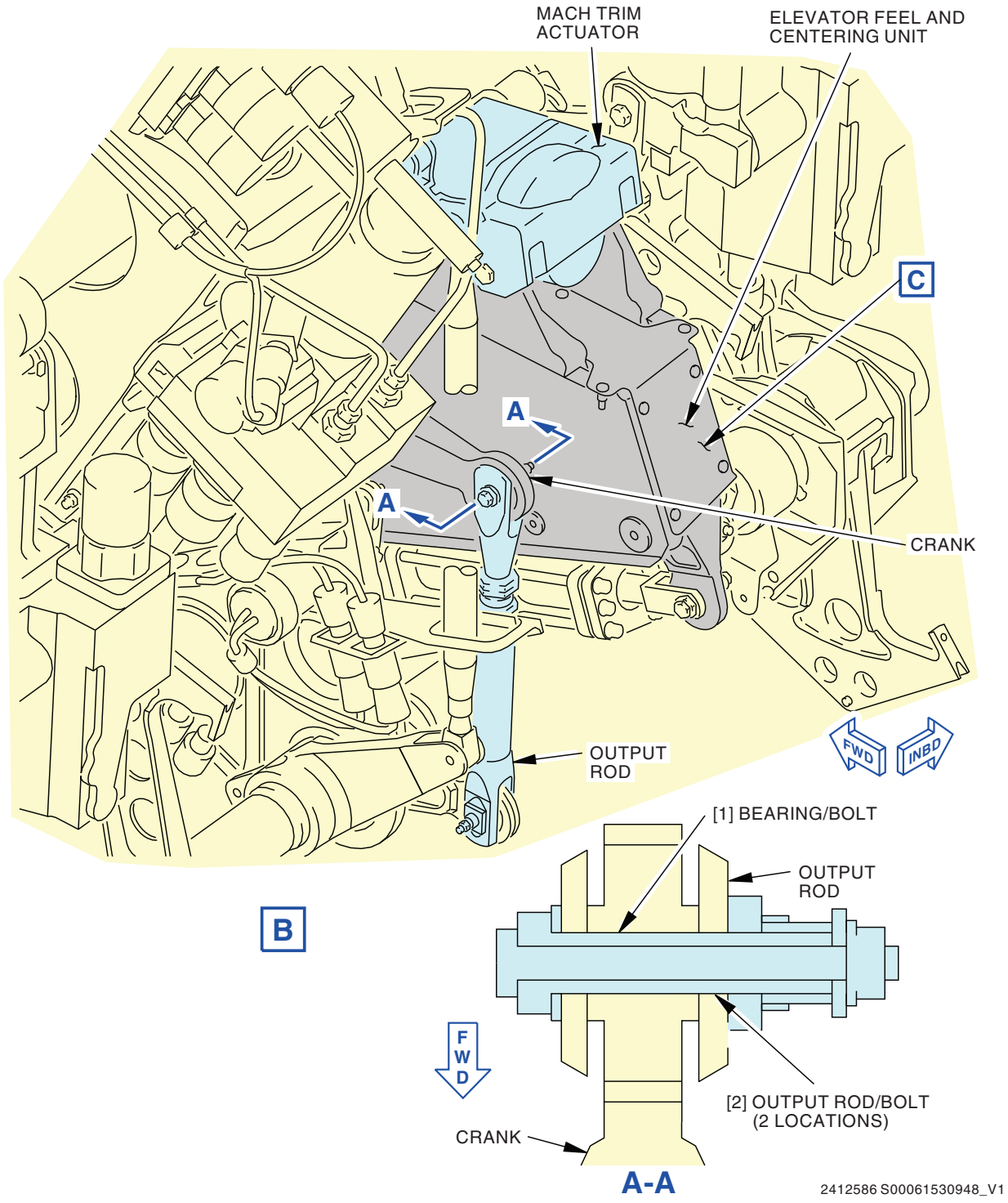
——— END OF TASK ———

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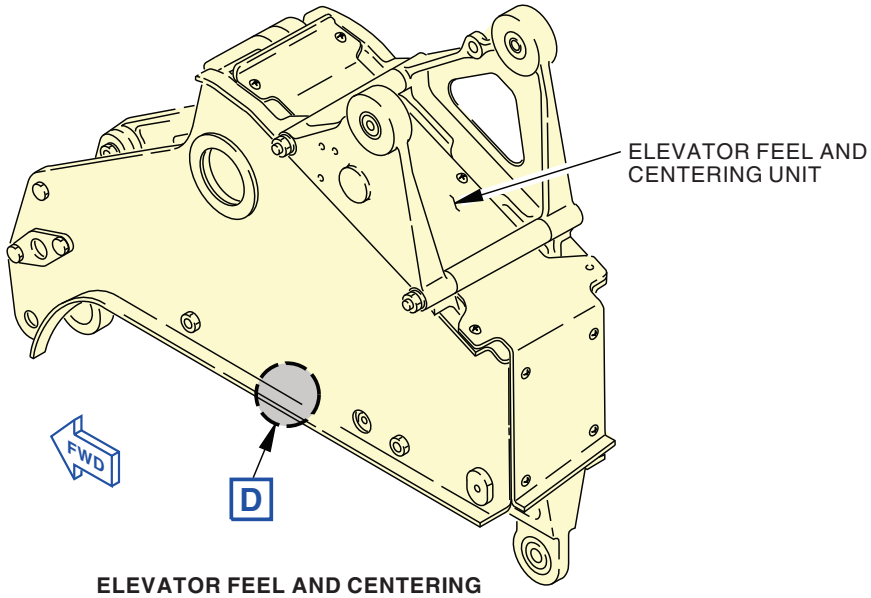
Elevator Feel and Centering Unit Wear Limits
Figure 601/27-31-64-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



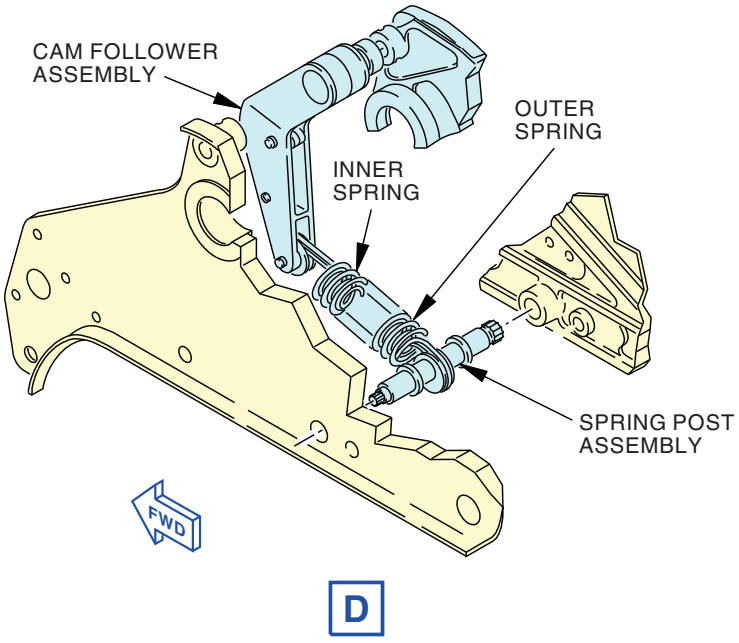
Elevator Feel and Centering Unit Wear Limits
Figure 601/27-31-64-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELEVATOR FEEL AND CENTERING UNIT

C



D

2412587 S00061530949_V1

Elevator Feel and Centering Unit Wear Limits
Figure 601/27-31-64-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR FEEL ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the elevator feel actuator

(2) An installation of the elevator feel actuator.

B. The elevator feel actuator is installed below the elevator feel and centering unit.

TASK 27-31-67-000-801

2. Elevator Feel Actuator Removal

(Figure 401)

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

D. Access Panels

Number	Name/Location
318BR	Tailcone Access Door

E. Prepare for the Removal

SUBTASK 27-31-67-860-001

(1) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-67-010-001

(2) Remove this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-67-480-001

(3) Install the rig pin E-5 from the rigging pin kit, SPL-1585, in the elevator aft control quadrant (View A, Figure 401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Elevator Feel Actuator Removal

SUBTASK 27-31-67-020-001

WARNING RELEASE THE CONNECTION SLOWLY AND LET THE PRESSURE DECREASE BEFORE YOU COMPLETELY REMOVE THE HYDRAULIC LINE. IF YOU DISCONNECT THE HYDRAULIC LINE WHEN IT HAS PRESSURE, A SPRAY OF HYDRAULIC FLUID CAN CAUSE INJURY TO PERSONS.

(1) Disconnect the hydraulic line [2], hydraulic line [25], hydraulic line [26], and hydraulic line [27] from the elevator feel actuator [3] (Figure 401).

(a) Remove the unions [12], reducer [13], and union [14].

(b) Discard the packings [11] and packings [15].

(c) Install the caps and plugs on the open ports and hydraulic line [2], hydraulic line [25], hydraulic line [26], hydraulic line [27].

SUBTASK 27-31-67-020-002

(2) Remove the elevator feel actuator [3] from the elevator feel and centering unit [1]:

(a) Remove the nut [20], washer [21], nut [22], washer [23], and bolt assembly [24] from the aft connection of the elevator feel actuator [3].

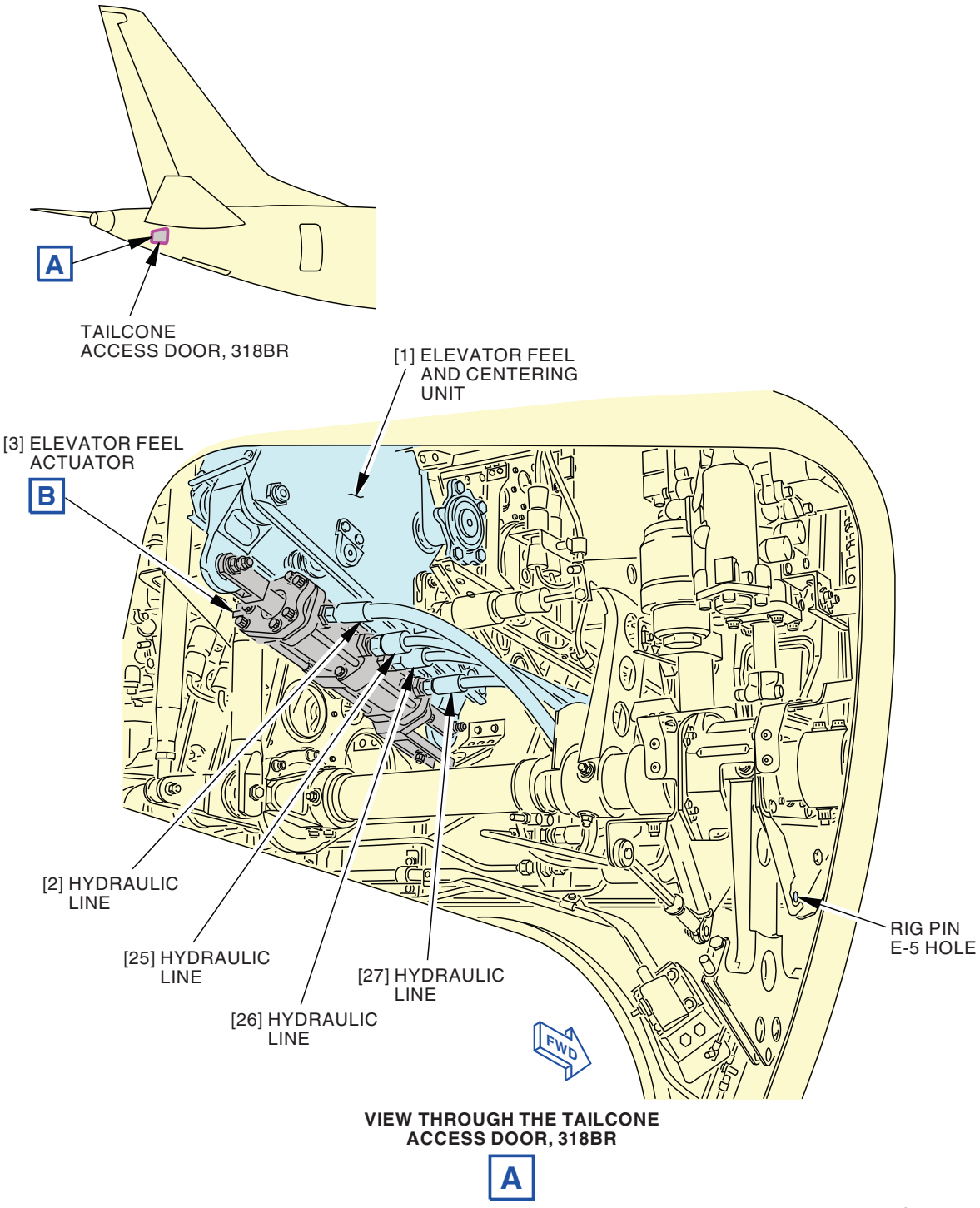
(b) Remove the nut [17], washer [18], bushing [19], and slide block [16] from the anti-rotation fitting [6].

(c) Remove the nut [10], washer [9], nut [8], washer [7], spacer [5], anti-rotation fitting [6], and bolt assembly [4] from the forward connection of the elevator feel actuator [3].

(d) Remove the elevator feel actuator [3].

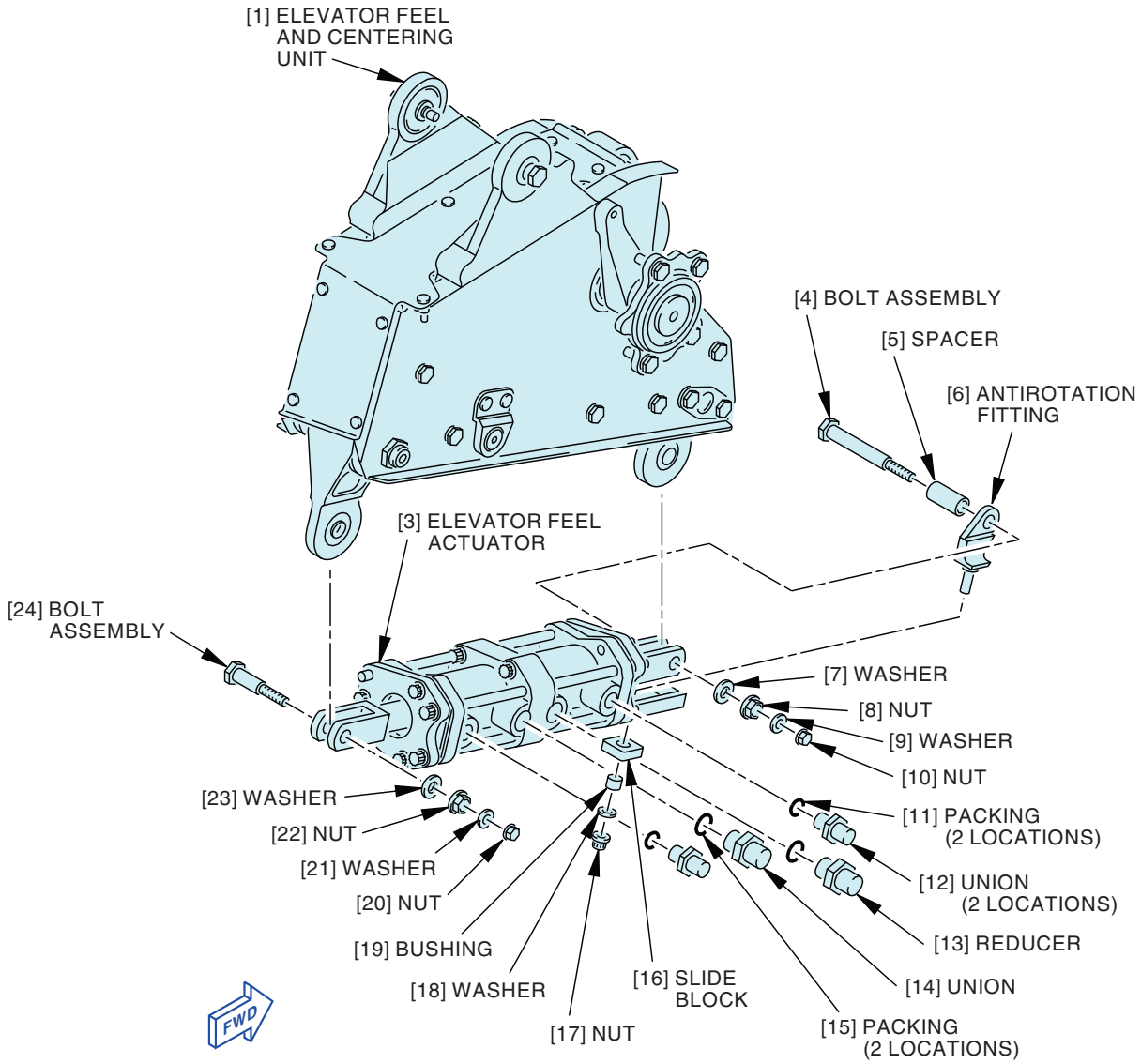
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Feel Actuator Installation
Figure 401/27-31-67-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELEVATOR FEEL ACTUATOR

B

2412589 S00061530954_V2

Elevator Feel Actuator Installation
Figure 401/27-31-67-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-67-400-801

3. Elevator Feel Actuator Installation

(Figure 401)

A. References

Reference	Title
27-31-00-700-813	Feel Force System - Test (P/B 501)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
29-11-00 P/B 201	HYDRAULIC SYSTEMS A AND B - MAINTENANCE PRACTICES
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
29-11-00-860-803	Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP) (P/B 201)
29-11-00-860-804	Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP) (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1914	Test Set - Air Data Model FLMTS (Flight Line Maintenance) Part #: ADSE 650 Supplier: 3BSK6 Part #: ADTS2000 Supplier: 52892 Part #: ADTS3350ER Supplier: 52892 Part #: ADTS405MKII Supplier: U0427 Part #: ADTS530 Supplier: U0427 Part #: ADTS552F Supplier: U0427 Part #: ADTS553F Supplier: U0427 Part #: ADTS554F Supplier: U0427 Part #: D60340MK Supplier: K1474 Part #: DPS1000 Supplier: 21844 Part #: MODEL 6300 Supplier: 0RDZ5 Part #: MPS34C Supplier: 48RQ2 Part #: MPS43B Supplier: A0197 Part #: MPS45 Supplier: 48RQ2 Part #: MPS49 Supplier: 48RQ2 Part #: TES9463 Supplier: 88277 Opt Part #: 01-0987-00 Supplier: 41364 Opt Part #: 18910480000 Supplier: 89944 Opt Part #: 18910920000 Supplier: 89944 Opt Part #: ADTS405F Supplier: U0427 Opt Part #: ADTS505 Supplier: U0427 Opt Part #: D60302 Supplier: K1474 Opt Part #: D60340 Supplier: K1474 Opt Part #: D60383 Supplier: K1474 Opt Part #: DPS350 Supplier: 21844 Opt Part #: DPS450 Supplier: 21844 Opt Part #: DPS500 Supplier: 21844 Opt Part #: MPS31C Supplier: 48RQ2 Opt Part #: MPS43 Supplier: A0197

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-1916	Adapter - Pitot Test (Typically included in Air Data Accessory Kit) Part #: CSA75700HT-3 Supplier: 3BSK6 Part #: P75701-4 Supplier: 38002 Part #: P75701M2-3 Supplier: 38002
COM-25177	Accessory Kit - Air Data, 737 (Pitot/Static adapters, hoses, etc.) Part #: ADA737-678 Supplier: 38002 Part #: AK737-900 Supplier: 3BSK6 Part #: PA-0120-A Supplier: 3BSK6
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1742	Regulator - Air Pressure, Elevator Feel Computer Part #: F72928-109 Supplier: 81205 Part #: F72928-110 Supplier: 81205 Part #: F72928-111 Supplier: 81205 Opt Part #: F72928-58 Supplier: 81205 Opt Part #: F72928-62 Supplier: 81205 Opt Part #: F72928-63 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Elevator feel actuator	27-31-67-01-090	PTF ALL
11	Packing	27-31-67-01-070	PTF ALL
15	Packing	27-31-67-01-085	PTF ALL

E. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

F. Access Panels

Number	Name/Location
318BR	Tailcone Access Door

G. Elevator Feel Actuator Installation

SUBTASK 27-31-67-420-001

(1) Install the new packings [11], new packings [15], unions [12], union [14], and reducer [13], do these steps:

(a) Lubricate the packings [11], packings [15], unions [12], union [14], and reducer [13] with hydraulic fluid, D00153.

(b) Install the packings [11], packings [15], unions [12], union [14], and reducer [13] to the elevator feel actuator [3].

SUBTASK 27-31-67-420-002

(2) Install the elevator feel actuator [3], do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put and hold the elevator feel actuator [3] in its position.

(b) Install the bolt assembly [24], washer [23], and nut [22] to attach the aft connection of the elevator feel actuator [3] to the elevator feel and centering unit [1].

1) Tighten the nut [22] to 85 in-lb (9.6 N·m) - 100 in-lb (11.3 N·m).

CAUTION

DO NOT TIGHTEN THE NUT [22] AGAIN AFTER YOU TIGHTEN THE NUT [20]. YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(c) Install the washer [21] and nut [20].

1) Tighten the nut [20] to 30 in-lb (3.4 N·m) - 35 in-lb (4 N·m).

(d) Install the bolt assembly [4], spacer [5], anti-rotation fitting [6], washer [7], and nut [8] to attach the forward connection of the elevator feel actuator [3] to the elevator feel and centering unit [1].

1) Tighten the nut [8] to 85 in-lb (9.6 N·m) - 100 in-lb (11.3 N·m).

CAUTION

DO NOT TIGHTEN THE NUT [8] AGAIN AFTER YOU TIGHTEN THE NUT [10]. YOU CAN CAUSE DAMAGE TO EQUIPMENT.

(e) Install the washer [9] and nut [10].

1) Tighten the nut [10] to 30 in-lb (3.4 N·m) - 35 in-lb (4 N·m).

(f) Install the slide block [16], bushing [19], washer [18], and nut [17] to anti-rotation fitting [6].

1) Tighten the nut [17] to 100 in-lb (11.3 N·m) - 125 in-lb (14.1 N·m).

SUBTASK 27-31-67-420-003

(3) Connect the hydraulic line [2], hydraulic line [25], hydraulic line [26], and hydraulic line [27] to the elevator feel actuator [3], do these steps:

(a) Remove the caps and plugs from the open ports, hydraulic line [2], hydraulic line [25], hydraulic line [26], and hydraulic line [27].

(b) Connect hydraulic line [2], hydraulic line [25], hydraulic line [26], and hydraulic line [27].

(c) Tighten the B-nuts on hydraulic line [2] and hydraulic line [27] to 140 ±7 in-lb (16 ±1 N·m).

(d) Tighten the B-nut on hydraulic line [25] to 170 ±9 in-lb (19 ±1 N·m).

(e) Tighten the B-nut on hydraulic line [26] to 140 ±7 in-lb (16 ±1 N·m).

SUBTASK 27-31-67-080-001

(4) Remove the rig pin E-5 of the rigging pin kit, SPL-1585, from the elevator aft control quadrant.

SUBTASK 27-31-67-860-002

(5) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-67-870-001

(6) Bleed the elevator control system, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER AND CONTROL CABLES IN THE FORWARD EQUIPMENT COMPARTMENT OR AREA AFT OF THE BULKHEAD. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Move the horizontal stabilizer to the maximum leading edge up (APL NOSE DOWN) position (the mechanical limits).

NOTE: The horizontal stabilizer can be moved with the control wheel STAB TRIM switches or the stabilizer trim control wheel.

1) Make sure that the ballnut is against the jackscrew upstop.

(b) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

CAUTION MAKE SURE THAT THE PITOT PROBE HAS NO ADDED WEIGHT ON IT FROM THE TEST HOSE. THE WEIGHT OF THE TEST HOSE CAN BEND OR TWIST THE PITOT PROBE OUT OF TOLERANCE.

(c) Connect the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to the two pitot probes on the vertical fin.

1) Do this step for the regulator, SPL-1742:

a) Connect the pressure lines from the regulator, SPL-1742, with the pitot test adapter, COM-1916, to the pitot probes.

2) Do this step for the air data model test set, COM-1914:

a) Use the air data accessory kit, COM-25177, to connect the air data model test set, COM-1914, to the pitot test adapter, COM-1916.

NOTE: The air data accessory kit, COM-25177, contains hoses to connect to the air data model test set, COM-1914, and the pitot test adapter, COM-1916.

(d) Use the regulator, SPL-1742, or air data model test set, COM-1914, to raise the airspeed to 2.46 psig (169.61 mBar) (300 knots).

1) Make sure that the rate is 2.46 psig (169.61 mBar) (300 knots) or less for each minute.

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(e) Pressurize Hydraulic System A and B, do one of these steps:

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1) If a Portable Hydraulic Cart is used, do this task: Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, TASK 29-11-00-860-802.

NOTE: Two carts are recommended.

a) If only one cart is available, bleed hydraulic A and B, as follows:

<1> Pressurize hydraulic system A with a cart and hydraulic system B with the Electric Motor-Driven Pump (EMDP).

<2> Pressurize hydraulic system B with a cart and hydraulic system A with the EMDP.

2) If a Portable Hydraulic Cart is not used, do these tasks:

NOTE: Both the EMDP and Engine Driven Pump (EDP) should be run at the same time.

- Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), TASK 29-11-00-860-803
- Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), TASK 29-11-00-860-804.

(f) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

(g) Wait for 10 seconds.

(h) Do these steps to cycle the power between Hydraulic Systems A and B:

1) Set the flight controls hydraulic power switches as follows:

a) Set the FLT CONTROL B switch to the OFF position.

b) Set the FLT CONTROL A switch to the A ON position.

c) Wait for 2 seconds.

d) Set the FLT CONTROL A switch to the OFF position.

e) Set the FLT CONTROL B switch to the B ON position.

f) Wait for 2 seconds.

2) Repeat the cycle 10 times.

(i) Slowly, do control column sweeps with pitot pressure applied.

(j) Do these steps to cycle pitot pressure:

1) Use the regulator, SPL-1742, or air data model test set, COM-1914, to lower the airspeed to 0 psig (0.00 mBar) (0 knots).

2) Use the regulator, SPL-1742, or air data model test set, COM-1914, to increase the airspeed to 2.46 psig (169.61 mBar) (300 knots).

3) Use the regulator, SPL-1742, or air data model test set, COM-1914, to lower the airspeed to 0 psig (0.00 mBar) (0 knots).

(k) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

(l) Wait for 10 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-67-860-003

(7) Remove the pressure from the Hydraulic Systems A and B (HYDRAULIC SYSTEMS A AND B - MAINTENANCE PRACTICES, PAGEBLOCK 29-11-00/201).

SUBTASK 27-31-67-790-001

(8) Make sure that there is no hydraulic leakage at the elevator feel actuator [3].

SUBTASK 27-31-67-020-003

(9) Do these steps at the two pitot probes on the vertical fin:

(a) Remove the pressure lines that connect the air data model test set, COM-1914, or regulator, SPL-1742, to the pitot probes.

(b) Remove the pitot test adapter, COM-1916, from the pitot probes.

SUBTASK 27-31-67-020-004

(10) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-31-67-860-004

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE AWAY FROM THE HORIZONTAL STABILIZER AND CONTROL CABLES IN THE FORWARD EQUIPMENT COMPARTMENT OR AREA AFT OF THE BULKHEAD. THE MOVEMENT OF THE HORIZONTAL STABILIZER DURING MAINTENANCE CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(11) Move the horizontal stabilizer to the neutral stabilizer position (zero degrees or 4 units of trim).

NOTE: The horizontal stabilizer can be moved with the control wheel STAB TRIM switches or the stabilizer trim control wheel.

SUBTASK 27-31-67-720-001

(12) Do this task: Feel Force System - Test, TASK 27-31-00-700-813.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-67-410-001

(1) Install this access panel:

Number	Name/Location
318BR	Tailcone Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR FEEL ACTUATOR - DIFFERENTIAL PRESSURE TEST

1. General

A. This task gives instructions on how to diagnose an Elevator Feel System differential pressure issue, and how to use the elevator feel computer test box, SPL-23206.

TASK 27-31-67-700-801

2. Elevator Feel Actuator - Differential Pressure Test

(Figure 501)

A. General

(1) This task gives instructions on how to diagnose an Elevator Feel System differential pressure issue, and how to use the elevator feel computer test box, SPL-23206.

(2) This Predictive Maintenance & Enhanced Troubleshooting (PMET) procedure is optional and the results are for analysis only. It is not necessary to do more steps as a result of this procedure if there is no flight deck effect or no faults. The recommendations show Boeing's best practices to make sure that the aircraft can get maximum reliability. If there is a flight deck effect or other faults, use accepted procedures to correct them.

B. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-37-000-801	Elevator Feel Computer Removal (P/B 401)
27-31-37-400-801	Elevator Feel Computer Installation (P/B 401)
27-31-67-400-801	Elevator Feel Actuator Installation (P/B 401)
27-41-00-700-805	Stabilizer Trim Limit Switches Test (P/B 501)
27-61-45-000-801	Spoiler Control Electronics Unit Removal (P/B 401)
29-11-00-860-803	Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP) (P/B 201)
29-11-00-860-804	Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP) (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
29-11-41-700-801	EMDP Case Drain Filter - Metal Contamination Check (P/B 401)
29-11-51-200-801	EDP Case Drain Filter Element and Bowl Inspection (P/B 601)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-23206	Test Kit - Elevator Feel System Differential Pressure Part #: 1055.0.00x Supplier: 3BSK6

D. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door

F. Prepare for Troubleshooting

SUBTASK 27-31-67-780-001

(1) Remove hydraulic power from the aircraft, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-31-67-010-002

(2) Open these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-67-860-007

(3) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

G. Elevator Feel System Troubleshooting

SUBTASK 27-31-67-480-002

(1) Place the test box in the aft galley and connect the power plug.

SUBTASK 27-31-67-480-003

(2) Install Elevator Pitot System test equipment:

(a) Install the pitot adaptors on each of the elevator pitot probes.

(b) Using the Y hose, connect a hose to each pitot adaptor.

(c) Install the remaining end of the Y hose to the REGULATED AIR connection on the test box.

SUBTASK 27-31-67-480-004

(3) Install the pressure return hose:

(a) Install one return line on the system A restrictor valve.

(b) Install one return line on the system B restrictor valve.

(c) Install the system A return line onto the PITOT A connection on the test box.

(d) Install the system B return line onto the PITOT B connection on the test box.

SUBTASK 27-31-67-410-002

(4) Install the pressure transducers:

(a) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Access the ELEVATOR FEEL ACTUATOR in the tail cone compartment.

(c) Remove the SYSTEM B METERED PRESSURE line from the ELEVATOR FEEL ACTUATOR.

(d) Install a cap on the ELEVATOR FEEL ACTUATOR where the SYSTEM B METERED PRESSURE line was removed.

(e) Install one of the pressure transducers from the kit on the SYSTEM B METERED PRESSURE line.

(f) Remove the SYSTEM A METERED PRESSURE line from the ELEVATOR FEEL ACTUATOR.

(g) Install a cap on the ELEVATOR FEEL ACTUATOR where the SYSTEM A METERED PRESSURE line was removed.

(h) Install the remaining pressure transducer from the kit on the SYSTEM A METERED PRESSURE line.

(i) Connect a wire harness to each of the pressure transducers.

(j) Connect the SYSTEM A METERED PRESSURE transducer wire harness to the TRANSMITTER A port on the test box.

(k) Connect the SYSTEM B METERED PRESSURE transducer wire harness to the TRANSMITTER B port on the test box.

SUBTASK 27-31-67-700-001

(5) Pitot Blockage Test:

(a) Pressurize the hydraulic systems, do this task: Hydraulic System Pressurization with an Electric Motor-Driven Pump (EMDP), TASK 29-11-00-860-803 or Hydraulic System A or B Pressurization with an Engine-Driven Pump (EDP), TASK 29-11-00-860-804.

NOTE: If you do this task because of the FEEL DIFF PRESS light illuminates on the ground, it is recommended to use Engine Driven Pump (EDP) to pressurize the system.

(b) Loosen the PRESSURE REGULATOR knob until it can spin freely.

(c) Push the SHUT OFF VALVE knob in.

(d) Turn on the test box with the POWER SUPPLY switch.

(e) Turn on the pitot pump with the PUMP switch.

(f) Monitor the PITOT PRESSURE A and B gages.

(g) Slowly tighten the PRESSURE REGULATOR knob until the PITOT PRESSURE gages read 2.5 psig (2.5 psi).

NOTE: The pitot pressure gauges will increase in pressure at an identical rate. If either gauge (A or B) lags behind, that is an indication of a restriction or blockage in that pitot system.

SUBTASK 27-31-67-790-002

(6) Pitot System Leak Check:

(a) Pull the SHUTOFF VALVE knob.

(b) Make sure that the PITOT PRESSURE does not decrease more than 0.2 psig (0.2 psi) during a 2-minute time.

NOTE: If the pressure in either gauge (A or B) decreases more than 0.2 psig (0.2 psi) in that time period, that system has a leak.

(c) Push the SHUT OFF VALVE knob in.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-67-700-003

(7) Hydraulic Blockage Test:

NOTE: It may be necessary to run the engines (EDP) to have the sufficient volume of hydraulic fluid that is needed to expose the hydraulic blockage.

(a) Adjust the pitot pressure to 2.5 psig (2.5 psi).

(b) Move the stabilizer to 0 units, full nose down (TASK 27-41-00-700-805).

(c) Record the hydraulic pressure reading from the test equipment into Table 501.

Table 501/27-31-67-993-801

Pitot Pressure	Stabilizer Units	Hydraulic Pressure Limits	System A Hydraulic Pressure	System B Hydraulic Pressure
2.5 +0.5 / -0.0 psig (2.5 +0.5 / -0.0 psi)	0	1400 psig (1400 psi) - 1495 psig (1495 psi)		

(d) The pressure difference between system A and system B must be less than 25%.

1) Use Table 501 to identify the system that is out of the hydraulic pressure limits.

a) The system that is out of limits must be checked for contamination (TASK 29-11-41-700-801 and TASK 29-11-51-200-801).

SUBTASK 27-31-67-700-004

(8) Feel Computer Integrity Test:

(a) Adjust the pitot pressure to 2.0 psig (2.0 psi).

(b) Move the stabilizer to 0 units, full nose down (TASK 27-41-00-700-805).

(c) Use Table 502 to observe the hydraulic pressures.

Table 502/27-31-67-993-802

Pitot Pressure	Stabilizer Units	Hydraulic Pressure Limits	System A Hydraulic Pressure	System B Hydraulic Pressure
2.0 +0.5 / -0.0 psig (2.0 +0.5 / -0.0 psi)	0	1400 psig (1400 psi) - 1495 psig (1495 psi)		
2.0 +0.5 / -0.0 psig (2.0 +0.5 / -0.0 psi)	4	850 psig (850 psi) - 1105 psig (1105 psi)		
2.0 +0.5 / -0.0 psig (2.0 +0.5 / -0.0 psi)	16	245 psig (245 psi) - 370 psig (370 psi)		

(d) Record the hydraulic pressure reading from the test equipment into Table 502.

(e) Move the stabilizer to 4 units (TASK 27-41-00-700-805).

(f) Record the hydraulic pressure reading from the test equipment into Table 502.

(g) Move the stabilizer to 16 units, full nose up (TASK 27-41-00-700-805).

(h) Record the hydraulic pressure reading from the test equipment into Table 502.

(i) The pressure difference between system A and system B must be less than 25%.

1) Use Table 502 to identify the system that is out of the hydraulic pressure limits.

2) If there is a system pressure difference of 25% or greater, replace the elevator computer (TASK 27-31-37-000-801 and TASK 27-31-37-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-67-710-001

(9) Solenoid and Relays Operation Test:

(a) Remove the Spoiler Control Electronics (SCE) Unit, M2940 (TASK 27-61-45-000-801).

(b) Install a jumper wire from the SCE rack connector D15998D, pin C14 to airframe ground to maintain a ground connection for R979 relay.

(c) Adjust the pitot pressure to 2.0 psig (2.0 psi).

(d) Move the stabilizer to 4 units.

(e) In the flight deck, monitor the FEEL DIFF PRESS light on the P5-3 panel.

(f) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

(g) After 7 seconds, the FEEL DIFF PRESS light will illuminate, that indicates the R979 has closed.

1) If the light does not illuminate, troubleshoot the R979 circuit.

(h) Simulate AIR MODE for SYS 1 and SYS 2 (TASK 32-09-00-860-801).

(i) After 7 seconds, the FEEL DIFF PRESS light will extinguish once the airplane is in AIR MODE, that indicates R594, R819 are de-energized and R980 has energized.

1) Monitor the FEEL DIFF PRESS light for one minute.

2) On the test box, both A and B Hydraulic Pressure will reduce to 210 psig (210 psi) - 260 psig (260 psi).

3) If there is a pressure split, the system that is above 210 psig (210 psi) - 260 psig (260 psi) is suspect.

4) If the FEEL DIFF PRESS light comes back on, troubleshoot the R980 and Elevator Feel Actuator Solenoid circuit.

5) If the FEEL DIFF PRESS light does not extinguish, troubleshoot the R594, R819, and R980 circuits.

(j) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-67-700-002

(1) Release the pressure in the pitot system.

(a) Push in the SHUTOFF Valve knob.

(b) Loosen the PRESSURE REGULATOR knob until it can spin freely.

(c) Once the pitot pressure gauges read 0 psig (0 psi), turn off the PUMP switch.

NOTE: Do not remove the pitot equipment at this time.

SUBTASK 27-31-67-020-005

(2) Remove the pressure transducers.

(a) Remove hydraulic power from the aircraft, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the pressure transducers from the SYSTEM B METERED PRESSURE line and the SYSTEM A METERED PRESSURE line.

(c) Remove the caps from the ELEVATOR FEEL ACTUATOR.

(d) Install the SYSTEM B METERED PRESSURE line and the SYSTEM A METERED PRESSURE line on to the corresponding ports of the ELEVATOR FEEL ACTUATOR.

SUBTASK 27-31-67-410-003

(3) Bleed the elevator control system (TASK 27-31-67-400-801).

SUBTASK 27-31-67-020-006

(4) Remove the Pitot Static Equipment.

(a) Remove the pitot adaptors from each of the elevator pitot probes.

(b) Disconnect the Y hose from the pressure adaptors.

(c) Remove the pressure return hoses from the restrictor valve drains.

SUBTASK 27-31-67-410-004

(5) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

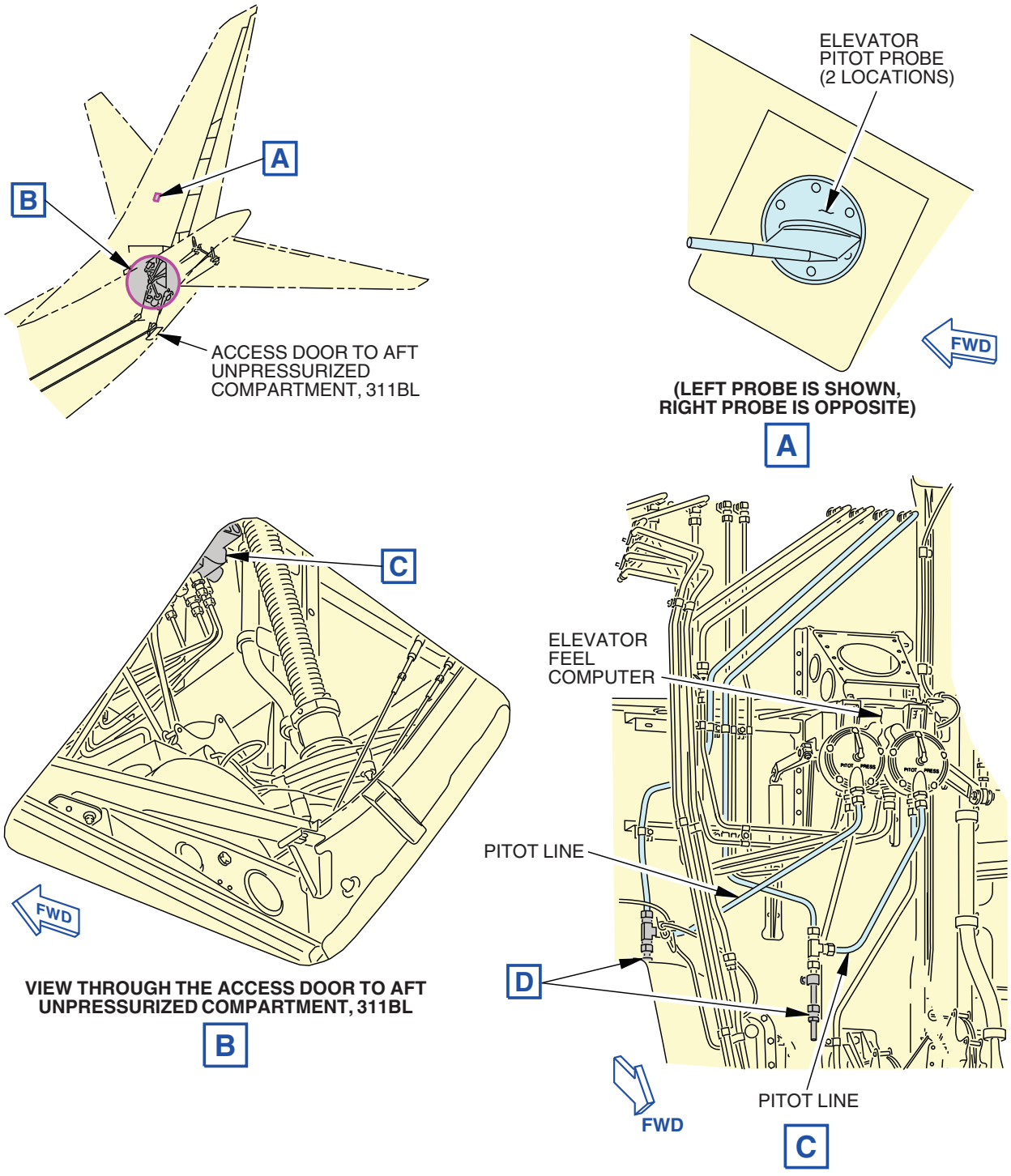
SUBTASK 27-31-67-410-005

(6) Close these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door

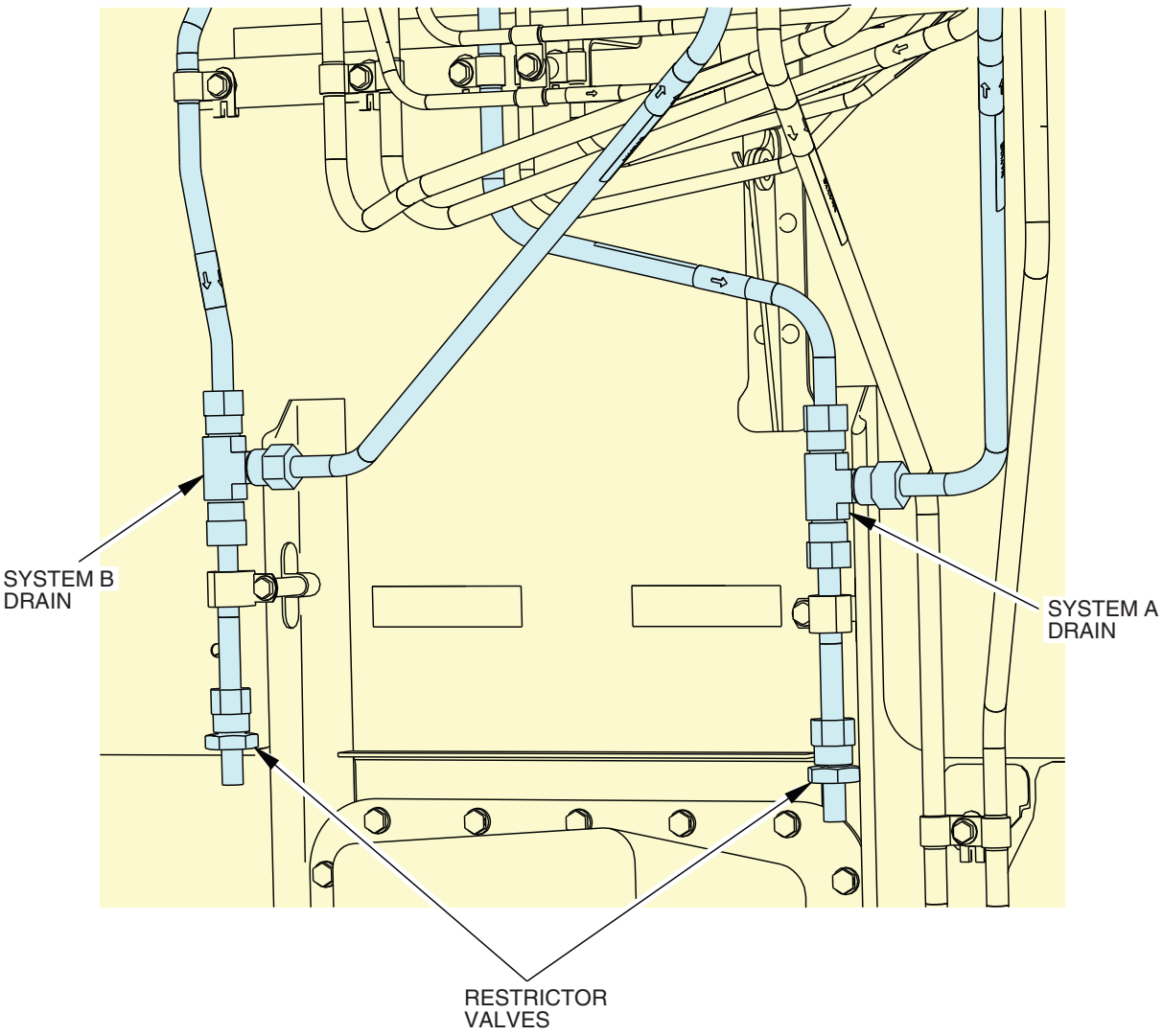
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Feel System Differential Pressure Test
Figure 501/27-31-67-990-802 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



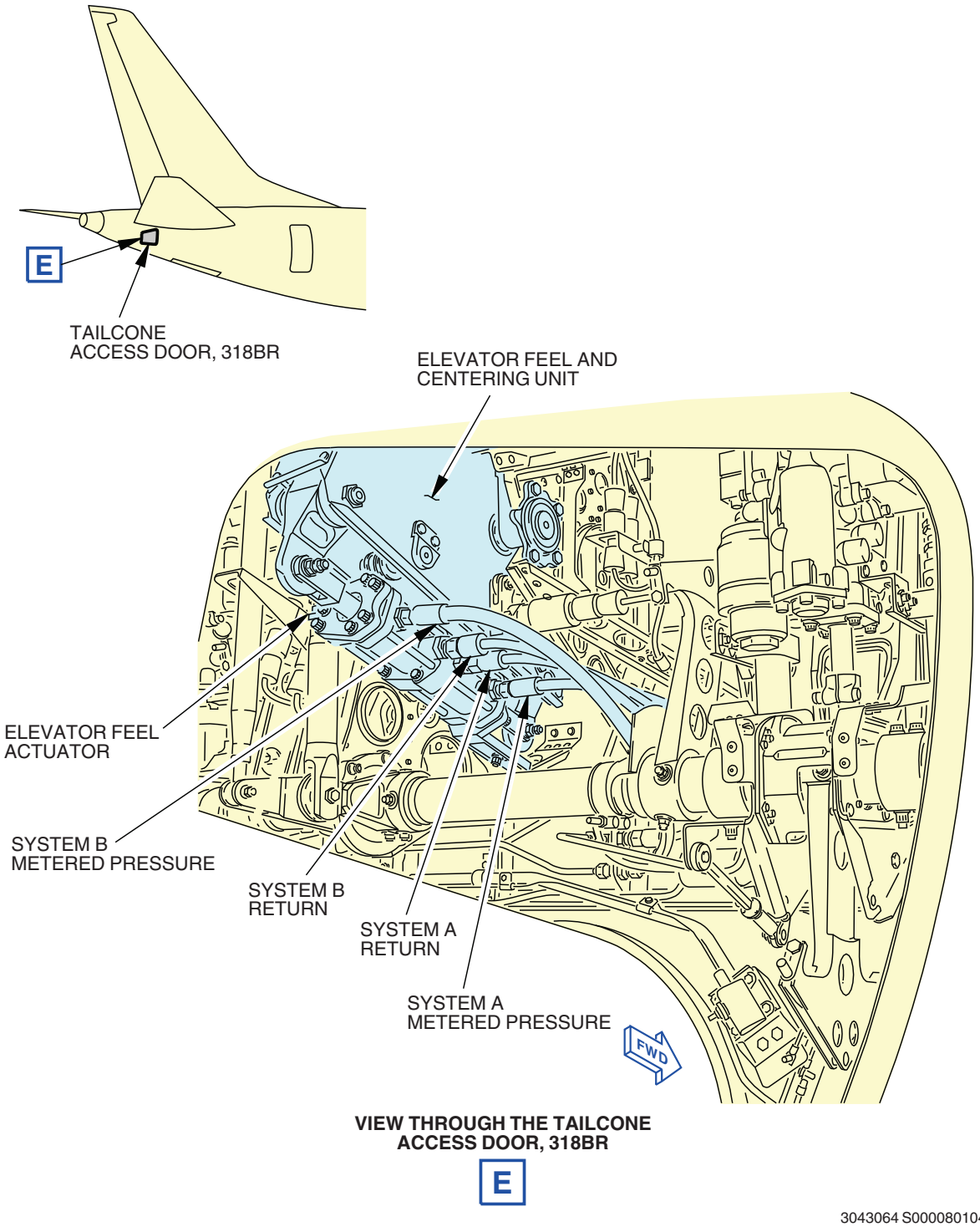
(VIEW IN THE AFT DIRECTION)

D

3043061 S0000801044_V2

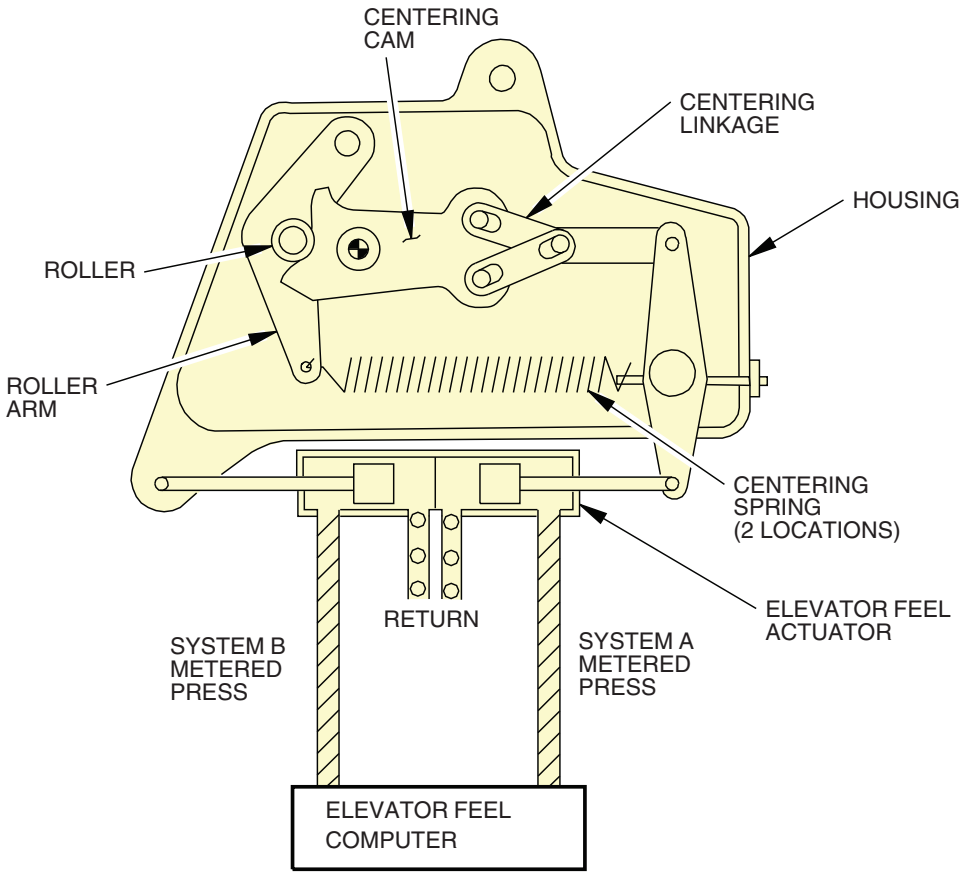
Elevator Feel System Differential Pressure Test
Figure 501/27-31-67-990-802 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Feel System Differential Pressure Test
Figure 501/27-31-67-990-802 (Sheet 3 of 5)

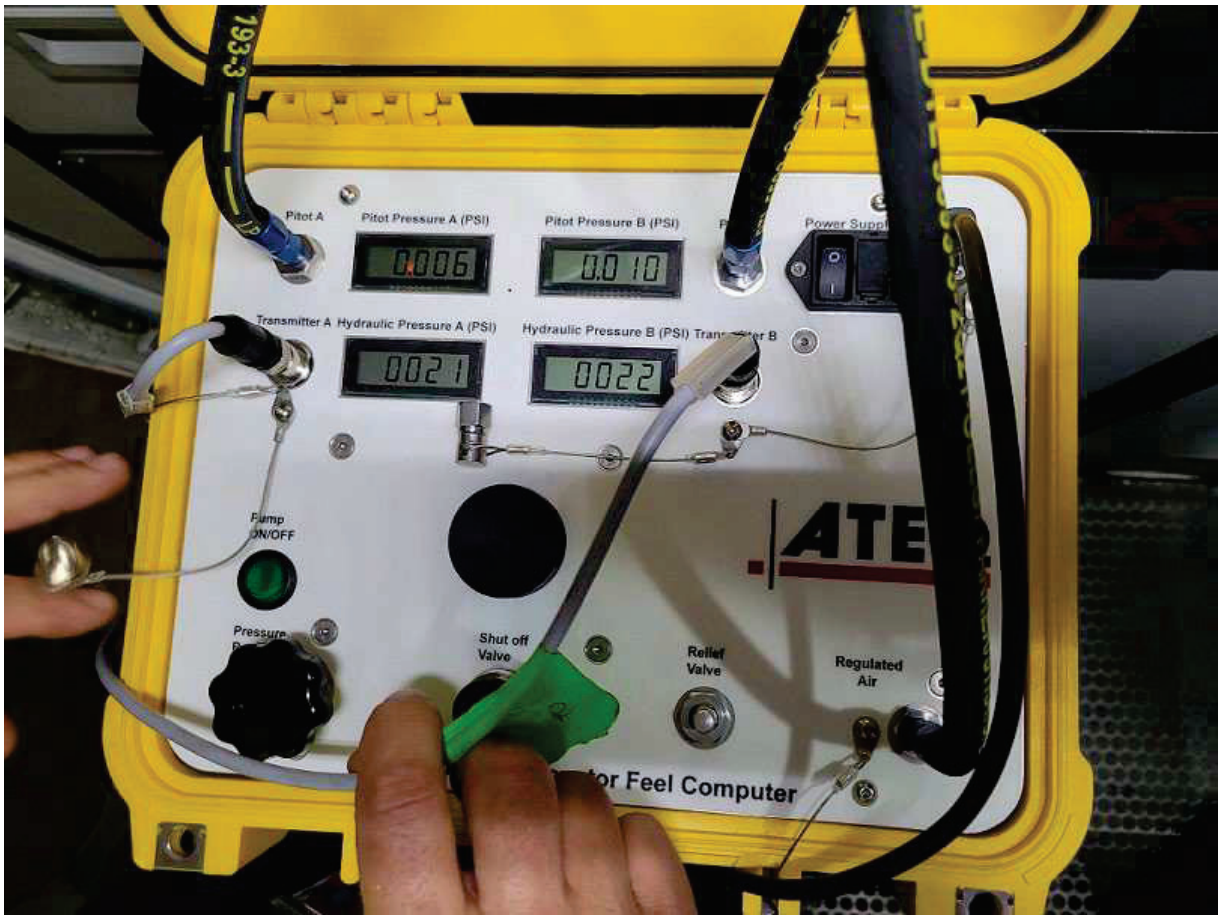
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3043062 S0000801046_V2

Elevator Feel System Differential Pressure Test
Figure 501/27-31-67-990-802 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



3043065 S0000801047_V2

Elevator Feel System Differential Pressure Test
Figure 501/27-31-67-990-802 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR INPUT TORQUE TUBE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the elevator input torque tube.

(2) An installation of the elevator input torque tube.

B. The elevator input torque tube is referred to as the "torque tube" in this procedure.

C. The torque tube includes four quadrants and six input cranks.

TASK 27-31-71-000-801

2. Elevator Input Torque Tube Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
53-53-00-000-801	Tailcone Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

C. Location Zones

Zone	Area
315	APU Compartment - Left
316	APU Compartment - Right

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

E. Prepare for the Removal

SUBTASK 27-31-71-860-001

(1) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-71-020-001

(2) If it is necessary, remove the tail cone to get access to the torque tube [18]:

(a) Use marking pens or masking tape to make marks on the airplane to show the tail cone locations for correct installation of the tail cone.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do this task: Tailcone Removal, TASK 53-53-00-000-801.

SUBTASK 27-31-71-010-002

(3) If the tail cone was not removed, do this step to gain access to the torque tube [18]:

(a) Open these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-71-860-002

(4) Use the trammel bar, SPL-1677, to measure the “B” dimension (Figure 401).

(a) Make sure to protect wire bundles when using the trammel bar, SPL-1677.

NOTE: The sharp edges on the trammel bar, SPL-1677, can cause chafing or damage to the wires.

SUBTASK 27-31-71-080-001

(5) Install the rig pin E-5 from the rigging pin kit, SPL-1585, on the elevator aft quadrant.

SUBTASK 27-31-71-010-001

(6) To get access to the turnbuckles, do this step:

(a) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Elevator Input Torque Tube Removal

SUBTASK 27-31-71-020-002

(1) Remove the cables EAL, EBL, EAR, and EBR from the quadrants [5]:

(a) Install the clamps on the cables so the cables will not go through the aft pressure bulkhead.

(b) Remove the turnbuckles locking clips.

(c) Loosen the turnbuckles for the elevator cables EAL, EBL, EAR, and EBR.

(d) Remove the nuts [4], washers [3], washers [55] (if applicable), bolts [1], and cable retainers [2] from the four quadrants [5].

(e) Remove the cables from the quadrants [5].

SUBTASK 27-31-71-020-003

(2) Remove the torque tube [18] (Figure 402):

(a) Remove the nut [16], washer [15], and bolt [14] to disconnect the elevator position sensor rod [13] from the torque tube [18].

(b) Remove the nut [29], washer [28], washer [27], and bolt [26] to disconnect the autopilot actuator rod [19] from the input crank [30].

(c) Remove the nuts [11], washers [10], washers [9], and bolts [8] to disconnect the four input rods [7] of the elevator power control units [6] from the four input cranks [12].

(d) Remove the nut [24], washer [23], washer [22], and bolt [21] to disconnect the feel and centering rod [20] from the input crank [25].

(e) Use the drill bit 0.247 +0.003 / -0.000 in. (6.27 +0.08 / -0.00 mm) and the drill centering guide to drill out the lockbolts [32] and collars [33] from the center bearing retainer [31].

1) Make sure that the holes are not out of tolerance.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Remove the rig pin E-5 from the elevator aft quadrant.

(g) Remove the nuts [43], washers [42], nuts [41], washers [40], bolt assemblies [38], and bushings [39] to disconnect the two elevator power control units [6] from the torque tube [18].

(h) Remove the bolt shrouds [54] from the left and right assemblies.

NOTE: There is a bolt shroud on the bolt head and the nut end of the bolts.

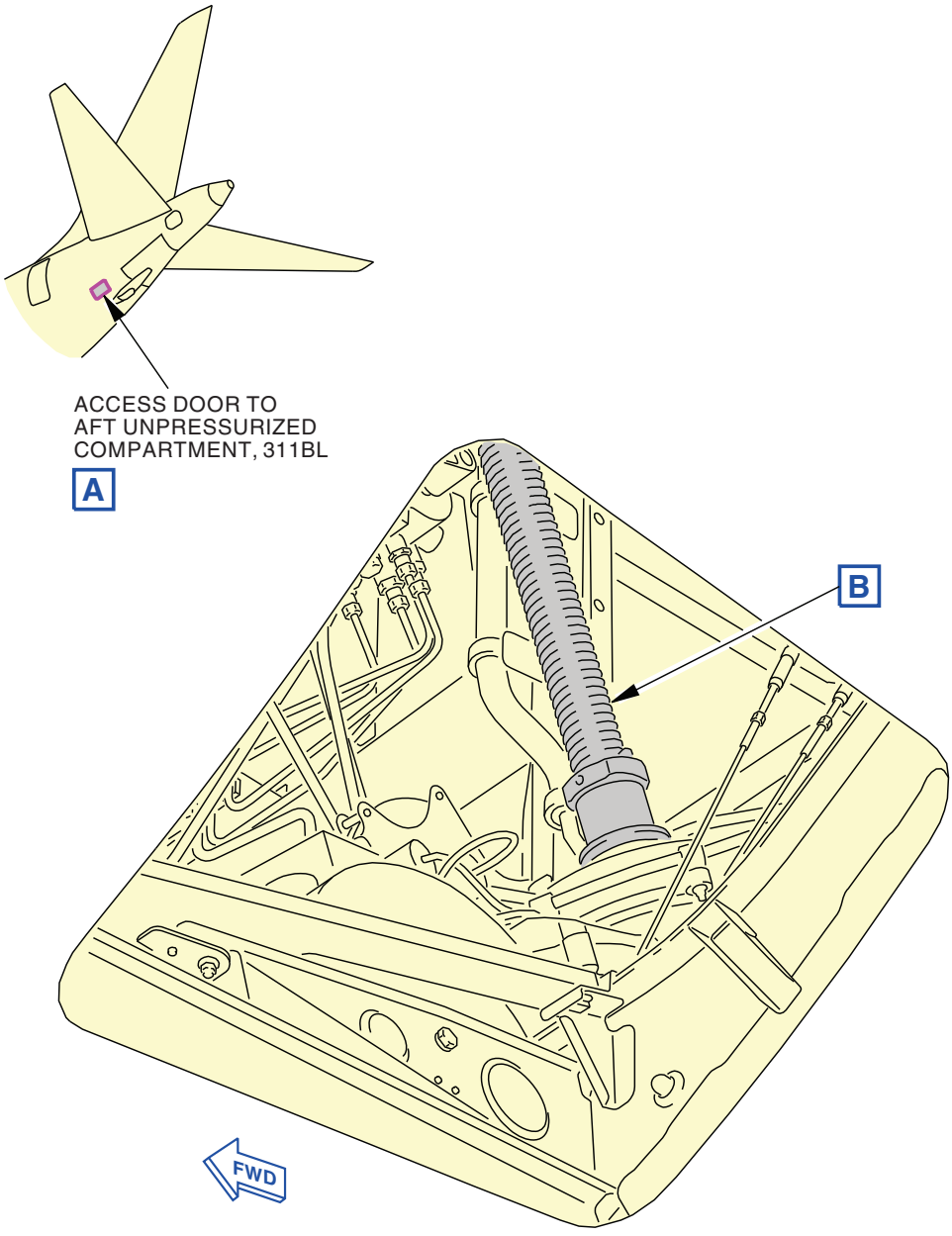
(i) Remove the nuts [47], washers [46], bolts [44], and bushings [45] to disconnect the two reaction links [37] from the torque tube [18].

(j) Remove the nuts [48], washers [49], nuts [50], washers [52], washers [51], washers [34], bolt assemblies [35], and bushings [53] to disconnect the torque tube [18] from the support structure.

(k) Remove the torque tube [18] from the airplane.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



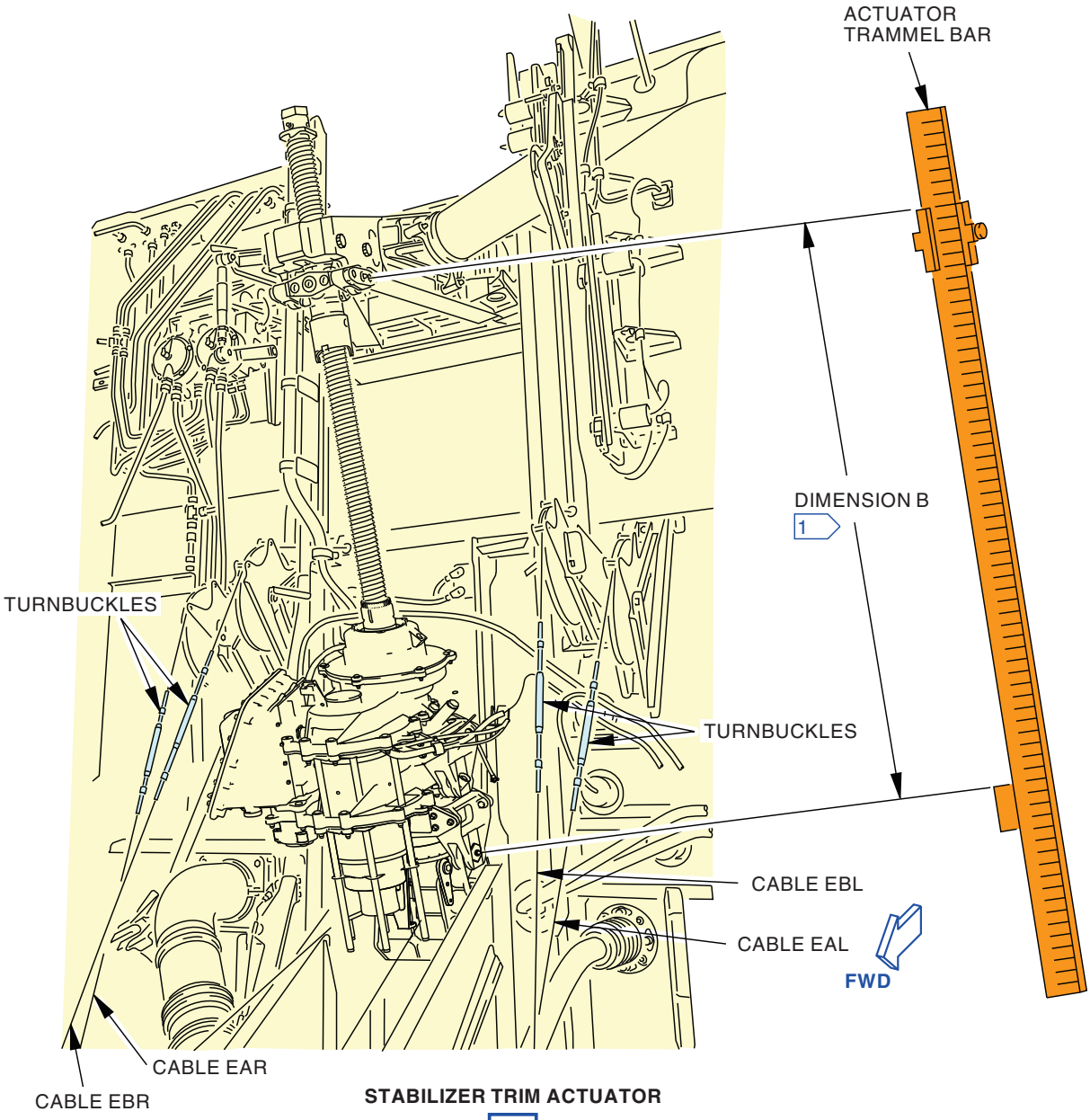
**VIEW THROUGH THE ACCESS DOOR TO
AFT UNPRESSURIZED COMPARTMENT, 311BL**

A

2412590 S00061530959_V1

**Elevator Cable and Turnbuckle Locations
Figure 401/27-31-71-990-801 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

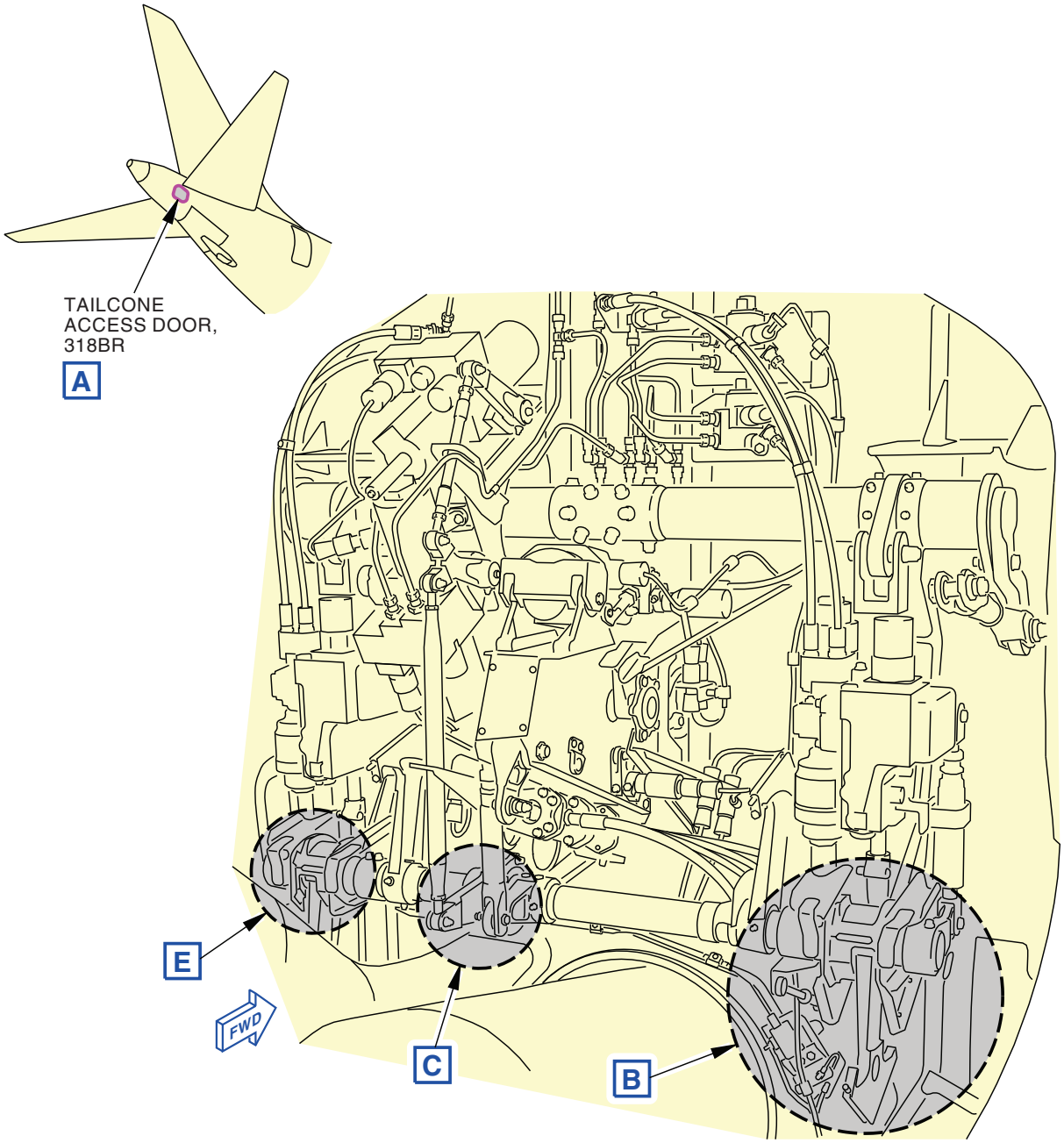
THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

2412591 S00061530960_V2

Elevator Cable and Turnbuckle Locations
Figure 401/27-31-71-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



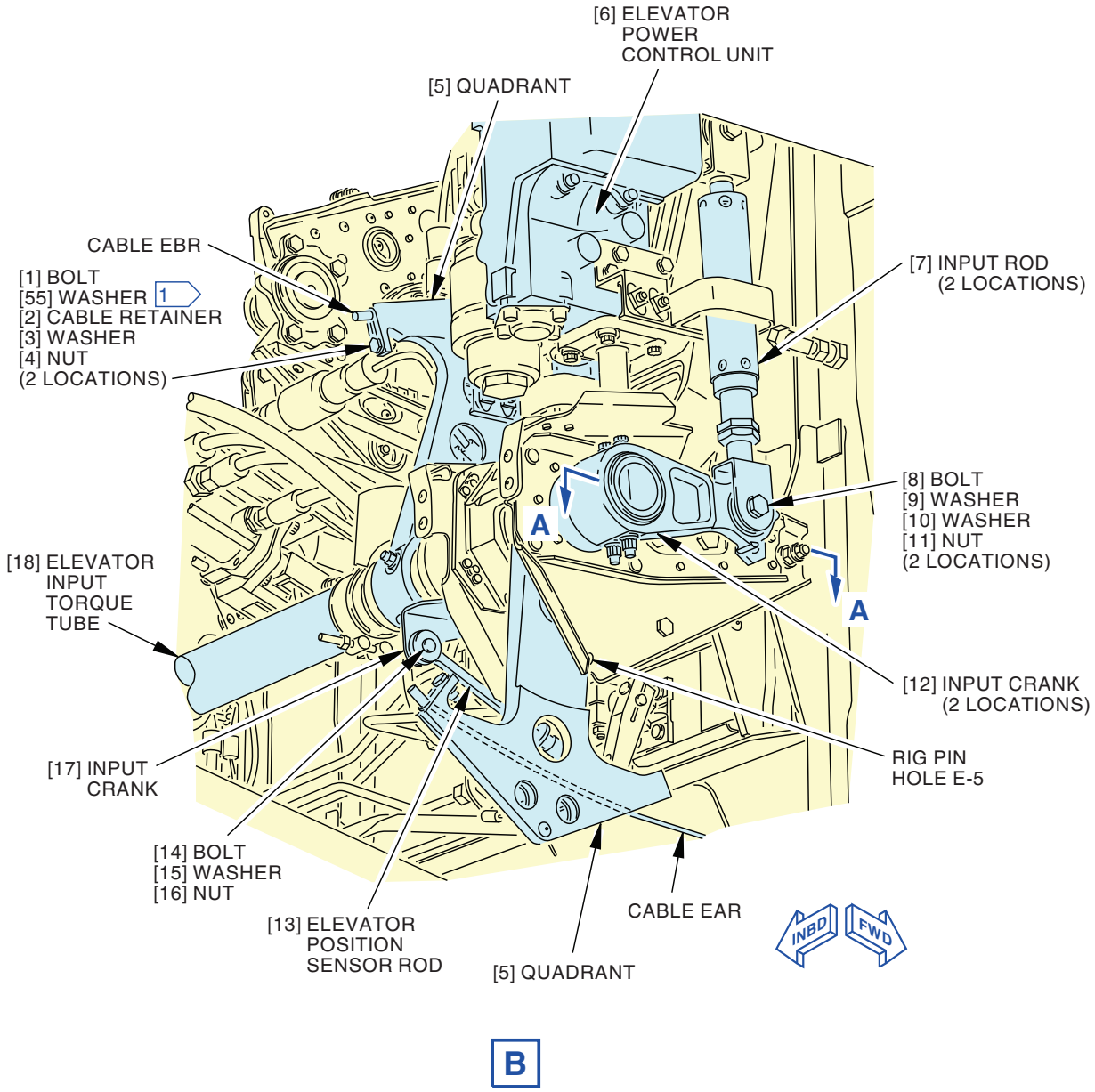
VIEW THROUGH THE TAILCONE ACCESS DOOR, 318BR

A

2412592 S00061530961_V1

Elevator Input Torque Tube Installation
Figure 402/27-31-71-990-802 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

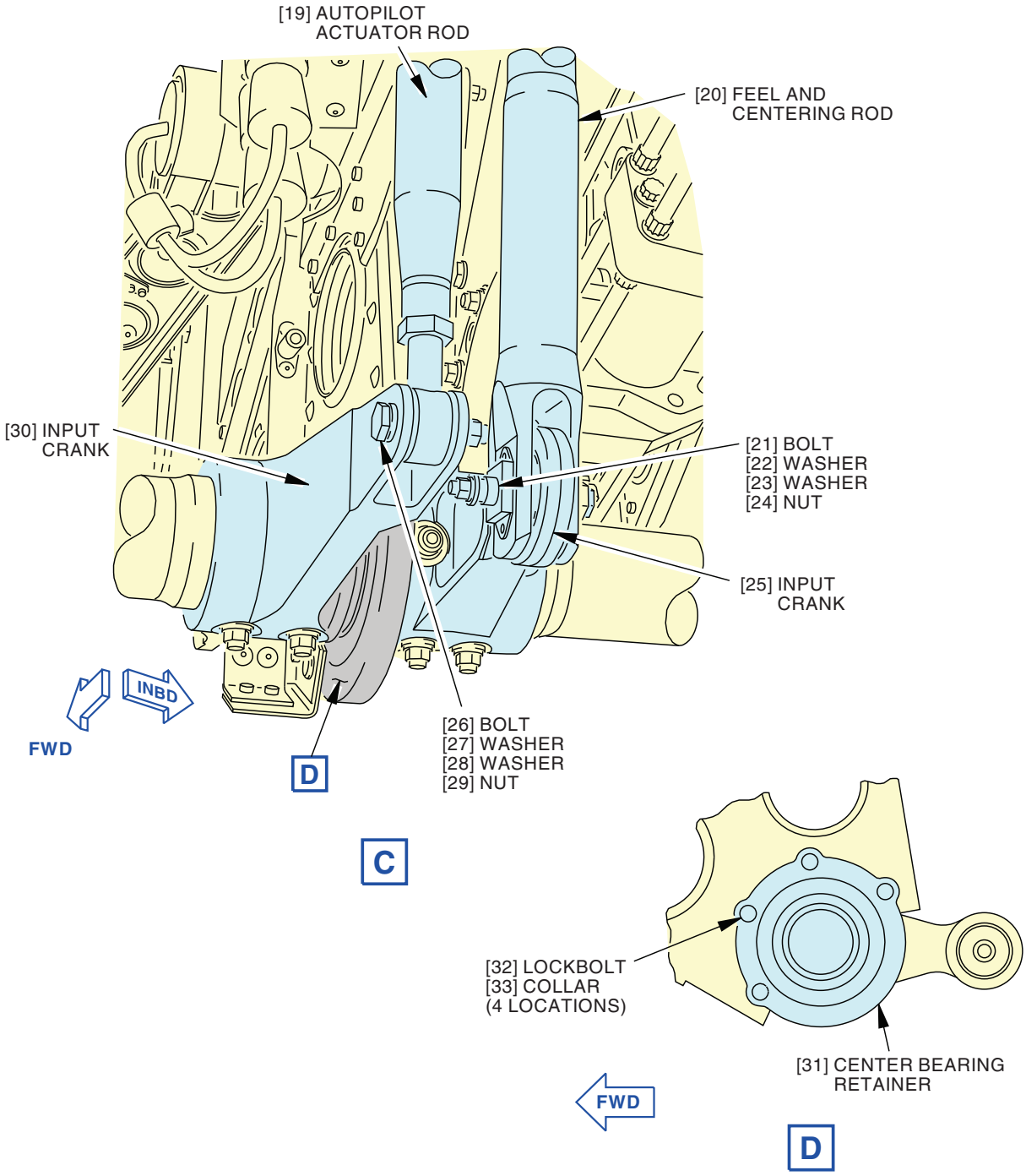


1 IF APPLICABLE

2412593 S00061530962_V3

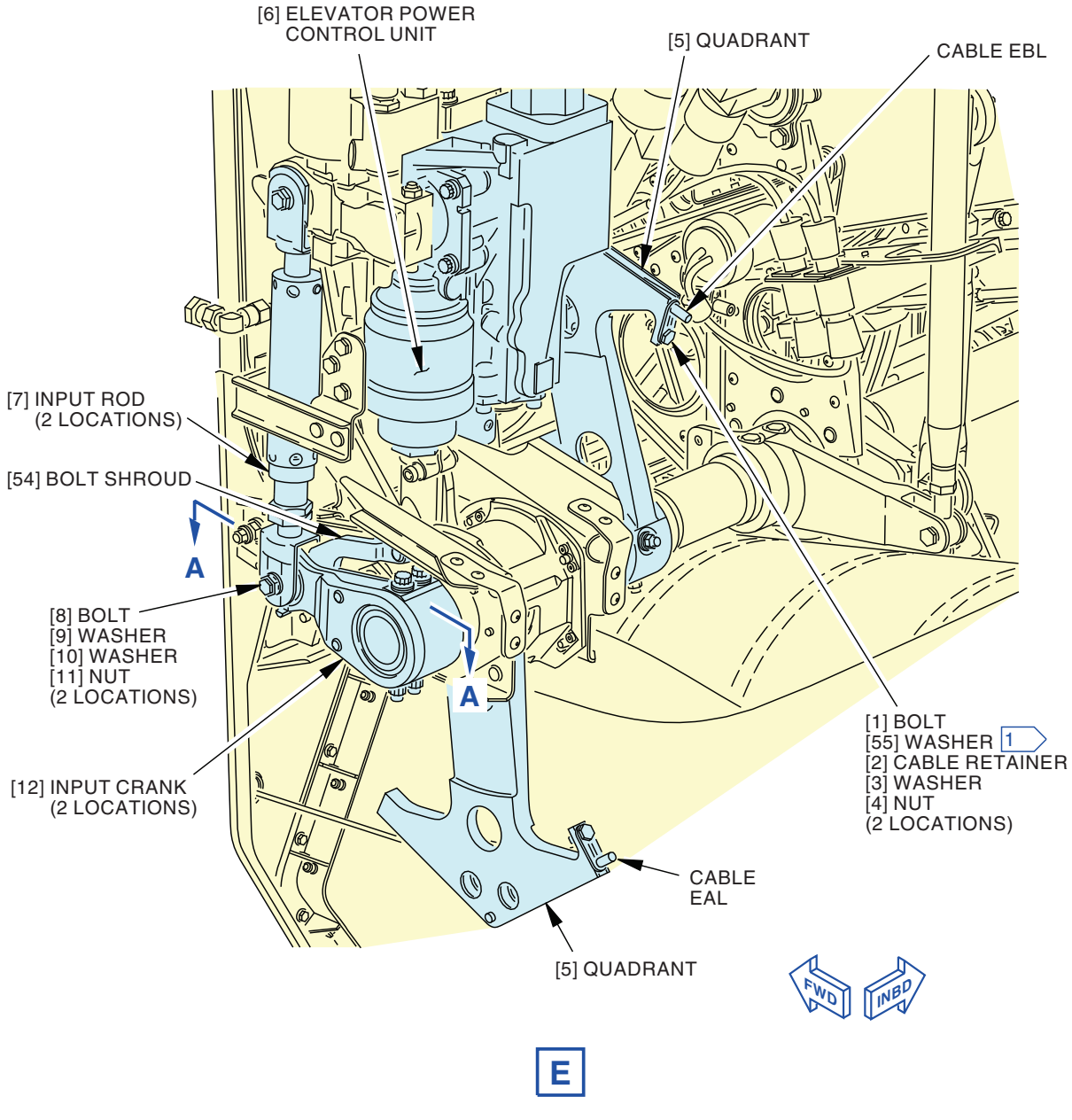
Elevator Input Torque Tube Installation
Figure 402/27-31-71-990-802 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



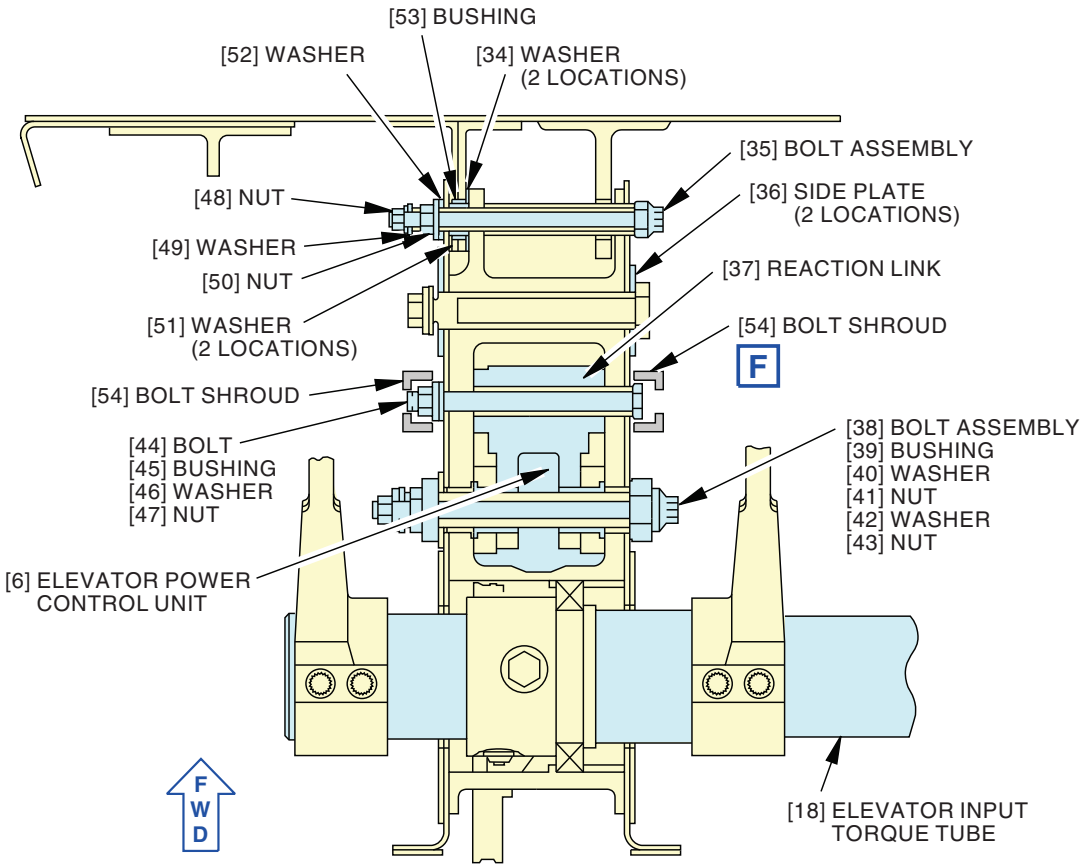
Elevator Input Torque Tube Installation
Figure 402/27-31-71-990-802 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



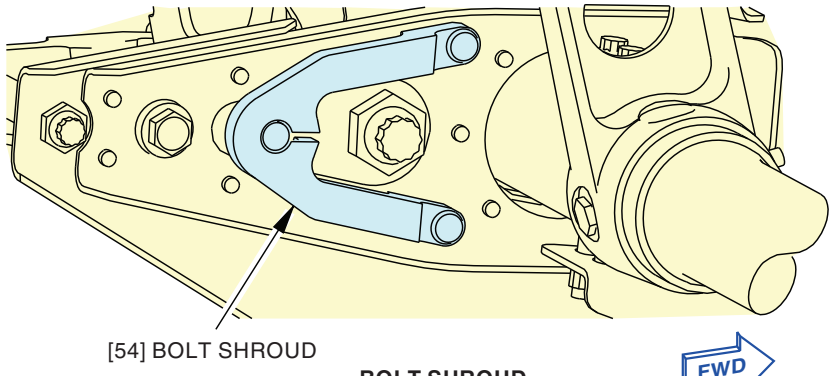
Elevator Input Torque Tube Installation
Figure 402/27-31-71-990-802 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A-A



BOLT SHROUD
(INBOARD SHOWN, TYPICAL)

F

2412596 S00061530965_V1

Elevator Input Torque Tube Installation
Figure 402/27-31-71-990-802 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-71-400-801

3. Elevator Input Torque Tube Installation

(Figure 401, Figure 402)

A. References

Reference	Title
20-10-93-400-801	Control Cable End Retaining Plate - Installation (P/B 401)
27-31-00-700-810	Control Column Power Force - Test (P/B 501)
27-31-00-700-813	Feel Force System - Test (P/B 501)
27-31-00-820-801	Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment (P/B 501)
27-31-00-820-802	Elevator Control Pushrods - Adjustment (P/B 501)
27-31-00-820-803	Input Rods of the Elevator Power Control Unit - Adjustment (P/B 501)
27-31-00-820-804	Input Rods of the Elevator Neutral Shift Mechanism and Output Rod of the Feel and Centering Unit - Adjustment (P/B 501)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
53-53-00-400-801	Tailcone Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
D00013	Grease - Aircraft And Instrument Grease	MIL-PRF-23827 (NATO G-354) (Supersedes MIL-G-23827)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
18	Torque tube	27-31-34-04-095	PTF ALL

D. Location Zones

Zone	Area
315	APU Compartment - Left
316	APU Compartment - Right

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
317BL	Tailcone Access Door
318BR	Tailcone Access Door

F. Elevator Input Torque Tube Installation

SUBTASK 27-31-71-420-001

(1) Install the torque tube [18] (Figure 402):

(a) Lubricate the bushings, bolts, and bearings with grease, D00013.

(b) Hold the torque tube [18] in its position.

(c) Attach the torque tube [18] to the support structure:

1) Install the bushings [53], bolt assemblies [35], washers [34], washers [51], washers [52], and nuts [50].

a) Tighten the nuts [50] to 90 in-lb (10.2 N·m) - 105 in-lb (11.9 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE THAT YOU DO NOT TIGHTEN THE INNER NUT AGAIN AFTER YOU TIGHTEN THE OUTER NUT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

2) Install the washers [49] and nuts [48].
   a) Tighten the nuts [48] to 30 in-lb (3.4 N·m) - 35 in-lb (4.0 N·m).

CAUTION EXAMINE FOR DAMAGE THE INNER/OUTER OF THE RODS/BOLTS BEFORE YOU INSTALL THE ROD END BOLTS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(d) Connect the torque tube [18] to the two reaction links [37]:
   1) Install the bushings [45], bolts [44], washers [46], and nuts [47].
      a) Install the left bolt [44] (on the left assembly) with the bolt head inboard and the right bolt [44] (on the right assembly) with the bolt head outboard.

(e) Install the bolt shrouds [54] on the left and right assemblies.

NOTE: There is a bolt shroud on the bolt head and the nut end of the bolts.

(f) Install the two elevator power control units [6] to the torque tube [18]:
   1) Install the bushings [39], bolt assemblies [38], washers [40], and nuts [41].
      a) Tighten the nuts [41] to 550 in-lb (62.1 N·m) - 600 in-lb (67.8 N·m).

CAUTION MAKE SURE THAT YOU DO NOT TIGHTEN THE INNER NUT AGAIN AFTER YOU TIGHTEN THE OUTER NUT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

   2) Install the washers [42] and nuts [43].
      a) Tighten the nuts [43] to 225 in-lb (25.4 N·m) - 250 in-lb (28.2 N·m).

(g) If it is necessary, drill the diameter of the 0.247 +0.003 / -0.000 in. (6.27 +0.08 / -0.00 mm) holes in the center bearing retainer [31].

(h) Use the drill centering guide adapter to make sure that the holes are not out of tolerance.

(i) Install the lockbolts [32] and collars [33] to attach the center bearing retainer [31] to the bracket.

(j) Connect the feel and centering rod [20] to the input crank [25]:
   1) Install the bolt [21], washer [22] (under the bolt [21]), washer [23] (under the nut [24]), and nut [24].

CAUTION DO NOT TRY TO TIGHTEN THE BOLT AGAIN AFTER YOU TIGHTEN THE NUT. YOU CAN CAUSE DAMAGE TO EQUIPMENT.

      a) Tighten the outer bolt of the bolt [21] to 75 in-lb (8.5 N·m) - 85 in-lb (9.6 N·m).
      b) Tighten the nut [24] to 25 in-lb (2.8 N·m) - 35 in-lb (4.0 N·m).

(k) Connect the four input rods [7] of the two elevator power control units [6] to the input cranks [12]:
   1) Install the bolts [8], washers [9], washers [10], and nuts [11].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(l) Install the bolt [26], washer [27], washer [28], and nut [29] to connect the autopilot actuator rod [19] to the input crank [30].

1) Tighten the nut [29] to 76 in-lb (8.6 N·m) - 80 in-lb (9.0 N·m).

(m) Install the bolt [14], washer [15], and nut [16] to connect the elevator position sensor rod [13] to the torque tube [18].

SUBTASK 27-31-71-420-002

(2) Install the cables EAL, EBL, EAR, and EBR to the quadrants [5]:

(a) Install the bolts [1], washers [55] (if applicable), washers [3], nuts [4], and cable retainers [2] to attach the cables to the four quadrants [5] (TASK 20-10-93-400-801).

(b) Remove the clamps from the cables.

(c) Install the turnbuckles locking clips.

SUBTASK 27-31-71-420-003

(3) If the tail cone was removed, do this task: Tailcone Installation, TASK 53-53-00-400-801.

G. Elevator Input Torque Tube Installation Test

SUBTASK 27-31-71-820-001

(1) Adjust the tension of the EAL, EBL, EAR, and EBR control cables (TASK 27-31-00-820-801).

SUBTASK 27-31-71-820-002

(2) Adjust the elevator pushrod (TASK 27-31-00-820-802).

SUBTASK 27-31-71-820-003

(3) Adjust the output rod of the elevator feel and centering unit (TASK 27-31-00-820-804).

SUBTASK 27-31-71-710-001

(4) Move the elevator slowly by hand through the full travel up and down.

(a) Make sure that the elevator moves freely.

SUBTASK 27-31-71-860-003

(5) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

SUBTASK 27-31-71-710-002

(6) Do this check of the synchronized operation of the elevator power control units:

(a) Put the FLT CONTROL A switch to the OFF position and FLT CONTROL B switch to the ON position.

1) Make sure that the elevator trailing edge moves less than 0.02 in. (0.5 mm).

2) If it is necessary, adjust the input rods of the power control units (TASK 27-31-00-820-803).

(b) Put the FLT CONTROL A switch to the ON position and FLT CONTROL B switch to the OFF position.

1) Make sure that the elevator trailing edge moves less than 0.02 in. (0.5 mm).

2) If it is necessary, adjust the input rods of the power control units (TASK 27-31-00-820-803).

SUBTASK 27-31-71-710-003

(7) Do this task: Control Column Power Force - Test, TASK 27-31-00-700-810.

SUBTASK 27-31-71-710-004

(8) Do this task: Feel Force System - Test, TASK 27-31-00-700-813.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-71-410-002

(1) If the tail cone was not removed, do this step:

(a) Close these access panels:

Number	Name/Location
317BL	Tailcone Access Door
318BR	Tailcone Access Door

SUBTASK 27-31-71-410-001

(2) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR OUTPUT TORQUE TUBE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the elevator output torque tube

(2) An installation of the elevator output torque tube.

B. The elevator output torque tube is referred to as the "output torque tube" in this procedure.

TASK 27-31-74-000-801

2. Elevator Output Torque Tube Removal

(Figure 401)

A. References

Reference	Title
22-11-26-000-801	Autopilot Elevator Actuator - Removal (P/B 401)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
53-53-00-000-801	Tailcone Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205

C. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

D. Prepare for the Removal

SUBTASK 27-31-74-860-001

(1) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-74-020-001

(2) Do this task: Tailcone Removal, TASK 53-53-00-000-801.

SUBTASK 27-31-74-860-002

(3) Use the trammel bar, SPL-1677, to set the horizontal stabilizer to 4 units of trim.

(a) Make sure to protect wire bundles when using the trammel trammel bar, SPL-1677.

NOTE: The sharp edges on the trammel trammel bar, SPL-1677, can cause chafing or damage to the wires.

SUBTASK 27-31-74-020-002

(4) Do this task: Autopilot Elevator Actuator - Removal, TASK 22-11-26-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Elevator Output Torque Tube Removal

CAUTION

YOU MUST BE CAREFUL WITH THE ELEVATOR AFTER YOU REMOVE THE TORQUE TUBES. IT IS POSSIBLE THAT THE ELEVATOR CAN HIT THE HORIZONTAL STABILIZER TRAILING EDGE LOWER BEAM AND ELEVATOR MAST ARM FITTING. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

SUBTASK 27-31-74-020-003

(1) Remove the mounting bracket [31] of the elevator autopilot actuators:

(a) Remove the bolts [27] and washers [28] from the mounting bracket [31].

(b) Remove the bolts [29] and washers [30] from the mounting bracket [31].

(c) Remove the nuts [35], washers [34], washers [33], and bolts [32] from the mounting bracket [31].

(d) Remove the nuts [39], washers [38], washers [37], and bolts [36] from the mounting bracket [31].

(e) Remove the mounting bracket [31] from the airplane.

SUBTASK 27-31-74-480-001

(2) Install the rig pin E-5 from the rigging pin kit, SPL-1585, in the elevator aft control quadrant.

SUBTASK 27-31-74-020-004

(3) Remove the output torque tube [1]:

(a) Remove the nuts [25], nuts [23], washers [22], bushings [24], and bolt assemblies [26] to disconnect the two elevator control pushrods [4] from the output torque tube [1].

(b) Remove the nuts [9], washers [8], washers [7], and bolts [6] to disconnect the two elevator PCUs [5] from the output torque tube [1].

(c) Move the elevator PCUs [5] away from the aft bulkhead until the input cranks are in the bottom.

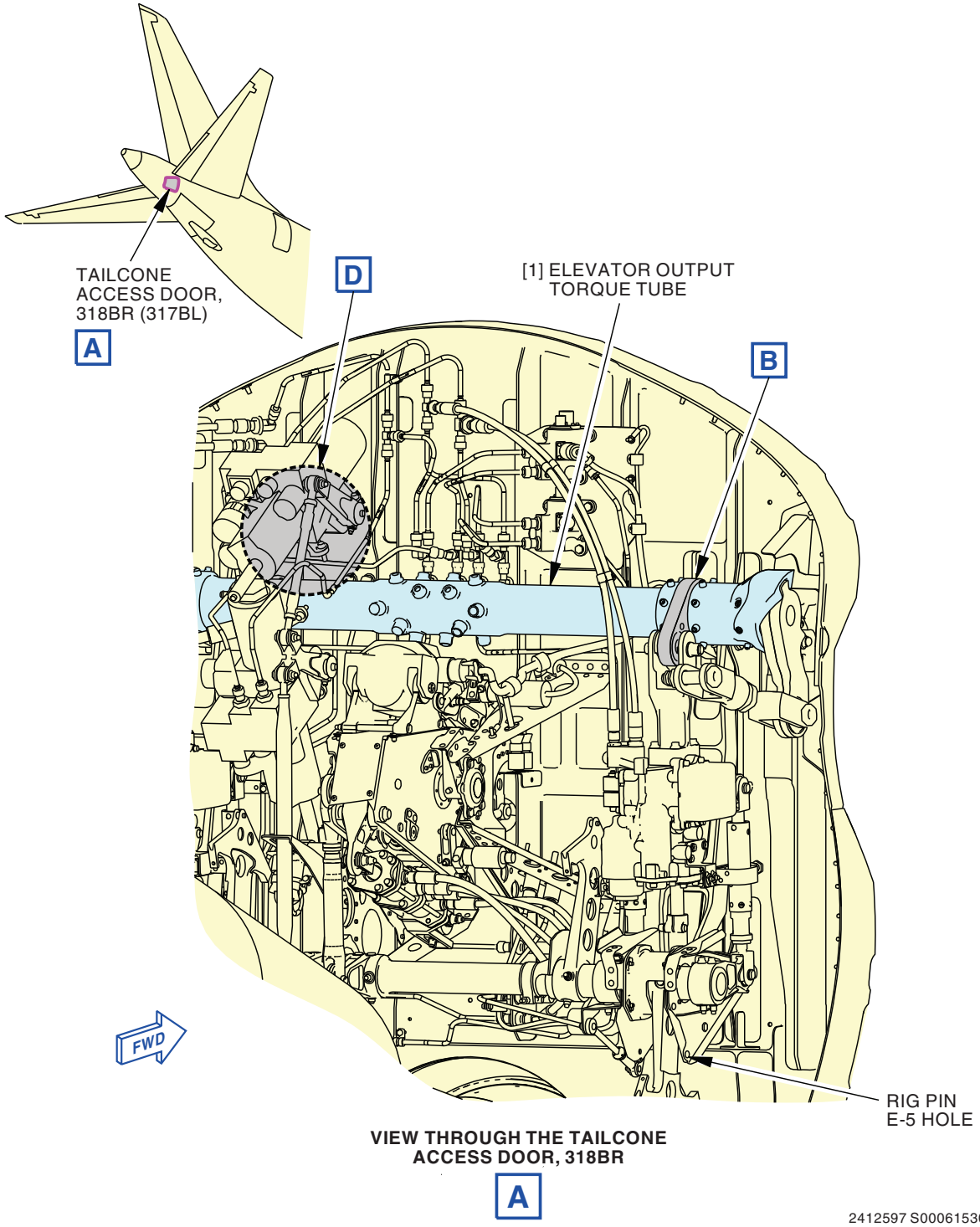
(d) At each end of the output torque tube [1], remove the nuts [20], washers [21], nuts [15], washers [14], bolts [11], bushing [12] (different on right and left sides), and bushings [13] to disconnect the output torque tube [1] from the inboard pivot brackets [2] and reaction links [10].

(e) At each end of the output torque tube [1], remove the nuts [17], washers [18], washers [19], and bolts [16] to disconnect the output torque tube [1] from the outboard pivot brackets [3].

(f) Remove the output torque tube [1] from the airplane.

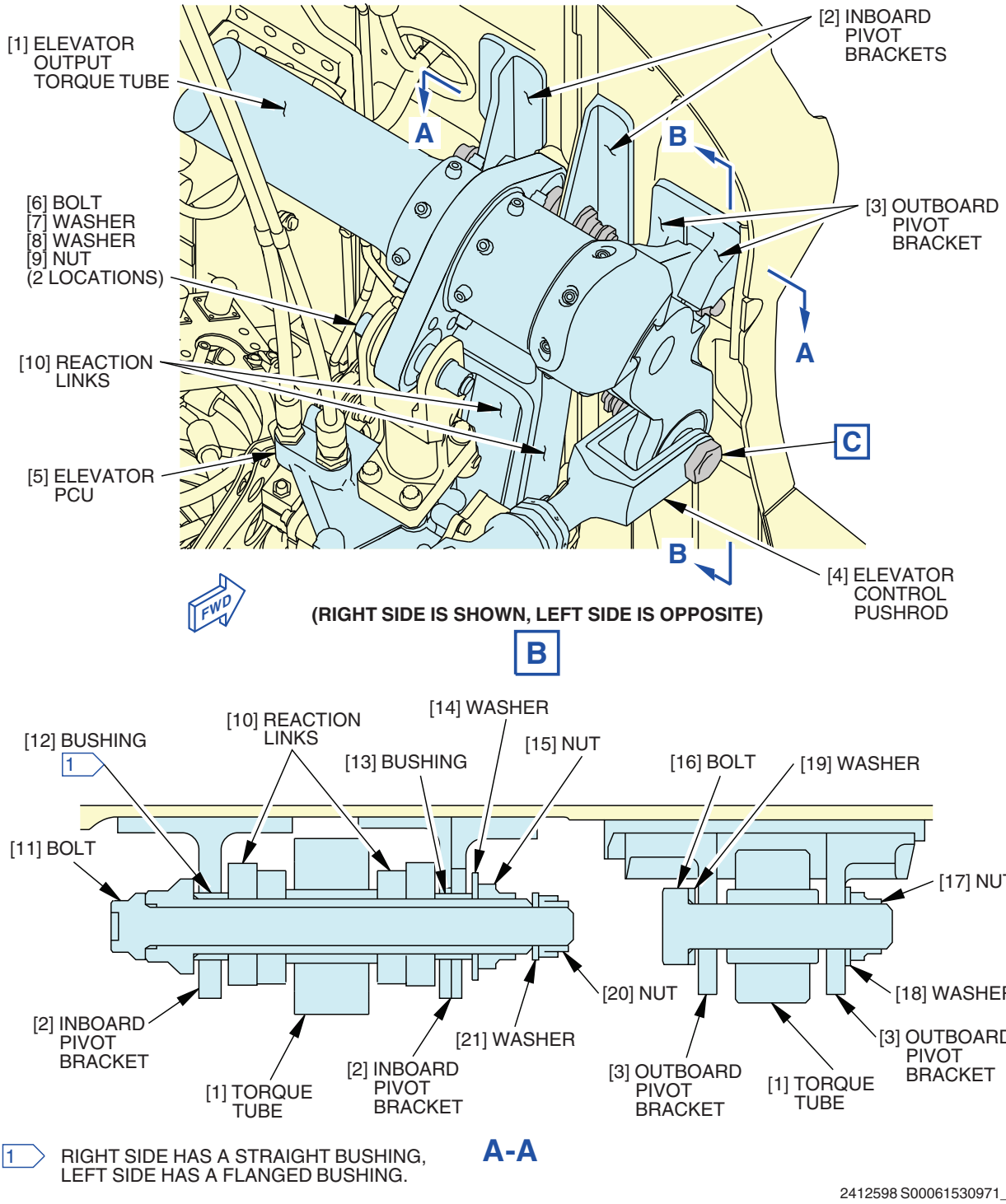
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



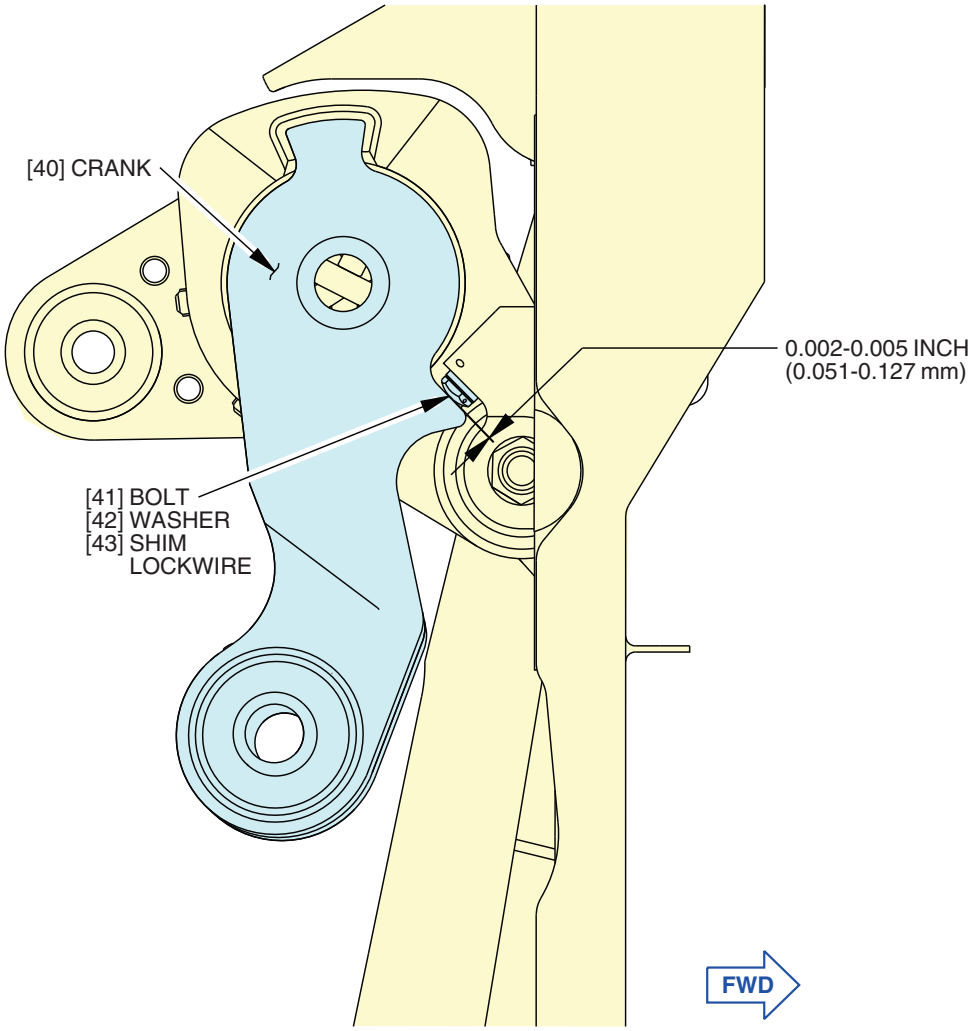
Elevator Output Torque Tube Installation
Figure 401/27-31-74-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Output Torque Tube Installation
Figure 401/27-31-74-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



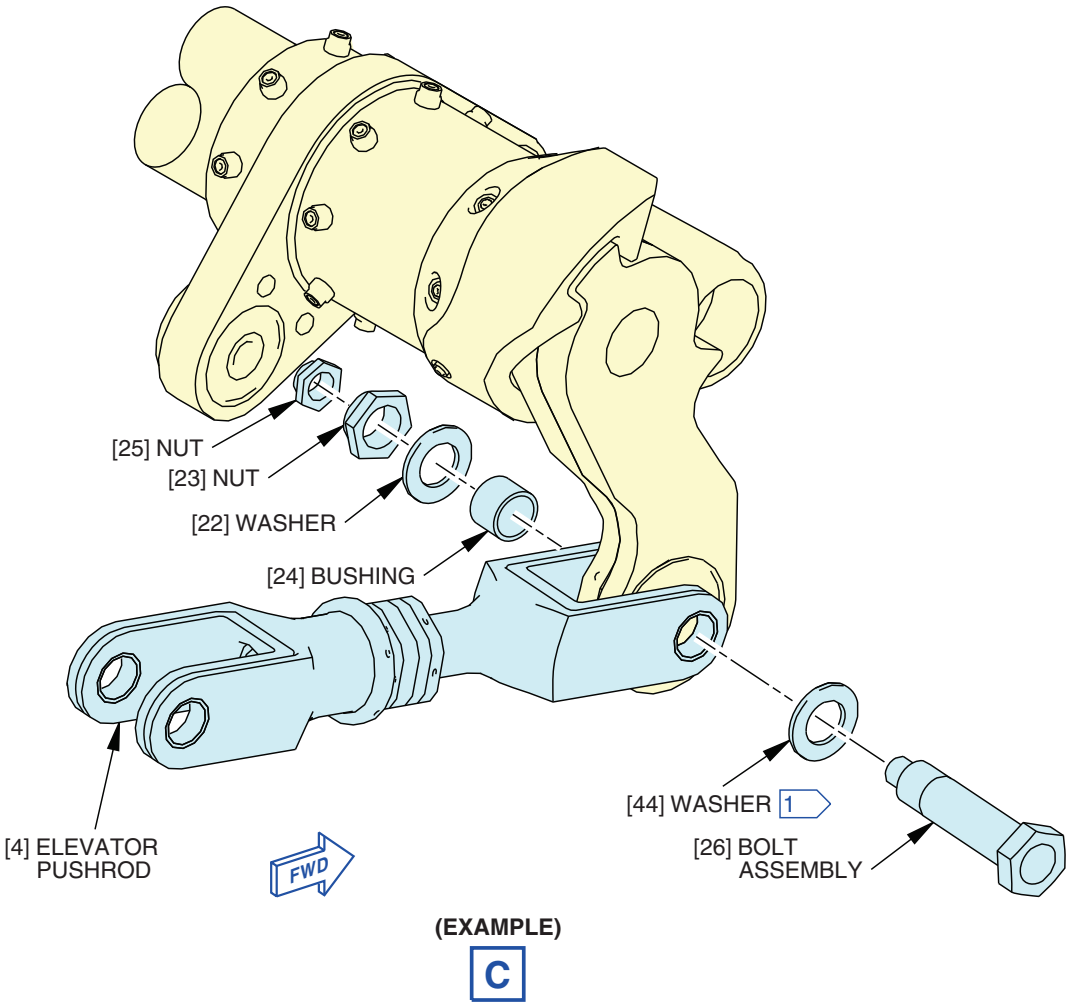
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)

B-B

3073205 S0000832614_V1

Elevator Output Torque Tube Installation
Figure 401/27-31-74-990-801 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

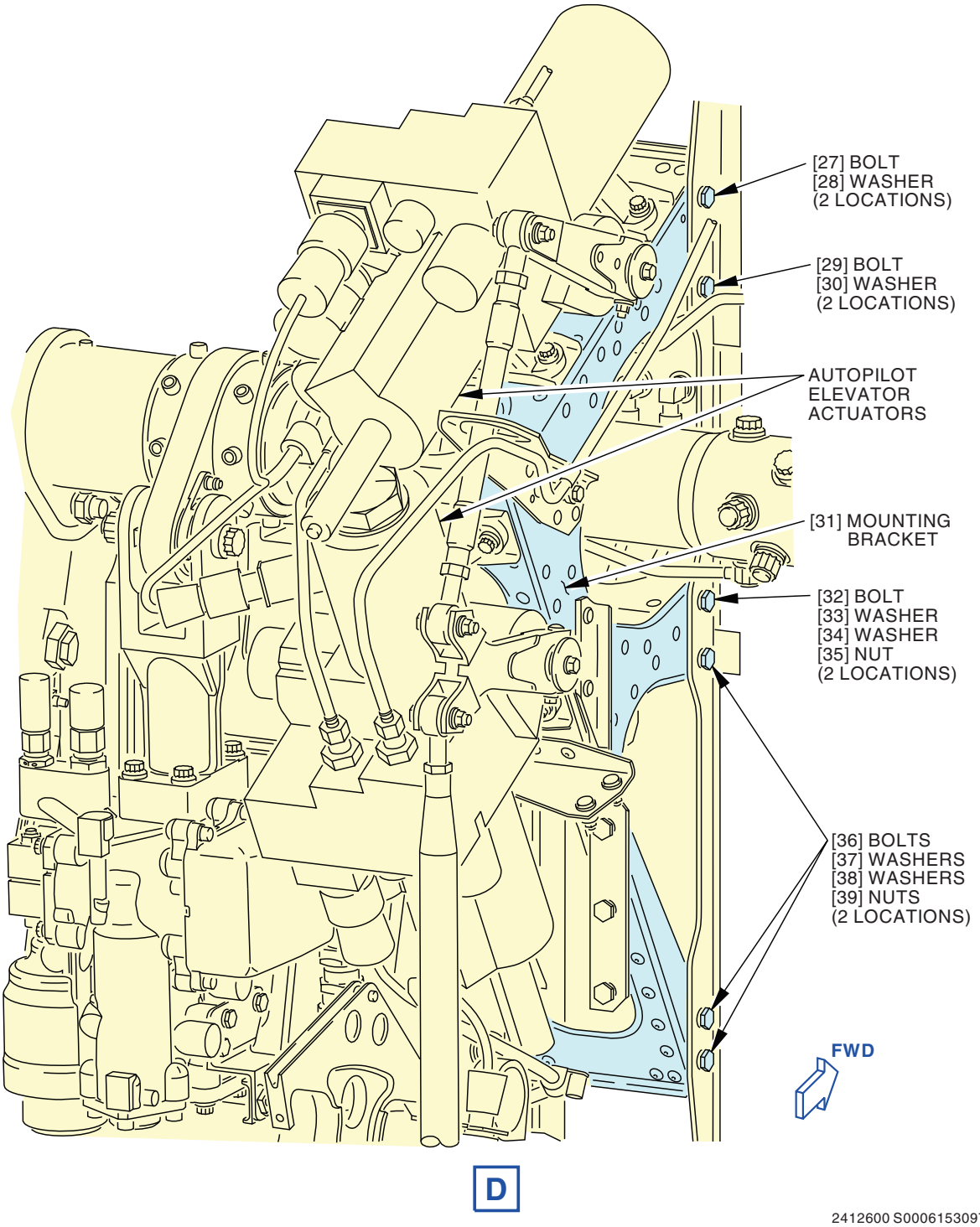


1 IF INSTALLED.

2412599 S00061530972_V5

Elevator Output Torque Tube Installation
Figure 401/27-31-74-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Elevator Output Torque Tube Installation
Figure 401/27-31-74-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-74-400-801

3. Elevator Output Torque Tube Installation

(Figure 401)

A. References

Reference	Title
22-11-26-400-801	Autopilot Elevator Actuator - Installation (P/B 401)
27-31-00-820-802	Elevator Control Pushrods - Adjustment (P/B 501)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)
53-53-00-400-801	Tailcone Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-405	Gauge - Feeler

C. Consumable Materials

Reference	Description	Specification
D00013	Grease - Aircraft And Instrument Grease	MIL-PRF-23827 (NATO G-354) (Supersedes MIL-G-23827)
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Output torque tube	27-31-00-17-110	PTF ALL

E. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

F. Elevator Output Torque Tube Installation

SUBTASK 27-31-74-420-001

(1) Install the output torque tube [1]:

(a) Lubricate the bushings, bolts, and bearings with grease, D00013.

(b) Put the output torque tube [1] in its position.

(c) Install the alignment pins to keep the bearing components in the inboard pivot clevis fittings of the output torque tube [1].

(d) Connect the output torque tube [1] to the outboard pivot brackets [3]:

1) At each of the output torque tube [1], install the bolts [16] with the bolt head inboard, washers [19], washers [18], and nuts [17].

a) Tighten the nuts [17] to 480 in-lb (54.2 N·m) - 790 in-lb (89.3 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

BE CAREFUL WHEN YOU INSTALL THE BOLTS [11]. YOU CAN CAUSE DAMAGE TO THE BEARING COMPONENTS IF YOU HIT THE BOLTS [11] DURING THE INSTALLATION.

(e) Connect the output torque tube [1] to the inboard pivot brackets [2] and reaction links [10]:

1) At each end of the output torque tube [1], install the bushing [12] (different on right and left sides), bushings [13], bolts [11], washers [14], and nuts [15].

a) Tighten the nuts [15] to 500 in-lb (56.5 N·m) - 600 in-lb (67.8 N·m).

CAUTION

DO NOT TIGHTEN THE NUT [15] AGAIN AFTER YOU TIGHTEN THE NUT [17]. YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

2) Install the washers [21] and nuts [20].

a) Tighten the nuts [20] to 150 in-lb (16.9 N·m) - 220 in-lb (24.9 N·m).

SUBTASK 27-31-74-420-007

(2) Install the elevator control pushrods [4] to the output torque tube [1]:

(a) Install the bolt assemblies [26] with the bolt heads facing outboard, bushings [24], washers [22], nuts [23], and nuts [25].

1) Do not tighten the nuts [23] and nuts [25].

NOTE: The nuts [23] and nuts [25] will be tightened after the adjustment.

SUBTASK 27-31-74-420-008

(3) Install the elevator PCUs [5] to the output torque tube [1]:

(a) Install the bolts [6], washers [7] (under the bolt), washers [8] (under the nut), and nuts [9].

SUBTASK 27-31-74-010-002

(4) Push down on the elevator trailing edge to fully extend the elevator PCUs [5].

SUBTASK 27-31-74-820-002

(5) Use the feeler gauge, STD-405, on both sides of the torque tube to measure the clearance between the crank [40] and head of the bolt [41].

(a) Make sure that the clearance value is 0.002 in. (0.051 mm) - 0.005 in. (0.127 mm).

1) If the clearance value does not align, do these steps:

2) Remove MS20995C32 lockwire, G01048.

3) Remove the bolt [41], washer [42], and shim [43].

4) Make sure that the bolt [41], washer [42], and shim [43] is not damaged.

5) Install the bolt [41], washer [42], and shim [43].

6) Make sure that the clearance value is 0.002 in. (0.051 mm) - 0.005 in. (0.127 mm).

a) If the clearance value is less than 0.002 in. (0.051 mm), adjust (peel) the shim [43] to get the clearance value in 0.002 in. (0.051 mm) - 0.005 in. (0.127 mm).

NOTE: If the clearance value is more than 0.005 in. (0.127 mm), it is acceptable to install an additional one shim.

7) Install MS20995C32 lockwire, G01048, on the bolts [41].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

8) Do these steps again for the opposite side.

SUBTASK 27-31-74-420-002

(6) Install the mounting bracket [31] for the autopilot elevator actuators:

(a) Put the mounting bracket [31] in position.

(b) Install bolts [27] and washers [28].

(c) Install bolts [29] and washers [30].

(d) Install the bolts [32], washers [33] (under the bolt), washers [34] (under the nut), and nuts [35].

(e) Install the bolts [36], washers [37] (under the bolt), washers [38] (under the nut), and nuts [39].

SUBTASK 27-31-74-420-003

(7) Install the autopilot elevator actuators on the mounting bracket [31] (TASK 22-11-26-400-801).

SUBTASK 27-31-74-080-001

(8) Remove rig pin E-5 of the rigging pin kit, SPL-1585, from the elevator aft quadrant.

SUBTASK 27-31-74-420-009

(9) Do this task: Tailcone Installation, TASK 53-53-00-400-801.

SUBTASK 27-31-74-820-001

(10) Adjust the elevator control pushrods (TASK 27-31-00-820-802).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-74-860-003

(1) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELEVATOR INDEX PLATE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of an elevator index plate

(2) An installation of an elevator index plate.

B. This procedure is applicable for the left and right elevator index plates.

C. You must do this procedure when the tail cone is removed and a new or different tail cone is installed.

TASK 27-31-81-000-801

2. Elevator Index Plate Removal

(Figure 401)

A. General

(1) This task includes the steps to remove the elevator index plate [1] from a removed tail cone.

B. Location Zones

Zone	Area
315	APU Compartment - Left
316	APU Compartment - Right

C. Elevator Index Plate Removal

SUBTASK 27-31-81-020-001

(1) Do these steps to remove the elevator index plate [1]:

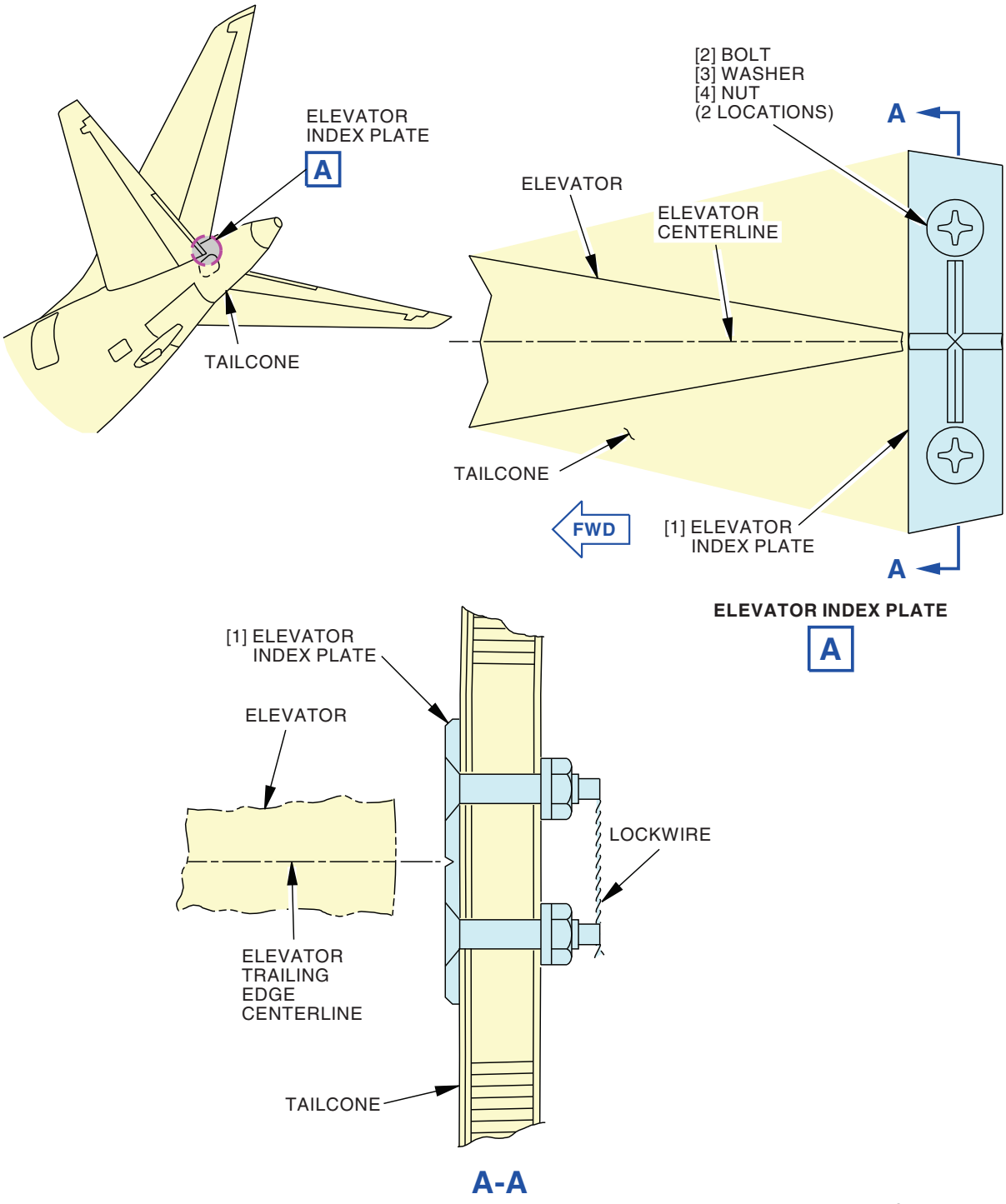
(a) Remove the lockwire.

(b) Remove the nuts [4], washers [3], and bolts [2] from the elevator index plate [1].

(c) Remove the elevator index plate [1] from the tail cone.

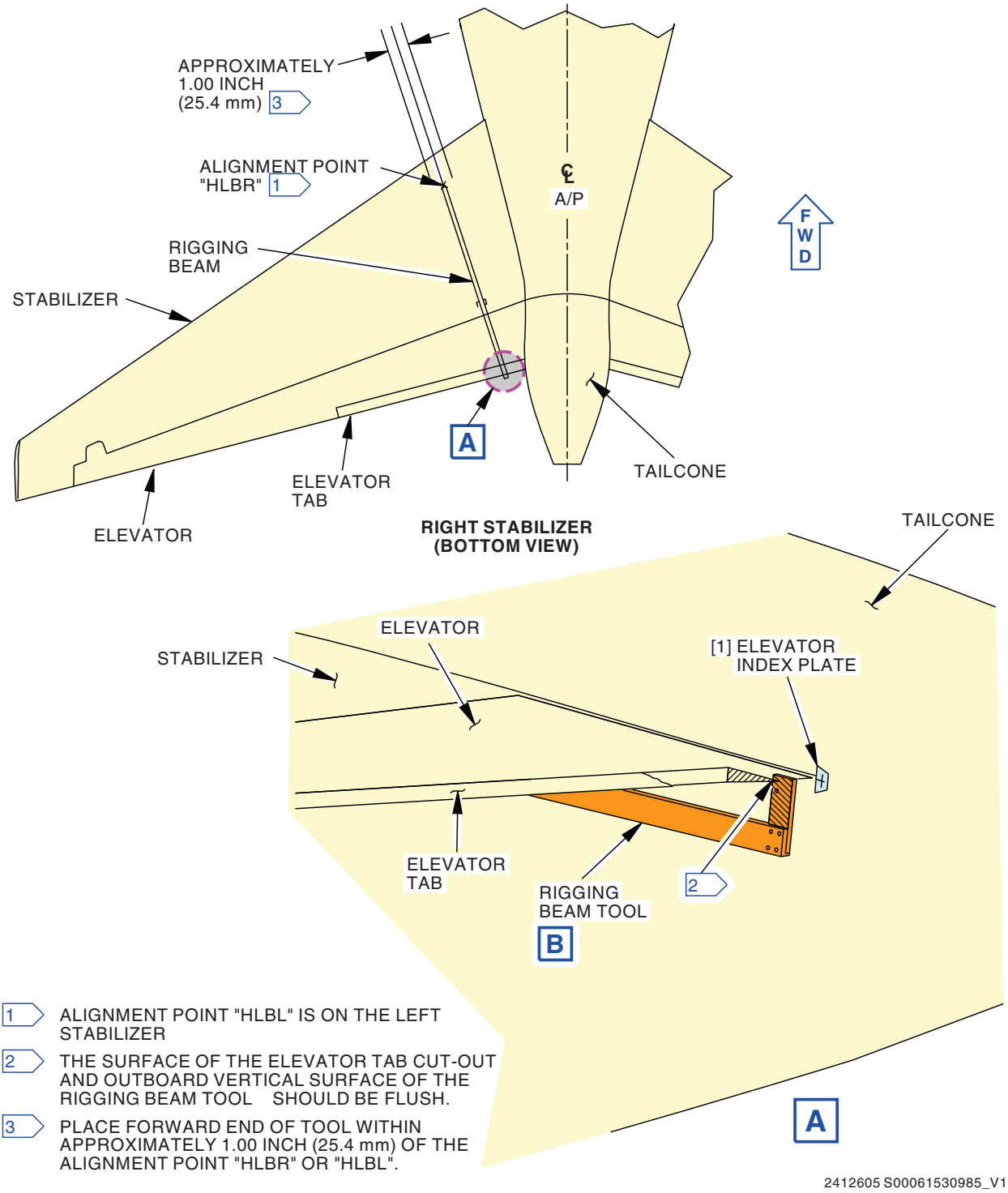
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



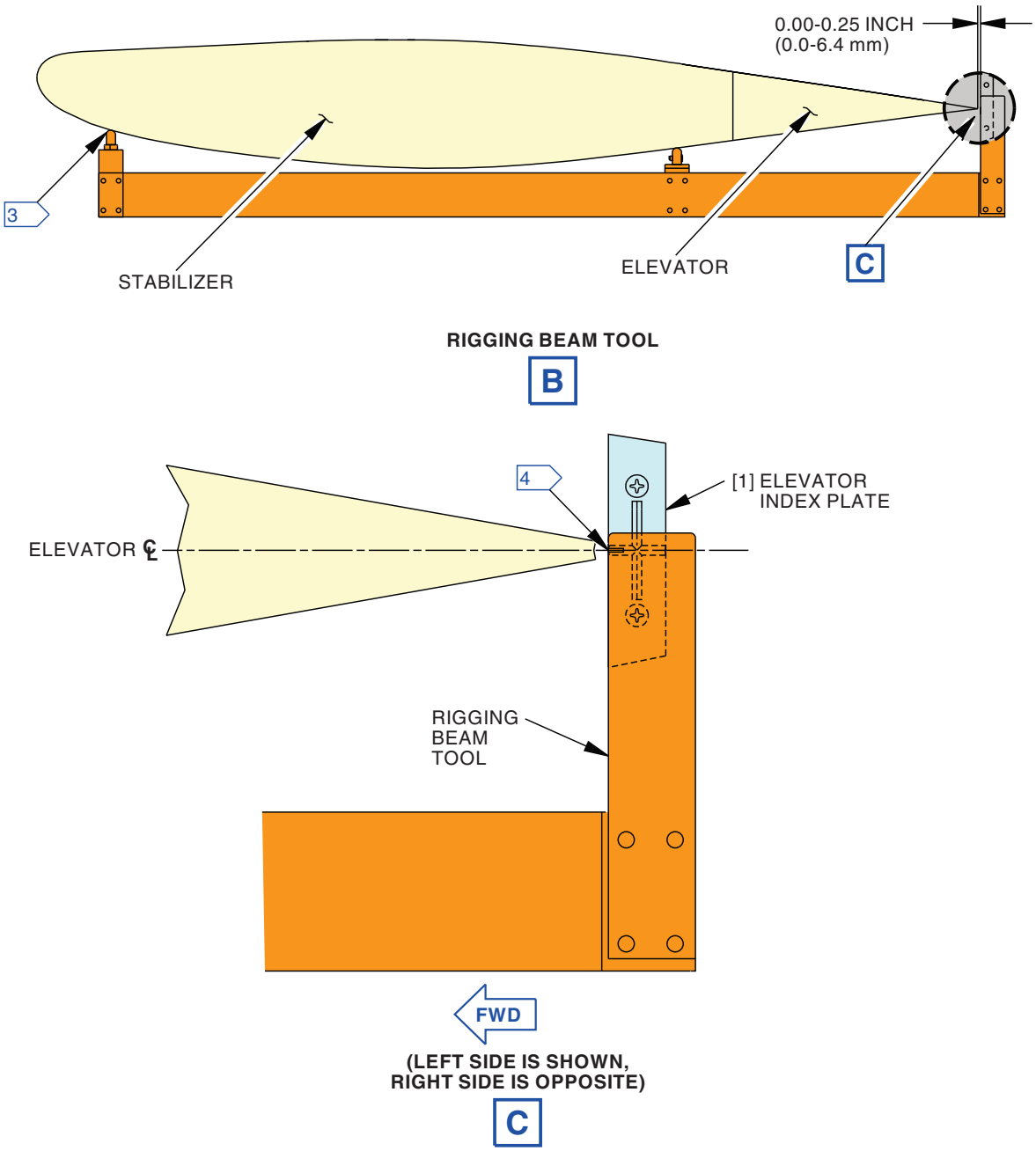
Elevator Index Plate Installation
Figure 401/27-31-81-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rigging Beam Installation
Figure 402/27-31-81-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

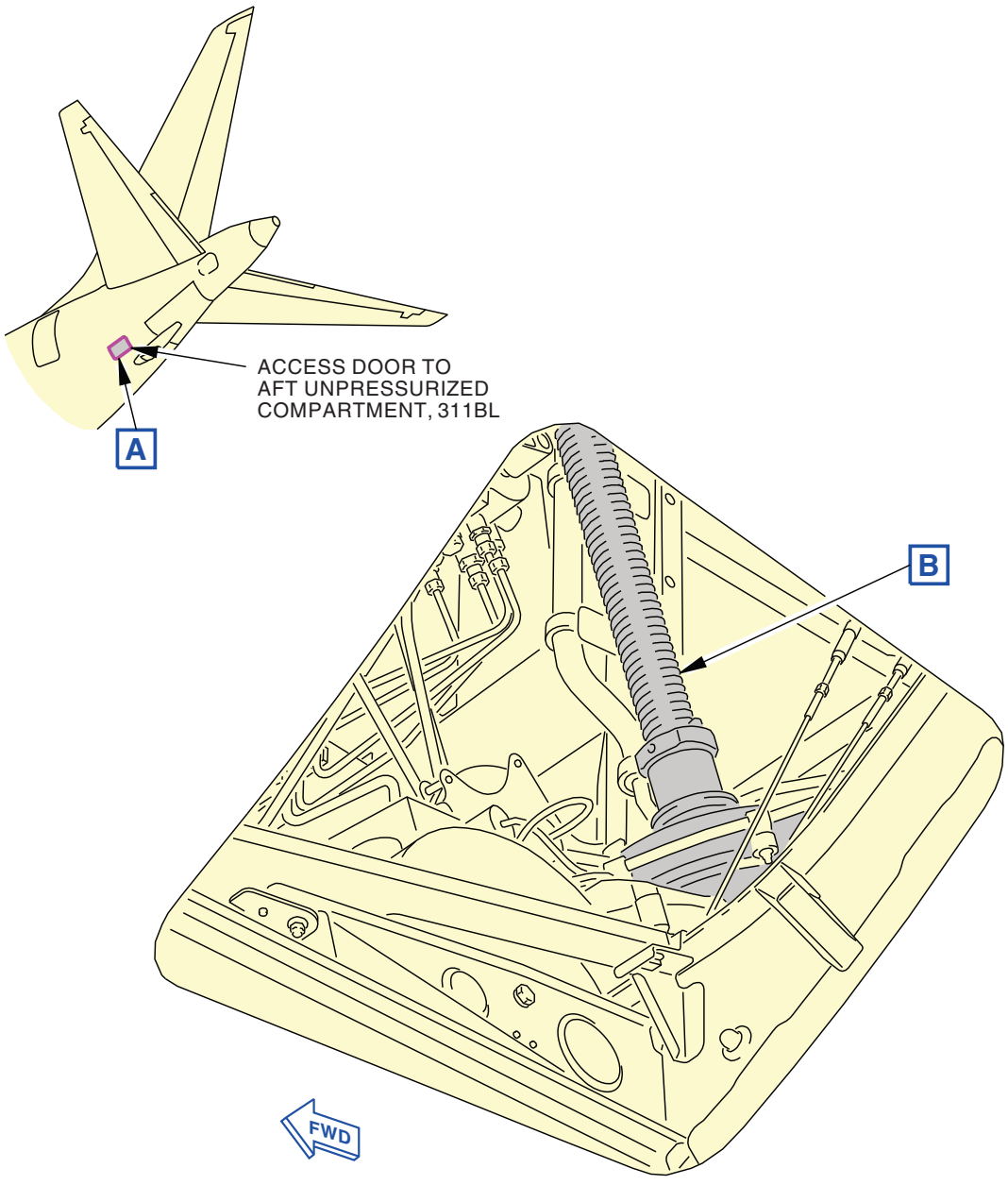


4 THE ELEVATOR CENTER LINE, THE RIGGING BEAM INDEX MARK, AND THE CENTER OF THE ELEVATOR INDEX PLATE GROOVE SHOULD ALL BE IN THE SAME PLANE FOR PERFECT ALIGNMENT OF THE ELEVATOR INDEX PLATE.

2412606 S00061530986_V1

Rigging Beam Installation
Figure 402/27-31-81-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

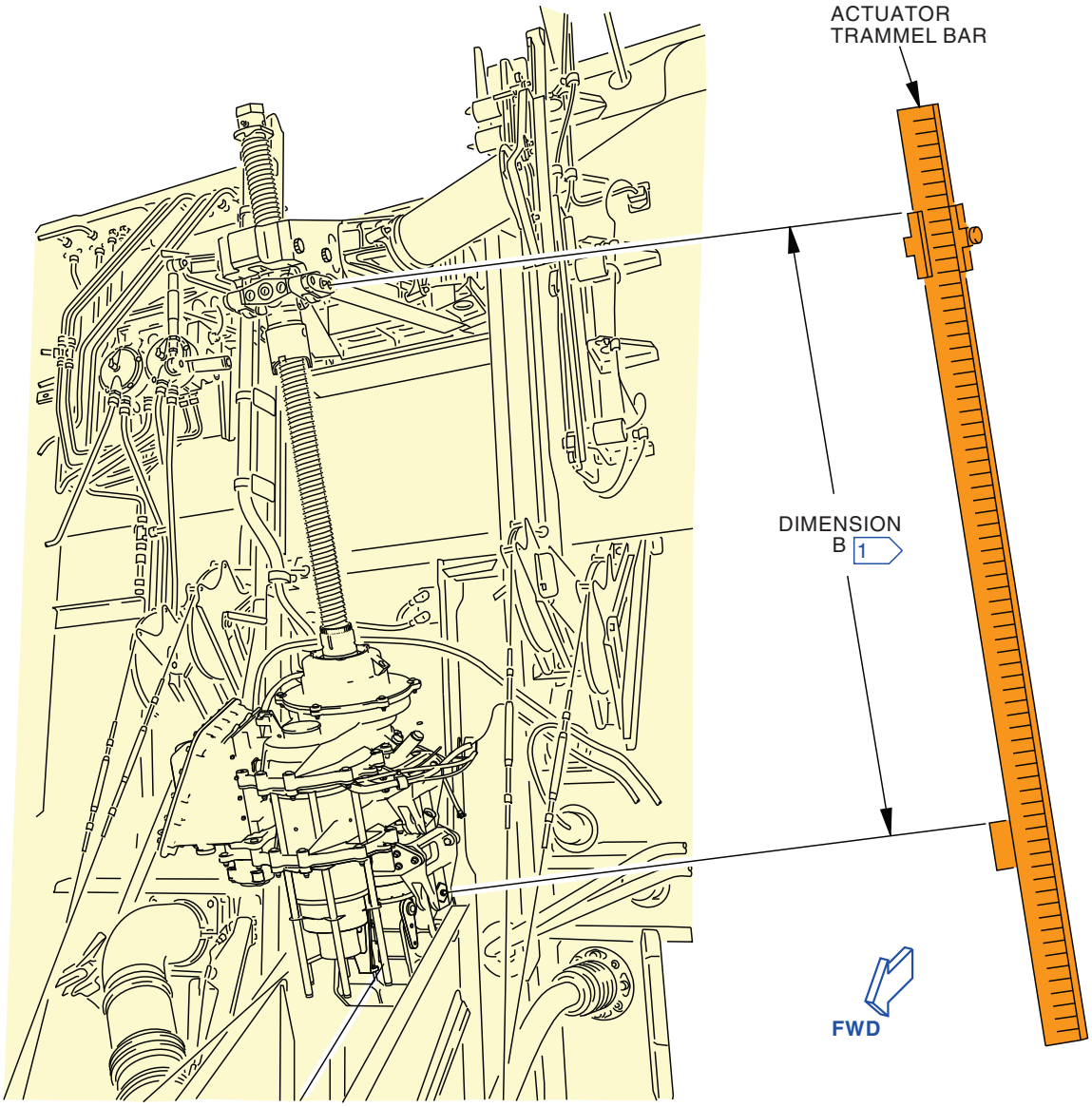


VIEW THROUGH THE ACCESS DOOR TO
AFT UNPRESSURIZED COMPARTMENT, 311BL
A

2412602 S00061530981_V1

Stabilizer Trim Jackscrew Adjustment
Figure 403/27-31-81-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACTUATOR

B

NOTE:

THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

2412603 S00061530982_V1

Stabilizer Trim Jackscrew Adjustment
Figure 403/27-31-81-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-31-81-400-801

3. Elevator Index Plate Installation

(Figure 401, Figure 402, Figure 403)

A. General

(1) This task includes the steps to install the elevator index plate [1] to a new or different installed tail cone.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-802	Pressure from the Elevator Hydraulic Systems A and B Removal (P/B 201)
27-31-00-840-802	Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-1730	Beam - Rigging, Elevator Index Plate Installation Part #: C27092-1 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Elevator index plate	27-31-81-01-005	PTF ALL

F. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
315	APU Compartment - Left
316	APU Compartment - Right

G. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door
318BR	Tailcone Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Prepare for the Installation

SUBTASK 27-31-81-010-001

(1) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-81-010-002

(2) Remove this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-81-480-001

CAUTION MAKE SURE TO PREVENT DAMAGE TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CHAF OR CAN CAUSE DAMAGE TO THE WIRES OR BUNDLE INSULATION.

(3) Use the trammel bar, SPL-1677, to set the B dimension at 39.17 ±0.01 in. (994.92 ±0.26 mm). (Figure 403).

SUBTASK 27-31-81-860-001

(4) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-31-81-860-002

(5) Do this task: Pressure from the Elevator Hydraulic Systems A and B Removal, TASK 27-31-00-800-802.

SUBTASK 27-31-81-480-002

(6) Attach DO NOT OPERATE tags, STD-858, to the control columns.

SUBTASK 27-31-81-480-003

(7) Use the beam, SPL-1730, to adjust the elevator trailing edge (Figure 402):

(a) Set the beam, SPL-1730, on the stabilizer lower surface.

(b) Set the elevator trailing edge to the index mark on the beam, SPL-1730.

(c) Remove the beam, SPL-1730.

I. Elevator Index Plate Installation

(Figure 401)

SUBTASK 27-31-81-420-001

(1) Do these steps to install the elevator index plate [1]:

(a) Align the center mark of the elevator index plate [1] with the elevator trailing edge.

(b) Drill two 0.190 +0.004 / -0.000 in. (4.826 +0.102 / -0.000 mm) diameter holes through the tail cone and the elevator index plate [1].

(c) Install the bolts [2], washers [3], and nuts [4] to attach the elevator index plate [1] to the tail cone.

(d) Install the MS20995NC32 lockwire, G01912.

SUBTASK 27-31-81-080-001

(2) Remove the DO NOT OPERATE tags, STD-858, from the control columns.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-31-81-210-001

(3) Make sure that the elevator trailing edge aligns with the center mark of the elevator index plate [1].

J. Put the Airplane Back to Its Usual Condition

SUBTASK 27-31-81-410-001

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-31-81-410-002

(2) Install this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 27-31-81-860-003

(3) Do this task: Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal, TASK 27-31-00-840-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details